

IBM Z NetView
Version 6 Release 3

Command Reference
Volume 2 (O – Z)



Note

Before using this information and the product it supports, read the information in [“Notices” on page 623.](#)

This edition applies to version 6, release 3 of IBM Z NetView (product number 5697-NV6) and to all subsequent versions, releases, and modifications until otherwise indicated in new editions.

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About this publication

The IBM Z® NetView® product provides advanced capabilities that you can use to maintain the highest degree of availability of your complex, multi-platform, multi-vendor networks and systems from a single point of control. This publication, the *IBM Z NetView Command Reference Volume 2 (O-Z)*, describes the commands and components of the NetView program that can be used for network operation. You can use many of these commands in command lists or command procedures. For more detailed information about specific functions, see the NetView management console online help.

Intended audience

This publication is for operators and system programmers. Before using this publication, you should be familiar with the basic functions described in the *IBM Z NetView User's Guide: NetView*.

Publications

This section lists publications in the IBM Z NetView library and related documents. It also describes how to access NetView publications online and how to order NetView publications.

IBM Z NetView library

The following documents are available in the IBM Z NetView library:

- *Administration Reference*, SC27-2869, describes the NetView program definition statements required for system administration.
- *Application Programmer's Guide*, SC27-2870, describes the NetView program-to-program interface (PPI) and how to use the NetView application programming interfaces (APIs).
- *Automation Guide*, SC27-2846, describes how to use automated operations to improve system and network efficiency and operator productivity.
- *Command Reference Volume 1 (A-N)*, SC27-2847, and *Command Reference Volume 2 (O-Z)*, SC27-2848, describe the NetView commands, which can be used for network and system operation and in command lists and command procedures.
- *Installation: Configuring Additional Components*, GC27-2851, describes how to configure NetView functions beyond the base functions.
- *Installation: Configuring the NetView Enterprise Management Agent*, GC27-2853, describes how to install and configure the IBM Z NetView Enterprise Management Agent.
- *Installation: Getting Started*, GI11-9443, describes how to install and configure the base NetView program.
- *Installation: Migration Guide*, GC27-2854, describes the new functions that are provided by the current release of the NetView product and the migration of the base functions from a previous release.
- *IP Management*, SC27-2855, describes how to use the NetView product to manage IP networks.
- *Messages and Codes Volume 1 (AAU-DSI)*, GC27-2856, and *Messages and Codes Volume 2 (DUI-IHS)*, GC27-2857, describe the messages for the NetView product, the NetView abend codes, the sense codes that are included in NetView messages, and generic alert code points.
- *Programming: Pipes*, SC27-2859, describes how to use the NetView pipelines to customize a NetView installation.
- *Programming: REXX and the NetView Command List Language*, SC27-2861, describes how to write command lists for the NetView product using the Restructured Extended Executor language (REXX) or the NetView command list language.

- *Security Reference*, SC27-2863, describes how to implement authorization checking for the NetView environment.
- *Troubleshooting Guide*, GC27-2865, provides information about documenting, diagnosing, and solving problems that occur in the NetView product.
- *Tuning Guide*, SC27-2874, provides tuning information to help achieve certain performance goals for the NetView product and the network environment.
- *User's Guide: Automated Operations Network*, SC27-2866, describes how to use the NetView Automated Operations Network (AON) component, which provides event-driven network automation, to improve system and network efficiency. It also describes how to tailor and extend the automated operations capabilities of the AON component.
- *User's Guide: NetView*, SC27-2867, describes how to use the NetView product to manage complex, multivendor networks and systems from a single point.
- *User's Guide: NetView Enterprise Management Agent*, SC27-2876, describes how to use the NetView Enterprise Management Agent.
- *Using Tivoli System Automation for GDPS/PPRC HyperSwap Manager with NetView*, GI11-4704, provides information about the Tivoli® System Automation for GDPS®/PPRC HyperSwap® Manager with NetView feature, which supports the GDPS and Peer-to-Peer Remote Copy (PPRC) HyperSwap Manager services offering.
- *Licensed Program Specifications*, GC31-8848, provides the license information for the NetView product.
- *Program Directory for IBM Z NetView US English*, GI11-9444, contains information about the material and procedures that are associated with installing the NetView product.
- *Program Directory for IBM Z NetView Japanese*, GI11-9445, contains information about the material and procedures that are associated with installing the NetView product.
- *Program Directory for IBM Z NetView Enterprise Management Agent*, GI11-9446, contains information about the material and procedures that are associated with installing the IBM Z NetView Enterprise Management Agent.

The following books are archived:

- *Customization Guide*, SC27-2849, describes how to customize the NetView product and points to sources of related information.
- *Data Model Reference*, SC27-2850, provides information about the Graphic Monitor Facility host subsystem (GMFHS), SNA topology manager, and MultiSystem Manager data models.
- *Installation: Configuring Graphical Components*, GC27-2852, describes how to install and configure the NetView graphics components.
- *Programming: Assembler*, SC27-2858, describes how to write exit routines, command processors, and subtasks for the NetView product using assembler language.
- *Programming: PL/I and C*, SC27-2860, describes how to write command processors and installation exit routines for the NetView product using PL/I or C.
- *Resource Object Data Manager and GMFHS Programmer's Guide*, SC27-2862, describes the NetView Resource Object Data Manager (RODM), including how to define your non-SNA network to RODM and use RODM for network automation and for application programming.
- *SNA Topology Manager Implementation Guide*, SC27-2864, describes planning for and implementing the NetView SNA topology manager, which can be used to manage subarea, Advanced Peer-to-Peer Networking, and TN3270 resources.
- *User's Guide: NetView Management Console*, SC27-2868, provides information about the NetView management console interface of the NetView product.

Related publications

You can find additional product information on the IBM Z NetView web site at <https://www.ibm.com/us-en/marketplace/ibm-tivoli-netview-for-zos>.

For information about the NetView Bridge function, see *Tivoli NetView for OS/390 Bridge Implementation*, SC31-8238-03 (available only in the V1R4 library).

Terminology in this Library

The following terms are used in this library:

CNMCMD

For the CNMCMD member and the members that are included in it using the %INCLUDE statement

CNMSTYLE

For the CNMSTYLE member and the members that are included in it using the %INCLUDE statement

DSIOPF

For the DSIOPF member and the members that are included in it using the %INCLUDE statement

IBM® Tivoli Netcool®/OMNIbus

For either of these products:

- IBM Tivoli Netcool/OMNIbus
- IBM Tivoli OMNIbus and Network Manager

MVS™

For z/OS® operating systems

MVS element

For the base control program (BCP) element of the z/OS operating system

NetView

For the following products:

- IBM Z NetView version 6 release 3
- IBM Tivoli NetView for z/OS version 6 release 2 modification 1
- NetView releases that are no longer supported

PARMLIB

For SYS1.PARMLIB and other data sets in the concatenation sequence

VTAM®

For Communications Server - SNA Services

Unless otherwise indicated, topics to programs indicate the latest version and release of the programs. If only a version is indicated, the topic is to all releases within that version.

When a topic is made about using a personal computer or workstation, any programmable workstation can be used.

Using IBM Z NetView online help

The following types of IBM Z NetView mainframe online help are available, depending on your installation and configuration:

- General help and component information
- Command help
- Message help
- Sense code information
- Recommended actions

Accessing publications online

IBM posts publications for this and all other products, as they become available and whenever they are updated, to the IBM Knowledge Center at <https://www.ibm.com/support/knowledgecenter>. You can find IBM Z NetView documentation on [IBM Z NetView Knowledge Center](#).

Note: If you print PDF documents on other than letter-sized paper, set the option in the **Print** window that enables Adobe Reader to print letter-sized pages on your local paper.

Ordering publications

You can order many Tivoli publications online at <http://www.ibm.com/e-business/linkweb/publications/servlet/pbi.wss>

You can also order by telephone by calling one of these numbers:

- In the United States: 800-426-4968
- In Canada: 800-879-2755

In other countries, contact your software account representative to order Tivoli publications. To locate the telephone number of your local representative, perform the following steps:

1. Go to <http://www.ibm.com/e-business/linkweb/publications/servlet/pbi.wss>.
2. Select your country from the list and click the grey arrow button beside the list.
3. Click **About this site** to see an information page that includes the telephone number of your local representative.

Accessibility

Accessibility features help users with a physical disability, such as restricted mobility or limited vision, to use software products successfully. Standard shortcut and accelerator keys are used by the product and are documented by the operating system. Refer to the documentation provided by your operating system for more information.

For additional information, see the Accessibility appendix in the *User's Guide: NetView*.

Tivoli user groups

Tivoli user groups are independent, user-run membership organizations that provide Tivoli users with information to assist them in the implementation of Tivoli Software solutions. Through these groups, members can share information and learn from the knowledge and experience of other Tivoli users.

Support information

If you have a problem with your IBM software, you want to resolve it quickly. IBM provides the following ways for you to obtain the support you need:

Online

Please follow the instructions located in the support guide entry: <https://www.ibm.com/support/home/pages/support-guide/?product=4429363>.

Troubleshooting information

For more information about resolving problems with the IBM Z NetView product, see the *IBM Z NetView Troubleshooting Guide*. You can also discuss technical issues about the IBM Z NetView product through the NetView user group located at <https://groups.io/g/NetView>. This user group is for IBM Z NetView customers only, and registration is required. This forum is also monitored by interested parties within IBM who answer questions and provide guidance about the NetView product. When a problem with the code is found, you are asked to open an official case to obtain resolution.

Conventions used in this publication

This section describes the conventions that are used in this publication.

Typeface conventions

This publication uses the following typeface conventions:

Bold

- Lowercase commands and mixed case commands that are otherwise difficult to distinguish from surrounding text
- Interface controls (check boxes, push buttons, radio buttons, spin buttons, fields, folders, icons, list boxes, items inside list boxes, multicolumn lists, containers, menu choices, menu names, tabs, property sheets), labels (such as **Tip:**, and **Operating system considerations:**)
- Keywords and parameters in text

Italic

- Citations (examples: titles of publications, diskettes, and CDs)
- Words defined in text (example: a nonswitched line is called a *point-to-point line*)
- Emphasis of words and letters (words as words example: "Use the word *that* to introduce a restrictive clause."; letters as letters example: "The LUN address must start with the letter *L*.")
- New terms in text (except in a definition list): a *view* is a frame in a workspace that contains data.
- Variables and values you must provide: ... where *myname* represents...

Monospace

- Examples and code examples
- File names, programming keywords, and other elements that are difficult to distinguish from surrounding text
- Message text and prompts addressed to the user
- Text that the user must type
- Values for arguments or command options

Operating system-dependent variables and paths

For workstation components, this publication uses the UNIX convention for specifying environment variables and for directory notation.

When using the Windows command line, replace *\$variable* with *%variable%* for environment variables and replace each forward slash (/) with a backslash (\) in directory paths. The names of environment variables are not always the same in the Windows and UNIX environments. For example, %TEMP% in Windows environments is equivalent to \$TMPDIR in UNIX environments.

Note: If you are using the bash shell on a Windows system, you can use the UNIX conventions.

Syntax diagrams

The following syntax elements are shown in syntax diagrams. Read syntax diagrams from left-to-right, top-to-bottom, following the horizontal line (the main path).

- [“Symbols” on page xv](#)
- [“Parameters” on page xvi](#)
- [“Punctuation and parentheses” on page xvi](#)
- [“Abbreviations” on page xvii](#)

For examples of syntax, see [“Syntax examples” on page xvii](#).

Symbols

The following symbols are used in syntax diagrams:



Marks the beginning of the command syntax.



Marks the end of the command syntax.



Indicates that the command syntax is continued on the next line.



Indicates that a statement is continued from the previous line.



Marks the beginning and end of a fragment or part of the command syntax.

Parameters

The following types of parameters are used in syntax diagrams:

Required

Required parameters are shown on the main path.

Optional

Optional parameters are shown below the main path.

Default

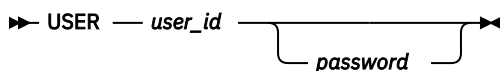
Default parameters are shown above the main path. In parameter descriptions, default parameters are underlined.

Syntax diagrams do not rely on highlighting, brackets, or braces. In syntax diagrams, the position of the elements relative to the main syntax line indicates whether an element is required, optional, or the default value.

When you issue a command, spaces are required between the parameters unless a different separator, such as a comma, is specified in the syntax.

Parameters are classified as keywords or variables. Keywords are shown in uppercase letters. Variables, which represent names or values that you supply, are shown in lowercase letters and are either italicized or, in NetView help, displayed in a differentiating color.

In the following example, the `USER` command is a keyword, the `user_id` parameter is a required variable, and the `password` parameter is an optional variable.



Punctuation and parentheses

You must include all punctuation that is shown in the syntax diagram, such as colons, semicolons, commas, minus signs, and both single and double quotation marks.

When an operand can have more than one value, the values are typically enclosed in parentheses and separated by commas. For a single value, the parentheses typically can be omitted. For more information, see [“Multiple operands or values” on page xviii](#).

If a command requires positional commas to separate keywords and variables, the commas are shown before the keywords or variables.

When examples of commands are shown, commas are also used to indicate the absence of a positional operand. For example, the second comma indicates that an optional operand is not being used:

```
COMMAND_NAME opt_variable_1,,opt_variable_3
```

You do not need to specify the trailing positional commas. Trailing positional and non-positional commas either are ignored or cause a command to be rejected. Restrictions for each command state whether trailing commas cause the command to be rejected.

Abbreviations

Command and keyword abbreviations are listed in synonym tables after each command description.

Syntax examples

The following examples show the different uses of syntax elements:

- “Required syntax elements” on page xvii
- “Optional syntax elements” on page xvii
- “Default keywords and values” on page xvii
- “Multiple operands or values” on page xviii
- “Syntax that is longer than one line” on page xviii
- “Syntax fragments” on page xviii

Required syntax elements

Required keywords and variables are shown on the main syntax line. You must code required keywords and variables.

➤ REQUIRED_KEYWORD — *required_variable* ➤

A required choice (two or more items) is shown in a vertical stack on the main path. The items are shown in alphanumeric order.

➤ REQUIRED_OPERAND_OR_VALUE_1
 REQUIRED_OPERAND_OR_VALUE_2 ➤

Optional syntax elements

Optional keywords and variables are shown below the main syntax line. You can choose not to code optional keywords and variables.

➤
 OPTIONAL_OPERAND ➤

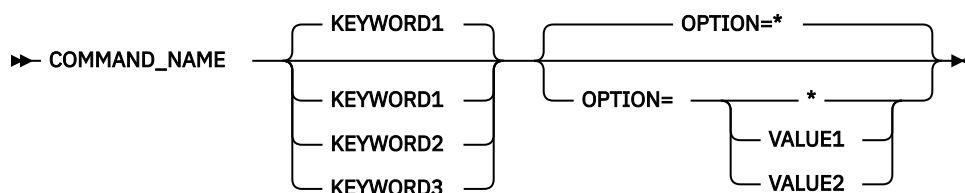
A required choice (two or more items) is shown in a vertical stack below the main path. The items are shown in alphanumeric order.

➤
 OPTIONAL_OPERAND_OR_VALUE_1
 OPTIONAL_OPERAND_OR_VALUE_2 ➤

Default keywords and values

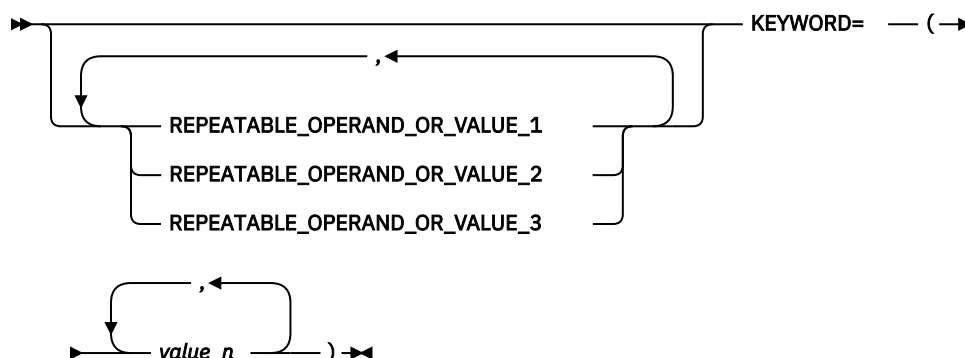
Default keywords and values are shown above the main syntax line in one of the following ways:

- A default keyword is shown only above the main syntax line. You can specify this keyword or allow it to default. The following syntax example shows the default keyword KEYWORD1 above the main syntax line and the rest of the optional keywords below the main syntax line.
- If an operand has a default value, the operand is shown both above and below the main syntax line. A value below the main syntax line indicates that if you specify the operand, you must also specify either the default value or another value shown. If you do not specify the operand, the default value above the main syntax line is used. The following syntax example shows the default values for operand OPTION=* above and below the main syntax line.



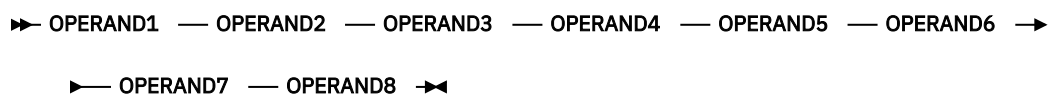
Multiple operands or values

An arrow returning to the left above a group of operands or values indicates that more than one can be selected or that a single one can be repeated.



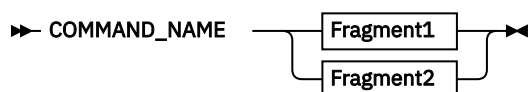
Syntax that is longer than one line

If a diagram is longer than one line, each line that is to be continued ends with a single arrowhead and the following line begins with a single arrowhead.



Syntax fragments

Some syntax diagrams contain syntax fragments, which are used for lengthy, complex, or repeated sections of syntax. Syntax fragments follow the main diagram. Each syntax fragment name is mixed case and is shown in the main diagram and in the heading of the fragment. The following syntax example shows a syntax diagram with two fragments that are identified as `Fragment1` and `Fragment2`.



Fragment1



Fragment2



Chapter 1. Using Commands

When using NetView commands, it is helpful to know that the NetView program is composed of the following components:

Automated Operations Network

Provides a set of programs that can be customized and extended to provide network automation.

Browse facility

Provides the capability to browse the network log or a data set member.

Command facility

Provides base service functions and automated operations. Included are the operator interface and network and trace logging facilities. The command facility can also operate as a subsystem of MVS.

The component identifier for the command facility is NCCF.

Event/Automation Service

Provides translation and forwarding services for alerts, messages, Event Integration Facility (EIF) events, and SNMP traps. The component identifier for the Event/Automation Service is EAS.

Graphic Monitor Facility host subsystem

Provides a link between RODM and the NetView management console to display Systems Network Architecture (SNA) and non-SNA resources.

The component identifier for the Graphic Monitor Facility host subsystem is GMFHS.

Hardware monitor

Provides information about physical network resources. This includes failure information that shows probable cause and suggested actions. The information can be grouped into the following categories:

- Events
- Statistics
- Alerts

Events are unusual conditions detected by a device about itself or on behalf of a device it controls. Events can be records of permanent errors and other warning and exception conditions. Statistics include information describing the number of transmissions and retransmissions for traffic on a line. An alert is an event that is considered critical and requires operator attention. Whether an event is important enough to be considered an alert is determined by a filter. This filtering decision is made using criteria set in your installation based on how you want to manage and control your network and what information the operators must see.

The component identifier for the hardware monitor is NPDA.

Help facility

Displays online help information for the NetView components, panels, messages, and commands. This facility includes a procedure that help desk personnel can use to help with problem determination. An index is provided for quick reference.

MultiSystem Manager

Simplifies the task of managing your network resources.

NetView management console

Monitors and graphically displays the resources that represent a network, a portion of a network, or a group of networks at various levels of detail.

The component identifier for the NetView management console is NMC.

Resource Object Data Manager

An object-oriented, high-speed, multithread in-memory data cache that provides application programs with a means of rapidly accessing or changing the status of data.

The component identifier for the Resource Object Data Manager is RODM.

Session monitor

Provides information about the logical network resources. This includes session-related information such as response time measurement and the components that make up a session. [Table 1 on page 2](#) shows how this information is grouped:

Table 1. Session Monitor Data Categories

Data Type	Data Description
Session awareness data	Information about session activity within the networks. This data identifies the partners of each session, which can be in the same domain, in different domains, or in different networks.
Session trace data	Consists of session activation parameters, VTAM path information unit (PIU) data, and network control program (NCP) data.
Session response time data	Measured response time broken down into ranges of time that are specified by the performance class definitions.
Route data	Includes a list of PUs and transmission groups (TGs) that make up the explicit route used by a session.
Network accounting and availability measurement data	Network availability data and distribution of use of network resources.

The component identifier for the session monitor is NLDM.

Status monitor

Displays the status of network resources in a hierarchical manner and enables you to browse the network log. This facility also automatically reactivates minor nodes except applications and cross-domain resources (CDRSC).

The component identifier for the status monitor is STATMON.

SNA Topology Manager

Performs dynamic collection and display of APPN, subarea, and logical unit (LU) topology and status. This topology and status are stored in RODM for use by the NetView management console.

The component identifier for the SNA Topology Manager is TOPOSNA.

The NetView program operates as a VTAM application or as a subsystem of MVS. It provides a network log to record information as necessary.

Table 2. Additional information

Topic	Refer to
Using commands	<i>IBM Z NetView User's Guide: NetView</i>
Changes to commands from previous releases	<i>IBM Z NetView Installation: Migration Guide</i>

Tasks

The NetView program can perform many functions. The NetView program controls these functions by defining units of work called *tasks*. The types of tasks are:

OST

Operator station task. There is one OST for each NetView operator. There are also automated OSTs (autotasks), which perform unattended operations functions related to automation. You can use OSTs to maintain the online sessions with the command facility terminal operators. The OST also analyzes commands as it receives them from the operator and invokes the appropriate command processors.

NNT

NetView-NetView task. When you have more than one NetView domain, each OST can have secondary sessions across domains with other NetView systems. The OST in the remote domain is called a NetView-NetView task (NNT). There is one NNT for each cross-domain NetView with which the local NetView domain communicates. This task controls communication with cross domain NetView programs to issue commands and receive responses. An NNT is not used by the RMTCMD command, which uses LU 6.2 to communicate and uses distributed automated OSTs. Use the RMTCMD for issuing cross domain commands, although NNTs are still supported.

PPT

Primary program operator interface task. This task processes commands and command lists that are performed on a system-level basis. The NetView program uses the PPT to carry out all timer management functions. It runs timer-initiated commands designed to run under the PPT, and it can expand and run command lists. The PPT also routes unsolicited VTAM messages to the authorized receiver.

DST

Data services task. This task processes requests for communication network management (CNM) or Virtual Storage Access Method (VSAM) data. This is the interface to network management data and VTAM.

MNT

Main task. The NetView main task loads and attaches other NetView tasks.

HCT

Hardcopy task. The hardcopy task logs messages received from or sent to a specified operator station. The HCT uses a specified printer to accomplish this.

OPT

Optional task. Optional tasks are user-defined subtasks that can provide increased flexibility beyond the subtasks that the NetView program provides.

Command Types

The NetView program processes different types of commands, including:

- Regular commands
- Immediate commands
- Data services task commands

Most commands and all command lists are **regular commands**. Regular commands can run concurrently with other regular commands. Regular commands can be interrupted by system routines or by immediate commands.

Immediate commands, such as RESET, GO, and AUTOWRAP, can interrupt or preempt regular commands. As their name implies, they run as soon as you enter the command. Only one immediate command runs at a time.

Data services commands run under a data services task (DST). These commands are internal to the NetView program. You cannot enter them at your terminal.

Some commands can run as either regular or immediate commands. If you enter a command at your terminal, it is treated as an immediate command. If the command is in a command list, it is treated as a regular command.

Some commands can run as either immediate or data services commands. If you enter a command at your terminal, it is treated as an immediate command. If the command is run under a DST, it is treated as a data services command.

Command Priority

A NetView OST, PPT, NNT, or autotask's command priority determines the order in which NetView commands and command lists are processed. You can set or modify command priority with the NetView DEFAULTS or OVERRIDE commands, or override it for one command with the CMD command.

Whether a command is entered by the operator or sent to the task, the command is queued on the message queue of that task.

Any command entered at a terminal or sent using an EXCMD command from another task is queued to the target task's message queue corresponding to the target task's command priority.

Additional Information

Topic	Refer to
Command priority	<i>IBM Z NetView User's Guide: NetView</i>

Entering Commands

You can enter NetView commands in a wide variety of ways, for example from the MVS console, the web command interface, the NetView management console, the Z NetView Enterprise Management Agent, and from REXX, TSO, and UNIX.

From the MVS console, you can use either the MVS MODIFY command or prefix the command with the NetView designator character.

Table 3. Additional Information

Topic	Refer to
Entering commands from the MVS console	<i>IBM Z NetView User's Guide: NetView</i>

Restricting Access to Resources and Commands

Both the commands that you can issue and the resources you can access are set for your operator ID. One such restriction is called *span of control*. Span of control restricts your control to select network resources. The span of control for which you are authorized is defined in either your operator profile or in an SAF product, depending on the method used for security authorization.

Another restriction is called *command authorization*. Command authorization restricts the use of commands, keywords, and values. This command authorization is defined in either the DSIPARM data set or in an SAF product.

Table 4. Additional Information

Topic	Refer to
Restricting command access	<i>IBM Z NetView Security Reference</i>
List of commands, keywords, and values that can be restricted	<i>IBM Z NetView Security Reference</i>

Syntax Conventions Used in This Document

The commands are listed in alphabetic order for easy reference. The NetView component, the equivalent NetView command list (if any), and any operating system restrictions are provided for each command. If

more than one component name is listed next to the command name, you can use the command in more than one NetView component.

The formats and operands of the NetView commands and command lists are described in the same notation throughout the document. Each command description includes the format and description of operands and, where applicable, restrictions, examples, and responses. The command syntax and examples shown assume that they are being entered from within the appropriate component.

Chapter 2. NetView commands and command descriptions

This chapter describes the formats of NetView commands and command lists. You can enter these commands from the command facility or from any other NetView component.

The commands are listed in alphabetic order. Each command description includes the format and description of operands and, where applicable, usage notes, responses, and examples.

To get online help for a specific NetView component, enter:

```
HELP component
```

Where *component* is the name of the NetView component. The possible values for *component* are as follows:

AON

Automated Operations Network components

BROWSE

Browse facility

EAS

Event/Automation Service

GMFHS

Graphic Monitor Facility host subsystem

HELP

Online help

MSM

MultiSystem Manager components

NCCF

Command facility

NLDM

Session monitor

NPDA

Hardware monitor

PIPE

PIPE command and its stages

RODM

Resource Object Data Manager

STATMON

Status monitor

TOPOSNA

SNA topology manager

WINDOW

Full screen application

For online help on a specific command, enter:

```
HELP command
```

Where *command* is the name of the command.

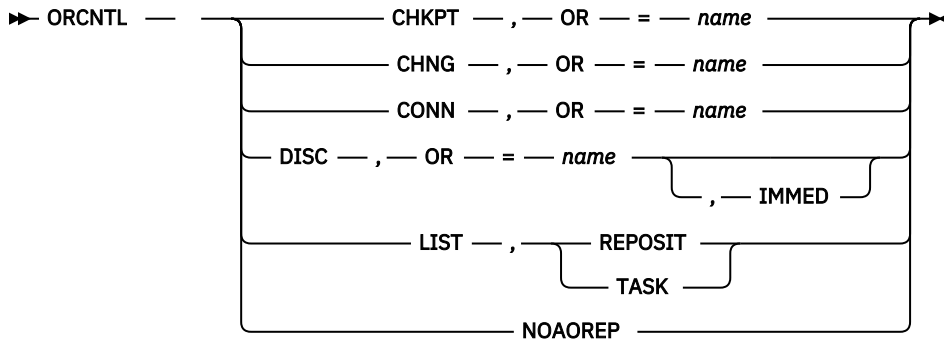
For online help on messages, enter:

Where *msgid* is the identifier of the NetView message for which a help panel is to be displayed.

ORCNTL (RODM)

Syntax

ORCNTL



IBM-Defined Synonyms

Command or Operand	Synonym
ORCNTL	ORC

Purpose of Command

The ORCNTL command assists in managing the RODM data caches and autotasks under the control of DSIQTSK.

This command also enables you to:

- Disconnect from a RODM
- Take a checkpoint on a RODM
- Change the current runtime RODM
- List the status of autotasks under the control of DSIQTSK
- List the status of all RODMs under the control of the DSIQTSK subtask

See the *IBM Z NetView Administration Reference* for information about DSIQTSKI.

Operand Descriptions

CHKPT

Indicates to take a checkpoint of the specified RODM. CHKPT stores RODM information to DASD storage.

OR=*name*

Specifies the name of the RODM on which you want to act. You must specify this operand with all keywords except LIST and NOAOREP.

CHNG

Changes the current runtime automation RODM to the one you specify in the command. The current runtime automation RODM is the default that automation applications access when using CNMQAPI, DSINOR, or ORCONV.

See *IBM Z NetView Programming: PL/I and C* or *IBM Z NetView Programming: Assembler* for more information.

If you defined an initialization command processor for RODM in DSIQTSKI, see the *IBM Z NetView Administration Reference* for more information.

CONN

Indicates to connect the specified RODM.

If you defined an initialization command processor for a RODM in DSIQTSKI, this keyword starts the command processor with a notice indicating that this RODM is now connected. However, if you issue an ORCNTL CONN command when you are already connected to the specified RODM, any defined initialization command is not processed. See the *IBM Z NetView Administration Reference* for information about DSIQTSKI.

DISC

Indicates to disconnect the specified RODM. If you do not specify IMMED with this keyword, a checkpoint occurs. If the RODM you are disconnecting is the current runtime RODM, there is not a default RODM. Automation using this RODM is effectively turned off; that is, DSINOR, CNMQAPI, and ORCONV requests that are using the default RODM fail.

IMMED

Specifies that DSIQTSK disconnects from the specified RODM without a checkpoint. IMMED is an optional operand.

LIST

Lists the state of all RODMs or autotasks defined to DSIQTSK, the current runtime RODM, and the NetView program-to-program interface receiver. The valid operands are:

REPOSIT

Lists all RODMs defined to DSIQTSK. The list also includes whether a connection exists to the DSIQTSK subtask and whether a checkpoint is currently in progress.

You can specify this operand as REPOSIT or REP.

TASK

Lists all the autotasks and operator station tasks defined to DSIQTSK. The list also includes whether a RODM is using the task.

NOAOREP

Specifies that you do not want a runtime automation RODM. The current runtime automation RODM is the default that automation applications access when using CNMQAPI, DSINOR, or ORCONV. You can use this keyword to stop automation processing when errors are occurring on RODM. This keyword does not cause RODM to be disconnected. Therefore, you can still access RODM to correct the errors.

Example: Taking a checkpoint of a specified RODM

To take a checkpoint of the RODM RODM1, enter:

```
ORCNTL CHKPT,OR=RODM1
```

Response

You receive a response from the DSIQTSK subtask similar to the following:

```
DW0718I DSIQTSK THE REQUEST TO CHECKPOINT THE RODM
        RODM1 HAS BEEN RECEIVED
DW0702I DSIQTSK A CHECKPOINT HAS BEEN ISSUED TO THE RODM RODM1
DW0731I DSIQTSK A CHECKPOINT HAS BEEN COMPLETED ON THE RODM RODM1
```

Example: Listing the status of databases maintained by the DSIQTSK task

To list the status of the databases maintained by the DSIQTSK subtask, enter:

```
ORCNTL LIST,REP
```

Response

You receive a response similar to:

```
DW0735I DSIQTSK RCVR = ACTIVE, AUTO REP = RODM1
DW0736I DSIQTSK RODM( RODM1 ) STATE( CONNECT ) CHKPT( ACTIVE )
DW0736I DSIQTSK RODM( RODM2 ) STATE( DISC ) CHKPT( INACTIVE)
DW0738I DSIQTSK END OF DISPLAY REQUEST.
```

The meanings of the operands that are displayed are:

RCVR

Indicates whether the status of the program-to-program interface subsystem is active or inactive, or whether a write-to-operator command (WTOR) was issued. If the status is INACTIVE, no commands from the RODMs are processed. If the status is WTOR, a WTOR was issued because no tasks were available to which to issue a command. When WTOR is outstanding, ORCNTL CHNG commands are processed, but no initialization commands are run. Attempts to connect to a RODM fail.

AUTO REP

Indicates the name of the current runtime RODM.

Note: The name of each RODM defined to the DSIQTSK subtask is listed, along with its current checkpoint status.

Example: Listing the status of tasks controlled by the DSIQTSK subtask

To list the status of the tasks controlled by the DSIQTSK subtask, enter:

```
ORCNTL LIST,TASK
```

Response

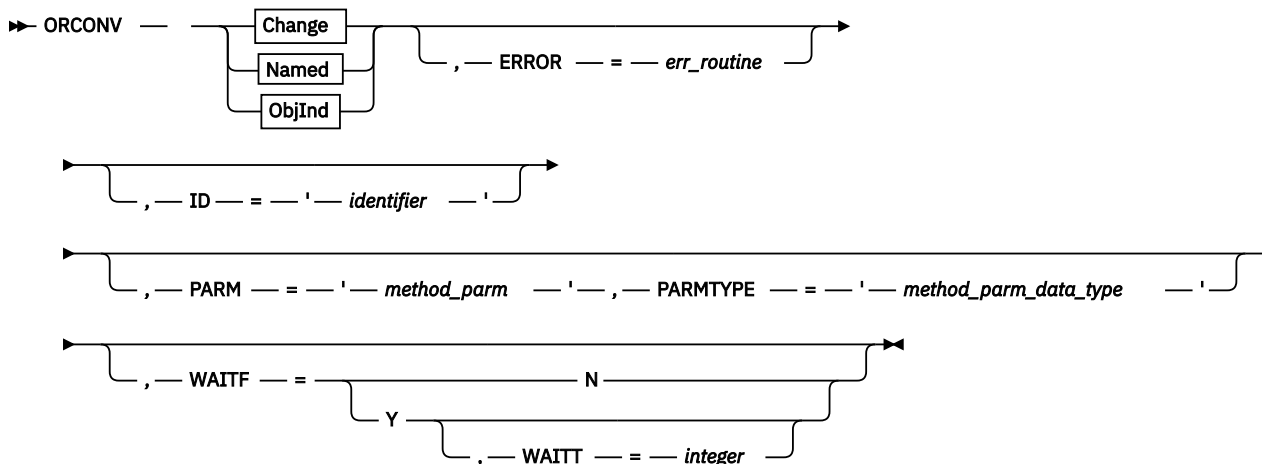
The response received from the DSIQTSK subtask is:

```
DW0735I DSIQTSK RCVR = ACTIVE, AUTO REP = RODM1
DW0737I DSIQTSK TASK( AUTO1 ) STATE( INUSE )
DW0737I DSIQTSK TASK( AUTO2 ) STATE( INUSE )
DW0738I DSIQTSK END OF DISPLAY REQUEST.
```

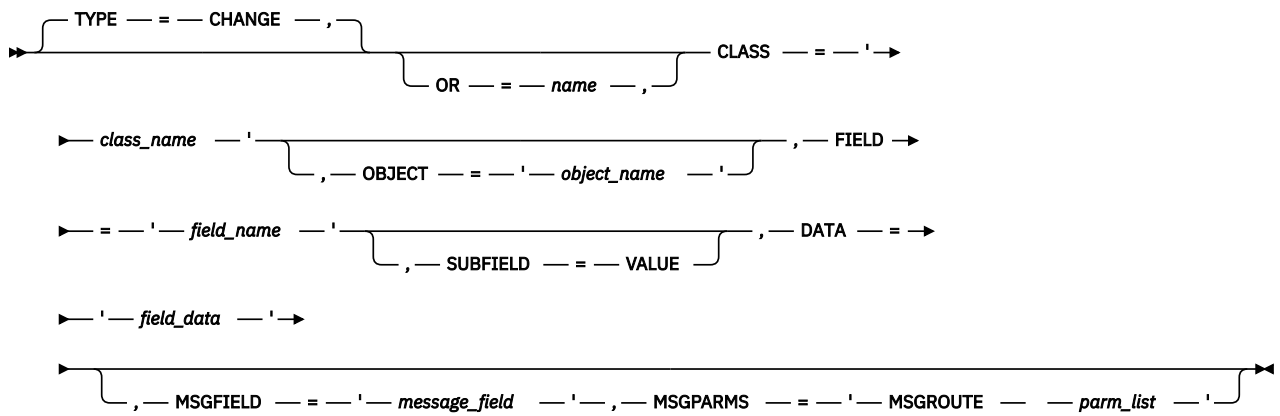
ORCONV (RODM)

Syntax

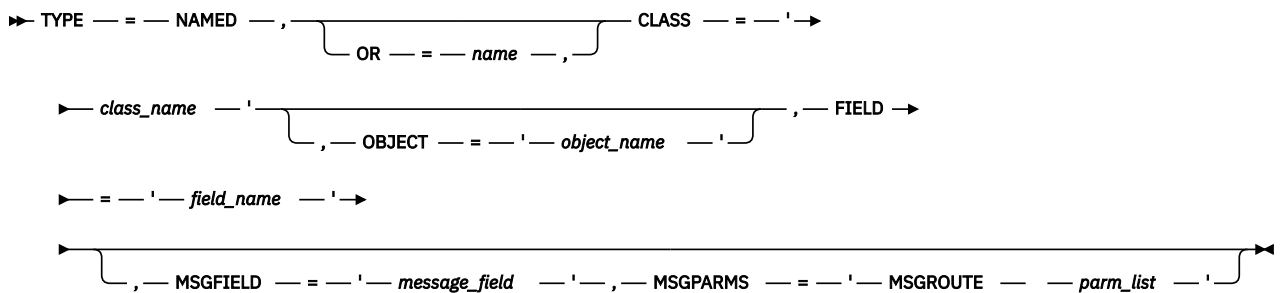
ORCONV



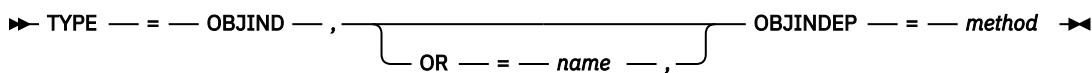
Change



Named



ObjInd



Purpose of Command

The ORCONV command changes fields and invokes methods in the RODM from the following sources:

- The NetView automation table
- Command lists
- The command facility
- Procedures written in the following languages:
 - REXX
 - PL/I
 - C

The ORCONV command can be used only with RODMs managed by the DSIQTSK task. See *IBM Z NetView Installation: Configuring Additional Components* for more information about the DSIQTSK subtask.

Operand Descriptions

TYPE

Indicates the type of RODM request. The valid values are:

CHANGE

Specifies that an object or class field change is performed. CHANGE is the default.

NAMED

Specifies that a named method is invoked.

OBJIND

Specifies that an object independent method is invoked.

OR=name

Specifies the name of the RODM on which you want to act. The default is the current runtime RODM. The *name* can be in the range of 1–8 bytes. See the description of the REP statement in the *IBM Z NetView Administration Reference* for information about setting the current runtime RODM.

CLASS='class_name'

Specifies the name of the class in RODM. The *classname* can be in the range of 1–64 bytes. See the *IBM Z NetView Resource Object Data Manager and GMFHS Programmer's Guide* for more information.

OBJECT='object_name'

Specifies the name of the RODM object that is to be altered. The *object_name* can be in the range of 1–254 bytes. See the *IBM Z NetView Resource Object Data Manager and GMFHS Programmer's Guide* for more information.

FIELD='field_name'

Specifies the name of the RODM object or class field on which you want to act. If you specify TYPE=NAMED, the field contains the name of the method to be invoked. The *field_name* can be in the range of 1–64 bytes. See the *IBM Z NetView Resource Object Data Manager and GMFHS Programmer's Guide* for more information.

If TYPE=CHANGE (default), the field that you specify must be one of the following data types:

- INTEGER
- SMALL INTEGER
- FLOATING POINT
- CHARACTER

SUBFIELD=VALUE

Specifies the subfield to be updated. You can use SUBFIELD to bypass the invocation of a change method for the specified field. If you do not specify SUBFIELD, the change method is started. If you specify SUBFIELD=VALUE, the method is not started. See the *IBM Z NetView Resource Object Data Manager and GMFHS Programmer's Guide* for more information.

DATA='field_data'

Specifies the data to be stored in the field you specified. The *field_data* name can be in the range of 1–254 bytes.

ID='identifier'

A user-defined string of characters that is logged in the network log when the ORCONV command is issued. You can define unique strings for each invocation of the ORCONV command to aid in debugging. The *identifier* name can be in the range of 1–254 bytes.

MSGFIELD='message_field'

Specifies the RODM field to be queried to determine whether to route a message. MSGFIELD must be of type *integer* or *smallint*. ORCONV checks the specified field in RODM and compares it to the DEFAULTS SENDMSG list. If a match is found, the message that drove the ORCONV command is routed to the operator IDs you specified in the MSGPARMS parameter list. If the ORCONV request specified by the TYPE keyword fails or the query of the field specified by the MSGFIELD parameter fails, the message is still routed. This precaution prevents the loss of an important message. If you specify MSGFIELD, MSGPARMS is required.

See the *IBM Z NetView Automation Guide* for a message-routing scenario. For information about setting the SENDMSG list, see the DEFAULTS command.

MSGPARMS='MSGROUTE parm_list'

Specifies the MSGROUTE parameter list used to route the message that drove ORCONV from the NetView automation table. If you specify MSGPARMS, also specify MSGFIELD.

Note: The field to be acted on by the ORCONV command, and the field to be queried for message routing by the ORCONV command must be in the same RODM class or object. The field to be queried for message routing must be *integer* or *smallint* and cannot be inherited.

OBJINDEP=method

Specifies the name of the object-independent method you want to start. The *method* name can be in the range of 1–8 bytes.

ERROR=err_routine

Specifies the name of a NetView application (command list or command processor) that is started if one of the following errors occurs:

- The specified object is not found.
- The specified field is not found.
- The type of data specified to be stored is not valid. This can occur when translation from character format to binary format takes place. If truncation of character data occurs, the event is logged in the error log.

The input to the started error routine is the input supplied to the ORCONV command processor, along with a return code. The *err_routine* name can be in the range of 1–8 bytes.

The return codes used with this keyword are:

Return Code**Meaning**

4	Syntax error
6	Truncation of data has occurred
8	RODM error
10	Translation binary/floating point failure
12	RODM is not available
16	Specified an ERROR command that is not valid
18	NetView program internal macro failure
20	Data type of MSGFIELD is not <i>integer</i> or <i>smallint</i>
22	The value of MSGFIELD is inherited
24	Message routing failed

PARM='method_parm'

Specifies the data to be passed to the started named method, change method, or object independent method. The *method_parm* name can be in the range of 1–254 bytes.

PARMTYPE='method_parm_data_type'

Specifies the data type of the parameter being passed to the started name method, change method, or object independent method. Valid values of the PARMTYPE parameter are INTEGER, SMALLINT, CHARVAR, and FLOAT.

WAITF=N|Y

Specifies whether to wait for a checkpoint completion.

N

Indicates that you do not want to wait for a checkpoint to complete if one is in progress. If a checkpoint is in progress, the request fails.

Y

Indicates that a checkpoint is to complete if one is in progress. If you specify Y, also specify WAITT to indicate the duration of the wait. If WAITF=Y and you did not specify the WAITT keyword, the wait time defaults to the time specified in the initialization file of DQIQTsk by the keyword T.

WAITT=integer

Specifies the amount of time to wait if a checkpoint is in progress. This operand is valid only if you specify WAITF=Y. The allowable range for *integer* is 10–3600 seconds (1 hour). If you specify a value greater than 3600, 3600 is used.

Restrictions

The following restrictions apply to the ORCONV command:

- You can specify the keywords in any order and you must specify only those keywords that are applicable to the request.
- If a keyword equal sign is followed directly by a comma, or you did not specify a keyword, it is a syntax error and the following rules apply:
 - For the keywords CLASS, OBJECT, ERROR, FIELD, OBJINDEP, and PARM, the value is set to null.
 - For the DATA keyword, the following apply:
 - If the field to be updated is defined as a character field, the value is set to blanks.
 - If the field to be updated is binary or floating point, the value is set to 0.
 - If you issue ORCONV from the NetView command facility, the command is converted to uppercase. This limits the RODM classes, fields, objects, and other elements that can be affected to only those that are entered in the RODM data cache in uppercase letters.

Example: Updating the status of a job

This example updates the status of a job, JOB1, to indicate an abend. This example assumes that RODM is customized to contain a class named JOB that has an object called JOB1 and a field called STATE with data type SMALLINT (FIXED BIN(15,0)).

Message trapped:

```
IEF450I JOB1 JOB1 - ABEND=S222 U0000 REASON=00000000
```

```
IF MSGID='IEF450I' & TOKEN(2)=JOBNAME THEN
  EXEC(CMD('ORCONV CLASS='JOB',OBJECT='' JOBNAME
    ',FIELD='STATE',DATA='256'))
  ROUTE(ONE *));
```

Example: Updating a field without method invocation

This example updates a field without method invocation. You can update a field in RODM without starting change and notification methods by specifying SUBFIELD=VALUE in the ORCONV command. In this example, an error routine, ERRROUT, and ID=JOBABEND are added. The update is performed on the current runtime RODM.

```
IF MSGID='IEF450I' & TOKEN(2)=JOBNAME THEN
  EXEC(CMD('ORCONV CLASS='JOB',OBJECT='' JOBNAME
    ',FIELD='STATE',SUBFIELD='VALUE'
    ',DATA='256',ERROR=ERRROUT, ID='JOBABEND'))
  ROUTE(ONE *));
```

Example: Starting an object-independent method

This example shows how you can start an object-independent method by specifying the name of the object-independent method using the OBJINDEP keyword. You can specify optional parameters with the

PARM keyword. Instead of updating RODM directly, you can trigger an object-independent method called VERIFY and pass it the jobname JOB1 to determine whether to recover this job. The command is as follows:

```
IF MSGID='IEF450I' & TOKEN(2)=JOBNAME THEN
  EXEC(CMD('ORCONV TYPE=OBJIND,OBJINDEP=VERIFY,PARM='' JOBNAME '''))
  ROUTE(ONE *));
```

Example: Starting a named method

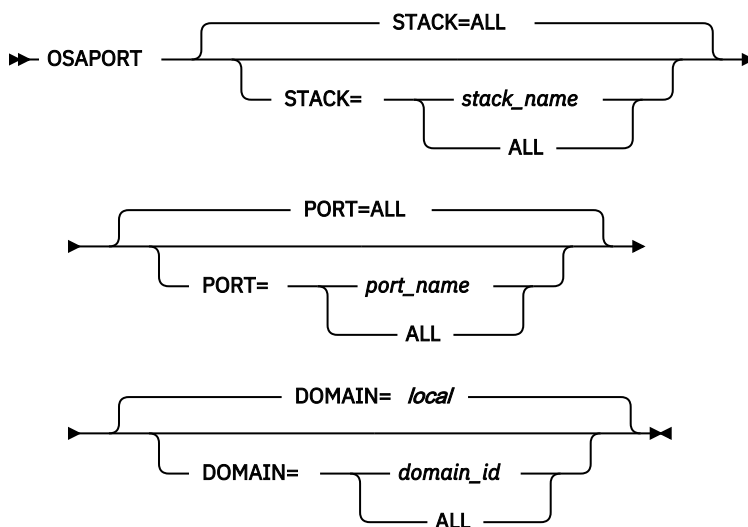
This example shows you how to start a named method. The difference between an object-independent method and a named method is that a named method is associated with a particular class or object and is referenced using a field name. The name of the method to be started is in the VERIFY field. The command is as follows:

```
IF MSGID='IEF450I' & TOKEN(2)=JOBNAME THEN
  EXEC(CMD('ORCONV TYPE=NAMED,CLASS='JOB',OBJECT='' JOBNAME
    ',FIELD='VERIFY'))
  ROUTE(ONE *));
```

OSAPORT (NCCF; CNME8236)

Syntax

OSAPORT



Purpose of Command

You can use the OSAPORT command to view OSA channel and port information from a 3270 console or from the Tivoli Enterprise Portal using the Z NetView Enterprise Management Agent,

OSA channel and port data is returned in the multilined BNH597I message. For the format of the data returned by the BNH597I message, see the online help for that command.

Note: This command is intended to be used as a REXX interface. For user interfaces, use the CNMSOSAP sample, the Tivoli Enterprise Portal, or the NetView management console.

Operand Descriptions

DOMAIN

The domain to which the request is sent. This keyword can have the following values:

ALL

Specifies all domains in the sysplex. This option can be issued only from the master NetView program.

Note: If you have many stacks or systems in your sysplex, issuing ALL can cause slow response times and high processor utilization.

domain_id

Specifies the ID of a specific domain. This option can be issued only from the master NetView program if the domain is not the local domain.

local

If the DOMAIN keyword is not specified, the domain name of the local NetView system is used. This is the default.

PORT

Specifies the port name assigned to the port on an IBM Open Systems Adapter. The default value is ALL, which indicates that data is requested from all OSA ports that are known to the specified stack.

STACK

Specifies the name of the stack for which data is requested. The default value is ALL, which indicates that data is requested from all stacks that are known to the specified domain.

Return Codes**Return Code****Meaning****0**

The command was processed successfully.

2

Help was issued.

3

No data is available to display.

4

The command syntax is in error, or the command contains parameters that are not valid.

5

A REXX Novalue error occurred.

6

A REXX syntax error occurred.

7

An internal command failed.

8

The DISCOVERY tower is disabled.

9

A signal halt occurred.

Usage Notes

- If message DSI047E is received, contact your system programmer to enable the appropriate tower or subtower. For information about data collection and display towers and subtowers, see *IBM Z NetView Installation: Configuring Additional Components*.

OUTPUT (EAS)

Syntax

EAS OUTPUT

➤ MODIFY — — *procname* — , — OUTPUT — — — — — TO — = — *destination* ➤

IBM-Defined Synonyms

Command or Operand	Synonym
MODIFY	F

Purpose of Command

The OUTPUT command controls the logical destination of the trace/error message output logged by all Event/Automation Service tasks.

Operand Descriptions

procname

Specifies the Event/Automation Service job name.

TO=*destination*

Specifies the logical Event/Automation Service output destination for all tasks. The *destination* can have the following values:

SYSOUT

Log trace and error messages to a designated output file for each Event/Automation Service task.

GTF

Log trace and error messages to the GTF trace facility.

ALL

Log trace and error messages to both the designated output file and the GTF trace facility.

Example: Setting output destinations

To send trace and error message information for all Event/Automation Service tasks to both the GTF trace facility and the designated output file for the Event/Automation Service job named IHSAEVNT, enter:

```
F IHSAEVNT,OUTPUT,TO=ALL
```

Response

A response similar to the following, depending on your destination output file and OUTSIZE parameter settings, is displayed:

```
IHS0165I The OUTPUT is set to ALL.
IHS0167I The OUTSIZE is set to 0 bytes.
IHS0168I CONTROL output file is //DD:IHSC
IHS0168I ALERTA output file is //DD:IHSA
IHS0168I MESSAGEA output file is //DD:IHSM
IHS0168I ALERTC output file is //DD:IHSB
IHS0168I MESSAGEC output file is //DD:IHSN
IHS0168I EVENTRCV output file is //DD:IHSE
IHS0168I TRAPALRT output file is //DD:IHST
IHS0168I ALRTTRAP output file is //DD:IHSL
```

Example: Displaying output destination settings

To display the current output destination for the Event/Automation Service job named IHSAEVNT, enter:

```
F IHSAEVNT,OUTPUT
```

Response

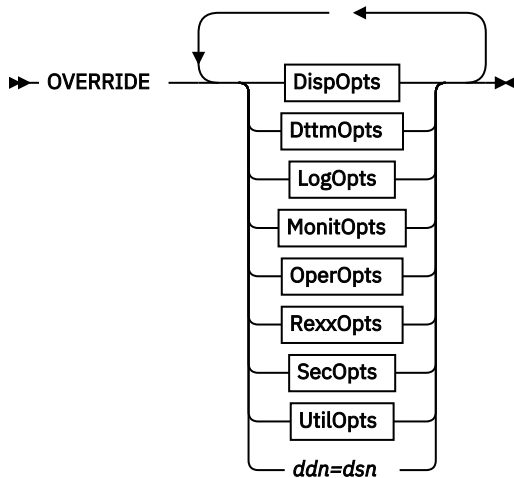
A response similar to the following, depending on your destination output file and OUTSIZE parameter settings, is displayed:

```
IHS0165I The OUTPUT is set to ALL.  
IHS0167I The OUTSIZE is set to 0 bytes.  
IHS0168I CONTROL output file is //DD:IHSC  
IHS0168I ALERTA output file is //DD:IHSA  
IHS0168I MESSAGEA output file is //DD:IHSM  
IHS0168I ALERTC output file is //DD:IHSB  
IHS0168I MESSAGEC output file is //DD:IHSN  
IHS0168I EVENTRCV output file is //DD:IHSE  
IHS0168I TRAPALRT output file is //DD:IHST  
IHS0168I ALRTTRAP output file is //DD:IHSL
```

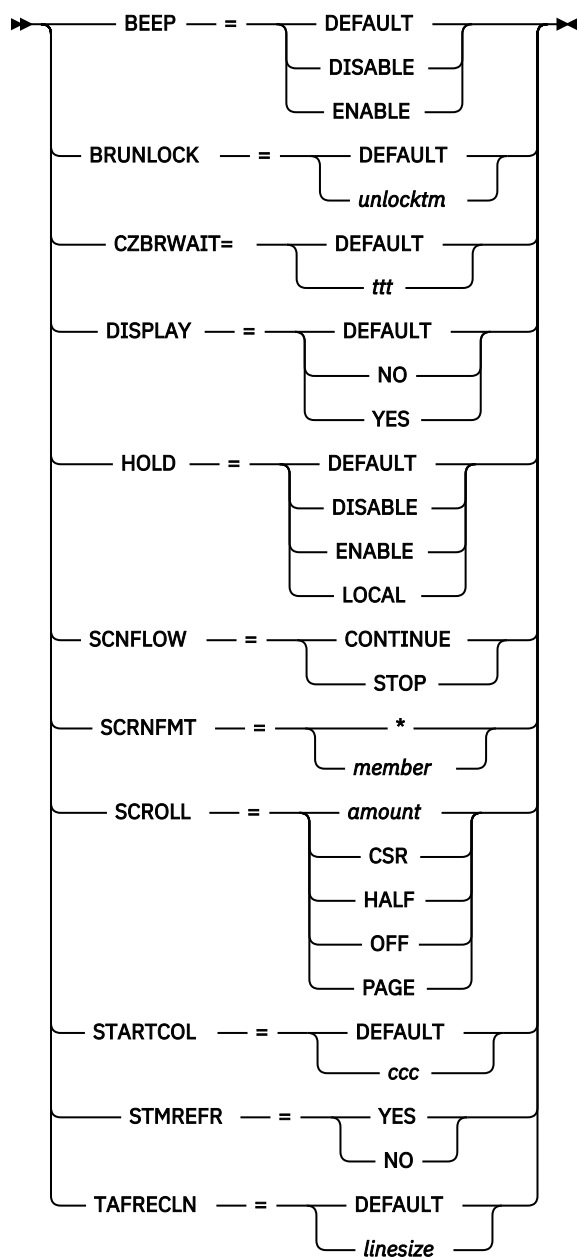
OVERRIDE (NCCF)

Syntax

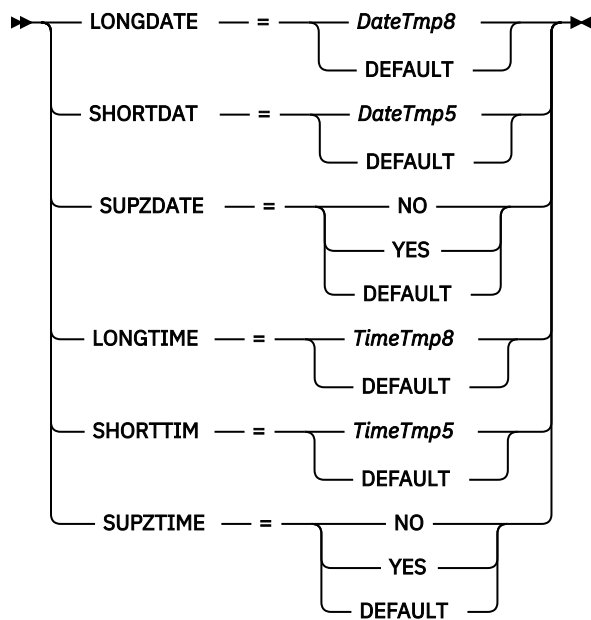
OVERRIDE



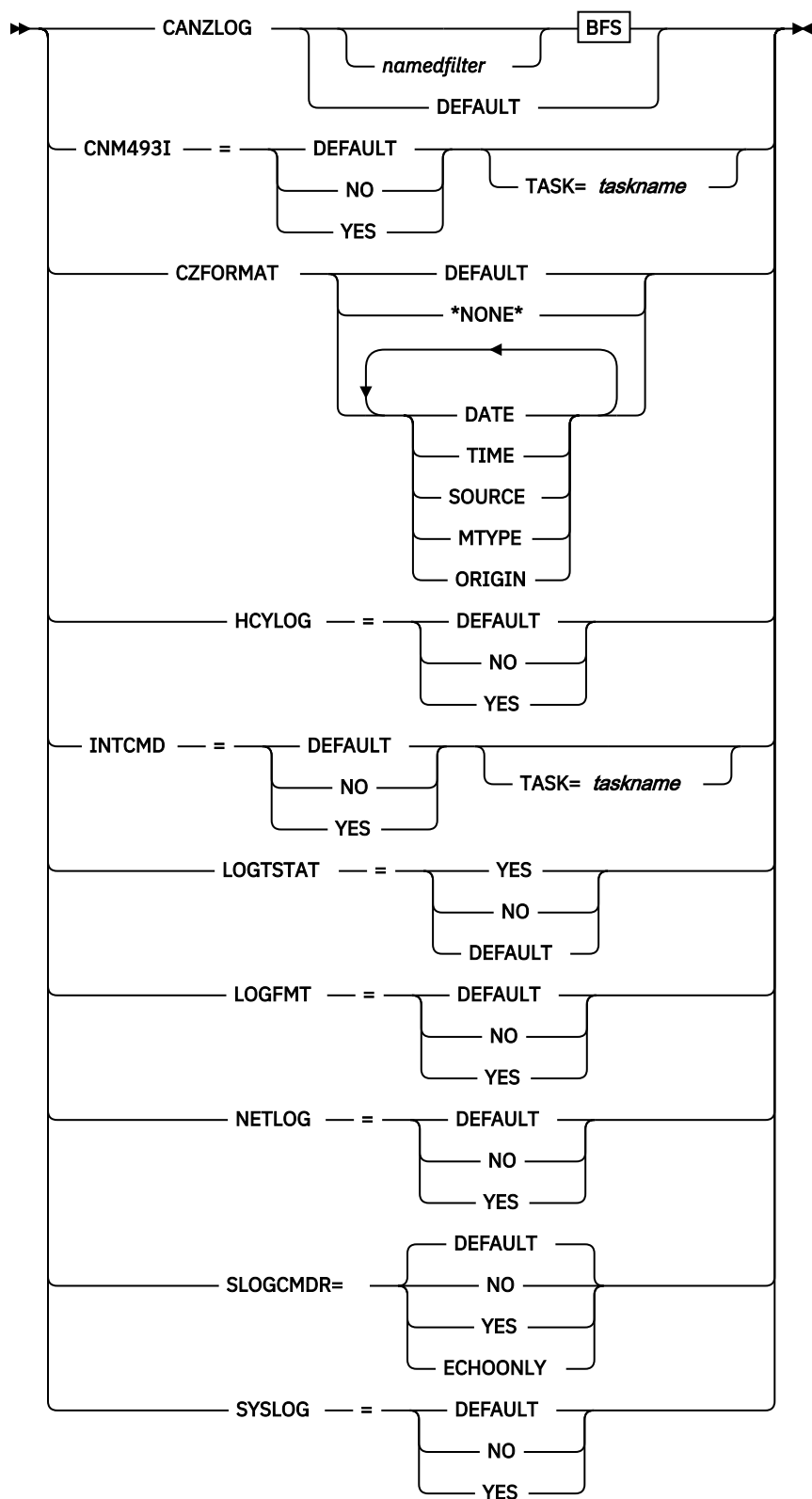
DispOpts



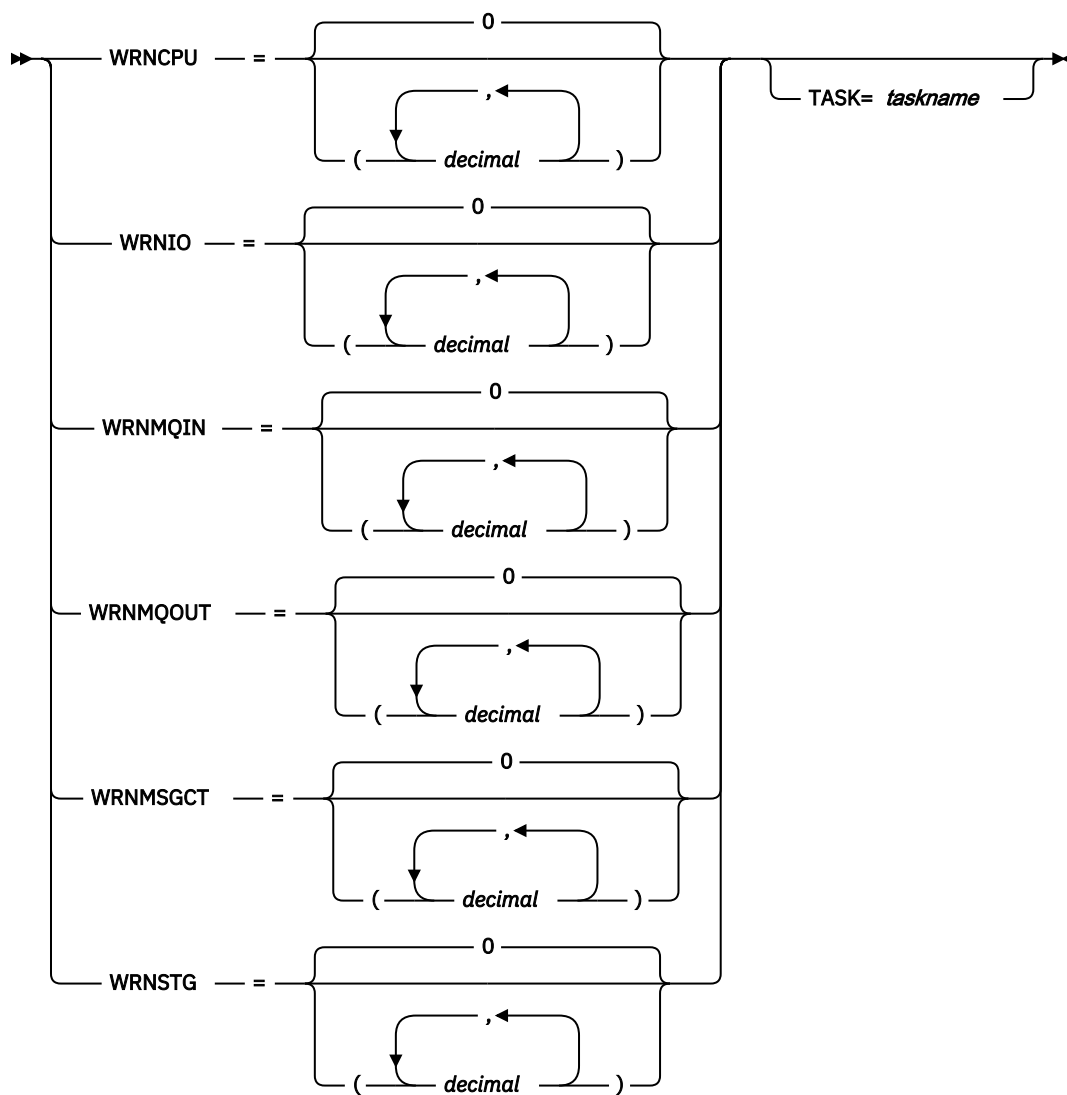
DttmOpts



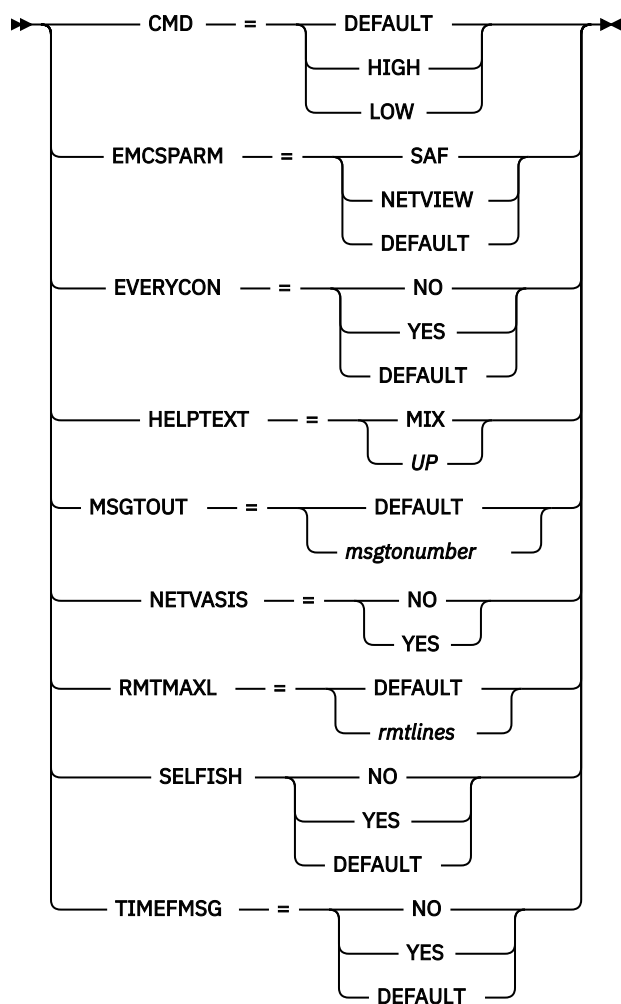
LogOpts



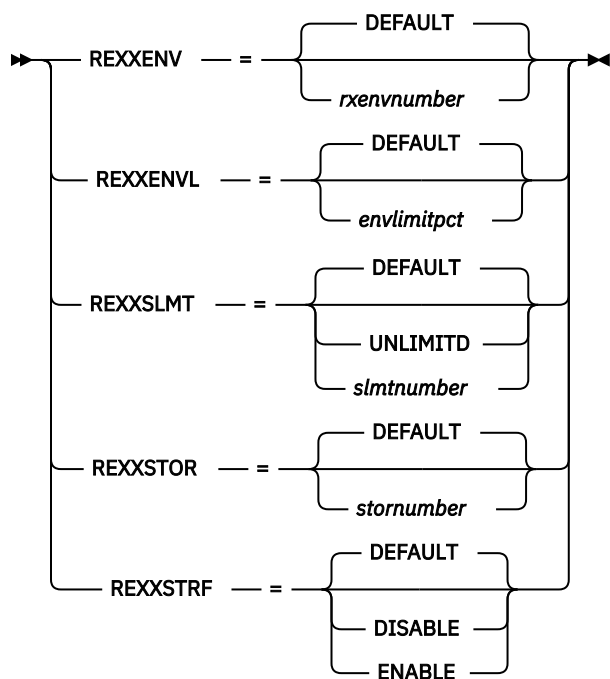
MonitOpts



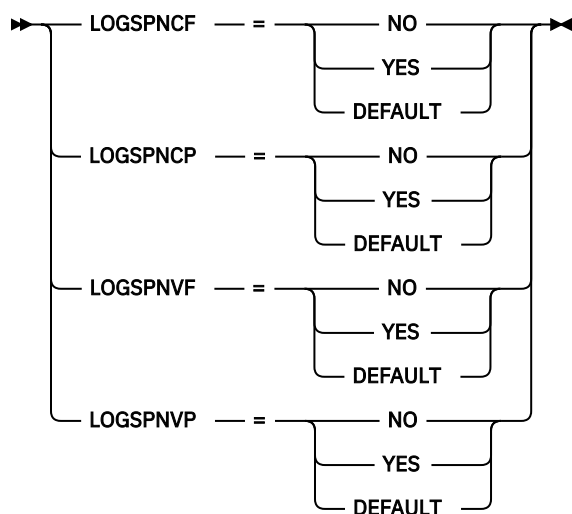
OperOpts



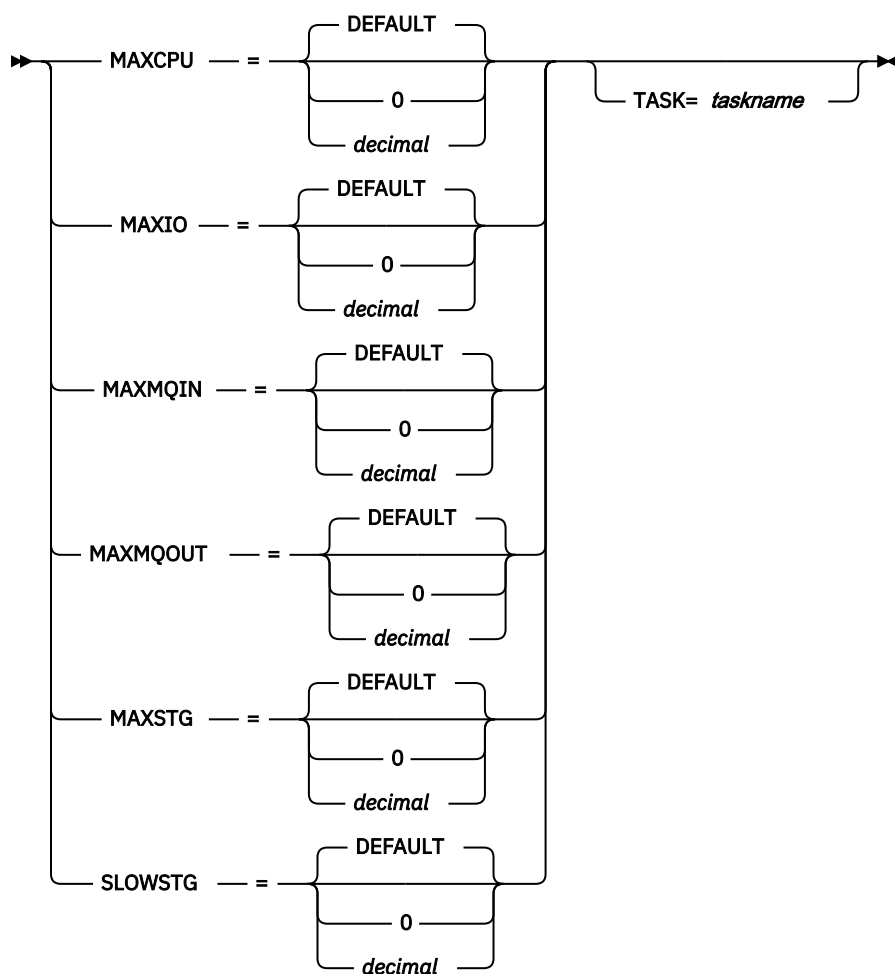
RexxOpts



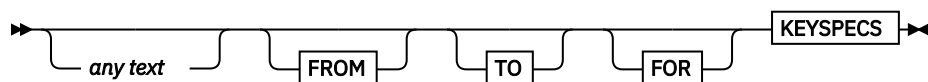
SecOpts



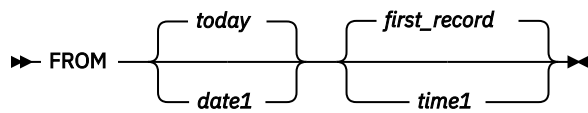
UtilOpts



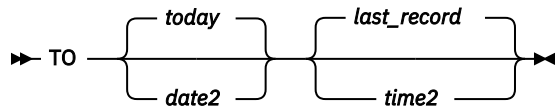
BFS (Basic Filter Syntax)



FROM



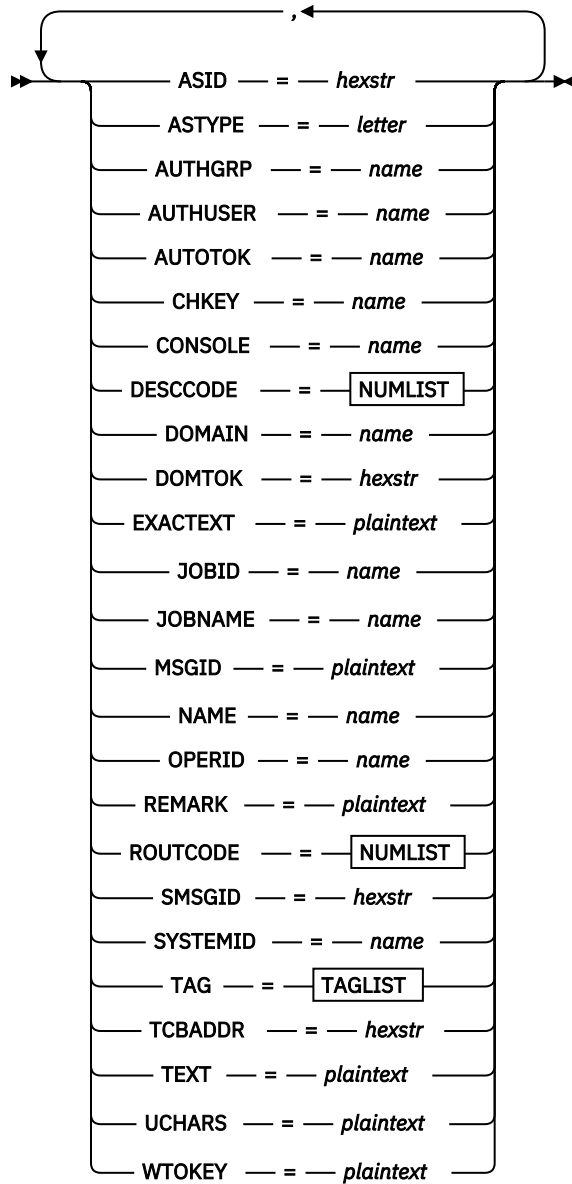
TO



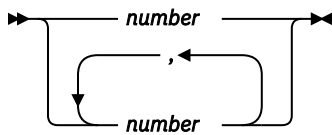
FOR

➤ FOR — *timespan* ➤

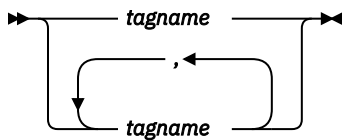
KEYSPECS



NUMLIST



TAGLIST



Purpose of Command

The OVERRIDE command can be used to specify options for a particular operator. The OVERRIDE options take precedence over the options specified by the DEFAULTS command. The OVERRIDE options that are related to message display (BEEP, DISPLAY, HOLD) apply to all messages that are to be displayed at the operator console. The OVERRIDE option settings for BEEP, DISPLAY, HCYLOG, HOLD, NETLOG, and SYSLOG take precedence over corresponding actions specified in the NetView automation table.

The OVERRIDE option settings for EVERYCON and TIMEFMSG affect AFTER, AT, and EVERY timer commands issued by these tasks.

Use the LIST OVERRIDE command to request a list of the current OVERRIDE settings.

Operand Descriptions

any text

Any alphanumeric text. Maximum size is 255 characters. If you use any delimiters (spaces, commas, or equal signs), enclose the text in quotation marks.

BEEP

Specifies whether the BEEP action is to be taken from the NetView automation table. Valid values are:

DEFAULT

Indicates that the option you specified on the DEFAULTS command is used. This is the initial value provided by the NetView program when you log on to the NetView system.

DISABLE

Indicates that the BEEP action is not taken from the NetView automation table.

ENABLE

Indicates that the BEEP action is taken from the NetView automation table.

BRUNLOCK

Specifies the number of seconds to wait for browse responses before unlocking the operator keyboard with message DSI360 indicating the request is in progress. This value does not apply to local member browse. For the first panel of a local network log browse, a value of 5 seconds is used regardless of the setting of BRUNLOCK. Valid values are:

DEFAULT

Indicates that the default value for the system is used.

unlocktm

The number of seconds to wait before unlocking the operator's keyboard. Valid values are 0 - 30.

CANZLOG

Specifies the filters that are used for a BROWSE LOG request. Valid values are:

BFS

The Basic Filter Syntax (BFS) where you can make specifications for the filter used.

DEFAULT

Used to specify a Basic Filter Syntax (BFS) to be the default value when BROWSE LOG is entered. The initial value supplied with the NetView product is NVMVS (MVS and local NetView messages). For examples of other values, enter the LIST CZFILTER command.

CMD

Specifies the task priority for operator commands. Valid values are:

DEFAULT

Indicates that the option you specified on the DEFAULTS command is used. This is the initial value provided by the NetView program when you log on to the NetView system.

HIGH

Causes operator commands to preempt- previous commands as soon as processing allows. Commands defined as TYPE=H (by CMDDEF or ADDCMD) are always queued at high priority.

LOW

Causes regular commands to be queued and processed in order. This is the safest option and is the setting in the CNMSTYLE member. This option allows you to be sure that commands (and command lists) that set status affecting subsequent commands are processed in the order you specify.

CNM493I

Specifies whether message CNM493I, which indicates automation table command processing, is sent to the network log. The task that CNM493I is logged under is determined by what task the message is automated on, not by what task the command or message is routed to by EXEC actions in the automation table. Valid values are:

DEFAULT

Indicates that the option you specified on the DEFAULTS command is used. This is the initial value provided by the NetView program when you log on to the NetView system.

YES

Causes message CNM493I to be sent to the network log.

NO

Prevents message CNM493I from being sent to the network log.

CZBRWAIT

Specifies the number of seconds to wait for Canzlog browse responses, such as BROWSE or ALL or FIND, before responding with status information.

DEFAULT

Indicates that the option that you specified on the DEFAULTS command is used. This is the initial value that is provided by the NetView program when you log on to the NetView system.

ttt

The number of seconds to wait for Canzlog browse responses. Valid values are in the range 1 - 1000000. The default value is 10.

CZFORMAT

Specifies the default elements of a prefix for Canzlog messages that are displayed by the BROWSE command. You can specify DEFAULT, *NONE*, or 1 - 5 of the following keywords after the CZFORMAT keyword:

- DATE
- TIME
- SOURCE
- MTYPE
- ORIGIN

Specify DATE, TIME, SOURCE, MTYPE, ORIGIN in the order in which they are to appear in the message prefix.

Instead of specifying any of those keywords, you can specify either of the following keywords:

- *NONE*
- DEFAULT

DATE

Specifies the formatted date of the message. For more information about the date format, see the help for the LONGDATE keyword.

TIME

Specifies the formatted time of the message. For more information about the time format, see the help for the LONGTIME keyword.

SOURCE

Specifies either the job name, if any, for an MVS message, or else it specifies the operator ID, if any, for a NetView message.

MTYPE

Specifies the NetView HDRMTYPE value.

ORIGIN

Specifies either the origin system for an MVS message or the origin domain for a NetView message.

DEFAULT

Specifies that you get the prefix that was specified by the DEFAULTS command.

NONE

Specifies that you do not get a prefix.

date1

Specifies the starting date of the time range. The format of *date1* is controlled by the setting of the date operands of the DEFAULTS and OVERRIDE commands. The specified date must be between 01/01/10 and 12/31/41.

Note: Wherever you can specify a date for the Canzlog log, you can substitute an asterisk (*). The asterisk is interpreted to be the date when the command is entered (today's date). You can use an asterisk for the TO and FROM values on the CANZLOG panel, with the BROWSE command, and with the FIND and ALL subcommands. Use of the asterisk for DEFAULTS and OVERRIDE specifications is discouraged because the value is not updated with the passage of local midnight. You can use an asterisk for the date and also specify a time.

date2

Specifies the end date of the time range. The format of *date2* is controlled by the setting of the date operands of the DEFAULTS and OVERRIDE commands. The specified date must be between 01/01/10 and 12/31/41.

ddn

A NetView DD name (for example, DSIPARM, DSIOPEN or CNMPNL1). For a complete list, see the BROWSE command.

Note: The following DD names are not allowed from the command line: DSIPARM, DSICLD, DSIVTAM and DSIPRF. They are only allowed from a command procedure, the PPT task, an optional task, or automation.

dsn

The name of a partitioned data set (PDS) which will be logically appended to the front of the concatenation for the specified DD, for all DSIDKS-based applications such as BROWSE, the < stage, or command list processing. Members in this PDS are then read by the OVERRIDE operator or operators.

Note:

1. Members in this operator data set are prioritized after those (with the same name) established by the INSTORE stage with the LOCAL operand, but before those established by the INSTORE stage with the COMMON operand.
2. This function works only if the operator has authority to read from the specified PDS.

3. When this command is issued, an internal list of members is created. When new members are created on this data set, the command must be reissued in order to read them.
4. A value of asterisk (*) removes any operator data set from the DD for this operator.

DISPLAY

Specifies whether all messages are displayed on NetView terminals. Valid values are:

DEFAULT

Indicates that the option you specified on the DEFAULTS command is used. This is the initial value provided by the NetView program when you log on to the NetView system.

NO

Indicates that no messages are displayed on your NetView terminal.

YES

Indicates that all messages are displayed on your NetView terminal.

Note: NetView messages that are sent to the immediate message area of the operator's screen will always be displayed, regardless of the setting for the DISPLAY action.

EMCSPARM

Specifies how the extended console attributes are determined when a task obtains an extended console. Extended consoles are obtained when operators or autotasks issue the GETCONID command, or when a task first issues an MVS command. The CNMCSSIR task also obtains an extended multiple console support (EMCS) console.

SAF

The console attributes for the EMCS console are obtained from the OPERPARM segment of the SAF product if the segment exists for the console name and can be accessed. This is the method by which extended console attributes were obtained in prior versions of the NetView program.

If the OPERPARM segment exists for the console name, and can be accessed, any console attributes that are not specified in the OPERPARM segment is given the MVS default value, which is not necessarily the same as the NetView default value.

Note: For the OPERPARM segment to be accessible, the operator must have READ access to the MVS.MCSOPER.console_name definition in the OPERCMDS class.

If the OPERPARM segment does not exist for the console name, or the segment cannot be accessed, the console attributes are determined as specified for EMCSPARM=NETVIEW.

NETVIEW

The console attributes for the EMCS console are a combination of the values specified on the MVSPARM statement in the CNMSTYLE member and the values specified on the GETCONID command. For a listing of the NetView default values, see the *IBM Z NetView Security Reference* for more information about setting EMCS console attributes.

DEFAULT

Indicates that the option you specified on the DEFAULTS command is used. This is the initial value provided by the NetView program when you log on to the NetView system.

EVERYCON

Specifies whether to continue to queue timed commands of type EVERY even after queuing failures occur. The valid values are:

NO

Indicates that queuing failures cause timed commands to be deleted (they are no longer queued).

YES

Indicates that such timed commands continue to be queued.

DEFAULT

Indicates that the option you specified on the DEFAULTS command is used. This is the initial value provided by the NetView program when you log on to the NetView system.

first_record

If you do not specify a starting time, the first record in the log with the specified date is used.

FOR

Specifies the duration of the span of time to be included. Use the FOR keyword if you want to specify the time span in terms of duration rather than specifying the start and end times. You can use the FOR keyword in the following ways:

- Use FOR with the FROM keyword to specify the beginning of the time span along with the duration.
- Use FOR with the TO keyword to specify the end of the time span along with the duration.
- Use FOR alone to specify a time span that ends at the current time.

You can specify a duration of up to 2 years. If you specify a larger value, a duration of 2 years is used.

Important: Do not specify both FROM and TO times if you are also specifying a duration with FOR.

FROM

Specifies the starting date and time. The specified date must be between 01/01/10 and 12/31/41. The format is controlled by the setting of the DATE and TIME operands of the DEFAULTS and OVERRIDE commands. This operand is optional.

HCYLOG

Specifies whether all messages are written to the hardcopy log if an operator has one active. Valid values are:

DEFAULT

Indicates that the option you specified on the DEFAULTS command is used. This is the initial value provided by the NetView program when you log on to the NetView system.

NO

Indicates that no messages are written to the hardcopy log.

YES

Indicates that all messages are written to the hardcopy log.

HELPTXT

Defines text for window-based help and help-desk help.

UP

Specifies that the text for window-based help and the HelpDesk will be uppercase.

MIX

Specifies that the text for window-based help and the HelpDesk will be mixed case.

HOLD

Defines whether the HOLD action is taken from the NetView automation table and whether held action messages are queued for rerouting to an authorized receiver upon logoff. Valid values are:

DEFAULT

Indicates that the option you specified on the DEFAULTS command is used. This is the initial value provided by the NetView program when you log on to the NetView system.

DISABLE

Indicates that the HOLD action is not taken from the NetView automation table. In addition, action messages are not rerouted to the authorized receiver upon logoff.

ENABLE

Indicates that the HOLD action is taken from the NetView automation table. In addition, action messages can be rerouted to the authorized receiver upon logoff.

LOCAL

Has the same function as ENABLE except that action messages are not rerouted to the authorized receiver upon logoff.

INTCMD

Specifies whether the task identified by *taskname* (by default, the task under which the OVERRIDE command is issued) logs NetView internally issued commands to the Canzlog log. Note that some commands, such as AFTER, AT, CHRON, EVERY, and PURGE are logged regardless of what you specify here. Valid values are:

DEFAULT

Indicates that the option you specified on the DEFAULTS command is used. This is the initial value provided by the NetView program when you log on to the NetView system.

YES

Causes internal commands under the given task to be logged to the Canzlog log.

NO

Prevents internal commands under the given task to be logged to the Canzlog log.

KEYSPECS**Notes:**

1. You can specify "not equal" by using `≠` for all KEYSPECS except NAME and REMARK. ("Not equal" is not valid with the FROM, TO, or FOR keywords.)
2. You can provide a single value or a list of values for any of these KEYSPECS, except for NAME and REMARK. (Specifying multiple values is not valid with the FROM, TO, or FOR keywords.) If you specify only one value for a KEYSPEC, parentheses are optional. These statements produce the same result:

```
Jobname=JOB9997
Jobname=(JOB9997)
```

If you specify more than one value for a KEYSPEC, the values must be enclosed in parentheses. The values can be separated by a blank or by a comma. These statements produce the same result:

```
Jobname=(JOB9997 JOB9998 JOB9999)
Jobname=(JOB9997, JOB9998, JOB9999)
```

If you specify either `Jobname=(JOB9997 JOB9998 JOB9999)` or `Jobname=(JOB9997, JOB9998, JOB9999)`, the logical operator OR evaluates the specifications, and any items with a Jobname of JOB9997 or JOB9998 or JOB9999 are displayed.

3. You can also specify the same KEYSPEC more than once. The logical operator AND evaluates multiple specification of KEYSPECS. This is useful when used with the "not equal" option. For example

```
Jobname='ABC', Jobname≠'ABC1'
```

matches every jobname that begins with ABC except those beginning with ABC1.

4. For the following KEYSPECS keywords, you can specify a shortened version of the matching value (for example, `OPERID=TOM` matches any operator ID beginning with TOM):

- AUTHGRP
- AUTHUSER
- AUTOTOK
- CHKEY
- CONSOLE
- DOMAIN
- JOBID
- JOBNAME
- MSGID
- OPERID
- SYSTEMID
- UCHARS
- WTOKEY

ASID

Address space ID.

ASTYPE

Address space type. Indicates how the address space was started (job type).

Value

Description

D

USS persistent procedure. The address space has a name for initiated programs, appropriate for a JOB. However, the existence of an OpenMVS address space block indicates a special purpose USS persistent procedure.

E

The address space is a system address space that is started before the NetView subsystem is initialized.

J

The address space is a JOB.

N

The address space is a system address space started during operating system initialization (NIP) processing.

S

The address space is a Started Task (STC).

Note: Because of the manner in which TN3270 is started, it might display as type S rather than type D.

T

The address space is a Time-Sharing User (TSO).

U

The address space is a USS forked or created procedure.

Error: the address space where the command originated has closed or else the message is not from the local LPAR.

?

Error: inconsistent data (might be a transient condition).

!

Error: inconsistent data.

>

Error: the supplied ASID is larger than the ASID limit for the system.

AUTHGRP

z/OS ACEE group ID (ACEEGRPN), if available.

AUTHUSER

z/OS ACEE user ID (ACEEUSRI), if available.

AUTOTOK

z/OS automation token.

CHKEY

z/OS CHKEY, as defined by system macro IEECHAIN; this is the step-name of a task or the job name of a job.

CONSOLE

z/OS destination console name.

DESCCODE

z/OS descriptor code.

DOMAIN

NetView domain name.

DOMTOK

A 4-byte token to identify a Delete Operator Message (DOM) or a token for a message for which a DOM was issued.

EXACTEXT

Specifies a comparison with message data that respects case and ignores national translation (if any). Search for EXACTEXT is faster than a search for TEXT.

JOBID

Identifier assigned by JES, also known as job number.

JOBNAME

z/OS job name.

MSGID

For DOMs with a MsgsMatch field of 1, the Canzlog ID of the associated message. The value specified cannot exceed 12 characters.

NAME

Specifies a 1 to 8 character value that is useful for saving a filter (see CANZLOG command) and as a reminder of the purpose of the filter. The NAME parameter has no effect on the operation of the filter.

OPERID

The NetView task/operator name that originated the message. A message that originates at a virtual OST (VOST) task is logged with OPERID set to the name of the VOST owner, if that information is available when the message is logged. The value that is specified cannot exceed eight characters.

REMARK

Species a 1 to 40 character value that can be useful as a reminder of the elements of a filter. You can read the remark value when editing a saved filter, in the output of LIST CZFILTER, as a result of the WHAT subcommand, and in some error messages. The REMARK parameter has no effect on the operation of the filter.

ROUTECD

z/OS route codes.

SMSGID

System message ID. This label is *SMSGID(s)*: for DOMS, which can have more than one. The value specified cannot exceed eight characters.

SYSTEMID

z/OS system ID. The value specified cannot exceed eight characters.

TAG (tagname)

Associated tags. You can specify more than one tag.

ALL

Matches any valid tagname.

AUDIT

Intended for audit purposes, such as internal commands.

BCAST

z/OS broadcast to active consoles applies.

CMDECHO

Command echo.

DELETED

Message was requested to be deleted. It is logged in the Canzlog log for automation purposes.

DOM

A Delete Operator Message (DOM) sent by the system to negate a previous message.

DOMEXP

Delete Operator Message (DOM) is expected for message, as defined by the WQEDOM flag.

MVSMMSG

Logged at the z/OS subsystem interface.

NVMSG

Originated in the NetView program.

TRACE

Intended for tracing purposes, such as debug messages.

TCBADDR

Task Control Block (TCB) address.

TEXT

Specifies a comparison with message data that is case-insensitive and occurs after national language translation, if any. Search for EXACTEXT is faster than a search for TEXT.

UCHARS

User-defined or installation-defined characters. The value specified cannot exceed 16 characters.

WTOKEY

Key field associated with the WTO system macro (also WQEKEY in system macro IHAWQE).

last_record

If you do not specify an end time, the last record in the log with the specified date is used.

LOGFMT=DEFAULT|NO|YES

Specifies whether the color and highlighting settings for operator screen formatting are used when browsing the network log. The valid values are:

DEFAULT

Specifies that the option you specified on the DEFAULTS command is to be used. A value of DEFAULT causes the override setting to be removed, and the applicable default value to become active. LOGFMT=DEFAULT is the initial value.

NO

Indicates that the operator screen formatting attributes are not to be used when browsing the network log.

YES

Indicates that messages that do not already have a specific color or highlighting attribute are subject to operator screen formatting color and highlighting when browsing the network log. The applicable screen format message types that are displayed using the screen format attributes are as follows:

HELD
ACTION
MVS

All other message types are considered normal. When no SCRNFMT member is in effect, the supplied screen format colors and attributes are used to display the messages. See the SCRNFMT operand for more information about operator screen formatting.

LOGSPNCF=NO|YES|DEFAULT

Specifies whether span of control FAILURES for requests from VTAM commands are to be logged. The valid values are:

NO

Indicates that no subsequent logging is to occur.

YES

Indicates data is to be logged.

DEFAULT

Specifies that the option you specified on the DEFAULTS command is to be used. A value of DEFAULT causes the override setting to be removed, and the applicable default value to become active.

LOGSPNCP=NO|YES|DEFAULT

Specifies whether span of control PASSes for requests from VTAM commands are to be logged. The valid values are:

NO

Indicates that no subsequent logging is to occur.

YES

Indicates data is to be logged.

DEFAULT

Specifies that the option you specified on the DEFAULTS command is to be used. A value of DEFAULT causes the override setting to be removed, and the applicable default value to become active.

LOGSPNVF=NO|YES|DEFAULT

Specifies whether span of control FAILURES for requests from NMC are to be logged. The valid values are:

NO

Indicates that no subsequent logging is to occur.

YES

Indicates data is to be logged.

DEFAULT

Specifies that the option you specified on the DEFAULTS command is to be used. A value of DEFAULT causes the override setting to be removed, and the applicable default value to become active.

LOGSPNVP=NO|YES|DEFAULT

Specifies whether span of control PASSes for requests from NMC are to be logged. The valid values are:

NO

Indicates that no subsequent logging is to occur.

YES

Indicates data is to be logged.

DEFAULT

Specifies that the option you specified on the DEFAULTS command is to be used. A value of DEFAULT causes the override setting to be removed, and the applicable default value to become active.

LOGTSTAT=YES|NO|DEFAULT

Specifies whether task resource utilization data is logged to the external System Management Facility (SMF) log. The valid values are:

YES

Indicates data is to be logged. Specifying the keyword without a value selects the default, which is YES.

NO

Indicates that no subsequent logging of resource data is to occur.

DEFAULT

Specifies that the option you specified on the DEFAULTS command is to be used. A value of DEFAULT causes the override setting to be removed, and the applicable default value to become active.

LONGDATE=*DateTmp8*|DEFAULT

Specifies a template describing the format used for dates when entered or presented in long form.

DateTmp8

The date template can contain up to eight characters, including delimiters, as follows:

Delimiter

Specifies the character used as a delimiter between components of the date. See [“Usage Notes” on page 46](#) for additional information.

DD

Specifies a two-digit day of the month.

DDD

Specifies a three-digit day of the year.

MM

Specifies a two-digit month of the year.

MMM

Specifies the first three characters of the month in uppercase (JAN, FEB, MAR,...DEC).

YY

Specifies the last two digits of the year.

YYYY

Specifies the complete four digits of the year.

The long form date template must specify a complete date including year and either month and day or day-of-year.

Some valid long form date templates are:

DD-MM-YY
DD/MM/YY
DDMMMYY
MM/DD/YY
YY.DDD
YYYY.DDD

DEFAULT

Specifies that the option you specified on the DEFAULTS command is to be used. A value of DEFAULT causes the override setting to be removed, and the applicable default value to become active.

LONGTIME=TimeTmp8|DEFAULT

Specifies a template describing the format used for times when entered or presented in long form.

TimeTmp8

The time template can contain up to eight characters, including delimiters, as follows:

DELIMITER

Specifies the character used as a delimiter between components of the time. See [“Usage Notes” on page 46](#) for additional information.

HH

Specifies the two-digit hour.

MM

Specifies the two-digit minutes.

SS

Specifies the two-digit seconds.

The long form time template must specify a complete time including hour, minutes, and seconds.

Some valid long form time templates are:

HH:MM:SS
HHMMSS
SS:MM:HH

DEFAULT

Specifies that the option you specified on the DEFAULTS command is to be used. A value of DEFAULT causes the override setting to be removed, and the applicable default value to become active.

MAXCPU=0|*decimal*|DEFAULT

Specifies percentage of processor utilization for the task in hundredths of a percent. A value of zero means no limit has been set and no slowdown occurs. The maximum value allowed is 99. When a task exceeds the limit, automatic task slowdown measures are used to bring the task back into the specified range. The task is suspended until enough time passes for the processor to be within limits. A value of DEFAULT removes the override setting and the applicable default value becomes active. Specifying the keyword without a value is equal to MAXCPU=DEFAULT.

When the NetView program is started, the MAXCPU default is set to 95% of one processor for each task. This value is intended to enable tasks to run at high workloads, but not prevent other tasks from running. You can revise these policies in the CNMSTYLE member using the DEFAULTS keywords.

Note: If you increase the value of this limit while a task is being delayed for an existing limit, the task delayed is canceled within approximately one second, and the task continues with the new limit in force.

MAXIO=0|*decimal*|DEFAULT

Specifies the number of I/O requests per minute allowed for the task. A value of zero means no limit has been set and no slowdown occurs. The maximum value allowed is 999999999 inputs and outputs per minute. When a task exceeds the limit, task slowdown measures are used to bring the task back into the specified range. A value of DEFAULT removes the override setting and the applicable default value becomes active. Specifying the keyword without a value is equal to MAXIO=DEFAULT.

Note: If you increase the value of this limit while a task is being delayed for an existing limit, the task delay is canceled within approximately one second, and the task continues with the new limit.

MAXMQIN=0|*decimal*|DEFAULT

Specifies the number of message kilobytes per minute that is allowed to be sent to the task from other tasks. A value of zero means no limit has been set and no slowdown occurs. The maximum value is 999999999 kilobytes per minute. When a task exceeds the limit, automatic task slowdown measures are used to bring the task back into the specified range. Other tasks that queue messages to the affected tasks slowed down until the rate is under control. A value of DEFAULT removes the override setting and the applicable default value becomes active. Specifying the keyword without a value is equal to MAXMQIN=DEFAULT.

Note: If you increase the value of this limit while a task is being delayed for an existing limit, the task delay is canceled within approximately one second, and the task continues with the new limit.

MAXMQOUT=0|*decimal*|DEFAULT

Specifies the number of message kilobytes per minute allowed for the task to send to another task. A value of zero means no limit has been set and no slowdown occurs. The maximum value allowed is 999999999 kilobytes per minute. When a task exceeds the limit, automatic task slowdown measures are used to bring the task back into the specified range. If the task attempts to queue a message to another task, it is slowed down until the rate is under the limit. A value of DEFAULT removes the override setting and the applicable default value becomes active. Specifying the keyword without a value is equal to MAXMQOUT=DEFAULT.

Note: If you increase the value of this limit while a task is being delayed for an existing limit, the task delay is canceled within approximately one second, and the task continues with the new limit.

MAXSTG=0|*decimal*|DEFAULT

Specifies the maximum amount of storage in kilobytes that a task can use. When this storage is reached the DSIGET macro returns an out of storage return code. In addition, queuing a message to a task that is over its MAXSTG limit results in the task not active condition, and the message is not transferred. The maximum value allowed is 999999K. A value of DEFAULT removes the override setting and the applicable default value becomes active. Specifying the keyword without a value is equal to MAXSTG=DEFAULT.

When the NetView program is started the MAXSTG default is set to 50% of the region size for each task other than the main task. The main task is preset to 80% of the region. (The main task serves as the manager for storage shared by all tasks, and thus needs a larger value.) You can revise these policies in the CNMSTYLE member using the DEFAULTS keywords.

MSGTOUT

Specifies the time interval that the NetView terminal access facility (TAF) waits before alerting an operator that an incoming message has not completed. When the time interval has passed, TAF displays message DWO310I with the fragment of data that is available. Valid values are:

DEFAULT

Indicates that the default value for the system is used initially. You can also return to this setting by specifying DEFAULT as the value of MSGTOUT.

msgtonumber

Specifies the value of the timeout in seconds. Valid values are in the range of 1 - 200 seconds.

namedfilter

Specifies any of the named filters that are supplied with the NetView product or created through use of the CANZLOG command. For information about named filters, issue the LIST CZFILTER command.

NETLOG

Specifies whether all messages are written to the network log. Valid values are:

DEFAULT

Indicates that the option you specified on the DEFAULTS command is used. This is the initial value provided by the NetView program when you log on to the NetView system.

NO

Indicates that no messages are written to the network log.

YES

Indicates that all messages are written to the network log.

NETVASIS

NETVASIS applies to NCCF, Window, and NMC. NETVASIS specifies whether lowercase characters in commands are converted to uppercase. Valid values are:

YES

Indicates that lowercase characters are not converted to uppercase. When OVERRIDE NETVASIS=YES is entered, the ??? at the bottom of the panel is replaced by >>>.

NO

Indicates that lowercase characters are converted to uppercase.

SELFISH

Specifies that a task (including PPT) might or might not be shared by other operators to process RMTCMDs or Web Browser functions.

NO

Specifies that this task allows sharing for RMTCMD and Web Browser functions.

YES

Specifying YES disallows task sharing by RMTCMDs and the Web Browser (except for Web Browser requests to autotasks started by the Web Browser.) Autotasks that are started by RMTCMD processing discontinue processing further RMTCMD commands if an operator logs on to the task by using the TAKEOVER=YES option.

DEFAULT

Indicates that the action specified on the DEFAULTS command is used.

REXXENV

Specifies the number of active and inactive, but initialized, REXX environments to be retained for this operator. Valid values are:

DEFAULT

Indicates that the option you specified on the DEFAULTS command is used. If you do not specify a value for REXXENV, this value is the default. This is the initial value provided by the NetView program when you log on to the NetView system.

rxenvnumber

Indicates the valid numeric values in the range of 0 - 250. If you set REXXENV higher than the current number of environments, storage is acquired only when a new environment is needed. If you set REXXENV lower than the current number of environments, storage for the excess environments is not freed until a REXX command list uses one of the environments and completes.

REXXENVL

Specifies the percentage of all REXX environments that are defined for the NetView program which are allowed for this task. See *IBM Z NetView Programming: REXX and the NetView Command List Language* for more information regarding the use of REXX environments. The valid values are as follows:

DEFAULT

Indicates that the option that you specified on the DEFAULTS command is used. If you do not specify a value for REXXENVL, this value is the default. This is the initial value that is provided by the NetView program when you log on to the NetView system.

envlimitpct

Specifies the percentage of REXX environments. The valid value range is 0 - 99 percent. If you set REXXENVL to a value lower than the current REXXENVL value and its current percentage of REXX initialized environments exceeds the new value, no new REXX environments can be initialized for this NetView task until the percentage of this task's initialized REXX environments drops below the new value.

REXXSLMT

Specifies the amount of storage, in 1K increments, that a REXX environment is allowed to accumulate before being ended after its current use is completed. Valid values are:

DEFAULT

Indicates that the option you specified on the DEFAULTS command is used. If you do not specify a value for REXXSLMT, this value is the default. This is the initial value provided by the NetView program when you log on to the NetView system.

slmtnumber

Indicates the valid value range, which is a number in the range of 0 - 10,000.

UNLIMITD

Indicates that there are no cumulative REXX environment storage restrictions.

REXXSTOR

Specifies the amount of storage, in 1 K increments, to be acquired by REXX environment initialization processing. Valid values are:

DEFAULT

Indicates that the option you specified on the DEFAULTS command is used. If you do not specify a value for REXXSTOR, this value is the default. This is the initial value provided by the NetView program when you log on to the NetView system.

stornumber

Indicates the valid numeric values, which are in the range of 0 - 250. Specify 12 K for each nesting level of REXX command lists.

REXXSTRF

Specifies whether the NetView operator can run REXX command lists that use the REXX STORAGE function. Valid values are:

DEFAULT

Indicates that the value you specified on the DEFAULTS command is used. This is the initial value provided by the NetView program when you log on to the NetView system.

DISABLE

Indicates that the NetView operator cannot run REXX command lists that use the REXX STORAGE function.

ENABLE

Indicates that the NetView operator can run REXX command lists that use the REXX STORAGE function.

Note: The OVERRIDE REXXSTRF command is effective the next time a REXX environment is initialized, which is a function of the current REXXENV value and the number of active REXX command lists. Thus, it is never effective in the same command list invocation from which it is issued.

RMTMAXL

Specifies the maximum number of lines transferred for a cross-domain member browse request. If the remote member contains more than the maximum number of lines, the BROWSE command continues with the permitted number of lines and message CNM206I is issued. The BROWSE command uses the RMTMAXL setting of the operator issuing the cross-domain browse request. A large value for RMTMAXL allows a cross-domain member browse request to return large amounts of data and can cause delays with other RMTCMD LU6.2 communication.

DEFAULT

Indicates that the default value for the system is used.

rmtlines

Specifies the maximum number of lines the remote NetView program transfers when a cross-domain member browse request is processed. The valid value range for *rmtlines* is 1 - 10 000 000.

Note: The RMTMAXL value does not apply to cross-domain netlog browse which can be used to browse network logs of unlimited size.

SCNFLOW

Controls the operation of the NetView console as data is entered. This option pertains only to an SNA attended operator station task (OST). It has no effect on other task types. Valid values are:

CONTINUE

Indicates that NetView continues sending screen updates as the operator types. This is the default value and produces the most efficient data flow.

STOP

Indicates that the NetView program stops sending screen updates until the operator finishes typing and presses an action key such as ENTER, PF key, or PA key. This option is related to possible terminal hardware limitations that cause loss of command-line input at a busy operator terminal. Do not use this option unless requested by IBM Software Support.

SCRNFMT

Specifies the screen format changes from the preset values or the current values. Valid values are:

Indicates a reserved notation that removes the current OVERRIDE screen format and resets the NetView preset values.

member

Indicates a member of DSIPARM containing command facility screen format definitions.

See the *IBM Z NetView Administration Reference* for more information.

SCROLL

Specifies whether a scroll field is present for the BROWSE command as well as the default scroll amount. Valid values are:

amount

Indicates a 1- to 4- character scroll amount which is used by the BACK, FORWARD, LEFT, and RIGHT commands. This value is placed in the scroll amount field on the BROWSE panel unless the value is OFF in which case the scroll amount field is not present.

CSR

Indicates that the cursor determines the amount to scroll forward or back.

HALF

Indicates that the scroll amount is half of a screen.

OFF

Indicates that no scroll field is displayed on the BROWSE panel. This setting makes BROWSE scrolling consistent with the scrolling in the WINDOW command.

PAGE

Indicates that the scroll amount is a full screen.

SHORTDAT=DateTmp5|DEFAULT

Specifies a template describing the format used for dates when entered or presented in short form.

DateTmp5

The date template can contain up to five characters, including delimiters, as follows:

Delimiter

Specifies the character used as a delimiter between components of the date. See [“Usage Notes” on page 46](#) for additional information.

DD

Specifies a two-digit day of the month.

DDD

Specifies a three-digit day of the year.

MM

Specifies a two-digit month of the year.

MMM

Specifies the first three characters of the month in uppercase (JAN, FEB, ... DEC).

YY

Specifies the last two digits of the year.

The short form date template must specify a date including either day and month or day-of-year.

Some valid short form date templates are:

DD-MM

DDD

DDMMM

MM/DD

MMMDD

YYDDD

DEFAULT

Specifies that the option you specified on the DEFAULTS command is to be used. A value of DEFAULT causes the override setting to be removed, and the applicable default value to become active.

SHORTTIM=TimeTmp5|DEFAULT

Specifies a template describing the format used for times when entered or presented in short form.

TimeTmp5

The time template can contain up to five characters, including delimiters, as follows:

Delimiter

Specifies the character used as a delimiter between components of the date. See [“Usage Notes” on page 46](#) for additional information.

HH

Specifies the two-digit hour.

MM

Specifies the two-digit minutes.

The short form date template must include hours and minutes.

Some valid short form time templates are:

HH:MM
HHMM
MMHH

DEFAULT

Specifies that the option you specified on the DEFAULTS command is to be used. A value of DEFAULT causes the override setting to be removed, and the applicable default value to become active.

SLOGCMDR

Specifies whether responses to MVS commands issued from within the NetView program or from other message traffic naming a console owned by a NetView operator are sent to the system log. System logging of all other messages is not affected.

Note: Beginning in NetView for z/OS Version 6.1, the SLOGCMDR option is supported regardless of whether the subsystem router task (CNMCSSIR) and the NetView subsystem interface (SSI) procedure are active. In releases before Version 6.1, the SLOGCMDR option was supported only when the subsystem router task (CNMCSSIR) and the NetView subsystem interface (SSI) procedure are active.

The valid values are as follows:

DEFAULT

The following is the order in which the value for DEFAULT is assigned:

1. The most current value specified with the DEFAULTS SLOGCMDR command
2. The DEFAULTS.SLOGCMDR value set in the CNMSTYLE member.
3. The default value of YES.

NO

Indicates that command responses are not sent to the system log. This option also suppresses command echoes. Use this value if you want to suppress lengthy responses or for commands that repeat at frequent intervals.

YES

Indicates that command responses are sent to the system log.

ECHOONLY

Indicates that command responses are not sent to the system log, but the echo of the command is logged. This is useful for tracking which MVS commands have been issued, but keeps the command results from filling up the system log.

SLOWSTG=0|decimal|DEFAULT

Specifies the maximum amount of storage in kilobytes that a task can use at which value slowdown measures are used. In addition, queuing a message to a task that is over its SLOWSTG limit results in that task having the same slowdown measures applied based on how much over the limit the receiving task gets. The maximum value allowed is 999999K. A value of DEFAULT removes the override setting and the applicable default value becomes active. Specifying the keyword without a value is equal to SLOWSTG=DEFAULT.

When the NetView program is started, the SLOWSTG default is set to 25% of the region size for each task other than the main task. The main task is preset to 72% of the region. (The main task serves as the manager for storage shared by all tasks, and thus needs a larger value.) You can revise these policies in the CNMSTYLE member using the DEFAULTS keywords.

Note: If you increase the value of this limit while a task is being delayed for an existing limit, the task delay is canceled within approximately one second, and the task continues with the new limit in force.

When a task exceeds SLOWSTG, all storage requests (using DSIGET) and message queueing to the affected task have a time delay. The time delay is microsecond for each byte of storage requested over the SLOWSTG limit value up to 110% of the limit, and doubled for each 10% over the SLOWSTG

value thereafter. This produces a slowdown effect that is proportional to the size of the request as the task use of storage grows.

The initial slowdown rate is equivalent to allow storage to grow at a rate of 1 MB per second, in the range 100-110% of the limit value. When you lower a SLOWSTG value, you lower both the point at which slowdown occurs, and the amount of growth allowed before the delay time is doubled. This allows you to tailor the exponential curve to fit the application. For example, SLOWSTG=100 will trigger at 100 KB limit value and double every 10 KB. Another example, SLOWSTG=500 will trigger at 500 KB limit value and double every 50 KB.

Note: The maximum doubler value is 2 raised to the 30th power.

STARTCOL

Specifies the starting column displayed when browsing the network log. The column indicator in the third line of the network log browse display can be used to determine column numbers. Valid values are:

DEFAULT

Indicates that the default value for the system is used.

ccc

Specifies the starting column used when browsing the network log. The valid range is 1 - 178.

STMREFR

Specifies whether the status monitor domain status summary panel starts in dynamic or static mode. For additional information about this setting, see the SREFRESH command. Valid values are:

YES

Indicates that the domain status summary panel starts in dynamic mode. This setting is indicated on the panel with REFRESH=ON. YES is the default.

NO

Indicates that the domain status summary panel starts in static mode. This setting is indicated on the panel with REFRESH=OFF.

SUPZDATE

Specifies whether leading zeros are to be suppressed when presenting dates. This applies only to the entire date string and not to each element of the string. The valid values are:

NO

Indicates that zeros are not to be suppressed. NO is the initial value provided by the NetView program.

YES

Indicates that leading zeros are to be suppressed.

DEFAULT

Specifies that the option you specified on the DEFAULTS command is to be used. A value of DEFAULT causes the override setting to be removed, and the applicable default value to become active.

SUPZTIME

Specifies whether leading zeros are to be suppressed when presenting times. This applies only to the entire time string and not to each element of the string. The valid values are:

NO

Indicates that zeros are not to be suppressed. NO is the initial value provided by the NetView program.

YES

Indicates that leading zeros are to be suppressed.

DEFAULT

Specifies that the option you specified on the DEFAULTS command is to be used. A value of DEFAULT causes the override setting to be removed, and the applicable default value to become active.

SYSLOG

Specifies whether all messages are written to the system log. Valid values are:

DEFAULT

Indicates that the option you specified on the DEFAULTS command is used. This is the initial value provided by the NetView program when you log on to the NetView system.

NO

Indicates that no messages are written to the system log.

YES

Indicates that all messages are written to the system log.

TASK=taskname

Indicates the task for which override value applies. If omitted, the settings apply to the task which the command runs. The TASK keyword can be used only in combination with the MAXCPU, MAXIO, MAXMQIN, MAXMQOUT, MAXSTG, SLOWSTG, WRNCPUR, WRNIO, WRNMQIN, WRNMQOUT, WRNMSGCT, WRNSTG, and CNM493I keywords. These values can be set by operators or automation even for tasks that do not run commands, such as CNMCSSIR.

TIMEFMSG

Specifies whether timed commands which cannot be queued to the target operator produce a BNH357E error message. The valid values are:

DEFAULT

Indicates that the option you specified on the DEFAULTS command is used. This is the initial value provided by the NetView program when you log on to the NetView system.

NO

Indicates that no error message will be issued.

YES

Indicates that the error message will be issued.

TAFRECLN

Specifies the maximum record length (line size) of lines returned by the session partner in a TAF OPCTL session. If a line is shorter or the same length as the specified value, the line is returned without changes. If it is longer, the line is split at the record length and the text is continued on the next line.

DEFAULT

Indicates that the default value for the system is used.

linesize

The maximum size of lines returned by a TAF OPCTL session partner. Valid values are 1 - 32000.

time1

Specifies the starting time of the time range. The format of *time1* is controlled by the setting of the time operands of the DEFAULTS and OVERRIDE commands.

time2

Specifies the end time of the time range. The format of *time2* is controlled by the setting of the time operands of the DEFAULTS and OVERRIDE commands.

timespan

Specifies the time span to be included. This parameter is a string in the following format:

```
ddDhhHmmM
```

where *dd* specifies the number of days, *hh* specifies the number of hours, and *mm* specifies the number of minutes. This string is not case sensitive. You do not need to specify the entire string:

- You can omit any of the three values, although you must specify at least one.
- You can omit the final trailing character (the next value in sequence is assumed).

The following examples show valid time span specifications:

Table 5. Time span examples	
String	Interpreted as
3D12H45 M	3 days, 12 hours, and 45 minutes
1d30m	1 day and 30 minutes
2D6	2 days and 6 hours
1h15	1 hour and 15 minutes
5	5 minutes

TO

Specifies the end date and time. The specified date must be between 01/01/10 and 12/31/41. The format is controlled by the setting of the DATE and TIME operands of the DEFAULTS and OVERRIDE commands. This operand is optional.

today

When specifying the FROM keyword, *date1* defaults to the current date if not specified. If *time1* is not specified, FROM defaults to the first record. When specifying the TO keyword, *date2* defaults to the current date if not specified. If *time2* is not specified, TO defaults to the last record.

WRNCPU=0|(decimal,decimal,decimal)|DEFAULT

Specifies the percentage of the processor utilization for the task. When the processor utilization reaches the percentage or percentages specified, a status change is to NetView Resource Manager.

Up to three percentage values can be specified. The parentheses are not required if only one value is specified. Multiple values must be enclosed in parentheses and separated by either blanks or commas. Zero cannot be specified with other values. A value of zero indicates that no status changes are sent. Valid values are any number in the range 1 - 99. These values are not positional. Specifying the keyword without a value is equal to WRNCPU=DEFAULT. WRNCPU is similar to the MAXCPU keyword.

WRNIO=0|(decimal,decimal,decimal)|DEFAULT

Specifies the number of I/O requests, per minute, allowed for the task. The *decimal* value or values specified indicate when status changes are sent to NetView Resource Manager.

Up to three values can be specified. The parentheses are not required if only one value is specified. Multiple values must be enclosed in parentheses and separated by either blanks or commas. Zero cannot be specified with other values. A value of zero indicates that no status changes are sent. Valid values are any number in the range 1 - 999999999. These values are not positional. Specifying the keyword without a value is equal to WRNIO=DEFAULT. WRNIO is similar to the MAXIO keyword.

WRNMQIN=0|(decimal,decimal,decimal)|DEFAULT

Specifies the number of message kilobytes, per minute, that can be sent to the task from other tasks. The *decimal* value or values specified indicate when status changes are sent to NetView Resource Manager.

Up to three values can be specified. The parentheses are not required if only one value is specified. Multiple values must be enclosed in parentheses and separated by either blanks or commas. Zero cannot be specified with other values. A value of zero indicates that no status changes are sent. Valid values are any number in the range 1 - 999999999. These values are not positional. Specifying the keyword without a value is equal to WRNMQIN=DEFAULT. WRNMQIN is similar to the MAXMQIN keyword.

WRNMQOUT=0|(decimal,decimal,decimal)|DEFAULT

Specifies the number of message kilobytes, per minute, that can be sent from this task to other tasks. The *decimal* value or values specified indicate when status changes are sent to NetView Resource Manager.

Up to three values can be specified. The parentheses are not required if only one value is specified. Multiple values must be enclosed in parentheses and separated by either blanks or commas. Zero cannot be specified with other values. A value of zero indicates that no status changes are sent. Valid values are any number in the range 1 - 999999999. These values are not positional. Specifying the keyword without a value is equal to WRNMQOUT=DEFAULT. WRNMQOUT is similar to the MAXMQOUT keyword.

WRNMSGCT=0|(decimal,decimal,decimal)|DEFAULT

Specifies the number of buffers on the message queue of the task. The *decimal* value or values specified indicate when status changes are sent to NetView Resource Manager.

Up to three values can be specified. The parentheses are not required if only one value is specified. Multiple values must be enclosed in parentheses and separated by either blanks or commas. Zero cannot be specified with other values. A value of zero indicates that no status changes are sent. Valid values are any number in the range 1 - 999999 kilobytes. These values are not positional. Specifying the keyword without a value is equal to WRNMSGCT=DEFAULT.

WRNSTG=0|(decimal,decimal,decimal)|DEFAULT

Specifies the number of kilobytes of storage in use by a task. The *decimal* value or values specified indicate when status changes are sent to NetView Resource Manager.

Up to three values can be specified. The parentheses are not required if only one value is specified. Multiple values must be enclosed in parentheses and separated by either blanks or commas. Zero cannot be specified with other values. A value of zero indicates that no status changes are sent. Valid values are any number in the range 1 - 999999 kilobytes. These values are not positional. Specifying the keyword without a value is equal to WRNSTG=DEFAULT. WRNSTG is similar to the MAXSTG keyword.

Usage Notes

The following usage notes apply to the OVERRIDE command:

- The commas between keywords are optional. You can use either commas or blank spaces to separate multiple keywords.
- If you omit a parameter of the OVERRIDE command, the current value of the parameter remains in effect. If you never changed a parameter of the OVERRIDE command, the initial value provided by the NetView program remains in effect.
- DISPLAY=NO enables an operator to suppress all messages.
- The default formats for the date and time operands are as follows:

LONGDATE

mm/dd/yy

LONGTIME

hh:mm:ss

SHORTDAT

mm/dd

SHORTTIM

hh:mm

- Delimiters for the LONGDATE, LONGTIME, SHORTDAT, and SHORTTIM operands can be any printable character except:
 - Alphanumeric characters
 - Apostrophes (')
 - Asterisks (*)
 - Blanks
 - Commas (,)
 - Equals (=)

- National characters (@, #, \$)
- Parentheses
- Underscores (_)

Restrictions

The following restrictions apply to the OVERRIDE command:

- The DEFAULT option sets the processing back to the environment created by the DEFAULTS command and the NetView automation table.
- Unsolicited messages received from the MVS subsystem interface are not written to the network log if they do not have an automation table entry and have not been assigned a task with the ASSIGN command. If the automation table entry used to automate an unsolicited message from the MVS subsystem interface contains an EXEC action with both the CMD and ROUTE parameters, only the command specified with the CMD keyword is routed. Routing of the message being processed by the automation table is not affected. To change the routing of the message, use an EXEC action with the ROUTE parameter and not the CMD parameter.
- No messages are produced if this command runs correctly.

Return Codes

Return Code Meaning

0

Processing successful

nonzero

Error in processing

Example: Changing the HCYLOG and DISPLAY options

To change the HCYLOG and DISPLAY options, enter:

```
OVERRIDE HCYLOG=YES,DISPLAY=NO
```

Messages are now written to the active hardcopy log but no longer displayed on the NetView terminal (*only* for the operator who entered the command). Other options are unaffected.

Response

You receive message DSI633I stating the OVERRIDE command completed successfully.

```
DSI633I OVERRIDE COMMAND SUCCESSFULLY COMPLETED
```

Example: Listing OPER4 DEFAULTS and OVERRIDE settings

To see a list of the DEFAULTS and OVERRIDE settings for OPER4, enter:

```
LIST OVERRIDE=OPER4
```

Example: Defining an operator-specific data set

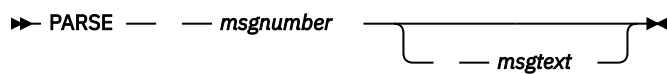
```
OVERRIDE DSIOPEN=NETV.OPDS.OPER1 DSICLD=NETV.OPDS.OPER1
```

This operator can now access members on NETV.OPDS.OPER1 as command lists or as PF key definitions. The PF key definitions can also be saved to NETV.OPDS.OPER1(CNMKEYSV). See the DISPFK command for information about saving PF key definitions.

PARSE (STATMON)

Syntax

PARSE



Purpose of Command

The PARSE command enables you to see how the status monitor parses a message.

Operand Descriptions

msgnumber

Identifies the number of the message.

msgtext

Specifies the full message text.

Example: Determining how a message is parsed

If you are writing a filter to set off message indicator 2 for certain nodes, you must know how the message is parsed. To find this information, enter:

```
PARSE IST105I nodename NODE NOW INACTIVE
```

Response

The following table is displayed:

CNM131I	ELEMENT	TEXT	DELIMITER
CNM132I	-----		
CNM133I	1	IST105I	' '
CNM133I	2	NODENAME	' '
CNM133I	3	NODE	' '
CNM133I	4	NOW	' '
CNM133I	5	INACTIVE	' '
CNM134I	END OF PARSE DISPLAY		

Example: Determining how a message is parsed

```
PARSE IST595I IRNLIMIT=NOLIMIT,CURRENT=0000000K,MAXIMUM=0000000K
```

Response:

The following table is displayed:

CNM131I	ELEMENT	TEXT	DELIMITER
CNM132I	-----		
CNM133I	1	IST595I	' '
CNM133I	2	IRNLIMIT	'='
CNM133I	3	NOLIMIT	' '
CNM133I	4	CURRENT	'='
CNM133I	5	0000000K	' '
CNM133I	6	MAXIMUM	'='
CNM133I	7	0000000K	' '
CNM134I	END OF PARSE DISPLAY		

PATHS (NCCF; CNME0026)

Syntax

PATHS

➤ PATHS — — *resname* ————— ➤
 ————— ↻ —————
 ————— , — *passthru* —————

Purpose of Command

The PATHS command list displays dial-out path information about a switched physical unit.

Operand Descriptions

rename

Specifies the name of a switched physical unit

passthru

Specifies up to 6 parameters which are appended unchanged to the VTAM DISPLAY command issued by the PATHS command. No validation for duplicate or conflicting parameters is performed.

Example: Displaying dial-out path information for a physical unit

To display dial-out path information for physical unit HD3790N1, enter:

PATHS HD3790N1

Response

A message similar to the following example is displayed:

```
IST097I DISPLAY ACCEPTED
IST148I DIAL OUT PATH INFORMATION FOR PHYSICAL UNIT HD3790N1
IST149I LINE GRP      TELEPHONE NUMBER OR LINE NAME    PID GID CNT
IST168I EGROUP40          4094                001 001 005 AVA AUT
IST168I EGROUP50          4094                002 002 001 AVA MAN
IST314I END
```

PID is the path identifier, GID is the group identifier, and CNT is the retry count. AVA means that the path is available, and AUT or MAN shows whether the dial-out is automatic or manual.

PDFILTER (NPDA; CNME3004)

Syntax

PDFFILTER

► PDFILTER ◄

Purpose of Command

The PDFILTER command list defines the recording filters to be used by the hardware monitor, thus controlling what data is recorded.

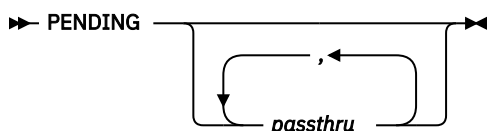
Usage Notes

The PDFILTER command list is called from a statement in the sample NetView automation table (DSITBL01) when the NetView BNJDSERV task completes initialization. You can customize the PDFILTER sample command list by using NPDA.PDFILTER statements in the CNMSTYLE member to set recording filters for the hardware monitor. For more information, see the CNMSTYLE member or *IBM Z NetView Administration Reference*.

PENDING (NCCF; CNME0028)

Syntax

PENDING



Purpose of Command

The PENDING command list displays information about nodes in the domain in a pending state.

Operand Descriptions

passthru

Specifies up to 6 parameters which are appended unchanged to the VTAM DISPLAY command issued by the PENDING command. No validation for duplicate or conflicting parameters is performed.

Example: Displaying nodes in a pending state

To display nodes in a pending state, enter:

```
PENDING
```

Response

Information similar to the following is displayed:

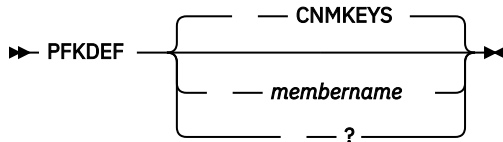
```
IST350I VTAM DISPLAY - DOMAIN TYPE = PENDING
IST159I THE FOLLOWING NODES ARE IN A PENDING STATE
IST080I M09      PACDR      M10      PACDR      H21CC94P  PCTD1
IST080I P668B    PCTD1      P1402C    PCTD1      P45A2      PCTD1
IST080I P45A5CE  PCTD2      T45A5E00  INOP       T45A5E01  INOP
IST314I END
```

In this example, node M09 is in the PACDR state. To display the meaning of PACDR, use the STATUS command list.

PFKDEF (NCCF; CNME1010)

Syntax

PFKDEF



Purpose of Command

The PFKDEF command list sets PF keys for an operator station task (OST).

Operand Descriptions

membername

Indicates the name of a member which contains operands of the SET command. The last 8 characters of each line are ignored. To continue an operand specification to another line, end the line to be continued with a comma and continue the operand on the following statement. See sample CNMKEYS. See the *IBM Z NetView Customization Guide* for more information.

?

Indicates to display help information.

Usage Notes

Consider the following in defining key definitions:

- If an operator data set has been defined for DSIOPEN and it contains a member called CNMKEYSV which contains key definitions, these will be defined after those in *membername*. If the same key is defined in both members, the definition in CNMKEYSV is used. This enables key definitions saved for specific operators by the DISPFK command. See the DISPFK and OVERRIDE commands in the NetView online help for information about overriding general key settings.

Note: The definitions in CNMKEYSV are different from those in *membername*. Do not invoke PFKDEF CNMKEYSV.

- OPID() and SUPPCHAR() are special names that are resolved as if they were REXX control variables.
- Do not use SUPPCHAR() with DELAY-type keys.

Return Codes

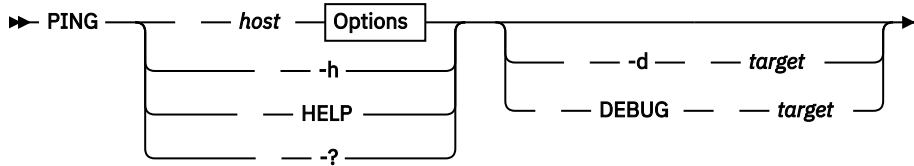
Return Code

Meaning

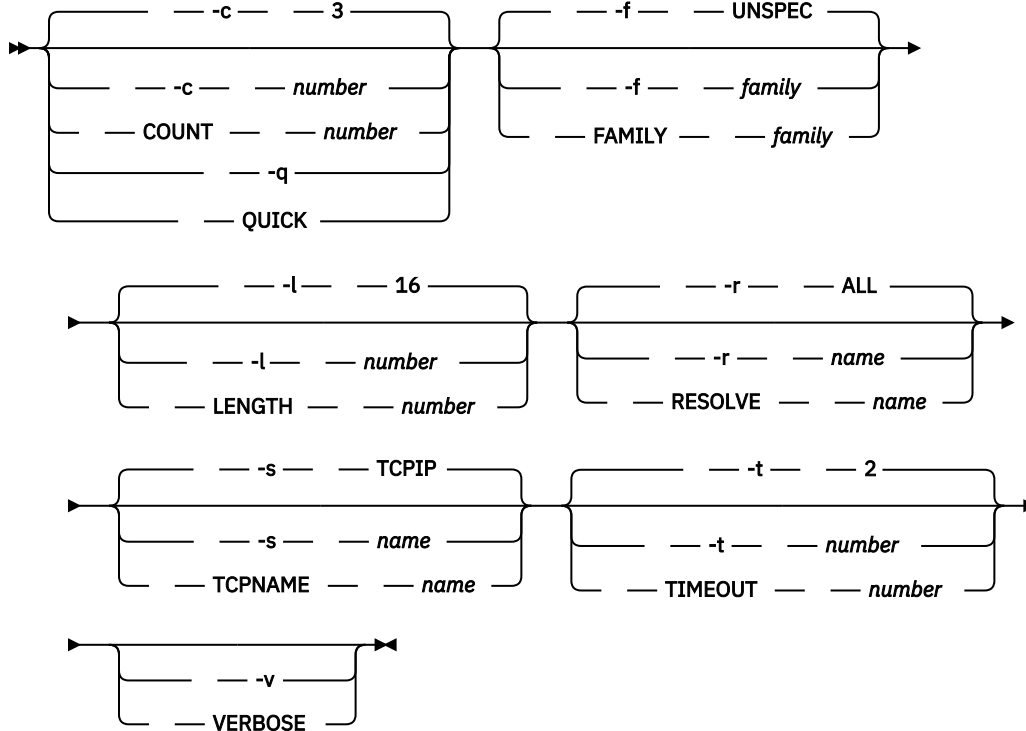
- | | |
|-----------|---|
| 0 | Successful completion. |
| 8 | Invoked from a task which does not support PF keys. |
| 12 | Extraneous parameters were given. |
| 16 | The <i>membername</i> is incorrect or contains incorrect key definitions. |

PING (NCCF; CNMEPING)

Syntax



Options



Purpose of Command

The PING command is used to send an ICMP echo request to an IP host, listen for responses, and report these responses.

Operand Descriptions

host

The IP host name or address of the target of the PING.

-c or COUNT

Specifies the number of ICMP echo requests (pings) to send to the target IP host. The default value is 3.

Note: If you are running this command on any platform other than NCCF, avoid using a value of 0 or using high values. If a value of 0 is specified, the PING command runs until it is reset. For large values, the output is displayed only when the command completes and this can be a very long time.

-d or DEBUG

Generates tracing information and details related to sent and received data packets.

Note: Previous releases of the NetView program also permitted DEBUG value specifications for ECHO, PING, and SOCKET. In this version, specification of any of these values does not cause an error condition, but results in a setting of **DEBUG ON**.

Valid values are as follows:

ALL or ON

Turns on debugging for all targets.

ARGS

Traces argument processing.

LOGECHO

Writes ping echoes to the NetView log. The message that is written to the log is BNH767I.

RESOLVE

Traces all operations relating to Domain Name Server (DNS) resolution of IP addresses and host names.

-f or FAMILY

Specifies the address family, indicating the IP networks in which the host running the NetView program and the target host are expected to reside. Valid values are as follows:

UNSPEC

Unspecified, meaning the NetView program and the target host can be in either an IPv4 network, an IPv6 network, or both. The PING command uses the first IP address returned by the resolver. UNSPEC is the default value.

When an IP address is specified for the target host, the PING command, for IPv4 addresses and IPv4-mapped IPv6 addresses, uses only an IPv4 network interface for determining connectivity. For an IPv6 address, only an IPv6 network interface is used for determining connectivity.

INET

Use only an IPv4 network interface to determine connectivity. This value is ignored if an IP address is supplied as the target host.

Note:

1. If an IPv4 address or IPv4-mapped IPv6 address cannot be found for the specified host name, the PING request fails.
2. If DEFAULT.IPV6ENV=ONLY, the PING request fails. See the *IBM Z NetView Administration Reference* for additional information about the DEFAULT.IPV6ENV statement.

INET6

Use only an IPv6 network interface to determine connectivity. This value is ignored if an IP address is supplied as the target host.

Note:

1. If an IPv6 address or IPv4-mapped IPv6 address cannot be found for the specified host name, the PING request fails.
2. If default DEFAULT.IPV6ENV=NONE, the PING request fails. See the *IBM Z NetView Administration Reference* for additional information about the DEFAULT.IPV6ENV statement.

-h or HELP or -?

Displays the online help for the PING command. All other operands are ignored.

-l or LENGTH

Specifies the length of ICMP data packets to be sent. The minimum length is 8 bytes and the maximum is 65487 bytes. The default is 16 bytes.

-q or QUICK

Specifies that only a single PING packet is to be sent to an IP host. See “Usage Notes” on page 54 for additional information. If either -q or QUICK is specified, a specification of -c or COUNT is ignored.

-r or RESOLVE

Specifies the level of DNS resolution to be employed when the command is started. Valid values are as follows:

ALL

DNS resolution of all IP host names or addresses is attempted. This is the default value.

NONE

No DNS resolution is ever attempted. The success of the command relies on the user entering a valid and reachable IP address.

-s or TCPNAME

Specifies the name of the IP stack in which to direct the PING request. The default value is specified by system z/OS Communications Server IP definitions.

-t or TIMEOUT

Specifies the effective time-to-live (TTL) in seconds on echo requests (pings) to be sent. The minimum value that can be specified is 1 second and the maximum value is 100 seconds. The default value is 2 seconds.

-v or VERBOSE

This option generates more output that can be useful when attempting to determine exactly why there are unexpected results from PING.

Usage Notes

Consider the following when using the PING command:

- The operands are not case-sensitive.
- The operands, including the host name or address, can be issued in any order.
- When you issue the PING command without any operands, a full-screen panel is displayed that you can use to ping a resource.
- If an IP address is specified and no host name can be determined through DNS resolution, the host name is specified in any output as *unresolved host name*. If a host name is specified which cannot be resolved to an address, the PING command ends.
- This command can be issued from the NetView command facility, from within a command list, or using CNMEPING as a REXX function or subroutine:
 - When issued from the command facility with the -q or QUICK option, a single echo request (ping) is sent and a single output message indicates the success or failure of that ping.
 - When run as a REXX external function or subroutine, the REXX result variable is set to 1 indicating success or 0 indicating failure. Therefore, CNMEPING() can be run as a REXX external Boolean function. In addition, no message is issued unless a syntax or other invocation error occurs.
 - Do not run PING commands in Data REXX.
- Running this command under a virtual OST (VOST) might result in a very long run time if there is a problem with IP domain name services. This delay is significantly reduced if a VOST is not used. Therefore, avoid running the PING command under a VOST when possible.

Return Codes

Return Code	Meaning
-------------	---------

0	The PING command completed successfully whether the IP host to which the echo request (ping) was sent replied or did not reply. However, if CNMEPING is called from a REXX command list as an external function or subroutine, a return code of 0 indicates that no valid echo reply was received from the target host.
---	---

1

Indicates the PING command completed successfully and the IP host to which an echo request (ping) has been sent has sent a valid echo reply. This return code only applies to calls of CNMEPING as an external function or subroutine from REXX.

4

Indicates that a processing error, such as a command syntax error, occurred. This return code is never returned when CNMEPING is invoked as an external function or subroutine from REXX.

Pinging an IP host name

The following example pings the fully qualified host name w3.olympus.ibm.com, using all defaults:

```
PING W3.OLYMPUS.IBM.COM
```

You receive a response similar to the following:

```
BNH765I Pinging OLYMPUS.TAMPA.ADVANTIS.COM at 164.120.77.170 with
3 packets of length 16 bytes
BNH767I 16 bytes received from 164.120.77.170: seq=0001 in 201ms
BNH767I 16 bytes received from 164.120.77.170: seq=0002 in 52ms
BNH767I 16 bytes received from 164.120.77.170: seq=0003 in 43ms
BNH769I 3 packets sent, 3 packets received, 0.00% packet loss
BNH770I Round trip times from 43 to 201 ms, averaging 122ms
```

Pinging an IP address with options

The following example pings an IP address ten times with 128-byte echo requests (pings), allowing five seconds for a valid echo reply to each request:

```
PING -C 10 9.8.0.1 -L 128 -T 5
```

You receive a response similar to the following:

```
BNH765I Pinging unresolved hostname at 9.8.0.1 with
10 packets of length 128 bytes
BNH767I 128 bytes received from 9.8.0.1: seq=0001 in 496ms
BNH767I 128 bytes received from 9.8.0.1: seq=0002 in 563ms
BNH767I 128 bytes received from 9.8.0.1: seq=0003 in 483ms
BNH767I 128 bytes received from 9.8.0.1: seq=0004 in 476ms
BNH767I 128 bytes received from 9.8.0.1: seq=0005 in 480ms
BNH767I 128 bytes received from 9.8.0.1: seq=0006 in 568ms
BNH767I 128 bytes received from 9.8.0.1: seq=0007 in 386ms
BNH767I 128 bytes received from 9.8.0.1: seq=0008 in 385ms
BNH767I 128 bytes received from 9.8.0.1: seq=0009 in 381ms
BNH767I 128 bytes received from 9.8.0.1: seq=0010 in 468ms
BNH769I 10 packets sent, 10 packets received, 0.00% packet loss
BNH770I Round trip times from 381 to 568 ms, averaging 474ms
```

Using PING in a REXX command list

In the following example, PING uses an external REXX Boolean function to test the availability of IP hosts 9.67.50.46 and 1.2.3.4:

```
/*TESTPING : Ping an IP host and see if it is up */
arg ipHost
if CNMEPING('-q' ipHost) then say 'OK'
  else address NETVIEW 'MESSAGE DSI042,' ipHost
exit 0
```

You receive a response similar to the following:

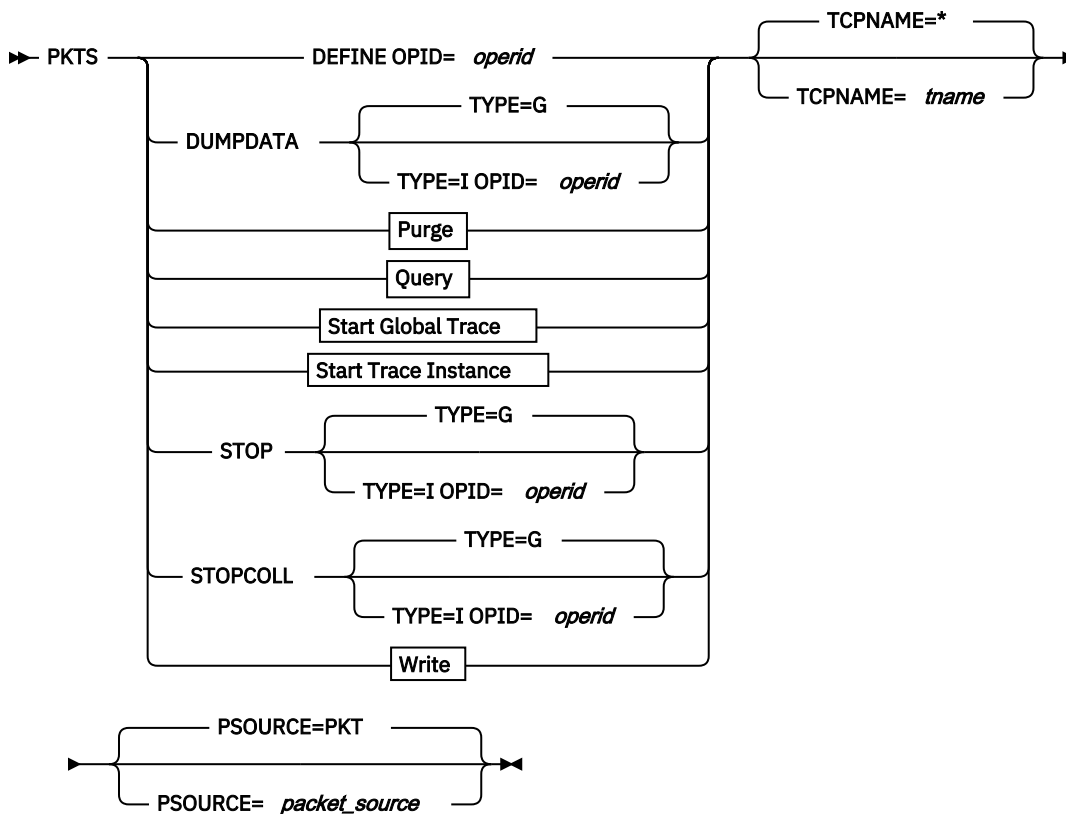
```
TESTPING 9.67.50.46
OK
```

Or:

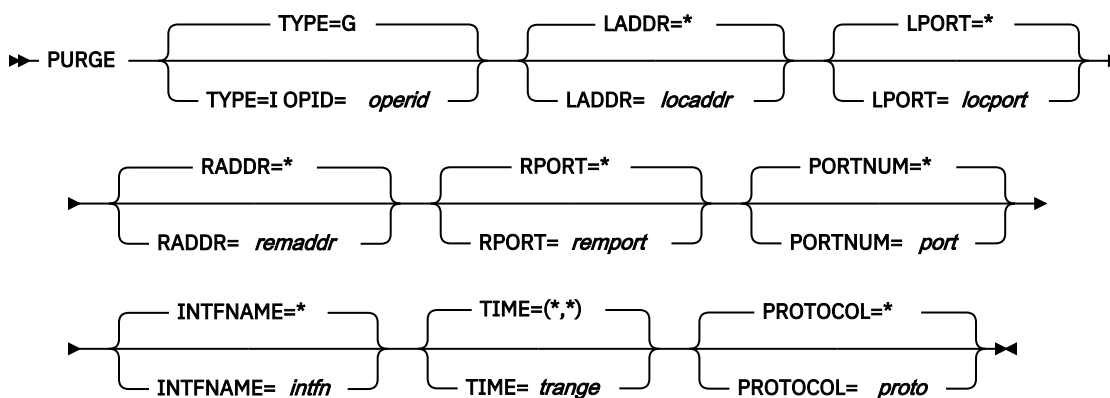
PKTS (NCCF)

Syntax

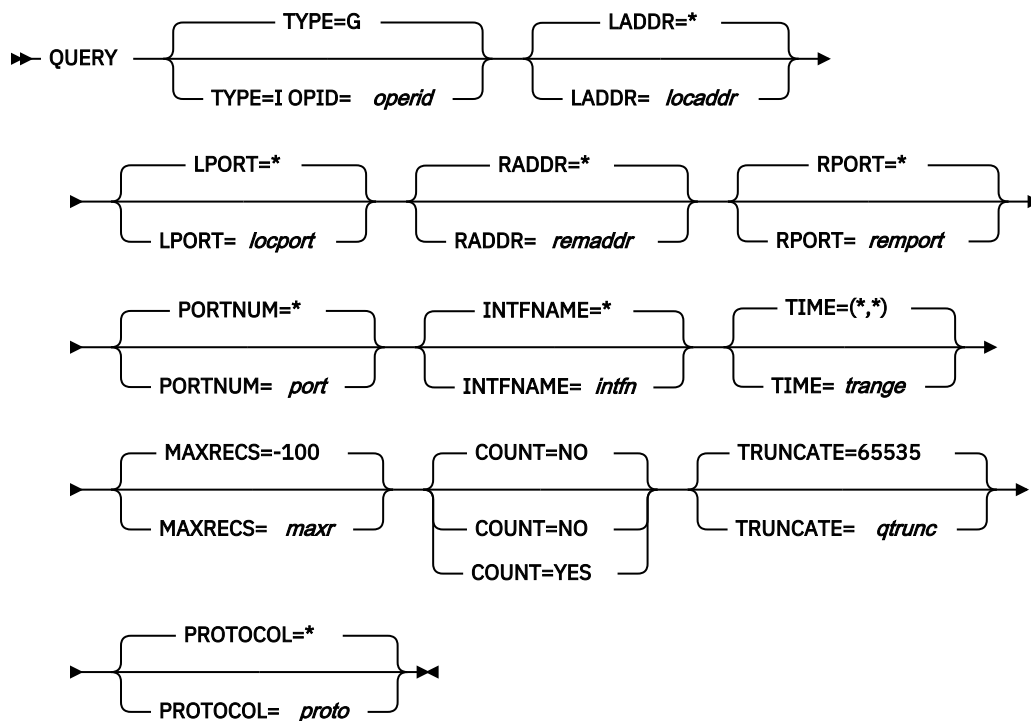
PKTS



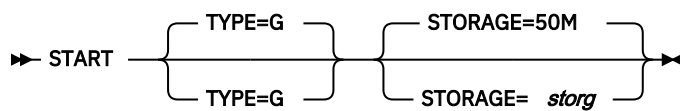
Purge



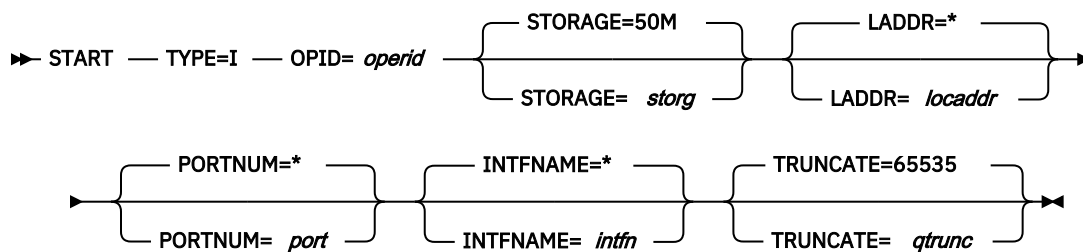
Query



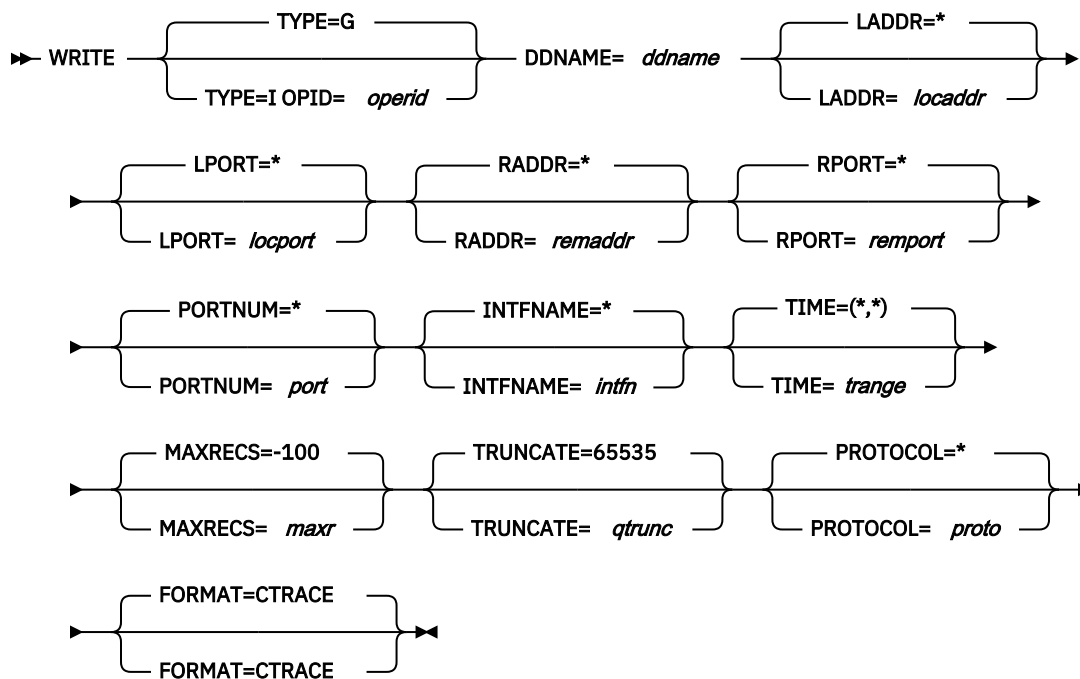
Start Global Trace



Start Trace Instance



Write



Purpose of Command

The PKTS command is used to control the collection of packet data and to view the collected data.

Operand Descriptions

DDNAME=*ddname*

Indicates the 1- to 8-character name that is associated with the data set to which the packet trace records are to be written. If you allocate this data set with the ALLOCATE command, either you can use the FILE parameter to specify the DD name or, if the FILE parameter is not specified, the DD name is assigned by the NetView program. This data set must be allocated before the PKTS command is issued to write the packet trace data. The following parameter values are recommended for the data set:

- Block size: BLKSIZE(27994)
- Logical record length: LRECL(27990)
- Record format: RECFM(VB)

If you use the IPTRACE panels to write the data, the data set is allocated automatically by the NetView program; for more information, see *IBM Z NetView IP Management*. To use the IPTRACE panels, enter the IPTRACE command without any parameters.

DEFINE

Associates a stack name and source type with the name of an autotask that collects packet data for the stack. This is the same function performed by the FUNCTION.AUTOTASK.PKTS or FUNCTION.AUTOTASK.OPKT statement in the CNMSTYLE member. The PKTS START command is then required to start the function. Data collection for the specified stack and source combination stops if it is redefined to another autotask. The same OPID operand can be defined only to one stack and source, and to either TCPCONN or PKTS.

A value of NONE undefines the specified stack and source combination.

DUMPDATA

Supports diagnostic information. Use this keyword when instructed to do so by IBM Software Support.

FORMAT=CTRACE

Indicates the format for writing the packet trace data to the specified data set. CTRACE is the only supported format.

INTFNAME=intfn

Specifies the interface name as defined by z/OS Communications Server. A single asterisk (*) indicates all interfaces. You can also use an asterisk as a wild card at the end of the interface name; for example, ABC* matches any interface name beginning with the letters ABC.

LADDR=locaddr

Specifies the local IP address (or set of addresses) for a QUERY or PURGE command. This address can be expressed in any of several possible formats:

- An IPv4 address in dotted-decimal format: ddd . ddd . ddd . ddd. Each ddd can be any of the following:
 - A decimal number from 0 to 255
 - A hyphen-separated range (such as 240-255)
 - An asterisk (*), representing the range 0 - 255

Leading zeros can be omitted. If the last ddd is an asterisk, and fewer than four ddd values are specified, the range 0 - 255 is assumed for each remaining ddd.

- An IPv6 address in colon-hexadecimal format: hhhh : hhhh : hhhh : hhhh : hhhh : hhhh : hhhh : hhhh, where each hhhh is a 0-4 digit hexadecimal number, or a hexadecimal range separated by a hyphen, such as FF00-FFFF. Consecutive groups of zeros can be replaced with a double colon (::). A double colon can be used to signify leading, trailing or embedded groups of zeros, and can be specified only once in an address. A single asterisk (*) can be used in place of hhhh to denote 0-FFFF. If the last hhhh is an asterisk and less than 8 hhhhs are specified, 0-FFFF is assumed for each remaining hhhh. If both an asterisk and a double colon are used in an address, the asterisk represents only a single hhhh group regardless of its position.
- A single asterisk (*), representing all local IP addresses.
- A TCP/IP symbolic host name. The NetView program will attempt to translate to an IP address using the TCPNAME operand, if the operand is specified without a wildcard. Otherwise, it uses the value for TCPNAME in the CNMSTYLE member. If a host name resolves to multiple IP addresses, the QUERY or PURGE command is attempted for all of these addresses.

Note: If an IPv6 address is specified in colon-hexadecimal format:

hhhh : hhhh : hhhh : hhhh : hhhh : hhhh : hhhh : hhhh, the NetView program inspects the high-order 12 bytes of the 16-byte address. If these bytes equate to either 0 or X'FFFF', the NetView program strips the high-order 12 bytes and processes the remaining 4 bytes as an IPv4 address of the dotted-decimal format ddd . ddd . ddd . ddd. If anything other than 0 or X'FFFF' is specified in the high-order 12 bytes of the IPv6 address, such a specification is treated as a host name.

These are examples of valid IPv4 address specifications:

```
::*
::FFFF:*
9.42.44.52
::FFFF:9.42.*
```

These are examples of valid IPv6 address specifications. Because the high-order 12 bytes are either 0 or X'FFFF', the NetView program recognizes these as an IPv6 specification and uses the low-order 4 bytes as an IP address.

```
00:000:0000:0:000:FFFF:9.42.44.52 (the high-order portion is
stripped off, and 9.42.44.52 is used as a valid IPv4) address
::0:0:FFFF:9.42.44.52 (the high-order portion is stripped off,
and 9.42.44.52 is used as a valid IPv4) address)
```

The NetView program treats these specifications as host name specifications:

```
myhostname-9-42-33-52 (valid host name specification)
::1:FFFF:9.42.44.52 (because this is not a valid IP address)
```

```
specification; NetView will treat this as a host name)
0:0:0:0:0:FFFF:9.42.33- (because this is not a valid IP address
specification; NetView will treat this as a host name)
```

The LADDR parameter is not valid if PSOURCE=OSA is specified.

LPORT=locport

Specifies the local port number. The *locport* value can be either a decimal number or a single asterisk (*), representing all ports.

The LPORT parameter is not valid if PSOURCE=OSA is specified.

MAXRECS=maxr

Specifies the maximum number of packet records (data lines) to return from PKTS QUERY. The *maxr* value is a number between -9999999 and 9999999 (do not insert commas or periods). Connections are always listed in chronological order. A positive value specifies the set of records beginning with the oldest matching connection; a negative value specifies the set of records ending with the most recent matching connection. The default value is -100.

OPID=operid

Specifies the name of the autotask that collects packet information for the associated stack and source combination.

For TYPE=I, this parameter is required.

PORTNUM=port

Specifies the port number, which is matched against both local and remote port numbers. The *port* value can be either a decimal number or a single asterisk (*), representing all ports.

The PORTNUM parameter is not valid if PSOURCE=OSA is specified.

PROTOCOL=proto

Specifies an IP protocol by name or number from 0 to 255 as defined in the IP architecture. Supported names are TCP, UDP, and OSPF. The default value is a single asterisk (*), meaning "all".

The PROTOCOL parameter is not valid if PSOURCE=OSA is specified.

PSOURCE=packet_source

Specifies the packet source as PKT (the default) or OSA. The z/OS Communications Server defines separate interfaces for each. For more information, see the documentation of the NETMONITOR profile statement in *z/OS Communications Server: IP Configuration Reference* at http://publib.boulder.ibm.com/infocenter/zos/v1r13/topic/com.ibm.zos.r13.halz001/netmon.htm?path=8_6_4_43#netmon. For this NetView command, PKT corresponds to NETMONITOR PKTTRCSERVICE, and OSA corresponds to NETMONITOR NTATRCSERVICE.

PURGE

Purges packet data records matching the input criteria. If the purge is successful, the BNH774I message is returned.

QUERY

Queries packet data records matching the input criteria. If the QUERY is successful, the BNH773I message is returned.

RADDR=remaddr

Specifies the remote IP address (or set of addresses) for a QUERY or PURGE. See the description of the LADDR keyword for information about how to specify an IP address.

The RADDR parameter is not valid if PSOURCE=OSA is specified.

RPORT=remport

Specifies the remote port number. *remport* can be either a decimal number or a single asterisk (*), representing all ports.

The RPORT parameter is not valid if PSOURCE=OSA is specified.

START

Starts a long-running process to collect packet data from the z/OS Communications Server for the specified stack and source combination. The data-collection process persists as an autotask defined

by a previous PKTS DEFINE command, or a FUNCTION.AUTOTASK.PKTS or FUNCTION.AUTOTASK.OPKT statement in the CNMSTYLE member.

STOP

Stops the collection of packet data for the specified stack and source combination. The PKTS STOP command is used to end the data collection started by PKTS START command. After the PKTS STOP command is issued, the QUERY and PURGE commands are no longer valid for the stack.

STOPCOLL

Stops the collection of packet data for the specified stack and source combination while keeping the data already collected available. Use PKTS STOPCOLL instead of PKTS STOP if you still want to be able to run PKTS QUERY or PKTS PURGE against the stack.

STORAGE=*storg*

The number of megabytes of data space storage to allocate. This value must be a positive or negative integer, followed by the letter M. A positive value indicates that the packet storage wraps. That is, when the storage area is full, arriving packets cause the oldest packets to be purged. A negative value indicates that the packet storage does not wrap. That is, when the storage area is full, the collection process stops, as in the STOPCOLL command. The minimum integer value is 16 MB, and the maximum integer value is 2047 MB. The default value is 50 MB.

TCPNAME=*tname*

Specifies the name of the TCP/IP stack that is associated with this request.

For TYPE=I, this parameter is required for the START, STOP, and STOPCOLL actions.

For TYPE=G, for requests other than DEFINE and QUERY, you can use an asterisk (*) as a wildcard character at the end of the name or as the only specified character, which indicates all stacks specified by a previous PKTS DEFINE command or in the CNMSTYLE member as a FUNCTION.AUTOTASK.PKTS or FUNCTION.AUTOTASK.OPKT statement.

A wildcard value, entered explicitly or by default, is supported for QUERY if it matches exactly one defined stack and source combination. Otherwise you must issue separate QUERY commands for each stack and source combination.

Wild cards are not supported for DEFINE or for TYPE=I traces.

TIME=*trange*

Specifies the range of times for packets to be included in a QUERY or PURGE command. *trange* consists of two values separated by a comma and enclosed in parentheses; the first value specifies the beginning date and time for the range, and the second value specifies the ending date and time for the range. The following rules apply to these values:

- To specify a date and time, type the date followed by the time, separated by a blank space. The formats of the date and time are controlled by the DEFAULTS and OVERRIDE commands.
- If either value consists only of a date, the current time for that date is used.
- If either value consists only of a time, the assumed date depends upon the specified time. If the time is later than the current time, yesterday's date is assumed; if the time is earlier than the current time, today's date is assumed.
- An asterisk before the comma will include all packets from the beginning of data collection. An asterisk after the comma will include all packets up to the present.
- Either value can also be specified as a 1-character hexadecimal store-clock value. This can be used to continue a previous query by specifying a previously returned store-clock value from the BNH773I message.

The following examples show valid values for *trange*:

- (04/05/04 10:00:00,04/06/04 17:00:00)
- (*,04/05/04)
- (B7ACDD41D4F00A01,*)

TRUNCATE=*qtrunc*

The maximum number of bytes output for each packet in this QUERY response. This includes the CTE and z/OS Communications Server headers (see message BNH7731). The smallest allowable *qtrunc* value is 144 (the length of the above headers plus an IPv6 header). The largest value is 65535, which means truncation is not to occur. This is the default value.

Note: If you intend to format the resulting packets (see HELP FMTPACKT) the default truncation value is recommended.

TYPE=G|I

The type of trace, either global (G) or a specific trace instance (I). The default is G.

WRITE

Saves the requested packet trace records in CTRACE format so that the saved records can be used as input to Interactive Problem Control System (IPCS). If the WRITE is successful, the DSI633I message is displayed.

Restrictions

Collection of trace data requires that all appropriate z/OS Communications Server definitions are in place, for example NETMONITOR PKTTRCSERVICE and NETMONITOR NTATRCSERVICE. (See *IBM Z NetView Installation: Configuring Additional Components* for more information.) Without this support enabled, the PKTS command returns an error message.

The WRITE option cannot be used with data from a NetView program that is earlier than Version 6.2.

Return Codes**Return Code****Meaning****0**

Successful processing.

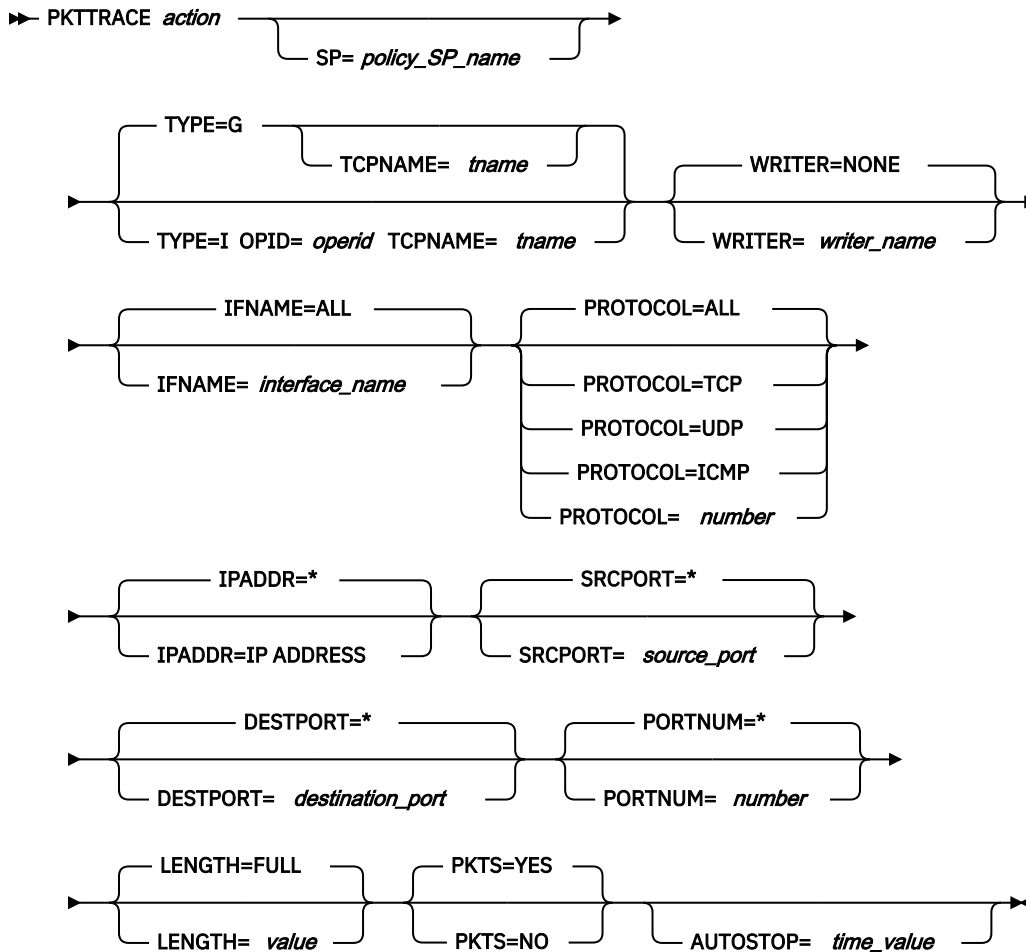
4

The command did not complete successfully. Check the accompanying messages for more information.

PKTTRACE

Syntax

PKTTRACE



IBM-Defined Synonyms

Command or Operand	Synonym
ALL	*
*	ALL

Purpose of Command

The PKTTRACE command starts or stops packet traces. This command supports two types of packet traces:

- Global trace on a stack, which uses the SYSTCPDA component trace and VARY TCPIP commands to set filters. This type of trace, which is the default, is specified with the TYPE=G parameter. The PKTTRACE TYPE=G command determines what components are inactive when a start request is made and starts the needed components.
- Multiple independent traces on a stack. This type of trace is specified with a TYPE=I parameter.

An AUTOSTOP option starts a timer that terminates the trace after a specified time. The PKTTRACE command can be called from a REXX program, from a timer, or automatically.

Operand Descriptions

action

Required. Must be the first parameter. These are the options that can be specified.

START

For TYPE=G: Starts the trace component SYSTCPDA if it is not already active and starts the requested Writer at the same time. If the PKTS task is not active, it becomes active unless PKTS=NO is specified. Issue the MVS VARY commands to set filters as requested.

For TYPE=I: Calls the PKTS command (with TYPE=I) to start a new trace instance. For the START action, the TCPNAME parameter is optional.

STOP

For TYPE=G: Issues the command to stop trace filters as requested.

For TYPE=I: Calls the PKTS command (with TYPE=I) to stop data collection for the trace. Data collection can be resumed with a new START action that specifies the OPID and TCPNAME values of the existing trace.

STOPALL

For TYPE=G: Issues the command to stop trace filters as requested and to stop the SYSTCPDA trace component, which also stops any active Writer.

For TYPE=I: Calls the PKTS command (with TYPE=I) to end the trace instance. After a trace is ended, the trace data is no longer available for viewing.

AUTOSTOP

When you start the packet trace using the START action, you can specify a value for this parameter to allow you to automatically set a timer to stop packet tracing after a specified time value.

These are valid time_value specifications:

- Minutes only. Must be 1 or 2 digits, in the range 1 - 59.

```
AUTOSTOP=30
```

- Seconds only. Must be 3 characters in length; the first character must be a colon (:) followed by 2 digits. The minimum value you can specify is 15.

```
AUTOSTOP=:45
```

- Minutes and Seconds, in the format *m:ss* or *mm:ss*, where *mm* must be in a range of 0 - 59 and *ss* must be in a range of 0 - 59. You cannot specify the value 00:00.

```
AUTOSTOP=1:20  
AUTOSTOP=15:30
```

- Hours, minutes, and seconds, in the format *hh:mm:ss*, where *hh* can be in a range 0 - 99, *mm* must be in a range of 0 - 59, and *ss* must be in a range of 0 - 59. The minimum value that you can specify is 00:00:15.

```
AUTOSTOP=1:10:10
```

DESTPORT

The destination (remote) port for which packets are to be collected. The default is * (all ports). For TYPE=I, this parameter cannot be specified.

IFNAME

Interface to trace packets for. The default is ALL or *.

IPADDR

The IPAddress (remote host) for which packets are to be collected. The default is * (all interfaces).

LENGTH

The record length of the packet data that is captured. The default is FULL.

OPID

The name of the autotask for which the trace was started. This parameter is used only for TYPE=I. For the START action, this parameter is optional; if it is not specified, one of the autotasks that is defined in the CNMSTYLE member is assigned.

PKTS

This parameter controls whether the PKTS task is to be started. If NO, the PKTS task is not started on a START request. For TYPE=I, this parameter cannot be specified. For TYPE=G, this parameter is ignored on STOP and STOPALL actions.

PORTNUM

Collect packets for this port regardless of whether it is the source or destination. This parameter overrides SRCPORT and DESTPORT. The default is * (all ports).

PROTOCOL

The protocol type for which to collect packets. You can specify a protocol name or a protocol number, where *number* is a valid IP Protocol 1 - 256. You can also specify TCP, UDP, or ICMP. The default value is ALL or asterisk (*).

SP

Name for the stack as defined in POLICY. This is used to determine Domain and TCPNAME. If the stack name is not specified, The PKTTRACE command attempts to use the LOCAL SP from CNMPOLCY.

SRCPORT

The source (local) port for which packets are to be collected. The default is * (all ports). For TYPE=I, this parameter cannot be specified.

TCPNAME

The name of the TCP/IP stack that is associated with this request. For TYPE=I, this parameter is optional for the START action.

TYPE

The type of trace, either global (G) or a specific trace instance (I). The default is G.

WRITER

Name of the writer job to be started when starting the SYSTCPDA packet trace component. The default is *NONE* (no writer). For TYPE=I, this parameter cannot be specified. For TYPE=G, this parameter is ignored on STOP and STOPALL actions.

Return Codes**Messages and Return Codes****Meaning****Message FKX400 Return Code 0**

Trace Schedule completed successfully

Message FKX403 Return Code 0

Trace stopped

Message FKX403 Return Code 1

The FKXETRS1 call failed (error message passed from FKXETRS1)

Message FKX437 Return Code 3

Invalid trace parameters

Message EZL203 Return Code 4

Invalid Action parameter or No Local Service Point found

Message FKX470 Return Code 5

Invalid Service Point

Message FKX413 Return Code 6

No TCPNAME defined on Service Point

Message EZL271 Return Code 7

REXX No Value error

Message EZL275 Return Code 8

REXX syntax Error

Message FKX414 Return Code 11

Trying to allocate writer but SYSTCPDA is already active

Message FKX402 Return Code 16

Trace command failed or Comm Server messages

Message EZL249 Return Code 18

Obeyfile processing failed

Message FKX462 Return Code 20

Trace started, but autostop failed

Message FKX461 Return Code 99

Trace started, but autostop was blocked by Security

Using a timer to schedule a packet trace

You can issue the PKTTRACE command from a timer to capture a packet trace at a specific time. This is an example:

```
CHRON AT=(23:00:00),ROUTE=AUTO1,ID=TESTIT2,COMMAND='PKTTRACE START
IPADDR=9.42.45.101 SRCPORT=4022 PROTOCOL=TCP AUTOSTOP=10'
```

This example creates a timer called TESTIT2 to run at 11:00 p.m. on operator AUTO1. When the timer expires, it runs the PKTTRACE command to start a packet trace for IP address 9.42.45.101, local port of 4022, and protocol type of TCP. When PKTTRACE runs at 11:00, it also schedules an autostop for the trace 10 minutes later.

Calling PKTTRACE from a REXX program

You call the PKTTRACE command from a REXX program by building the command parameters and issuing the PKTTRACE command directly or in a NetView Pipeline. When the PKTTRACE command (FKXETRP1) is called from a REXX program, all messages are returned to the calling program in a NetView safe named PKTT01. This is a partial example:

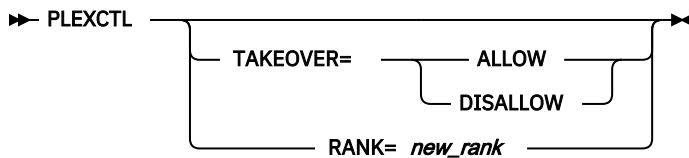
```
/* */
.
.
.
SP = 'THISHOST'
Protocol = 'TCP'
.
.
.
PKTTRACE START SP=SP,
'PROTOCOL=Protocol,
'IPADDR=RemIP,
'SRCPORT=LPort,
'DESTPORT=RPort

If RC = 0 then
Do /* the trace is started, perform additional processing */
.
.
.
End
Else /* a non zero RC means there was a problem starting */
Do /* trace, show the errors and leave */
'WINDOW TITLELINE %UNABLE TO START PACKET TRACE% ',
'PIPE Safe PKTT01 SEIZE | CONS'
End
```

FKXE221C is a NetView REXX function that also contains an example of how to call this command.

PLEXCTL (NCCF)

Syntax



Purpose of Command

You can use the PLEXCTL command to change the rank of the NetView program in the DSIPLXnn group in which it participates. If the NetView program is the group master, the command can also be used to control whether the master NetView program will allow another NetView program to assume the master role in the sysplex group.

Command processing will check that the requested change is not already in effect. For example, if PLEXCTL RANK=MASTER is specified and the NetView program is already a master NetView program, command processing stops and a BNH559E message with a reason code of 1 results.

Operand Descriptions

TAKEOVER

Specifies whether a master NetView program will allow another NetView program to take over as master, either because of an operator command or because a system-defined master NetView program became active. The TAKEOVER option is only valid when issued on a master NetView program. The following values are valid:

ALLOW

Specifies that another NetView program can take over. This is the normal setting when a NetView program assumes the master role.

DISALLOW

Specifies that another NetView program cannot take over.

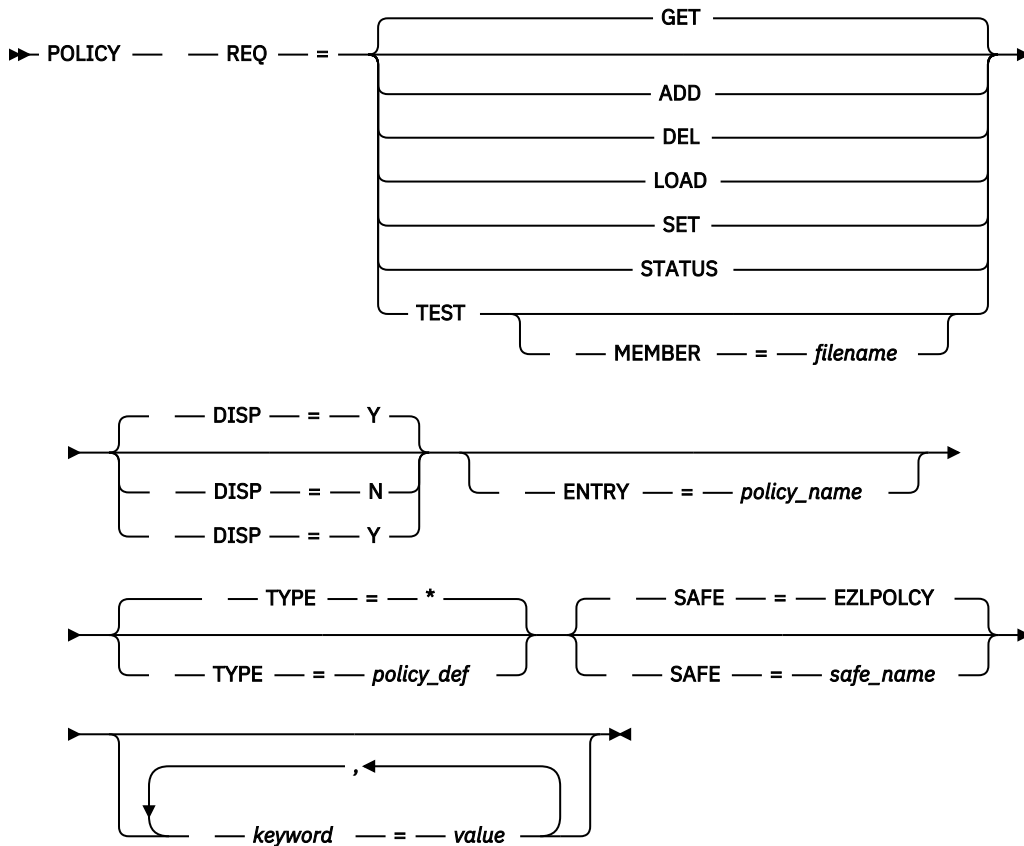
RANK

Changes the current rank of this NetView program in the sysplex. The value can range from 0 (basic NetView) to 250 (master NetView). The change in rank results in the user state field being updated, which drives exits at the other NetView programs in the group. If the specified rank is 250, this NetView program takes over as master assuming that there is not another master in the group that is disallowing takeovers. The character string MASTER can be used as a synonym for 250, and the character string BASIC can be used as a synonym for 0. The character string MCAP can be used as a synonym for 1.

POLICY (NCCF)

Syntax

POLICY



Purpose of Command

The POLICY command manages which policy files are loaded into the Policy Repository and performs actions on those policy definitions. The POLICY command is a multi-purpose generic Application Programming Interface (API) into the Policy Repository. The POLICY command provides standardized access to all policy loaded into the Policy Repository. Some applications might ship more specific interfaces. For example, you can use the SETAUTO command to manage just the AON RECOVERY policy. When using SETAUTO, you do not see any other policy definitions even if they are loaded. Application-specific interfaces are documented in the appropriate user's guide.

Operand Descriptions

ADD

Creates a new policy definition in the Policy Repository with provided keywords and values.

DEL

Deletes a policy definition from the Policy Repository.

GET

Retrieves the requested policy definition from the Policy Repository. This is the default.

LOAD

Loads the Policy Repository based on the CNMSTYLE member definitions.

SET

Updates the requested policy definition keyword with a value.

STATUS

Queries which policy file or files have been loaded in the Policy Repository.

TEST

Performs a syntax check of the policy file or files.

MEMBER=filename

The name of the actual policy file to test. If not specified on a TEST request, then all of the currently active policy files are tested.

DISP

Whether to display messages at the user console.

N

Does not display messages.

Y

Displays messages. This is the default.

ENTRY=policy_name

Any valid policy name, such as RECOVERY or NMCSTATUS, as defined in the *IBM Z NetView Administration Reference* or by other applications.

TYPE

The type of policy definition to return.

Returns all policy definitions for a given *policy_name*. This is the default.

policy_def

Any valid policy definition, such as HOLIDAY, as defined in the *IBM Z NetView Administration Reference* or by other applications.

SAFE=safe_name

The name of a safe containing the output from the request.

keyword

Any policy keyword allowed by the policy application, such as NOAUTO.

value

Any keyword value allowed by the policy application.

Usage Notes

- You can have one or more *keyword=value* pairs.
- You can specify TYPE=* to retrieve all policy definitions for a given policy grouping. For example, POLICY REQ=GET ENTRY=RECOVERY TYPE=* will return all RECOVERY policy definitions from all policies.
- You cannot delete (REQ=DEL) keywords or keyword values, only specific policy definitions.
- You cannot query (REQ=GET) keywords or keyword values, only specific policy definitions.
- No parameters are allowed with REQ=STATUS or REQ=LOAD.
- If MEMBER= is not specified for a REQ=TEST then all currently active policy files are syntax tested based on the current policy loaded in the Policy Repository.
- User-written applications that drive 3270 panels must use DISP=N to avoid messages interrupting the application.
- User-written applications that are querying or changing policy definitions can use the default value of DISP=Y.

Return Codes**-1**

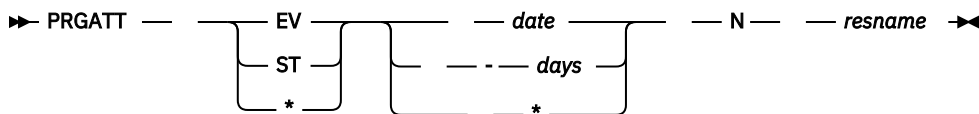
SIGNAL FAILURE

- 5 SIGNAL HALT
- 0 Request was successful.
- 1 Requested policy definition not found (GET/ADD/SET).
- 3 Missing Parameters — look for message EZL203I
- 4 Parameters not valid - look for message EZL204I.
- 7 SIGNAL NOVALUE- look for message EZL271E.
- 8 SIGNAL SYNTAX - look for message EZL275E.
- 9 Security Authorization Failure - look for message EZL228E.
- 10 Request not processed - other error encountered.

PRGATT (NPDA)

Syntax

PRGATT



Purpose of Command

The PRGATT command removes event or statistical data for the specified resource and all the resources that are attached to it and known to the hardware monitor.

If you issue this command from a non-hardware monitor screen or from a command list, the response to this command is sent to the command facility screen and the messages are logged. The messages received are the same as if the command was issued under the PPT.

This command can be useful if you rename your network control program (NCP). When generating a new NCP, a new set of hardware monitor database records is recorded for each resource attached to the NCP. With this command, you can purge all database records for the specified resource and all attached resources. You cannot process multiple PRGATT commands concurrently.

If the name of the resource is not associated with a unique resource configuration on the database, a selection panel is displayed on which you can choose which configuration is relevant.

Operand Descriptions

EV

Specifies event data.

ST

Specifies statistical data.

*

Specifies both event and statistical data.

date

Deletes all data recorded before this date. The format of *date* is controlled by the setting of the date operands of the DEFAULTS and OVERRIDE commands.

-days

Deletes data that is older than the specified number of days (0–365).

Deletes data regardless of age.

N

Specifies that the operand that follows is a resource name.

resname

Specifies the symbolic name of the resource. You can specify up to five resource names to fully qualify the resource for which data is to be displayed.

Restrictions

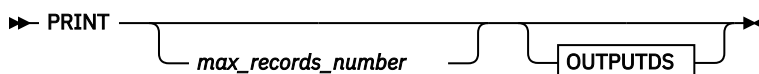
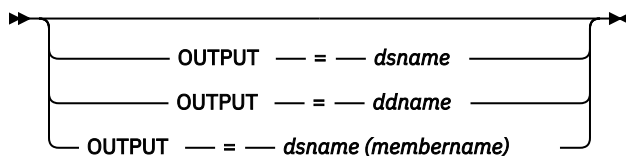
To reclaim the space made available by the PRGATT command, use the DBAUTO NPDA,REORG command. If the REORG is not done, the free space cannot be reused.

Example: Erasing statistical data for NCP001 and attached resources

To erase all statistical data for NCP001 and all resources attached to it, enter:

```
PRGATT ST * N NCP001
```

PRINT (BROWSE)

Syntax**OUTPUTDS****Purpose of Command**

The Canzlog BROWSE subcommand PRINT enables you to print Canzlog messages by using the filter specification that is specified on the Canzlog BROWSE command.

Operand Descriptions**max_records_number**

Specifies the maximum number of messages that can be printed to an output file. The valid range is 1 to 4000000.

OUTPUT

Specifies the data set name, data set and member names, or DD name to which the Canzlog messages will be printed.

This operation will overwrite the original content of the sequential data set or member in a partitioned data set. If a data set is full, after you execute the PRINT command, data will fail to be re-written into

the data set in use. In this case, use the `FREE FILE()` command to release the data set. You can use the `LISTA` command to obtain the DD name as the value for `FILE()`.

ddname

Specifies a DD name that is associated with a data set or spool file. The DD name may have been defined as a part of customizing NetView startup JCL or defined with a Command Facility `ALLOCATE` command. The *ddname* can be 1 - 8 characters long.

dsname

Specifies the location to which the Canzlog PRINT output will be written. The data set must observe the following rules:

- Pre-allocated by the user before issuing the PRINT command
- Fully qualified sequential data set name
- Fully qualified partitioned or partitioned extended data set if a member name is to be specified

For more information about allocating the data set, see the usage notes for an example of those parameters.

membername

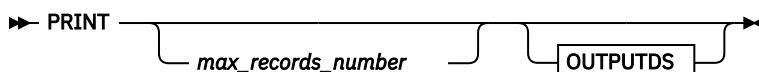
Specifies the member name of the data set to print Canzlog messages.

Usage Notes:

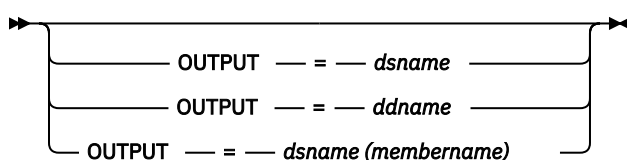
- If any parameter other than *max_records_number* or the output keyword is specified, the PRINT (NCCF) command will be issued as if it is invoked from the NetView 3270 console. For more information about the PRINT (NCCF) command, see the NetView online help or the *IBM Z NetView Command Reference Volume 2 (O-Z)*.
- If the *max_records_number* parameter is not specified on the command, and is specified on the `CZ.PRINT.MAXRECS CNMSTYLE` statement, the value on the CNMSTYLE statement will be used. If no value is specified at all, the filter specification will determine the number of messages that are printed. If the *OUTPUT* parameter is not specified on the command, and is specified on the `CZ.PRINT.OUTPUT CNMSTYLE` statement, the value on the CNMSTYLE statement will be used. If no value is specified at all, the error message DSI649I will be displayed.
 - All other PRINT parameter specifications are defined by the CZ CNMSTYLE statements. For more information, see the *IBM Z NetView Administration Reference*.
- Single-line messages are automatically wrapped based on the record length of the data set. The maximum record length that is accepted by the PRINT command is 32,000. If the record length that the users set exceeds 32,000, the PRINT command uses 32,000 as its record length to wrap the messages. The PRINT command does not wrap multi-line messages.
- The PRINT command is only supported for Canzlog data that is displayed by the BROWSE command.

PRINT (CANZLOG)

Syntax



OUTPUTDS



Purpose of Command

The CANZLOG subcommand PRINT enables you to use the CANZLOG panel filter specification to print Canzlog messages.

Operand Descriptions

max_records_number

Specifies the maximum number of messages that can be printed to an output file. The valid range is 1 to 4000000.

OUTPUT

Specifies the data set name, data set and member names, or DD name to which the Canzlog messages will be printed.

This operation will overwrite the original content of the sequential data set or member in a partitioned data set. If a data set is full, after you execute the PRINT command, data will fail to be re-written into the data set in use. In this case, use the `FREE FILE()` command to release the data set. You can use the `LISTA` command to obtain the DD name as the value for `FILE(ddname)`. For more information about `FREE FILE()`, see *FREE (NCCF)* in the *IBM Z NetView Command Reference Volume 1 (A-N)*.

ddname

Specifies a DD name that is associated with a data set or spool file. The DD name may have been defined as a part of customizing NetView startup JCL or defined with a Command Facility `ALLOCATE` command. The DD name can be in the range 1 - 8 characters.

dsname

Specifies the location to which the Canzlog PRINT output will be written. The data set must observe the following rules:

- Pre-allocated by the user before issuing the PRINT command
- Fully qualified sequential data set name
- Fully qualified partitioned or partitioned extended data set if a member name is to be specified

For more information about allocating the data set, see the usage notes for an example of those parameters.

membername

Specifies the member name of the data set to print Canzlog messages.

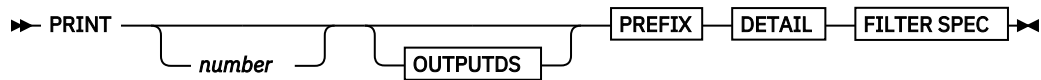
Usage Notes:

- If any parameter other than *max_records_number* or the output keyword is specified, the PRINT (NCCF) command will be issued as if it is invoked from the NetView 3270 console. For more information about the PRINT (NCCF) command, see the NetView online help or the *IBM Z NetView Command Reference Volume 2 (O-Z)*.
- If the *max_records_number* parameter is not specified on the command, and is specified on the `CZ.PRINT.MAXRECS CNMSTYLE` statement, the value on the `CNMSTYLE` statement will be used. If no value is specified at all, the filter specification will determine the number of messages that are printed. If the `OUTPUT` parameter is not specified on the command, and is specified on the `CZ.PRINT.OUTPUT CNMSTYLE` statement, the value on the `CNMSTYLE` statement will be used. If no value is specified at all, the error message DSI649I will be displayed.
 - All other PRINT parameter specifications are defined by the `CZ CNMSTYLE` statements. For more information, see the *IBM Z NetView Administration Reference*.
- Single-line messages are automatically wrapped based on the record length of the data set. The maximum record length that is accepted by the PRINT command is 32,000. If the record length that the users set exceeds 32,000, the PRINT command uses 32,000 as its record length to wrap the messages. The PRINT command does not wrap multi-line messages.

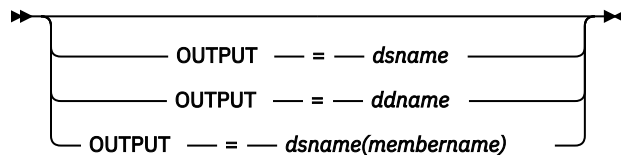
PRINT (NCCF)

Syntax

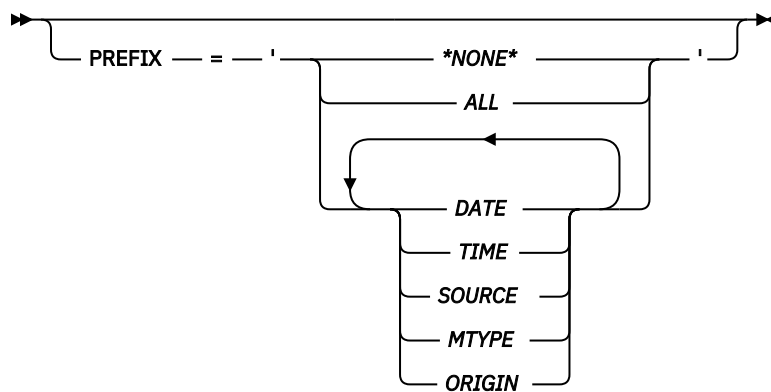
PRINT



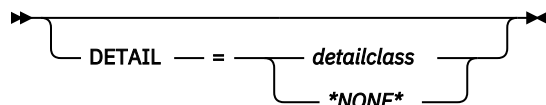
OUTPUTDS



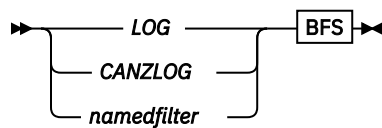
PREFIX



DETAIL



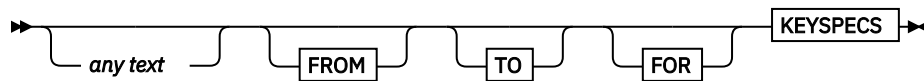
FILTER SPEC



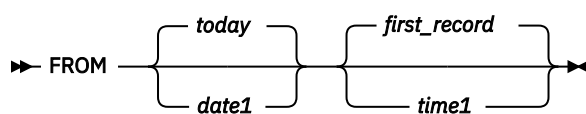
BFS

Diagram illustrating the BFS (Basic Filter Syntax) structure:

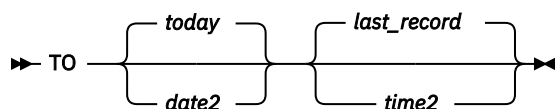
BFS (Basic Filter Syntax)



FROM



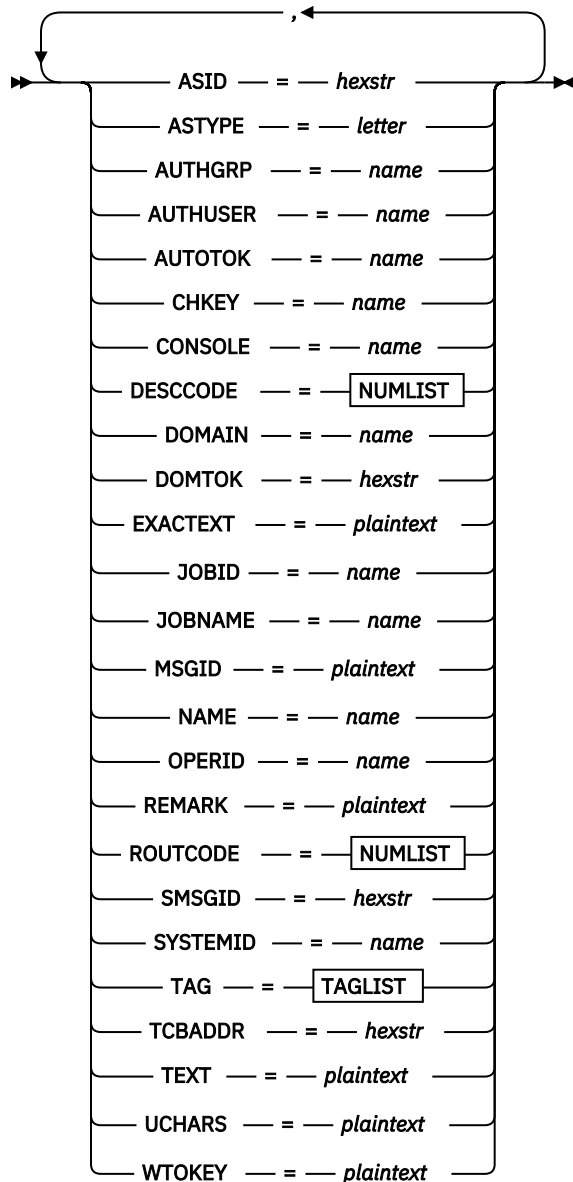
TO



FOR

➤ FOR — *timespan* ➤

KEYSPECS



IBM-Defined Synonym

Command or Operand	Synonym
PRINT	CNMECZPR

Purpose of Command

The PRINT command enables you to print the specified Canzlog messages to a sequential data set or a member of a partitioned or partitioned extended data set.

Canzlog messages to be printed are specified by the following parameters:

FOR, TO, or FROM keywords

When calling PRINT, use FOR, TO, and FROM to set a starting time and range for the print operation. Otherwise, printing will begin with the oldest message that is allowed by your DEFAULTS CZTOPDAT or CZTOPAGE, and continue to the limits described by MAXRECS in the stylesheet. If CZTOPDAT and CZTOPAGE are set to *NONE*, printing begins with the oldest message in the Canzlog log. FOR can be used with either FROM or TO to establish a range. When used with TO, FOR establishes the start of the range. When used with FROM, FOR sets the end of the range.

Filter specification

When the PRINT command is issued from the **Canzlog BROWSE** window or **CANZLOG** filter panel, the filter specification is optional and positional.

Operand Descriptions

CANZLOG

Specifies that the consolidated audit, NetView, and z/OS log (Canzlog) log is to be printed. If you specify the keyword CANZLOG, the filter specification is positional. The default filters for the BROWSE command will be used to the PRINT command.

DETAIL

detailclass

Specifies any of the message detail values that are supported by the NetView program. The detail class is defined in the CNMSTYLE member. For more information about message detail values, see the CNMSTYLE member or the *IBM Z NetView Administration Reference*.

Note: CZ.DETAIL.MVMSG.DEFAULT and CZ.DETAIL.NVMSG.DEFAULT are used by default. Otherwise, you need to specify them.

NONE

Specifies that no named details are provided. If you specify the *NONE* value, do not specify any other CZ.DETAIL CNMSTYLE statements in your CxxSTGEN or CNMSTUSR members.

FOR

Specifies the duration of the time span to be included. Use the FOR keyword if you want to specify the time span in terms of duration rather than specifying the start and end times. You can use the FOR keyword in the following ways:

- Use FOR with the FROM keyword to specify the beginning of the time span along with the duration.
- Use FOR with the TO keyword to specify the end of the time span along with the duration.
- Use FOR alone to specify a time span that ends at the current time.

You can specify a duration of up to 2 years. If you specify a larger value, a duration of 2 years is used.

Important: Do not specify both FROM and TO times if you are also specifying a duration with FOR.

FROM

Specifies the starting date and time. The specified date must be in the range 01/01/10 - 12/31/41. For more information about the date and time format, see the help for LONGDATE and LONGTIME keywords. This operand is optional.

KEYSPECS

Notes:

1. You can specify not equal by using $\neq$ for all KEYSPECS except NAME and REMARK. Not equal is not valid with the FROM, TO, or FOR keywords.
2. You can provide a single value or a list of values for any of these KEYSPECS, except for NAME and REMARK. Specifying multiple values is not valid with the FROM, TO, or FOR keywords. If you specify only one value for a KEYSPECS, parentheses are optional. These statements produce the same result:

```
Jobname=JOB9997  
Jobname=(JOB9997)
```

If you specify more than one value for a KEYSPECS, the values must be enclosed in parentheses. The values can be separated by a blank or by a comma. These statements produce the same result:

```
Jobname=(JOB9997 JOB9998 JOB9999)  
Jobname=(JOB9997, JOB9998, JOB9999)
```

If you specify either `Jobname=(JOB9997 JOB9998 JOB9999)` or `Jobname=(JOB9997, JOB9998, JOB9999)`, the logical operator OR evaluates the specifications, and any items with a Jobname of JOB9997 or JOB9998 or JOB9999 are printed.

3. You can also specify the same KEYSPECS more than once. The logical operator AND evaluates multiple specification of KEYSPECS. This is useful when used with the "not equal" option. For example

```
Jobname='ABC', Jobname¬='ABC1'
```

matches every jobname that begins with ABC except those beginning with ABC1.

4. For the following KEYSPECS keywords, you can specify a shortened version of the matching value. For example, `OPERID=TOM` matches any operator ID beginning with TOM:

- AUTHGRP
- AUTHUSER
- AUTOTOK
- CHKEY
- CONSOLE
- DOMAIN
- JOBID
- JOBNAME
- MSGID
- OPERID
- SYSTEMID
- UCHARS
- WTOKEY

ASID

Address space ID.

ASTYPE

Address space type. Indicates how the address space was started (job type).

Value

Description

D

USS persistent procedure. The address space has a name for initiated programs, appropriate for a JOB. However, the existence of an OpenMVS address space block indicates a special purpose USS persistent procedure.

E

The address space is a system address space that is started before the NetView subsystem is initialized.

J

The address space is a JOB.

N

The address space is a system address space started during operating system initialization (NIP) processing.

S

The address space is a Started Task (STC).

Note: Because of the manner in which TN3270 is started, it might display as type S rather than type D.

T

The address space is a Time-Sharing User (TSO).

U

The address space is a USS forked or created procedure.

Error: the address space where the command originated has closed or else the message is not from the local LPAR.

?

Error: inconsistent data (might be a transient condition).

!

Error: inconsistent data.

>

Error: the supplied ASID is larger than the ASID limit for the system.

AUTHGRP

z/OS ACEE group ID (ACEEGRPN), if available.

AUTHUSER

z/OS ACEE user ID (ACEEUSRI), if available.

AUTOTOK

z/OS automation token.

CHKEY

z/OS CHKEY, as defined by system macro IEECHAIN; this is the step-name of a task or the job name of a job.

CONSOLE

z/OS destination console name.

DESCCODE

z/OS descriptor code.

DOMAIN

NetView domain name.

DOMTOK

A 4-byte token to identify a Delete Operator Message (DOM) or a token for a message for which a DOM was issued.

EXACTEXT

Specifies a comparison with message data that respects case and ignores national translation, if any. Search for EXACTEXT is faster than a search for TEXT.

JOBID

Identifier assigned by JES, also known as job number.

JOBNAME

z/OS job name.

MSGID

For DOMs with a MsgsMatch field of 1, the Canzlog ID of the associated message. The value specified cannot exceed 12 characters.

NAME

Specifies a 1 to 8 character value that is useful for saving a filter (see CANZLOG command) and as a reminder of the purpose of the filter. The NAME parameter has no effect on the operation of the filter.

OPERID

The NetView task/operator name that originated the message. A message that originates at a virtual OST (VOST) task is logged with OPERID set to the name of the VOST owner, if that information is available when the message is logged. The value that is specified cannot exceed 8 characters.

REMARK

Species a 1 - 40 character value that can be useful as a reminder of the elements of a filter. You can read the remark value when editing a saved filter, in the output of LIST CZFILTER, as a result of the WHAT subcommand, and in some error messages. The REMARK parameter has no effect on the operation of the filter.

ROUTCODE

z/OS route codes.

SMSGID

System message ID. This label is *SMSGID(s)*: for DOMS, which can have more than one. The value specified cannot exceed 8 characters.

SYSTEMID

z/OS system ID. The value specified cannot exceed 8 characters.

TAG (tagname)

Associated tags. You can specify more than one tag.

ALL

Matches any valid tagname.

AUDIT

Intended for audit purposes, such as internal commands.

BCAST

z/OS broadcast to active consoles applies.

CMDECHO

Command echo.

DELETED

Message was requested to be deleted. It is logged in the Canzlog log for automation purposes.

DOM

A Delete Operator Message (DOM) sent by the system to negate a previous message.

DOMEXP

Delete Operator Message (DOM) is expected for message, as defined by the WQEDOM flag.

MVSMMSG

Logged at the z/OS subsystem interface.

NVMSG

Originated in the NetView program.

TRACE

Intended for tracing purposes, such as debug messages.

TCBADDR

Task Control Block (TCB) address.

TEXT

Specifies a comparison with message data that is case-insensitive and occurs after national language translation, if any. Search for EXACTEXT is faster than a search for TEXT.

UCHARS

User-defined or installation-defined characters. The value specified cannot exceed 16 characters.

WTOKEY

Key field associated with the WTO system macro (also WQEKEY in system macro IHAWQE).

LOG

Specifies that the consolidated audit, NetView, and z/OS log (Canzlog) log is to be printed. This keyword indicates that the current OVERRIDE or DEFAULTS specification for the CANZLOG parameter is to be used. The filter specification for the LOG keyword is positional.

namedfilter

Specifies any of the named filters that are supplied with the NetView product or created through use of the CANZLOG command or defined in CNMSTYLE. The filter specification for the *namedfilter* value is positional. For more information about named filters, issue the LIST CZFILTER command or browse CNMSTYLE for defined filters.

number

Specifies the number of Canzlog messages to print. The valid range is 1 to 4,000,000.

This variable is optional and positional, and must closely follow the PRINT keyword if it is specified.

If this variable is not specified, the value specified on the CZ.PRINT.MAXRECS CNMSTYLE statement is used.

OUTPUT

Specifies the data set name, data set and member names, or dd name to which the Canzlog messages will be printed.

- This operation will overwrite the original content of the sequential data set or member in a partitioned data set.
- After you execute the PRINT command, if a data set is full, re-writing to the data set causes the failure of the data set in use. In this case, use the FREE FILE() command to release the data set. You can use the LISTA command to obtain the DD name to supply as the value for FILE().

ddname

Specifies a DD name that is associated with a data set or spool file. The DD name may have been defined as part of customizing NetView startup JCL or defined with a Command Facility ALLOCATE command. The DD name can be in the range 1 - 8 characters.

dsname

Specifies the location to which the Canzlog PRINT output will be written. The data set must observe the following rules:

- Pre-allocated by the user before issuing the PRINT command
- Fully qualified sequential data set name
- Fully qualified partitioned or partitioned extended data set if a member name is to be specified

See the usage notes for an example of parameters for allocating the data set.

membername

Specifies the member name of the data set to print Canzlog messages.

PREFIX

Specifies the default elements of a prefix for Canzlog messages that are to be printed.

This keyword is optional. If the PREFIX keyword is not specified on the PRINT command, the values on the CZ.PRINT.PREFIX CNMSTYLE statement will be used.

At least one of the following variables must follow the PREFIX keyword, in the order in which you want them to appear in the message prefix.

Note: You can provide a single value or a list of values for PREFIX. If you specify only one value for PREFIX, parentheses are optional. If you specify more than one value for PREFIX, the values must be enclosed in parentheses. The values can be separated by a blank.

These statements all establish DATE as one of the prefixes. The last one also includes TIME and MTYPE as prefixes.

- PREFIX=DATE
- PREFIX=(DATE)
- PREFIX=(DATE TIME MTYPE)

NONE

Specifies that no message prefix is to be included. If you specify the **NONE** value, do not specify any others.

ALL

Specifies that all supported prefixes are to be included. If you specify the *ALL* value, do not specify any others.

DATE

The 8-character standard date on which the message was issued. For more information about the date format, see the help for the LONGDATE keyword.

TIME

The 8-character standard time at which the message was issued. For more information about the time format, see the help for the LONGTIME keyword.

SOURCE

The source of the message. For an MVS message, the source is an MVS job name. For a NetView message, the source is a NetView operator ID.

MTYPE

For NetView messages, a single character specifying the message type.

ORIGIN

The name of the MVS system or NetView domain from which the message originated.

TO

Specifies the end date and time. The specified date must be between 01/01/10 and 12/31/41. For more information about the date and time format, see the help for LONGDATE and LONGTIME keywords. This operand is optional.

Usage notes:

• **Allocating the output data set**

Before specifying the OUTPUT variable or the CZ.PRINT.OUTPUT CNMSTYLE statement for the PRINT command to collect log, for sequential data set, you must preallocate the output data set.

Here lists an example for the sequential data set that you can use:

```
Organization . . . : PS
Record format . . . : FB
Record length . . . : 134
Block size . . . . : 27872
1st extent cyls . . : 15
Secondary cyls . . . : 2
```

Here lists an example for the partitioned data set that you can use:

```
Device type . . . . : 3390
Organization . . . . : P0
Record format . . . : FB
Record length . . . : 80
Block size . . . . : 27920
1st extent cylinders: 15
Secondary cylinders : 2
```

dsname can be 1 – 44 characters. If a PDS or PDSE is used, the member name can be 1 – 8 characters.

• **Issuing PRINT from Canzlog BROWSE window**

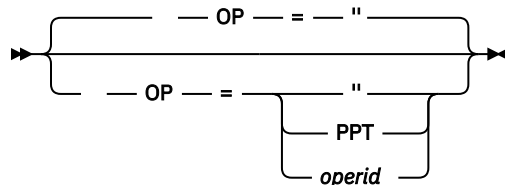
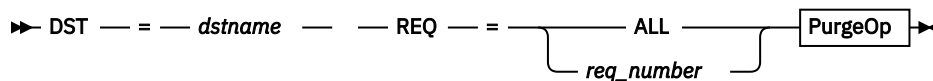
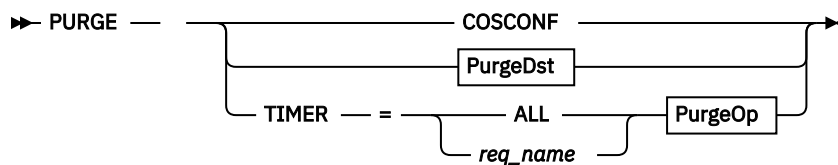
When PRINT is issued from the **Canzlog BROWSE** window or **CANZLOG** panel, the filter specification is optional. If the filter specification is omitted, PRINT will inherit its filter from the calling environment.

• **Message wrapping in the output data set**

- **Spool Dataset Support**

```
ALLOCATE FILE(XMPFILE) SYSOUT(A) RECFM(FB) LRECL(80)
PRINT OUTPUT=XMPFILE .....
```

Syntax



REQ

Specifies that DST requests are to be purged.

ALL

Indicates that all DST requests for the *operid* specified on the OP operand are to be purged (maximum value is 999 requests).

req_number

Specifies the DST request number you want purged. The DST numbers are obtained with the command LIST DST=*dstname*.

OP

Purges timer requests or timer elements for the designated operator.

..

Specifies to purge non-PPT timer elements, or DST requests sent by you. This is the default value.

PPT

Purges primary program operator interface task (PPT) timer elements.

operid

Specifies the operator whose timer or DST requests are to be purged.

TIMER

Specifies that timer requests are to be purged, depending on the OP operand.

ALL

Indicates that all pending timer requests are to be purged. If you do not specify the OP operand, all pending timer elements you entered without the PPT operand are purged.

req_name

Indicates that the named timer request sent under the operator or task specified on the OP operand is to be purged.

Restrictions

The PURGE command is asynchronous and requires a CORRWAIT stage if used in a PIPE.

Example: Purging all outstanding DST requests

To purge all outstanding DST requests for BNJDSESV that are from operator HELPDISK1, enter:

```
PURGE DST=BNJDSESV REQ=ALL OP=HELPDSK1
```

Example: Purging all timer requests entered without the PPT operand

To purge all timer requests that you entered without the PPT operand, enter:

```
PURGE TIMER=ALL
```

Example: Purging timer request SYS0001

To purge the timer request SYS0001, enter:

```
PURGE TIMER=SYS0001
```

Example: Purging all timer requests sent with the PPT operand

To purge all timer requests that you sent with the PPT operand, enter:

```
PURGE TIMER=ALL OP=PPT
```

Example: Purging all requests made by OPER7 to the DST task DSIGDS

To purge all requests made by OPER7 to the DST task DSIGDS, enter:

```
PURGE DST=DSIGDS REQ=ALL OP=OPER7
```

Response

If the purge is successful, the following is displayed:

```
DSI205I   nnn TIMER ELEMENTS PURGED   OP = operid
```

```
DSI510I   taskname:   rrr REQUESTS PURGED
```

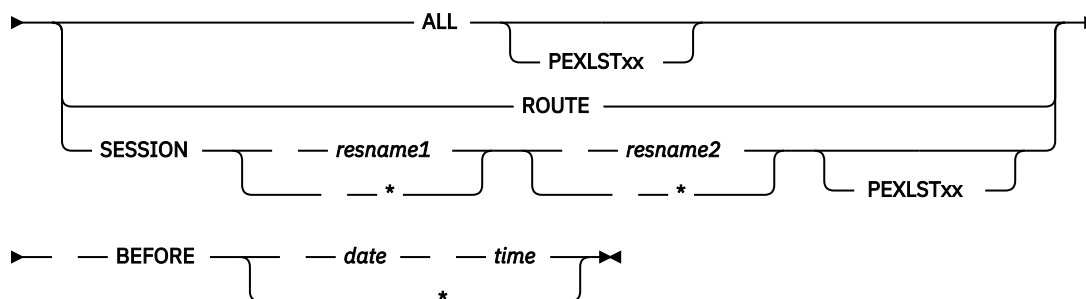
In the example, *rrr* is the decimal number of requests purged.

PURGE (NLDM)

Syntax

NLDM PURGE

➡ PURGE — ➡



Purpose of Command

The NLDM PURGE command deletes route data, session data, or both from the session monitor database.

The session monitor purge, when it is initiated by the PURGE command, runs asynchronously. When the network operator receives the PURGE STARTED message indicating that the session monitor purge has started, the operator can continue using the session monitor. When the session monitor purge has completed, the authorized receiver receives a PURGE COMPLETED message. This message indicates the type and count of the data purged. If a network operator issues a session monitor PURGE and a PURGE COMPLETED message has not been received by the authorized receiver for a prior session monitor PURGE command, the network operator still receives a PURGE STARTED message for the PURGE command. However, the new session monitor PURGE does not begin until the prior purge has completed.

Operand Descriptions

ALL

Deletes all session and route data. This operand deletes route and session data that ended prior to the specified BEFORE date and time.

PEXLSTxx

Specifies a purge exception list that is defined in the CNMSTYLE member.

The xx corresponds to the CNMSTYLE member exception list definition (NLDM.PEXLSTxx.suffix).

ROUTE

Deletes route data only. This operand deletes only route data that ended before the specified BEFORE date and time.

SESSION

Deletes session data only. This operand deletes only session data that ended before the specified BEFORE date and time. If you specify SESSION, also specify *resname1* and *resname2*.

For the SESSION operand, the resource names limit the purge to only the sessions that had session partners of *resname1* and *resname2*. You can use the wildcard characters * and ? within *resname1* and *resname2* to specify more generic resource names.

To vary one character, use a question mark (?). For example, A?B matches any name that begins with A, ends with B, and has one character between, such as AAB, ABB, AXB, and so on. A??B matches any name that begins with A, ends with B, and has any two characters between, such as AAXB. A character must always appear in the position of a ?; that is, A?B is not matched by AB because there is no character replacing the character ?.

To vary a string of characters at the end of a group of resources, use an asterisk (*). For example, TSO* matches any name that begins with the letters TSO, such as TSOXYZ, TSOB2219, and so on. You can use the * only at the end of a character string. You cannot use it between characters.

resname1 resname2

Specifies the resource names of the session partner data to be purged. These names must be 1–8 alphanumeric characters and can contain the wildcard characters * and ?.

*

An asterisk (*) alone is an alternative way of specifying either *resname1* or *resname2*.

Note: Specify either two resource names, a resource name and an asterisk, or two asterisks.

BEFORE

Deletes the specified data that has an ending time stamp that is less than or equal to the specified date and time.

date

Specifies the date before which data is to be purged. The format of *date* is controlled by the setting of the date operands of the DEFAULTS and OVERRIDE commands.

time

Specifies the time before which data is to be purged. The format of *time* is controlled by the setting of the time operands of the DEFAULTS and OVERRIDE commands.

*

Specifies that all data up to the current date and time is to be purged.

Restrictions

When you purge using wild cards (*) for both *resname1* and *resname2*, you might have to issue the PURGE command twice to achieve the desired results if the database has become corrupted with mixed session data. Corruption to the database can occur if you change the PURGE=SPEED|DASD parameter of the AAUPRMLP and do not delete and redefine the database.

Examples

The format of dates and times specified in the following examples assumes the default setting for date and time formats on the DEFAULTS and OVERRIDE commands.

Example: Purging all sessions by resource name pairs and session end time

To purge all sessions that have resource name pairs of LCL3278A and L51R79M and a session end time before (or at) 23:59 on 3/21/19, enter:

```
PURGE SESSION LCL3278A L51R79M BEFORE 3/21/19 23:59
```

Example: Purging all route data by end time

To purge all route data with an end time before (or at) the current system date/time, enter:

```
PURGE ROUTE BEFORE *
```

Example: Purging all session and route data by current system date and time

To purge all session data and all route data before (or at) the current system date/time, enter:

```
PURGE ALL BEFORE *
```

Example: Purging all sessions according to name pair patterns and current system date and time

To purge all sessions that have resource name pairs matching the patterns A?C*, * and a session end time before (or at) the current system date/time, enter:

```
PURGE SESSION A?C* * BEFORE *
```

Example: Purging all sessions between two resources according to session end time

To purge all sessions between resources APPL01 and LU01 with a session end time at or before 23:59 on 3/21/19, enter:

```
PURGE SESSION APPL01 LU01 BEFORE 3/21/19 23:59
```

Example: Purging sessions using an exception list

To purge all session data except for entries with session information between HOST1 and NCP resources or CP-CP sessions, enter:

```
PURGE SESSION * * PEXLST02 BEFORE * *
```

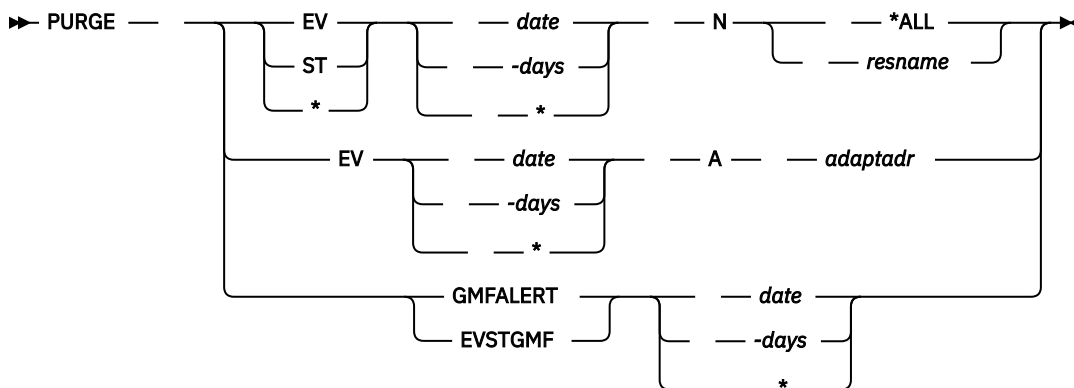
The exception lists are located in the CNMSTYLE member:

```
NLDM.PEXLST02.A = HOST1 NCP*  
NLDM.PEXLST02.B = CP-CP
```

PURGE (NPDA)

Syntax

NPDA PURGE



IBM-Defined Synonyms

Command or Operand	Synonym
PURGE	PRG

Purpose of Command

The NPDA PURGE command removes event or statistical data for a specified resource or for all resources from the database. In addition, it can purge all information based on the adapter address. If you PURGE events, statistics, or GMFALERT records, they are purged from the hardware monitor database. Specify GMFALERT to purge GMFALERT records from the hardware monitor database.

If you issue this command from a non-hardware monitor screen or from a command list, such as the PURGEDB command list, the response to this command is sent to the command facility screen and the messages are logged. The messages received are the same as if the command was issued under the PPT.

Operand Descriptions

EV

Specifies event data.

ST

Specifies statistical data.

*

Specifies both event and statistical data.

date

Deletes all the data before this date. The format of *date* is controlled by the setting of the date operands of the DEFAULTS and OVERRIDE commands.

-days

Deletes data that is older than the specified number of days (0–365).

*

Deletes data regardless of age.

N

Identifies the operand that follows as a resource name.

*ALL

Deletes data for the date previously noted for all resources.

resname

Specifies the symbolic name of the resource. You can specify up to five resource names to fully qualify the resource for which data is to be purged.

A

Identifies the operand that follows as an adapter address.

adaptadr

Specifies the 12-hexadecimal-digit adapter address. The A (adapter) address is not a valid option for a resource type of CBUS.

GMFALERT

Purges the GMFALERT records from the hardware monitor database.

EVSTGMF

Purges the events, statistics, and GMFALERT records. Note that if the target domain is prior to NetView for OS/390® V1R1 (for example, if you issue an NPDA SDOMAIN command to the hardware monitor of an earlier NetView release), the PURGE EVSTGMF command purges only events and statistics records because the target domain contains no GMFALERT records.

Restrictions

The following restrictions apply to the PURGE command:

- This command does not purge records for attached resources.
- The recording function is suspended when a purge of the entire database is in progress, and incoming data can be lost. You cannot process multiple PURGE commands concurrently.
- If the name of the resource is not associated with a unique resource configuration on the database, a selection panel is displayed on which you can choose which configuration is relevant.
- To reclaim the space available as a result of the PURGE command, use the "DBAUTO NPDA,REORG" command. If the REORG command is not issued, the free space cannot be reused.

Example: Erasing all records on an event database recorded prior to a specified date

To erase all records on an event database that were recorded before March 1, 2010, enter either command:

```
PURGE EV 030110 N *ALL
```

```
PRG EV 030110 N *ALL
```

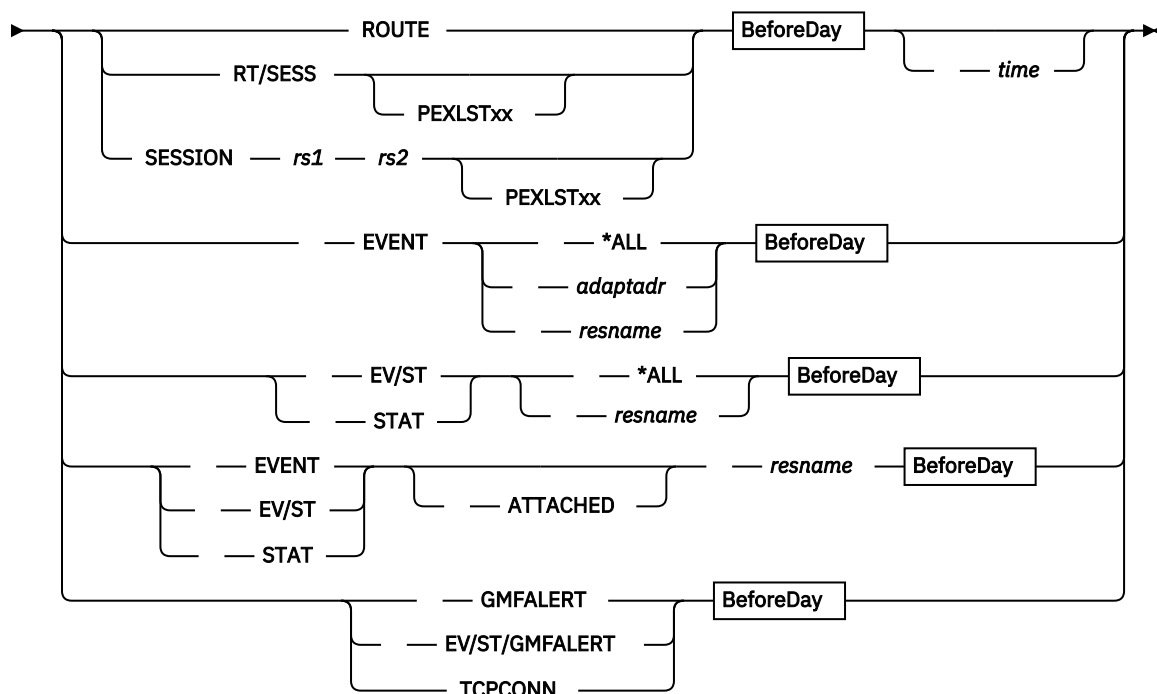
The format of specified dates assumes the default setting for date formats on the DEFAULTS and OVERRIDE commands.

PURGEDB (NLDM, NPDA; CNME2007)

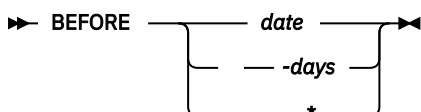
Syntax

PURGEDB

➤ PURGEDB ➤



BeforeDay



IBM-Defined Synonyms

Command or Operand	Synonym
ROUTE	RT
SESSION	SESS
EVENT	EV
STAT	ST
ATTACHED	ATT
EV/ST/GMFALERT	EVSTGMF and EV/ST/GMF

Purpose of Command

The PURGEDB command list deletes hardware monitor, session monitor, and TCP/IP connection data from the database.

If you purge events, statistics, or GMFALERT records, they are purged from the hardware monitor database. Specify GMFALERT to purge GMFALERT records from the hardware monitor database. Specify EV/ST/GMFALERT to purge events, statistics, and GMFALERT records from the hardware monitor database.

The session monitor purge, when initiated by the PURGE command, runs asynchronously. When you receive the PURGE STARTED message indicating that the session monitor purge began, you can continue using the session monitor. When the session monitor purge ends, you and an authorized receiver receive a PURGE COMPLETED message. This message indicates the type and count of the data purged.

Operand Descriptions

ROUTE

Indicates that session monitor route data is to be purged.

The ROUTE or RT operand deletes session monitor route data that ended before the specified BEFORE date and time.

RT/SESS

Indicates that route data and session monitor data are to be purged.

The RT/SESS operand deletes session monitor session data and route data that ended before the specified BEFORE date and time.

PEXLSTxx

Specifies a purge exception list that is defined in the CNMSTYLE member.

The xx corresponds to the CNMSTYLE member exception list definition (NLDM.PEXLSTxx.suffix).

SESSION

Indicates that session monitor session data is to be purged.

The SESSION or SESS operand deletes session monitor session data that ended before the specified BEFORE date and time.

For the session monitor SESSION or SESS operand, the resource name limits the purge to only those sessions that had session partners of resource *rs1* and *rs2*. Use the wildcard characters * and ? within *rs1* and *rs2* to specify more generic resource names.

To vary one character, use a question mark (?). For example, A?B matches any name that begins with A, ends with B, and has one character between, such as AAB, ABB, AXB, and so on. A??B matches any name that begins with A, ends with B, and has any two characters between, such as AAXB. A character must always appear in the position of a ?; that is, A?B is not matched by AB because there is no character replacing the character ?.

To vary a string of characters at the end of a group of resources, use an asterisk (*). For example, TSO* matches any name that begins with the letters TSO, such as TSOXYZ, TSOB2219, and so on. You can use the * only at the end of a character string. You cannot use it between characters.

rs1 rs2

Specifies the name of the resource for which data is to be purged. The resource name can contain the wildcard character ? for session monitor data.

BEFORE

Identifies the operands that follow as the starting date and time.

date

Deletes all the data before this date. The format of *date* is controlled by the setting of the date operands of the DEFAULTS and OVERRIDE commands.

-days

Deletes data that is older than the specified number of days (0–365).

Deletes data that is older than the current date (and current time unless time is explicitly specified following the date).

time

Time before which the specified data is to be purged from the session monitor database only. The format of *time* is controlled by the setting of the time operands of the DEFAULTS and OVERRIDE commands.

EVENT

Indicates hardware monitor event data is to be purged.

The EVENT or EV operand deletes hardware monitor event data logged before the specified BEFORE date.

***ALL**

Indicates all hardware monitor data is to be purged (for hardware monitor purge only).

adaptadr

Specifies a 12-hexadecimal-digit adapter address in place of a resource name (for hardware monitor purge only). The A (adapter) address is not a valid option for a resource type of CBUS.

resname

Specifies the symbolic name of the resource. You can specify up to five resource names to fully qualify the resource for which data is to be purged.

EV/ST

Indicates hardware monitor event data and statistical data are to be purged.

The EV/ST operand deletes hardware monitor event and statistical data logged before the specified BEFORE date.

STAT

Indicates hardware monitor statistical data is to be purged.

The STAT or ST operand deletes hardware monitor statistical data logged before the specified BEFORE date.

ATTACHED

Indicates all hardware monitor data associated with *resname* is to be purged (for hardware monitor purge only).

GMFALERT

Indicates GMFALERT records are to be purged from the hardware monitor.

EV/ST/GMFALERT

Indicates events, statistics, and GMFALERT records are to be purged. Note that if the target domain is prior to NetView for OS/390 V1R1 (for example, if you issue an NPDA SDOMAIN command to the hardware monitor of an earlier NetView release), the PURGEDB EVSTGMF command purges only events and statistics records because the target domain contains no GMFALERT records.

TCPCONN

Indicates that TCP/IP connection data is to be purged.

Restrictions

The following restrictions apply to the PURGEDB command:

- When you purge using wildcards (*) for both *rs1* and *rs2*, it might be necessary to issue the PURGE command twice to achieve the desired results if the database has become corrupted with mixed session data. Corruption to the database can occur if you change the PURGE=SPEED|DASD parameter of the AAUPRMLP and do not delete and redefine the database.
- For the hardware monitor EVENT, EV, STAT, ST, or EV/ST operand, the resource name limits the purge to only the event data or statistical data that used resource *resname*.
- If you issue PURGEDB for hardware monitor or session monitor data and PURGE, PRGATT, or PURGEDB command is currently running, the new PURGEDB request is denied. You cannot concurrently process multiple PURGEDB, PURGE, or PRGATT commands for hardware monitor or session monitor.
- The PURGEDB command list issues NPDA PURGE, NPDA PRGATT, and NLDM PURGE commands. If these commands are individually checked for authorization, the use of the PURGEDB command list is restricted as if it were also checked for authorization.
- To clear an entire hardware monitor database or session monitor database, use the IDCAMS purge database facility or the RESETDB command.
- To reclaim the free space made available by the PURGEDB command, use the "DBAUTO NPDA,REORG" command or the "DBAUTO NLDM,REORG" command, as appropriate. If the REORG is not done, the free space cannot be reused.

Examples

The format of dates and times specified in the following examples assumes the default setting for date and time formats on the DEFAULTS and OVERRIDE commands.

Example: Purging all session monitor route data before a specified time of the current date

To purge all session monitor route data before a specified time of the current date, enter:

```
PURGEDB ROUTE BEFORE * 16:32
```

Response

All session monitor route data before 16:32:59 of the current date are purged.

Example: Purging all session monitor data using wildcard characters for a specified date

To purge all session monitor session data using the wildcard characters asterisk (*) and question mark (?), and for a specified date, enter:

```
PURGEDB SESSION * ABC?E BEFORE 7/1/19
```

Response

All session monitor session data matching * ABC?E before 7/1/19 23:59:09 are purged.

Example: Purging session monitor data using an exception list

To purge all session monitor session data except for entries with session information between HOST1 and an NCP resource or any CP-CP session, enter:

```
PURGEDB SESSION * * PEXLST02 BEFORE * *
```

The exception lists are located in the CNMSTYLE member:

```
NLDM.PEXLST02.A = HOST1 NCP*  
NLDM.PEXLST02.B = CP-CP
```

Response

All session monitor session data except for sessions between HOST1 and NCP resources or CP-CP sessions are purged.

Example: Purging all session monitor session data and route data before the current date and time

To purge all session monitor session data and route data before the current date and time, enter:

```
PURGEDB RT/SESS BEFORE *
```

Response

All session monitor session data and route data before the current date and time are purged.

Example: Purging all hardware monitor event data for a resource before the current date

To purge all hardware monitor event data for resource RES25 before the current date, enter:

```
PURGEDB EVENT RES25 BEFORE *
```

Response

All hardware monitor event data for resource RES25 before the current date and time are purged.

Example: Purging all hardware monitor statistical data before a specified date

To purge all hardware monitor statistical data before March 1, 2010, enter:

```
PURGEDB ST *ALL BEFORE 3/1/10
```

Response

All hardware monitor statistical data before March 1, 2010, are purged.

Example: Purging all hardware monitor event data attached to a resource before the current date

To purge all hardware monitor event data attached to resource NAME2 before the current date, enter:

```
PURGEDB EV ATT NAME2 BEFORE *
```

Response

All hardware monitor event data attached to resource NAME2 before the current date are purged.

Example: Purging all session monitor session data for specified resources before the current date and time

To purge all session monitor session data for sessions between resources specified by A and B* before the current system date and time, enter:

```
PURGEDB SESS A B* BEFORE * *
```

Example: Purging session monitor route data

To purge session monitor route data for all routes that have a session end date before May 26, 2010, enter:

```
PURGEDB RT BEFORE 5/26/10
```

Example: Purging all session monitor data for sessions with an end date 5 days before the current date

To purge all session monitor data for sessions with an end date of 5 days prior to the current system date and all routes with this end date, enter:

```
PURGEDB RT/SESS BEFORE -5
```

Example: Purging all hardware monitor event data for a resource before the current system date

To purge all hardware monitor event data for resource RES1 before the current system date, enter:

```
PURGEDB EV RES1 BEFORE *
```

Example: Purging all event and statistical data before the current system date

To purge all event and statistics data before the current system date, enter:

```
PURGEDB EV/ST *ALL BEFORE *
```

PWDSEC (NCCF)

Syntax

➤ PWDSEC — — *resourcename* ➤

Purpose of Command

The PWDSEC command checks the authorization for an operator to view a password or password phrase.

Operand Descriptions

resourcename

Specifies the item for which a password or password phrase has been assigned.

Usage Notes

PWDSEC serves as a central point to control access to passwords used by AON, including IPMAN, in which an operator might view password data.

Return Codes

Return Code

Meaning

0

User is authorized to view the password or password phrase.

4

User is not authorized to view the password or password phrase.

Determining password authority

To determine if an operator is allowed to access a password for a resource named, for example, TVT2011, enter the following:

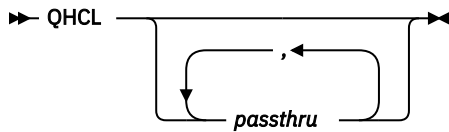
```
PWDSEC TVT2011
```

If the operator is authorized, message DSI633I is issued. Otherwise, the operator receives security-generated messages stating why they are not authorized.

QHCL (NCCF; CNME1011)

Syntax

QHCL



Purpose of Command

The QHCL command list displays information about the hardcopy log (printer), if one exists.

Operand Descriptions

passthru

Specifies up to six parameters which are appended unchanged to the VTAM DISPLAY command issued by the QHCL command. No validation for duplicate or conflicting parameters is performed.

Example: Querying a printer

To query the printer, enter:

```
QHCL
```

Response

If no printer exists, the following is displayed:

```
CNM393I  QHCL: NO ACTIVE HARD-COPY LOG FOR THIS SESSION
```

QOS (NCCF)

Syntax

```
➔ QOS — — OP — = — operid ➔
```

Purpose of Command

The QOS command displays information about whether an operator is defined to the NetView program and whether the operator is currently logged on.

Operand Descriptions

OP=*operid*

Specifies the operator ID whose status is being queried. The response indicates whether the operator is defined to the NetView program and whether it is logged on. Optionally, the value can be a question mark character together with a NetView nickname defined using the function.autotask statement. For example: OP=?POLICY

Return Codes

Return Code	Meaning
-------------	---------

0

The operator is defined to the NetView program and is currently logged on.

4

The operator is defined to the NetView program but is not currently logged on.

8

The operator is not defined to the NetView program but is currently logged on.

12

The operator is not defined to the NetView program.

104

The command did not complete successfully. Check the accompanying messages for more information.

Usage Notes

The NetView program considers operators defined in SAF to be defined to the NetView program if the current OPERSEC value is SAFDEF, or if there is currently a default profile (see HELP for the DEFAULTS command LOGPROF keyword) and the OPERSEC value is SAFPW or SAFCHECK.

Example: Querying the status of an operator

If the command is issued from an operator terminal, the command format to query whether operator BOB is defined to the NetView program and is logged on is:

```
QOS OP=BOB
```

Response

One of the following messages is returned:

```
DW0837I BOB IS DEFINED AND LOGGED ON TO NETVIEW
DW0838I BOB IS DEFINED TO NETVIEW BUT IS NOT LOGGED ON
DW0839I BOB IS NOT DEFINED TO NETVIEW BUT IS LOGGED ON
DW0840I BOB IS NOT DEFINED TO NETVIEW
```

QREXX (NCCF; CNME8002)

Syntax

QREXX

➡ QREXX ⬅

Purpose of Command

The QREXX command list queries whether the REXX interpreter is installed and available. You enter the QREXX command list from the command line or a command list.

Example: Querying the availability of the REXX Interpreter

To query the availability of the REXX interpreter, enter:

```
QREXX
```

Response

If the REXX interpreter is available, the following message is displayed:

```
CNM228I QREXX : REXX INTERPRETER INSTALLED AND AVAILABLE
```

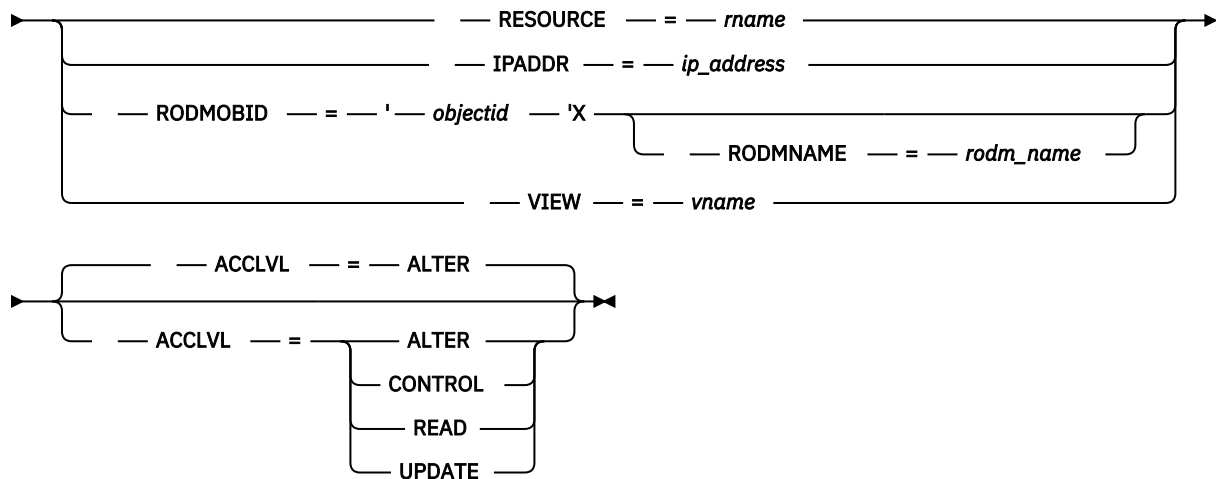
If the REXX interpreter is unavailable, the following message is displayed:

```
CNM229I QREXX : REXX INTERPRETER NOT AVAILABLE
```

QRS (NCCF)

Syntax

➔ QRS — — OP — = — *operid* ➔



Purpose of Command

The QRS command indicates whether an IP address, a resource, or a view is contained in an operator's span of control. An IP address, a resource, or a view is contained in an operator's span of control under the following conditions:

- When CTL=SPECIFIC or CTL=GENERAL and the IP address, resource, or view is contained in an active span for the operator.
- When CTL=GENERAL and the IP address, resource, or view is not contained in any span.
- When CTL=GLOBAL is in effect.

Operand Descriptions

OP=*operid*

Specifies the operator ID whose span authority is being checked.

RESOURCE=*rname*

The name of the resource to be verified for span authorization. The name can be 1–255 characters in length, can contain DBCS characters, can be network qualified, and is case-sensitive. You can use the NETVASIS command to keep the NetView program from translating the resource name to uppercase when the QRS command is entered on the NetView command line, and the NETVASIS environment when the QRS command is used in a REXX command list. The value of *rname* cannot contain wildcard characters. An asterisk (*) or question mark (?) is used literally.

IPADDR=*ip_address*

Specifies an IPv4 or IPv6 address. If necessary, this IP address changes to a standard format before security and span checking is performed. See *IBM Z NetView Security Reference* for a description of the IP address standard format.

RODMOBID='objectid'X

Specifies the hexadecimal RODM object ID of the resource to be checked for span authorization. The length must be 16 hexadecimal characters. The RODM that is used by GMFHS will be queried unless *rodm_name* specifies a different RODM. The RODM must be active when using this keyword.

RODMNAME=rodm_name

Specifies the name of the RODM containing the resource to be checked for span authorization. The length can be 1–8 characters in length, cannot contain DBCS characters, and must be in uppercase. The default is the RODM used by GMFHS.

VIEW=vname

The name of the view to be checked for span authorization. The name can be 1–255 characters in length, can contain DBCS characters, and is case-sensitive. You can use the NETVASIS command to keep the NetView program from translating the view name to uppercase when the QRS command is entered on the NetView command line, and the NETVASIS environment when the QRS command is used in a REXX command list. The value of *vname* cannot contain wildcard characters. An asterisk (*) or question mark (?) is used literally.

NMC restricts view names to 32 characters. See the MyName field in the *IBM Z NetView Data Model Reference* for information about the maximum length for view names. You can also see the *IBM Z NetView Resource Object Data Manager and GMFHS Programmer's Guide*.

ACCLVL=

Specifies the access level against which to check the operator's authority. This is the access level given to the operator when the operator ID is permitted to access the span. For more information, see the *IBM Z NetView Security Reference*.

The ACCLVL value can be one of the following:

ALTER

Multiwrite access. This is the default.

CONTROL

Multiread and single-write access

UPDATE

Change access. This is the access level of commands such as VARY, MODIFY, REPLY, or generic actions such as activate and deactivate.

READ

Information-only access. This is the access level of commands such as LIST or DISPLAY.

Return Codes**Return Code****Meaning****0**

The operator can access the IP address, resource, or view at the specified access level.

64

The operator cannot access the IP address, resource, or view at the specified level.

101

A RODM query failed.

104

The command syntax is not valid, or an authorization error has occurred.

128

The operator logged off.

160

The operator has no active spans.

200

An internal storage error occurred.

Example: Querying the availability of a resource for an operator

The QRS command format to query whether resource A01CNM01 is in a span that is active for operator BOB is:

```
QRS OP=BOB RESOURCE=A01CNM01 ACCLVL=READ
```

Response

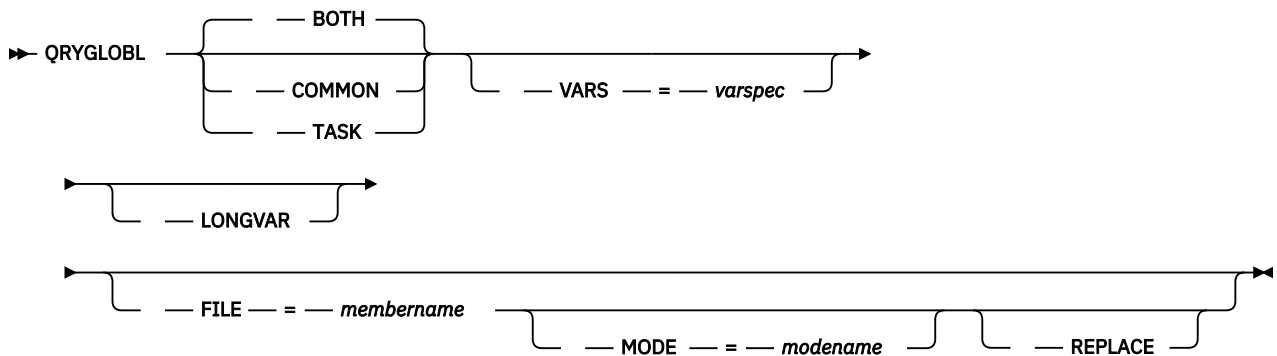
One of the following messages is received:

```
BNH224I BOB IS ALLOWED ACCESS TO A01CNM01 AT ACCESS LEVEL READ
```

```
BNH225I BOB IS NOT ALLOWED ACCESS TO A01CNM01 AT ACCESS LEVEL READ
```

QRYGLOBL (NCCF)

Syntax



Purpose of Command

The QRYGLOBL command displays information about NetView global variables. QRYGLOBL performs the following tasks:

- Displays the number of either task or common global variables, or both
- Displays either task or common global variables, or both
- Optionally, directs all of the preceding information to an output file

Operand Descriptions

BOTH

Displays requested information about both common and task global variables.

COMMON

Displays requested information about only the common global variables.

LONGVAR

Causes each variable that matches a VARS specification to display the entire variable name up to 250 characters, followed by one blank space and the entire value up to 31,000 characters, without consideration for the starting column. Lines with values that exceed 255 characters are displayed in turquoise; other lines are displayed in green. LONGVAR is designed as a programming interface.

If you do not specify the LONGVAR operand, values start in column 45, variable names longer than 31 characters are truncated (denoted by the >> characters after the truncated name) and values longer than 255 characters are truncated.

TASK

Displays requested information about only the task global variables.

VARs

Specifies to display the specific variables for either the common global variable dictionary, the task global variable dictionary, or both.

When the VARs keyword is omitted, both messages BNH034I and BNH035I are generated for the output. When the VARs keyword is specified, only message BNH035I is generated.

varspec

Specifies the variables that are displayed or written to the output file. The *varspec* supports an asterisk (*) as a multicharacter wildcard and a percent sign (%) as a single-character wildcard.

FILE

Directs the output to the specified output file.

membername

Specifies the member or file in which the NetView program places the output it creates. It creates or replaces *membername* in a DSILIST DD. If the DSILIST DD is a concatenation of partitioned data sets, the member created is placed in the first concatenated data set.

MODE

Specifies the mode on which the output file resides. It is valid only on VM systems.

modename

Specifies the minidisk where the output file is written. This operand is used only with the VM operating system. You specify this operand as **Az**, where:

A =

A letter from A-Z indicating the minidisk where the information is written. The default is A.

z =

A number in the range of 0-6 indicating the use characteristics of the file created. The default is 1.

You can specify the minidisk identifier without the created file's characteristics, but you cannot specify the use characteristic without a minidisk identifier. For example, MODE=Y is valid and defaults to a file mode of Y1. But MODE=3 is not valid.

The minidisk you select must be in the CMS format. If you select a DOS formatted minidisk, you receive an ABEND013 error code.

REPLACE

Specifies whether the NetView program replaces a preexisting output file with any newly created output file. The default value is to not replace.

If you specify REPLACE and the report *membername* file does not exist, the file is created. If the member being replaced is not an automation table usage report, the member is not replaced and message DWO826 is issued.

Restrictions

The following restrictions apply to the QRYGLOBL command:

- When specifying a variable or group of variables to display, do not include the ampersand (&) associated with NetView command list language variables.
- The information displayed at the operator console is in a multiline write-to-operator (MLWTO) message. To determine how many variables exist, run the QRYGLOBL command without the VARs keyword. To see many variables, direct the output to an output file. Using an output file avoids potential storage or performance problems that can result from displaying a large MLWTO message.
- When a QRYGLOBL output file is created, a 2-character key field (&*) is placed on the first line of the file. This key field identifies the file as a QRYGLOBL output file and stops the output file from replacing non-QRYGLOBL files. If you attempt to replace a member or file that does not have the key field in the correct location, you receive an error message. You can prevent a QRYGLOBL file from being overwritten if you delete the key field from the first line of the file.

- The DSILIST DD has a record length of 80 characters. Anything beyond that is truncated. However, the result from a QRYGLOBL command can be written to a data set that has a larger record size using the PIPE QSAM stage.

Return Codes

Return Code Meaning

0

The command completed successfully.

4

The command did not complete successfully. Check the accompanying messages for more information.

Example: Determining the number of task and common global variables

To determine the number of tasks and common global variables, enter:

```
QRYGLOBL
```

Response

The NetView program responds with the following multiline write-to-operator (MLWTO):

```
BNH031I NETVIEW GLOBAL VARIABLE INFORMATION
BNH103I COMMAND ISSUED AT: 03/20/19 21:02:51
BNH061I
BNH032I COMMON GLOBAL VARIABLES
BNH061I -----
BNH034I EXPECTED NUMBER OF VARIABLES: 100
BNH035I NUMBER OF VARIABLES FOUND: 5
BNH061I
BNH033I TASK GLOBAL VARIABLES FOR OPER1
BNH061I -----
BNH034I EXPECTED NUMBER OF VARIABLES: 100
BNH035I NUMBER OF VARIABLES FOUND: 2
BNH061I
BNH037I NETVIEW GLOBAL VARIABLE INFORMATION COMPLETE
```

Example: Displaying the value of a single common global variable

To display the value of a single common global variable, enter:

```
QRYGLOBL COMMON VARS=CGAUTHID1
```

Response

The NetView program responds with the following multiline write-to-operator (MLWTO):

```
BNH031I NETVIEW GLOBAL VARIABLE INFORMATION
BNH103I COMMAND ISSUED AT: 03/20/19 21:04:44
BNH061I
BNH032I COMMON GLOBAL VARIABLES
BNH036I GLOBAL VARIABLE NAME:          GLOBAL VARIABLE VALUE:
BNH061I -----
BNH039I CGAUTHID1                      AUT01
BNH035I NUMBER OF VARIABLES FOUND: 1
BNH061I
BNH037I NETVIEW GLOBAL VARIABLE INFORMATION COMPLETE
```

Example: Displaying all global variables starting with CGAUTH

To display all global variables starting with CGAUTH, enter:

```
QRYGLOBL VARS=CGAUTH*
```

Response

The NetView program responds with the following multiline write-to-operator (MLWTO):

```
BNH031I NETVIEW GLOBAL VARIABLE INFORMATION
BNH103I COMMAND ISSUED AT: 03/20/19 21:06:14
BNH061I
BNH032I COMMON GLOBAL VARIABLES
BNH036I GLOBAL VARIABLE NAME:          GLOBAL VARIABLE VALUE:
BNH061I -----
BNH039I CGAUTHID2                      AUTO2
BNH039I CGAUTHID1                      AUTO1
BNH039I CGAUTHID4                      NETOP2
BNH039I CGAUTHID3                      NETOP1
BNH035I NUMBER OF VARIABLES FOUND: 4
BNH061I
BNH033I TASK GLOBAL VARIABLES FOR OPER1
BNH036I GLOBAL VARIABLE NAME:          GLOBAL VARIABLE VALUE:
BNH061I -----
BNH035I NUMBER OF VARIABLES FOUND: 0
BNH061I
BNH037I NETVIEW GLOBAL VARIABLE INFORMATION COMPLETE
```

Example: Directing a QRYGLOBL output to a file

To direct a QRYGLOBL output file to a specific file, enter:

```
QRYGLOBL FILE=QRYG1 REPLACE
```

Response

The NetView program responds by writing the information to an output file. The file name is DSILIST(QRYG1).

The following information is written to the output file:

```
&*BNH031I NETVIEW GLOBAL VARIABLE INFORMATION
BNH103I COMMAND ISSUED AT: 03/06/19 21:22:22
BNH061I
BNH032I COMMON GLOBAL VARIABLES
BNH061I -----
BNH034I EXPECTED NUMBER OF VARIABLES: 100
BNH035I NUMBER OF VARIABLES FOUND: 5
BNH061I
BNH033I TASK GLOBAL VARIABLES FOR NETOP1
BNH061I -----
BNH034I EXPECTED NUMBER OF VARIABLES: 100
BNH035I NUMBER OF VARIABLES FOUND: 2
BNH061I
BNH037I NETVIEW GLOBAL VARIABLE INFORMATION COMPLETE
```

Example: Avoiding QRYGLOBL output data truncation

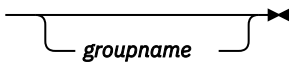
To direct QRYGLOBL output to a data set other than DSILIST, enter the following command:

```
PIPE NETV QRYGLOBL BOTH LONGVAR VARS=* |
  QSAM (DSN) GLOBAL.DATA(VARS)
```

This PIPE command writes all COMMON and TASK global variables to a member named VARS in the GLOBAL.DATA data set using the QSAM stage. To accommodate the message ID, longest global variable name, longest global variable value and intervening spaces, the data set should be pre-allocated with a record length of at least 31259.

QRYGROUP

Syntax

➤ QRYGROUP 

Purpose of Command

The QRYGROUP command shows information about the groups that are defined for browsing Canzlog data from multiple domains. You can use this command to list the available groups, or to list the members of a specific group.

Operand Descriptions

groupname

The name of the group whose members you want to list. The name can be any of the following:

- The name of a group defined using the ENT.GROUP CNMSTYLE statement
- A sysplex name
- The name of a remote alias defined using the RMTSYN and RMTALIAS CNMSTYLE statements
- The name of a NetView domain

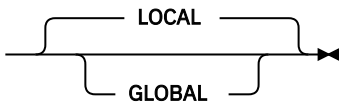
This parameter is optional. If you do not specify a group name, the QRYGROUP command displays a list of all defined groups.

Usage Notes

The output from the QRYGROUP command is appended to the CNM100I message.

QRYKEEP (NCCF)

Syntax

➤ QRYKEEP 

Purpose of Command

The QRYKEEP command reports the name, number of messages and the total storage of each active PIPE KEEP stage command on the task where it is issued. See *IBM Z NetView Programming: Pipes* for more information about the PIPE KEEP stage.

Operand Descriptions

LOCAL

Indicates to generate a report for KEEPs that are only visible to the task under which the command is issued.

GLOBAL

Indicates to generate a report for KEEPs that are visible by all tasks.

Examples

The following is a sample output:

BNH560I	KEEP	Status for taskName		
Keep Name	Number	Messages	Total Storage	Time Out
KEEP1	40		18212	1000
KEEP2	1		1040	*

QUERY RANGE (NLDM)

Syntax

QUERY RANGE

►► QUERY — — RANGE ►►

IBM-Defined Synonyms

Command or Operand	Synonym
QUERY RANGE	Q RANGE

Purpose of Command

The QUERY RANGE command displays the active time range from either the last SET RANGE command or the default time range.

Example: Displaying the current time range

To display the time range set on the last SET RANGE command, enter either command:

QUERY RANGE

Q RANGE

QUEUE (NCCF)

Syntax

QUEUE

►► QUEUE — — *text* ►►

IBM-Defined Synonyms

Command or Operand	Synonym
QUEUE	Q

Purpose of Command

The QUEUE command adds a text message to the operator input queue of a high-level language (HLL) command processor or installation exit routine running with the HLL_QUEUED_INPUT bit of HLLOPTS turned on. See *IBM Z NetView Programming: PL/I and C* for more information about HLLOPTS.

Use the QUEUE command when it is not possible for an operator to determine whether an HLL command processor or installation exit routine is ready to accept input. One example is when an HLL command processor or installation exit routine is running under an autotask.

Use the EXCMD command to route text to an HLL command processor or installation exit routine running under a different task.

After the text message has been added to the HLL command processor or installation exit routine's operator input queue, the message can be read from this queue by the HLL service routine, CNMGETD. See *IBM Z NetView Programming: PL/I and C* for more information.

Operand Descriptions

text

Specifies the message to be added to the operator input queue.

Restrictions

Use the QUEUE command only with tasks running a single HLL command processor or installation exit routine.

RCFB (NCCF; CNME0029)

Syntax

RCFB

➤ RCFB — — *return_code* — , — *feedback_code* ➤

Purpose of Command

The RCFB command list displays information describing the specified return code and feedback code. This command list handles only VTAM return codes and feedback codes.

Operand Descriptions

return_code

Specifies the 1- or 2-digit hexadecimal return code.

feedback_code

Specifies the 1- or 2-digit hexadecimal feedback code.

Leading zeros are not required for the return or feedback codes.

Example: Displaying the meaning of return and feedback codes

To display the meaning of return code 8 and feedback code 0, enter:

```
RCFB 8,0
```

REACC (NCCF)

Syntax

REACC

➤ REACC *ddname* ➤

Purpose of Command

Use the REACC command to regain access to members of the NCCF DD name that have been placed in dynamic extents.

Operand Descriptions

ddname

Name of one of the preset NCCF input DD names. Examples are DSIPARM and DSICLD.

Restrictions

If REACC is issued while another operator or task is reading the same *ddname*, the other operator or task experiences I/O errors.

Example: Regaining use of WINDOW

After changing WINDOW (CNME1505) several times, the following error occurs when you attempt to use WINDOW: DSI030I I/O ERROR READING CNME1505. Issue the following to regain the use of WINDOW:

```
REACC DSICLD
```

READSEC (NCCF)

Syntax

READSEC

➤ READSEC — DD — = — *ddname* — (— *membername* —) — DSN — = — *dsname* —

Purpose of Command

The READSEC command checks an operator's read authority to a data set or DD name and optionally checks a member.

Operand Descriptions

ddname

Specifies the DD name against which a read authority check is performed. If *ddname* is a NetView standard partitioned data set, authority checking is done for that DD name and, optionally, for member *membername*. See the BROWSE command help for a list of valid DDNAMES.

For nonstandard DD names, READSEC converts *ddname* to its underlying data set name and then checks for read authority for that data set name.

For VSAM files, READSEC also converts *ddname* to its underlying data set name and checks for read authority for that data set name.

dsname

Specifies the data set name against which read authority checking is done.

(*membername*)

Specifies the member name of the data set against which read authority is checked.

Note: A VSAM file does not have a *membername*.

Usage Notes

Consider the following when using the READSEC command:

- The READSEC command acts as a central point to control read access by all methods by which an operator might display the contents of a data set. These methods are:
 - QSAM PIPE stage
 - < PIPE stage

- CLIST and PROFILE keywords of the LIST command
- Member BROWSE
- DSIVSMX (VSAM access through REXX and CLIST)
- When the READSEC command is entered by an operator, it results in one of the following messages:
 - DSI633I for successful read access
 - DSI213I for unsuccessful read access

The actual data is not accessed nor its existence verified.

- When protecting specific data set names using CMDAUTH=TABLE or CMDAUTH=SAF, slashes (/) must be substituted for periods (.) in the data set name.

Restrictions

The READSEC command does not resolve member names to their potential synonym names. It is therefore possible that the BROWSE command will disallow access to a given member after resolving it to a synonym name, even if READSEC indicates that the member name is authorized.

Return Codes

Return Code Meaning

- | | |
|-----------|--|
| 0 | Authorization check was successful. |
| 4 | Not authorized to issue the command. |
| 8 | Syntax error. |
| 12 | Unknown DD name. |
| 16 | Dynamic names not allowed under the PPT. |

Example: Checking read access to the DSIPARM data set

To determine whether read access authority exists for the DSIPARM data set, enter:

```
READSEC DD=DSIPARM
```

Response

If read access is valid, the response is message DSI633I. Otherwise, the response is message DSI213I.

RECORD (NLDM)

Syntax

RECORD

```
➔ RECORD — — SESSTATS — — resname1 — — resname2 ➔
           — — STRGDATA — —
```

Purpose of Command

The RECORD command writes accounting and resource statistics data, or storage and processor utilization data, to the external log. After you issue the RECORD command, the counters are reset; they are not cumulative.

If you request that only availability data be collected (SESSTATS=AVAIL on an INITMOD statement), the session monitor does not collect any accounting data and records zeros for accounting data in the external log.

Also, if you request that availability data not be kept for the session (AVAIL=NO on a KCLASS statement), the session monitor does not record any data in the external log.

Operand Descriptions

SESSTATS

Writes accounting and resource statistics for *resname1* and *resname2* to the external log.

resname1

Specifies the resource name of the primary session partner.

You can use specific resource names for *resname1*, or you can use one of the following forms:

Represents all resource names.

abc*

Represents all resource names beginning with abc.

ab?????c

Represents all 8-character resource names, beginning with ab and ending with c.

Note: Record data in either the VTAM program or the session monitor. If both the VTAM program and the session monitor record data, the counter is inaccurate.

resname2

Specifies the resource name of the secondary session partner.

You can use specific resource names for *resname2*, or you can use one of the following forms:

Represents all resource names.

abc*

Represents all resource names beginning with abc.

ab?????c

Represents all 8-character resource names, beginning with ab and ending with c.

Note: Record data in either the VTAM program or the session monitor. If both the VTAM program and the session monitor record data, the counter is inaccurate.

STRGDATA

Writes session monitor storage data statistics to the external log.

Example: Writing accounting and resource statistics for all sessions

To write accounting and resource statistics for all sessions, use the following command:

```
RECORD SESSTATS * *
```

Example: Writing accounting and resource statistics for all sessions with specified resources

To write accounting and resource statistics for all sessions between resources beginning with TSO and LU, use the following command:

```
RECORD SESSTATS TSO* LU*
```

Example: Writing accounting and resource statistics for all sessions with specified resources and names

To write accounting and resource statistics for all sessions between resources with 8-character names, beginning with a T and ending with 1, and all resources beginning with LU2, use the following command:

```
RECORD SESSTATS T??????1 LU2*
```

Example: Writing accounting and resource statistical data to the external log

To write accounting and resource statistical data to the external log for session partners NY37 and L01, use the following command:

```
RECORD SESSTATS NY37 L01
```

Example: Writing data to the log for sessions with specified resources

To write data to the log for sessions between resources beginning with T2 and all LU2 resources, use the following command:

```
RECORD SESSTATS T2?????? LU2*
```

Example: Writing storage statistics to the external log

To write storage statistics to the external log, use the following command:

```
RECORD STRGDATA
```

RECYCLE (EAS)

Syntax

EAS RECYCLE



IBM-Defined Synonyms

Command or Operand	Synonym
MODIFY	F

Purpose of Command

The RECYCLE command stops any Event/Automation Service (E/AS) task that is not already stopped, and then recycles the task.

Note: If an attempt is made to recycle a task that is already stopped, a warning console message is issued and the recycle attempt proceeds.

Operand Descriptions

procname

Specifies the E/AS job name.

TASK=*taskid*

Specifies the E/AS service tasks to be recycled. The *taskid* can have the following values:

ALERTA

The alert adapter service task

ALERTC

The confirmed alert adapter service task

MESSAGEA

The message adapter service task

MESSAGEC

The confirmed message adapter service task

EVENTRCV

The event receiver service task

TRAPALRT

The trap to alert conversion task

ALRTTRAP

The alert to trap conversion task

ALL

All service tasks

Restrictions

You can specify only one TASK operand for each RECYCLE command. If you want to specify more than one service task, separate each *taskid* with a comma and enclose the *taskids* string within parentheses.

Example: Recycling a service task

To recycle the alert adapter service task for the E/AS job named IHSAEVNT, enter:

```
F IHSAEVNT,RECYCLE,TASK=ALERTA
```

Response

The following response is received:

```
IHS0118I Alert Adapter task has terminated.  
IHS0124I Alert Adapter task initialization complete.
```

Example: Recycling multiple service tasks

To recycle the alert adapter and message adapter service tasks for the E/AS job named IHSAEVNT, enter:

```
F IHSAEVNT,RECYCLE,TASK=(ALERTA,MESSAGEA)
```

Response

The following response is received:

```
IHS0118I Alert Adapter task has terminated.  
IHS0118I Message Adapter task has terminated.  
IHS0124I Alert Adapter task initialization complete.  
IHS0124I Message Adapter task initialization complete.
```


Operand Descriptions

taskname

Specifies the name of the task to be started or restarted. This name corresponds with a name specified on a TSKID keyword on a TASK definition in the CNMSTYLE member.

parameter

Specifies a 1- to 8-character token to be passed to the task when it is started or restarted. For data services tasks (DSTs), this is the name of the initialization member. This operand is optional.

Restrictions

Any authorization checking performed on the START and STOP commands is in force for the RECYCLET command list.

Example: Restarting a task

If you determine that a primary status focal point cannot be activated in a reasonable period of time, log on to another node, and enter:

```
RECYCLET CNMTAMEL DUIISFP
```

Response

Issue multiple CHANGEFP or FOCALPT commands to switch data server support to your new focal point when the RECYCLET command list finishes processing.

Example: Re-enabling a status collection point

If you determine that your status collection point can be re-enabled, enter:

```
RECYCLET CNMTAMEL DUIISC
```

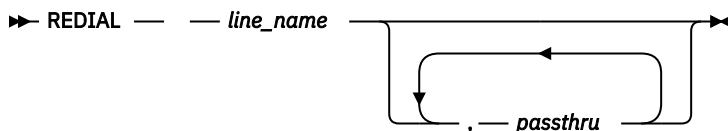
Response

Issue CHANGEFP or FOCALPT commands from the new status focal point for all resource status collectors when the RECYCLET commands finish processing.

REDIAL (NCCF; CNME0031)

Syntax

REDIAL



Purpose of Command

The REDIAL command list requests that the VTAM program search for an alternative path, if a dial-out attempt is unsuccessful, or end a session request without searching for an alternative path.

Operand Descriptions

line_name

Specifies the resource name of a switched line.

passthru

Specifies up to 6 parameters which are appended unchanged to the VTAM VARY command issued by the REDIAL command. No validation for duplicate or conflicting parameters is performed.

Example: Searching for an alternative path

To search for an alternative path for LINE27, enter:

```
REDIAL LINE27
```

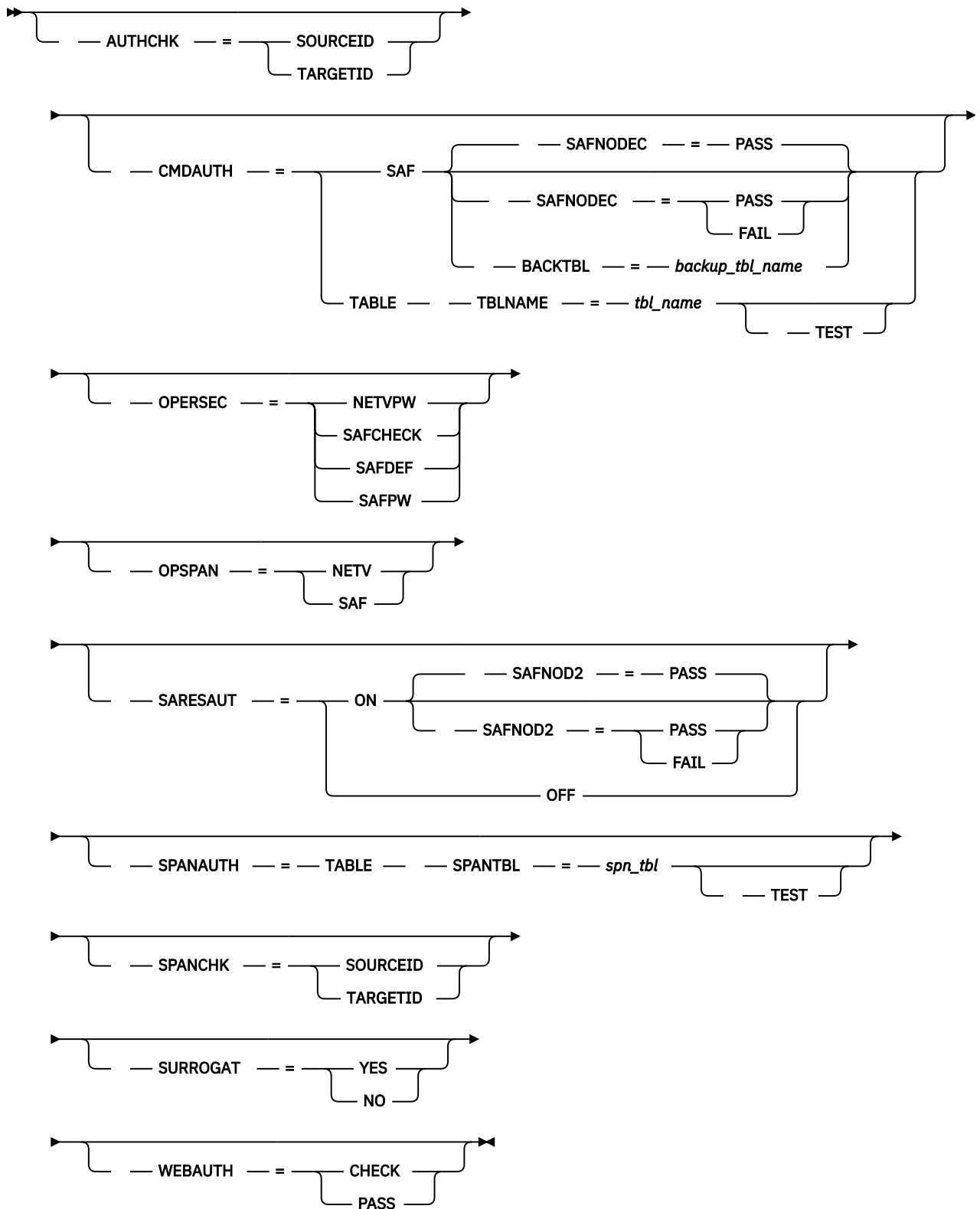
Example: Ending a session without searching for an alternative path

To end the session request for LINE1 without searching for an alternative path, enter:

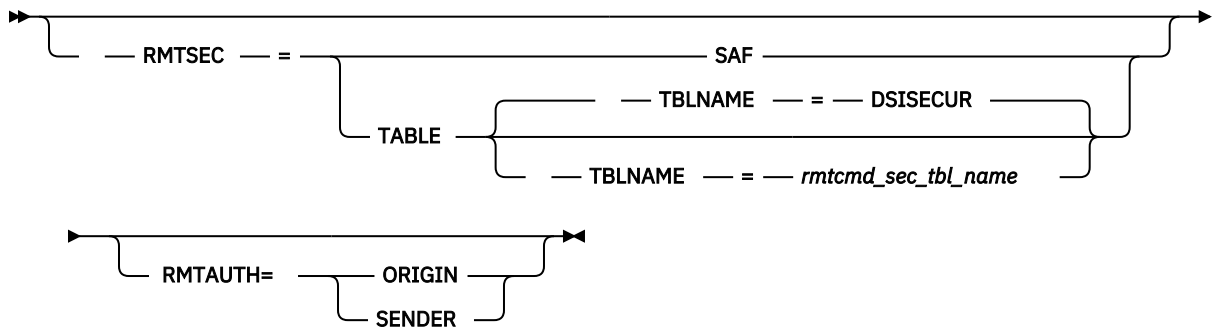
```
REDIAL LINE1,END
```

REFRESH (NCCF)

Syntax**REFRESH****SecOpts**



RmtOpts



Purpose of Command

The REFRESH command dynamically updates operator definitions and security options. The REFRESH command performs system symbolic substitution on records read from the DSIOPF member in the DSIPARM data set, profile members in the DSIPRF data set, the DSISECUR data sets, the command authorization tables in the DSIPARM data set, and a NetView span table. The &DOMAIN symbolic that is supplied with the NetView program is also included in the substitution process. The substitution is performed after comment removal but prior to record processing. This command also removes comments after substitution. Substitution is always performed on the &DOMAIN symbolic, unless substitution was disabled when the NetView program was started. For MVS and user-defined system symbolics, substitution is not performed if you are not running on an MVS system, you are running on an MVS system prior to MVS Version 5 Release 2, substitution was disabled when the NetView program was started, or you have not defined an MVS system symbolic on your MVS system.

Operand Descriptions

OPERS

Specifies that the NetView operator definitions in DSIOPF are to be refreshed. Operators that are logged on when the REFRESH OPERS command is issued do not have their profile information refreshed until they log off and then log on again.

TEST

Indicates that you want to see a list of the specified changes to the operator definitions in DSIOPF before you issue the REFRESH OPERS command to make the changes effective.

The OPERS operand will fail if there are no valid operator definitions in DSIOPF.

The OPERS operand is not available when OPERSEC=SAFDEF.

If you issue the REFRESH OPERS command and the value of OPERSEC is SAFCHECK or SAFPW, you might also need to add, delete, or change operator definitions in your SAF security product.

AUTHCHK=

Specifies whether a command's authorization is checked against the original issuer of the command or the final issuer of the command when they are different. You can specify the following values:

SOURCEID

Specifies that the command is checked against the authority of the original issuer of the command. When this information is not available, the identity of the task where the command first entered the NetView program is checked.

TARGETID

Specifies that the command is checked against the authority of the final issuer of the command.

For more information about the prerequisites and security considerations of this keyword, see the *IBM Z NetView Security Reference*.

CMDAUTH=

Specifies which method must be used for command authority checking. You can specify the following values:

SAF

Specifies that command authorization must be checked in the NETCMDS class of the SAF product. This value is only allowed when OPERSEC=SAFCHECK or OPERSEC=SAFDEF. Specifying CMDAUTH=SAF without specifying BACKTBL results in no backup command authority checking.

The SAF product is not called for immediate commands. If a backup command authorization table exists, immediate commands will be checked in the backup table. The backup table will also be checked if the SAF product is not able to make a security decision. This process can happen when:

- No definition in the NETCMDS class applies to the command.
- The NETCMDS class is not active.
- The SAF product is not active.

In these cases, you can define another method for command authority checking by specifying a backup command authorization table.

If you choose not to specify a backup table, you can specify the SAFNODEC option to pass or fail command authorization when the SAF product does not make a security decision. It is recommended that a backup table always be specified when using CMDAUTH=SAF to provide a more controlled security environment when the SAF product is unable to make a decision.

SAFNODEC=

Specifies the action to take during command authorization when no security decision has been made by the SAF product for the command and no backup command authorization table has been specified. SAFNODEC and BACKTBL are mutually exclusive. Valid values are:

PASS

Specifies that command authorization is to pass for the command. This is the default value.

Note: Specifying SAFNODEC=PASS allows any command to process when the SAF product cannot make a decision on the authority to issue the command. See the *IBM Z NetView Security Reference* for information about the effects of not using BACKTBL.

FAIL

Specifies that command authorization is to fail for the command.

Note: If you specify SAFNODEC=FAIL and a problem occurs in the SAF product, an operator will not be able to issue any command except immediate commands and commands with SEC=BY specified on the CMDDEF statement.

BACKTBL=

Specifies the 1-8 character name of the backup command authorization table in DSIPARM to be used when the SAF product cannot be called for immediate commands, or the SAF product does not yield a security decision. For further information about the command authorization table, see the *IBM Z NetView Security Reference*.

TABLE

Specifies that a command authorization table is to be read and built into a dynamic internal table to be used for any subsequent command authorization checks.

TBLNAME=

Specifies the 1-8 character name of the command authorization table in DSIPARM to be used when CMDAUTH=TABLE is specified. For further information about the command authorization table, see the *IBM Z NetView Security Reference*.

TEST

Tests the syntax of the command authorization table without enabling it.

Commands entered before the REFRESH command completes are checked using the authorization method in effect before the REFRESH command was issued.

OPERSEC=

Specifies the method used for operator identifier checking. You can specify the following values:

NETVPW

Specifies that NetView operators are defined by a list of operator identifiers in DSIOPF. The identification is validated with a password associated with the identifier in DSIOPF. The profile, read from DSIPRF at logon, contains the operator attributes.

SAFCHECK

Specifies that operator identification and password or password phrase checking is done using an SAF security product. Attributes given to the operator at logon are taken from the operator's profile in DSIPRF which can be specified by OPERATOR and PROFILEN statements in DSIOPF, or by default (see HELP for the DEFAULTS command LOGPROF keyword).

Other SAF checks, such as data set access and MVS system command authorization through the OPERCMDS class, are done using NetView task user IDs.

SAFDEF

Specifies that operator identification and password or password phrase checking is done using an SAF security product. Authority to log on as a NetView operator is controlled through the APPL class in the SAF product. To log on, the operator must be permitted to access the resource name that represents your NetView application in the APPL class. The resource name is the domain name.

The attributes given to the operator at logon are defined in the NetView segment of the operator's user profile.

Other SAF checks, such as data set access and MVS system command authorization through the OPERCMDS class, are done using NetView task user IDs.

SAFPW

Specifies that operator identification and password or password phrase checking is done using an SAF security product. Attributes given to the operator at logon are taken from the operator's specified profile in DSIPRF which can be specified by OPERATOR and PROFILEN statements in DSIOPF, or by default (see HELP for the DEFAULTS command - LOGPROF keyword).

Other SAF checks, such as data set access and MVS system command authorization through the OPERCMDS class, are done against the NetView startup procedure name.

For more information about operator attributes, see the *IBM Z NetView Administration Reference*.

Before changing from NETVPW to SAFPW, SAFCHECK, or SAFDEF, ensure that your operators are defined to the SAF product.

If you specify a value for SECOPTS.OPERSEC that uses DSIOPF definitions, the DSIOPF operator definitions are refreshed from DSIPARM. Although MINIMAL is a valid SECOPTS.OPERSEC value in the CNMSTYLE member, you cannot use the REFRESH command to specify OPERSEC=MINIMAL.

OPSPAN=

Specifies whether the operators ability to start spans is controlled through DSIPRF or by the NETSPAN class of an SAF product. You can specify the following values:

NETV

Specifies that authority checking for starting spans is controlled by SPAN and ISPAN statements in operator profiles defined in DSIPRF. You cannot specify this value when OPERSEC=SAFDEF.

SAF

Specifies that authority checking for starting spans is controlled by the NETSPAN class of an SAF product. This value is only valid when OPERSEC=SAFCHECK or OPERSEC=SAFDEF.

For more information about span of control, see the *IBM Z NetView Security Reference*.

When changing from SAF to NETV, any currently active spans that were defined by logging on with an operator profile (OPSPAN=NETV) will have their access level reset to the authority permitted by the operator profile. Any active spans known only to SAF will retain only the access level permitted by the SAF product. If these spans are stopped, they cannot be restarted.

When changing from NETV to SAF, any currently active spans will have their access level reset to the authority permitted by the SAF product. You might lose some currently active spans unless they are defined in an SAF product and the operator is permitted to the span.

SARESAUT=

Specifies whether to check resource level security for IBM Z System Automation resources. You can specify the following values:

ON

Specifies that System Automation resource level security checking is active. Security authorization is made by using profiles that are specified for specific System Automation resources (for example, subsystems or groups). For information about resource level security checking, refer to the IBM Z System Automation library.

SAFNOD2=

Specifies the action to be taken when the following conditions occur:

- The SAF product (such as RACF®) is not active
- A resource profile that matches the resource that is being checked is not found

PASS

Specifies that the resource security request is granted. This is the default value.

FAIL

Specifies that the resource security request is denied.

OFF

Specifies that System Automation resource level security checking is not active. No authorization checks are made.

SPANAUTH=

Defines the span of control used by the NetView program to secure resource and view access.

SPANAUTH cannot be specified when OPERSEC=MINIMAL. You can specify the following values:

TABLE

Specifies that the NetView program is to verify authorization for resources and views using a NetView span table specified by the SPANTBL keyword. The table can be modified and reloaded using the REFRESH command without requiring the NetView program to be recycled.

SPANTBL=

Specifies the 1-8 character name of the NetView span table in DSIPARM to be used when SPANAUTH=TABLE is specified.

TEST

Tests the syntax of the NetView span table without enabling it.

SPANCHK=

Specifies whether a VTAM command is span checked against the original issuer of the command or the final issuer of the command. You can specify the following values:

SOURCEID

Specifies that span checking is done using the source operator ID. The original issuer must be logged on to the NetView system when the VTAM command is run, except for a VTAM command that is issued from a system console where no logon has been performed (source ID is *BYPASS*). This VTAM command is assumed to be fully authorized and no span checking is performed.

TARGETID

Specifies that span checking is done using the target operator ID.

SURROGAT=

YES

Specifies that SAF surrogate checking is performed for the START TSOSERV, STOP TSOSERV, and the TSO PIPE stage.

NO

Specifies that SAF surrogate checking is not performed for the START TSOSERV, STOP TSOSERV, and the TSO PIPE stage.

WEBAUTH=CHECK|PASS

Specifies whether authorization checking is to be performed for operator access to the NetView web server.

CHECK

Perform authorization checking for access to the NetView web server.

PASS

Do not perform authorization checking for access to the NetView web server. For performance reasons, if all NetView operators are to be granted access to the NetView web server, specify a value of PASS.

RMTSEC=

Specifies the method to use to determine which incoming RMTCMD and ENDTASK requests are accepted. You can use this security function to specify which operators from which domains are authorized to start, stop, and share RMTCMD sessions with autotasks in your NetView program.

Note:

1. The REFRESH RMTSEC=SAF command differs from the REFRESH OPSPAN=SAF and REFRESH CMDAUTH=SAF in that the RMTSEC operator will complete if the RMTOPS class is not active, but the NETSPAN class and NETCMDS classes must be active for the other two REFRESH commands noted to complete.
2. This RMTCMD security is in addition to the normal command authority checking for RMTCMD and the command being issued within the RMTCMD command.

You can specify the following values:

SAF

Specifies that the RMTOPS class of the SAF product is to be called for authorization checking of the initial RMTCMD request and ENDTASK requests. If an RMTCMD security table was previously in use, the storage for the table is reclaimed.

TABLE

Specifies that a RMTCMD security table is to be read and built into an internal table to be used for any subsequent RMTCMD or ENDTASK authorization checks.

TBLNAME=

Specifies the 1-8 character name of the RMTCMD authorization table in DSIPARM to be used when RMTSEC=TABLE is specified. If TBLNAME is not specified for RMTSEC=TABLE, the DSISECUR table is used.

For more information about the RMTCMD authorization table, see the *IBM Z NetView Security Reference*. A sample table is provided as member DSISECUR in CNMSAMP.

RMTAUTH

Specifies which operator ID is used in the comparisons generated by the REFRESH RMTSEC or RMTINIT.SECOPT setting. This is useful when a RMTCMD or ENDTASK command is invoked as a result of routing by another command such as EXCMD or another RMTCMD.

You can specify the following values:

ORIGIN

Use the operator ID where the process was started which later resulted in a RMTCMD command.

SENDER

Use the operator ID under which the RMTCMD command was invoked. This is the default value.

The RMTAUTH comparisons use the fully qualified operator identification including NETID and DOMAIN, as well as user name. You can change the value at any time.

Usage notes

The following usage notes apply to the REFRESH command:

- After issuing the REFRESH command, you can use the LIST SECOPTS command to review the security options in effect.
- For RMTCMD security, if the NetView program cannot create an SAF security environment during the initialization of the DSIUDST task, REFRESH RMTSEC=SAF is rejected. You can issue a REFRESH RMTSEC=TABLE command or fix the SAF error and recycle the DSIUDST task.
- Have a VTAM APPL statement in your VTAM applications major node member (for example, A01APPLS) for each defined NetView operator. If you do not have enough APPL statements for dynamically added operators, define a new VTAM application major node with a VTAM APPL statement for each additional operator.

For more information about creating VTAM APPL definitions for NetView operators, see *IBM Z NetView Installation: Getting Started*.

- If you want to retrieve operator availability and span of control information for dynamically added operators, use the QOS and QRS commands or use the service macros DSIQOS and DSIQRS in any user-written applications. For more information about the DSIQOS and DSIQRS service macros, see *IBM Z NetView Programming: Assembler*.

Restrictions

The following restrictions apply to the REFRESH command:

- If RMTINIT.RMTSECUR=*NONE* was specified in the CNMSTYLE member or if RMTSECUR NONE was specified in DSIUINIT at installation, no RMTCMD security checking is performed. You can access the RMTCMD security table by issuing the REFRESH command. You can also use the REFRESH command to access a security product if RMTINIT.SAFrefresh=Yes was specified in the CNMSTYLE member or if SAFREFSH=YES was specified in DSIUINIT.
- If SECOPTS.OPERSEC=MINIMAL is specified in the CNMSTYLE member, the REFRESH command cannot be used to change any of the security settings. However, the REFRESH command can be used to test the validity of the syntax of a command authorization table.
- The following conditions exist for operators that you dynamically delete using the REFRESH OPERS command:
 - If an operator is logged on when you issue the REFRESH OPERS command, the operator session continues until the operator logs off. However, the operator can no longer issue the DISPLAY, MODIFY, or VARY commands for any resource that is defined in any span of control, unless OPSPAN=SAF is being used for span definitions. If you do not want a deleted operator to remain logged on after issuing the REFRESH OPERS command, issue the STOP FORCE command to end the session.
 - If the operator is not logged on when you issue the REFRESH OPERS command, the operator can no longer log on.
- Only one task at a time can issue the REFRESH command. If you issue a REFRESH command while another REFRESH command is processing under another task, your task waits until the other REFRESH command completes before your command proceeds.

Return Codes

Return Code	Meaning
-------------	---------

0	The function was successful.
4	The syntax of the REFRESH command was not valid.

- 8**
Incompatible keyword/value combinations were entered on the REFRESH command, or keyword/value combinations conflict with current settings.
- 12**
Required task DSIUDST is not active.
- 16**
The RMTCMD SAF security environment is not available.
- 20**
REFRESH to SAF is not allowed as specified in the initialization member for task DSIUDST.
- 24**
The RMTCMD security table member is not found in DSIPARM.
- 28**
The RMTCMD security table has no valid statements.
- 32**
Internal error in processing the RMTCMD security table.
- 36**
Storage exhausted building the RMTCMD security table.
- 40**
Internal error encountered by REFRESH OPERS.
- 44**
No valid operator definitions in DSIOPF.
- 48**
Syntax error encountered in DSIOPF for the REFRESH command.
- 52**
Syntax error encountered in DSIOPF for the REFRESH OPERS,TEST command.
- 56**
DSIPRF is not allocated for refreshing operator definitions.
- 60**
The NETCMDS class, NETSPAN class, or SAF is unavailable.
- 64**
The FASTAUTH service for the NETCMDS class is unavailable.
- 68**
Internal dynamic resource error.
- 72**
The security environment for the NetView program's main task does not exist.
- 104**
The command authorization table or NetView span table has syntax errors.
- 108**
The command authorization table member or NetView span table member was not found in DSIPARM.
- 112**
An error was found in a %INCLUDE statement contained in a command authorization table or NetView span table.
- 120**
An error occurred while reading a command authorization table member or NetView span table member from DSIPARM.
- 124**
Storage was exhausted when building a command authorization table or NetView span table.

Example: Consulting an SAF product for remote security authorization decisions

If RMTSECUR TABLE TBLNAME=*tblname* was specified in the DSIUINIT member of DSIPARM at installation, the *tblname* member of DSIPARM is consulted for authorization checking during RMTCMD and ENDTASK processing. To consult an SAF product for authorization decisions, enter:

```
REFRESH RMTSEC=SAF
```

Response

```
DSI633I REFRESH COMMAND SUCCESSFULLY COMPLETED
```

Example: Using the LCLSECUR table in DSIPARM for all RMTCMD security checks

If the NetView program is using an SAF product for authorization decisions during RMTCMD and ENDTASK processing, you can use the REFRESH command to switch to the LCLSECUR table. To use the LCLSECUR table in DSIPARM for all subsequent checks, enter:

```
REFRESH RMTSEC=TABLE TBLNAME=LCLSECUR
```

Response

```
DW0330I RMTSEC SECURITY TABLE HAS BEEN INITIALIZED  
DSI633I REFRESH COMMAND SUCCESSFULLY COMPLETED
```

Example: Dynamically updating the group of valid NetView operators

You can add or delete operators in DSIOPF and dynamically update their ability to log on by using the REFRESH OPERS command. To add OPERADD and delete OPERDEL you can change OPERDEL to OPERADD in DSIOPF. To update the group of valid NetView operators, enter:

```
REFRESH OPERS
```

Response

```
DW0831I OPERATOR OPERADD IS ADDED TO THE GROUP OF VALID NETVIEW OPERATORS  
DW0830I OPERATOR OPERDEL IS DELETED FROM THE GROUP OF VALID NETVIEW  
OPERATORS  
DSI633I REFRESH COMMAND SUCCESSFULLY COMPLETED
```

Note: This example assumes that the value of SECOPTS.OPERSEC in the CNMSTYLE member is neither SAFDEF nor MINIMAL.

Example: Checking updates in the DSIOPF table

You can add or delete operators in DSIOPF and check these changes by using the REFRESH OPERS TEST command before changing the group of valid NetView operators. To check the changes in the previous example before issuing the REFRESH OPERS command, enter:

```
REFRESH OPERS TEST
```

Response

```
DW0834I NEW OPERATOR DEFINITION FOR OPERADD IS FOUND IN DSIOPF  
DW0833I CURRENTLY DEFINED OPERATOR OPERDEL IS NOT FOUND IN DSIOPF  
DW0835I TEST OF REFRESH OPERS COMMAND SUCCESSFULLY COMPLETED
```

This example assumes that the value of SECOPTS.OPERSEC in the CNMSTYLE member is neither SAFDEF nor MINIMAL.

Example: Setting SAF command authority checking

You can specify that SAF command authority checking is the normal method for command authorization. You can also specify a backup command authorization table to be used for immediate commands and when the SAF product yields no security decision.

To set SAF security checking for command authorization, and to specify a backup command authorization table of BKUPTBL, enter:

```
REFRESH CMDAUTH=SAF BACKTBL=BKUPTBL
```

Response

```
BNH211I BACKUP COMMAND AUTHORIZATION TABLE BKUPTBL HAS BEEN INITIALIZED
DSI633I REFRESH COMMAND SUCCESSFULLY COMPLETED
```

The previous example assumes that the value of SECOPTS.OPERSEC is either SAFDEF or SAFCHECK as defined in either the CNMSTYLE member or in a previous REFRESH command.

Example: Specifying that origin operators are to be used for all RMTCMD security checks

If the NetView program is using the sending operator for authorization decisions during RMTCMD and ENDTASK processing, you can use the REFRESH command to use the origin operator instead. To use the origin operator, enter the following command:

```
REFRESH RMTAUTH=ORIGIN
```

Response

```
DSI633I REFRESH COMMAND SUCCESSFULLY COMPLETED
```

REFRESH (TARGET)

Syntax

➤ REFRESH ➤

Purpose of Command

Use the REFRESH subcommand within the panel that is displayed by the TARGET command. This command sends a data request to all inactive remote NetView domains included in the current session. You can use this command to attempt to refresh data from domains that are shown as inactive. (For more information about the included data, see the TARGET command.)

REGIP (NCCF)

Syntax

REGIP

➤ REGIP — — *host* — —

ADD

DELETE

LIST

 ➤

Purpose of Command

The REGIP command grants a remote host permission to send syslog messages to the NetView syslog server. The command can be used to add remote hosts, delete them or list the registered hosts. NetView task DSIIPLOG must be active for the command to work.

Operand Descriptions

host

Specifies the remote host to register. It can be specified as a TCP/IP host name or as an IP address in dotted notation (127.44.44.44 for example).

ADD

Registers the specified host.

DELETE

Deletes the specified host.

LIST

Outputs an MWLTO listing the registered hosts. If no hosts are registered, BNH733I is issued instead.

Usage Notes

The following restrictions apply to the REGIP command:

- TCP/IP can resolve host names and addresses (the GETHOSTBYNAME and GETHOSTBYADDR functions must work)
- Hosts can be deleted using either host names or IP addresses.
- When using the LIST option, the host operand is not valid. The only valid form of the command is "REGIP * LIST".

Example: Registering a host using host name

To register host HOST1.RALEIGH.IBM.COM, enter:

```
REGIP HOST1.RALEIGH.IBM.COM
```

Example: Registering a host using IP address

To register host with IP address 146.84.158.83, enter:

```
REGIP 146.84.158.83
```

Example: Deleting a host

To delete HOST1, enter:

```
REGIP HOST1 DELETE
```

Example: Listing registered hosts

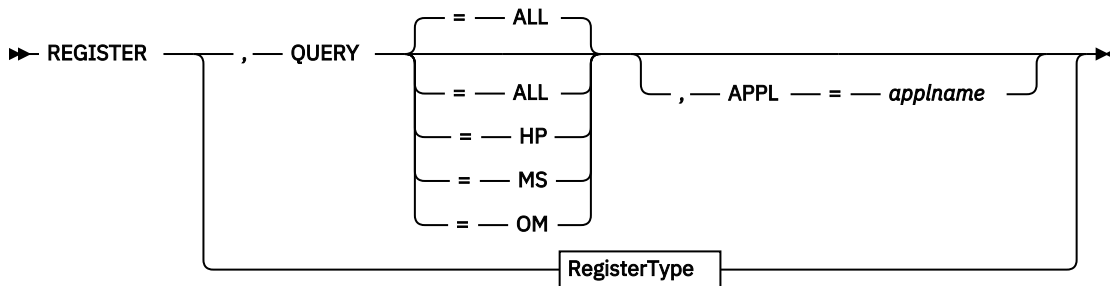
To list currently registered hosts, enter:

```
REGIP * LIST
```

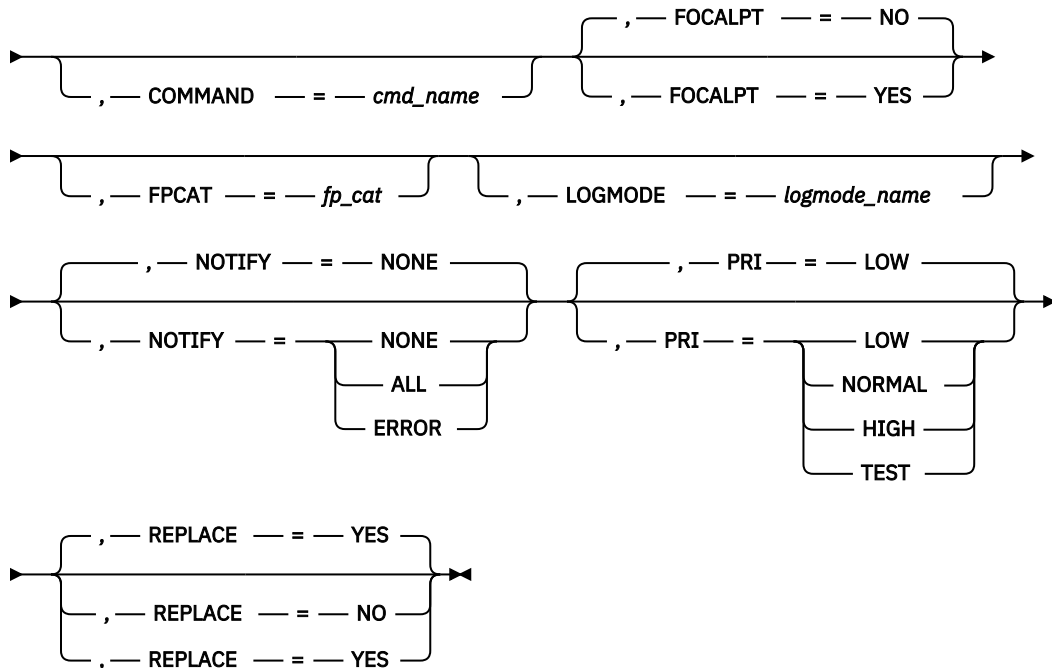
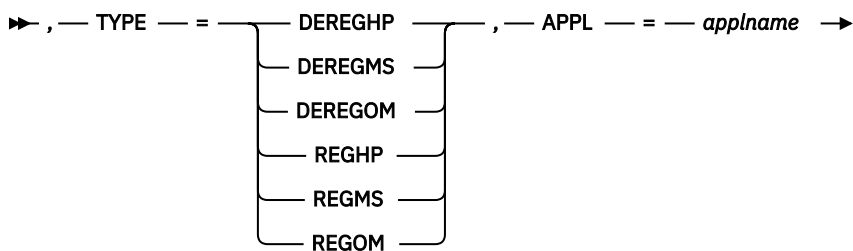
REGISTER (NCCF)

Syntax

REGISTER



RegisterType



Purpose of Command

The REGISTER command registers or deregisters the following:

- A management services (MS) application with:
 - The NetView MS transport
 - The MS_CAPS focal point application
- An operations management served application with the NetView operations management application

- A high performance application with the NetView high performance transport

Before accepting the registration request, the NetView program verifies that the task issuing the REGISTER command has the authority to issue the command specified on the COMMAND=*command_name* keyword.

The current task where the command is issued is registered as the task where the application receives unsolicited data.

An application can register as an MS application, as an operations-management-served application, and as a high performance application. A registration of one type does not affect the registration of the other type.

You can use the CNMREGIST and CNMHREGIST service routines to register user-written applications. For more information about the CNMREGIST and CNMHREGIST service routines, see *IBM Z NetView Programming: PL/I and C, SC27-2860*.

You can use the DSI6REGS and DSIHREGS macros to register user-written applications. See *IBM Z NetView Programming: Assembler* for more information about the DSI6REGS and DSIHREGS macros.

If the REGISTER command runs more than one line, you can use the INPUT command to add up to two more lines.

Operand Descriptions

QUERY

Specifies which registered applications are displayed.

ALL

Specifies that all registered MS applications, operations management served applications, and high performance applications are displayed. ALL is the default value if you do not specify any value.

HP

Specifies that registered high performance applications are displayed.

MS

Specifies that registered MS applications are displayed.

OM

Specifies that registered operations management served applications are displayed.

APPL=*applname*

Specifies the name of the MS application, operations management served application, or high performance application being registered, deregistered, or queried. The application name is either one of the architecturally defined hexadecimal values or a 1–8 character user-defined name.

User-defined names must use only the EBCDIC characters 0–9 and A–Z (uppercase only).

If an architected name is used, it is specified as X'*hhhhhhhh*', where *hhhhhhhh* is the architecturally defined hexadecimal value. The application names that are supplied by IBM are restricted so that they can be registered and deregistered only from code that is supplied by IBM. The application names that are supplied by IBM include the following names:

Restricted Application

Hex Equivalent

ALERT

X'23F0F3F1'

EP_OPS

X'23F0F1F6'

EP_SPCS

X'23F0F1F4'

HMON_DST

X'30F0F8F5'

HMON_OST

X'30F0F8F4'

LINKSERV

X'23F0F3F5'

MDS_ROUT

X'23F0F1F0'

MS_CAPS

X'23F0F1F1'

OPS_MGMT

X'23F0F1F7'

R_BRIDGE

X'30F0F5F9'

RMTCMD_O

X'30F0F7F2'

RMTCMD_R

X'30F0F5F5'

RMTCMD_S

X'30F0F7F0'

SPCS

X'23F0F1F5'

STATUS

No hexadecimal equivalent

No character equivalent

X'23F0F0F1'

No character equivalent

X'30F0F7F3'

TYPE

Specifies the type of application to register or unregister. Valid operands are:

DEREGHP

Deregisters a high performance application.

DEREGMS

Deregisters an MS application.

DEREGOM

Deregisters an operations management served application.

REGHP

Registers a high performance application to the NetView high performance transport.

REGMS

Registers an MS application to the NetView MS transport.

REGOM

Registers a second-level application to the operations management application, making the application a served application of operations management.

COMMAND=cmd_name

Specifies the name of the command processor that is driven with any data that has a destination name equal to the one in the *applname* operand. The only exception is for replies to requests that specified a command on the CNMSENDMU invocation. See *IBM Z NetView Programming: PL/I and C* for more information about the CNMSENDMU invocation. The data is received as an MDS-MU.

This operand is required for registration requests and not valid for deregistration requests.

FOCALPT

Specifies whether the MS application is a focal point application.

NO

Specifies that the MS application is not a focal point application. NO is the default.

Note: If an MS application registers as a focal point application and later the same MS application registers again with FOCALPT=NO, the registered application still receives focal point data from any node that did not attempt to send while the application was unregistered. This operand is not valid for operations management served applications or high performance applications.

YES

Specifies that the MS application is a focal point application. YES is allowed only for an MS application registration.

Note: When you specify YES, the focal point category name is the application name specified in *applname*.

This operand is optional for registration requests and not valid for deregistration requests.

FPCAT=*fp_cat*

Specifies the focal point category for which the registering application receives focal point information. Focal point category names are the *applnames* of MS focal point applications (applications registered with FOCALPT=YES). The NetView program supplies three focal point categories:

- Alerts — ALERT (X'23F0F3F1')
- Operations management — OPS_MGMT (X'23F0F1F7')
- Common Operations Services — SPCS (X'23F0F1F5')
- LINKSERV (X'23F0F3F5')

This operand is optional for registration requests and not valid for deregistration requests.

LOGMODE=*logmode_name*

Specifies the logmode that the registered high performance application uses when sending data. The *logmode_name* is matched with an entry in the VTAM logmode table to obtain session parameters. If no matching *logmode_name* is found, VTAM uses its own default session parameters for any session established for the registered high performance application.

Note: Specify the *logmode_name* operand for REPLACE=NO. If you specify the *logmode_name* for REPLACE=YES, the name must match the logmode of the currently registered application of the same name. If the application is not currently registered, specify *logmode_name*. This operand is not valid for MS applications or operations management served applications.

This operand is optional for registration requests and is not valid for deregistration requests.

NOTIFY

Specifies the type of session outage notification requested.

NONE

The application does not receive session outage notification.

ALL

The application receives notification of all session outages.

ERROR

The application receives session outage notification when there is an abnormal loss of connectivity to another node and the transport cannot re-establish connectivity.

PRI

Specifies the MQS priority for incoming requests. The MQS priority is used when the high performance transport uses the MQS for processing any unsolicited MDS-MU. Valid values are:

LOW

Processing is preempted by HIGH and NORMAL priority requests. This is the default.

NORMAL

Processing preempts a queue of LOW priority requests.

HIGH

Processing begins after any NORMAL requests currently in progress completes, but before queued NORMAL or LOW requests.

TEST

Queues the request based on the priority of the receiving task. The command priority can be set using the DEFAULTS or OVERRIDE commands.

REPLACE

Specifies whether the registration supersedes any previous registration for the same application.

YES

Specifies that this registration supersedes any previous registration for the same application. YES is the default. Use this operand to:

- Change the name of the command driven to process data received.
- Change the task where the command is driven with unsolicited data
- Specify whether the application receives focal point information.

Note: You cannot change the logmode of a high performance application using the REPLACE=YES option. To change the logmode of a high performance application, unregister the application and register it again with the new logmode value.

NO

Specifies that this registration does not supersede any current registration for the same application. If the same application is already registered, the command fails and a message is displayed.

This operand is optional for registration requests and not valid for deregistration requests.

Restrictions

The following restrictions apply to the REGISTER command:

- The FOCALPT STATUS category is restricted. You cannot register MS applications as interested in an *fp_cat* of STATUS.
- You cannot authority check architected application names because their length exceeds 8 characters.
- Deregistration stops the NetView MS transport's awareness of the MS application, operations management's awareness of the served application, or the NetView high performance transport's awareness of the high performance application. No further data can be sent or received by the application using the NetView transports.

Example: Registering an operations-management-served application

To register an operations-management-served application named ACTIVATR, with ACTCMD as the name of the command for receiving data and OPS_MGMT as the focal point category, enter:

```
REGISTER TYPE=REGOM,APPL=ACTIVATR,  
COMMAND=ACTCMD,FPCAT=OPS_MGMT
```

Response

The task under which this command is processed also receives unsolicited data. The ACTIVATR application receives notification of any focal point for operations management.

Example: Registering a high-performance application

To register a high performance application named HPOPS, with HPCMD as the name of the command for receiving data and DSIL6MOD as the logmode to be used for session parameters, enter:

```
REGISTER TYPE=REGHP,APPL=HPOPS,  
COMMAND=HPCMD,LOGMODE=DSIL6MOD
```

Response

The HPOPS application is ready to send and receive data. For sends, the DSIL6MOD logmode is used (if it exists).

Example: Displaying all registered applications

To display all registered applications, enter:

```
REGISTER QUERY
```

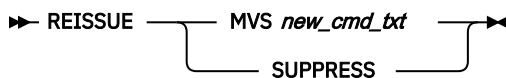
Response

A panel that lists all registered applications and their associated data is displayed.

REISSUE (NCCF)

Syntax

REISSUE



Purpose of Command

The REISSUE command is used for the MVS Command Revision function. It determines whether a revised command is to be issued or suppressed.

Operand Descriptions

MVS *new_cmd_txt*

Specifies that the new command text is issued under identical MVS command authority conditions as the original command triggering the NETVONLY action. The command is resubmitted to MVS, but is not subject to command revision.

new_cmd_txt

Specifies the command text.

SUPPRESS

Specifies that the command is not to be reissued.

Usage Notes

If TESTMODE was requested when the Command Revision Table (CRT) was compiled, the REISSUE command does not invoke the *new_cmd_txt*, but rather issues a WTO indicating the changed command text.

If the procedure invoked by the NETVONLY action ends without issuing a valid REISSUE command, the NetView program issues a WTO (CNM017E) to the original console indicating that the command was not acted upon (programming error).

Restrictions

The REISSUE command can only be used in a procedure driven by a NETVONLY action.

Return Codes

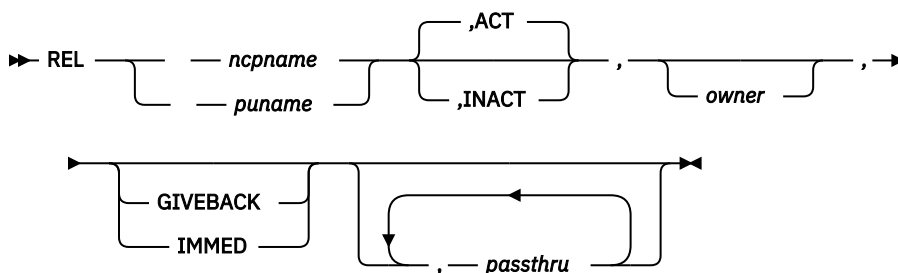
Return Code Meaning

- 4**
Keyword missing
- 16**
A REISSUE SUPPRESS command was previously issued in the current revision command environment.
- 69**
The MVS command text is missing.
- 75**
The MVS command text exceeded the maximum length allowed for commands to be sent to the MVS system.
- 290**
The REISSUE command was used outside of the revision command environment.

REL (NCCF; CNME0032)

Syntax

REL



IBM-Defined Synonyms

Command or Operand	Synonym
GIVEBACK	G
IMMED	I

Purpose of Command

The REL command list releases a previously acquired network control program (NCP) or releases a physical unit attached by a nonswitched line to an NCP.

Operand Descriptions

ncpname

Specifies the name of the NCP to be released.

puname

Specifies the name of the physical unit to be released.

ACT

Specifies that active cross-domain links and link stations are to remain active after they are released. ACT is the default.

INACT

Specifies that cross-domain links and link stations within the scope of the release are to be deactivated as part of the release processing.

owner

Specifies the owner of the NCP named by *ncpname* or *puname*. You can specify this keyword only if the NCP has previously been activated.

GIVEBACK

Specifies that all subordinate resources that are capable of nondisruptive deactivation are to be released without disruption to the LU-LU sessions. This applies only to an NCP.

IMMED

Specifies that the sessions using the released resources are to be ended immediately.

Note: If you do not specify this keyword, normal release occurs. Normal release prevents the establishment of new sessions, but does not end existing sessions.

passthru

Specifies up to 6 parameters which are appended unchanged to the VTAM VARY command issued by the REL command. No validation for duplicate or conflicting parameters is performed.

Restrictions

The following restrictions apply to the REL command:

- The commas between operands are required. If you omit a positional operand, indicate its absence with a comma. You do not need to specify trailing commas.
- If you specify ACT or INACT for a non-NCP PU, it is ignored.

Return Codes

Return Code	Meaning
0	Functioned normally
100	Internal failure; see message DWO050 in the NetView log for more information.

Example: Releasing a previously acquired specified NCP

To release a previously acquired NCP1, use the following command:

```
REL NCP1
```

Example: Releasing a specified NCP and leaving links and link stations active

To release NCP1 and leave links and link stations active, enter:

```
REL NCP1,ACT
```

Example: Specifying immediate release of a specified NCP

To specify immediate release of NCP1, and to default the status of links and link stations to active, enter:

```
REL NCP1,,,I
```

Example: Specifying immediate release of a specified NCP and deactivating links and link stations

To specify immediate release of NCP1 and deactivate links and link stations, enter:

```
REL NCP1,INACT,,I
```

Example: Releasing a specified PU

To release PU2, enter:

```
REL PU2
```

Example: Having a normal release of a specified NCP and deactivating cross-domain links and link stations

To have a normal release of NCP1 and deactivate the cross-domain links and link stations whose owning SSCP is SSCP1, enter:

```
REL NCP1,INACT,SSCP1
```

Example: Releasing a specified NCP and leaving cross-domain links and link stations active

To release NCP1 and leave the cross-domain links and link stations active to the NCP (the default is ACT), enter:

```
REL NCP1,,,G
```

RELCONID (NCCF)

Syntax

RELCONID

➤ RELCONID ➤

Purpose of Command

The RELCONID command releases any MVS console that was previously obtained using an MVS or GETCONID command.

You obtain an MVS console when you issue an MVS or GETCONID command. For normal operation, you do not need the RELCONID command because an operator authorized to issue the MVS command needs to retain a console to receive command response messages and to issue more MVS commands. Your console is released when you log off.

Restrictions

RELCONID cannot be used to release a console that was previously associated using an AUTOTASK command. However, the AUTOTASK command can be used with the DROP option to drop the previously associated console.

Return Codes

Return Code

Meaning

0

Successful processing.

4

Error in processing.

213

The command is rejected by the NetView program.

Example: Releasing a specified MVS console

To release the MVS console, enter:

```
RELCONID
```

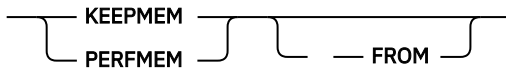
Response

The specified console is released from the task.

RELOAD (NLDM)

Syntax

RELOAD

➤ RELOAD —  — *membername* ➤

Purpose of Command

The RELOAD command reloads the response time monitor (RTM) PCLASS and MAPSESS or the KCLASS and MAPSESS definition statements.

The RELOAD command does not affect current sessions.

Operand Descriptions

KEEPMEM

Reloads keep class definitions.

PERFMEM

Reloads RTM performance class definitions.

FROM

Identifies the operand that follows as the name of a member in DSIPARM from which definitions are loaded. This operand is optional.

membername

Specifies the name of a member in DSIPARM (a data set) from which definitions are reloaded.

Example: Reloading the RTM performance class definitions

To reload the RTM performance class definitions from member RTMCLASS of DSIPARM, enter:

```
RELOAD PERFMEM RTMCLASS
```

Example: Reloading performance class definitions from a file

To reload performance class definitions from file RTMCLASS (use of FROM is optional), enter:

```
RELOAD PERFMEM FROM RTMCLASS
```

RELOAD (RODM)

Syntax

From an MVS console:

RELOAD

➤ MODIFY — — *name* — , — RELOAD — { , — MEMBER — = — EKGCUST — }
— { , — MEMBER — = — *name* — } ➤

From a NetView terminal:

RELOAD

➤ RODM — — RELOAD — { , — MEMBER — = — EKGCUST — }
— { , — MEMBER — = — *name* — } ➤

IBM-Defined Synonyms

Command or Operand	Synonym
MODIFY	F

Purpose of Command

The RELOAD command reloads the RODM customization member.

Operand Descriptions

name

Specifies the RODM MVS job name.

MEMBER=*name*

Specifies the name of the customization member to be loaded. This member must exist in the EKGCUST partitioned data set as specified in the RODM start JCL. The default customization member is EKGCUST.

Restrictions

Only the following parameters of EKGCUST are reloadable:

- IO_QUEUE_THRESHOLD
- LOCK_SLEEPTIME

For additional information, see the *IBM Z NetView Administration Reference*.

Example: Reloading a customization member

To reload the customization member EKGRMCT, enter the following command from a NetView terminal:

```
RODM RELOAD, MEMBER=EKGRMCT
```

Response

The RODM customization member EKGRMCT is reloaded.

REMOTEBR (NCCF)

Syntax

REMOTEBR

➤ REMOTEBR ➤

Purpose of Command

The REMOTEBR command initializes the autotask that receives the cross-domain or cross-network requests and replies. The REMOTEBR command enables the NetView Bridge remote dispatcher to register as an application on the high performance transport. This command is driven by the profile of the NetView Bridge remote dispatcher autotask. Set up the autotask in the host NetView system sending transactions and receiving replies and the host NetView system that contains the database server.

This autotask can be an autotask that is already being used by the NetView Bridge to communicate with a local database server, but this setup is not recommended. Issue the REMOTEBR command from a new autotask.

When the REMOTEBR command issues an error message, the autotask in which the command is running is still active. You can do one of the following:

- Recycle the autotask as follows:
 1. Issue EXCMD *autotaskname*,LOGOFF to log off the autotask from the NetView operator terminal.
 2. Correct the error.
 3. Issue AUTOTASK OPID=*autotaskname* to bring up the autotask.
- Correct the error, then issue EXCMD *autotaskname*,REMOTEBR from the NetView operator terminal.

When the REMOTEBR command has completed successfully, the task can be ended under severe errors. You can recycle the autotask as follows:

1. Correct the error.
2. Issue AUTOTASK OPID=*autotaskname*.

Return Codes

Return Code

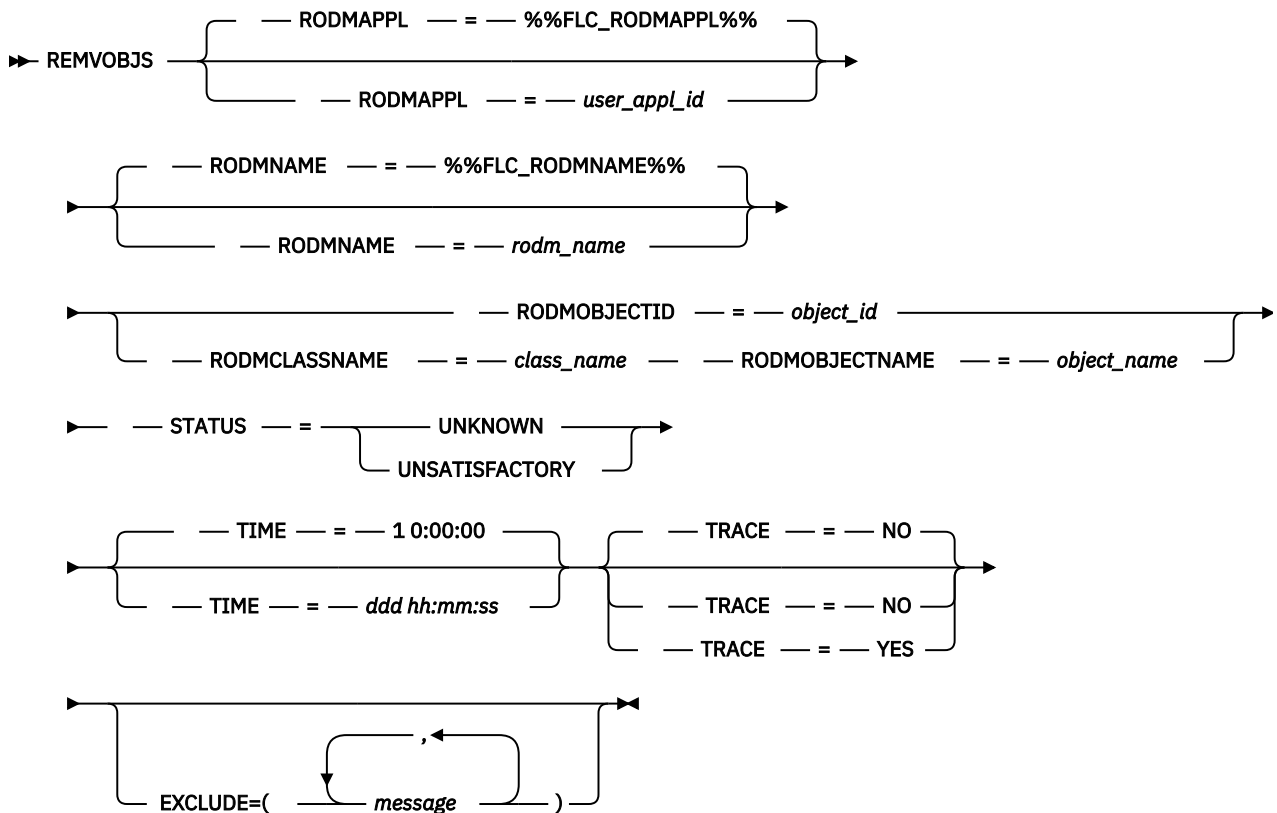
Meaning

- | | |
|-----------|-----------------------|
| 0 | Successful processing |
| 20 | Error in processing |

REMOBJS (MSM)

Syntax

REMOBJS



Purpose of Command

The **REMOBS** command removes objects from MultiSystem Manager views, provided they meet specific criteria. The criteria for removal is a combination of the **PURGE** attribute value and status of the object, and the length of time the object has maintained that status.

Operand Descriptions

RODMAPPL

The user application ID used to access the RODM that contains the topology and status data. The ID is case-sensitive and must be entered exactly as created with RACF.

The application ID for MultiSystem Manager is set during NetView initialization. Specify this keyword only if you are overriding the ID used by MultiSystem Manager, or if the **COMMON.FLC_RODMAPPL** statement in the **CNMSTYLE** member has not been coded.

RODMCLASSNAME

The RODM class name of the object that is the target of the command.

RODMNAME

The name of the RODM where topology and status information is stored. This is normally the RODM used by MultiSystem Manager. For MultiSystem Manager, this is the 1- to 8-character name that was specified on the **COMMON.FLC_RODMNAME** statement in the **CNMSTYLE** member. Specify this keyword only if you are overriding the ID used by MultiSystem Manager, or if the statement in the **CNMSTYLE** member has not been coded.

RODMOBJECTID

The RODM object ID (ObjectID) of the object. The object ID is 16 characters that represent an 8-byte hexadecimal field; for example: 00010027EC211161.

You can enter either the RODM object ID or the RODM class name and object name combination. MultiSystem Manager uses the object ID in the commands that it builds during command facility processing, because it can determine the ID based on the object you selected in the view. If you are entering this command from the command line or an automated procedure, use the RODM class name and object name combination.

RODMOBJECTNAME

The RODM object name of the object that is the target of the command.

STATUS

The status of the real object you are removing. You can enter UNKNOWN or UNSATISFACTORY.

An aggregate object's status is a reflection of the real objects that it contains, so if you issue this command on an aggregate object, this is the status of the real objects in the aggregate that you want removed.

TIME

The minimum time interval an object must have been in the specified status. You can specify *days* and *time*, or just *time*. If you want days without time, enter **0** for *time*. Specify the interval in the format: *ddd hh:mm:ss*, where:

ddd

0–365 days

hh

0–23 hours

mm

00–59 minutes

ss

00–59 seconds

Minimum

:00 (0 seconds)

Default

1 day (24 hours)

For example:

2 full days:

2 0

0 days, 5 minutes:

5

0 days, 5 minutes, 3 seconds:

0:05:03

2 days, 1 hour, 10 minutes, 30 seconds:

2 1:10:30

0 days, 30 seconds:

:30

TRACE

Determines whether a trace is generated.

YES

MultiSystem Manager generates messages FLC040I, FLC041I, and FLC042I, which contain a running account of the objects that were removed. If an aggregate object is selected for removal, the trace messages indicate the real objects in the aggregate that are removed.

NO

No account of the objects removed is generated. This is the default.

EXCLUDE

The EXCLUDE option can be specified only when TRACE=YES is specified. The EXCLUDE option indicates which TRACE messages (FLC040I, FLC041I, and FLC042I) must not be issued during removal processing. If you attempt to remove an aggregate object that has many objects beneath it, you might want to receive the FLC040I and FLC042I successful removal messages and suppress the FLC041I unsuccessful removal messages (for example, EXCLUDE=FLC041I). Otherwise, you might receive many unwanted FLC041I messages. One to three of these TRACEmessages can be specified. No other messages are permitted.

Usage Notes

- The MSM tower must be enabled in the CNMSTYLE member to successfully run this command.
- The display of special connectivity relationships, such as in an IBM Token-Ring, might be affected by the removal of objects in the view. If the workstation will not display the view after the REMVOBJS command runs, you must rebuild the view by issuing the appropriate GETTOPO command.
- You can issue the REMVOBJS command on an aggregate or a real object. When you issue this command on an aggregate object, the objects that make up the aggregate are also affected. Removal of the real objects that make up an aggregate depends on their PURGE attributes, statuses, and length of time at those statuses.
- When you issue this command on a real object, only the real object is affected. Removal of a real object depends on its PURGE attribute, status, and the length of time in that status.
 - If you are removing a real object:
 - Ensure that it has a status of *unknown* or *unsatisfactory*.
 - Determine the minimum time interval that will allow removal of the object.
 - If you are removing an aggregate object:
 - Determine whether you want to remove its composite real objects that have statuses of *unknown* or *unsatisfactory*.
 - Determine the minimum time interval that allows removal of the real objects.

Because an aggregate reflects the status of its composite real objects, the status and time do not apply directly to the aggregate, but to the real objects in it.

- Ensure that the object's PURGE attribute permits it to be removed. You can use the NetView RODMVIEW command to determine the current value of the PURGE attribute. You can use the SETREMV command to change the value. The PURGE attribute values are:

0

You can remove the object and all its links from the views and RODM if all other criteria are met.

This value is valid for real or aggregate objects. Real objects are removed if they have a status of *unsatisfactory* or *unknown* for the time specified in the REMVOBJS command. Aggregate objects are removed if all real objects in them have been removed.

All MultiSystem Manager objects have an initial PURGE attribute of 0.

1

You cannot remove the object from RODM. However, any of the links that connect the object to an object with a PURGE attribute of 0 can be removed. For example, if an aggregate object has a PURGE attribute of 0, the real object with a PURGE attribute of 1 can be unlinked and removed from the view.

This value is valid only for real objects. If you specify a value of 1 for an aggregate object, MultiSystem Manager treats it as if it has a value of 2.

2

You cannot remove the object or its links from the views or RODM.

This value is valid for real and aggregate objects. If a REMVOBJS command is issued on aggregate objects, this value shields the aggregate and the real objects in it from removal. If an aggregate

has a value of 2, the real objects in the aggregate are not removed, regardless of their PURGE attribute values.

If a PURGE attribute was never defined for an object, MultiSystem Manager treats the object as if it has a value of 2.

Command Responses

The REMVOBJS command returns one or more of the following messages when TRACE=YES. See the NetView online help for a complete description of MultiSystem Manager messages.

- FLC040I OBJECT *object_name* AND ALL OF ITS LINKS WERE REMOVED.
- FLC041I OBJECT *object_name* AND ITS LINKS WERE NOT REMOVED.
- FLC042I THE LINK BETWEEN *object_name1* FIELD *field_name1* AND *object_name2* FIELD *field_name2* WAS REMOVED.

Examples

This example removes the composite real objects of aggregate object number 00010027EC211161 that have been in unknown status for more than 29 hours. If all of the composite real objects are removed, the aggregate is removed. The name of the RODM that is being used is RODMNAME and the application name is MYAPPL.

```
REMOBJS RODMNAME=RODMNAME,RODMAPPL=MYAPPL,  
RODMOBJECTID=00010027EC211161,STATUS=UNKNOWN,  
TIME=1 5:00:00
```

REPEAT (NETLOG BROWSE)

Syntax

REPEAT

➤ REPEAT — — *findp* ➤

IBM-Defined Synonyms

Command or Operand	Synonym
REPEAT	R or RFIND

Purpose of Command

The REPEAT command reissues the last FIND command while you are browsing the network log or a member of a partitioned data set. Because the REPEAT command is sensitive to the current position of the cursor, it is normally entered using a PF key.

By repeatedly pressing the PF key set to REPEAT, you can find successive occurrences of a specified character string. When the first occurrence of a character string has been found, the REPEAT key will find the next occurrence. When the last occurrence of a character string has been found, the REPEAT key can be used to continue the search, wrapping around from the bottom line to the top line (or from the top line to the bottom line if the FIND command included the PREV parameter).

Operand Descriptions

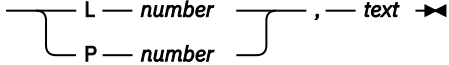
findp

Specifies the same set of parameters found on the FIND command. *findp* is only supported if the REPEAT command is being issued from a web browser.

REPLY (NCCF)

Syntax

REPLY

➤ REPLY —  , — *text* ➤

Purpose of Command

The REPLY command responds to outstanding VTAM requests.

Note: The REPLY command is actually a z/OS command, which is used by the z/OS Communications Server and other products or system components in specific ways. This documentation refers only to the NetView front-end command, which passes the REPLY command on to the VTAM program.

Operand Descriptions

Pnumber|*Lnumber*

Is the 2 or 4 digit reply number that is displayed in a message.

When a VTAM message is displayed with one of the following formats, the message requests a network operator reply:

```
Pnn messagenumber messagetext
```

```
Lnn messagenumber messagetext
```

Enter the NetView reply command using the reply ID (*Pnumber* or *Lnumber*) from the message.

text

The information you want to enter in the VTAM program.

Example: Replying to a message

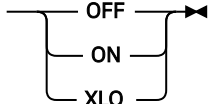
If you receive the message P45 IST272A NO INITIAL TEST FOR A08NV6 - REPLY 'U' TO BYPASS - OR CANCEL, you can respond using the P45 reply number. You enter:

```
REPLY P45,CANCEL
```

REPORTS (NPDA)

Syntax

REPORTS

➤ REPORTS — 

Purpose of Command

The REPORTS command controls the logging of report records by causing them to be written to a system management facilities (SMF) log. See the *IBM Z NetView Administration Reference* for the format of this record.

Operand Descriptions

OFF

Stops external logging.

ON

Starts external logging.

XLO

Indicates that if a record is external log only (XLO), based on BNJDSERV/XITCI return code or automation table setting, it is recorded on the external log.

Restrictions

Report records are not created in focal point domains for alert records that have been forwarded to the focal point from distributed host domains.

REQMS (NPDA)

Syntax

REQMS



Purpose of Command

The REQMS command requests Systems Network Architecture (SNA) summary error counts from a physical unit and stores the data on the hardware monitor database.

Operand Descriptions

pname

Specifies the physical unit name.

N

Specifies that you are to be notified only of a negative response to the request. N is the default.

Y

Specifies that you are to be notified of both positive and negative responses.

Restrictions

The following restrictions apply to the REQMS command:

- The REQMS command is not supported by locally attached 3174 controllers. You receive sense code X'080C' and the command is rejected.
- You can also use the REQMS command in a command list. However, do not use this command in a command list before you start the hardware monitor.

Example: Requesting error data and statistics from a specified PU

To request error data and statistics from PU04, enter:

```
REQMS PU04
```

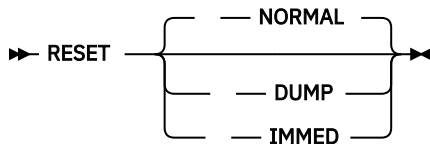
Example: Requesting error data from a specified PU

To request error data from PU99, enter:

RESET (NCCF)

Syntax

RESET



Purpose of Command

The RESET command ends the command or command procedure that is running.

Operand Descriptions

NORMAL

Causes the NetView program to stop the active command or command procedure at the next breakpoint even if work remains to be done. NORMAL is the default.

RESET NORMAL might not work for a command that has no breakpoint. You might have to use RESET IMMED instead.

DUMP

Causes the same processing as IMMED. Additionally, a memory dump is requested.

IMMED

Causes the system to end the command immediately, the NetView program issues user abend 257. All currently active command procedures, NetView components, cross-domain sessions, and TAF sessions are ended on the task being RESET. Spans are not affected. If the task has not exceeded the MAXABEND count (specified in the CNMSTYLE member), the task is reinstated and processing of queued and new commands continues. If the abend causes the reinstate count of the task to exceed MAXABEND, the task is ended. The MAXABEND count for a task will be reset to zero if the task has run for at least 1 hour since the last abend.

Usage Notes

The following usage notes apply to the RESET command:

- If you issue RESET NORMAL and no command or command procedure is active, no message is issued.
- If you send the RESET IMMED or RESET DUMP command from an NNT/OST cross-domain session to the other domain using the ROUTE command, the cross-domain session ends. This does not apply to RMTCMD cross-domain sessions.
- RESET IMMED closes any open files (for example, EXECIO files) that are associated with your operator station task (OST). If you are using a function that runs partially under your OST and partially under a data services task (DST), the associated DST (for example, the session monitor or hardware monitor) is not affected by the RESET command. Data returned from the DST is discarded or ignored because any correlation data kept under the OST has been lost. Because the OST part of the function has dropped correlation, it can send duplicate requests to the DST and cause problems in the DST part of the function. If an error occurs while the system is attempting to close any open files, messages indicating the error are sent to the system operator's console.
- You cannot retrieve RESET IMMED and RESET DUMP using the RETRIEVE command. You can retrieve RESET or RESET NORMAL.

The following usage notes are for high-level language (HLL):

- If a command procedure is cancelable, it behaves according to the rules specified above. A command procedure ends at its next breakpoint whenever RESET NORMAL is issued. A breakpoint occurs in an HLL command procedure whenever an HLL service routine is invoked.
- If the command procedure is not cancelable, it is reset only when RESET IMMED or RESET DUMP is issued. If you issue RESET NORMAL, the command procedure might or might not be reset. When an HLL command procedure is defined as not cancelable, the system programmer can determine what action is taken when RESET NORMAL is issued. Because the system programmer has control over not cancelable command procedures, the RESET command can appear to behave erratically to a NetView operator. The following results can occur when RESET NORMAL is issued:
 - The command procedure might or might not be reset.
 - A long delay can result before the command procedure is reset.
 - If the command procedure is reset, the operator gets a NetView cancelation message (CNM982E) for each command procedure that is canceled. This includes command procedures that were called from within the command procedure invoked by the operator.

For more detail on how HLL command procedures respond to the RESET command, see *IBM Z NetView Programming: PL/I and C*.

Example: Ending a currently running command or command procedure

To end the command or command procedure that is currently running, enter:

```
RESET
```

Response

The following message is displayed only if a command or command list is running; otherwise, there is no response.

```
DSI052I  command COMMAND SELF-TERMINATED BY OPERATOR REQUEST
```

Note: Message CNM982E is displayed for an HLL command procedure.

The normal responses to RESET IMMED or RESET DUMP are these messages:

```
DSI131I  COMMANDS ABENDED BY RESET COMMAND.  STATION HAS BEEN RESET.  
DSI172I  SUBTASK OPER1 ABENDED WITH CODE X'000101'
```

RESETDB (NCCF, NLDM, NPDA)

Syntax

RESETDB

➤ RESETDB — — *ddname* ➤

Purpose of Command

The RESETDB command clears the hardware monitor or session monitor VSAM database while the NetView program is active.

The RESETDB command clears the specified hardware monitor or session monitor VSAM database by opening the VSAM database with the RST option, then closing it. Before you issue the RESETDB command, define the VSAM database with the Access Method Services Define Cluster parameter REUSE and deactivate the database. You can do this while the NetView program is active, and while the hardware monitor or session monitor is active, as long as the intended VSAM database is inactive.

Operand Descriptions

ddname

Specifies the ddname of the VSAM database you want to clear. This field is required.

Restrictions



Attention: Unpredictable results can occur if you use RESETDB to reset NetView VSAM databases other than the hardware monitor or the session monitor databases.

Example: Clearing the session monitor primary VSAM database

To clear the session monitor primary VSAM database, enter:

```
RESETDB AAUVSPL
```

Response

```
DSI387I DATA BASE ALLOCATED TO 'AAUVSPL' HAS BEEN CLEARED
```

RESETSRV (EAS)

Syntax

EAS RESETSRV

➤ MODIFY — — *procname* — , — RESETSRV — — TASK — = — (— *taskid* —) — ➤

IBM-Defined Synonyms

Command or Operand	Synonym
MODIFY	F

Purpose of Command

The RESETSRV command resets the connection attempt to the beginning of the ServerLocation list for alert and message adapter services. This command has no effect if the last connection used was the first, or primary, server in the list.

The RESETSRV command does not have an immediate effect. It is used to control the next event sent. It does not disconnect a non-primary connection-oriented connection until the next event is processed by the adapter service.

Operand Descriptions

procname

Specifies the E/AS job name.

TASK=taskid

Specifies the E/AS service tasks to be reset. The *taskid* can have the following values:

ALERTA

The alert adapter service task

ALERTC

The confirmed alert adapter task

MESSAGEA

The message adapter service task

MESSAGEC

The confirmed message adapter task

ALL

All E/AS service tasks: ALERTA, ALERTC, MESSAGEA, and MESSAGEC.

Restrictions

You can specify only one TASK operand for each RESETSRV command. If you want to specify more than one service task, separate each *taskid* with a comma and enclose the *taskids* string within parentheses.

Example: Resetting the server connections for a service task

To reset the alert adapter service task server connections for the E/AS job named IHSAEVNT, enter the following command:

```
F IHSAEVNT,RESETSRV,TASK=ALERTA
```

Response

The following response is received:

```
IHS0121I Alert Adapter: Reset of backup server connection was
successful.
```

Example: Resetting the server connections for multiple service tasks

To recycle the alert adapter and message adapter service tasks for the E/AS job named IHSAEVNT, enter the following command:

```
F IHSAEVNT,RESETSRV,TASK=ALL
```

Response

The following response is received:

```
IHS0121I Alert Adapter: Reset of backup server connection
was successful.
IHS0121I Message Adapter: Reset of backup server connection
was successful.
IHS0121I Confirmed Message Adapter: Reset of backup server connection
was successful.
IHS0121I Confirmed Alert Adapter: Reset of backup server connection
was successful.
```

The IHS0121I message is issued regardless of whether a backup server is currently in use.

RESOLVE (NCCF)

Syntax

RESOLVE

►► RESOLVE — — *cmdverb* ◄◄

Purpose of Command

The RESOLVE command displays information about other commands.

Operand Descriptions

cmdverb

The *cmdverb* parameter is a token that might be a valid NetView command verb. RESOLVE determines its validity and the type of command verb.

If the *cmdverb* parameter is not a valid NetView command verb, RESOLVE produces the DSI002I message.

If *cmdverb* is prohibited by command authority in the current environment, RESOLVE produces the DSI213I message before continuing. Information about the command is displayed in the DWO081I message.

Usage Note:

Any text typed after the command verb is ignored.

RESOURCE (NCCF)

Syntax

RESOURCE

➡ RESOURCE ➡

IBM-Defined Synonyms

Command or Operand	Synonym
RESOURCE	RES

Purpose of Command

The RESOURCE command displays system resource use by the NetView program. The resources displayed are processor utilization, processor time used, and storage use.

The following information is displayed for the NetView address space:

Total processor Utilization

Total complex processor utilization based on a maximum of 100%. This utilization is calculated over the most recent 1-second interval.

NetView processor Utilization

NetView processor utilization based on a maximum of 100%. This utilization is calculated over the most recent 1-second interval.

NetView processor Time Used

The combination of task control block (TCB) and service request block (SRB) processor time used since the NetView program was started.

Real Storage Usage

The amount of storage (shown in kilobytes) that is currently in use in real storage frames.

This value includes real storage frames that are in use for both the storage that is allocated in the NetView address space and the data spaces that are owned by the NetView program.

Private Below

The amount of virtual storage allocated below the 16M line.

Private Above

The amount of virtual storage allocated above the 16M line.

Private Below Region

The total amount of virtual storage below the 16M line.

Private Above Region

The total amount of virtual storage above the 16M line.

The RESOURCE command is useful in determining how much resource the NetView program is using. By having your NetView automation table activate a command list when message DSI386I is generated in response to the RESOURCE command, you can compare current resource utilization to previously established thresholds.

Example: Displaying NetView resource use

To display NetView resource use, enter:

```
RESOURCE
```

Response

```
DSI386I NETVIEW RESOURCE UTILIZATION 09:41:36
TOTAL CPU %                =      3.28
E330EGNV CPU %             =      0.04
E330EGNV CPU TIME USED     =      6.43 SEC.
REAL STORAGE IN USE        =     7872K
PRIVATE ALLOCATED < 16M    =      876K
PRIVATE ALLOCATED > 16M    =    49064K
PRIVATE REGION < 16M      =     8160K
PRIVATE REGION > 16M      =    71680K
END OF DISPLAY
```

RESTOPO (MSM)

Syntax

RESTOPO

➤ RESTOPO ➤

Purpose of Command

The RESTOPO command resumes processing of MultiSystem Manager GETTOPO commands for all NetView operators and sets the status of the topology manager objects displayed in the Graphic Monitor-Details window to *AVAILABLE*.

The MultiSystem Manager program must be in *SUSPENDED* status to resume processing. After successful completion of the RESTOPO command, the status is set to *enabled*.

Usage Notes

The MSM tower must be enabled in the CNMSTYLE member to successfully run this command.

RESTORE (NCCF)

Syntax

RESTORE

➤ RESTORE — — TIMER — — DELETE ➤

Purpose of Command

The RESTORE command restores the timer command data that has been saved in the Save/Restore database.

Operand Descriptions

TIMER

Performs the requested operation on all TIMER type records in the Save/Restore database. These are events that have been saved by the SAVE parameter on CHRON, AT, EVERY, or AFTER commands.

DELETE

Erases all TIMER type records.

Restrictions

The following restrictions apply to the RESTORE command:

- RESTORE TIMER DELETE does not affect the internal active timer list. It causes all TIMER records to be deleted from the currently active Save/Restore database.

Note: The RESTORE command reads timer events from the Save/Restore database and activates them for processing. Invoking the command more than once between NetView outages activates duplicate timer events. Use the RESTORE command only when starting the NetView program.

- If the local system time is changed and the NetView program is recycled between the time an event is saved and the time when the RESTORE command is issued, the scheduled execution of the restored timer is adjusted according to the GMT or LOCAL option you specified at the time of the SAVE. For EVERY timers, GMT or LOCAL affects only the next processing time and not the continuing execution interval.
- Any AT or AFTER timer event whose execution time is prior to the restore time is not rescheduled or processed. The event is listed with message CNM465I and is deleted from the Save/Restore database. CNM465I tells you the command that was not processed and the information in this multiline message can be used to issue the command from the timer event.
- For timer events from EVERY commands, the next processing time is adjusted based on the GMT or LOCAL operand you specified at the time of the SAVE. Subsequent processing times are computed so that the events are scheduled to process on their original intervals. For example, at 12:00, a user requested that every hour a message be sent to an operator. If the NetView program is recycled at 13:30 after one processing of the command, the system clock is not reset, and a RESTORE occurs at 13:35, the message continues to be sent on the hour (at 14:00, 15:00, and so on).

Note: See the CHRON, AFTER, AT, and EVERY commands for examples of the effect of GMT and LOCAL on the scheduled execution timers.

- The CHRON command has an option that allows past due timers to be issued when timers are restored.
- The NetView program assigns each timer event an ID equal to SYSxxxxx, where xxxxx is a number in the range of 1–99999. Timer events that were originally scheduled without an ID are assigned a new ID, and their image in the Save/Restore database is updated. The new ID is a RESTORE ID (RSTxxxxx).
- Any timer event that is being restored and has a user-selected ID that is a duplicate of a currently active timer is not restored. The event is also listed with message CNM465I.

Example: Restoring all timer requests that were saved

To restore all timer requests that were saved, enter:

```
RESTORE TIMER
```

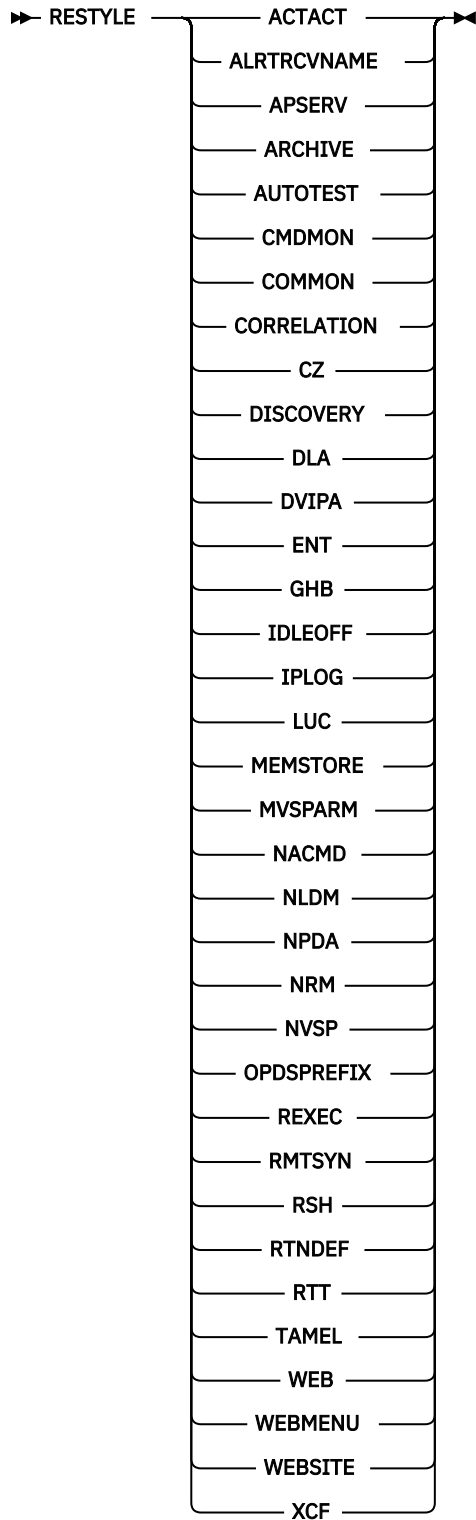
Example: Deleting all timer requests that were saved

To delete all timer requests that were saved, enter:

RESTYLE (NCCF)

Syntax

RESTYLE



Purpose of Command

The RESTYLE command reads the CNMSTYLE member and sets new values related to the keyword specified.

Operand Descriptions

ACTACT

Rereads all ACTACT statements.

If replication data collection is stopped for a subtower, the RESTYLE ACTACT command processes any socket path names and starts replication data collection for that subtower.

ALRTRCVNAME

Rereads the alert receiver name and recycles the CNMCALRT task.

APSERV

Rereads the APSERV definitions from CNMSTYLE and automatically enables the service accordingly. Note that APSERV.PREFIX statements are required for successful operation after RESTYLE. For further information about the APSERV.PREFIX Style statement, see *APSERV.PREFIX* in the *Administration Reference*.

ARCHIVE

Rereads all ARCHIVE statements.

AUTOTEST

Rereads all AUTOTEST statements.

CMDMON

Rereads all CMDMON CNMSTYLE statements and the Command Statistics Inclusion/Exclusion (CNMSCSIE) sample.

The command statistics function will be restarted, or not, based on the CMDMON.INIT.STATS CNMSTYLE statement.

COMMON

All common global variables defined with the COMMON keyword are redefined. No variable definitions are removed.

CORRELATION

Rereads all CORRELATION stem values and recycles the DSICORSV task.

CZ

Rereads all CZ statements from CNMSTYLE.

DISCOVERY

Rereads DISCOVERY variables. This includes the intervals for the COLLECTL command for the APPL, INTERFACES, and TELNET sub towers.

DLA

Rereads all DLA stem values.

DVIPA

Rereads all DVIPA statements.

ENT

Updates the global variables associated with Enterprise support. The ENT definitions in the CNMSTYLE member are reprocessed. New systems are discovered. Systems are dropped when CNMSTYLE is processed and systems have been removed from CNMSTYLE.

GHB

Rereads all GHB stem values and recycles DUIDGHB.

IDLEOFF

Reads or rereads all IDLEOFF stem values except for the autotask name, and resets any IDLEOFF settings. Starts IDLEOFF autotask if it is currently inactive.

IPLOG

Rereads all IPLOG stem values and recycles the DSIIPLOG task.

LUC

Rereads all LUC stem values and recycles the *domid* LUC task.

MEMSTORE

Refreshes the MEMSTORE definitions. Removes from storage all members controlled by the MEMSTORE component.

MVSPARM

Rereads the MVSPARM definitions from CNMSTYLE and recycles the CNMCSSIR task without loss of any automatable MVS messages.

NACMD

Rereads the parameters for the NACMD command. The NACMD command does not end and restart.

If the values for the NACMD.ROWS statement in the CNMSTYLE member are changed, the NetView program must be recycled before the changes take effect.

NLDM

Rereads all NLDM stem values and recycles the three NLDM tasks.

NPDA

Rereads all NPDA stem values and recycles the BNJDSERV task.

NRM

Rereads the NetView Resource Manager parameters, and shuts down NetView Resource Manager. NetView Resource Manager is restarted if INIT.NRM is Yes.

NVSP

Rereads the NetView Web Services server (NVSOA) stem values and recycles the AUTONVSP task.

OPDSPREFIX

Rereads the operator data set prefix. You must reissue OVERRIDE commands for data sets using the prefix (set in LOGPROF1).

REXEC

Rereads all REXEC stem values and recycles the DSIRXEXC task.

RMTSYN

Rereads all RMTSYN and RMTALIAS stem values and restarts the process of resolving any domain names in those definitions.

RSH

Rereads all RSH stem values and recycles the DSIRSH task.

RTNDEF

Rereads the RTNDEF statements.

RTT

Rereads all RTT stem values and recycles the DSIRTTR task.

TAMEL

Rereads all TAMEL stem values and recycles the CNMTAMEL task.

WEB

Rereads all WEB stem values and recycles the DSIWBTSK task.

WEBMENU

Reprocesses the definitions in the CNMSTWBM member.

WEBSITE

Rereads the definitions for the WEBSITES command.

XCF

Updates global variables pertaining to XCF services. The state field is updated for any variables that are exposed in the user state field. Many of the XCF variables allow customization of the actions that the NetView program takes when it assumes the role of master NetView program. Takeover-related changes take effect the next time the NetView program assumes the master role in the sysplex.

If the XCF.RANK statement in the CNMSTYLE member is changed such that a NetView program is to enter or leave a group, the START XCFGROUP or STOP XCFGROUP command is issued for that

DSIPLXnn group. A change in rank results in an internal PLEXCTL RANK=*new_rank* command being issued.

If the XCF.GROUPNUM statement in the CNMSTYLE member is changed, the NetView program might generate STOP XCFGROUP and START XCFGROUP commands.

Example: Reloading the NPDA specifications

You can change the characteristics of the hardware monitor by changing NPDA statements in the CNMSTUSR or CxxSTGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*. After you make your changes, enter the following command to set the new values and to restart the hardware monitor:

```
RESTYLE NPDA
```

RESUME (NCCF)

Syntax

RESUME

➤ RESUME — — *component* ➤

Purpose of Command

The RESUME command returns you to the component from which you issued a command that took you to another NetView component.

Operand Descriptions

component

Specifies the 1–8 character name of the component to which you want to return. This is the command verb that was used to start a rollable element.

If RESUME is used from a command list to redisplay a VIEW panel, you do not need to specify the *component*.

Usage Notes

To list the active components, enter the LIST ROLL command.

Example: Returning to the command facility from another component

To return to the command facility from another component, enter:

```
RESUME NCCF
```

RETRIEVE (NCCF)

Syntax

RETRIEVE

➤ RETRIEVE — { — EDIT — } — { — EXECUTE — } ➤

IBM-Defined Synonyms

Command or Operand	Synonym
RETRIEVE EXECUTE	=

Purpose of Command

The RETRIEVE command places the last command you issued in the command input area. This command gives you a convenient method to review, rerun, or edit and rerun commands you have recently entered.

Operand Descriptions

EDIT

Places the most recently processed command on the command line, ready to be reprocessed, or altered and processed. EDIT also advances a pointer, so repeated processing of RETRIEVE EDIT retrieves successively older commands. EDIT is the default.

EXECUTE

Processes, without redisplaying, the most recent command, unless RETRIEVE EDIT is the most recent. RETRIEVE EXECUTE processes the next command on the retrieve stack (the one to have been displayed by the RETRIEVE EDIT command).

Restrictions

The following restrictions apply to the RETRIEVE command:

- The NetView program maintains a stack of operator commands. Each command you enter is added to the top of the stack. However, suppressed commands and commands processed with the RETRIEVE command are not added to the stack. Only commands you enter are saved, including commands entered with a PF key. Commands issued through automation, or those that are changed or suppressed by DSIEX01 are not saved.
- If you enter a command other than the RETRIEVE command, the pointer is reset, so that the most recent command is invoked by RETRIEVE.
- You can define and use synonyms for RETRIEVE and its keywords. When the symbol = is used as the RETRIEVE synonym, the default is EXECUTE. With any other symbol, the default is EDIT.
- You cannot use the RETRIEVE command when accessing the NetView system through an MVS system console.
- The number of commands you can retrieve depends on the length of the commands you entered. Use the following as guidelines:
 - If you enter commands that are 240 characters long, you can retrieve approximately the last three commands that you entered.
 - If you enter commands that are 80 characters long, you can retrieve approximately the last nine commands that you entered.
 - If you enter commands that are 40 characters long, you can retrieve approximately the last 18 commands that you entered.
- You can use the RETRIEVE command from any NetView fullscreen application that uses DSIPSS TYPE=ASYPANEL (PANEL parmlist-2). This includes:
 - Log browse
 - Member browse
 - Command facility
 - Session monitor
 - Status monitor
 - Hardware monitor
 - VIEW command using the NOINPUT option

- VIEW command using the INPUT option and special variables.

For additional information about the VIEW command, see the *IBM Z NetView Customization Guide*.

- You cannot retrieve RESET IMMED and RESET DUMP using the RETRIEVE command. However, you can retrieve RESET, used without operands.

RETURN

Syntax

RETURN

➤ RETURN ➤

IBM-Defined Synonyms

Command or Operand	Synonym
RETURN	• RET (for BROWSE, HELP, HELPDESK, STATMON, and VIEW) • R (for NLDM and NPDA)

Note: The command facility has no synonym for the RETURN command.

Purpose of Command

The RETURN command returns you to the previous component or the last selection panel that you used.

Do not issue this command from a command list.

Restrictions

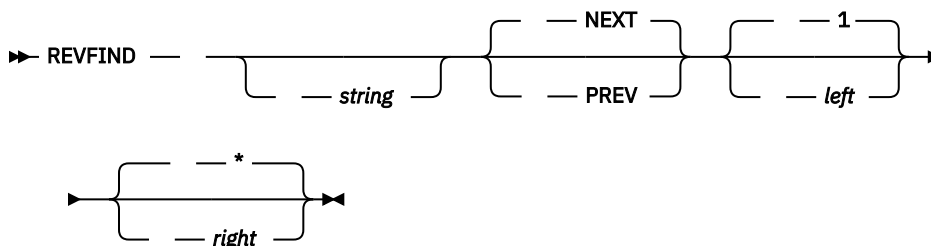
The following restrictions apply to the RETURN command:

- The hardware monitor maintains a table (called the hierarchy table) to track the sequence of panels you have viewed. When you issue an explicit hardware monitor command, this table is reset and the panel sequence is lost. Therefore, if you enter RETURN from a panel that is presented as the result of an explicit hardware monitor command, the NetView program takes you back to the hardware monitor main menu, not to the panel you were viewing before you issued the explicit command. If you issued the RETURN command from the help panel, the panel from which you requested help is displayed.

REVFIND (WINDOW)

Syntax

REVFIND



Purpose of Command

The REVFIND command reissues the last FIND command and searches for a previous occurrence of the specified character string. By repeatedly pressing the PF key set to REVFIND, you can continue finding previous occurrences of the specified character string.

Operand Descriptions

string

Specifies the information for which you are searching. If the information contains blanks or single quotation marks, enclose the information in single quotation marks. If the information contains single quotation marks, each must be entered as two single quotation marks. If the information contains neither blanks nor single quotation marks, single quotation marks are not necessary.

NEXT

Searches backward to find the previous entry. This is the default.

PREV

Searches forward to find the next entry.

1

Begins the search in column 1 of the left column. This is the default.

left

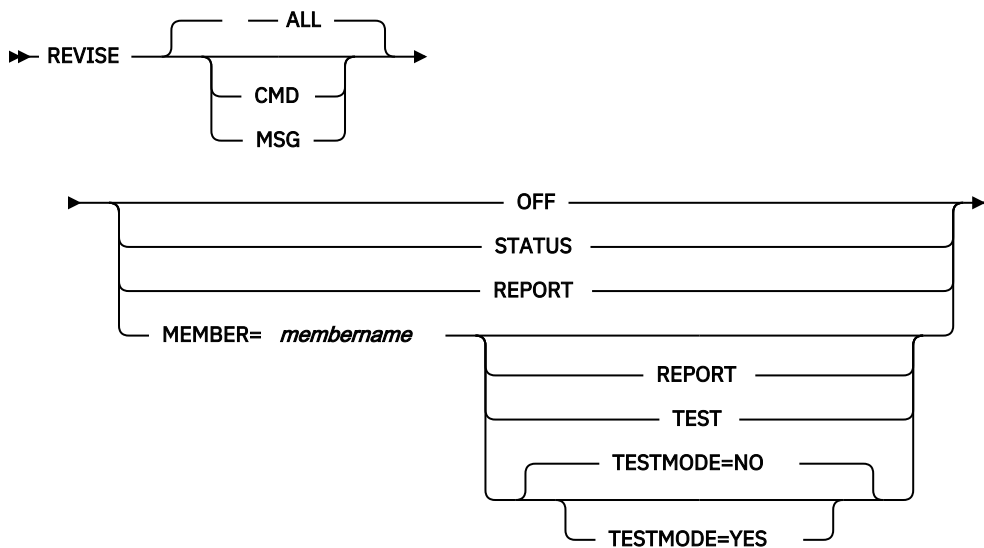
Specifies the left column where the search is to be started.

Searches all the data to the right of the left limit.

REVISE (NCCF)

Syntax

REVISE



Purpose of Command

The REVISE command is used to control or get information about a NetView command or message revision table. The revision table provides a set of conditions and actions that change MVS messages (all WTOs and WTORs) or commands before they are presented to the system log, console, or both.

Operand Descriptions

ALL

For stopping or querying revision table processing, sets the scope of the REVISE command to both command and message revision tables. For creating or updating revision tables, the result depends upon the statements found in the specified member:

- If the member contains only UPON statements related to message revision, only a message revision table is created and loaded.
- If the member contains only UPON statements related to command revision, only a command revision table is created and loaded.
- If the member contains both types of UPON statements, then both a message revision table and a command revision table are created and loaded.

This is the default unless the command definition for REVISMSG is referenced.

CMD

Limits the scope of the REVISE command to command revision statements or tables.

MSG

Limits the scope of the REVISE command to message revision statements or tables.

OFF

Requests to end the command or message revision process.

STATUS

Requests information about how and when the active revision table was loaded.

REPORT

Specifies that statistics and usage information about the active revision table is to be displayed. If successful, the CNM012I message is followed by several CNM015I multiline messages detailing the usage of each part of the active revision table.

When this keyword is specified with the MEMBER operand, the information displayed is about the table being replaced at the time it was replaced.

MEMBER

Specifies that a revision table specification is to be read. The *membername* value specifies a DSIPARM member in which a revision table specification can be found. Specification details for the member can be found in *IBM Z NetView Automation Guide*.

TEST

Specifies that the revision table member specified is to be tested for proper syntax and structure, but not loaded.

TESTMODE=NO|YES

Indicates whether to change the MVS command or list the changes in a WTO message (CNM018I) in syslog for evaluation. This keyword is ignored for message revisions.

NO

Indicates that changes to the MVS command are to be made. This is the default.

YES

Indicates that for a REVISE or NETVONLY action, changes to the MVS command are to be presented by a WTO message for evaluation only.

Usage Notes

- The REVISE MSG command functionally replaces the REVISMSG command, which is deprecated.
- The most recent message revision table and command revision table remain active even if the NetView program is restarted. If the SSI address space is restarted while the NetView program is running, any active revision tables and revision variables that were active at the time of the restart are reinstated.

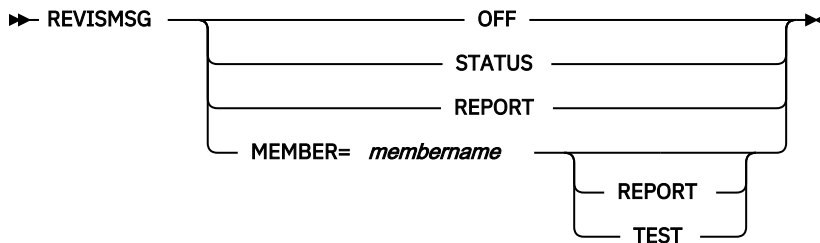
Restrictions

- Except for the TEST keyword, the subsystem routing task (CNMCSSIR) must be active and the associated subsystem interface address space must be active when you issue the REVISE command.
- If two or more NetView programs run in the same LPAR, only one can load a command revision table.

REVISMSG (NCCF)

Syntax

REVISMSG



Purpose of Command

The REVISMSG command is deprecated. The REVISE command replaces the REVISMSG command. Consider finding and replacing all instances of the REVISMSG command and replacing each occurrence with the REVISE MSG syntax. For more information, see the help for the REVISE command.

The REVISMSG command is used to control or get information about the NetView message revision table. This table is a set of conditions and actions that change MVS messages (all WTOs and WTORs) before they are presented to the system log, console, or both.

Operand Descriptions

OFF

Requests to end message revision process.

STATUS

Requests information about how and when the active revision table was loaded.

REPORT

Specifies that statistics and usage information about the active revision table is to be displayed. If successful, the CNM012I message is followed by several CNM015I multiline messages detailing the usage of each part of the active revision table.

When this keyword is specified with the MEMBER operand, the information displayed is about the table being replaced at the time it was replaced.

MEMBER

Specifies that a revision table specification is to be read. The *membername* value specifies a DSIPARM member in which a revision table specification can be found.

TEST

Specifies that the revision table member specified is to be tested for proper syntax and structure, but not loaded.

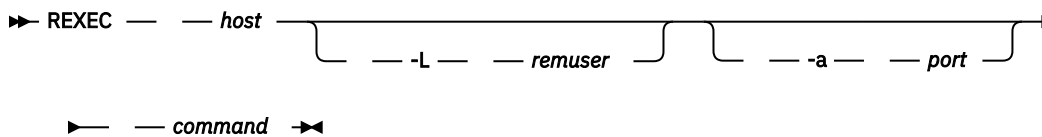
Restrictions

Except for the TEST keyword, the subsystem routing task (CNMCSSIR) must be active and the associated subsystem interface address space must be active when you issue the REVISMSG command.

REXEC (NCCF)

Syntax

REXEC



Purpose of Command

The REXEC command sends a command to remote host over IP for execution and displays the resulting output. Standard UNIX RSH protocol is used. The remote host must have an REXEC server listening at the specified (or defaulted) port for the command to work.

Operand Descriptions

host

Specifies the name of the remote host. It can be specified as a host name, or in the dotted IP address format.

-L *remuser*

Specifies a username on the remote system. The command runs on the remote system under this user. It can be a value of 1–16 characters. When REXEC is entered from the terminal, a screen is presented prompting the operator for the password or password phrase of the remuser on the remote system. If REXEC is driven from an environment in which interactive panels might cause problems, such as a CLIST or an automated environment, you must specify *remuser* as *remuser/password* or password phrase. If specified from the operator's terminal, *remuser* can be omitted, in which case it defaults to the operator ID of the operator issuing the command.

-a *port*

Specifies a port on the remote server. The default port is 512.

command

Specifies the name of the command to be sent to the remote host for processing.

Usage Notes

The following restrictions apply to the REXEC command:

- The command sent to the remote host does not require interactive input (such as prompting the operator for information) and it produces only line-mode output.
- When sending a command to a system that supports mixed-case commands, such as a UNIX system, prefix REXEC with NETVASIS. This respects the case of the username and command.
- The command might hang because TCP/IP is not responding or there is a problem on the remote host. If that happens, RESET IMMED can be used to cancel the command. In some cases, the operator task might need to be recycled.

Sending an LS command to a UNIX system:

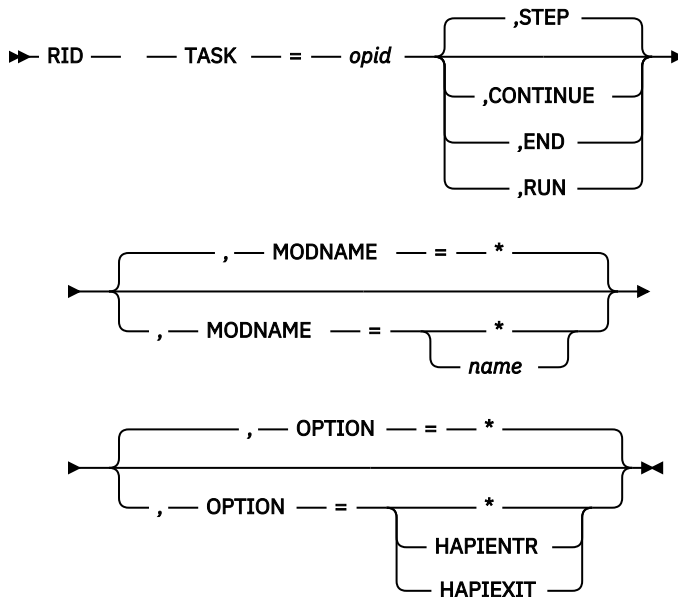
To list the contents of the home directory on user Testuser on the UNIX host HOST1, enter:

```
NETVASIS REXEC HOST1 -l Testuser ls
```


RID (NCCF)

Syntax

RID



Purpose of Command

The RID command controls the debugging of high-level language (HLL) programs running under a NetView subtask.

Operand Descriptions

TASK=*opid*

Specifies that the target task is to be debugged.

STEP

Specifies that the target task is to stop whenever control is given to a debug point that matches the criteria specified by the MODNAME or OPTIONS keyword (see the following for details about these keywords). Messages providing data captured at the debug point are displayed at the operator station that invoked RID to monitor the target task. STEP is the default.

CONTINUE

Resumes the processing of a task that was stopped by the STEP option of RID. You can specify new debug point match criteria in conjunction with the CONTINUE option. The CONTINUE keyword is provided for readability only. To resume processing, reissue the RID command with its original operands.

END

Causes debugging of a task to cease and allows other operators to invoke RID for the target task. If the target task is stopped when the END option is invoked, the high-level language (HLL) command processor or installation exit running under the target task resumes.

RUN

Specifies that the target task is to continue to process after issuing the messages at the debug points. The RUN option resumes processing a task stopped in STEP mode.

MODNAME

Specifies the name of the module being monitored:

Specifies that the RID command is to monitor all HLL command processors and installation exits running under the target task. The asterisk (*) is the default.

name

The name of the module being monitored by the RID command.

OPTION

Specifies the type of debug points to display and trace:

All debug points are displayed. The asterisk (*) is the default.

HAPIENTR

Entry to HLL API service routines.

HAPIEXIT

Exit from HLL API service routines.

Restrictions

The following restrictions apply to the RID command:

- The RID command is useful in debugging HLL programs running under a NetView subtask. To use this facility, you must have a NetView or MVS operator console to display the RID output. You can run the target program under any subtask.
- For more information about the RID command, see *IBM Z NetView Programming: PL/I and C*.

Example: Debugging HLL programs from an operator station task

To debug HLL programs running under OPER1 from an operator station task (OST) other than OPER1, enter:

```
RID TASK=OPER1
```

Response

If OPER1 is logged on, the following message is displayed:

```
CNM986I RID FUNCTION 'STEP' COMPLETED FOR TASK OPER1
```

If OPER1 is not logged on, the following message is displayed:

```
DSI031I SPECIFIED NAME 'OPER1' INVALID
```

RIGHT

Syntax

RIGHT

►► RIGHT — amount ►►

IBM-Defined Synonyms

Command or Operand	Synonym
RIGHT	R

Purpose of Command

The RIGHT command displays session configuration data in the secondary direction from the Session Configuration Data panel.

The RIGHT command enables you to view columns of data that are not currently visible on the Log-Browse screen. The data moves to the left a specified number of positions to display information to the right of the last column.

Operand Descriptions

amount

For session monitor, this operand is not allowed. For Log-Browse, this operand specifies the amount to scroll to the right. The possible values are:

PAGE or P

Scroll right one screen

HALF or H

Scroll right half a screen

CSR or C

Scroll right making the column indicated by the cursor the left column

MAX or M

Scroll to the rightmost column of the data

number

Scroll right a specific number of columns

The default is CSR if the cursor is located in the data display area; otherwise, the default is PAGE.

Usage Notes

Consider the following when using the RIGHT command:

- When you have issued the OVERRIDE command with the SCROLL keyword specifying a value other than OFF, the LOG-BROWSE panel displays a scroll amount in the upper right area of the panel.
- When you issue the RIGHT command, the number of columns scrolled is determined in the following order:
 1. The explicit scroll amount specified on either the RIGHT command or on the command line when the RIGHT PF key is pressed.
 2. The scroll amount displayed in the message area at the bottom of the LOG-BROWSE screen as message BNH183I indicating the last scroll amount.
 3. The implicit scroll amount specified in the scroll amount area in the upper right area of the panel.
 4. The cursor position when the scroll amount area indicates CSR.
 5. The cursor position when there is no scroll field or BNH183I message displayed.
- You can change the scroll amount in the scroll amount area by entering any portion of CSR, HALF, OFF, PAGE, or a numeric scroll amount. Typing over the remaining contents of the field is not necessary unless you are changing a numeric value to another numeric value.

Example: Moving in the secondary direction from the configuration data panel

To move in the secondary direction from the Session Configuration Data panel, enter either command:

```
RIGHT
RI
```

Example: Issuing several RIGHT commands from the Log-Browse screen

RIGHT commands are cumulative. For example, entering the commands:

```
RI 10  
RI 10
```

Is equivalent to entering:

```
RI 20
```

Example: Displaying the rightmost margin of the Log-Browse screen

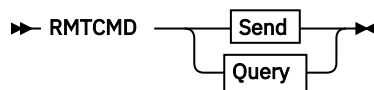
If you have issued several RIGHT or LEFT commands and want to view the last column, enter:

```
RI MAX
```

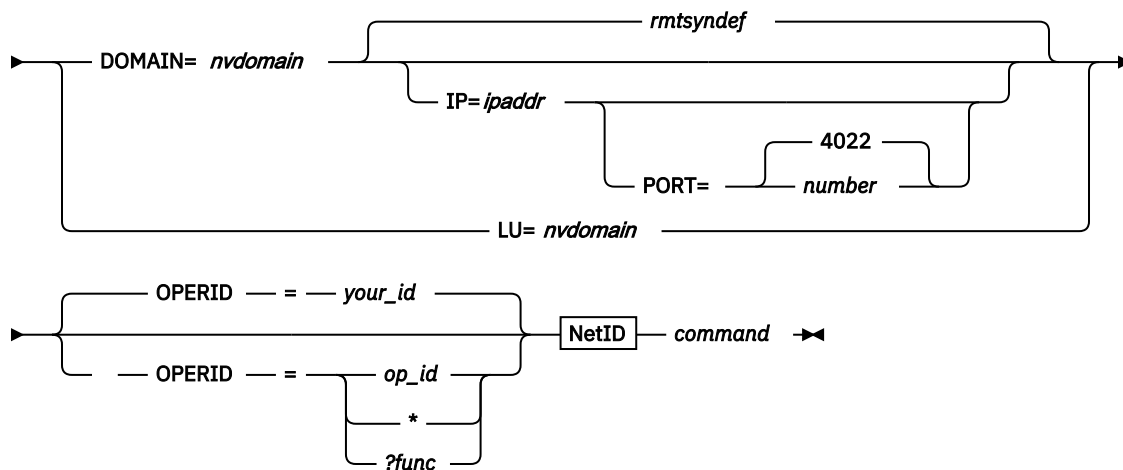
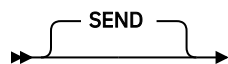
RMTCMD (NCCF)

Syntax

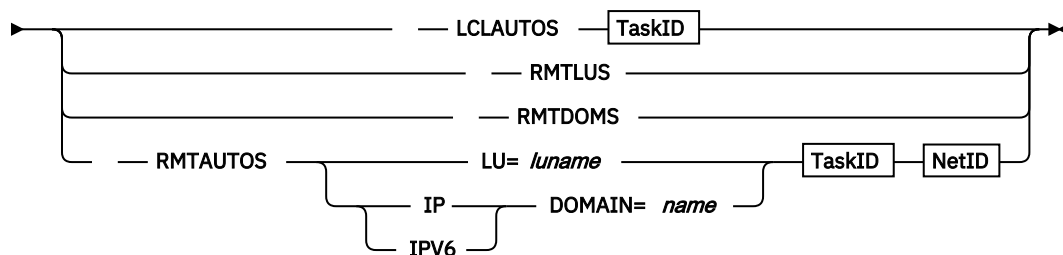
RMTCMD



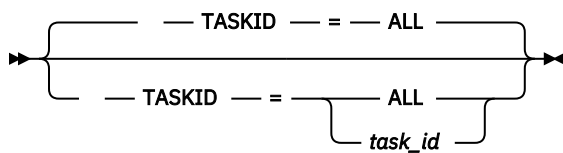
Send



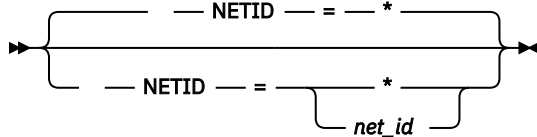
Query



TaskID



NetId



Purpose of Command

The RMTCMD command processor sends system, subsystem, and network commands to a remote NetView system for processing. The responses to these commands are returned to the RMTCMD issuer. To do this, RMTCMD can use an existing task, if already active. If the task specified by the OPERID value is not active, RMTCMD processing automatically starts the task. The command specified with RMTCMD is processed by the task. Any responses from the command are returned to the RMTCMD issuer across an LU 6.2 session or over TCP/IP, whichever is used to send the request.

To use RMTCMD over an LU 6.2 session, tasks DSIHPDST and DSIUDST must be active on both NetView systems. To use RMTCMD over TCP/IP, only the DSIUDST task must be active.

When you send a command to a task that is disconnected or was inactive prior to the command, RMTCMD establishes an ownership relationship between you and that task. Because of this relationship, the task is then known as a *distributed autotask*. A distributed autotask forwards all unsolicited messages it receives to the owner. These include messages sent to the task by ASSIGN routing, message automation routing, the MSG command, the MSGROUTE command, and all other messages that can arrive at an operator task, other than as a direct, correlated response to a command.

To end the relationship with a distributed autotask, issue a DISC command at the autotask. You can also end the task using the LOGOFF command or the ENDTASK command. When an owner logs off, all owned distributed autotasks will also end. For more information about these commands, see the NetView online help.

When a message is automated on a distributed autotask and then routed to the owner, the message is not considered to have been automated on the domain that receives the message if the receiving domain is not the same as the sending domain. This prevents double automation of messages in a NetView system.

The command text specified in the RMTCMD command processor runs in a remote NetView program with the same command type as it has in the driving RMTCMD. When RMTCMD is driven as terminal input, the command in the remote NetView program runs as terminal input and echoes and logs in the remote NetView program. When RMTCMD is driven as an internal function request (such as through an automation table or the PIPE command), the command on the remote NetView program runs as an internal function request and does not echo or log.

When RMTCMD is driven with a command that has message buffers associated with it (such as a command driven from the automation table), the message buffers are sent to the remote NetView program along with the command text specified on the RMTCMD command processor. When that command runs on the remote NetView program, you can access the message buffers by issuing PIPE SAFE * and by using the GETMSIZE or GETMLINE facility.

When RMTCMD is driven by a PIPE NETVIEW stage command, the input message buffer to the NETVIEW RMTCMD stage command is sent to the remote NetView program along with the command text specified on the RMTCMD command processor. Use the COLLECT stage command before the NETVIEW RMTCMD stage to combine multiple messages of an input stream into one message. The collected messages are sent to the remote NetView program along with the command text. For more information, see PIPE CORRCMD in *IBM Z NetView Programming: Pipes*.

Operand Descriptions

SEND

Enables you to send a command to a remote NetView system and receive the responses. SEND is the default.

QUERY

Enables you to query RMTCMD details.

Note: Use the RMTSESS command for an easier method to determine all the distributed autotasks you have activated. The RMTSESS command uses this QUERY function to determine your distributed autotask information.

LU=nvdomain

Specifies the NetView domain ID where the command is to be sent. Use LU, instead of DOMAIN, when you want to require that the command be sent over LU 6.2. This overrides any RMTALIAS and RMTSYN definitions set in the CNMSTYLE member.

DOMAIN=nvdomain

Specifies the NetView domain ID where the command is to be sent. When DOMAIN is specified, but IP is not, the command is sent by either TCP/IP or LU 6.2, as specified with the RMTALIAS and RMTSYN definitions in the CNMSTYLE member.

rmtsyndef

If the IP address and port number are omitted, the values default to the values specified with the RMTALIAS and RMTSYN definitions in the CNMSTYLE member.

IP=ipaddr

Specifies the host name or IP address for remote NetView operations over TCP/IP.

If you specify a host name, ensure that the DUIDGHB task is started. When you specify a host name, the IP address (IPv4 or IPv6) is determined based on the setting of the IPv6Env statement in the CNMSTYLE member:

- NONE - IPv4 address is used, if available
- ONLY - IPv6 address is used, if available
- MIXED - first address returned is used (either IPv4 or IPv6)

If you specify an IPv4 address, you can use standard dotted decimal notation or the IPv4-mapped IPv6 address form. If you specify an IPv6 address, use the colon-hexadecimal format.

The default value is the host name or IP address specified with the RMTALIAS and RMTSYN definitions in the CNMSTYLE member.

PORT=number

Specifies the port number of a RMTCMD server for remote NetView operations over TCP/IP. The default is either 4022 or the value provided with applicable RMTALIAS and RMTSYN definitions, if any, in the CNMSTYLE member. PORT is used only when IP is specified.

NETID

Specifies the network ID.

net_id

Specifies the remote network identifier for the NetView system on which you want to run the specified command.

Specifies that the network identifier is to be determined dynamically. This is the default value.

If IP is specified, the local network ID is used for communication to the remote host.

If DOMAIN is specified but IP is not specified, RMTCMD processing tries to find matching RMTALIAS and RMTSYN statements in the CNMSTYLE member by using the following steps, in order:

1. If the specified domain is an alias (meaning there is a RMTALIAS/RMTSYN statement pair mapping the alias to a real domain ID), then the network ID used is provided by the ON *netid* segment of the RMTSYN value.
2. If all RMTSYN statements applicable to a particular domain (including those used to define aliases) see the same network ID, then that network ID is used for communication with that remote domain.
3. If a RMTSYN statement containing the specified NetView domain name and the local network ID is found, then the local network ID and the value specified on the matching RMTSYN statement are used for communication to the remote host.
4. Otherwise, SNA is used as the transport.

If LU is specified, or else if DOMAIN is specified and matching RMTALIAS and RMTSYN statements are not found, VTAM determines the network ID value to use. If two NetView systems in different networks have the same domain name, the one that VTAM finds can vary depending on the configuration of nodes that are active at any given time.

OPERID

Specifies the autotask for processing the commands. This is an optional operand. If you do not specify this operand, your operator ID is used as a default at the remote NetView system.

Command authorization checking is performed on the OPERID keyword except when:

- OPERID is not specified and one of the following conditions is true:
 - The RMTCMD command was not forwarded from another operator (using EXCMD or another RMTCMD)
 - The setting of AUTHCHK is TARGETID
- OPERID=* is specified

Command authorization checking is bypassed when OPERID=* is specified to allow applications to use OPERID=* as a default value, even when command authorization checking for OPERID is in effect.

Command authorization checking is performed for the OPERID keyword and a value of your operator ID when all of the following conditions are true:

- The OPERID keyword is not specified
- The RMTCMD request was forwarded from another operator
- The setting of AUTHCHK is SOURCEID

This is done to protect the operator being used for RMTCMD from other operators trying to gain access to the same operator ID on the remote host.

your_id

Specifies your operator ID to be started as the autotask on the remote NetView system for processing commands. This is the default.

op_id

Specifies the autotask on the remote NetView system for processing the commands.

Specifies that the *op_id* defaults to an autotask that exists for the requesting operator on the remote NetView system. If no autotask exists for the requesting operator, your operator ID is used as the default. If more than one autotask exists for the requesting operator, the first active autotask found processes the command.

?func

Specifies the autotask defined in the CNMSTYLE member by the global variable function.autotask.*func*. The autotask that is retrieved from the global variable is treated as a keyword when the command is checked for authorized use.

command

Is the command and parameters to be sent to the remote NetView system for processing. If the command parameters include a date or time specification, the format must match the format required by the domain and task where the command is to run.

The maximum length of a command string that can be transported to a NetView domain by RMTCMD is 15200 bytes.

LCLAUTOS

Queries distributed autotasks active on your NetView system. The output includes the distributed autotask origin information.

TASKID

An optional parameter specifying the distributed autotask on the NetView system you are querying. ALL is the default.

ALL

Specifies that all distributed autotasks on this NetView system are to be queried. ALL is the default.

task_id

Specifies the distributed autotask on the NetView system which you are querying.

RMTLUS

Lists the remote NetView systems on which you have started distributed autotasks. Use the RMTCMD QUERY RMTAUTOS command against entries in this list to determine the distributed autotasks you have active on them. A RMTLUS remote NetView program list is kept for each operator. An entry is added to your list when you successfully start a distributed autotask on a remote NetView program. This entry remains in the list until the end of your session, even if the distributed autotasks you started on a remote NetView are stopped.

The returned list includes data for remote NetView systems contacted by way of LU6.2 only.

When you log off, your remote NetView program list is removed and all your distributed autotasks are ended.

RMTDOMS

Lists the remote NetView systems on which you have started distributed autotasks. Use the RMTCMD QUERY RMTAUTOS command against entries in this list to determine the distributed autotasks you have active on them. A RMTDOMS remote NetView program list is kept for each operator. An entry is added to your list when you successfully start a distributed autotask on a remote NetView program. This entry remains in the list until the end of your session, even if the distributed autotasks you started on a remote NetView are ended.

The returned list includes data for all remote NetView systems.

When you log off, your remote NetView program list is removed and all your distributed autotasks are ended.

RMTAUTOS

Determines distributed autotasks you have started on a remote NetView program. The request is sent to the remote NetView program only if you have already started a distributed autotask on it. If you issue the RMTCMD QUERY RMTDOMS command, you can see the list of remote NetView programs to which you can send the RMTCMD QUERY RMTAUTOS request.

Note: If the request is sent to the remote NetView program, the response to this query is asynchronous. When using the PIPE command to trap the output of this command, use the CORRWAIT stage to trap the asynchronous responses.

LU=luname

Specifies a remote NetView domain name (VTAM application name). The connection between the current and target NetView domains is SNA LU6.2.

IP

Specifies the query is for remote operators owned by the current operator. The connection between the current and target NetView domains is TCP/IP (IPv4).

The IPV6 and IP keywords are mutually exclusive.

IPV6

Specifies the query is for remote operators owned by the current operator. The connection between the current and target NetView domains is TCP6 (TCP over IPv6).

The IPV6 and IP keywords are mutually exclusive.

DOMAIN=*name*

Specifies the 1- to 5-character target NetView domain identifier. DOMAIN is required when IP or IPV6 is specified.

TASKID

The distributed autotask on the remote NetView program you are querying to determine if it has started.

ALL

Specifies that all distributed autotasks you have started in the remote NetView program are queried. The first operator ID in the message is the operator which is used when the RMTCMD command is issued to that NetView system with the OPERID=* specification. ALL is the default.

task_id

Specifies the distributed autotask on the remote NetView program you are querying.

Specifies that the distributed autotask used with the RMTCMD OPERID=* for sending commands is to be queried.

Note: If you have started more than one distributed autotask on the remote NetView program, the first operator in the list is used if you specify OPERID=* for a RMTCMD send command.

Usage Notes

The following considerations apply to the RMTCMD command:

- If the task name you specified is already logged on or actively associated with another operator, your command is invoked and correlated responses are returned to you. In this case, you do not become the owner of the task unless the task was previously in a disconnected state.
- The remote NetView program can control which operators at which remote NetView nodes can initiate or share a distributed autotask. Security filters can be set for one or more of the following:
 - Remote *net_id*
 - Remote *luname*
 - Remote *op_id*

These values reflect either the sender of the RMTCMD or the originator of the request (such as if the RMTCMD was forwarded using EXCMD or another RMTCMD), depending on a setting in the receiving NetView program.

This control is available by using an SAF product such as RACF or a table in DSIPARM. See the *IBM Z NetView Security Reference* for more information about RMTCMD security.

- If you send commands to a V2R3 domain by the RMTCMD command in a NetView PIPE stage, use the regular RMTCMD command format and construct an appropriate PIPE command to serialize the output of the command.
- When you are using RMTCMD in a NetView PIPE stage such as CORR CMD, consider using the label syntax to route the command rather than RMTCMD. The label syntax provides automatic serialization at the target NetView program. See the *IBM Z NetView User's Guide: NetView* for more information.
- For more information about protecting your NetView application from receiving RMTCMD requests from other domains, see the *IBM Z NetView Security Reference*.

Restrictions

The following restrictions apply to the RMTCMD command:

- When specified, the SEND or QUERY operand must directly follow the RMTCMD command. SEND is the default.
- You can send commands that produce single or multiline messages, or commands that do not produce output. RMTCMD does not support commands that produce full-screen output. However, the BROWSE command uses the RMTCMD command to provide a full-screen cross-domain member and netlog browse facility. See the BROWSE command for details.
- You can specify the operands in any order with the following exceptions:
 - The SEND and QUERY operands must be specified first.
 - The text of the command to be sent to the remote NetView system must be specified last.
- Do not send commands to optional tasks unless the documentation for the task provides information about how to do so.

Return Codes

Return Code	Meaning
-------------	---------

0	Successful processing. The request was sent to the target system.
8	Task not found.
12	Error in processing.

Note: Although a return code of zero indicates that the RMTCMD command ran successfully, it does not guarantee that the remote NetView program processed the request correctly. You can trap the output of the RMTCMD request to determine if the remote NetView program successfully processed the request.

Example: Sending a message to an operator

To send a message to an operator with an ID of OP01 in a NetView system, with a resource name of NETV11, enter:

```
RMTCMD SEND LU=NETV11,MSG OP01 HELLO
```

Response

The following message is displayed:

```
DSI001I MESSAGE SENT TO OP01
```

You now have an association with an active task on NETV11 with the same operator ID as yours. You can use this task to issue additional commands routed with RMTCMD commands. To end the task, issue the ENDTASK command, log off, or issue an RMTCMD command containing a LOGOFF.

Example: Sending a DIS APPLS command to be started

To send a DIS APPLS command to be started on task OPER2 in NetView domain CNM02, by way of OPER4, in domain CNM03, enter:

```
RMTCMD SEND LU=CNM03,OPERID=OPER4,RMTCMD  
SEND LU=CNM02,OPERID=OPER2,DIS APPLS
```

Response

You now have an active association with task OPER4 in domain CNM03, and OPER4 has an active association with OPER2 in CNM02. The response to the DIS APPLS command is returned to the task where you issued the RMTCMD command. The messages are processed by OPER4 in CNM03 and can be automated there. These messages are expedited back to your task and can be automated in your domain, CNM02, or CNM03.

Example: Collecting the lines of the DSIUINIT member and sending the resulting message to a remote NetView program

To collect the lines of the DSIUINIT member and send the resulting multiline message to a remote NetView program named CNM01, enter:

```
PIPE < DSIUINIT
| COLLECT
| NETVIEW RMTCMD SEND LU=CNM01, cmd
```

Where:

cmd

Specifies the command and parameters to be sent to the remote NetView system.

Response

When *cmd* runs on the remote NetView program (CNM01), it has access to the multiline message containing the lines of the DSIUINIT member. For information about data buffers, see PIPE CORR CMD in *IBM Z NetView Programming: Pipes*.

Example: Determining whether a task is an RMTCMD autotask

To determine if OPER6 is a distributed autotask, enter:

```
RMTCMD QUERY LCLAUTOS TASKID=OPER6
```

Response

The following multiline response shows that OPER6 is a distributed autotask and the origin operator is OPER6 on NETB.CNM02. The origin NetView program level is V5R2:

```
BNH060I RMTCMD QUERY INFORMATION
BNH061I -----
BNH064I DISTRIBUTED ORIGIN      ORIGIN  ORIGIN
BNH065I AUTOTASK   NETVIEW      OPERATOR VERSION  TRANSPORT
BNH061I -----
BNH067I OPER6      NETB.CNM02    OPER6   V5R2     SNA
```

The following response implies that OPER6 is a valid operator and is active; however, it is not a distributed autotask:

```
BNH062I OPER6 ON NETA.CNM01 IS NOT A DISTRIBUTED AUTOTASK
```

See also REXX functions DISTAUTO(), DOMAIN('R'), and OPID('R').

Example: Listing all active distributed autotasks for a NetView system

To list all active distributed autotasks on this NetView system, enter:

```
RMTCMD QUERY LCLAUTOS
```

By not specifying the TASKID parameter on the query, the default is that ALL distributed autotasks are queried.

Response

The following response lists all active distributed autotasks on this NetView system and the details of each autotask:

```
BNH060I  RMTCMD QUERY INFORMATION
BNH061I  -----
BNH064I  DISTRIBUTED  ORIGIN      ORIGIN  ORIGIN
BNH065I  AUTOTASK     NETVIEW      OPERATOR VERSION  TRANSPORT
BNH061I  -----
BNH067I  OPER2        NETZ.DOM42    PAUL     V5R2     SNA
BNH066I  OPER3        NETK.LUONE    MONITOR  V5R2     SNA
BNH067I  OPER6        NETB.CNM02    OPER5    V5R2     SNA
BNH066I  NETOP1       NETB.CNM55    OPER55   V3R1     SNA
BNH066I  NETOP2       NETB.CNM55    OPER55   V3R1     SNA
```

Example: Sending a command to a remote NetView system in an IPv6 network

For the following example, assume that the NetView program is enabled for IPv6 networking. To use TCP6 transport service over IPv6 to send the GO command to the NTVXX remote NetView domain, enter:

```
RMTCMD SEND,IP=2002:92A:111:501:10:10:163:7,DOMAIN=NTVXX,PORT=4022,
OPERID=OPER1,GO
```

By using an IPv6 target host address, the TCP6 transport service is used. The OPER1 operator ID in the NTVXX domain runs the GO command. This command also indicates that the DSIUDST task in the NTVXX domain is listening for connection requests on port 4022.

You can also send a command using the TCP6 transport service to a remote NetView system if the value of the IP keyword is a host name that resolves to an IPv6 target address as in the following example:

```
RMTCMD SEND,IP=host6.yourcompany.com,DOMAIN=NTVXX,PORT=4022,OPERID=OPER1,GO
```

Example: Using the RMTSYN statement to enable sending commands to remote NetView systems in an IPv6 network

You can specify the transport to use when you send a command to a remote NetView domain by using a RMTSYN statement in the CNMSTYLE member.

Assume that this RMTSYN statement defines the local NetView domain:

```
RMTSYN.USIBMNT.NTVXX = 2002:92A:111:501:10:10:163:7/4022
```

The following example sends the GO command to remote domain NTVXX over an IPv6 network:

```
RMTCMD SEND,DOMAIN=NTVXX,OPERID=OPER1,GO
```

Example: Querying remote LUs

To list the remote NetView programs to which you have started one or more distributed autotasks, enter:

```
RMTCMD QUERY RMTLUS
```

Response

In the following response, the message indicates that OPER1 has no remote NetView programs in its remote NetView program list, hence the operator currently has no active distributed autotasks.

```
BNH063I NO RMTCMD QUERY INFORMATION EXISTS FOR REQUEST
```

This problem can occur if OPER1 has not issued an RMTCMD, or has logged off then logged on again.

For the following response, assume that the operator has issued RMTCMDs successfully to NETB.CNM02 and NETC.CNM03.

```
BNH060I  RMTCMD QUERY INFORMATION
BNH061I  -----
BNH068I  REMOTE NETVIEW      VERSION  TRANSPORT
```

```
BNH061I -----
BNH069I NETB.CNM02      V2R4      SNA
BNH069I NETC.CNM03      V2R4      SNA
```

This response shows that the remote domains on which OPER1 has started distributed autotasks are NETB.CNM02 and NETC.CNM03. The versions of the remote NetView programs are included.

Example: Querying remote autotasks

To query remote autotasks which you have started on NETA.CNM02, enter:

```
RMTCMD QUERY RMTAUTOS,LU=CNM02,TASKID=ALL
```

Response

For the following response, assume OPER1 has started OPER1 and OPER2 on CNM02.

```
BNH072I RMTCMD QUERY INFORMATION FROM NETA.CNM02
BNH061I -----
BNH070I OPER1
BNH070I OPER2
```

The response shows that OPER1 on NETA.CNM01 has active RMTCMD associations with OPER1 and OPER2 on NETB.CNM02. This response is asynchronous because it was processed by the remote NetView program; however, the response can be correlated and trapped by using PIPE with RMTCMD and CORRWAIT. In addition, if you issue an RMTCMD SEND specifying OPERID=* on the QUERY, remote operator OPER1 is used.

To query remote autotasks that you started on X.N90 via TCP/IP, enter:

```
RMTCMD QUERY IP NETID=X DOMAIN=N90 RMTAUTOS
```

Response

```
BNH072I RMTCMD QUERY INFORMATION FROM X.N90
BNH690I IP ADDRESS: 69.97.5.90 PORT: 4022
BNH061I -----
BNH070I NETOP2
```

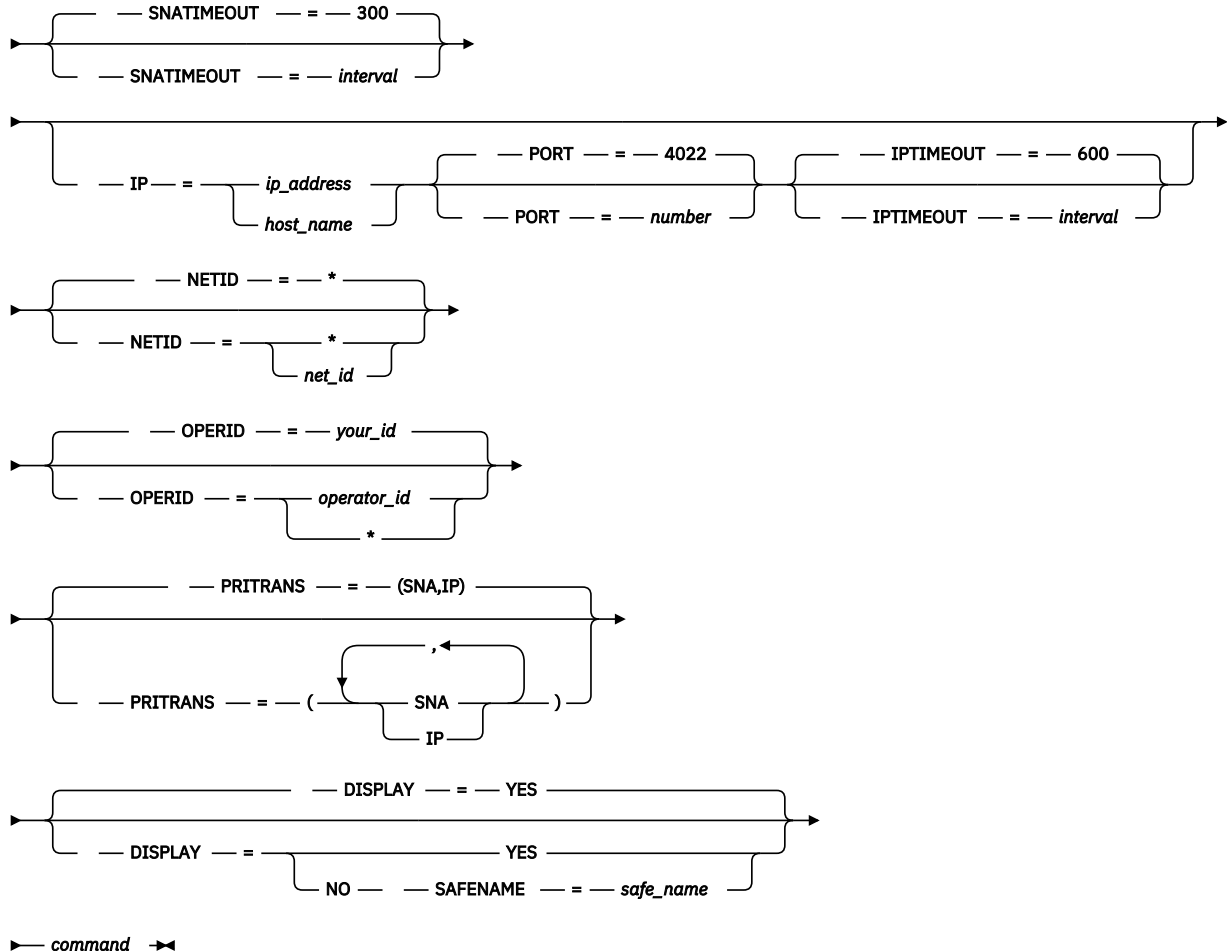
This response has an additional line to show the IP address and port for the target NetView system.

RMTSEND (NCCF)

Syntax

RMTSEND

➤ RMTSEND — — DOMAIN — = — *domain_name* ➤



Purpose of Command

The RMTSEND command enables operators and command procedures to issue one command to send RMTCMDs using SNA or TCP/IP. All RMTCMD SEND restrictions and notes apply to this command.

Operand Descriptions

DOMAIN=*domain_name*

Specifies a remote NetView domain name (VTAM application name) or the name of the NetView domain to be found at the specified IP address and port. If a question mark (?) is found in the domain name, a selection panel is displayed from which you can select a remote target.

For information about how the order is determined for the transport method, see [“Usage Notes” on page 174](#).

SNATIMEOUT=*interval*

Specifies the timeout value to be used for SNA RMTCMD requests. If responses are not received in this period of time, the command ends. The default is 300 seconds.

IP=*ip_address*

Specifies the host name or IP address for remote NetView operations over TCP/IP. When specifying a host name, ensure that the DUIDGHB task is started. If you specify an IPv4 address, use standard dotted decimal notation or the IPv4-mapped IPv6 address form. If you specify an IPv6 address, use the colon-hexadecimal format. If a question mark (?) is found in the IP address, a selection panel is displayed from which you can select a remote target.

PORT=*number*

For remote NetView operations over TCP/IP, specifies the port number to use for a RMTCMD server. PORT is only valid when IP is specified. The default is 4022.

IPTIMEOUT=*interval*

Specifies the timeout value to be used for IP RMTCMD requests. If responses are not received in this period of time, the command ends. The default is 600 seconds.

NETID

Specifies the network ID.

net_id

Specifies the remote network identifier for the NetView system on which you want to run the specified command.

Specifies that the network identifier is the one determined by VTAM based on the LU name of the remote node. This is the default.

Note: If two NetView systems in different networks have the same domain name, the one that VTAM finds can vary depending on the configuration of nodes that are active at any given time.

OPERID

Specifies the autotask for processing the commands. This is an optional operand. If you do not specify this operand, your operator ID is used as a default at the remote NetView system.

Command authorization checking is performed on the OPERID keyword except when:

- OPERID is not specified and one of the following conditions is true:
 - The RMTCMD command was not forwarded from another operator (using EXCMD or another RMTCMD)
 - The setting of AUTHCHK is TARGETID
- OPERID=* is specified

Command authorization checking is bypassed when OPERID=* is specified to allow applications to use OPERID=* as a default value, even when command authorization checking for OPERID is in effect.

Command authorization checking is performed for the OPERID keyword and a value of your operator ID when all of the following conditions are true:

- The OPERID keyword is not specified
- The RMTCMD request was forwarded from another operator
- The setting of AUTHCHK is SOURCEID

This is done to protect the operator being used for RMTCMD from other operators trying to gain access to the same operator ID on the remote host.

your_id

Specifies your operator ID to be started as the autotask on the remote NetView system for processing commands. This is the default.

op_id

Specifies the autotask on the remote NetView system for processing the commands.

Specifies that the *op_id* defaults to an autotask that exists for the requesting operator on the remote NetView system. If no autotask exists for the requesting operator, your operator ID is used

as the default. If more than one autotask exists for the requesting operator, the first active autotask found processes the command.

PRITRANS=(*transport_method*)

Specifies which transport method to use. If both SNA and IP are specified, the second method is attempted *only* if there is a failure using the first method.

For information about how the order is determined for the transport method, see [“Usage Notes” on page 174](#).

DISPLAY

Specifies whether the results of the command are displayed.

YES

Display the results of the command for the user

NO

Return the results of the command in a PIPE safe.

SAFENAME

Specifies the name of the safe in which the command results are returned.

command

Specifies the command and parameters to be sent to the remote NetView system for processing. If the command parameters include a date or time specification, the format must match the format required by the domain and task where the command is to run.

Usage Notes

The following considerations apply to the RMTSEND command:

- The transport method for the specified domain is determined as follows:
 1. If a RMTSYN statement in the CNMSTYLE member is found for the specified domain, the specification in that statement is attempted. If that fails, the IP and PRITRANS values specified on the RMTSEND command are attempted.
 2. If a RMTSYN statement does not exist for the specified domain and a SNA connection is available, the SNA connection is attempted. If that fails, the IP and PRITRANS values specified on the RMTSEND command are attempted.
 3. When IP and PRITRANS values are used:
 - If you specify both IP and SNA for PRITRANS and also provide target information for both IP and SNA, the RMTSEND command issues the RMTCMD using the second transport method only if the first RMTCMD fails.
 - If you specify PRITRANS=(SNA , IP) and only provide SNA target information, the RMTCMD over SNA is attempted.
 - If you specify PRITRANS=(IP , SNA) and only provide SNA target information, message DSI651I is displayed, indicating that the IP keyword is missing.
 - If you specify PRITRANS=(IP) and only provide SNA target information, message DSI651I is displayed, indicating that the IP keyword is missing.
- Use caution when specifying the command target. If you are using the IP address, it is possible for the NETID and DOMAIN of the local system to be identical to the NETID and DOMAIN of a remote system. If this occurs and you specify both IP and SNA for PRITRANS, the SNA request is sent to the local VTAM and the IP request is sent to the remote system.
- Validity checks are not made for the values of DOMAIN, IP, PORT, NETID, and OPERID.
- Validity or syntax checks are not made for the specified *command*.
- If the task name you specified is already logged on or actively associated with another operator, your command is invoked and correlated responses are returned to you. In this case, you do not become the owner of the task unless the task was previously in a disconnected state.

- The remote NetView program can control which operators at which remote NetView nodes can initiate or share a distributed autotask. Security filters can be set for one or more of the following:
 - Remote *net_id*
 - Remote *domain_name*
 - Remote *operator_id*

These values reflect either the sender of the RMTCMD or the originator of the request (such as if the RMTCMD was forwarded using EXCMD or another RMTCMD), depending on a setting in the receiving NetView program.

This control is available by using an SAF product such as RACF or a table in DSIPARM. See the *IBM Z NetView Security Reference* for more information about RMTCMD security.

- For more information about protecting your NetView application from receiving RMTSEND requests from other domains, see the *IBM Z NetView Security Reference*.
- The output received from the RMTCMD is not reformatted by the RMTSEND command.

The following restrictions apply to the RMTSEND command:

- You can send commands that produce single or multiline messages, or commands that do not produce output. RMTSEND does not support commands that produce full-screen output.
- You can specify the keyword operands in any order. The text of the command to be sent to the remote NetView must be specified last.
- Do not send commands to the following tasks:
 - DSIACBMT
 - DSIDCBMT
 - DSIHLLMT
 - DSILOGMT
 - DSISTMMT
 - DSITIMMT
 - DSIWTOMT

Return Codes

Return Code Meaning

0	Successful completion
4	Required parameter missing
8	Incorrect keyword specified
12	Incorrect value specified
16	Parameter conflict
20	RMTCMD command failed

Note: Although a return code of zero indicates that the RMTSEND command ran successfully, it does not guarantee that the remote NetView program processed the request correctly. You can trap the output of the RMTSEND request to determine if the remote NetView program successfully processed the request.

Example: Issuing a command over two possible paths

The following example sends a message to an operator with an ID of OP01 in a remote NetView system. The remote NetView system can be reached using an LU name of NETV1, or by using a TCP/IP host name of NETVIEW1. The preferred method for sending the command is over TCP/IP. To send the message, enter:

```
RMTSEND DOMAIN=NETV1,IP=NETVIEW1,PRITRANS=IP,MSG OP01 HELLO
```

RMTSESS (NCCF; CNME1092)

Syntax

RMTSESS

➤ RMTSESS ➤

Purpose of Command

The RMTSESS command list displays the distributed autotasks an operator has activated.

Note: When a remote NetView program is slow in responding to an RMTSESS request, RMTSESS waits up to 30 seconds for a reply. You can enter the GO command to end the wait. RMTSESS ends and only the RMTSESS information gathered up to this point is displayed.

Restrictions

The following restrictions apply to the RMTSESS command:

- Remote NetView V2R2 or V2R3 systems cannot be queried and the distributed autotask query information is listed as *UNKNOWN*. If a NetView V2R2 or V2R3 system is listed as *UNKNOWN*, there was at one time active RMTCMD associations, but there is no way of determining the specific distributed autotask associations that might currently exist.
- The RMTSESS command will display only the first level of detail for nested RMTCMD associations. A nested RMTCMD association is when you issue a RMTCMD that issues another RMTCMD, giving two distributed autotask associations.
- The RMTSESS command issues the RMTCMD QUERY RMTLUS and RMTCMD QUERY RMTAUTOS commands to gather the autotask association details. The operator must be authorized to issue these commands for the RMTSESS command to gather the details. The RMTSESS command will issue message BNH285I when a problem is encountered issuing the RMTCMD command.

Example: Listing RMTCMD associations

To list the RMTCMD associations that you started, enter:

```
RMTSESS
```

Response

```
C CNM99
BNH060I RMTCMD QUERY INFORMATION
BNH061I -----
BNH083I REMOTE          RMTCMD      REMOTE
BNH084I NETVIEW        AUTOTASK    VERSION
BNH061I -----
BNH085I NETA.CNM01      OPER1       V2R4
BNH085I NETA.CNM01      OPER5       V2R4
BNH085I NETB.CNM02      *UNKNOWN*   V2R3
```

This example shows that the operator has two active distributed autotasks on NETA.CNM01 (OPER1 and OPER5). In addition, the operator started an RMTCMD autotask on NETB.CNM02 but the specific autotask

details cannot be queried because that NetView program is version V2R3. The autotask query function is available on NetView V2R4 and later releases.

RODM (RODM; CNME1098)

Syntax

RODM

➤ RODM — — *command* ➤

Purpose of Command

The RODM command list processes NetView RODM commands. You can also process RODM commands using the MVS MODIFY command.

command

Specifies the RODM command to process:

- CHKPT
- LOGF
- LOGP
- LOGQ
- LOGS
- LOGT
- RELOAD
- START
- STATAPI
- STATCELL
- TERM

Some of these commands support additional parameters that can be appended to the end of the command. Additional parameters must be separated by commas or spaces.

Restrictions

The following restriction applies to the RODM command:

- The global variables EKGHPRC and EKGHNAM, defined in the CNMSTYLE member, must be set to your RODM procedure name and your RODM nickname respectively. If you do not use a RODM nickname, do not use the EKGHNAM global variable.

Example: Warm-starting RODM

To warm-start RODM, enter:

```
RODM START,TYPE=WARM
```

See the RODM START command for additional information.

Example: Writing API statistics to the RODM log

To write API statistics to the RODM log and clear the statistics enter:

```
RODM STATAPI,CLEAR
```

See the STATAPI command for additional information.

RODMVIEW (RODM)

Syntax

RODMVIEW

➤ RODMVIEW ➤

Purpose of Command

The RODMVIEW command starts a full-screen application, which provides a series of menus, and displays as a front-end to the RODMVIEW command processors. The RODMVIEW panels enable you to display, create, update, and delete classes, objects, fields, and relationships in RODM.

Usage Notes

For additional information about the classes, objects, attributes, and relationships, see the *IBM Z NetView Resource Object Data Manager and GMFHS Programmer's Guide*. For additional usage information, see the *IBM Z NetView User's Guide: NetView*.

Restrictions

This command can only be run under the control of an OST.

ROLL (NCCF)

Syntax

ROLL

➤ ROLL ➤

Purpose of Command

The ROLL command returns to a previous component and the last panel that you used in that component.

The system remembers the sequence in which you go from one component to another. When you use the ROLL command, the system moves the name of your current component to the beginning of the sequence of components, and starts the component at the end of the sequence, displaying the panel that was displayed when you left that component.

Example: Re-entering the Help facility

If you previously entered the session monitor, followed by the help component, followed by the hardware monitor, the sequence remembered is:

```
NLDM - HELP - NPDA
```

To return to the place within the help facility from where you exited, enter:

```
ROLL
```

Response

The last panel you had accessed within the help facility is displayed.

ROUTE (NCCF)

Syntax

ROUTE

➤ ROUTE — — *domainid* — , — *command* ➤

IBM-Defined Synonyms

Command or Operand	Synonym
ROUTE	RT

Purpose of Command

The ROUTE command sends NetView and VTAM commands to other domains.

You can use the ROUTE command to send logon information to other domains after these domains have started.

Messages associated with the command being sent are returned to the sending terminal.

For information about authority checking of this command and the effect of SOURCEID and TARGETID, see the *IBM Z NetView Security Reference*.

Operand Descriptions

domainid

The name of the NetView domain where the command is sent

command

The command that is sent

Restrictions

The following restrictions apply to the ROUTE command:

- Do not use the ROUTE command to route commands that produce full-screen output such as HELP or HELPDESK.
- Some multiline messages might become single-line messages when transmitted across a NetView-to-NetView session.
- If you use the suppression character defined by your system programmer as a prefix with the ROUTE command, neither the ROUTE command nor the command that is sent is echoed to the screen or logged.

Example: Displaying the status of a resource

If you want to display the status of a resource named LU2024A1 in domain D2, enter:

```
ROUTE D2,DISPLAY NET,ID=LU2024A1
```

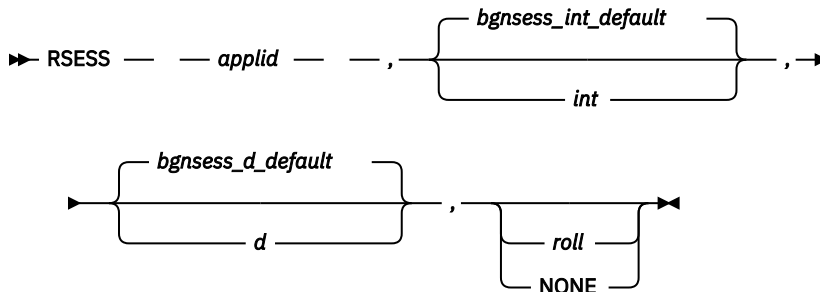
Example: Displaying the status of a resource

To display the status of LU2A in DOM2, enter:

```
ROUTE DOM2,DISPLAY NET,ID=LU2A
```

Syntax

RSESS



Purpose of Command

The RSESS command list returns to a previously disconnected full-screen terminal access facility (TAF) session.

This command list generates a BGNSSESS command.

Operand Descriptions

applid

Specifies the logical unit name of the subsystem to which you want to return. This name must match the *luname* specified on the APPLID parameter in a previous BGNSSESS command.

bgnsess_int_default

If you do not specify whether most messages interrupt your session (with the *int* operand), the value specified on the previous BGNSSESS command for this session is used.

int

Specifies whether most messages interrupt your session. This operand can be either Y (yes) or N (no). If omitted, *int* defaults to the selection made in the previous BGNSSESS command for this session. If no selection was made in the previous BGNSSESS command, the default is N.

bgnsess_d_default

If you do not specify a Disconnect key for this full-screen session (with the *d* operand), the value specified on the previous BGNSSESS command for this session is used.

d

Specifies the Disconnect key for this full-screen session. Valid values for *d* are:

CLR

Specifies that the Clear key disconnects the full-screen session.

PAkey

Specifies the PA key that disconnects the full-screen session. Valid values for *key* are 1–3.

PFkey

Specifies the PF key that disconnects the full-screen session. Valid values for *key* are 1–24.

If omitted, *d* defaults to the selection made in the previous BGNSSESS command for this session. If no selection was made in the previous BGNSSESS command, PA1 is the default.

roll

Specifies the Roll key for this session. Valid values for *roll* are:

PAkey

Specifies the PA key that disconnects the full-screen session. Valid values for *key* are 1–3.

PFkey

Specifies the PF key that disconnects the full-screen session. Valid values for *key* are 1–24.

If you are resuming a session and you omit roll, the Roll Intercept key remains unchanged. If you use NONE to begin or resume a session, issue a BGNSSESS command to resume that session after disconnecting.

Restrictions

If you omit a positional operand, indicate its absence with a comma. You do not need to specify trailing positional commas.

Example: Returning to a previously disconnected full-screen TSO1 session

To return to a previously disconnected full-screen TSO1 session using PF12 as the Disconnect key and PF6 as the Roll key, enter:

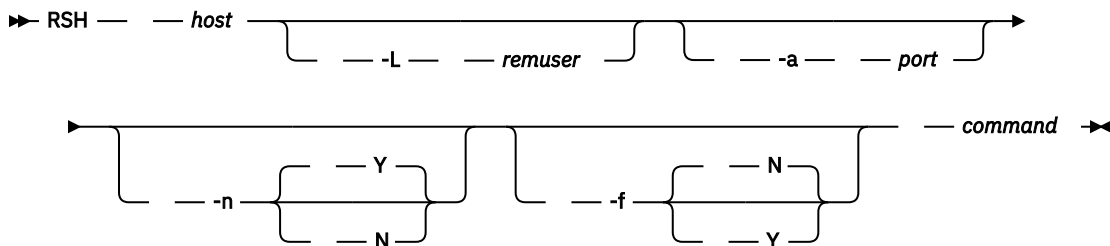
```
RSESS TSO1,N,PF12,PF6
```

Because N is specified, you are not interrupted with messages. The new Disconnect key is PF12, and the new Roll key is PF6.

RSH (NCCF)

Syntax

RSH



Purpose of Command

The RSH command sends a command to a remote host over IP for execution. The output can be displayed as line-mode output or in a panel. The remote host must have an RSH server listening at the specified (or defaulted) port for the command to work, and the NetView/NetView operator ID combination must be authorized at the remote host. If the remote host supports it, additional commands can be issued from the panel where the output is displayed. The panel is placed on the NetView roll stack.

Operand Descriptions

host

Specifies the remote host to run the command. It can be specified as a host name or in the dotted IP address format.

-L *remuser*

Specifies the name of the user on the remote host that runs the command. It can be a value of 1-16 characters. It defaults to the operator ID of the operator issuing the command.

-a *port*

Specifies a port on the remote server. This defaults to port 514.

-n

Specifies whether the remote host is a NetView host. This option can be specified as Y (NetView host) or N (not a NetView host). The Y option is the default value.

-f

Specifies that a full-screen panel is used to display the output. This option can be specified as Y or N. The N option is the default value. The Y option is ignored if specified in an environment that does not support full-screen processing.

command

The name of the command sent to the remote host.

Usage Notes

The following restrictions apply to the RSH command:

- The command sent to the remote host does not require interactive input (such as prompting the operator for information) and it only produces line-mode output.
- When sending a command to a system that supports mixed-case commands, such as a UNIX system, prefix RSH with NETVASIS. This respects the case of the username and command.
- Security is the responsibility of the remote host. The command assumes processing such as /etc/hosts.equiv and .rhosts occurs on the remote host. In addition to the username on the remote host, a username from the NetView system must be supplied. The remote host then grants or denies access based on the combination of requesting host and requesting username. The NetView program uses the OPID of the operator issuing the RSH command as the requesting username.
- The command might hang because of TCP/IP not responding or a problem on the remote host. If that happens, RESET IMMED can be used to cancel the command. In some cases, it might be necessary to recycle the operator task.
- The -f option used with the RSH command is for interactive commands and is not used on commands scheduled to run by way of the TIMER command unless it runs on an operator's task with an operator present.

Example: Sending an LS command to a UNIX system

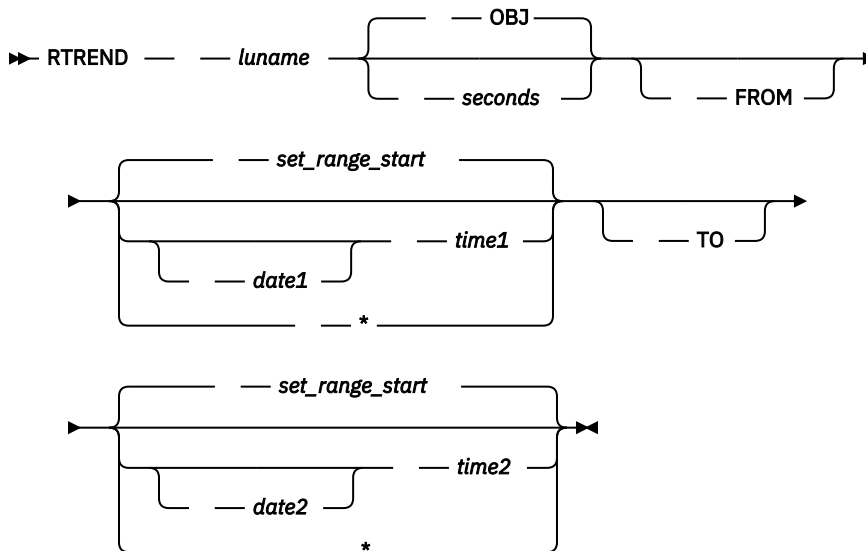
To list the contents of the home directory on user Testuser on the UNIX host HOST1, enter:

```
NETVASIS RSH HOST1 -L Testuser ls
```


RTREND (NLDM)

Syntax

RTREND



Purpose of Command

The RTREND command displays the response-time trend over a specified range of time for a terminal LU connected to a cluster controller that supports the response time monitor (RTM) feature.

This command displays the Response Time Trend panel.

Operand Descriptions

luname

Specifies the name of the terminal for which response time is measured.

OBJ

Specifies the response time objective set for this LU. OBJ is the default.

seconds

Specifies the maximum response time, in seconds, to be used for the graph. The format is ssss.s. The maximum value allowed is 1800 seconds.

FROM

Identifies the operands that follow as the starting date and time. This operand is optional.

set_range_start

If you do not specify a starting date and time, the starting time set by the most recent SET RANGE command is used.

date1

Specifies the starting date of the time range. The format of *date1* is controlled by the setting of the date operands of the DEFAULTS and OVERRIDE commands. If you specify a starting time but omit the date, the current date is used.

time1

Specifies the starting time of the time range. The format of *time1* is controlled by the setting of the time operands of the DEFAULTS and OVERRIDE commands. If you do not specify a starting time, the default is the starting time set by the most recent SET RANGE command. If no SET RANGE command has been issued, the default for *time1* is two hours earlier than the time specified on the ending time (*time2*).

Specifies to use the default start and end times and not the start and end times specified by the SET RANGE command. A single asterisk (*) in place of both *date1* and *time1* specifies a starting time of two hours before the ending time. A single * in place of both *date2* and *time2* specifies an ending time of the current date and time.

TO

Identifies the operands that follow as the ending date and time. This operand is optional.

set_range_end

If you do not specify an ending date and time, the ending time set by the most recent SET RANGE command is used.

date2

Specifies the ending date of the time range. The format of *date2* is controlled by the setting of the date operands of the DEFAULTS and OVERRIDE commands. If you specify an ending time but omit the date, the current date is used.

time2

Specifies the ending time of the time range. The format of *time2* is controlled by the setting of the time operands of the DEFAULTS and OVERRIDE commands. If you do not specify an ending time, the default is the ending time set by the most recent SET RANGE command. If no SET RANGE command has been issued, the default for *time2* is the current time. If you issue a command that has an end time past the time the last command was issued, the latest time is used.

Restrictions

If the LU is in another domain, first use the SDOMAIN command to specify the domain that contains the LU.

Examples

The examples shown assume that the following conditions are true:

- You are collecting data every 30 minutes on the quarter hour (07:45, 08:15, 08:45, and so on) and a user with terminal lname LU3270A logged on at 08:43 (March 14, 2011).
- The user is still logged on and the time is now 14:17.
- The user's session was mapped into a performance class with counter boundaries of .5, 2, 5, and 10 seconds, and an objective that 80% of the transactions takes less than 5 seconds.
- The format of dates and times specified use the default setting for date and time formats on the DEFAULTS and OVERRIDE commands.

Example: Displaying response time trends

To display the response time trend from 09:00 today to the current date and time (14:17), enter either command:

```
RTREND LU3270A FROM 9:00 TO *
RTREND LU3270A 9:00 *
```

Response

A panel of two pages (because there are more bars than can be shown on one page) with percentages of transactions under 5 seconds from 09:00 to 14:17 is displayed.

Example: Displaying response time trends meeting a specified performance objective

To display the response time trend for transactions that met the performance objective of 80% (less than 5 seconds), for the period of 2 hours before the current date and time (12:17 on March 14, 2011) to 15:00 today, enter:

Response

A panel with four bars representing the percentage of transactions under 5 seconds from 12:45 to 14:17 is displayed.

Example: Displaying response time trends by percentages

To display the response time trend showing the percentage of transactions under 1.5 seconds from 08:00 on March 14, 2019 to 13:30 on March 14, 2019, enter either command:

```
RTREND LU3270A 1.5 FROM 3/14/19 08:00
TO 3/14/19 13:30
```

```
RTREND LU3270A 1.5 3/14/19 08:00
3/14/19 13:30
```

Response

A one-page panel showing the percentage of transactions under 2 seconds from 08:43 to 13:15 is displayed; 1.5 seconds were rounded to the nearest counter boundary, which is 2 seconds.

RTRINIT (NCCF)

Syntax**RTRINIT**

➤ RTRINIT — — *hqueue* — , — *rqueue* — , — *oqueue* — , — *hqueue1* ➤

Purpose of Command

The RTRINIT command initializes the interface between a NetView autotask and a specific set of database servers. This command must be run under an autotask.

The RTRINIT command is driven by the profile of a NetView autotask. This autotask serves as the interface between a NetView autotask and a specific set of database servers, and connects three user-provided queue names to the NetView program-to-program interface (PPI). For more information about PPI queues, see the *IBM Z NetView Application Programmer's Guide*.

Operand Descriptions***hqueue***

Identifies the name of the hold queue. A PPI queue is created to save transactions that have not been dispatched to a database server. This 1–8 character field is required and can contain uppercase alphabetic characters, numeric characters, or \$, %, &, @, and #.

rqueue

Specifies the name of the ready queue. A PPI queue is created to save the READY tokens generated by the database servers. This 1–8 character field is required and can contain uppercase alphabetic characters, numeric characters, or \$, %, &, @, and #.

oqueue

Specifies the name of the output queue. A PPI queue is created to receive transaction replies and control messages from the database servers. This queue has the same name as the *tioutq* queue name defined in the CNMETIN service routine of the server support API.

This 1–8 character field is required and can contain uppercase alphabetic characters, numeric characters, or \$, %, &, @, and #.

hqueue

Defines the limit of the hold queue to the PPI. This parameter is required, and must be an integer. Its value must be 50 - 2000, inclusive.

Restrictions

The following restrictions apply to the RTRINIT command:

- The operands are positional and must be placed in the order shown.
- To activate NetView Bridge, use the RTRINIT command. If you attempt to route transactions first use the SDOMAIN command to specify through the bridge without issuing the RTRINIT command, several error messages are displayed. Repeated attempts do not produce error messages, however.
- If you issue the RTRINIT command from a NetView command list, do not use the ampersand (&) as part of a queue name. The ampersand is defined as a special character in the NetView command list language.
- When the RTRINIT command issues an error message, the autotask in which the command is running is still active. You can do one of the following:
 - Recycle the autotask as follows:
 1. Issue EXCMD *autotaskname*,LOGOFF to log off the autotask from the NetView operator terminal.
 2. Correct the error.
 3. Issue AUTOTASK OPID=*autotaskname* to bring up the autotask.
 - Correct the error. Then issue EXCMD*autotaskname*,RTRINIT *hqueue,rqueue,oqueue,hqueue* from the NetView operator terminal.
- When the RTRINIT command completes successfully, the task can be ended under severe errors. You can recycle the autotask as follows:
 1. Correct the error
 2. Issue AUTOTASK OPID=*autotaskname*
- The queue limit for *rqueue*, *hqueue*, and *oqueue* is 2000. If this limit is exceeded, NetView Bridge message DWO550I is issued.
- You can increase the limit of *hqueue* by changing the value specified by *hqueue* in the RTRINIT command and issuing AUTOTASK OPID=*autotaskname* to start the autotask.
- If the NetView subsystem interface address space is down, use the same subsystem interface procedure to bring it up. If you use a different subsystem interface procedure, recycle the autotask.

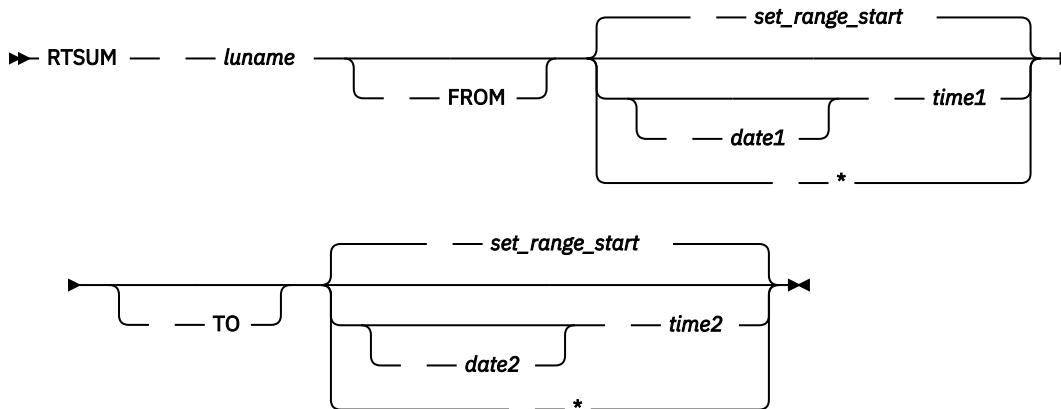
Return Codes**Return Code
Meaning**

- | | |
|-----------|-----------------------|
| 0 | Successful processing |
| 20 | Error in processing |

RTSUM (NLDM)

Syntax

RTSUM



Purpose of Command

The RTSUM command displays the response-time summary for a terminal LU connected to a cluster controller that supports the response time monitor (RTM) feature.

This command displays the Response Time Summary panel.

Operand Descriptions

luname

Specifies the name of the terminal for which response time is measured.

FROM

Identifies the operands that follow as the starting date and time. This operand is optional.

set_range_start

If you do not specify a starting date and time, the starting time set by the most recent SET RANGE command is used.

date1

Specifies the starting date of the time range. The format of *date1* is controlled by the setting of the date operands of the DEFAULTS and OVERRIDE commands. If you specify a starting time but omit the date, the current date is used.

time1

Specifies the starting time of the time range. The format of *time1* is controlled by the setting of the time operands of the DEFAULTS and OVERRIDE commands. If you do not specify a starting time, the default is the starting time set by the most recent SET RANGE command. If no SET RANGE command has been issued, the default for *time1* is one hour earlier than the time specified on the ending time (*time2*).

Specifies to use the default start and end times and not the start and end times specified by the SET RANGE command. A single asterisk (*) in place of both *date1* and *time1* specifies a starting time of one hour before the ending time. A single * in place of both *date2* and *time2* specifies an ending time of the current date and time.

TO

Identifies the operands that follow as the ending date and time. This operand is optional.

set_range_end

If you do not specify an ending date and time, the ending time set by the most recent SET RANGE command is used.

date2

Specifies the ending date of the time range. The format of *date2* is controlled by the setting of the date operands of the DEFAULTS and OVERRIDE commands. If you specify an ending time but omit the date, the current date is used.

time2

Specifies the ending time of the time range. The format of *time2* is controlled by the setting of the time operands of the DEFAULTS and OVERRIDE commands. If you do not specify an ending time, the default is the ending time set by the most recent SET RANGE command. If no SET RANGE command has been issued, the default for *time2* is the current time. If you issue a command that has an end time past the time the last command was issued, the latest time is used.

Restrictions

If the LU is in another domain, first use the SDOMAIN command to specify the domain that contains the LU.

Examples

The examples shown assume that the following conditions are true:

- You are collecting data every 30 minutes on the quarter hour (07:45, 08:15, 08:45, and so on) and a user with terminal LU name LU3270B logged on at 08:12 (March 15, 2011).
- The user is still logged on and the time is now 13:24.
- The format of dates and times specified use the default setting for date and time formats on the DEFAULTS and OVERRIDE commands.

Example: Displaying the response time summary**Example 1**

To display the response time summary from 8:00 today to the current date and time, enter either command:

```
RTSUM LU3270B 8:00 *  
  
RTSUM LU3270B FROM 3/15/19 8:00 TO 3/15/19  
13:30
```

Response

A panel showing a summary of the user's response times between 08:12 and 13:24 is displayed.

Example 2

To display the response time summary from 9:00 to 10:00 today, enter:

```
RTSUM LU3270B 9:00 10:00
```

Response

A panel showing a summary of the user's response times between 08:45 and 10:15 is displayed.

Example 3

To display the response time summary from one hour before the current date and time and the current date and time, enter:

```
RTSUM LU3270B * *
```

Response

A panel showing a summary of the user's response times between 12:15 and 13:24 is displayed.

RTTBL (NCCF)

Syntax

▶ RTTBL ◀

Purpose of Command

The RTTBL command dynamically updates member BNJRESTY. For additional information about BNJRESTY, see the *IBM Z NetView Customization Guide*.

RUNCMD (NCCF)

Syntax

RUNCMD

► RUNCMD — — SP — = — *spname* { — NETID — = — *local_network* — , —
 { — NETID — = — *net_id* —
 { * — }

► APPL — = — *applname* { , — CLISTVAR — = — NO — , —
 { , — CLISTVAR — = — NO —
 { YES — }

◄ *command* ►

Purpose of Command

The RUNCMD command routes commands to service points for processing by one of the service point applications.

For information about screens and messages that this command generates, enter:

HELP SPECS

Operand Descriptions

SP=spname

Specifies the name of the service point to process the command.

NETID

Specifies the network identifier of the network in which the service point is located. If there is another node or logical unit in any connected network with the same name as the service point you specified on the SP operand, communication is allowed only if VTAM locates that service point based solely on the LU name (*spname*) of the NETID. NETID can be specified as one of the following:

local network

Specifies to search for the target service point only in the local network. This is the default if NETID is not specified.

net_id

Specifies which network to search for the target service point. The *net_id* must be a 1–8 character value using only the EBCDIC characters 0–9 and A–Z. At least one of the characters must be alphabetic.

Specifies to search for the target service point in any network.

APPL=applname

Specifies the name of the link connection subsystem manager (LCSM) to process the command.

CLISTVAR

Specifies whether to save replies in command list variables. You can only use CLISTVAR when coding the RUNCMD command in a command list. For more information, see *IBM Z NetView Programming: REXX and the NetView Command List Language*.

NO

Does not save replies in command list variables. NO is the default.

YES

Saves replies in command list variables.

You cannot use the CLISTVAR=YES option in a pipeline even if the pipeline is issued from a NetView command list. You receive message BNH074I if you try to use CLISTVAR=YES in a pipeline. See *IBM Z NetView Programming: REXX and the NetView Command List Language* for more information about using RUNCMD in a pipeline.

command

Specifies the command to be run.

Usage Notes

The following considerations apply to the RUNCMD command:

- If the RUNCMD command is invoked from a command list, the operator's low-priority command queue is serviced after the command has completed. To prevent commands from remaining in an outstanding status, implement a time-out value.

See the COSTIME operand of the DEFAULTS command for more information.

Alternatively, you can periodically issue the DISPCMD command to display outstanding COS commands and then issue the CANCEMD command for each COS command that must be canceled.

- The RUNCMD command calls installation exit DSIEX19, which can be used to perform command authority checking for the service point application commands. For more information, see *IBM Z NetView Programming: Assembler*.

Restrictions

The following restrictions apply to the RUNCMD command:

- The limit on the length of the RUNCMD is 253 characters.
- The given command string must be the last operand. It can be in any format. Sample command lists are provided with the NetView program to simplify the specification of the parameters for this command. These command lists are described in the *IBM Z NetView Application Programmer's Guide, SC27-2870*.
- The RUNCMD command builds, as part of its outgoing record, an unformatted subvector 31, which has been retired. This deviates from the current architecture.
- Do not use the WAIT function with this command. Use the NetView automation table to trap these messages to command list variables or to have them returned to command list variables.

Example: Routing commands to service points

```
RUNCMD SP=SP01,APPL=APPL02,DISPLAY LINES
```


Response

The normal response to RUNCMD command is messages from the service point application, or message DSI268I RUNCMD COMPLETE when no messages are returned from the service point application. The messages returned can be command facility or service point application messages. If you specify CLISTVAR, the messages are returned in command list variables. See *IBM Z NetView Programming: REXX and the NetView Command List Language* for more information. If the RUNCMD is issued from within a PIPE stage command, message DSI268I will be issued in addition to any messages from the service point application.

Example: Sending the SWITCH_LINES command to a service point application

To send the SWITCH_LINES command to service point application APPL07, enter:

```
RUNCMD SP=NMWS1,APPL=APPL07,SWITCH_LINES OLD=LINE1,NEW=LINE2
```

Response

The response is the messages from the service point application or command facility messages that the service point application wants to display.

RXTRACE (NCCF)

Syntax

RXTRACE

➤ RXTRACE ➤

Purpose of Command

The RXTRACE command assists in problem determination. Use the RXTRACE command to set Program tracing, to set Entry/Exit tracing, or to set both Program tracing and Entry/Exit tracing.

Restrictions

The following restriction applies to the RXTRACE command:

- Not all NetView command lists and not all REXX programs support tracing.

Displaying the Set Trace panel

To display the Set Trace panel, enter RXTRACE.

This is the Set Trace panel.

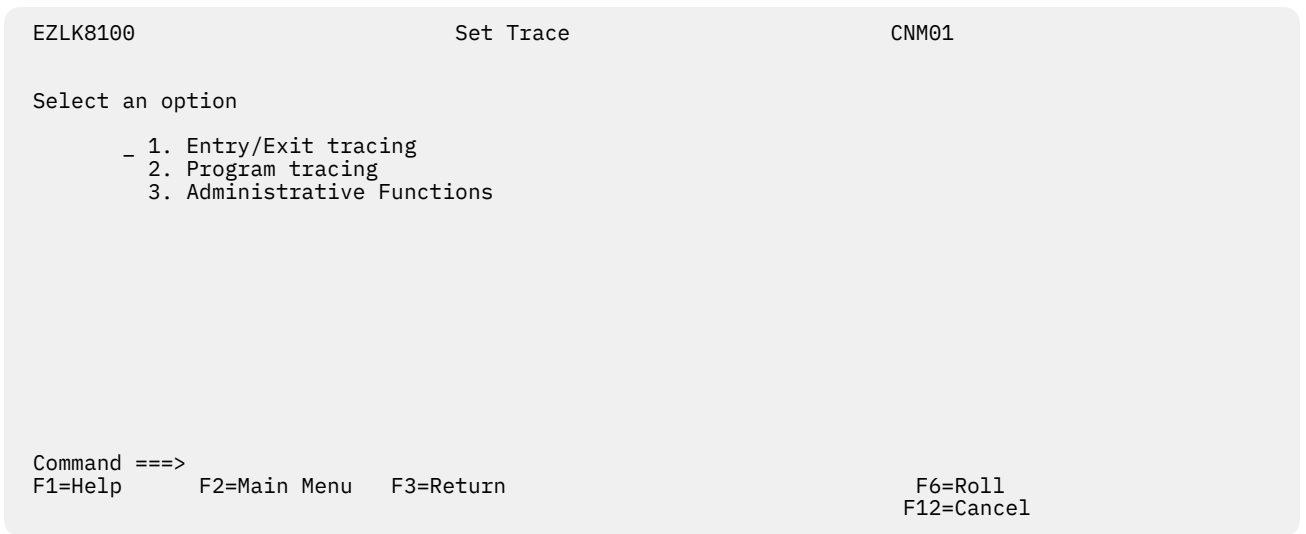


Figure 1. Set Trace Panel

The Set Trace panel provides the capability of doing the following:

- Entry/Exit tracing, which traces all of the entry and exit parameters of the code.
- Program tracing, which traces the command lists and interpreted REXX programs you specify.
- Administrative Functions, which provide the capability of enabling Entry/Exit tracing and Program tracing.

Setting Entry/Exit traces

Traces all the entry and exit parameters of the code. This includes command lists, REXX programs, and some command processors. You can trace the programs running on an operator ID, a NetView domain, or both. To trace all the programs:

1. Display the Set Trace panel.
2. Type **1** in the entry field on the Set Trace panel.
3. Press **Enter**.

This is the Set Entry/Exit Tracing panel.

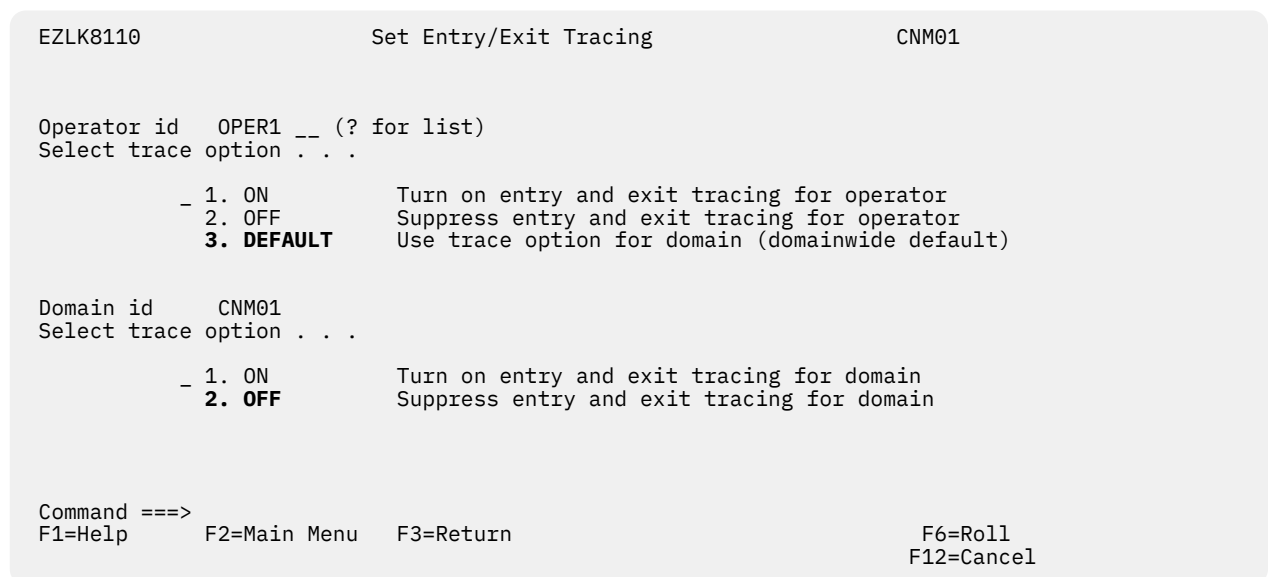


Figure 2. Set Entry/Exit Tracing Panel

4. Select Entry/Exit tracing for an operator ID, a domain, or both on the Set Entry/Exit Tracing panel. Domain tracing occurs only on the local domain. The current settings are highlighted.
- To select Entry/Exit tracing for a domain only, use the following steps:
- a. Type **1** for ON or **2** for OFF in the **Domain id Select trace option** entry field.
 - b. Press **Enter**.

This message is displayed:

```
EZL908I SETTINGS REPLACED
```

Setting program traces

Traces the command lists and interpreted REXX programs you specify. Also, specify a trace option that limits the trace. Use the Program tracing option for the operator ID or the domain. To set a trace for a particular program:

1. Display the Set Trace panel.
2. Type **2** in the entry field on the Set Trace panel.
3. Press **Enter**. This is the Set Program Tracing panel.

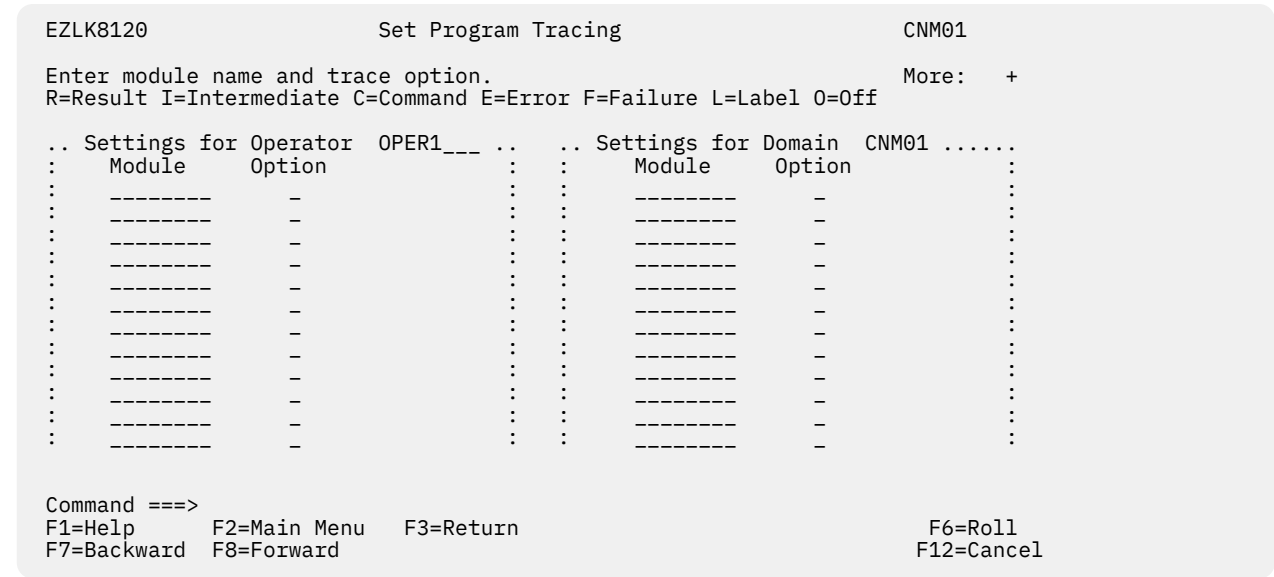


Figure 3. Set Program Tracing Panel

- You can trace the programs by operator ID or domain ID. If you trace by domain ID, tracing is done on the programs that are in the domain that the operator is logged on to. You can also select a tracing option on this panel that enables you to limit the trace.
4. Type the name of the program or programs you want to trace in the **Module** column for either the operator ID or domain ID.
 5. Type the letter for the trace option you want to use in the **Option** column. You can select one of the following trace options:

R (Result)

Use this option for general debugging. Tracing is done on all the clauses before running them and tracing is done on the final results of evaluating expressions.

I (Intermediate)

Use this option to trace all clauses before they are run and trace any intermediate results during expression evaluation and substitution.

C (Command)

Use this option to trace all commands before running them and display any error return codes from the commands.

E (Error)

Use this option to trace any command that has an error or fails after it is run. This option also displays the return codes.

F (Failure)

Use this option to trace any command that fails after it is issued. This option is the same as the Trace Normal command.

L (Label)

Use this option to trace all labels passed. Make a note of all subroutine calls and signals when you use this option.

O (Off)

Use this option to turn off all traces and reset any previous trace settings.

Note: If the program being traced is a NetView command list, the C, E, and O options are valid and all other selections result in a trace ALL.

6. Press **Enter**.

This message is displayed:

```
EZL908I SETTINGS REPLACED
```

Administrative functions

Authorized operators can enable Entry/Exit tracing and Program tracing. By default, tracing is disabled and can be enabled only through this option. By default, entry/exit and program tracing is disabled for performance purposes. This is defined on your environment setup policy definition statement. To enable tracing:

1. Display the Set Trace panel.
2. Type **3** in the entry field on the Set Trace panel.
3. Press **Enter**.

This is the Trace Administrative Functions panel.

EZLK8130	Trace Administrative Functions	CNM01
Current Trace Setting is . . . NONE		
To modify the trace setting select an option:		
Trace Setting	Entry/Exit Trace	Program Trace
- 1. ON	Yes	Yes
2. OFF	No	Yes
3. NONE	Disabled	Disabled
Selecting NONE will disable both Entry/Exit and Program tracing.		
You should select NONE for best overall performance and only select ON or OFF when you need to debug a possible problem.		
Command ==>		
F2=Main Menu	F3=Return	F6=Roll F12=Cancel

Figure 4. Trace Administrative Functions Panel

If the Trace Administrative Functions panel indicates that tracing is currently set to NONE, then all tracing is disabled for this domain. To enable tracing, select Options 1 or 2. Option 1 enables both entry/exit and program tracing. Option 2 enables only Program tracing.

When you select options 1 or 2 you can turn on program tracing for any AON program by going back to the Trace Menu panel and selecting the **Program Trace** option.

SCLIST (STATMON)

Syntax

SCLIST

➤ SCLIST ➤

IBM-Defined Synonyms

Command or Operand	Synonym
SCLIST	SC

Purpose of Command

The SCLIST command displays the command lists that you can run against one or more of the displayed resources.

Example: Displaying allowed command lists from a status monitor panel

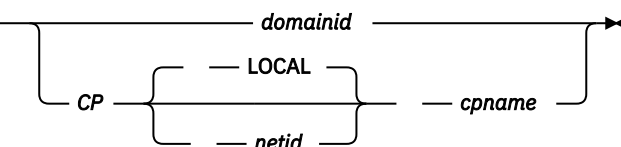
If you are displaying a status monitor panel containing resources, you can show a list of allowed command lists that you can run against one or more displayed resources by entering:

```
sclist
```

SDOMAIN (NLDM)

Syntax

NLDM SDOMAIN

➤ SDOMAIN — 

IBM-Defined Synonyms

Command or Operand	Synonym
SDOMAIN	SD

Purpose of Command

The NLDM SDOMAIN command specifies the domain from which session data is to be displayed.

Operand Descriptions

domainid

Specifies the name of the desired domain.

cpname

Indicates the name of the CP or SSCP associated with the NetView domain from which session monitor collects the data.

netid

Specifies the network in which *cpname* is defined. If not entered, the default value is the network of the NetView system to which you are logged on.

Restrictions

Before using the SDOMAIN command, define the domains to each other in DSILUCTD and DSIAMLTD. You can also allow or prohibit operator access to the specified domain by using the appropriate initialization statements for that domain. For more information, see the *IBM Z NetView Administration Reference*.

Example: Displaying session data

To specify that session data is to be displayed from DOM3, enter either command:

```
SDOMAIN DOM3
SD DOM3
```

Example: Specifying where session data is to be obtained

To specify that session data is to be obtained from domain DOM1, enter either command:

```
SDOMAIN DOM1
SD DOM1
```

To specify that session data is to be obtained from CP1, in the local network, enter either command:

```
SDOMAIN CP CP1
SD CP CP1
```

SDOMAIN (NPDA)

Syntax

NPDA SDOMAIN

➡ SDOMAIN — — *domainid* —————> QUIET

IBM-Defined Synonyms

Command or Operand	Synonym
--------------------	---------

Purpose of Command

The NPDA SDOMAIN command establishes a cross-domain session with the specified hardware monitor domains.

Operand Descriptions

domainid

Specifies the network-qualified name of the NetView domain where you want to view data. The *domainid* can be of the form *netid.domainid* or be specified as the unqualified domain name, in which case the *netid* defaults to *. In either case, the actual domain name is limited to 5 characters in length. The network-qualified form is ignored when an LU 6.2 session cannot be established with the target node.

QUIET

Sets domain and returns a message for automation (used from a command list).

Restrictions

The following restrictions apply to the SDOMAIN command:

- You cannot run this command from a multiple-entries panel.
- For the SDOMAIN command, the NetView program tries to establish the cross-domain session in the following order:
 1. Over the LU 6.2 transport. If unsuccessful, message BNJ923I or BNH094I is sent to the NetView log.
 2. Over the LUC transport. If unsuccessful, message BNJ70nI (where n is a number 0–9) is sent to the authorized receiver.
 3. Over the LU0 transport (OST-NNT session). If unsuccessful, message BNJ924I is sent to the NetView console and message BNJ926I is displayed on the hardware monitor panel message line.

Note: For the hardware monitor SDOMAIN command to successfully establish a session over the LU0 transport, issue the NCCF START DOMAIN command prior to issuing the NPDA SDOMAIN command.

When the cross-domain session is established, message BNJ911I is displayed on the hardware monitor panel message line.

- If you use the SDOMAIN command from an alert focal point to establish a cross-domain session with an entry point, and one or more intermediate nodes separate the focal point and entry point, then the SDOMAIN command might fail. The focal point NetView program might be unable to establish a session directly with the entry point.
- If you establish a cross-domain session with a focal point domain and request data from one of the focal point's distributed hosts, the request fails if the NetView program cannot establish a session between your host domain and the distributed (owning) host domain.

Example: Viewing data in the NCCF2 domain

From NCCF1, to view data in the NCCF2 domain, enter either command:

```
SDOMAIN NCCF2
SDOMAIN *.NCCF2
```

Response

The usual response is:

```
BNJ911I SESSION DOMAIN NOW NETA.NCCF2, WAS NETA.NCCF1
```

Example: Invoking SDOMAIN with the QUIET option

You can issue the hardware monitor SDOMAIN command with the QUIET option from a command list to set the domain and return a message for automation.

To trap the message in a REXX command list, issue the SDOMAIN command after issuing a TRAP instruction, but before issuing a WAIT command.

To trap the message in a command list written in the NetView command list language, issue the SDOMAIN command from the &WAIT statement.

The NetView program supplies a sample command list written in the NetView command list language that issues this command. In this command list, whenever an SDOMAIN message occurs that is not tested on the &WAIT statement, the message is written to the command facility panel and the command list stops running.

The following example shows how to invoke the SDOMAIN command with the QUIET option from a command list:

```
CLIST
&CONTROL ERR
*****
* 5655-007 (C) COPYRIGHT IBM CORP. 1987, 2011 *
* LAST CHANGE: *
* *
* NAME(CNME0044) SAMPLE(CNME0044) RELATED-TO() *
* *
* DESCRIPTION: THIS CLIST ISSUES THE SDOMAIN QUIET COMMAND WHICH *
* INITIATES A CROSS DOMAIN SESSION WITHOUT DISPLAYING *
* THE NPDA MAIN MENU. IF THE SDOMAIN QUIET COMMAND IS *
* SUCCESSFUL, THE ALERTSD COMMAND IS ISSUED. *
* *
* CNME0044 CHANGED ACTIVITY: *
* CHANGE CODE DATE DESCRIPTION *
* ----- *
*****
* THE FIRST (AND ONLY) PARAMETER EXPECTED BY THIS CLIST IS THE DOMAIN *
* NAME FOR WHICH THE ALERTSD INFORMATION IS DESIRED. *
*****
&DOMAINID = &1
*****
* IF A DOMAIN NAME IS NOT PASSED TO THE CLIST, THEN SET THE DOMAIN *
* NAME TO THE DOMAIN THE USER IS LOGGED ONTO. *
*****
&IF .&DOMAINID NE . &THEN &GOTO -XDOMAIN
&DOPART = &LENGTH &APPLID
&DOPART = &DOPART - 3
&DOMAINID = &SUBSTR &APPLID 1 &DOPART
*****
* INVOKE THE SDOMAIN COMMAND WITHIN THE &WAIT STATEMENT TO TRAP THE *
* MESSAGES PUT OUT BY HARDWARE MONITOR. *
*****
-XDOMAIN
&WAIT CONTWAIT SUPPRESS
&WAIT 'NPDA SDOMAIN &DOMAINID QUIET' +
      BNJ711I=-NPDATL +
      BNJ911I=-NPDACM +
      BNJ912I=-INCOMPAT +
      BNJ924I=-BADXDOM +
      BNJ926I=-SDFAIL +
      BNJ1303I=-NPDANA +
      DSI210I=-WRITE210 +
      *ERROR=-ERROR +
      *10=-TIMEOUT
&GOTO -ERROR
*****
* DISPLAY APPROPRIATE MESSAGE *
*****
**
** SD/SDOMAIN OPERAND domain IS TOO LONG, GREATER THAN FIVE CHARACTERS
**
-NPDATL
GO
&WRITE IN CNME0044: &MSGID &MSGSTR
&GOTO -END
**
** CROSS DOMAIN SESSION TO AN INCOMPATIBLE LVL OF NETVIEW WAS
** ATTEMPTED
**
-INCOMPAT
GO
&WRITE IN CNME0044: &MSGID &MSGSTR
&GOTO -END
**
** CROSS DOMAIN SESSION TO AN UNDEFINED DOMAIN WAS ATTEMPTED
**
-BADXDOM
```



```

&WRITE IN CNME0044: &MSGID &MSGSTR
&WAIT CONTINUE
**
** THE SDOMAIN COMMAND FAILED
**
-SDFAIL
GO
&WRITE IN CNME0044: &MSGID &MSGSTR
&GOTO -END
**
** USER NOT AUTHORIZED TO ISSUE COMMAND
**
-NPDANA
GO
&WRITE IN CNME0044: &MSGID &MSGSTR
&GOTO -END
**
** AN UNEXPECTED ERROR OCCURRED
**
-ERROR
&WRITE IN CNME0044: UNDETERMINED ERROR OCCURRED
&GOTO -END
**
** &WAIT TIMED OUT BEFORE ANY MESSAGES IT WAS TESTING FOR OCCURRED
**
-TIMEOUT
&WRITE IN CNME0044: TIME OUT ON SDOMAIN COMMAND
&GOTO -END
*****WHILE IN A WAIT, DSI210I WAS RECEIVED *****
-WRITE210
&WAIT DISPLAY
&WRITE &MSGID &MSGSTR
&WRITE TERMINATING CLIST
&GOTO -END
*****
* SDOMAIN COMMAND WORKED CORRECTLY, ISSUE THE ALERTS DYNAMIC COMMAND *
*****
-NPDACM
NPDA ALERTSD
&GOTO -END
*****
*                               END OF CLIST                               *
*****
-END
&CONTROL ERR

```

The most common messages produced by the SDOMAIN command are:

BNJ911I

SESSION DOMAIN NOW *netid1.nau1*, WAS *netid2.nau2*.

BNJ912I

RELEASE LVLS INCOMPATIBLE BETWEEN DOMAINS *domain1* AND *domain2*.

BNJ924I

CANNOT SEND TO SPECIFIED DOMAIN.

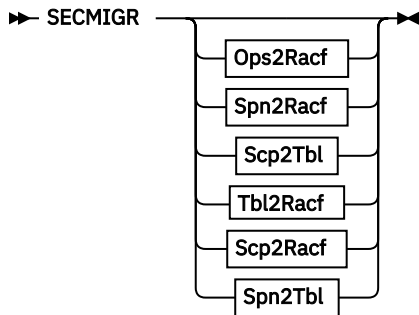
BNJ926I

SD/SDOMAIN COMMAND FAILED. SESSION DOMAIN IS UNCHANGED.

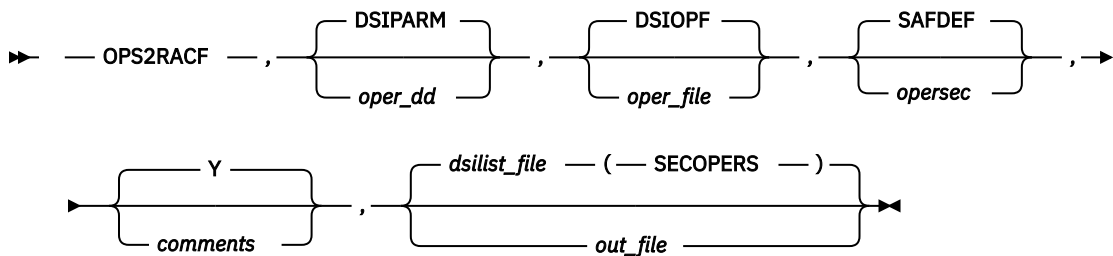
SECMIGR (NCCF; CNME8004)

Syntax

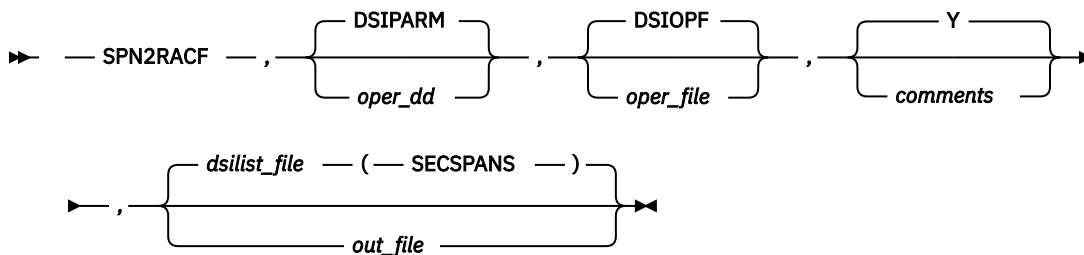
SECMIGR



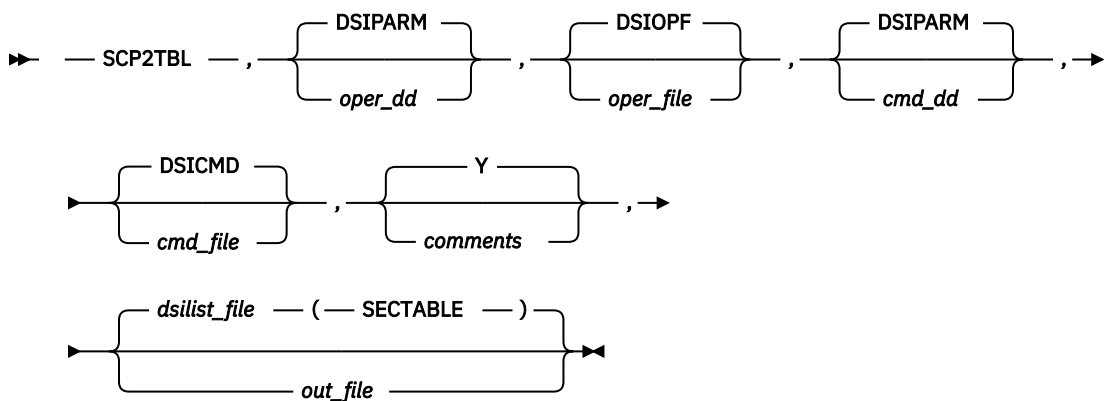
Ops2Racf



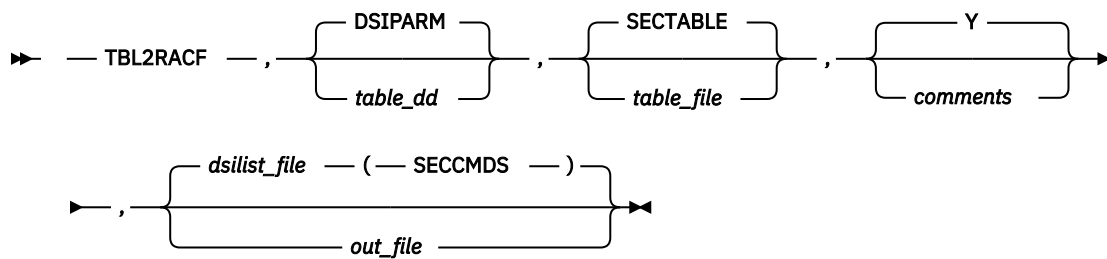
Spn2Racf



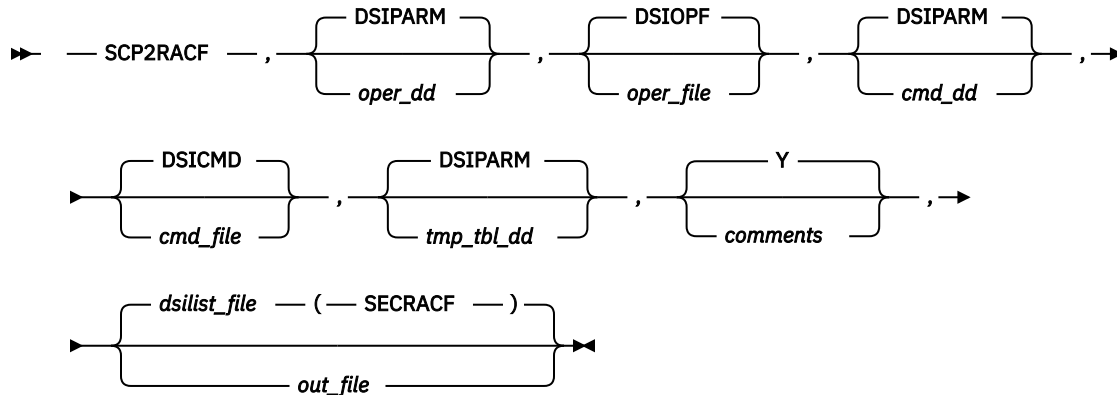
Scp2Tbl



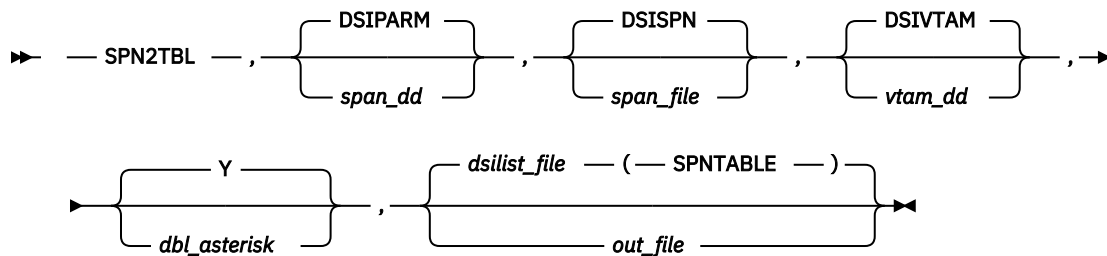
Tbl2Racf



Scp2Racf



Spn2Tbl



Purpose of Command

The SECMIGR command assists you in converting your current security settings and definitions. The command produces output members which can be used to define the desired security settings. In most cases SECMIGR converts existing security statements into statements that provide equivalent security. Review the generated statements to validate that they provide the desired security before enabling them in your installation.

The SECMIGR command will perform system symbolic substitution on records read from disk. The NetView SUBSYM PIPE stage is added to all PIPE commands that have disk read stages.

Substitution is always performed on the &DOMAIN symbolic, unless substitution was disabled when the NetView program was started. For MVS and user-defined system symbolics, substitution is not performed if substitution was disabled when the NetView program was started, or if you have not defined an MVS system symbolic on your system.

Operand Descriptions

Entering the SECMIGR command with no operands initiates a full-screen interface to prompt you for your processing options. To bypass this set of panels, enter the options on the command line.

cmd_dd

Specifies the DD statement containing the member specified by *cmd_file*. The default value is DSIPARM. This specification is limited to a subset of the DD statements in the NetView procedure. See [“Restrictions” on page 204](#) for the DD statement specifications that are valid.

cmd_file

Specifies the member containing the CMDMDL statements as well as CMDCLASS, KEYCLASS, and VALCLASS statements that you are migrating. This member can contain %INCLUDE statements to embed other members. The default value is DSICMD.

comments

Specifies that comments are to be copied from the input file to the output file. A value other than Y causes comments not to be copied. The default value is Y. This keyword is not valid for the SPN2TBL function.

dbl_asterisk

Specifies that a double asterisk and a period (**) are prefixed to each resource name as it is added to the list of resources for a span. This list of resources is then used to create the SPANDEF statement. A value other than Y prevents a double asterisk from being added. The default value is Y. Creating identifiers in the NetView span table that consist of resource names prefixed with a double asterisk ensures that a match will be found for not only the resource name but the fully qualified resource name as well.

oper_dd

Specifies the DD statement containing the member specified by *oper_file*. The default value is DSIPARM. This specification is limited to a subset of the DD statements in the NetView procedure. See [“Restrictions” on page 204](#) for the DD statement specifications that are valid.

oper_file

Specifies the member containing the OPERATOR statements to be migrated. This member might contain %INCLUDE statements to embed other members. The default value is DSIOPF.

opersec

Specifies the desired OPERSEC setting to be used with the generated RACF statements.

Note: Information about password checking can be found in the *IBM Z NetView Administration Reference*.

Valid values are:

SAFPW

Causes RACF operator and password definition statements to be generated.

SAFCHECK

Causes RACF operator and password definition statements to be generated.

SAFDEF

Causes RACF operator and password definition statements as well as NETVIEW segment definition statements to be generated. SAFDEF is the default.

OPS2RACF

Converts operator definitions in the NetView program to a list of statements to define them in RACF based on the *opersec* setting.

out_file

Specifies the output file to contain the generated statements. The *out_file* specification is in the form of *data_set(member)*. The default value is the first data set in the DSILIST concatenation. The default member name is determined by the migration function you are using, as follows:

OPS2RACF

Member = SECOPERS

SPN2RACF

Member = SECSPANS

SCP2TBL

Member = SECTABLE

TBL2RACF

Member = SECCMDS

SCP2RACF

Member = SECRACT

SPN2TBL

Member = SPNTABLE

SCP2RACF

Converts scope statements and operator classes into a list of RACF statements required to provide equivalent security. This migration is performed by converting the scope statements to NetView command authorization table statements and then converting the command authorization statements to RACF definition statements.

Note: If you plan to use CMDAUTH=SAF and SAFNODEC=FAIL, there will be commands that are not protected under the current CMDAUTH, that are now protected because they were not defined in RACF with your SECMIGR output. SAFNODEC=FAIL specifies that the undefined commands, and their keywords and values, will fail the authority check. Add more definitions in RACF to allow access. The default value is not to allow access. If you are sure that you have protected all sensitive commands, add an RDEFINE command for netid.luname.* to force a match on undefined commands.

SCP2TBL

Converts obsolete scope statements and operator classes into NetView command authorization table statements required to provide equivalent security.

span_dd

Specifies the DD statement that contains the member specified by *span_file*. The default value is DSIPARM. This specification is limited to a subset of the DD statements in the NetView procedure. See [“Restrictions” on page 204](#) for the DD statement.

span_file

Specifies the member containing the SPANLIST statements to be migrated. The default value is DSISPN.

SPN2RACF

Converts the spans defined in each of the operator profiles to a list of RACF statements necessary to define them. SPN2RACF does not create statements that define operators to RACF. You can use the OPS2RACF function to do this.

SPN2TBL

Converts the spans defined in the DSISPN and VTAMLST statements to NetView span table statements. Span names and resource names are alphabetically sorted. Resources are grouped with their associated spans.

table_dd

Specifies the DD statement containing the member specified by *table_file*. This member might contain %INCLUDE statements to embed other members. The default value is DSIPARM. This specification is limited to a subset of the DD statements in the NetView procedure. See [“Restrictions” on page 204](#) for the DD statement specifications that are valid.

table_file

Specifies the member containing the NetView command authorization statements to be migrated to RACF statements. The default value is SECTABLE.

tmp_tbl_dd

Specifies to which DD concatenation the temporary NetView command authorization table is written. The first data set in the concatenation is used for this purpose. The valid values are:

DSILIST

If DSILIST is selected, the temporary command authorization table is not syntax checked prior to converting the table to RACF statements.

DSIPARM

If DSIPARM is selected, the temporary command authorization table is syntax checked prior to converting the table to RACF statements. DSIPARM is the NetView-defined default specified in the SECMIGR command. The operator needs write-authority to the first data set in the DSIPARM concatenation.

TBL2RACF

Converts NetView command authorization table statements to the statements required to define equivalent command security with RACF.

Note:

1. Immediate commands in the NetView program are not checked by RACF. You can specify a backup table on either the SECOPTS.CMDAUTH statement in the CNMSTYLE member, or on the REFRESH command to provide protection for these commands.
2. If you plan to use CMDAUTH=SAF and SAFNODEC=FAIL, there will be commands that are not protected under the current CMDAUTH, that are now protected because they were not defined in RACF with your SECMIGR output. SAFNODEC=FAIL specifies that the undefined commands, and their keywords and values, will fail the authority check. Add more definitions in RACF to allow access. The default value is not to allow access. If you are sure that you have protected all sensitive commands, add an RDEFINE command for netid.luname.* to force a match on undefined commands.

vtam_dd

Specifies the DD statement containing the VTAMLST statements. The default value is DSIVTAM. This specification is limited to a subset of the DD statements in the NetView procedure. See [“Restrictions” on page 204](#) for the DD statement specifications that are valid.

Usage Notes

- When SECMIGR creates output members, it places a 2-character key field (>S) in the first line of the report. This key field identifies the file as SECMIGR output and stops SECMIGR from replacing members it did not create. If you attempt to replace a member or file that does not have the key field in the correct location, you will receive an error message. You can prevent a SECMIGR output member from being overwritten by removing the write protect key from the first line of the member.
- The output member of the SECMIGR command using either the SPN2TBL or the SCP2TBL function must be placed in a data set in the DSIPARM concatenation before it can be used by the NetView program.
- When the SECMIGR command creates an output member, it includes comments that include the operator ID that issued the command, the date and time it was run, the input DD name and member names, and the output data set and member name.
- For SECMIGR requests using the OPS2RACF, SCP2RACF, SPN2RACF, and TBL2RACF functions, the output member contains RACF commands. The output member formatting must be altered in order to place the statements into effect using a batch job or TSO CLIST.

Restrictions

The following restrictions apply to the SECMIGR command:

- The DD statement specifications for the SECMIGR command are restricted to a subset of the DD statements in the NetView procedure. See the BROWSE command help for a list of valid DDNAMEs.
- The SECMIGR command assumes that the input control statements that you are converting are syntactically correct and performs minimal syntax checking.
- The SECMIGR command uses only the first profile specified on the PROFILEN statement in the operator definition member. Profiles are read from DSIPRF DD members.
- The SECMIGR command does not support comments on the same line with keywords in DSIOPF and DSIPRF. They must be removed before running the SECMIGR command. Whole line comments are allowed.
- When using the SCP2RACF function, a member of the first data set in the DSILIST or DSIPARM concatenation is used to hold the intermediate table during the conversion. The SECMIGR command list contains a constant which is assigned the value of this member name. The value of the constant as shipped is SECMTEMP. If this data set and member combination is not available, the command fails.

Return Codes**Return Code
Meaning**

- 0** Functioned normally.
- 1** Non-valid argument.
- 2** Error reading input file.
- 4** Error reading profile.
- 6** Error returned from call to allocate output or temporary file.
- 8** Error writing to output file.
- 10** SECMIGR invoked outside the NetView system.
- 12** Halt condition encountered.
- 14** DSIPRF DD not available.
- 16** REFRESH TABLE=*membername*,TEST failed for TBL2RACF function.
- 18** VIEW was unable to find a required SECMIGR panel.
- 20** An attempt was made to overwrite a file which was not created by SECMIGR.
- 22** Command authorization table DD specified was not valid.

Example: Migrating Operator IDs to RACF

The following example converts existing operator definitions and passwords from member DSIOPF in DSIPARM to RACF operator definition statements. This also includes NetView attributes and places the RACF statements in member SECOPERS of the first data set in the DSILIST concatenation:

```
SECMIGR OPS2RACF , , , SAFDEF
```

The input data set and member names, as well as the output data set and members names, are determined by the command defaults.

Example: Migrating DSISPN and VTAMLST to NetView span table

The following example converts existing span of control definitions from member DSISPN in DSIPARM and the VTAMLST members in DSIVTAM to NetView span table statements in member SPNTABLE of the USER.DSIPARM data set:

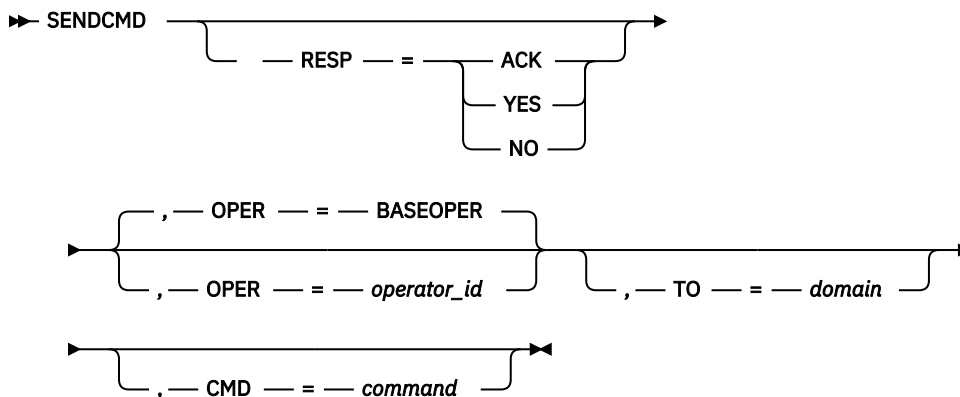
```
SECMIGR SPN2TBL , , , USER.DSIPARM(SPNTABLE)
```

The input data sets and member names as well as the output data set and member name are determined by the command defaults.

SENDCMD (AON)

Syntax

SENDCMD



Purpose of Command

The SENDCMD command allows you to route commands to other domains using the gateway sessions. When you use gateway sessions, an automation operator known as a gateway operator logs on to the other domain, so it is unnecessary for you to have your own session with that domain. To send commands from the operator interface, see *IBM Z NetView User's Guide: Automated Operations Network*.

Operand Descriptions

RESP=

Parameters are defined as follows:

ACK

Specifies that a notification is required indicating whether the command was run on the target domain or not.

YES

Specifies that the output from the command being issued is to be displayed on the current domain.

NO

Specifies that neither a response nor acknowledgment is required for the command that is being issued.

OPER=

The command is run on this operator ID at the target domain. If *operator_id* is not specified, the command is run by the AON automation operator, BASEOPER, on the target domain.

TO=domain

Specifies the domain to which the command is being issued.

CMD=command

Any valid NetView command or AON command except LOGOFF.

Usage Notes

- The AON tower must be enabled in the CNMSTYLE member to successfully run this command.
- If the command you are sending to another domain using the SENDCMD command contains a comma, you must use the delimiters ' or " for full-screen mode or / if you are using line mode. For example:

```
SENDCMD RESP=YES,TO=CNM02,CMD='D NET,CDRMS'
```


SENSE (NLDM; CNME2003)

Syntax

SENSE

► SENSE — — *code* —————►
 └— M —┘

Purpose of Command

The SENSE command list displays help for the SNA sense codes set by VTAM. See the appropriate SNA manual for more information about SNA (System Network Architecture) sense code.

Operand Descriptions

code

The sense code in hexadecimal.

M

Indicates that message output (AAU977I) is desired. The M option is only supported by the SENSE command list, not the NLDM SENSE command.

Usage Notes

The SENSE command is an NLDM command and thus requires TOWER=NLDM to be defined in the CNMSTYLE member. The NLDM tasks (DSIAMLUT, AAUTSKLP, and AAUTCNMI) do not need to be started for the command to work. See the *IBM Z NetView Administration Reference* for more information about the CNMSTYLE member, and about towers and how to enable them.

Example: Getting a description of a 2-byte sense code

To get a description of the 2-byte sense code X'0806', enter:

```
SENSE 0806
```

Example: Displaying a description of a 4-byte sense code

To display a description of 4-byte sense code X'087D0005', enter:

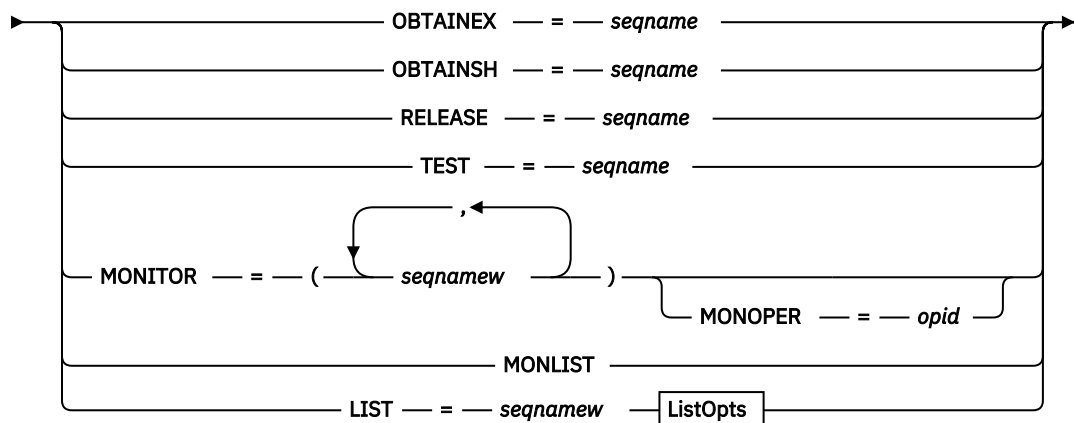
```
SENSE 087D0005
```

SEQUENT

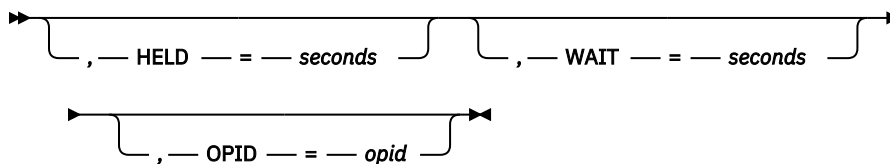
Syntax

SEQUENT

➤ SEQUENT ➔



ListOpts



Purpose of Command

The SEQUENT command allows for the serialization of operations in a programmatic environment, similar to the ENQ and DEQ assembler macros in MVS. A named resource is specified to obtain, release, monitor, or list. Obtaining a resource as exclusive prevents other SEQUENT obtain requests for the same resource name to be satisfied, until the resource is released by the program that obtained it. Obtain requests that cannot be immediately satisfied cause the running program to be suspended. At the instant when the resource is released, the next waiting obtain request is satisfied, and the requesting program resumes. If a resource is requested to be obtained as shared, other shared requests can also be satisfied, as long as an exclusive request is not queued before the shared request.

Obtain requests are satisfied in the order they are requested, unless the program environment gets in a state where it cannot take ownership of the resource, in which case the next waiting request is offered ownership of the resource. An example of this would be when a high priority command interrupts the program requesting the SEQUENT resource.

The SEQUENT command can also be used to perform the following tasks:

- Test the current state of a SEQUENT resource name in a programmatic environment. The state of the resource is indicated by a return code.
- Monitor actions taken upon one or more resource names, which include obtaining, releasing, and waiting for ownership of a resource. Each of these actions is indicated by a message.
- List one or more resource names to show the current status of ownership or request, and how long it has been in that state. This list can be filtered by amount of time held (obtained), waiting, or by requesting operator ID.

Operand Descriptions

HELD

Limits the display of SEQUENT names that are listed to those names that are currently held and owned by some task, and for at least the amount of time specified. To list all that are held, specify HELD with a value of 0 or no value at all.

LIST

Causes information about the specified SEQUENT names to be displayed, which could be more than one if wild cards are specified. Messages that contain the following information are presented to the operator who issues the command:

- SEQUENT name
- Task name that is currently holding the SEQUENT resource or blank if currently not being held
- Task names that are currently waiting for the SEQUENT resource (on separate lines)
- Indication as to whether the SEQUENT resource is being held or being waited upon or available
- Date or time when the SEQUENT resource was obtained or started waiting for the SEQUENT resource or became available
- The length of time that the SEQUENT resource has been obtained, waiting, or been available

This option is not restricted to procedure environments.

MONITOR

Specifies a list of SEQUENT resource filters, each of which specifies that a set of SEQUENT names is or is not to be monitored for activity. The following actions will cause a message to be issued, which is also automatable, for SEQUENT names that are being monitored:

- Successfully obtaining a SEQUENT resource
- Being queued up on an OBTAINEX or OBTAINSH request
- Successfully releasing a SEQUENT resource

Multiple filters, with or without wild cards, and with or without a 'not' sign (' $\neg$ ') as the first character, can be specified on a single command. Successive SEQUENT commands with MONITOR result in a cumulative effect of the filters, which are scanned in the order specified to determine whether a SEQUENT name matches. The last filter that a SEQUENT name matches determines whether a monitoring message is issued for it. To remove all monitoring, specify MONITOR= $\neg$ *. A filter specified following the $\neg$ *, even on the same invocation of MONITOR, behaves as if there were no filters in place before it. This option is not restricted to procedure environments.

MONLIST

Causes the filters in the current MONITOR list, as well as the current MONOPER value, to be displayed. This option is not restricted to procedure environments.

MONOPER

Specifies an operator ID to which monitoring messages are sent, in addition to being logged to NETLOG and Canzlog. If * is specified, the current operator ID is used. If *NONE* is specified, monitoring messages are only logged to NETLOG and Canzlog. If MONOPER is not specified together with MONITOR, the monitoring operator ID remains the same as it had been set from a previous specification of MONOPER. If no previous MONOPER had been specified, and MONOPER is not specified, the operator that issues the SEQUENT MONITOR command is sent the monitoring messages.

OBTAINEX

Requests exclusive ownership of the resource specified on *seqname*. If the current command group currently owns the resource as shared, ownership is converted to exclusive, if no other task request currently owns it as shared; otherwise, the request fails with a specific return code. Control is not returned to the issuing code until the resource is available or an error is detected. Ownership is assigned based on the order of each request, as long as the command group is available to take ownership. This option may only be used when running in a procedure environment. Unless the option is not issued from within a procedure environment, in which case DSI290I is issued and the command

fails, or a syntax or environmental error is detected, no message is issued, and the only error indication is in the form of a return code. See the list of [return codes](#) for details.

OBTAINSH

Requests shared ownership of the resource specified on *seqname*. Control is not returned to the issuing code until the resource is available or an error is detected. For this request, other command groups in the same or other tasks may also own the resource as shared. Ownership is assigned based on the order of each request, as long as the command group is available to take ownership. This option may only be used when running in a procedure environment. Unless the option is not issued from within a procedure environment, in which case DSI290I is issued and the command fails, or a syntax or environmental error is detected, no message is issued, and the only error indication is in the form of a return code. See the list of [return codes](#) for details.

OPID

Limits the display of SEQUENT names that are listed to those that are either held by or being waited upon by the specified operator ID.

opid

A 1 - 8 character operator ID or function name that is preceded by a question mark (?) that resolves to an operator ID. The ID may not contain wild card characters. When the operator ID is specified on the OPID keyword with the LIST action, the specified operator ID limits the output to SEQUENT resources that are holding or waiting for a matching name. When the operator ID is specified on the MONOPER keyword with the MONITOR action, monitoring messages are sent to the specified operator ID, or are only logged if *NONE* is specified for *opid*.

RELEASE

Releases the resource specified on *seqname*, if it is currently owned by the current command group. This option may only be used when running in a procedure environment. Unless the option is not issued from within a procedure environment, in which case DSI290I is issued and the command fails, or a syntax or environmental error is detected, no message is issued, and the only error indication is in the form of a return code. See the list of [return codes](#) for details. If a SEQUENT resource had not been released by a SEQUENT RELEASE command when the command group ends or the operator logs off, the resource is automatically released, and a warning message is issued.

seconds

The number of seconds to limit the LIST output to. Only those SEQUENT resources that are held or waiting at least the number of seconds specified will be listed. The format of seconds can be one of the following formats:

- ss (number of seconds, up to 3599999)
- mm:ss (number of minutes up to 59999 and seconds up to 59)
- hh:mm:ss (number of hours up to 999, minutes up to 59 and seconds up to 59)

seqname

The 1 - 16 character name of a SEQUENT resource to obtain, release, or test. Only the following characters are allowed in the name:

```
A-Z a-z 0-9 . # @ ¢ $ _ ! ?
```

The name is case-sensitive, which means that a resource name of ABC is a different resource from one named abc.

seqnamew

A SEQUENT name that follows the character limitations that are documented for *seqname* that can also contain wild cards: asterisk (*) for multicharacter and percent sign (%) for single character.

TEST

Tests whether the resource specified on *seqname* is known to the NetView program, and if so, whether it is currently owned, and in what manner. This option may only be used when running in a procedure environment. Unless the option is not issued from within a procedure environment, in which case DSI290I is issued and the command fails, or a syntax or environmental error is detected, no message is issued, and the current status of the SEQUENT resource is returned in the form of a return code. See the list of [return codes](#) for details.

WAIT

Limits the display of SEQUENT names that are listed to those names that are currently being waited upon and requested to be obtained, but the OBTAINEX or OBTAINSH request hasn't been satisfied yet, and for at least the amount of time specified. To list all that are being waited upon, specify WAIT with a value of 0 or no value at all.

Usage Notes

- Obtain requests are satisfied in the order that they are requested, unless the requesting command procedure group is not in control when the requested resource becomes available. This case can occur, for example, when a high priority command interrupts the command group that is waiting for a resource.
- When you specify a SEQUENT name that contains lower-case characters, ensure that the SEQUENT command is executed in a NETVASIS environment.
- Use caution when requesting ownership (with OBTAINEX or OBTAINSH) when running under a high priority command. Doing so can result in a lockout condition if the high priority command cannot obtain the resource immediately and the command group (running under a different task) that owns the resource is dependent on an action taken by a command that is interrupted by the high priority command.
- For a more in-depth discussion of various considerations for using the SEQUENT command, see *Using the SEQUENT Command to Serialize Access to Resources* in the IBM Z NetView Application Programmer's Guide.

Return Codes

For all requests, the return codes are as follows:

Return code	Meaning
-1	Returned when an invalid parameter is specified for any type of request. This is handled as failure error if the SEQUENT command was executed within a command procedure.
0	The request completed successfully.
20	Environment error when one of the following conditions occurs: • If OBTAINEX, OBTAINSH, RELEASE, or TEST is requested in a non-procedure environment (in addition to issuing DSI290I). • If SEQUENT is issued in an IBM Automation Control for z/OS or GDPS/PPRC HyperSwap Manager environment in addition to issuing DWO038I. • If SEQUENT is issued in an immediate command environment or VTAM exit.
99	Internal error.
999	Storage failure.

For OBTAINEX requests, the return codes are as follows:

Return code	Meaning
0	You obtained the resource as exclusive, which might have been converted from shared ownership.
7	You already own the resource as exclusive.
12	The resource is currently owned as shared by another command group under your task.
13	The resource is currently owned as exclusive by another command group under your task.

For OBTAINSH requests, the return codes are as follows:

Return code	Meaning
0	You obtained the resource as shared.
5	You already own the resource as shared, and you are the only one who owns it.
6	You already own the resource as shared, and there are other shared owners of it.
7	You already own the resource as exclusive.
13	The resource is currently owned as exclusive by another command group under your task.

For RELEASE requests, the return codes are as follows:

Return code	Meaning
0	The resource was successfully released.
1	You do not currently own the resource.
12	The resource is currently owned as shared by another command group under your task.
13	The resource is currently owned as exclusive by another command group under your task.
16	The specified SEQUENT name is not known.

For TEST requests, the return codes are as follows:

Return code	Meaning
0	The resource is currently not owned at all.
2	One or more tasks other than your own currently own the resource as shared, and the resource is currently available for a shared request.
3	One or more tasks other than your own currently own the resource as shared, but an exclusive request is waiting for it, so it is not immediately available for a shared request.
4	The resource is currently owned as exclusive by another task.

Return code	Meaning
5	You own the resource as shared, and you are the only one who owns it, so it could be immediately converted to exclusive use.
6	You own the resource as shared, and there are other shared owners of it, so it is not immediately available to be converted to exclusive use.
7	You own the resource as exclusive.
12	The resource is currently owned as shared by another command group under your task.
13	The resource is currently owned as exclusive by another command group under your task.
16	The specified SEQUENT name is not known.

Example: Requesting exclusive use of resource MySeqName

To request exclusive ownership and subsequent release of SEQUENT name MySeqName in a REXX program, issue the following commands:

```
'ADDRESS NETVASIS SEQUENT OBTAINEX=MySeqName'
<Code that cannot execute on more than one task at a time>
'ADDRESS NETVASIS SEQUENT RELEASE=MySeqName'
```

Example: Requesting shared use of resource MySeqName

To request shared ownership and subsequent release of SEQUENT name MySeqName in a REXX program, issue the following commands:

```
'ADDRESS NETVASIS SEQUENT OBTAINSH=MySeqName'
<Code that cannot run while another task owns the resource exclusively>
'ADDRESS NETVASIS SEQUENT RELEASE=MySeqName'
```

Example: Updating a global KEEP in REXX code

To safely update a global KEEP that can be updated and referenced by multiple tasks in a REXX program, issue the following commands:

```
'ADDRESS NETVASIS SEQUENT OBTAINEX=GlobalKeepABC'
'PIPE KEEP GLOBAL /ABC/ | VAR ABCVAL'
abcVal = abcVal 'more data'
'PIPE VAR ABCVAL | KEEP GLOBAL /ABC/'
'ADDRESS NETVASIS SEQUENT RELEASE=GlobalKeepABC'
```

Example: Testing the status of SEQUENT resource MySeqName in REXX code

To find out whether resource MySeqName is owned or not, and in what manner, in a REXX program, code and execute the following commands:

```
'ADDRESS NETVASIS SEQUENT TEST=MySeqName'
SAY 'The status of SEQUENT name MySeqName is' rc
```

Example: Monitoring requests for SEQUENT resources beginning with MySeq

To set up monitoring of resources that begins with "MySeq", and have the monitoring messages appear on operator task SEQMON, enter the following command from a NCCF screen:

```
NETVASIS SEQUENT MONITOR=MySeq*,MONOPER=SEQMON
```


Example: Displaying SEQUENT resources

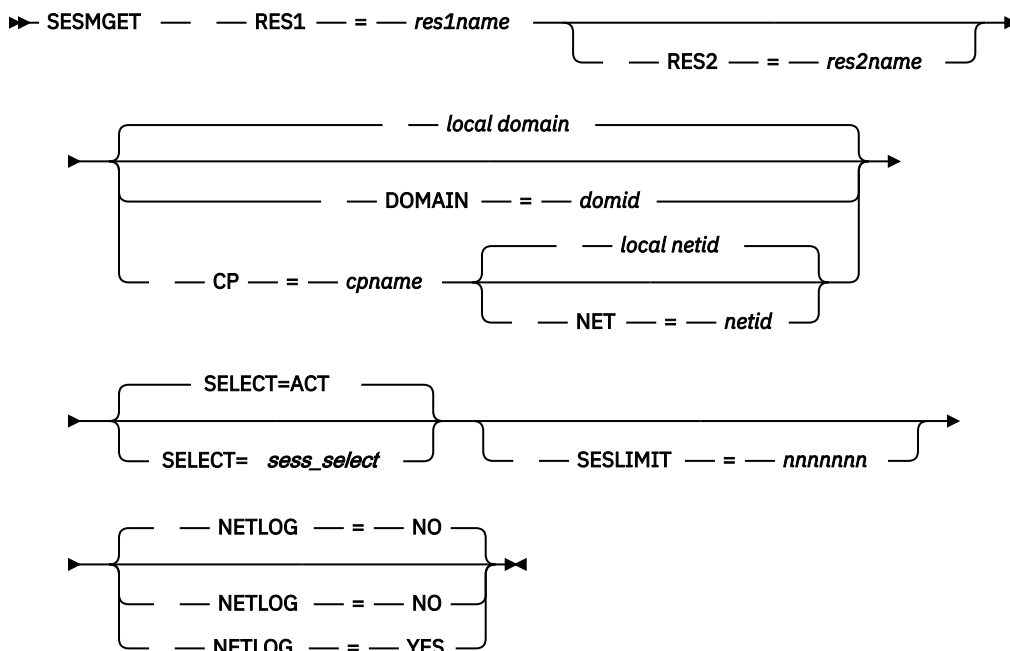
To display a list of SEQUENT resource names that contain the string "Seq" that are currently owned by any task for at least 5 seconds, enter the following command from a NCCF screen:

```
NETVASIS SEQUENT LIST=*Seq*,HELD=5
```

SESMGET (NLDM; CNME2011)

Syntax

SESMGET



IBM-Defined Synonyms

Command or Operand	Synonym
ACT	ACTIVE

Purpose of Command

The SESMGET command displays session monitor data.

Operand Descriptions

RES1=*res1name*

Specifies the resource for which you want session data. You can use an asterisk (*) as a wildcard character at the end of the resource name or as the only specified character.

When RES2 is not present, you receive sessions for which *res1name* is either the primary or secondary endpoint.

RES2=*res2name*

Specifies the name of the second endpoint. You can use an asterisk (*) as a wildcard character at the end of the resource name or as the only specified character.

When present, you get sessions between the two named endpoints. If RES2 is not present, you will receive sessions for which *res1name* is either the primary or secondary endpoint.

DOMAIN=domid

Specifies the NetView domain from which the session monitor collects the data. When neither DOMAIN nor CP is present, the local domain is used. DOMAIN and CP are mutually exclusive.

CP=cpname

Indicates the name of the CP or SSCP associated with the NetView domain from which session monitor collects the data. DOMAIN and CP are mutually exclusive. If CP is specified without NET the default value is the local network.

NET=netid

Specifies the network in which *cpname* is defined.

SELECT=sess_select

Selects the sessions for the resource or resource pair. See “SESS (NLDM; CNME2004)” on page 217 for a description of the valid selection values, such as ALL, ACT, INACT, or ERR.

The default value is ACT.

Note: Some values for the SESS command require multiple words, such as ACTREF ABC*. To use such values from SESMGET, code a forward slash (/) between the words. For example:

```
SELECT=ACTREF/ABC*
```

SESLIMIT=nnnnnnnn

Specifies the maximum number of sessions for which you want responses. If not specified, SESMGET will attempt to provide data for all sessions specified by the other parameters, except when invoked through a web browser, in which case it defaults to 200.

NETLOG**NO**

Indicates that the output messages are not put in the network log. This is the default.

YES

Indicates that the output messages are put in the network log.

Successful response from a SESMGET command

A successful response to a SESMGET command is derived from the session monitor Session List panel (NLDM.SESS) in the same order as the corresponding NLDM.SESS entries. The response is returned as a multiline message consisting of one AAU975I message, followed by one or more AAU976I messages. See the message help for AAU975I and AAU976I.

When invoked from a web browser, these messages are presented on a web page that consists of a list of the specified sessions. The configuration information is formatted from AAU978I messages, which contain most, but not all, of the configuration information known to session monitor. You can view the configuration of any session. The primary resource name, session start time, and PCID fields are all selectable. If there is a sense code, it is also selectable.

Example: SESMGET command

Enter:

```
SESMGET RES1=NTVFE SELECT=ALL
```

Requests a data message for each session known to the local session monitor for which NTVFE is an endpoint. The response is similar to the following messages, which are displayed as one multiline message:

Response:

Message 1:

```
|---+---1---+---2---+---3---+---4---+---5---+---6---+---7--
AAU975I NTVAA
```

Message 2:

```
|---+---1---+---2---+---3---+---4---+---5---+---6---+---7--
AAU976I USIBMNT.NTVFE LU USIBMNT.NTVE8 LU 081596
--+---8---+---9---+---10---+---11---+---12---+---13---+---14---
054255INITF USIBMNT.NTFEMVS.F56B5A6E977D7FC7 087D0001 C
```

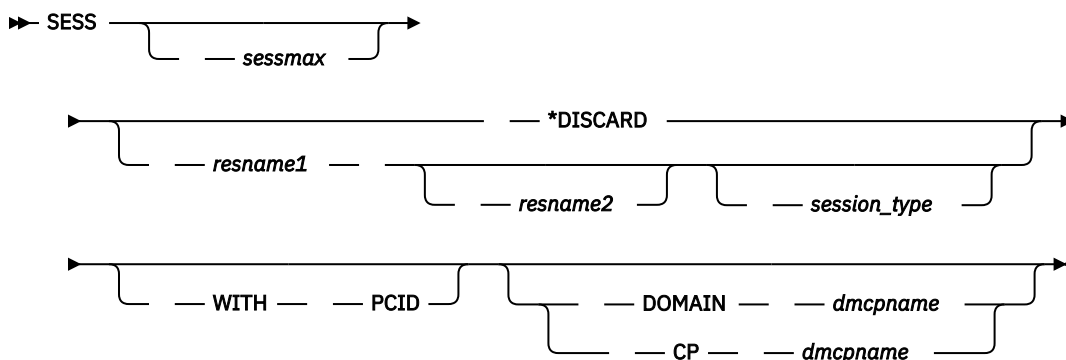
Message 3:

```
|---+---1---+---2---+---3---+---4---+---5---+---6---+---7--
AAU976I USIBMNT.NTVFE LU N/A USIBMNT.NTAAL703 LU NTVAATOV081496
--+---8---+---9---+---10---+---11---+---12---+---13---+---14---
154259081496154259USIBMNT.NTFEMVS.F5B31C2E7EE86147 D
```

SESS (NLDM; CNME2004)

Syntax

SESS



Purpose of Command

The SESS command lists sessions that match the specified operands.

This command displays the Session List panel.

Operand Descriptions

sessmax

Specifies the maximum number of sessions to be displayed. If not specified, the NLDM.SESSMAX setting specified in the CNMSTYLE member is used. If this limit is reached, message DWO979I is displayed.

Note: If *resname1* and *resname2* both contain wild cards, and the *sessmax* limit is reached while collecting inactive sessions, the inactive sessions displayed are not necessarily the most recent.

**DISCARD*

Selects the **DISCARD* pseudosession (the session to which all discarded PIUs are associated).

resname1

Selects only those sessions in which one of the session partners is indicated by *resname1*. *resname1* must be 1 - 8 alphanumeric characters. You can use an asterisk (*) as a wildcard character at the end of the resource name or as the only specified character.

resname2

Selects only those sessions between *resname1* and *resname2*. *resname2* must be 1 - 8 alphanumeric characters. You can use an asterisk (*) as a wildcard character at the end of the resource name or as the only specified character.

session_type

Selects the following sessions for the resource or resource pair:

ALL

All sessions. This is the default.

ACT

Only active sessions.

ACTREF *resname3*

Active sessions that traverse a given resource, as known by the local VTAM. The specified name must directly follow the ACTREF keyword. It is matched against those resource names on the local session configuration panel (NLDM.CON series) that were provided by the local VTAM, for example an RTP PU name. It is not matched against CP names as seen on APPN route panels. You can use an asterisk (*) as a wildcard character at the end of the resource name.

This option requires that *resname2* be specified.

ACTREFCP *resname3*

Active sessions that traverse a given resource, as known by the local VTAM. The specified name must directly follow the ACTREFCP keyword. It is matched against those resource names on the local session configuration panel (NLDM.CON series) or APPN route panels (NLDM.AR or NLDM.OAR) that were provided by the local VTAM. You can use an asterisk (*) as a wildcard character at the end of the resource name.

This option requires that *resname2* be specified.

INACT

Only inactive sessions.

ERR

Only inactive sessions that ended in error (bind-failure, initialization failure, or unbind with reason code greater than 2).

sensecd

Only inactive sessions that ended with the specified sense code or set of sense codes. The parameter *sensecd* consists of up to 8 characters, each of which is either a hexadecimal character or a question mark (?); the question mark represents any hexadecimal character. This option requires that *resname2* be specified. An asterisk (*) can be used.

WITH PCID

Displays the fully qualified procedure-correlation identifier (FQPCID) associated with each session.

dmcpname

The domain or CP name of the NetView program for which the list of sessions is requested. The *dmcpname* must be preceded by either DOMAIN or CP to indicate whether it is a domain or a CP. An NLDM SDOMAIN command is driven internally, followed by the remainder of the SESS command. If the SDOMAIN command fails, the SESS command is not attempted, and the NLDM MENU screen is displayed and might contain an error message. For additional information, see [“SDOMAIN \(NLDM\)” on page 195](#).

Restrictions

The following restrictions apply to the SESS command:

- Session monitor trace data cannot be viewed for sessions that are active unless the session monitor trace is active for those sessions.
- This command requires TOWER NLDM to be defined in the CNMSTYLE member.

Example: Displaying a list of active and inactive sessions for a specified resource

To display a list of active and inactive sessions associated with resource L51R79M1 enter:

```
SESS L51R79M1
```

Example: Displaying a list of active and inactive sessions between specified resources

To display a list of active and inactive sessions between the resources LCL3278A and L51R79M, along with their PCIDs, enter:

```
SESS LCL3278A L51R79M WITH PCID
```

Example: Displaying a list of active sessions between resources with wildcards

To display a list of active sessions between resources beginning with LCL and resources beginning with L5, enter:

```
SESS LCL* L5* ACT
```

Example: Displaying a list of inactive sessions between specified resources

To display a list of inactive sessions between resources L51R79M1 and AP01, along with their PCIDs, enter:

```
SESS L51R79M1 AP01 INACT WITH PCID
```

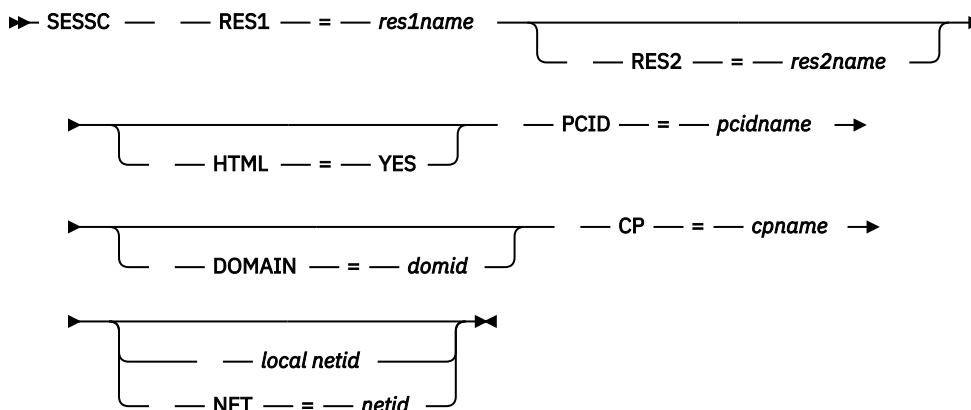
Example: Displaying a list of inactive sessions between specified resources on another NetView domain

To display a list of inactive sessions between resources L51R79M1 and AP01, along with their PCIDs, on NetView domain NTVAA, enter:

```
SESS L51R79M1 AP01 INACT WITH PCID DOMAIN NTVAA
```

SESSC (NLDM; CNME2012)

SESSC



Purpose of Command

The SESSC command displays session monitor configuration data.

Operand Descriptions

RES1=res1name

Specifies the resource for which you want session configuration data. If RES2 is not present, the list of sessions searched internally for a matching PCID are those for which *res1name* is either the primary or secondary endpoint.

RES2=res2name

Specifies the name of the second endpoint. When present, the list of sessions searched internally for a matching PCID are those for which *res1name* and *res2name* are the endpoints.

HTML=YES

Specifies that the output is HTML. This is used by NMC and is not needed for web browser requests.

PCID=pcidname

Specifies the PCID (or partial PCID) which is matched against the session list described by the endpoint or endpoints. The first one that matches is the one for which the configuration is gathered.

DOMAIN=domid

Specifies the NetView domain from which the session monitor collects the data. When neither DOMAIN nor CP is present, the local domain is used. DOMAIN and CP are mutually exclusive.

CP=cpname

Indicates the name of the CP or SSCP associated with the NetView domain from which session monitor collects the data. DOMAIN and CP are mutually exclusive. If CP is specified without NET, the default value is the local network.

NET=netid

Specifies the network in which *cpname* is defined.

Example: Request configuration for a session

The following example requests configuration information for a particular session:

```
SESSC RES1=NTVFELUC RES2=NTVAALUC PCID=F56B5A6E489CFFBB
```

For an example of a successful response, see the help for message AAU978I.

SESSDGRP (NLDM)

Syntax**SESSDGRP**

```
➤ SESSDGRP — — dgroup_name ————— WITH — — PCID —
```

Purpose of Command

The SESSDGRP command displays session history (on the NLDM.SESS panel) for all sessions that belong to the specified direct access storage device (DASD) group name.

Operand Descriptions**dgroup_name**

Specifies the DASD group name for which the session history of all sessions in the DASD group is listed.

WITH PCID

Displays the fully qualified procedure-correlation identifier (FQPCID) associated with each session.

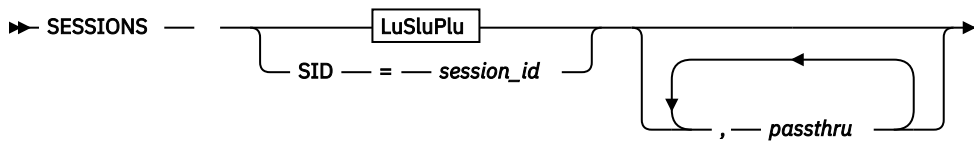
Restrictions

Define the DASD group name in the KEEPMEM initialization member with the DGROUP operand of the KCLASS statements.

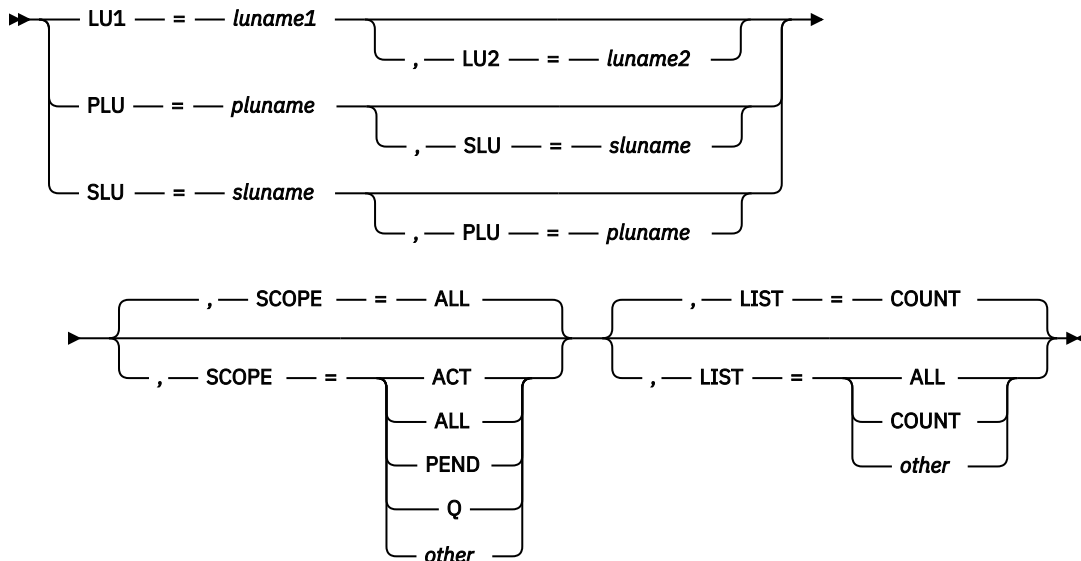
SESSIONS (NCCF; CNME0048)

Syntax

SESSIONS



LuStuPlu



Purpose of Command

The **SESSIONS** command list displays session status information. To display sessions between specified logical units (LUs), one of the session partners must reside in the host VTAM network.

Operand Descriptions

LU1=luname1

Specifies the LU for which sessions are to be displayed. You can specify *luname1* as a network-qualified name. If you specify **LU2**, only sessions between **LU1** and **LU2** are displayed. Do not specify the **PLU**, **SLU**, and **SID** operands with the **LU1** operand.

LU2=luname2

Specifies the LU for which sessions are to be displayed. Specify *luname2* as a network-qualified name. If you specify **LU1**, only sessions between **LU2** and **LU1** are displayed. You cannot specify the **PLU**, **SLU**, and **SID** operands with the **LU2** operand.

PLU=pluname

Specifies the logical unit that is the primary session partner. Specify *pluname* as a network-qualified name. If you also specify **SLU**, only sessions in which the **PLU**'s *pluname* is the primary logical unit and the **SLU**'s *pluname* is the secondary logical unit are displayed. You cannot specify the **LU1**, **LU2**, and **SID** operands with the **PLU** operand.

SLU=sluname

Specifies the logical unit that is the secondary session partner. You can specify *sluname* as a network-qualified name. If you also specify **PLU**, only sessions in which the **PLU**'s *sluname* is the primary logical unit and the **SLU**'s *sluname* is the secondary logical unit are displayed. You cannot specify the **LU1**, **LU2**, and **SID** operands with the **SLU** operand.

SCOPE

Specifies the status of the sessions to be displayed. You cannot issue the SID operand with the SCOPE operand. Parameters are:

ALL

Displays all sessions, regardless of session status. ALL is the default.

ACT

Displays only active sessions.

PEND

Displays only pending sessions.

Q

Displays only queued sessions.

other

Specifies a value for the SCOPE keyword used by the VTAM DISPLAY SESSIONS command.

LIST

Specifies the amount of detail to be displayed. You cannot issue the SID operand with the LIST operand. Parameters are:

COUNT

Displays the total number of sessions whose status has been specified by the SCOPE operand. COUNT is the default.

ALL

Displays all session status information for those sessions whose status has been specified by the SCOPE operand.

other

Specifies a value for the LIST keyword used by the VTAM DISPLAY SESSIONS command.

SID=*session_id*

Specifies the VTAM session to be displayed. This *session_id* cannot be more than 16 characters long. You cannot specify the LU1, LU2, PLU, SLU, SCOPE, and LIST operands with the SID operand.

passthru

Specifies additional parameters which are appended unchanged to the VTAM DISPLAY command issued by the SESSIONS command. You can specify up to 6 additional parameters on the SESSIONS command. No validation for duplicate or conflicting parameters is performed.

Restrictions

The following restrictions apply to the SESSIONS command:

- If you specify LU1 and LU2, only sessions involving both LUs are displayed.
- If you specify PLU and SLU, only sessions involving both named LUs in the primary/secondary relationship are displayed.
- This command list is supported by VTAM Version 3 Release 2 and later releases only.

Return Codes

Return Code	Meaning
-------------	---------

0	Functioned normally
---	---------------------

Example: Displaying pending sessions for a specified LU

To display pending sessions for a specific LU (ECHO01), enter:

```
SESSIONS LU1=ECHO01,SCOPE=PEND,LIST=ALL
```


Response

If the command list has processed successfully, the following message is displayed:

```
IST873I          PLU          SLU          SID          STATUS
IST874I NETA.ECH001  NETA.ECH0099  DF279FE40944400D  PSEST
IST874I NETA.ECH001  NETA.ECH0099  DF279FE40944400E  PSEST
IST874I NUMBER OF PENDING SESSIONS  2
IST314I END
```

Example: Displaying number of active sessions in which ECH099 Is the primary partner

To display the number of active sessions in which ECH099 is the primary partner, enter:

```
SESSIONS PLU=ECH099,SCOPE=ACT,LIST=COUNT
```

Example: Displaying active sessions in which A01A741 is the secondary partner

To display the active sessions in which A01A741 is the secondary partner, enter:

```
SESSIONS SLU=A01A741,SCOPE=ACT,LIST=ALL
```

SESSMDIS (NCCF)

Syntax

SESSMDIS



Purpose of Command

The SESSMDIS command displays session monitor session counts, storage use, and traffic rates.

You can use the SESSMDIS command to monitor and tune session monitor storage and VSAM usage. You can also use this command to determine the size of the network (as known by the session monitor) by using the session counts.

The following information is displayed:

- Session monitor options in effect:
 - SAW (Yes/No)
 - LU trace (Yes/No)
 - CP/SSCP trace (Yes/No)
 - SESSTATS (Avail/Yes/No)
- Session counts (current and high water marks):
 - CP - CP
 - SSCP - SSCP
 - SSCP - PU
 - SSCP - LU
 - LU - LU
 - Filtered

Note: Filtered session counts reflect the number of sessions filtered by VTAM and the session monitor.

- Session monitor storage use:
 - Resource storage
 - Session storage
 - Session parameter storage
 - PIU trace storage
 - SESSTATS storage
 - RTM storage
 - RSCV storage
 - OTHER storage (internal control blocks, work storage, and others)
 - TOTAL storage
- VSAM recording queue (current and high water mark):
 - Number of sessions waiting to be recorded to VSAM.
- Session monitor workload (total and 4-second rate):
 - Number of SAW buffers sent from VTAM to the session monitor.
 - Number of session starts sent from VTAM to the session monitor.
 - Number of session ends sent from VTAM to the session monitor.
 - Number of PIU trace buffers sent from VTAM to the session monitor.
 - Number of sessions recorded to the session monitor.

Note: These totals represent the values since the session monitor was started or the session monitor record STRGDATA command was entered.
- Total current explicit routes (SARTs). This number is intended as a guide in setting the value of NLDM.ERCOUNT in the CNMSTYLE member.

The above data is output in multiline message DSI378I. If SESSMDIS is invoked in a full-screen environment, this message is displayed through the WINDOW command. Use the WINDOW REFRESH subcommand (typically F2) to refresh the data. Each REFRESH will also log the new data if it was logged the first time. See the descriptions for the LOG and NOLOG keywords.

For system and subsystem consoles, in multiline messages, the title line is truncated at 34 characters, while the remaining lines are truncated at 68 characters.

Operand Descriptions

LOG

Log the resulting message lines in the network log. This is the default.

NOLOG

Do not log the resulting message lines.

Restrictions

The following restrictions apply to the SESSMDIS command:

- The AAUTSKLP data services task must be active.
- SESSMDIS does not run on the primary program operator interface task (PPT).

Example: Displaying session counts, storage usage, and statistics

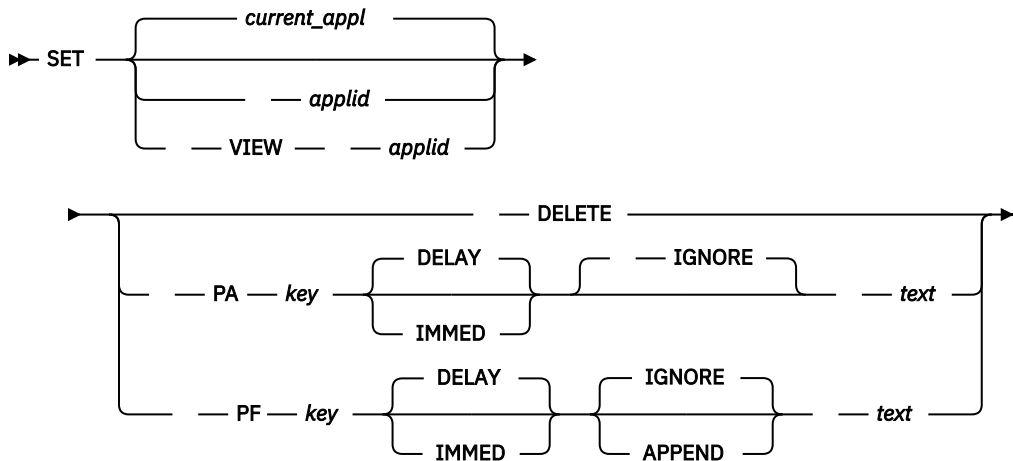
To display the session counts, storage usage, and statistics for the session monitor, enter:

```
SESSMDIS
```

SET (NCCF)

Syntax

NCCF SET



Purpose of Command

The NCCF SET command defines PA and PF keys for the command facility or a full-screen application that supports its PF or PA settings. These settings remain valid until you delete them or log off.

Operand Descriptions

applid

Indicates the application to which the specified PF or PA key setting applies. Application IDs that are supplied with the NetView program and that support their own PA and PF keys are LBROWSE, MAINMENU, MBROWSE, NCCF, NETVIEW, NLD, NPDA, STATMON, VIEW, and WINDOW. Also, other applications using the first parameter on the VIEW command to specify an application name can have their own PF key settings, which can be specified using SET.

If *applid* is omitted from the SET command, the default value is the current application, or if that one is neither one of the above nor an application ID specified using DSIPSS, the default value is NCCF.

DELETE

The system deletes all PA and PF key definitions for the specified *applid*.

PAkey

Specifies which program attention key you want to set. You can specify 1-3 for PA keys. (Your keyboard might not have that many PA keys. Some keyboards have only 2 PA keys.) When you type the key number, do not leave a space between the PA and the number.

PFkey

Specifies which program function key you want to set. You cannot set PF keys that are set to a command module statement. You can specify 1 - 24 for PF keys. (Your keyboard might not have that many PF keys. Some keyboards have only 12 PF keys.) When you type the key number, do not leave a space between the PF and the number.

DELAY

Indicates that the command is written into the input area of the screen with a blank space between the last character and the cursor. The command runs when you press the ENTER key. This enables you to modify the command before it is started. DELAY is the default.

IMMED

The system will run this command immediately when you press the key. Use IMMEDIATE for commands you want to enter the same way each time. Do not use IMMEDIATE to assign a PA key to a command that is

sensitive to the position of the cursor, because the cursor position is unknown when a PA key is pressed.

IGNORE

Specifies that the input area for this key is to be ignored. Type IGNORE after the PF or PA key and before the command text. IGNORE is the default.

APPEND

Appends the data in the input area to the end of the command text from the SET command when you press the PF key.

text

Specifies the command or command list for the PF or PA key. A comma or blanks must come before the message text. All characters after the comma or blanks are considered part of the text. You can use blanks and commas in the text. Quotation marks (") or apostrophes (') are not required; if they are in the text, they are treated as part of the text.

Restrictions

The following restrictions apply to the SET command:

- For migration, LOG is supplied as a PARMSYN for LBROWSE so that many existing SET LOG definitions continue to work, although some abbreviated key texts previously supported by SET LOG must be updated (H, HL, HLP, HE, HEL, E, EN, RO, ROL, BC, BCK, F, RP*, REPEATFIN*, LF, LFT, RGT, RG, ALL, ONE). Other applications that use the first parameter on the VIEW command to specify an application name, for example MAINMENU and ACTION command lists, can have their own key settings, which are set by the SET command. For such VIEW applications, add the VIEW keyword before the application ID on the SET command so that the LIST KEY command can list VIEW defaults for this applid. Set default keys by using application ID NETVIEW; set default keys for VIEW applications by using application ID VIEW.
- If you define a key for a specific application ID, the definition takes effect for that application. Otherwise, if that application uses VIEW to output its panels, and a given key is defined for application ID VIEW, the VIEW definition takes effect. If the key is defined for application ID NETVIEW, the NETVIEW definition takes effect. If you press a key that is not defined in this hierarchy, a message is displayed.
- If you attempt the SET command on an NetView-NetView task (NNT) or the primary program operator interface task (PPT), a warning message is issued and no action is taken. The NNT and PPT do not have terminals, and therefore do not have PF or PA keys to set.
- PA keys cannot send data; therefore, do not use the PA key and the APPEND operand together.
- If you start a cross-domain session and specify a profile that has an initial command or command list, a SET command might unintentionally be attempted. An initial command list checks the task before attempting SET.

Return Codes

Return Code	Meaning
--------------------	----------------

0	Operation successful if not accompanied by a storage error message.
4	Syntax error.
12	NetView service failure, for example, storage.

Example: Setting PF12 to retrieve your last command

To set the PF12 key to retrieve your last command, enter:

```
SET PF12 IMMED RETRIEVE
```

Response

Press PF12 while in any application for which you have not set PF12 to something else. The previous command is placed in the command area.

Example: Setting PA2 to write a partial command to the command line

To set PA2 to write the partial command ID=MODEM1, STATION=JAT, BROWSE=NOCHANGE to the command line, enter:

```
SET PA2 ID=MODEM1, STATION=JAT, BROWSE=NOCHANGE
```

Response

Press PA2 while in NCCF. The cursor is positioned immediately after the last character of the command text. You can then modify the command before pressing ENTER to process it.

Example: Setting PF10 to run the SET HEX command immediately

To set PF10 to process the SET HEX command immediately, enter:

```
SET NCCF PF10 IMMED APPEND SET HEX
```

Response

Press PF10 while in NCCF. Data from the command input line is appended to the end of the command. For example, if either ON or OFF were appended, then pressing PF10 enters SET HEX ON or SET HEX OFF.

Example: Setting PF2 to end NPDA

To set PF2 to run the END command from NPDA, enter:

```
SET NPDA PF2 IMMED END
```

Response

Pressing PF2 while in NPDA causes you to exit from NPDA.

Example: Setting PF14 to FIND a user-entered string

To set PF14 to run FIND followed by your command line input, enter:

```
SET MBROWSE PF14 IMMED APPEND FIND  
SET LBROWSE PF14 IMMED APPEND FIND
```

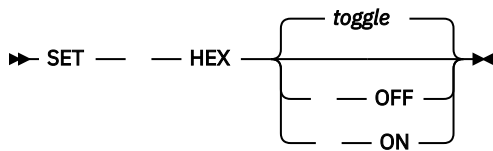
Response

Pressing PF14 while in Member-browse or Log-browse causes your command line input to be appended to FIND and processed.

SET HEX (NLDM)

Syntax

NLDM SET HEX



Purpose of Command

The SET HEX command sets hexadecimal display mode on or off.

Operand Descriptions

toggle

If you do not specify ON or OFF, the SET HEX command toggles between the current setting and the reverse setting.

OFF

Specifies that hexadecimal mode is to be turned off.

ON

Specifies that hexadecimal mode is to be turned on.

Restrictions

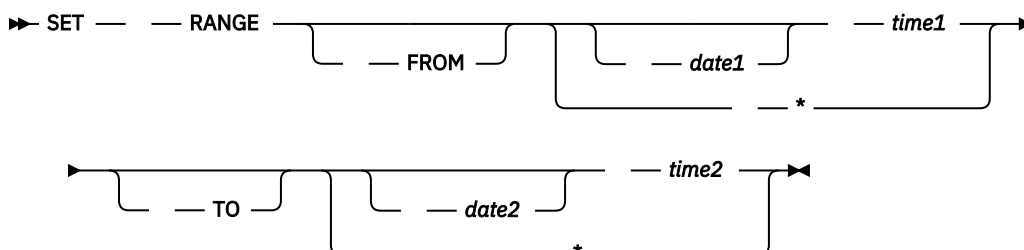
The following restrictions apply to the SET HEX command:

- When hexadecimal mode is on, the trace data and the session activation parameters are displayed in hexadecimal. When hexadecimal mode is off, the trace data and the session activation parameters are displayed as text.
- This command affects the following panels only:
 - PIU Trace Data
 - NCP Session Trace Data
 - Session Parameters
- The hexadecimal mode setting remains in effect until you leave the session monitor with the END command. Hexadecimal mode is turned off the next time you enter the session monitor.

SET RANGE (NLDM)

Syntax

NLDM SET RANGE



Purpose of Command

The SET RANGE command sets the default time range used for all session monitor RTM commands that display data by time.

Operand Descriptions

FROM

Identifies the operands that follow as the starting date and time. This operand is optional.

Uses the default time and date for the date display command being issued.

date1

Specifies the starting date of the time range. If you specify the time but omit the date, the date defaults to the current date. The format of *date1* is controlled by the setting of the date operands of the DEFAULTS and OVERRIDE commands.

time1

Specifies the starting time of the time range. The format of *time1* is controlled by the setting of the time operands of the DEFAULTS and OVERRIDE commands.

TO

Identifies the operands that follow as the ending date and time. This operand is optional.

date2

Specifies the ending date of the time range. The format of *date2* is controlled by the setting of the date operands of the DEFAULTS and OVERRIDE commands. If you specify the time but omit the date, the date defaults to the current date.

time2

Specifies the ending time of the time range. The format of *time2* is controlled by the setting of the time operands of the DEFAULTS and OVERRIDE commands.

Restrictions

The time range setting remains in effect until you leave the session monitor with the END command. The time range returns to its default setting the next time you enter the session monitor. See *IBM Z NetView Installation: Getting Started* for a description of the default setting.

Example: Setting a default range of 8 a.m. to 5 p.m.

The format of times specified in the following example assumes the default setting for time formats on the DEFAULTS and OVERRIDE commands.

To set a default range of 8 a.m. to 5 p.m., enter:

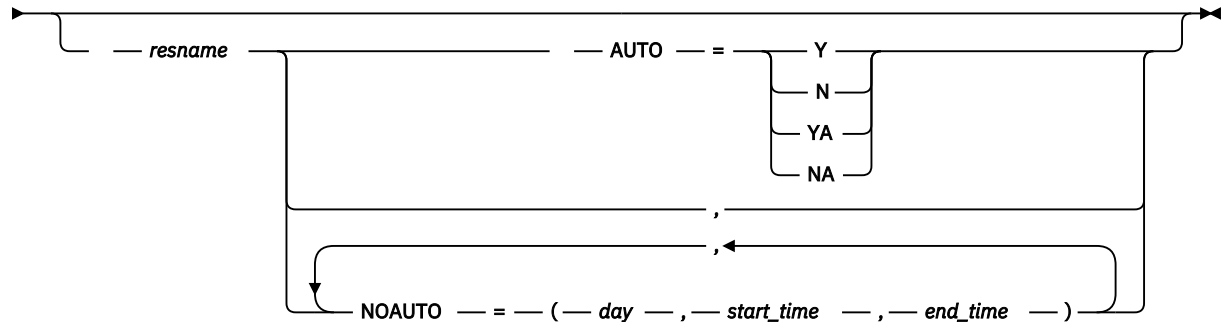
```
SET RANGE FROM 08:00 TO 17:00
```

SETAUTO (AON)

Syntax

SETAUTO

➡ SETAUTO ➡



Purpose of Command

The SETAUTO command sets recovery automation for a specific resource or a group of resources. To use the SETAUTO command from the operator interface, see *IBM Z NetView User's Guide: Automated Operations Network*.

Operand Descriptions

resname

The name of the resource for which you are changing automation settings. The resource name can be the name of one resource, DEFAULTS, or a generic resource type. If you specify DEFAULTS, the settings affect all resources that do not have other settings explicitly set for them. You can set automation for entire groups of resources by type such as LU, PU, LINE, and NCP, or for entire days such as MONDAY, HOLIDAY, or JAN/1/2011. You can also use the wildcard characters * and %. The most specific setting coded for a resource issued to control whether automation recovery is on or off for the resource.

AUTO

Defines whether automation is allowed for the resource. Use the following values with the AUTO keyword to determine whether automation is running:

Y

Sets recovery on.

N

Sets recovery off.

YA

Sets recovery on for the specified resource and its lower nodes.

NA

Sets recovery off for the specified resource and its lower nodes.

Note: YA and NA are valid only for SNA resources that do not contain wildcard characters (* and %). If entered, recovery is set on or off, but the AON ignores the lower nodes.

NOAUTO

Defines specific times when automation does not occur. Contrast with AUTO=N, which sets automation to off all the time. Define all of the following values when you use the NOAUTO keyword:

day

Specifies the days when recovery is set to off and is one of the following:

MONDAY or MON
TUESDAY or TUE
WEDNESDAY or WED
THURSDAY or THU
FRIDAY or FRI
SATURDAY or SAT
SUNDAY or SUN

Month/date_spec/year (examples: JAN/1/2000 or JAN/LAST-30/2000)

Month/day_spec/year (examples: JAN/SAT/1ST/2000 or JAN/SAT/LAST-4/2000) In this case SAT cannot be SATURDAY because only three digits are allowed.

Special_day_name (example: My_Birthday or HOLIDAY or NEW_YEARS_DAY)

Use an asterisk (*) to set automation to off during some interval every day.

start_time

Sets automation to off starting at this time. Specify the time in the 24-hour format *hh:mm* where *hh* is a number 00 - 23 and *mm* is a number 00 - 59.

end_time

Determines the end of the interval when automation is not active for the resource. Specify the time in the 24-hour format *hh:mm* where *hh* is a number 00 - 23 and *mm* is a number 00 - 59.

You can specify up to five settings for NOAUTO to begin. To enter more NOAUTO settings, press **F10** to save the settings, then press PF8 to add more settings.

Usage Notes

- The AON tower must be enabled in the CNMSTYLE member to successfully run this command.
- You must specify an AUTO or NOAUTO keyword on the SETAUTO command. If you do not specify AUTO, you must schedule at least one NOAUTO interval for the resource.
- It is possible for automation to always be inactive for a resource even if you specify AUTO=Y or AUTO=YA, if you also specify NOAUTO=*,00:00,23:59. Pay close attention to the intervals you specify when you use the NOAUTO keyword.
- This command can be issued in line mode if all parameters are correctly entered. Therefore, you can issue it from within your own routines. If no parameters are specified, a full-screen interface is displayed.

Examples

To have automation active for the resource, TA1TT167, every day *except*:

- Saturday and Sunday from 8:00 a.m. to 10:00 a.m.
- Every day from 3:00 p.m. (15:00) to 5:00 p.m. (17:00)

issue:

```
SETAUTO TA1TT167 AUTO=Y,NOAUTO=(WEEKEND,08:00,10:00),  
NOAUTO=(*,15:00,17:00)
```

To set automation on for all NCPs *except* during the interval between 1:00 p.m. and 2:00 p.m. on Saturday and Sunday, issue:

```
SETAUTO NCP NOAUTO=(WEEKEND,13:00,14:00)
```

SETBQL (NCCF)

Syntax

SETBQL

➤ SETBQL — — *receiver_id* — — *qlimit* ➤

Purpose of Command

The SETBQL command resets a receiver's buffer queue limit.

Operand Descriptions

receiver_id

Specifies the receiver ID that was defined to the NetView program-to-program interface.

qlimit

Specifies a new buffer queue limit in the range of 0–4294967295.

Restrictions

The following restrictions apply to the SETBQL command:

- The new buffer queue limit can be up to 10 digits.
- You can use the SETBQL command to adjust the buffer queue limit for the NetView alert receiver (*receiverid* is NETVALRT).
- You can use the SETBQL command on the NetView procedure that has the same first 4 characters as the NetView application.
- You can use the DISBQL command to display information about receivers.
- If the NetView subsystem interface (SSI) is inactive, the NetView program-to-program interface is also inactive. You receive message CNM563I as well as the information about all the receivers. If you recycle the subsystem address space and the NetView program-to-program interface is started, the receivers that were previously defined in the NetView program-to-program interface are still defined, but the receivers' buffer queue is lost.

Return Codes

Return Code

Meaning

0

Successful processing

8

Error in processing

Example: Resetting the buffer queue limit for CNMRCV

To reset the buffer queue limit for CNMRCV, enter:

```
SETBQL CNMRCV 100
```

SETCGLOB (NCCF; CNME1081)

Syntax

SETCGLOB

➤ SETCGLOB — — *varname* — — TO — — *value* ➤

Purpose of Command

Use SETCGLOB to set the value of the specified common global variable.

Operand Descriptions

varname

Specifies the common global variable for the value which is updated.

value

Specifies the new value assigned to the specified common global variable. A null value (blank) is acceptable.

Restrictions

The following restrictions apply to the SETCGLOB command:

- The use of SETCGLOB is limited to the command procedure and automation environments (command must originate in REXX, HLL, automation, or an optional task). Use of SETCGLOB directly from the operator's command line results in message DSI290I and return code 8.
- The SETCGLOB command list sets a common global variable to any value. This command list is appropriate to use in any situation where it is not important to serialize the access between multiple tasks. If serialization of updates is important, use PIPE VARLOAD. If the value is numeric, you can use the UPDCGLOB command list. The *value* cannot be longer than 255 characters.

Return Codes

0

The common global variable was set as requested.

8

The operator who issued the command list is not authorized to use SETCGLOB.

16

No variable name was specified or no value was specified.

Example: Setting the NetView common global variable &TASKCOUNT to 1

To set the NetView common global variable &TASKCOUNT to a value of 1, enter:

```
SETCGLOB TASKCOUNT TO 1
```

SETCONID (NCCF)

Syntax

NCCF SETCONID

➤ SETCONID — — CONSOLE — = — *consolename* ➤

Purpose of Command

The SETCONID command enables you to change the default console name used by an operator or autotask for submitting MVS commands. See the CNMSTYLE member for a description about how the default console name is determined. Use SETCONID in the initial command list for an operator or an autotask. Program the initial command list to choose a unique console name based upon the naming convention you decide to use.

Unlike the GETCONID command, SETCONID does not allocate the console when the CLIST runs. Instead, it sets the name of the console that is later used when the GETCONID command is entered with no console name specified, or if an MVS command is issued by the operator or autotask and a console is already allocated. SETCONID is similar to setting the console name in the user profile using the CONSNAME keyword (when the value of ConsMask in the CNMSTYLE member is the asterisk (*) character or is not specified). To display the current value for this console name, use the LIST *operatorid* command.

Operand Descriptions

consolename

Specifies the default console name for this operator or task. If an asterisk (*) character is specified, SETCONID resets the console name to the system default. This is derived from the CONSMASK value if that is in use or to the operator's ID, otherwise.

Return Codes

The following are return code values for the SETCONID command:

Return Code

Meaning

- | | |
|-----------|--|
| 0 | Completed successfully |
| 4 | Syntax error, operand in error |
| 24 | The value for console name is not valid |
| 44 | This task has already obtained a console |

Example: Issuing SETCONID

To issue the SETCONID command, enter:

```
SETCONID  CONSOLE=GOODNAME
```

The following message is received:

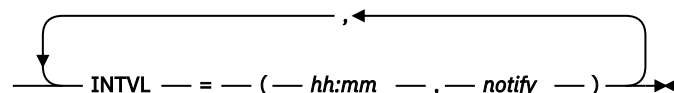
```
DSI633I SETCONID COMMAND SUCCESSFULLY COMPLETED
```

SETMONIT (AON)

Syntax

SETMONIT

➡ SETMONIT — — *resname* — — INTVL — = — (— *hh:mm* — , — *notify* —) — ➡



Purpose of Command

The SETMONIT command sets the intervals AON uses during recovery monitoring. At these recovery monitoring intervals, AON monitors the failed resource to see if it has recovered. For SNA resources, AON also attempts to reactivate the failed resource at each recovery monitoring interval. Finally, you use the SETMONIT command to specify whether AON sends messages to the notification operators at recovery monitoring intervals. To use the DELTHRES command from the operator interface, see *IBM Z NetView User's Guide: Automated Operations Network*.

Operand Descriptions

resname

The name of the resource to be recovered.

INTVL

The reactivation interval setting. The variables on this setting are:

hh:mm

The length of the interval between reactivation attempts expressed as hours (*hh*) and minutes (*mm*).

notify

The setting that determines whether messages are sent to the notification operators when AON attempts to reactivate the resource. The settings can be:

Y

Send messages to the notification operators.

N

Do not send messages to the notification operators.

YF

Send messages to the notification operators and repeat recovery monitoring at the last interval specified until the resource recovers or the control file is reloaded.

NF

Repeat recovery monitoring at the last interval specified, but do not send messages to the notification operators.

Usage Notes

- The AON tower must be enabled in the CNMSTYLE member to successfully run this command.
- Using the SETMONIT command, you can specify up to 12 recovery monitoring intervals for a resource.
- If the recovery monitoring intervals exist for the resource, the SETMONIT command replaces the existing intervals.
- Use the YF and NF settings on the last interval only. If you use YF or NF for any interval before the last one, AON ignores that interval.
- The SETMONIT command can be issued in line mode. Therefore, you can issue it from within your own routines.

Examples

To set recovery monitoring for all CDRMs, type:

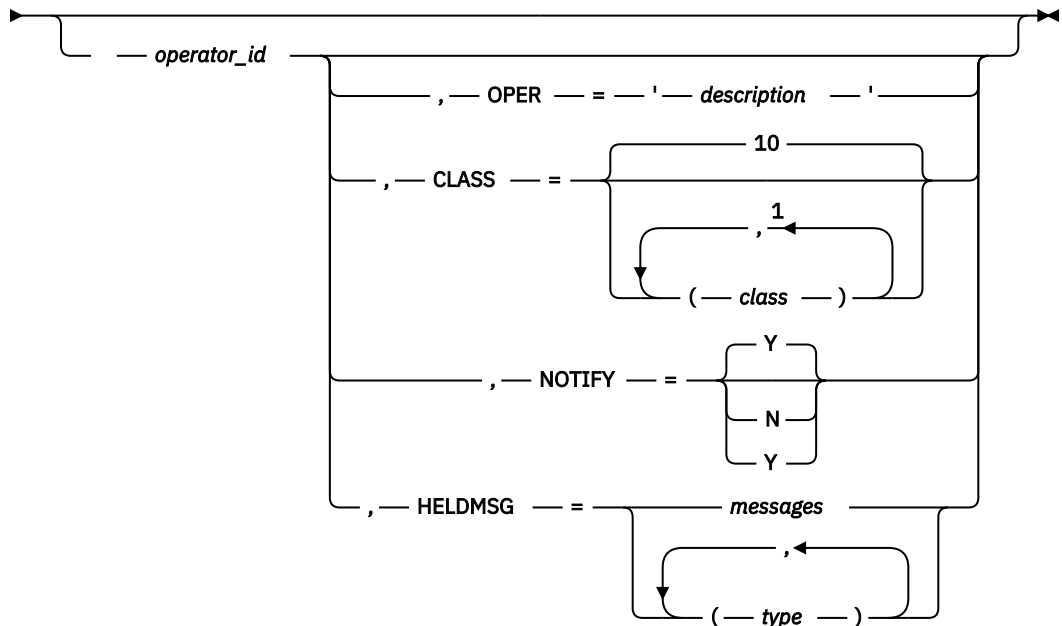
```
SETMONIT CDRM INTVL=(02:10,Y),  
              INTVL=(01:10,N),  
              INTVL=(01:20,Y),  
              INTVL=(01:45,YF)
```

SETNTFY (AON)

Syntax

SETNTFY

➤ SETNTFY ➡



Notes:

¹ You can specify up to 10 variables.

IBM-Defined Synonyms

Command or Operand	Synonym
SETNTFY	SETALERT

Purpose of Command

The SETNTFY command controls the settings for notification operators. With the SETNTFY command, you can add a notification operator, turn messages to the notification operator on and off without deleting the notification operator, select message classes for the notification operator, and determine whether automation messages sent to the notification operator's command facility are held. To use the SETNTFY command from the operator interface, see *IBM Z NetView User's Guide: Automated Operations Network*.

Operand Descriptions

operator_id

The operator ID of the notification operator.

CLASS

The classes of messages this operator is to receive. The parentheses are optional if you code one value for *class*, but are required if you code two or more values. You can define up to 10 message classes for an operator. Default class is 10.

NOTIFY

Specifies whether messages are actually sent to this notification operator. You can use this parameter to temporarily stop the notification messages without actually deleting the notification operator. The default is Y.

description

A brief description (20 characters or shorter) for the notification operator, which is usually the name of the operator. You must enclose the description inside single quotation marks.

HELDMSG

The types of messages that are to be held on the notification operator's command facility. Held messages remain on the command facility until you clear them. You can specify multiple types, as follows:

I or INFO

Informational messages

W or WARN

Warning messages

E or ERROR

Error messages

A or ACTION

Action messages

Usage Notes

- The AON tower must be enabled in the CNMSTYLE member to successfully run this command.
- You can specify the keywords CLASS=, OPER=, NOTIFY=, and HELDMSG= in any order. Use only the keywords you need.
- You must put the parameters for the HELDMSG keyword in parentheses, for example, HELDMSG=(I).
- The SETNTFY command can be issued in line mode if the resource name is provided. Therefore, you can issue it from within your own routines. If the resource name is provided with no other parameters specified, all defaults are used (CLASS=10, NOTIFY=Y).
- If no parameters are issued with this command, a full-screen panel is displayed showing all valid notification operators.

Examples

To add the operator ID, OPER1, to the list of valid notification operators using the default settings (CLASS=10, no description, no held messages), type:

```
SETNTFY OPER1
```

To hold all error messages for OPER1, type:

```
SETNTFY OPER1 HELDMSG=(E)
```

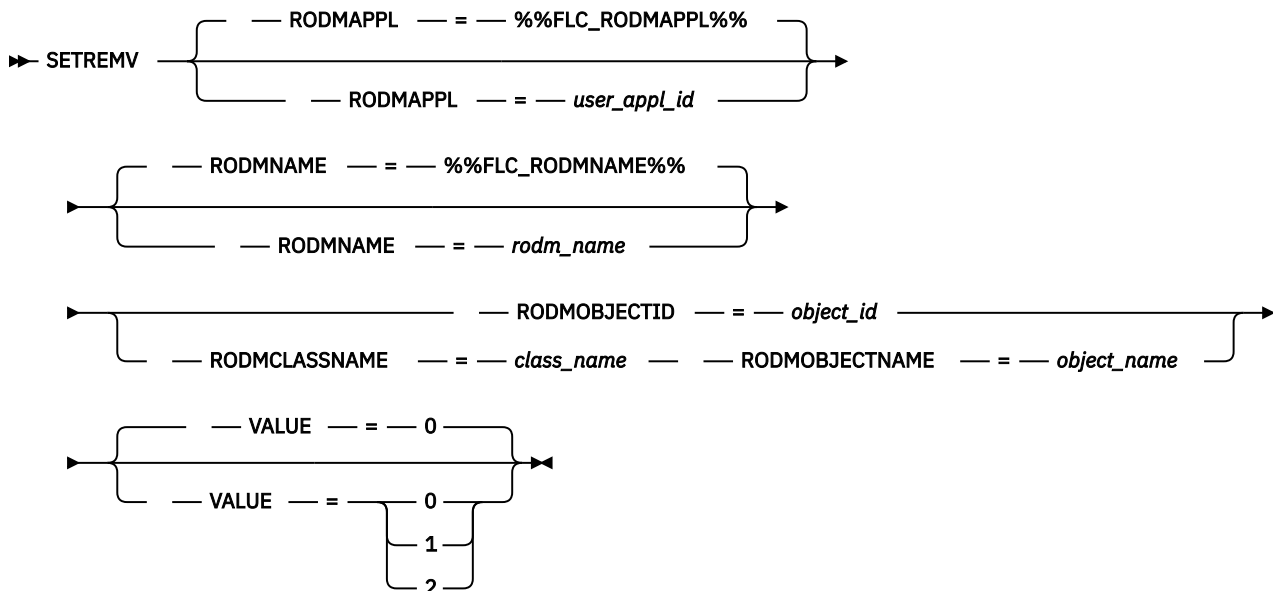
To set OPER1 as a valid notification operator, but specify that AON does not send notifications, type:

```
SETNTFY OPER1 CLASS=10,NOTIFY=N,HELDMSG=(I,E,W,A),OPER='J Smith'
```

SETREMV (MSM)

Syntax

SETREMV



Purpose of Command

The SETREMV command sets the value of an object's PURGE attribute in RODM to indicate whether the object and its links can be removed. You can issue this command to an aggregate or a real object. If you issue it to an aggregate object, the current PURGE attributes of the objects that make up the aggregate are not changed.

Operand Descriptions

RODMAPPL

The user application ID used to access the RODM that contains the topology and status data. The ID is case-sensitive and must be entered exactly as created with RACF.

The application ID for MultiSystem Manager is set during NetView initialization. Specify this keyword only if you are overriding the ID used by MultiSystem Manager, or if the `COMMON.FLC_RODMAPPL` statement in the CNMSTYLE member has not been coded.

RODMCLASSNAME

The RODM class name of the object that is the target of the command.

RODMNAME

The name of the RODM where topology and status information is stored. This is normally the RODM used by MultiSystem Manager. For MultiSystem Manager, this is the 1- to 8-character name that was specified on the `COMMON.FLC_RODMNAME` statement in the CNMSTYLE member. Specify this keyword only if you are overriding the ID used by MultiSystem Manager, or if the statement in the CNMSTYLE member has not been coded.

RODMOBJECTID

The RODM object ID (ObjectID) of the object. The object ID is 16 characters that represent an 8-byte hexadecimal field. An example object ID is: 00010027EC211161.

You can enter either the RODM object ID or the RODM class name and object name combination. MultiSystem Manager uses the object ID in the commands that it builds during command facility processing, because it can determine the ID based on the object you selected from the view. If you

are entering this command from the command line or an automated procedure, use the RODM class name and object name combination.

RODMOBJECTNAME

The RODM object name of the object that is the target of the command.

VALUE

The value for the object's PURGE attribute, indicating what you can remove. Valid values are:

0

You can remove the object and all its links from the views and RODM if all other criteria are met.

This value is valid for real or aggregate objects. Real objects are removed if they have a status of *unsatisfactory* or *unknown* for the time specified in the REMVOBJS command. Aggregate objects are removed if all real objects in them have been removed.

All MultiSystem Manager objects have an initial PURGE attribute of 0.

1

You cannot remove the object from RODM. However, any of the links that connect the object to an object with a PURGE attribute of 0 can be removed. For example, if an aggregate object has a PURGE attribute of 0, the real object with a PURGE attribute of 1 can be unlinked and removed from the view.

This value is valid only for real objects. If you specify a value of 1 for an aggregate object, MultiSystem Manager treats it as if it has a value of 2.

2

You cannot remove the object or its links from the views or RODM.

This value is valid for real and aggregate objects. If a REMVOBJS command is issued on aggregate objects, this value shields the aggregate and the real objects in it from removal. If an aggregate has a value of 2, the real objects in the aggregate are not removed, regardless of their PURGE attribute values.

If a PURGE attribute was never defined for an object, MultiSystem Manager treats the object as if it has a value of 2.

Usage Notes

The MSM tower must be enabled in the CNMSTYLE member to successfully run this command.

Examples

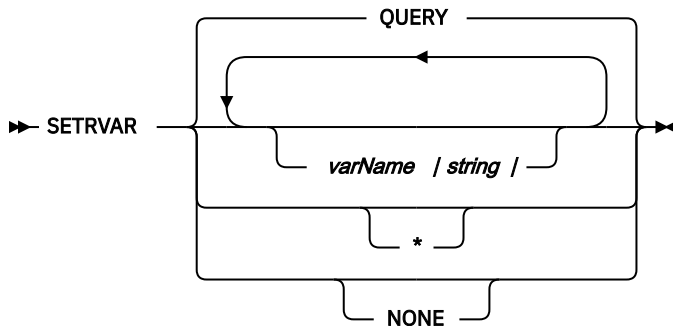
This example changes the PURGE attribute so that the object ID 00010027EC11161 and all of its links are subject to removal. The name of the RODM that is used by MultiSystem Manager is EKGXRODM and the application name is MYAPPL.

```
SETREMV  
RODMNAME=EKGXRODM,RODMAPPL=MYAPPL,RODMOBJECTID=00010027EC11161,VALUE=0
```

SETRVAR (NCCF)

Syntax

SETRVAR



Purpose of Command

The SETRVAR command creates a table of variable names and values that is accessible from revision edit scripts.

Operand Descriptions

QUERY

Generates a report showing the values of all revision variables currently defined. This is the default value.

varname

Specifies a variable name that can be 1 - 12 alphanumeric characters. The variable name is not case-sensitive.

/string/

Specifies a delimited string that can be 1 - 16 characters. This character string is case-sensitive. The character string is associated with the *varname* variable.

*

Indicates that variable names and values are read from the current message. In each line, the variable name is read from column 1 and the hex representation of the value is read from column 33. The variable name can be 1 to 12 alphanumeric characters, and is not case-sensitive. The value can be 2 to 32 hexadecimal characters, representing a 1- to 16-character value.

Note: Only line types of DATA and END are taken as name/value pairs. All other line types are ignored. Therefore, if the COLLECT pipe stage is used to build a multiline message as input to SETRVAR, use the FREEFORM option on COLLECT to allow all lines to be used as input to SETRVAR. See the examples below, as well as the examples in the CNMSRVAR sample, for examples of usage.

NONE

Causes the revision variable table to be deleted.

Usage Notes

- The subsystem router and the NetView SSI must be active when you use the SETRVAR command.
- If the table of variable names and values exists when you run the SETRVAR command, it is replaced.
- The *varname* variable is used by the RVAR edit order to retrieve the value specified in the associated *string*. The RVAR edit order can be used with a command or message revision table.
- To add, modify, or delete a single variable in the revision variable table without changing the other variables, see the CNMSRVAR sample.

- The most recent revision variable definitions remain active even if the NetView program is restarted. If the SSI address space is restarted while the NetView program is running, any active revision variable definitions that existed at the time of the restart are reinstated.

Return Codes

Return Code Meaning

410

Value of */string/* is too long

486

Syntax error, unsupported delimiter

496

Variable name too long

608

SSI or CNMCSSIR is not active

647

Duplicate variable name encountered

653

Value missing for the variable name

741

Internal error

Examples

Example 1: Replacing the revision variable table with a single variable.

Use the following command to define a single variable of SHIFT to have a value of NORMAL to be used in revision tables:

```
SETRVAR SHIFT /NORMAL/
```

Example 2: Creating a set of revision variables.

One way to create a set of revision variables and their values is to pass a REXX stem to SETRVAR as a multiline message. The following REXX code shows how this can be done:

```

rvars = 0
CALL AddRvar 'SHIFT NORMAL'
CALL AddRvar 'DAY' DATE('W')
CALL AddRvar 'DATE' DATE('U')
rvar.0 = rvars
'PIPE STEM RVAR.' ,
'| COLLECT FREEFORM' ,
'| CC SETRVAR *' ,
'| CONSOLE'
EXIT
AddRvar:
PARSE ARG RvarVar RvarVal
rvars = rvars + 1
rvar.rvars = LEFT(RvarVar,32) || ,
              C2X(RvarVal)
RETURN

```

SETTHRES (AON)

Syntax

SETTHRES

```
➤ SETTHRES — — resname { , — CRIT — = — ( — nn — , — hh — : — mm — ) —  
  , — FREQ — = — ( — nn — , — hh — : — mm — ) —  
  , — INFR — = — ( — nn — , — hh — : — mm — ) — } ➤
```

Purpose of Command

The SETTHRES command sets thresholds for a specific resource or a group of resources. When AON detects messages that indicate resource problems through passive monitoring, AON attempts to recover the failed resource. You can use the SETTHRES command to specify whether AON sends messages to the notification operators when the infrequent, frequent, or critical thresholds are reached. To use the SETTHRES command from the operator interface, see *IBM Z NetView User's Guide: Automated Operations Network*.

Operand Descriptions

resname

The name of the resource for which you are setting thresholds. The resource can be a specific resource name or, more frequently, a generic resource type (for example, DEFAULTS, LU, PU, CDRM, APPL, NCP). The three types of thresholds are:

CRIT

The critical threshold. AON stop reactivation attempts for the resource when the critical threshold is reached.

FREQ

The frequent threshold. The frequent threshold indicates that the resource is having errors often enough that action is required.

INFR

The infrequent threshold. The infrequent threshold provides early warning of intermittent resource errors.

The thresholds are defined as a number of errors within a given time span as follows:

nn

The number of errors before threshold is reached. The number must be 0 - 12.

hh:mm

The time span before threshold is reached (hours:minutes), where *hh* is a number 0 - 99 and *mm* is a number 0 - 59.

Usage Notes

- The AON tower must be enabled in the CNMSTYLE member to successfully run this command.
- AON uses the thresholds coded for the resource name, DEFAULTS, as the default threshold settings.
- AON requires thresholds settings for the resource, DEFAULTS. Therefore, AON does not let you delete the DEFAULTS settings.
- You can code all three thresholds values (CRIT, FREQ and INFR) on one invocation of the SETTHRES command.
- AON keeps date and time stamps for all failures of a resource in the status file, so threshold analysis is based on the actual time span between outages (without rounding off to the whole hour or minute).

- This command can be issued in line mode if all parameters are correctly entered. Therefore, you can issue it from within your own routines. For the command to be considered correct, you must specify at least one of the CRIT, INFR, or FREQ parameters with the resource name.
- If no parameters are specified, a full-screen interface is displayed.

Examples

To set the default thresholds settings, type:

```
SETTHRES DEFAULTS,CRIT=(2,00:14),FREQ=(2,01:00),INFR=(4,04:00)
```

To replace the critical threshold setting for the resource type, NCP, type:

```
SETTHRES NCP,CRIT=(2,02:00)
```

The values for the frequent and infrequent thresholds remain unchanged.

SETTINGS (EAS)

Syntax

EAS SETTINGS



IBM-Defined Synonyms

Command or Operand	Synonym
MODIFY	F

Purpose of Command

The SETTINGS command displays the current configuration settings for the requested Event/Automation Service task. The settings are displayed as console messages on the system console.

Operand Descriptions

procname

Specifies the Event/Automation Service job name.

TASK=taskid

Specifies the service task for which configuration settings are to be displayed. The *taskid* can have the following values:

ALERTA

The alert adapter service task.

ALERTC

The confirmed alert adapter service task.

MESSAGEA

The message adapter service task.

MESSAGEC

The confirmed message adapter service task.

EVENTRCV

The event receiver service task.

TRAPALRT

The trap to alert conversion task.

ALRTTRAP

The alert to trap conversion task.

GLOBAL

Display settings that are common for all services.

ALL

Display all settings.

Usage Notes

Consider the following when using the SETTINGS command:

- The information returned for each service is dependent upon the service itself. Nearly all of these settings are provided by default if they are not explicitly provided through a startup parameter or a configuration file statement. If the following immediately follows the setting:
 - (C) The setting was taken from a configuration file statement.
 - (D) The setting was taken from the default.
 - (P) The setting was taken from a startup parameter.
- The settings for each service are derived from the statements in the service configuration file. All of the service configuration file statements have default values, except for the ServerLocation statement.
- The configuration file used by each service can be specified either as a startup parameter or in the global configuration, or IHSAINIT, file.
- The global Event/Automation Service settings include the name of the global configuration file, the PPI value, and the OUTSIZE parameter value.
- Any settings that specify file names use the actual file name instead of the exact value from the configuration file statement. For example, if you have the AdapterCdsFile=IHSAACDS statement in the alert adapter configuration file, and data set NETVIEW.SCNMUXCL is provided on the IHSSMP3 data set definition statement in the Event/Automation Service startup procedure and contains member IHSAACDS, the actual value displayed by the SETTINGS command is as follows:


```
NETVIEW.SCNMUXCL (IHSAACDS) (C)
```
- The settings provided on the Filter and FilterCache statements are not followed by an indication of how the setting was provided. These settings can be provided only from an adapter service configuration file. If there are no Filter or FilterCache statements in this file, these settings are not displayed.

Restrictions

You can specify only one TASK operand. If you want to specify more than one service task, separate each task ID with a comma and enclose the string within parentheses.

Example: Showing the alert adapter settings

To show the current alert adapter service settings, enter:

```
F IHSAEVNT,SETTINGS,TASK=ALERTA
```

Response

You see a response similar to the following:

```
IHS0182I <==Current Alert Adapter Service Settings==>
IHS0183I   CFG file : NETVIEW.SCNMUXCL(IHSAACDS) (D)
IHS0185I   ServerLocation 1 = my.ip.hostname.com (C)
```

```

IHS0185I ServerPort 1 = 0 (D)
IHS0185I ConnectionMode = connection_oriented (D)
IHS0185I TestMode = no (D)
IHS0185I RetryInterval = 120 (D)
IHS0185I BufferEvents = yes (D)
IHS0185I BufferFlushRate = 0 (D)
IHS0185I BufEvtPath = /etc/Tivoli/tec/cache_nv390alt (C)
IHS0185I BufferEventsLimit = 0 (D)
IHS0185I BufEvtMaxSize = 64 (D)
IHS0185I BufEvtShrinkSize = 8 (D)
IHS0185I BufEvtRdblkLen = 64 (D)
IHS0185I EventMaxSize = 4096 (D)
IHS0185I AdapterCdsFile = NETVIEW.SCNMUXCL(IHSAACDS) (D)
IHS0185I FilterMode = out (D)
IHS0186I Filter 1 slots:
IHS0187I     Class=SNA_Equipment_Malfunction;
IHS0187I     source=filtersource;
IHS0187I     severity=WARNING;
IHS0188I FilterCache 1 slots:
IHS0187I     Class=SNA_Equipment_Malfunction;
IHS0187I     source=filtersource;
IHS0187I     severity=WARNING;

```

The numeric value associated with Filter or FilterCache settings is equivalent to the position of the statement in the adapter configuration file, relative to the other Filter or FilterCache statements in the file.

Example: Showing the trap-to- alert service settings

To show the current trap-to-alert service settings, enter:

```
F IHSAEVNT,SETTINGS,TASK=TRAPALRT
```

Response

You see a response similar to the following:

```

IHS0182I <==Current Trap to Alert Conversion Service Settings==>
IHS0183I   CFG file : NETVIEW.SCNMUXCL(IHSATCFG) (D)
IHS0185I   NetViewAlertReceiver = NETVALRT (D)
IHS0185I   PortNumber = 162 (C)
IHS0185I   AdapterCdsFile = NETVIEW.SCNMUXCL(IHSATCDS) (D)

```

Example: Showing the global Event/Automation Service settings

To show the global Event/Automation Service settings, enter:

```
F IHSAEVNT,SETTINGS,TASK=GLOBAL
```

Response

You see a response similar to the following:

```

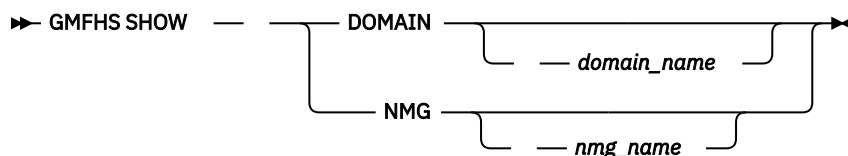
IHS0182I <==Current E/AS Global Service Settings==>
IHS0183I   CFG file : NETVIEW.SCNMUXCL(IHSAINIT) (D)
IHS0185I   PPI = IHSATEC (D)
IHS0185I   OUTSIZE = 0 (D)
IHS0185I   ROUTECDE = 1 (D)

```

SHOW (GMFHS)

Syntax

SHOW



Purpose of Command

The SHOW command provides a report with an entry for a specified network management gateway (NMG) or domain, or all NMGs or domains, defined to the NetView GMFHS. The report includes the display name, type, and status of the NMG or domain it represents.

You can enter the SHOW command from the MVS console using the MVS MODIFY command or from a NetView terminal by using the GMFHS command list.

Operand Descriptions

DOMAIN

Specifies to provide a report with an entry for each network management domain defined to the GMFHS.

domain_name

Specifies the network management domain for which to provide a report. *domain_name* must be the MyName attribute value of an SNA_Domain_Class instance in the RODM data cache or the EMDomain attribute value of a Non_SNA_Domain_Class instance.

This is an optional operand.

NMG

Specifies to provide a report with an entry for each NMG defined to the GMFHS.

nmg_name

Specifies the NMG for which to provide a report. *nmg_name* must be the value of the MyName attribute of an NMG_Class instance.

This is an optional operand.

Example: Displaying all network management domains

To display all network management domains, enter:

```
GMFHS SHOW DOMAIN
```

Response

You receive a response similar to the following:

```
DUI4035I  NETWORK MANAGEMENT DOMAIN DISPLAY
DUI4036I  NAME = CNM01      TYPE = SNA      STATE = COMPLETE  CFGTM =
02/23/19  08:36  SESS = N/A  NMG = N/A
DUI4036I  NAME = B01NV      TYPE = SNA      STATE = COMPLETE  CFGTM =
02/23/19  08:36  SESS = N/A  NMG = N/A
DUI4036I  NAME = LANMGR     TYPE = NON-SNA   STATE = COMPLETE  CFGTM =
02/23/19  08:36  SESS = YES  NMG = A0488P21
DUI4036I  NAME = LATTVIEW   TYPE = NON-SNA   STATE = COMPLETE  CFGTM =
02/23/19  08:36  SESS = YES  NMG = B3088P1
DUI4036I  NAME = DECNET     TYPE = NON-SNA   STATE = WAITING   CFGTM =
02/23/19  08:36  SESS = YES  NMG = B3088P2
DUI4036I  NAME = NETVIEW    TYPE = NON-SNA   STATE = COMPLETE  CFGTM =
```



```
02/23/19 08:36 SESS = YES NMG = A0488P31
DUI4037I END
```

For each domain, the displayed information includes:

- Name
- Type (SNA or non-SNA)
- State of domain configuration process
- Date and time that domain configuration started
- For non-SNA domains, whether a session exists with the service point application
- For non-SNA domains, the network management gateway to which the domain reports.

Example: Displaying all network management gateways

To display all network management gateways, enter:

```
GMFHS SHOW NMG
```

Response

You receive a response similar to the following:

```
DUI4038I NETWORK MANAGEMENT GATEWAY DISPLAY
DUI4039I NMG = A0488PB4 STATUS = UNKNOWN TRAN = COS WINDOW = 1
          OUT = 0 SENT = 0
DUI4039I NMG = A0488P21 STATUS = UNKNOWN TRAN = COS WINDOW = 1
          OUT = 0 SENT = 0
DUI4039I NMG = B3088P2 STATUS = UNKNOWN TRAN = COS WINDOW = 1
          OUT = 0 SENT = 0
DUI4039I NMG = B3088P1 STATUS = UNKNOWN TRAN = COS WINDOW = 1
          OUT = 0 SENT = 0
DUI4037I END
```

For each network management gateway (NMG) the displayed information includes:

- Name
- Status of the NMG
- Transport type
- Window size
- Number of commands outstanding and awaiting response
- Number of commands sent through the NMG

Example: Obtaining a report on a specified NMG

To obtain a report on the NMG named B3088P1, enter:

```
GMFHS SHOW NMG B3088P1
```

Response

A response similar to the following is displayed:

```
DUI4038I NETWORK MANAGEMENT GATEWAY DISPLAY
DUI4039I NMG = B3088P1 STATUS = UNKNOWN TRAN = COS WINDOW = 1
          OUT = 0 SENT = 0
DUI4037I END
```

SHOWTEXT (BROWSE CANZLOG, DISPMMSG)

Syntax

SHOWTEXT

➤ SHOWTEXT ➤

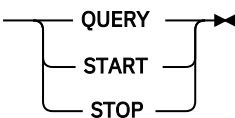
Purpose of Command

The SHOWTEXT subcommand presents a single message from the CANZLOG or detail display that allows you to more easily read and examine a message. If a single-line message is too wide for your display, SHOWTEXT wraps the text so that you can see the entire message. If a multiline message is too long for easy examination, SHOWTEXT presents it in a WINDOW where you can use the WINDOW text-based FIND and ALL commands.

SMDR (NLDM)

Syntax

SMDR

➤ SMDR —  ➤

Purpose of Command

The SMDR command restarts, stops, or queries the status of session monitor data recording.

Operand Descriptions

QUERY

Indicates to display the current data recording status.

START

Indicates to restart session monitor data recording.

STOP

Indicates to stop session monitor data recording. All current and future sessions on the data recording queue are discarded without being recorded to the session monitor database. Explicit route data is no longer recorded to the session monitor database. Session data continues to be recorded to the external log.

Restrictions

The following restrictions apply to the SMDR command:

- If data recording is already active when you issue the SMDR START command, you receive message AAU274I. If data recording is not active when you issue the SMDR START command, message AAU273I is sent to the authorized receiver. This message indicates the number of sessions that were suppressed while data recording was inactive.
- Message AAU274I is sent in response to the SMDR STOP and SMDR QUERY commands. For SMDR STOP, the message tells you whether data recording has stopped or is already inactive. For SMDR QUERY, the message tells you whether data recording is active or inactive.

Example: Displaying the status of session monitor data recording

To display the status of session monitor data recording, enter:

```
SMDR QUERY
```

Response

A message similar to one of the following is displayed:

```
AAU274I VSAM SESSION RECORDING IS ACTIVE
AAU274I VSAM SESSION RECORDING IS INACTIVE
```

SMENU (STATMON)

Syntax

SMENU

► SMENU ◀

IBM-Defined Synonyms

Command or Operand	Synonym
SMENU	SM

Purpose of Command

The SMENU command displays activity and analysis information for the selected resources displayed on the status monitor screen.

Example: Displaying activity and analysis information from a status monitor panel

If you are displaying a status monitor panel containing resources, you can display a menu from which you can select activity or analysis information for selected resources. To display the menu, enter:

```
smenu
```

SNAHD (AON)

Syntax

SNAHD

► SNAHD — *resname* , — *option* ◀

Purpose of Command

The SNAHD command provides access to a full-screen help desk to guide you through problems with SNA resources and NetView Access Services user IDs. See *IBM Z NetView User's Guide: Automated Operations Network* for more information.

Operand Descriptions

resname

The name of the SNA resource or the NetView Access Services user ID you want to investigate.

option

option can do one of the following:

- 1 (Recycle resource)
- 2 (Problem determination)
- 3 (NetView Access Services *userid*)

Restrictions

- The AON tower and the SNA subtower must be enabled in the CNMSTYLE member to successfully run this command.
- This command operates in full-screen mode only.

SNAMAP (AON)

Syntax

SNAMAP

➤ SNAMAP — *option* , — *resname* ➤

Purpose of Command

The SNAMAP command provides the ability to map a resource to its lower resources.

Operand Descriptions

option

One of the following:

- 1 (MAJNODES)
- 2 (APPLS)
- 3 (CDRMS)
- 4 (CDRSCS)
- 5 (LINKSTA)
- 6 (CLSTRS)
- 7 (TERMS)
- 8 (user-provided *resname*)

resname

The name of the SNA resource. This is used with Option 8.

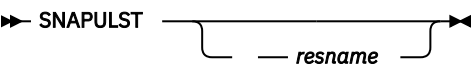
Restrictions

- The AON tower and the SNA subtower must be enabled in the CNMSTYLE member to successfully run this command.
- This command operates in full-screen mode only.

SNAPULST (AON)

Syntax

SNAPULST



Purpose of Command

The SNAPULST command provides a list display of the LUs and CPs that belong to a PU.

Operand Descriptions

resname

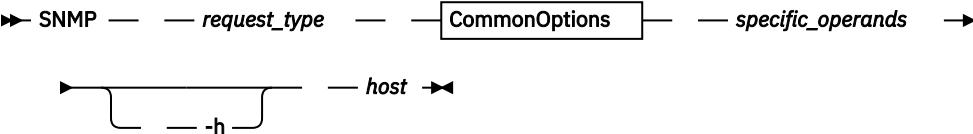
The name of the PU, LU, or CP. If the name is an LU or CP, a pop-up window is displayed with the option to either show the PU list on the PU, or enter the SNA Help Desk on the LU or CP. If *resname* is not specified, a panel is displayed to prompt you to enter the name of the resource.

Restrictions

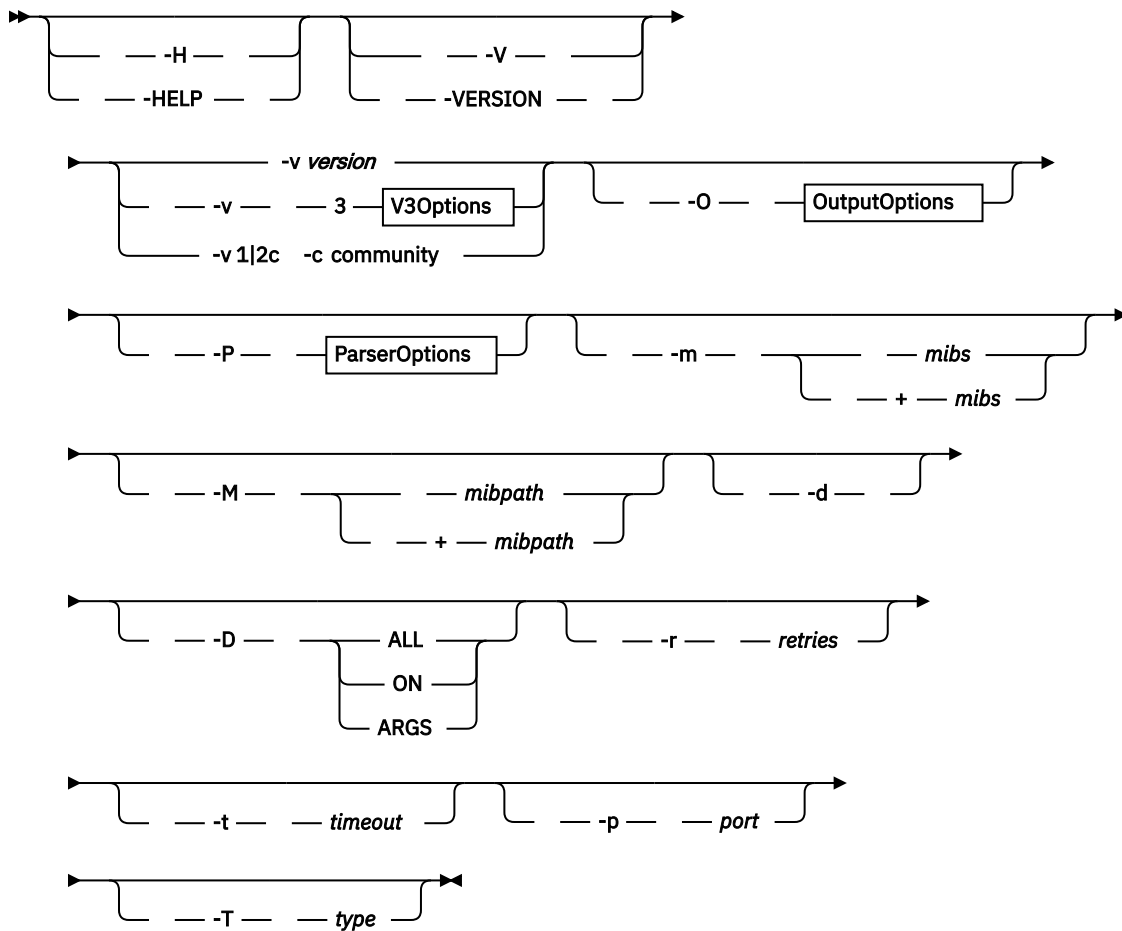
- The AON tower and the SNA subtower must be enabled in the CNMSTYLE member to successfully run this command.
- This command operates in full-screen mode only.

SNMP (NCCF; CNMESNMP)

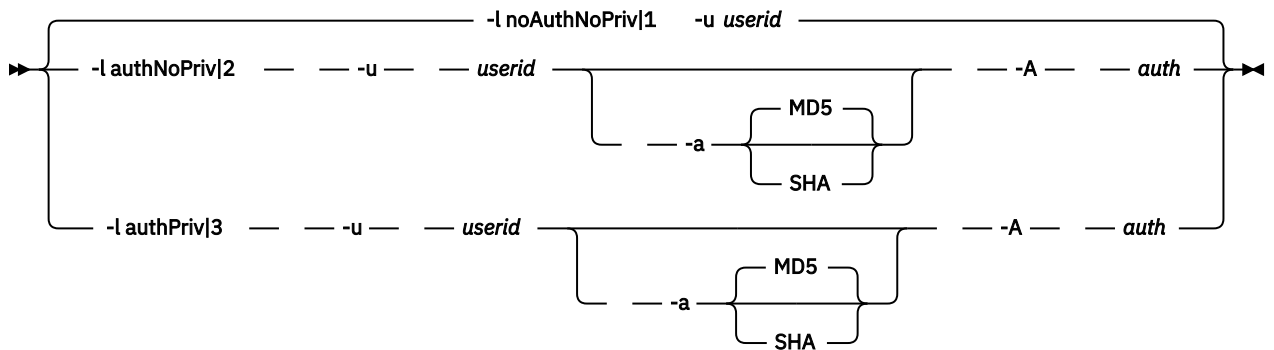
Syntax



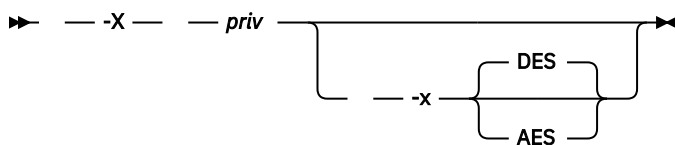
CommonOptions



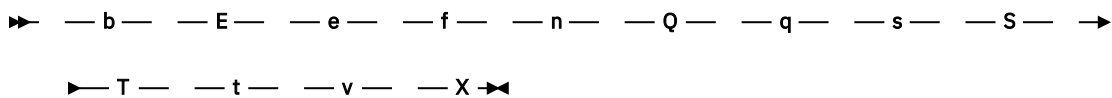
V3Options



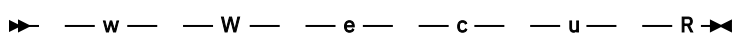
V3Pr



OutputOptions



ParserOptions



Purpose of Command

The SNMP command is used to send an SNMP request to a network device to set or obtain information about that device.

Note:

1. The NetView SNMP command syntax and keywords are different from the OMVS SNMP command.
2. The OMVS SNMP commands can be entered from NetView by using the NetView PIPE UNIX stage after the NetView UNIX command is started. Refer to the *IBM Z NetView Installation: Configuring Additional Components* for more information about the UNIX Command Server.

Operand Descriptions

-h

Optionally specifies the destination IP host.

host

Specifies the destination IP host.

request_type

The type of SNMP command. Valid values are:

- BULKWALK
- GET
- GETBULK
- GETNEXT
- INFORM
- SET
- TRAP
- WALK

Each SNMP request has online help. For example, issuing HELP SNMP SET provides the syntax options and usage information specific to the SET request.

specific_operands

See the online help for a specific request type.

The following parameter descriptions are for the common options:

-a

Sets the authentication protocol used for authenticating SNMPv3 messages. The default value is MD5. When MD5 is used, the -a MD5 parameter can be omitted.

-A

Sets the authentication pass phrase used for authenticated SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.

-c

Specifies the community name. This parameter is required for SNMPv1 and SNMPv2c.

-d

Specifies to include the contents of the input and output data packets with the output.

-D

Used for debugging purposes.

ON

Turns on debugging of the command line interface.

ALL

Is the same as ON but includes debugging of command processing.

ARGS

Traces argument handling.

-H|HELP

Displays the SNMP Command Menu. If other options are specified, they are ignored.

-l

Specifies the level of authentication and encryption.

noAuthNoPriv|1

No authentication and no encryption. The **-u** parameter is required.

authNoPriv|2

MD5 or SHA1 authentication, with no encryption. The **-u** and **-A** parameters are required. MD5 authentication is the default. SHA1 authentication requires the **-a SHA** parameter.

authPriv|3

MD5 or SHA1 authentication and DES or AES encryption. The **-u** and **-A** and **-X** parameters are required. If the **-a** parameter is not specified, MD5 is the default value for authentication. If the **-x** parameter is not specified, DES is the default value for encryption. SHA1 authentication requires the **-a SHA** parameter. AES encryption requires the **-x AES** parameter.

-m

Specifies the MIBs to parse for symbolic names. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.

-M

Specifies the directories in the UNIX System Services HFS, from where the NetView program is to search for MIB source files. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.

-O

Specifies the output options:

b

Prevents attempts to resolve index elements to names.

E

Modifies the input strings to include a backslash (\) before the quotation mark.

e

Removes any symbolic labels from values.

f

Prints the complete OID.

n

Prints the OID in its fully-specified numeric form.

Q

Removes the type information but keeps the equal sign (=).

q

Removes the equal sign (=) and type information.

s

Prints only the last symbolic part of the OID.

S

Prints only the last symbolic part of the OID and precedes it with the name of the MIB which defines the object.

T

Prints any printable characters enclosed in brackets [] after the hexadecimal encoding.

t

Prints timeticks values as raw numbers.

v

Prints only the value or values.

X

Indicates the index components of OIDs with brackets [].

-p
Specifies the port on the SNMP agent to send this request. The default value is 161 for all requests except INFORM and TRAP, which default to 162.

-P
Specifies the parser options:

w
Displays warning messages while parsing MIB source files.

W
Displays additional warning messages while parsing MIB source files.

e
Displays MIB errors.

c
Disables ASN.1 comments to extend to the end of the MIB source comment lines.

u
Enables underscores in symbols.

R
Replaces MIB objects using the MIB file that was read last.

-r
Specifies the number of retries. The default value is 1.

-t
Specifies the timeout value between retries. The default value is 1 second.

-T
Specifies the transport type. These are valid values:

- UDP
- UDP6
- UDPV6
- UDPIPV6
- TCP
- TCP6
- TCPV6
- TCPIPV6

Notes:

1. If IPv6Env=MIXED was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address. To use a host name that resolves to an IPv6 address, specify a transport type appropriate for IPv6 networking (any of the types that end with 6).
2. If IPv6Env=ONLY was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP6. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP6 transport type, namely, an IPv6 address.
3. If IPv6Env=NONE was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address.

-u
The security name. This option is only valid for SNMPv3.

-v
Specifies which SNMP Protocol version to use. Valid values are 1, 2c, and 3. The default value is 1.

Note: For SNMP BULKWALK, SNMP GETBULK, and SNMP INFORM, valid values are 2c and 3.

-V|VERSION

Displays the NetView SNMP command version information. If other options are specified, they are ignored.

-x

Sets the encryption algorithm that is used for SNMPv3 messages. The default value is DES. When DES encryption is used, the -x parameter can be omitted. A value of AES indicates that AES 128-bit encryption is used; the synonym AES128 can be used as a valid keyword for the -x parameter.

-X

Sets the privacy pass phrase used for encrypted SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.

Usage Notes

Consider the following when using the SNMP command:

- The command-line interface is case-sensitive. Use the NETVASIS command or set OVERRIDE NETVASIS to YES.
- Do not run SNMP commands in NetView preinitialization command lists.
- Various NetView SNMP defaults can be modified with the COMMON.CNMSNMP definitions in the CNMSTYLE member. See the instructions in the CNMSTYLE member for more information.
- If AES encryption is being used for SNMPv3 requests, the z/OS Cryptographic Services Integrated Cryptography Service Facility (ICSF) must be configured and running on the z/OS host on which NetView is also running.

Return Codes

Return Code	Meaning
--------------------	----------------

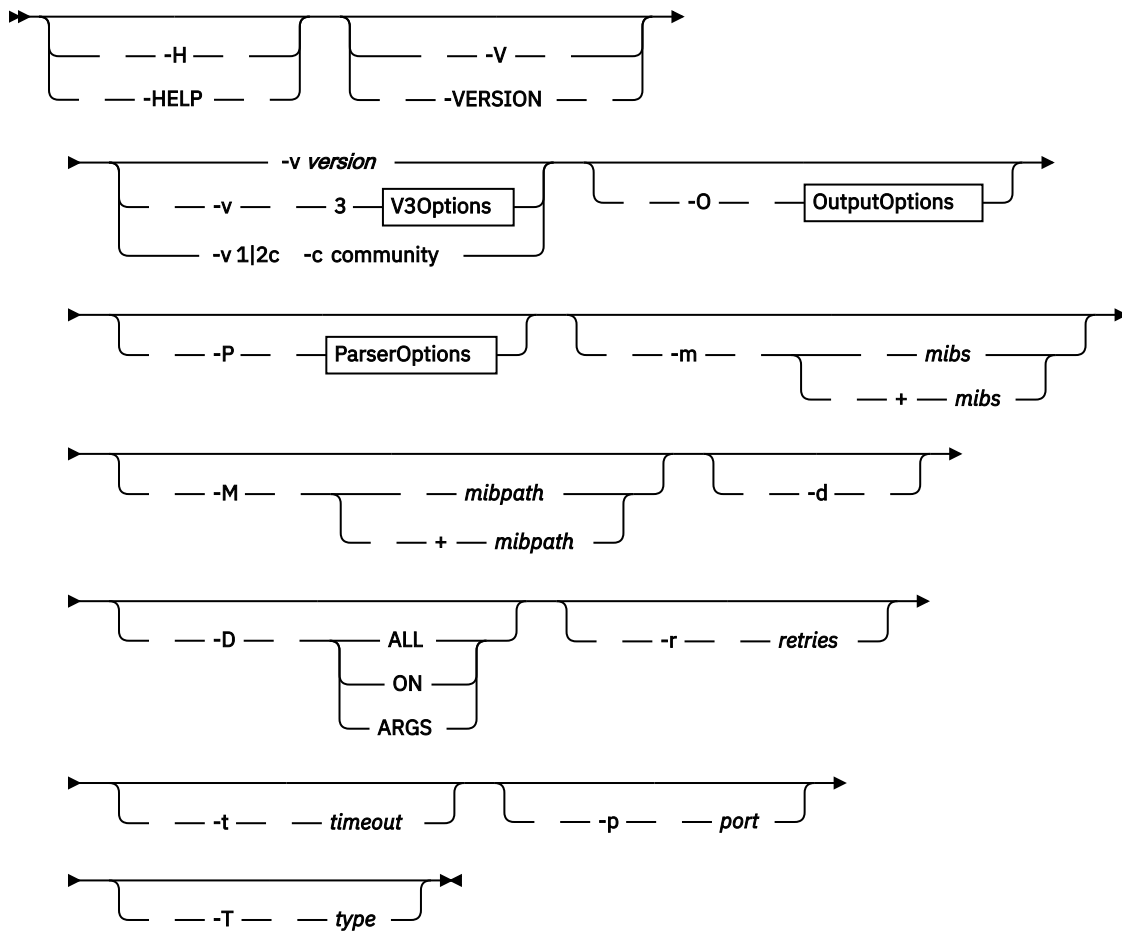
- | | |
|----------|---|
| 0 | Processing ended without errors. |
| 1 | Processing ended with unrecoverable errors. |
| 2 | Processing ended with general errors. |
| 5 | Processing ended abnormally. |

SNMP BULKWALK (NCCF; CNMESNMP)

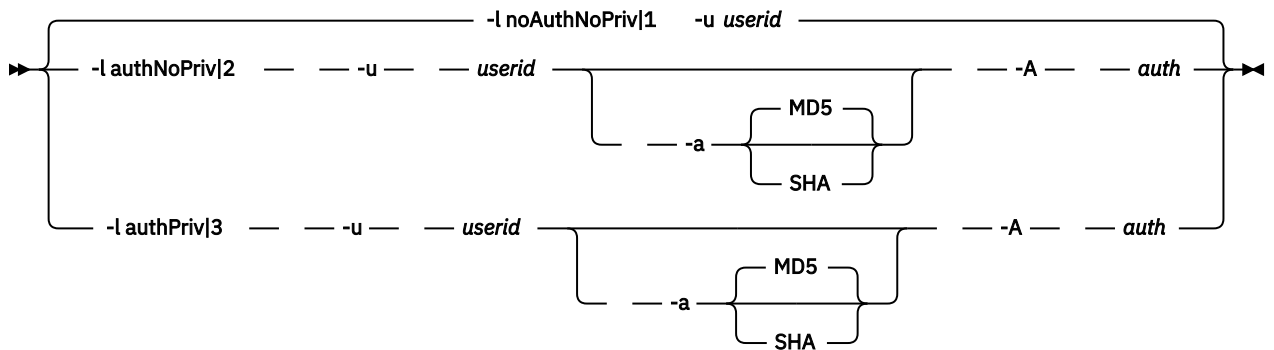
Syntax

```
➤ SNMP — — BULKWALK — CommonOptions — — -Cc — — — -Ci — — — -Cn ➤  
  
➤ < — nonrep — > — — -Cp — — — -Cr — < — maxrep — > ➤  
  
➤ ————— — host — — oid ➤  
    |—————|  
    — -h —————
```

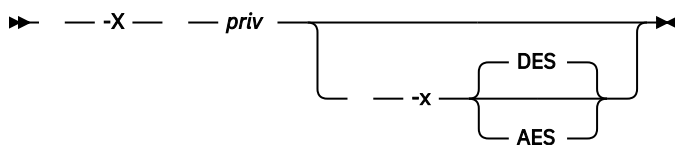
CommonOptions



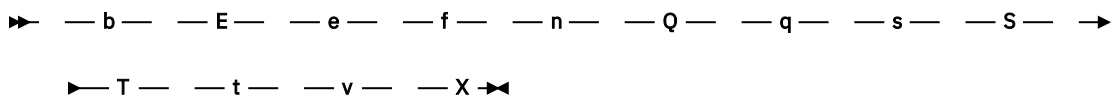
V3Options



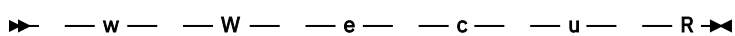
V3Pr



OutputOptions



ParserOptions



Purpose of Command

The SNMP BULKWALK command enables you to use SNMP GETBULK requests to query a network entity efficiently for a tree of information. An object identifier (OID) can be specified on the command line. This OID identifies which portion of the object identifier space will be searched using GETBULK requests. All variables in the subtree below the given OID are queried and their values presented to the user. If an OID argument is not specified, SNMP BULKWALK will search MIB-2.

If the tree search causes attempts to search beyond the end of the MIB, a message "End of MIB" will be displayed.

Operand Descriptions

-Cc

Specify -Cc to indicate that checking is not to be done on returned OIDs. Some agents (LaserJets, for example) return OIDs out of order, but can complete the walk nonetheless. Other agents return OIDs that are out of order and can cause BULKWALK to loop indefinitely. BULKWALK tries to detect this behavior and warns you when it finds an agent acting illegally. If you specify -Cc the volume of returned OIDs is not checked.

-Ci

Include the OID specified on the command line in the search range (assuming that the OID is a valid OID in the tree itself). BULKWALK uses GETBULK requests, starting with the specified OID, and returns all results in the MIB tree that occur after that OID.

-Cn

Set the non-repeaters field in the GETBULK PDUs.

nonrep

The number *nonrep* of supplied variables that are not iterated. Note that there is not a space between -Cn and the value that you specify for *nonrep*. For example, to specify a value of 5, specify -Cn5 without an intervening space. The default value is 0.

-Cp

Print the number of variables found upon completion of the walk.

-Cr

Set the max-repetitions field in the GETBULK PDUs.

maxrep

The number *maxrep* specifies the maximum number of iterations over the repeating variables. Note that there is not a space between -Cr and the value that you specify for *maxrep*. For example, to specify a value of 5, specify -Cr5 without an intervening space. The default value is 10.

-h

Optionally specifies the destination IP host. When -h is specified for an IP host name, name server resolution to an IP address is performed in a separate process.

host

Specifies the destination IP host.

oid

Specifies the object ID (OID) of the MIB variable.

The following parameter descriptions are for the common options:

-a

Sets the authentication protocol used for authenticating SNMPv3 messages. The default value is MD5. When MD5 is used, the -a MD5 parameter can be omitted.

-A

Sets the authentication pass phrase used for authenticated SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.

-c

Specifies the community name. This parameter is required for SNMPv1 and SNMPv2c.

- d**
Specifies to include the contents of the input and output data packets with the output.
- D**
Used for debugging purposes.
 - ON**
Turns on debugging of the command line interface.
 - ALL**
Is the same as ON but includes debugging of command processing.
 - ARGS**
Traces argument handling.
- H|HELP**
Displays the SNMP Command Menu. If other options are specified, they are ignored.
- l**
Specifies the level of authentication and encryption.
 - noAuthNoPriv|1**
No authentication and no encryption. The **-u** parameter is required.
 - authNoPriv|2**
MD5 or SHA1 authentication, with no encryption. The **-u** and **-A** parameters are required. MD5 authentication is the default. SHA1 authentication requires the **-a SHA** parameter.
 - authPriv|3**
MD5 or SHA1 authentication and DES or AES encryption. The **-u** and **-A** and **-X** parameters are required. If the **-a** parameter is not specified, MD5 is the default value for authentication. If the **-x** parameter is not specified, DES is the default value for encryption. SHA1 authentication requires the **-a SHA** parameter. AES encryption requires the **-x AES** parameter.
- m**
Specifies the MIBs to parse for symbolic names. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.
- M**
Specifies the directories in the UNIX System Services HFS, from where the NetView program is to search for MIB source files. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.
- O**
Specifies the output options:
 - b**
Prevents attempts to resolve index elements to names.
 - E**
Modifies the input strings to include a backslash (\) before the quotation mark.
 - e**
Removes any symbolic labels from values.
 - f**
Prints the complete OID.
 - n**
Prints the OID in its fully-specified numeric form.
 - Q**
Removes the type information but keeps the equal sign (=).
 - q**
Removes the equal sign (=) and type information.
 - s**
Prints only the last symbolic part of the OID.

- S** Prints only the last symbolic part of the OID and precedes it with the name of the MIB which defines the object.
- T** Prints any printable characters enclosed in brackets [] after the hexadecimal encoding.
- t** Prints timeticks values as raw numbers.
- v** Prints only the value or values.
- X** Indicates the index components of OIDs with brackets [].
- p** Specifies the port on the SNMP agent to send this request. The default value is 161 for all requests except INFORM and TRAP, which default to 162.
- P** Specifies the parser options:
- w** Displays warning messages while parsing MIB source files.
- W** Displays additional warning messages while parsing MIB source files.
- e** Displays MIB errors.
- c** Disables ASN.1 comments to extend to the end of the MIB source comment lines.
- u** Enables underscores in symbols.
- R** Replaces MIB objects using the MIB file that was read last.
- r** Specifies the number of retries. The default value is 1.
- t** Specifies the timeout value between retries. The default value is 1 second.
- T** Specifies the transport type. These are valid values:
- UDP
 - UDP6
 - UDPV6
 - UDPIPV6
 - TCP
 - TCP6
 - TCPV6
 - TCPIPV6

Notes:

1. If IPv6Env=MIXED was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address. To use a host name that resolves to an IPv6 address, specify a transport type appropriate for IPv6 networking (any of the types that end with 6).

2. If IPv6Env=ONLY was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP6. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP6 transport type, namely, an IPv6 address.
3. If IPv6Env=NONE was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address.

-u

The security name. This option is only valid for SNMPv3.

-v

Specifies which SNMP Protocol version to use. Valid values are 1, 2c, and 3. The default value is 1.

Note: For SNMP BULKWALK, SNMP GETBULK, and SNMP INFORM, valid values are 2c and 3.

-V|VERSION

Displays the NetView SNMP command version information. If other options are specified, they are ignored.

-x

Sets the encryption algorithm that is used for SNMPv3 messages. The default value is DES. When DES encryption is used, the -x parameter can be omitted. A value of AES indicates that AES 128-bit encryption is used; the synonym AES128 can be used as a valid keyword for the -x parameter.

-X

Sets the privacy pass phrase used for encrypted SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.

Usage Notes

Consider the following when using the SNMP command:

- The command-line interface is case-sensitive. Use the NETVASIS command or set OVERRIDE NETVASIS to YES.
- Do not run SNMP commands in NetView preinitialization command lists.
- Various NetView SNMP defaults can be modified with the COMMON.CNMSNMP definitions in the CNMSTYLE member. See the instructions in the CNMSTYLE member for more information.
- If AES encryption is being used for SNMPv3 requests, the z/OS Cryptographic Services Integrated Cryptography Service Facility (ICSF) must be configured and running on the z/OS host on which the NetView program is also running.

Return Codes

0

Processing ended without errors.

1

Processing ended with unrecoverable errors.

2

Processing ended with general errors.

5

Processing ended abnormally.

Example: Sending a BULKWALK request

This example sends an SNMPv2c BULKWALK request to an SNMP agent with a community string of public, specifying an output option that will remove the type information but keep the equal sign (=) and

printing only the last symbolic part of the OID, limiting the output to 0 input variables that are not to repeat, and specifying a maximum of 10 repetitions of the rest of the variables.

```
snmp bulkwalk -v2c -c public -Oqs -Cpn0 -Cr10  
development.tivlab.raleigh.ibm system
```

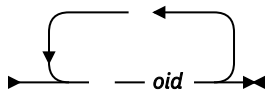
You receive a response similar to the following:

```
CNM005I sysDescr.0 = NetView Development  
CNM005I sysObjectID.0 = ibm.3.13  
CNM005I sysUpTime.0 = 1:0:11:37.00  
CNM005I sysContact.0 = SNMPBASE - Unspecified  
CNM005I sysName.0 = SNMPBASE - Unspecified  
CNM005I sysLocation.0 = SNMPBASE - Unspecified  
CNM005I sysServices.0 = 0  
CNM005I sysORLastChange.0 = 0:0:00:17.00  
CNM005I sysORID.1 = ibmAgentCaps.7.1  
CNM005I sysORID.2 = ibmAgentCaps.7.2  
CNM005I sysORDescr.1 = z/OS SNMP Agent  
CNM005I sysORDescr.2 = z/OS TCP/IP SNMP Subagent  
CNM005I sysORUpTime.1 = 0:0:00:00.00  
CNM005I sysORUpTime.2 = 0:0:00:17.00  
CNM005I Variables found: 14
```

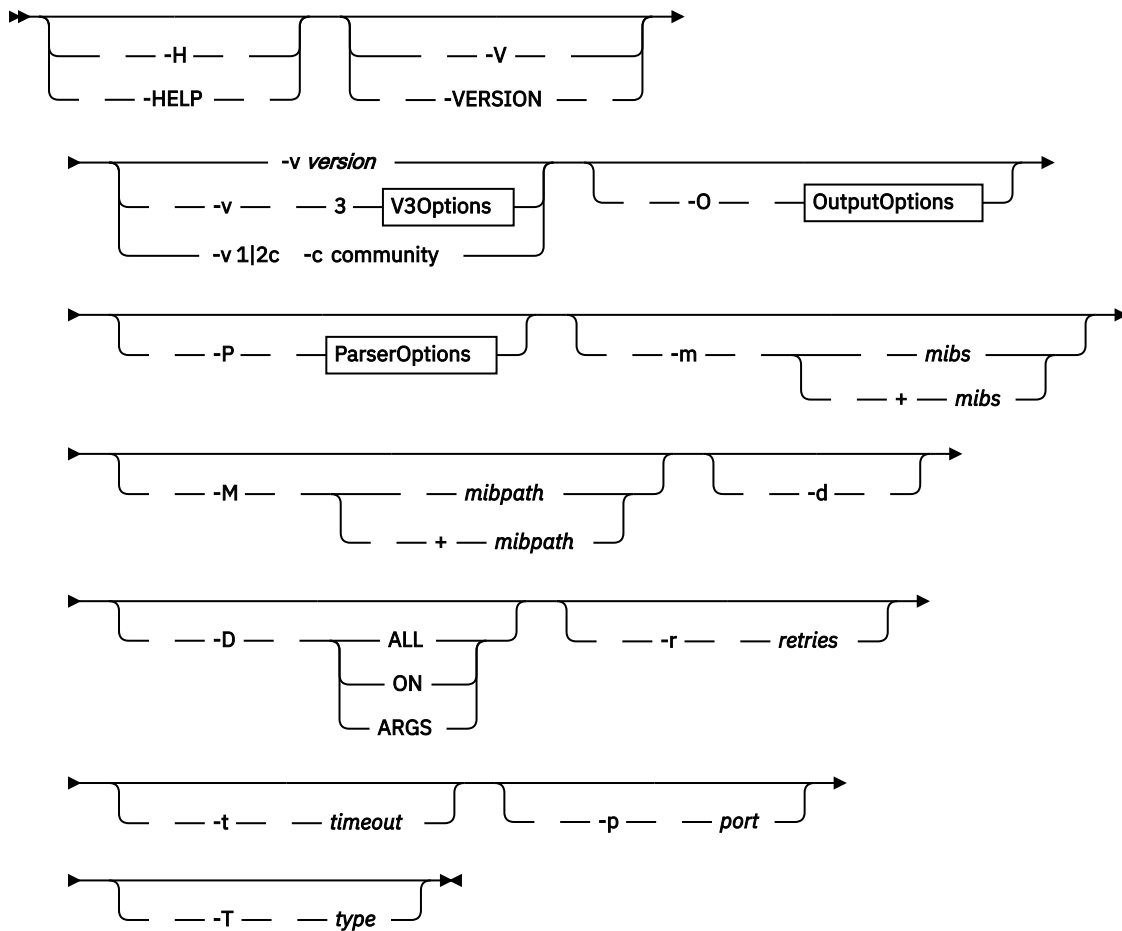
SNMP GET (NCCF; CNMESNMP)

Syntax

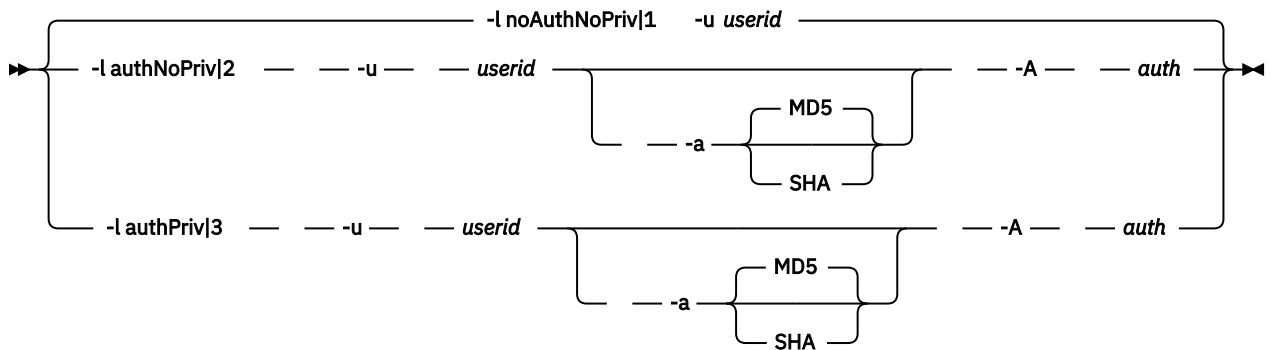
➔ SNMP — GET — CommonOptions — -Cf — -h — host ➔



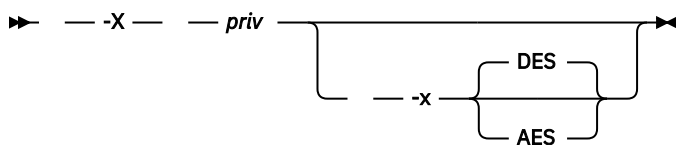
CommonOptions



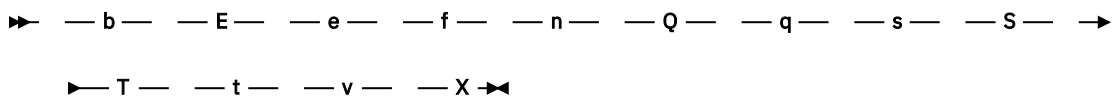
V3Options



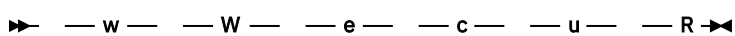
V3Pr



OutputOptions



ParserOptions



Purpose of Command

The SNMP GET command enables you to retrieve the value of one or more MIB variables.

Operand Descriptions

-Cf

Specifies to not attempt to fix errors returned by the agent on an errant request.

-h

Optionally specifies the destination IP host. When -h is specified for an IP host name, name server resolution to an IP address is performed in a separate process.

host

Specifies the destination IP host.

oid

Specifies the object ID (OID) of the MIB variable.

The following parameter descriptions are for the common options:

-a

Sets the authentication protocol used for authenticating SNMPv3 messages. The default value is MD5. When MD5 is used, the -a MD5 parameter can be omitted.

-A

Sets the authentication pass phrase used for authenticated SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.

-c

Specifies the community name. This parameter is required for SNMPv1 and SNMPv2c.

-d

Specifies to include the contents of the input and output data packets with the output.

-D

Used for debugging purposes.

ON

Turns on debugging of the command line interface.

ALL

Is the same as ON but includes debugging of command processing.

ARGS

Traces argument handling.

-H|HELP

Displays the SNMP Command Menu. If other options are specified, they are ignored.

-l

Specifies the level of authentication and encryption.

noAuthNoPriv|1

No authentication and no encryption. The -u parameter is required.

authNoPriv|2

MD5 or SHA1 authentication, with no encryption. The -u and -A parameters are required. MD5 authentication is the default. SHA1 authentication requires the -a SHA parameter.

authPriv|3

MD5 or SHA1 authentication and DES or AES encryption. The -u and -A and -X parameters are required. If the -a parameter is not specified, MD5 is the default value for authentication. If the -x parameter is not specified, DES is the default value for encryption. SHA1 authentication requires the -a SHA parameter. AES encryption requires the -x AES parameter.

-m

Specifies the MIBs to parse for symbolic names. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.

-M

Specifies the directories in the UNIX System Services HFS, from where the NetView program is to search for MIB source files. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.

-O

Specifies the output options:

b

Prevents attempts to resolve index elements to names.

E

Modifies the input strings to include a backslash (\) before the quotation mark.

e

Removes any symbolic labels from values.

f

Prints the complete OID.

n

Prints the OID in its fully-specified numeric form.

Q

Removes the type information but keeps the equal sign (=).

q

Removes the equal sign (=) and type information.

s

Prints only the last symbolic part of the OID.

S

Prints only the last symbolic part of the OID and precedes it with the name of the MIB which defines the object.

T

Prints any printable characters enclosed in brackets [] after the hexadecimal encoding.

t

Prints timeticks values as raw numbers.

v

Prints only the value or values.

X

Indicates the index components of OIDs with brackets [].

-p

Specifies the port on the SNMP agent to send this request. The default value is 161 for all requests except INFORM and TRAP, which default to 162.

-P

Specifies the parser options:

w

Displays warning messages while parsing MIB source files.

W

Displays additional warning messages while parsing MIB source files.

e

Displays MIB errors.

c

Disables ASN.1 comments to extend to the end of the MIB source comment lines.

u

Enables underscores in symbols.

R

Replaces MIB objects using the MIB file that was read last.

-r

Specifies the number of retries. The default value is 1.

-t

Specifies the timeout value between retries. The default value is 1 second.

-T

Specifies the transport type. These are valid values:

- UDP
- UDP6
- UDPV6
- UDPIPV6
- TCP
- TCP6
- TCPV6
- TCPIPV6

Notes:

1. If IPv6Env=MIXED was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address. To use a host name that resolves to an IPv6 address, specify a transport type appropriate for IPv6 networking (any of the types that end with 6).
2. If IPv6Env=ONLY was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP6. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP6 transport type, namely, an IPv6 address.
3. If IPv6Env=NONE was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address.

-u

The security name. This option is only valid for SNMPv3.

-v

Specifies which SNMP Protocol version to use. Valid values are 1, 2c, and 3. The default value is 1.

Note: For SNMP BULKWALK, SNMP GETBULK, and SNMP INFORM, valid values are 2c and 3.

-V|VERSION

Displays the NetView SNMP command version information. If other options are specified, they are ignored.

-x

Sets the encryption algorithm that is used for SNMPv3 messages. The default value is DES. When DES encryption is used, the -x parameter can be omitted. A value of AES indicates that AES 128-bit encryption is used; the synonym AES128 can be used as a valid keyword for the -x parameter.

-X

Sets the privacy pass phrase used for encrypted SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.

Usage Notes

Consider the following when using the SNMP command:

- The command-line interface is case-sensitive. Use the NETVASIS command or set OVERRIDE NETVASIS to YES.

- A maximum of 128 MIB variables is supported.
- Do not run SNMP commands in NetView preinitialization command lists.
- Various NetView SNMP defaults can be modified with the COMMON.CNMSNMP definitions in the CNMSTYLE member. See the instructions in the CNMSTYLE member for more information.
- If AES encryption is being used for SNMPv3 requests, the z/OS Cryptographic Services Integrated Cryptography Service Facility (ICSF) must be configured and running on the z/OS host on which the NetView program is also running.

Return Codes

- 0** Processing ended without errors.
- 1** Processing ended with unrecoverable errors.
- 2** Processing ended with general errors.
- 5** Processing ended abnormally.

Sending an SNMP GET request

The following example sends an SNMP GET request to tv2009 to retrieve the values of MIB variables sysDescr.0 and sysUpTime.0 and sysObjectID.0:

```
snmp get -c public tv2009
iso.org.dod.internet.mgmt.mib-2.system.sysDescr.0
.1.3.6.1.2.1.1.3.0 sysObjectID.0
```

You receive a response similar to the following:

```
CNM005I SNMPv2-MIB::sysDescr.0=STRING: Sysname: OS/390 Nodename:
TVT2009 Release: 14.00 Version: 03 Machine: 2064
CNM005I SNMPv2-MIB::sysUpTime.0=Timeticks: (1375000) 3:49:10.00
CNM005I SNMPv2-MIB::sysObjectID.0=OID:SUBAGENT-MIB::ibm.3.13
```

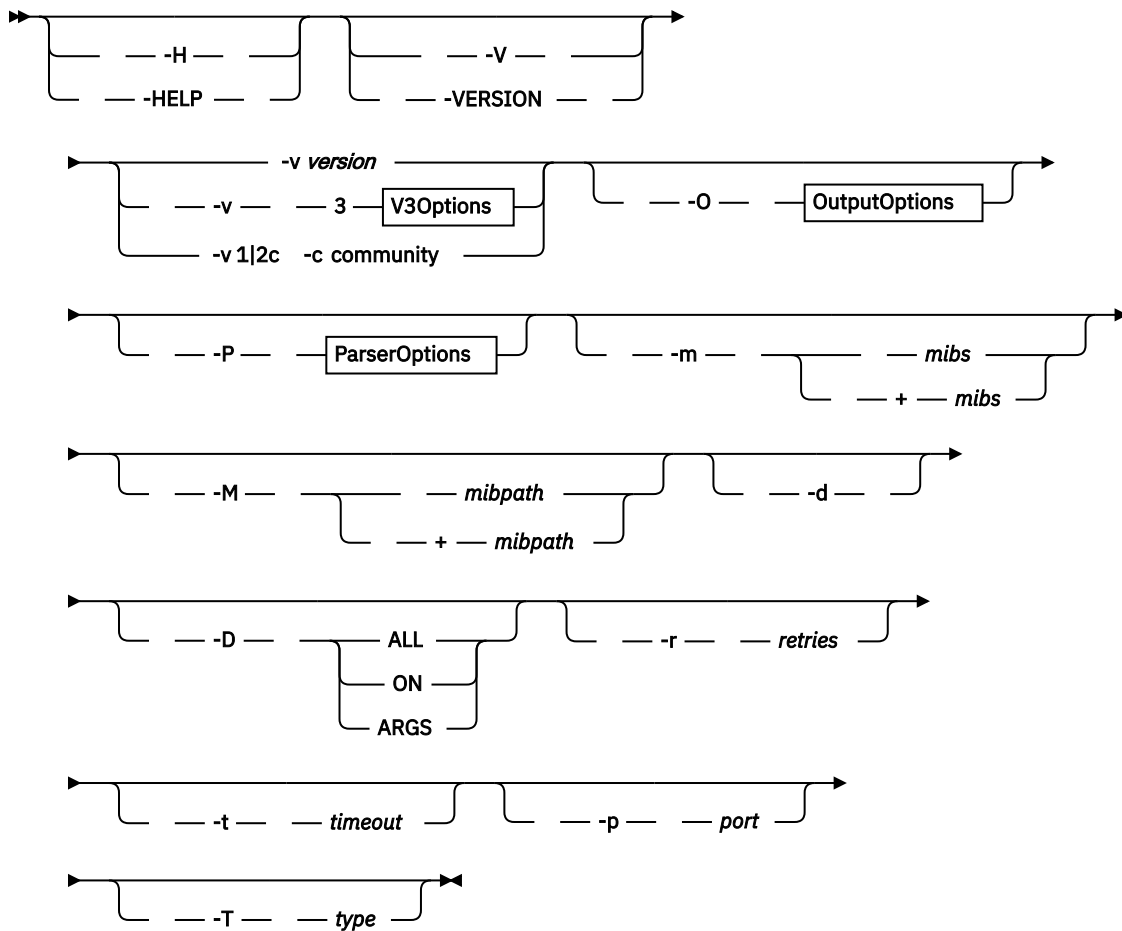
SNMP GETBULK (NCCF; CNMESNMP)

Syntax

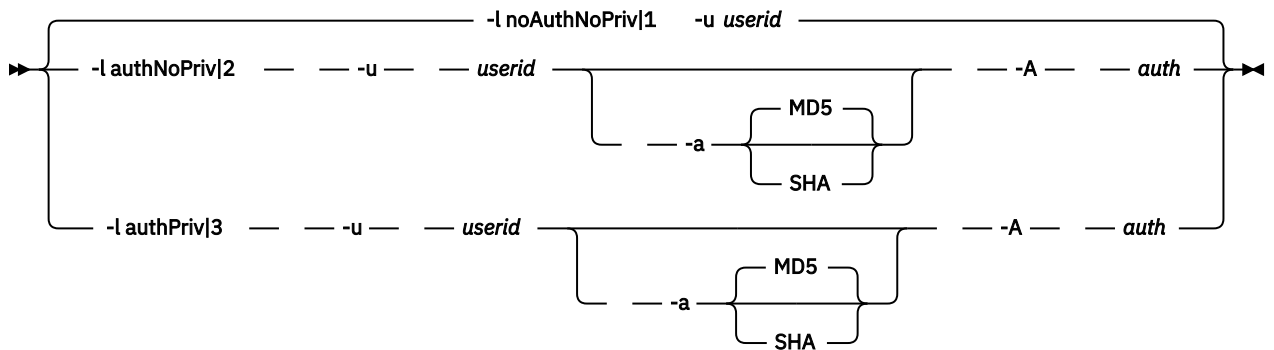
```
➤ SNMP — — GETBULK — CommonOptions — — -Cn — < — nonrep — > — — -Cr ➤
```

➤ < — *maxrep* — > — -h — *host* — *oid* ➤

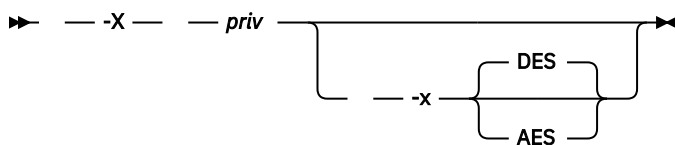
CommonOptions



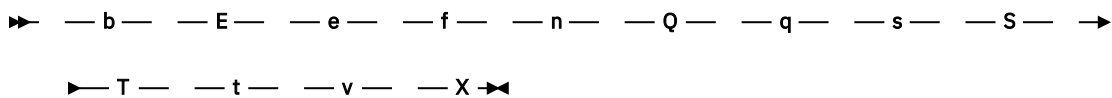
V3Options



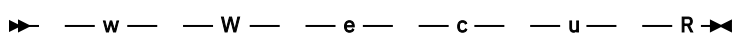
V3Pr



OutputOptions



ParserOptions



Purpose of Command

The SNMP GETBULK command enables you to get values for all of the MIB variables in a particular branch or in a single leaf node.

Operand Descriptions

-Cn

Set the non-repeaters field in the GETBULK PDU.

nonrep

The number *nonrep* of supplied variables that are not iterated. Note that there is not a space between -Cn and the value that you specify for *nonrep*. For example, to specify a value of 5, specify -Cn5 without an intervening space. The default value is 0.

-Cr

Set the maximum number of repetitions field in the GETBULK PDU.

maxrep

The number *maxrep* of iterations over the repeating variables. Note that there is not a space between -Cr and the value that you specify for *maxrep*. For example, to specify a value of 5, specify -Cr5 without an intervening space. The default value is 10.

-h

Optionally specifies the destination IP host. When -h is specified for an IP host name, name server resolution to an IP address is performed in a separate process.

host

Specifies the destination IP host.

oid

Specifies the object ID (OID) of the MIB variable.

The following parameter descriptions are for the common options:

-a

Sets the authentication protocol used for authenticating SNMPv3 messages. The default value is MD5. When MD5 is used, the -a MD5 parameter can be omitted.

-A

Sets the authentication pass phrase used for authenticated SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.

-c

Specifies the community name. This parameter is required for SNMPv1 and SNMPv2c.

-d

Specifies to include the contents of the input and output data packets with the output.

-D

Used for debugging purposes.

ON

Turns on debugging of the command line interface.

ALL

Is the same as ON but includes debugging of command processing.

ARGS

Traces argument handling.

-H|HELP

Displays the SNMP Command Menu. If other options are specified, they are ignored.

-l

Specifies the level of authentication and encryption.

noAuthNoPriv|1

No authentication and no encryption. The -u parameter is required.

authNoPriv|2

MD5 or SHA1 authentication, with no encryption. The **-u** and **-A** parameters are required. MD5 authentication is the default. SHA1 authentication requires the **-a SHA** parameter.

authPriv|3

MD5 or SHA1 authentication and DES or AES encryption. The **-u** and **-A** and **-X** parameters are required. If the **-a** parameter is not specified, MD5 is the default value for authentication. If the **-x** parameter is not specified, DES is the default value for encryption. SHA1 authentication requires the **-a SHA** parameter. AES encryption requires the **-x AES** parameter.

-m

Specifies the MIBs to parse for symbolic names. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.

-M

Specifies the directories in the UNIX System Services HFS, from where the NetView program is to search for MIB source files. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.

-O

Specifies the output options:

b

Prevents attempts to resolve index elements to names.

E

Modifies the input strings to include a backslash (\) before the quotation mark.

e

Removes any symbolic labels from values.

f

Prints the complete OID.

n

Prints the OID in its fully-specified numeric form.

Q

Removes the type information but keeps the equal sign (=).

q

Removes the equal sign (=) and type information.

s

Prints only the last symbolic part of the OID.

S

Prints only the last symbolic part of the OID and precedes it with the name of the MIB which defines the object.

T

Prints any printable characters enclosed in brackets [] after the hexadecimal encoding.

t

Prints timeticks values as raw numbers.

v

Prints only the value or values.

X

Indicates the index components of OIDs with brackets [].

-p

Specifies the port on the SNMP agent to send this request. The default value is 161 for all requests except INFORM and TRAP, which default to 162.

-P

Specifies the parser options:

w

Displays warning messages while parsing MIB source files.

W

Displays additional warning messages while parsing MIB source files.

e

Displays MIB errors.

c

Disables ASN.1 comments to extend to the end of the MIB source comment lines.

u

Enables underscores in symbols.

R

Replaces MIB objects using the MIB file that was read last.

-r

Specifies the number of retries. The default value is 1.

-t

Specifies the timeout value between retries. The default value is 1 second.

-T

Specifies the transport type. These are valid values:

- UDP
- UDP6
- UDPV6
- UDPIPV6
- TCP
- TCP6
- TCPV6
- TCPIPV6

Notes:

1. If IPv6Env=MIXED was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address. To use a host name that resolves to an IPv6 address, specify a transport type appropriate for IPv6 networking (any of the types that end with 6).
2. If IPv6Env=ONLY was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP6. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP6 transport type, namely, an IPv6 address.
3. If IPv6Env=NONE was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address.

-u

The security name. This option is only valid for SNMPv3.

-v

Specifies which SNMP Protocol version to use. Valid values are 1, 2c, and 3. The default value is 1.

Note: For SNMP BULKWALK, SNMP GETBULK, and SNMP INFORM, valid values are 2c and 3.

-V|VERSION

Displays the NetView SNMP command version information. If other options are specified, they are ignored.

-x

Sets the encryption algorithm that is used for SNMPv3 messages. The default value is DES. When DES encryption is used, the -x parameter can be omitted. A value of AES indicates that AES 128-bit encryption is used; the synonym AES128 can be used as a valid keyword for the -x parameter.

-X

Sets the privacy pass phrase used for encrypted SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.

Usage Notes

Consider the following when using the SNMP command:

- The command-line interface is case-sensitive. Use the NETVASIS command or set OVERRIDE NETVASIS to YES.
- A maximum of 128 MIB variables is supported.
- Do not run SNMP commands in NetView preinitialization command lists.
- Various NetView SNMP defaults can be modified with the COMMON.CNMSNMP definitions in the CNMSTYLE member. See the instructions in the CNMSTYLE member for more information.
- If AES encryption is being used for SNMPv3 requests, the z/OS Cryptographic Services Integrated Cryptography Service Facility (ICSF) must be configured and running on the z/OS host on which the NetView program is also running.

Return Codes**0**

Processing ended without errors.

1

Processing ended with unrecoverable errors.

2

Processing ended with general errors.

5

Processing ended abnormally.

Sending an SNMP GETBULK request

This example sends an SNMPv2c GETBULK request to an SNMP agent with a community string of public, specifying an output option that will remove the type information but keep the equal sign (=) and printing only the last symbolic part of the OID, limiting the output to 0 input variables that are not to repeat, and specifying a maximum of 5 repetitions of the rest of the variables.

```
snmp getbulk -v2c -c public -Oqs -Cn0 -Cr5
development.tivlab.raleigh.ibm system snmp
```

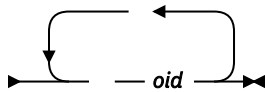
You receive a response similar to the following:

```
CNM005I sysDescr.0 = NetView Development
CNM005I snmpInPkts.0 = 263
CNM005I sysObjectID.0 = ibm.3.13
CNM005I snmpInBadVersions.0 = 0
CNM005I sysUpTime.0 = 1:0:18:36.00
CNM005I snmpInBadCommunityNames.0 = 46
CNM005I sysContact.0 = SNMPBASE - Unspecified
CNM005I snmpInBadCommunityUses.0 = 0
CNM005I sysName.0 = SNMPBASE - Unspecified
CNM005I snmpInASNParseErrs.0 = 0
```

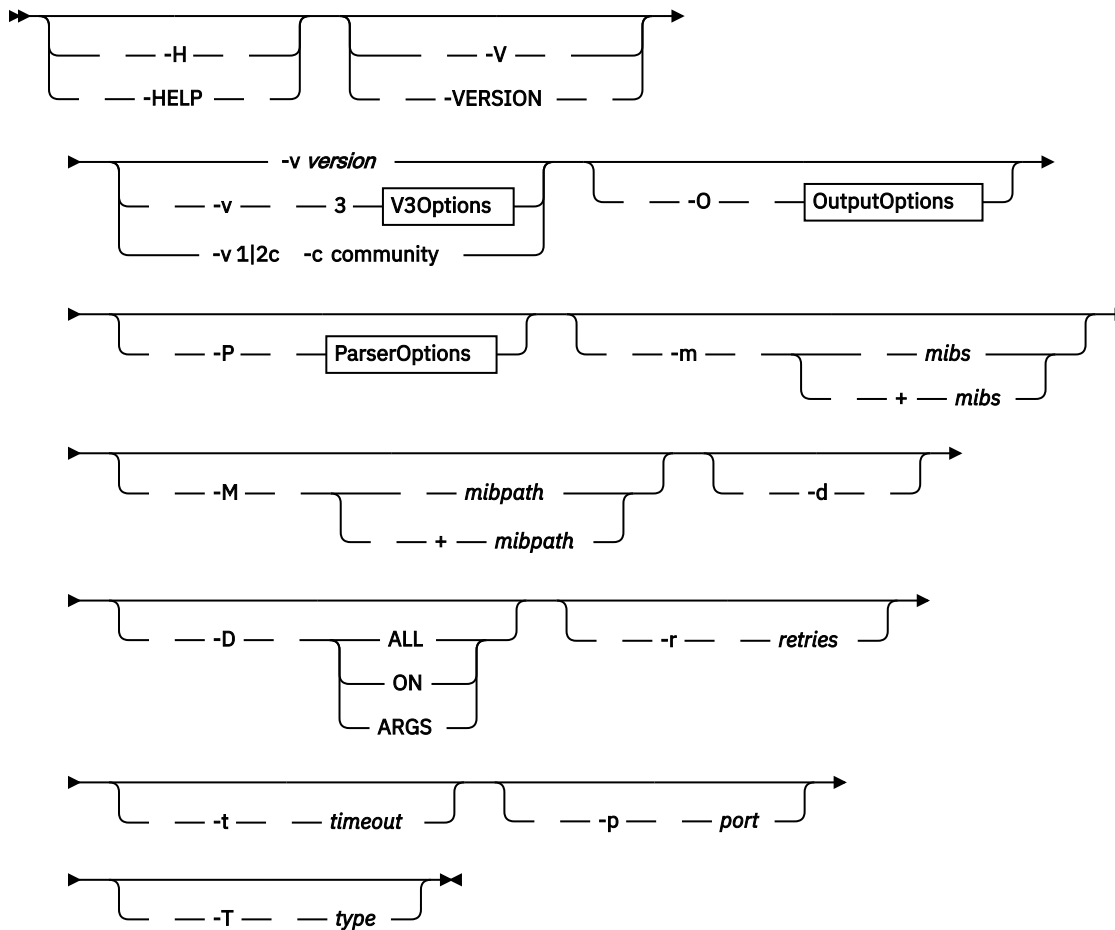
SNMP GETNEXT (NCCF; CNMESNMP)

Syntax

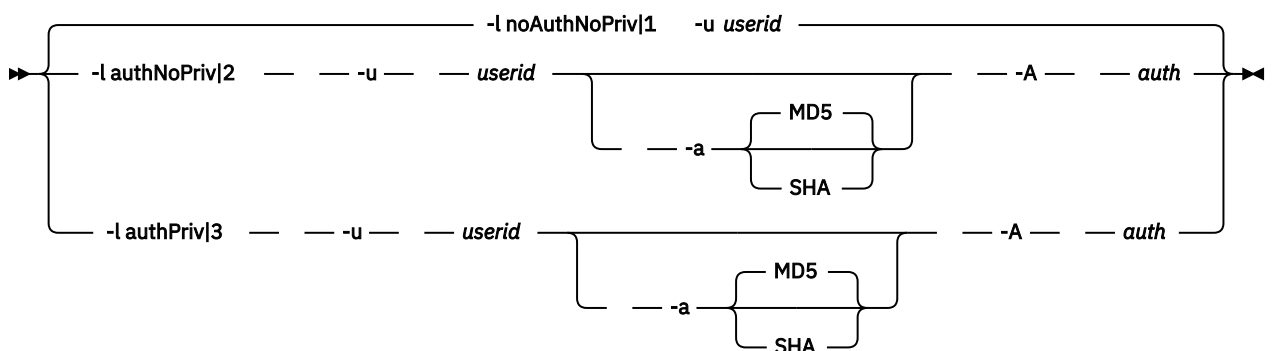
➤ SNMP — GETNEXT — CommonOptions — *host* ➤



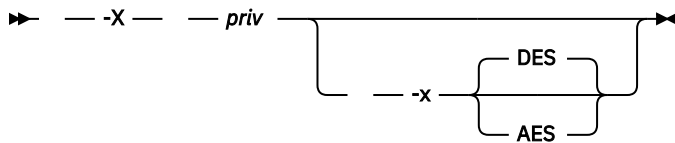
CommonOptions



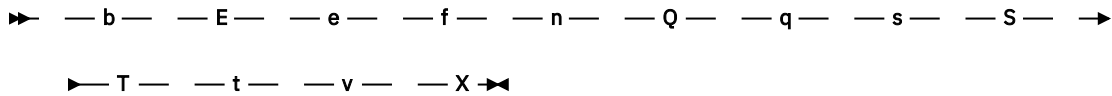
V3Options



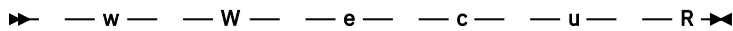
V3Pr



OutputOptions



ParserOptions



Purpose of Command

The SNMP GETNEXT command enables you to discover the value of the MIB variable after the one specified.

Operand Descriptions

-h

Optionally specifies the destination IP host. When -h is specified for an IP host name, name server resolution to an IP address is performed in a separate process.

host

Specifies the destination IP host.

oid

Specifies the object ID (OID) of the MIB variable.

The following parameter descriptions are for the common options:

-a

Sets the authentication protocol used for authenticating SNMPv3 messages. The default value is MD5. When MD5 is used, the -a MD5 parameter can be omitted.

-A

Sets the authentication pass phrase used for authenticated SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.

-c

Specifies the community name. This parameter is required for SNMPv1 and SNMPv2c.

-d

Specifies to include the contents of the input and output data packets with the output.

-D

Used for debugging purposes.

ON

Turns on debugging of the command line interface.

ALL

Is the same as ON but includes debugging of command processing.

ARGS

Traces argument handling.

-H|HELP

Displays the SNMP Command Menu. If other options are specified, they are ignored.

-l

Specifies the level of authentication and encryption.

noAuthNoPriv|1

No authentication and no encryption. The **-u** parameter is required.

authNoPriv|2

MD5 or SHA1 authentication, with no encryption. The **-u** and **-A** parameters are required. MD5 authentication is the default. SHA1 authentication requires the **-a SHA** parameter.

authPriv|3

MD5 or SHA1 authentication and DES or AES encryption. The **-u** and **-A** and **-X** parameters are required. If the **-a** parameter is not specified, MD5 is the default value for authentication. If the **-x** parameter is not specified, DES is the default value for encryption. SHA1 authentication requires the **-a SHA** parameter. AES encryption requires the **-x AES** parameter.

-m

Specifies the MIBs to parse for symbolic names. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.

-M

Specifies the directories in the UNIX System Services HFS, from where the NetView program is to search for MIB source files. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.

-O

Specifies the output options:

b

Prevents attempts to resolve index elements to names.

E

Modifies the input strings to include a backslash (\) before the quotation mark.

e

Removes any symbolic labels from values.

f

Prints the complete OID.

n

Prints the OID in its fully-specified numeric form.

Q

Removes the type information but keeps the equal sign (=).

q

Removes the equal sign (=) and type information.

s

Prints only the last symbolic part of the OID.

S

Prints only the last symbolic part of the OID and precedes it with the name of the MIB which defines the object.

T

Prints any printable characters enclosed in brackets [] after the hexadecimal encoding.

t

Prints timeticks values as raw numbers.

v

Prints only the value or values.

X

Indicates the index components of OIDs with brackets [].

-p

Specifies the port on the SNMP agent to send this request. The default value is 161 for all requests except INFORM and TRAP, which default to 162.

-P

Specifies the parser options:

- w**
Displays warning messages while parsing MIB source files.
- W**
Displays additional warning messages while parsing MIB source files.
- e**
Displays MIB errors.
- c**
Disables ASN.1 comments to extend to the end of the MIB source comment lines.
- u**
Enables underscores in symbols.
- R**
Replaces MIB objects using the MIB file that was read last.
- r**
Specifies the number of retries. The default value is 1.
- t**
Specifies the timeout value between retries. The default value is 1 second.
- T**
Specifies the transport type. These are valid values:
- UDP
 - UDP6
 - UDPV6
 - UDPIPV6
 - TCP
 - TCP6
 - TCPV6
 - TCPIPV6
- Notes:**
1. If IPv6Env=MIXED was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address. To use a host name that resolves to an IPv6 address, specify a transport type appropriate for IPv6 networking (any of the types that end with 6).
 2. If IPv6Env=ONLY was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP6. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP6 transport type, namely, an IPv6 address.
 3. If IPv6Env=NONE was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address.
- u**
The security name. This option is only valid for SNMPv3.
- v**
Specifies which SNMP Protocol version to use. Valid values are 1, 2c, and 3. The default value is 1.
Note: For SNMP BULKWALK, SNMP GETBULK, and SNMP INFORM, valid values are 2c and 3.
- V|VERSION**
Displays the NetView SNMP command version information. If other options are specified, they are ignored.

-x

Sets the encryption algorithm that is used for SNMPv3 messages. The default value is DES. When DES encryption is used, the -x parameter can be omitted. A value of AES indicates that AES 128-bit encryption is used; the synonym AES128 can be used as a valid keyword for the -x parameter.

-X

Sets the privacy pass phrase used for encrypted SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.

Usage Notes

Consider the following when using the SNMP command:

- The command-line interface is case-sensitive. Use the NETVASIS command or set OVERRIDE NETVASIS to YES.
- A maximum of 128 MIB variables is supported.
- Do not run SNMP commands in NetView preinitialization command lists.
- Various NetView SNMP defaults can be modified with the COMMON.CNMSNMP definitions in the CNMSTYLE member. See the instructions in the CNMSTYLE member for more information.
- If AES encryption is being used for SNMPv3 requests, the z/OS Cryptographic Services Integrated Cryptography Service Facility (ICSF) must be configured and running on the z/OS host on which the NetView program is also running.

Return Codes

0

Processing ended without errors.

1

Processing ended with unrecoverable errors.

2

Processing ended with general errors.

5

Processing ended abnormally.

Sending an SNMP GETNEXT request

The following example sends a GETNEXT request to an SNMP agent to retrieve the MIB variable following sysDescr.0:

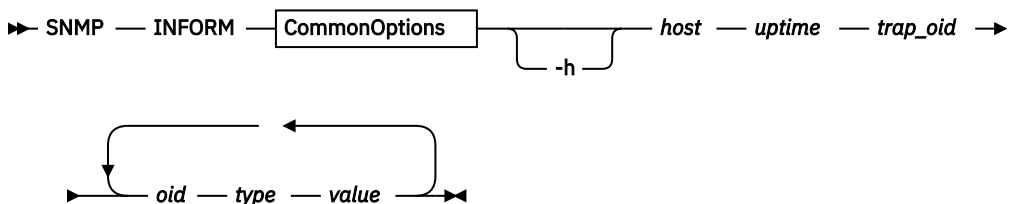
```
snmp getnext -c public 9.67.50.52 sysDescr.0
```

You receive a response similar to the following:

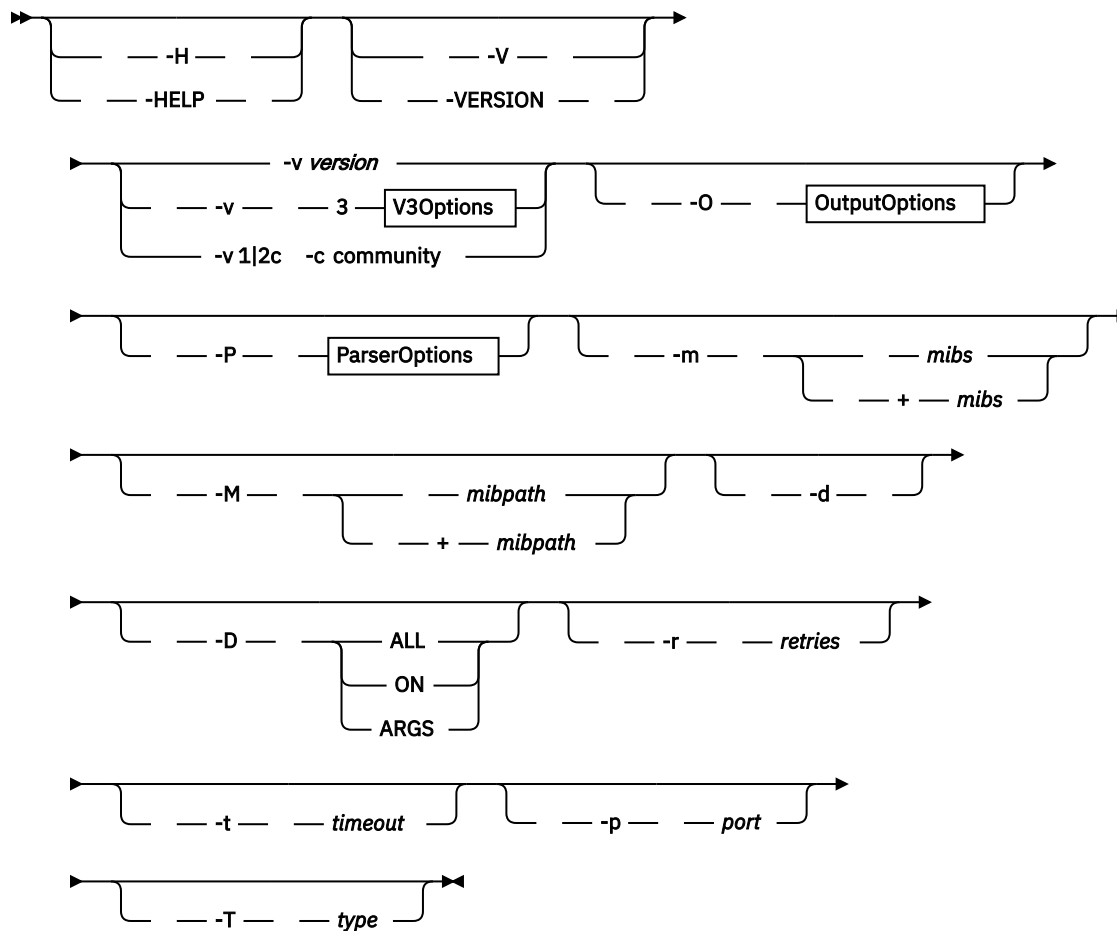
```
CNM005I SNMPv2-MIB::sysObjectID.0=OID: SUBAGENT-MIB::ibm.3.13
```

SNMP INFORM (NCCF; CNMESNMP)

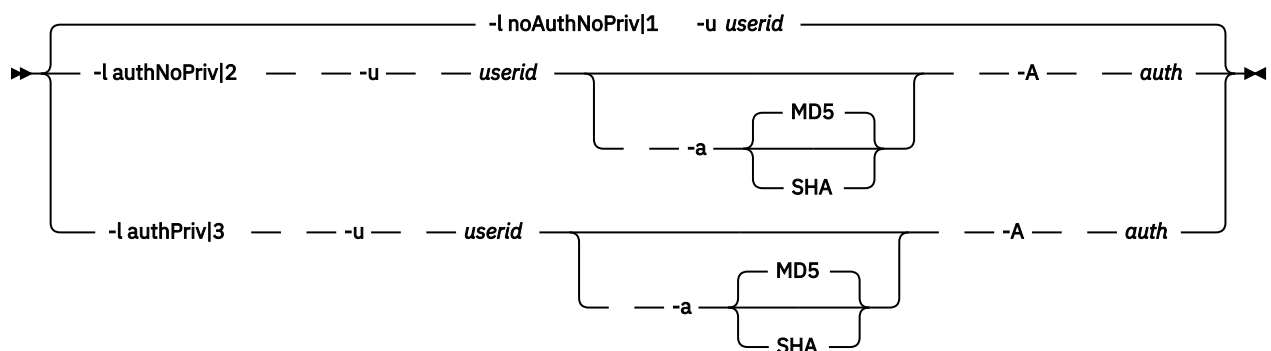
Syntax



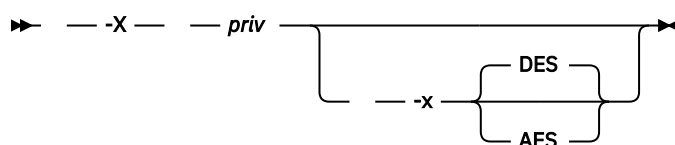
CommonOptions



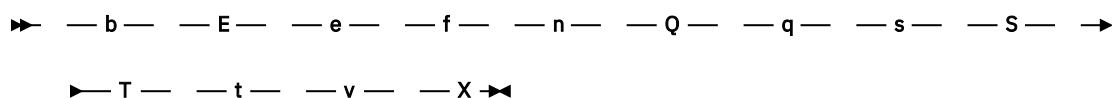
V3Options



V3Pr



OutputOptions



ParserOptions

Purpose of Command

The SNMP INFORM command sends an INFORM request PDU to an SNMP agent or manager.

Operand Descriptions

-h

Optionally specifies the destination IP host. When -h is specified for an IP host name, name server resolution to an IP address is performed in a separate process.

host

Specifies the destination IP host.

uptime

Indicates a time stamp representing the amount of time between the latest initialization of the agent and the trap.

trap_oid

Specifies the assigned name for the notification.

oid

Specifies the object ID (OID) of the MIB variable related to the event.

type

The *type* variable can be as follows:

i

Integer

u

Unsigned Integer

c

Counter32

s

Character String

x

Hexadecimal String

d

Decimal String

n

Null Object

o

OID

t

Timeticks

a

IP Address

b

Bits

value

Specifies the value to be assigned to the given MIB variable.

The following parameter descriptions are for the common options:

-a

Sets the authentication protocol used for authenticating SNMPv3 messages. The default value is MD5. When MD5 is used, the -a MD5 parameter can be omitted.

- A**
Sets the authentication pass phrase used for authenticated SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.
- c**
Specifies the community name. This parameter is required for SNMPv1 and SNMPv2c.
- d**
Specifies to include the contents of the input and output data packets with the output.
- D**
Used for debugging purposes.
- ON**
Turns on debugging of the command line interface.
- ALL**
Is the same as ON but includes debugging of command processing.
- ARGS**
Traces argument handling.
- H|HELP**
Displays the SNMP Command Menu. If other options are specified, they are ignored.
- l**
Specifies the level of authentication and encryption.
- noAuthNoPriv|1**
No authentication and no encryption. The **-u** parameter is required.
- authNoPriv|2**
MD5 or SHA1 authentication, with no encryption. The **-u** and **-A** parameters are required. MD5 authentication is the default. SHA1 authentication requires the **-a SHA** parameter.
- authPriv|3**
MD5 or SHA1 authentication and DES or AES encryption. The **-u** and **-A** and **-X** parameters are required. If the **-a** parameter is not specified, MD5 is the default value for authentication. If the **-x** parameter is not specified, DES is the default value for encryption. SHA1 authentication requires the **-a SHA** parameter. AES encryption requires the **-x AES** parameter.
- m**
Specifies the MIBs to parse for symbolic names. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.
- M**
Specifies the directories in the UNIX System Services HFS, from where the NetView program is to search for MIB source files. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.
- O**
Specifies the output options:
 - b**
Prevents attempts to resolve index elements to names.
 - E**
Modifies the input strings to include a backslash (\) before the quotation mark.
 - e**
Removes any symbolic labels from values.
 - f**
Prints the complete OID.
 - n**
Prints the OID in its fully-specified numeric form.
 - Q**
Removes the type information but keeps the equal sign (=).

- q** Removes the equal sign (=) and type information.
- s** Prints only the last symbolic part of the OID.
- S** Prints only the last symbolic part of the OID and precedes it with the name of the MIB which defines the object.
- T** Prints any printable characters enclosed in brackets [] after the hexadecimal encoding.
- t** Prints timeticks values as raw numbers.
- v** Prints only the value or values.
- X** Indicates the index components of OIDs with brackets [].
- p** Specifies the port on the SNMP agent to send this request. The default value is 161 for all requests except INFORM and TRAP, which default to 162.
- P** Specifies the parser options:
 - w** Displays warning messages while parsing MIB source files.
 - W** Displays additional warning messages while parsing MIB source files.
 - e** Displays MIB errors.
 - c** Disables ASN.1 comments to extend to the end of the MIB source comment lines.
 - u** Enables underscores in symbols.
 - R** Replaces MIB objects using the MIB file that was read last.
- r** Specifies the number of retries. The default value is 1.
- t** Specifies the timeout value between retries. The default value is 1 second.
- T** Specifies the transport type. These are valid values:
 - UDP
 - UDP6
 - UDPV6
 - UDPIPV6
 - TCP
 - TCP6
 - TCPV6
 - TCPIPV6

Notes:

1. If IPv6Env=MIXED was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address. To use a host name that resolves to an IPv6 address, specify a transport type appropriate for IPv6 networking (any of the types that end with 6).
2. If IPv6Env=ONLY was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP6. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP6 transport type, namely, an IPv6 address.
3. If IPv6Env=NONE was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address.

-u

The security name. This option is only valid for SNMPv3.

-v

Specifies which SNMP Protocol version to use. Valid values are 1, 2c, and 3. The default value is 1.

Note: For SNMP BULKWALK, SNMP GETBULK, and SNMP INFORM, valid values are 2c and 3.

-V|VERSION

Displays the NetView SNMP command version information. If other options are specified, they are ignored.

-x

Sets the encryption algorithm that is used for SNMPv3 messages. The default value is DES. When DES encryption is used, the -x parameter can be omitted. A value of AES indicates that AES 128-bit encryption is used; the synonym AES128 can be used as a valid keyword for the -x parameter.

-X

Sets the privacy pass phrase used for encrypted SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.

Usage Notes

Consider the following when using the SNMP command:

- The command-line interface is case-sensitive. Use the NETVASIS command or set OVERRIDE NETVASIS to YES.
- A maximum of 128 MIB variables is supported.
- Do not run SNMP commands in NetView preinitialization command lists.
- Values specified with INFORM that contain spaces must be enclosed in single or double quotation marks.
- Various NetView SNMP defaults can be modified with the COMMON.CNMSNMP definitions in the CNMSTYLE member. See the instructions in the CNMSTYLE member for more information.
- If AES encryption is being used for SNMPv3 requests, the z/OS Cryptographic Services Integrated Cryptography Service Facility (ICSF) must be configured and running on the z/OS host on which the NetView program is also running.

Return Codes

0

Processing ended without errors.

1

Processing ended with unrecoverable errors.

2

Processing ended with general errors.

5

Processing ended abnormally.

Example: Sending an SNMP INFORM request

The following example sends an INFORM PDU request:

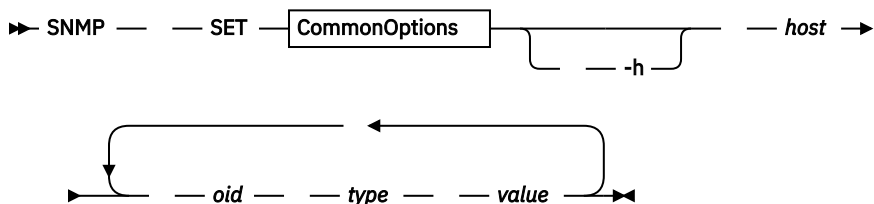
```
snmp inform -v2c -p 2005 -c public tvt2010 99 1.3.6 .1.3.6.1.2.1.1.6.0 s
'this is an inform2_pdu'
```

You receive a response similar to the following:

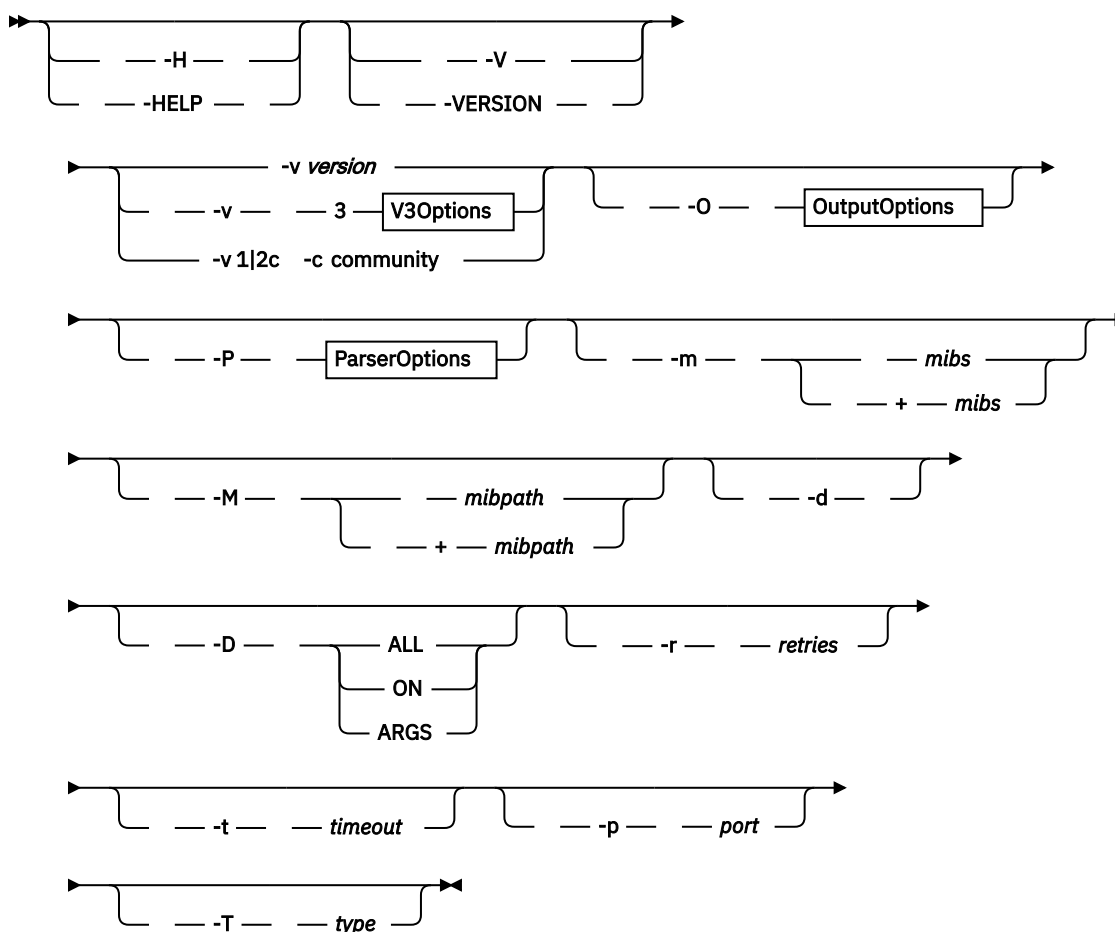
```
CNM007I SNMP INFORM request PDU sent successfully
```

SNMP SET (NCCF; CNMESNMP)

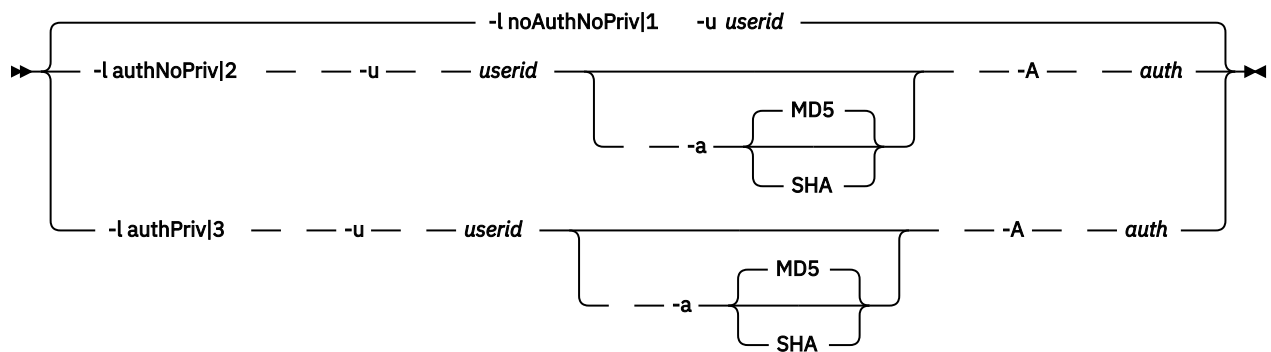
Syntax



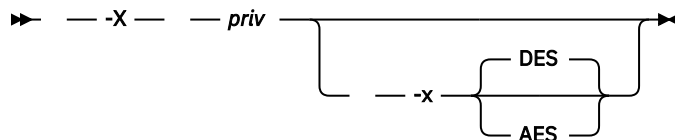
CommonOptions



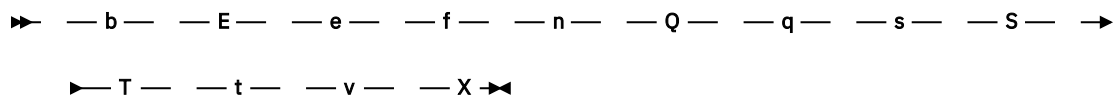
V3Options



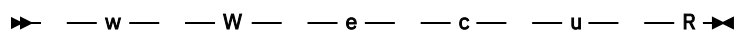
V3Pr



OutputOptions



ParserOptions



Purpose of Command

The SNMP SET command enables you to set one or more MIB variable values.

Operand Descriptions

-h

Optionally specifies the destination IP host. When `-h` is specified for an IP host name, name server resolution to an IP address is performed in a separate process.

host

Specifies the destination IP host.

oid

Specifies the object ID (OID) of the MIB variable related to the event.

type

The *type* variable can be as follows:

i

Integer

u

Unsigned Integer

s

Character String

x

Hexadecimal String

d

Decimal String

n

Null Object

- o** OID
- t** Timeticks
- a** IP Address
- b** Bits
- =** Use the type specified in the MIB file

value

Specifies the value to be assigned to the given MIB variable.

The following parameter descriptions are for the common options:

- a** Sets the authentication protocol used for authenticating SNMPv3 messages. The default value is MD5. When MD5 is used, the **-a MD5** parameter can be omitted.
- A** Sets the authentication pass phrase used for authenticated SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.
- c** Specifies the community name. This parameter is required for SNMPv1 and SNMPv2c.
- d** Specifies to include the contents of the input and output data packets with the output.
- D** Used for debugging purposes.
- ON** Turns on debugging of the command line interface.
- ALL** Is the same as ON but includes debugging of command processing.
- ARGS** Traces argument handling.
- H|HELP** Displays the SNMP Command Menu. If other options are specified, they are ignored.
- l** Specifies the level of authentication and encryption.
- noAuthNoPriv|1** No authentication and no encryption. The **-u** parameter is required.
- authNoPriv|2** MD5 or SHA1 authentication, with no encryption. The **-u** and **-A** parameters are required. MD5 authentication is the default. SHA1 authentication requires the **-a SHA** parameter.
- authPriv|3** MD5 or SHA1 authentication and DES or AES encryption. The **-u** and **-A** and **-X** parameters are required. If the **-a** parameter is not specified, MD5 is the default value for authentication. If the **-x** parameter is not specified, DES is the default value for encryption. SHA1 authentication requires the **-a SHA** parameter. AES encryption requires the **-x AES** parameter.
- m** Specifies the MIBs to parse for symbolic names. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.

-M

Specifies the directories in the UNIX System Services HFS, from where the NetView program is to search for MIB source files. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.

-O

Specifies the output options:

b

Prevents attempts to resolve index elements to names.

E

Modifies the input strings to include a backslash (\) before the quotation mark.

e

Removes any symbolic labels from values.

f

Prints the complete OID.

n

Prints the OID in its fully-specified numeric form.

Q

Removes the type information but keeps the equal sign (=).

q

Removes the equal sign (=) and type information.

s

Prints only the last symbolic part of the OID.

S

Prints only the last symbolic part of the OID and precedes it with the name of the MIB which defines the object.

T

Prints any printable characters enclosed in brackets [] after the hexadecimal encoding.

t

Prints timeticks values as raw numbers.

v

Prints only the value or values.

X

Indicates the index components of OIDs with brackets [].

-p

Specifies the port on the SNMP agent to send this request. The default value is 161 for all requests except INFORM and TRAP, which default to 162.

-P

Specifies the parser options:

w

Displays warning messages while parsing MIB source files.

W

Displays additional warning messages while parsing MIB source files.

e

Displays MIB errors.

c

Disables ASN.1 comments to extend to the end of the MIB source comment lines.

u

Enables underscores in symbols.

R

Replaces MIB objects using the MIB file that was read last.

- r**
Specifies the number of retries. The default value is 1.
- t**
Specifies the timeout value between retries. The default value is 1 second.
- T**
Specifies the transport type. These are valid values:
- UDP
 - UDP6
 - UDPV6
 - UDPIPV6
 - TCP
 - TCP6
 - TCPV6
 - TCPIPV6

Notes:

1. If IPv6Env=MIXED was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address. To use a host name that resolves to an IPv6 address, specify a transport type appropriate for IPv6 networking (any of the types that end with 6).
2. If IPv6Env=ONLY was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP6. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP6 transport type, namely, an IPv6 address.
3. If IPv6Env=NONE was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address.

- u**
The security name. This option is only valid for SNMPv3.

- v**
Specifies which SNMP Protocol version to use. Valid values are 1, 2c, and 3. The default value is 1.

Note: For SNMP BULKWALK, SNMP GETBULK, and SNMP INFORM, valid values are 2c and 3.

-V|VERSION

Displays the NetView SNMP command version information. If other options are specified, they are ignored.

- x**
Sets the encryption algorithm that is used for SNMPv3 messages. The default value is DES. When DES encryption is used, the -x parameter can be omitted. A value of AES indicates that AES 128-bit encryption is used; the synonym AES128 can be used as a valid keyword for the -x parameter.

- X**
Sets the privacy pass phrase used for encrypted SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.

Usage Notes

Consider the following when using the SNMP command:

- The command-line interface is case-sensitive. Use the NETVASIS command or set OVERRIDE NETVASIS to YES.

- A maximum of 128 MIB variables is supported.
- Do not run SNMP commands in NetView preinitialization command lists.
- Values specified with SET that contain spaces must be enclosed in single or double quotation marks.
- Various NetView SNMP defaults can be modified with the COMMON.CNMSNMP definitions in the CNMSTYLE member. See the instructions in the CNMSTYLE member for more information.
- If AES encryption is being used for SNMPv3 requests, the z/OS Cryptographic Services Integrated Cryptography Service Facility (ICSF) must be configured and running on the z/OS host on which the NetView program is also running.

Return Codes

- 0**
Processing ended without errors.
- 1**
Processing ended with unrecoverable errors.
- 2**
Processing ended with general errors.
- 5**
Processing ended abnormally.

Sending an SNMP SET request

The following example sends a SET request to an SNMP agent to change the value of `sysContact.0`:

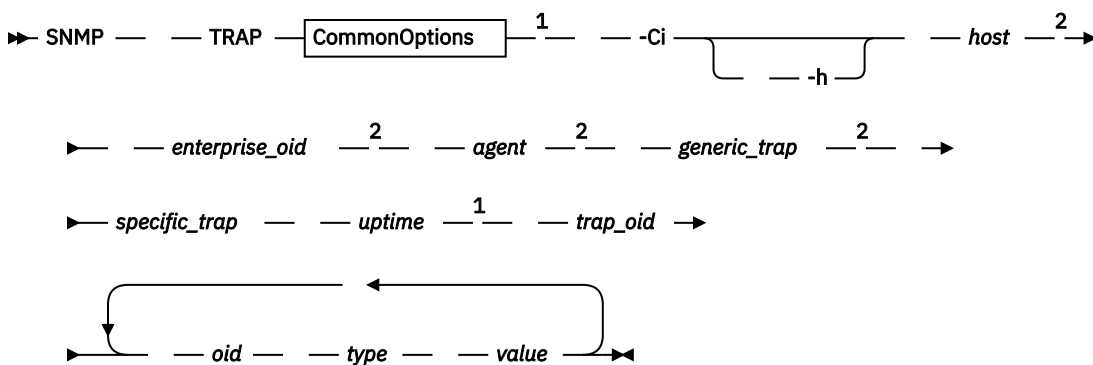
```
snmp set -c public tvt2009.raleigh.ibm.com sysContact.0
s "Nino Culotta x1251"
```

You receive a response similar to the following:

```
CNM005I SNMPv2-MIB::sysContact.0=STRING: Nino Culotta x1251
```

SNMP TRAP (NCCF; CNMESNMP)

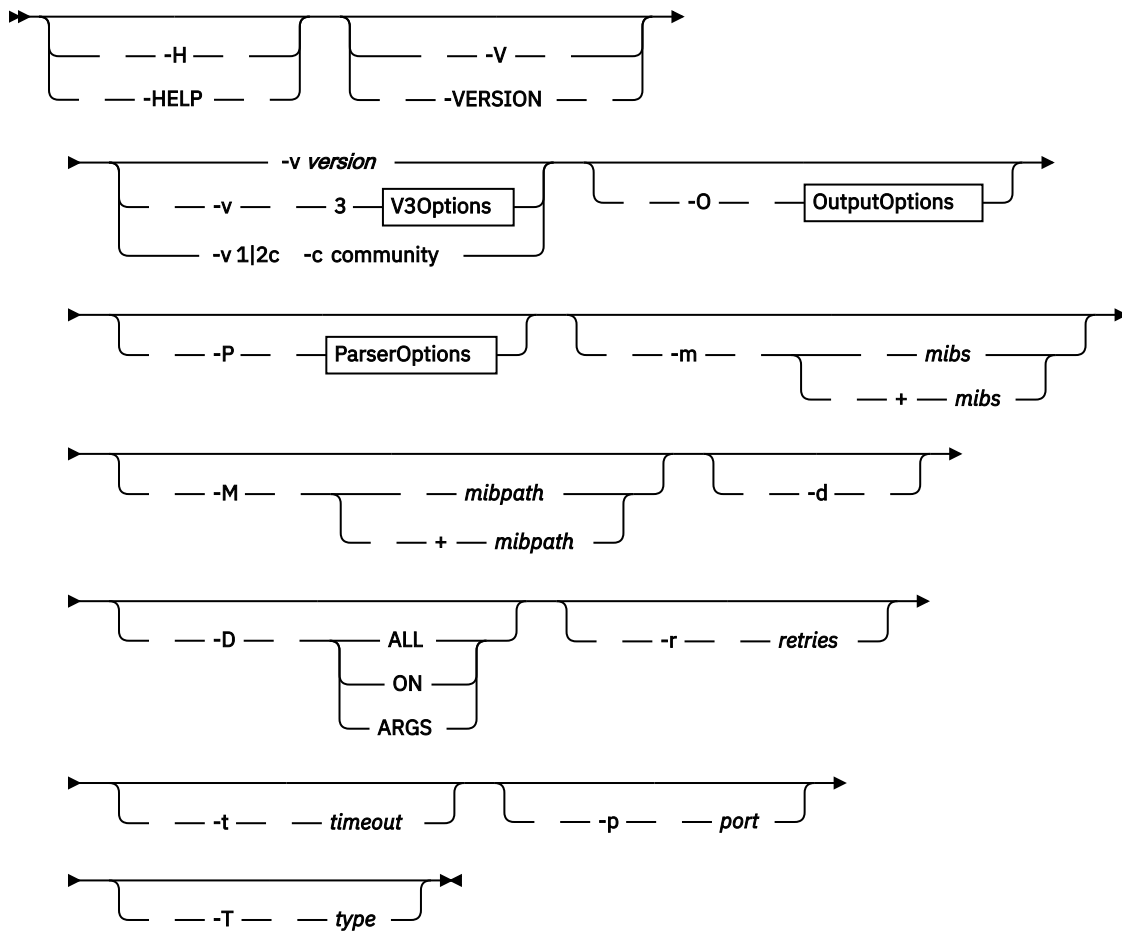
Syntax



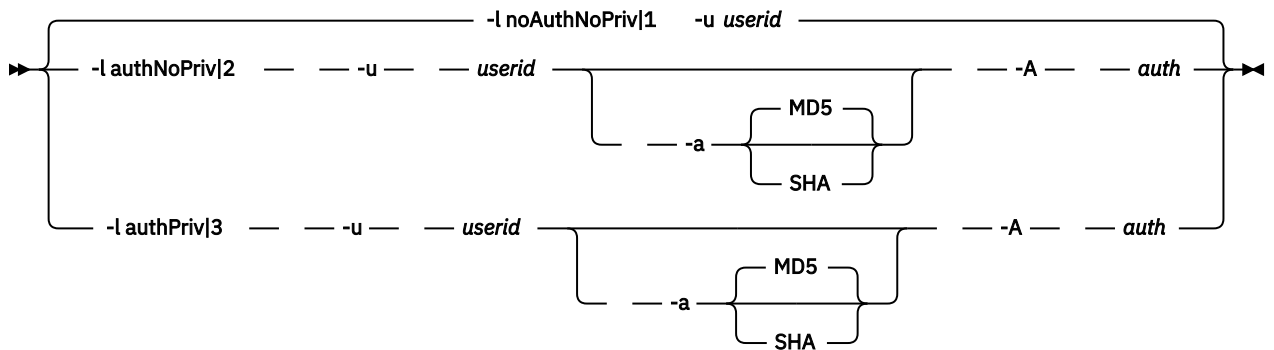
Notes:

- ¹ This option is only valid for SNMPv2c and SNMPv3.
- ² This option is only valid for SNMPv1.

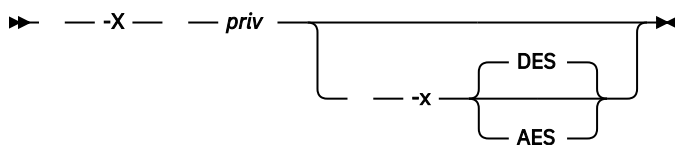
CommonOptions



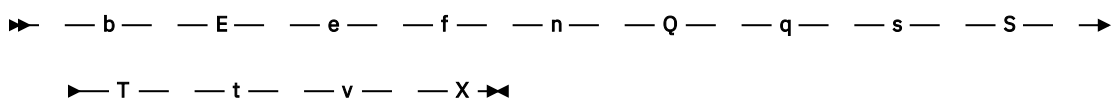
V3Options



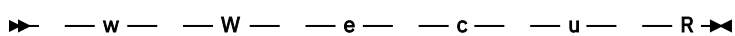
V3Pr



OutputOptions



ParserOptions



Purpose of Command

The SNMP TRAP command sends a TRAP PDU to an SNMP agent.

Operand Descriptions

-Ci

Specifies to send an INFORM request PDU to an SNMP agent or manager instead of a TRAP request PDU. This option is only valid for SNMP Version 2c and 3.

-h

Optionally specifies the destination IP host. When -h is specified for an IP host name, name server resolution to an IP address is performed in a separate process.

host

Specifies the destination IP host.

enterprise_oid

Identifies the management enterprise under whose registration authority the trap was defined. This option is only valid for SNMP Version 1.

agent

Specifies the IP host name or address of the SNMP agent. This option is only valid for SNMP Version 1.

generic_trap

Specifies a generic event being reported. This option is only valid for SNMP Version 1. Possible events are listed as follows:

Trap Type/Name	Description
----------------	-------------

0/coldStart	Agent is starting
--------------------	-------------------

1/warmStart	Agent is restarting
--------------------	---------------------

2/linkDown	Status of an interface has changed from Up to Down
-------------------	--

3/linkUp	Status of an interface has changed from Down to UP
-----------------	--

4/authenticationFailure	Message received from an SNMP manager with a non-valid community name specified
--------------------------------	---

5/egpNeighborLoss	Status of an EGP peer changed to Down
--------------------------	---------------------------------------

6/enterpriseSpecific	<i>Specific_trap</i> defines the information for this TRAP
-----------------------------	--

specific_trap

Specifies a more specific indication of the event being reported. This option is only valid for SNMP Version 1.

uptime

Indicates a time stamp representing the amount of time between the latest initialization of the agent and the trap.

trap_oid

Specifies the assigned name for the notification. This option is only valid for SNMP Version 2c and 3.

oid

Specifies the object ID (OID) of the MIB variable related to the event.

type

The *type* variable can be as follows:

i

Integer

u	Unsigned Integer
c	Counter32
s	Character String
x	Hexadecimal String
d	Decimal String
n	Null Object
o	OID
t	Timeticks
a	IP Address
b	Bits

value

Specifies the value to be assigned to the given MIB variable.

The following parameter descriptions are for the common options:

-a

Sets the authentication protocol used for authenticating SNMPv3 messages. The default value is MD5. When MD5 is used, the -a MD5 parameter can be omitted.

-A

Sets the authentication pass phrase used for authenticated SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.

-c

Specifies the community name. This parameter is required for SNMPv1 and SNMPv2c.

-d

Specifies to include the contents of the input and output data packets with the output.

-D

Used for debugging purposes.

ON

Turns on debugging of the command line interface.

ALL

Is the same as ON but includes debugging of command processing.

ARGS

Traces argument handling.

-H|HELP

Displays the SNMP Command Menu. If other options are specified, they are ignored.

-l

Specifies the level of authentication and encryption.

noAuthNoPriv|1

No authentication and no encryption. The -u parameter is required.

authNoPriv|2

MD5 or SHA1 authentication, with no encryption. The **-u** and **-A** parameters are required. MD5 authentication is the default. SHA1 authentication requires the **-a SHA** parameter.

authPriv|3

MD5 or SHA1 authentication and DES or AES encryption. The **-u** and **-A** and **-X** parameters are required. If the **-a** parameter is not specified, MD5 is the default value for authentication. If the **-x** parameter is not specified, DES is the default value for encryption. SHA1 authentication requires the **-a SHA** parameter. AES encryption requires the **-x AES** parameter.

-m

Specifies the MIBs to parse for symbolic names. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.

-M

Specifies the directories in the UNIX System Services HFS, from where the NetView program is to search for MIB source files. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.

-O

Specifies the output options:

b

Prevents attempts to resolve index elements to names.

E

Modifies the input strings to include a backslash (\) before the quotation mark.

e

Removes any symbolic labels from values.

f

Prints the complete OID.

n

Prints the OID in its fully-specified numeric form.

Q

Removes the type information but keeps the equal sign (=).

q

Removes the equal sign (=) and type information.

s

Prints only the last symbolic part of the OID.

S

Prints only the last symbolic part of the OID and precedes it with the name of the MIB which defines the object.

T

Prints any printable characters enclosed in brackets [] after the hexadecimal encoding.

t

Prints timeticks values as raw numbers.

v

Prints only the value or values.

X

Indicates the index components of OIDs with brackets [].

-p

Specifies the port on the SNMP agent to send this request. The default value is 161 for all requests except INFORM and TRAP, which default to 162.

-P

Specifies the parser options:

w

Displays warning messages while parsing MIB source files.

W

Displays additional warning messages while parsing MIB source files.

e

Displays MIB errors.

c

Disables ASN.1 comments to extend to the end of the MIB source comment lines.

u

Enables underscores in symbols.

R

Replaces MIB objects using the MIB file that was read last.

-r

Specifies the number of retries. The default value is 1.

-t

Specifies the timeout value between retries. The default value is 1 second.

-T

Specifies the transport type. These are valid values:

- UDP
- UDP6
- UDPV6
- UDPIPV6
- TCP
- TCP6
- TCPV6
- TCPIPV6

Notes:

1. If IPv6Env=MIXED was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address. To use a host name that resolves to an IPv6 address, specify a transport type appropriate for IPv6 networking (any of the types that end with 6).
2. If IPv6Env=ONLY was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP6. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP6 transport type, namely, an IPv6 address.
3. If IPv6Env=NONE was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address.

-u

The security name. This option is only valid for SNMPv3.

-v

Specifies which SNMP Protocol version to use. Valid values are 1, 2c, and 3. The default value is 1.

Note: For SNMP BULKWALK, SNMP GETBULK, and SNMP INFORM, valid values are 2c and 3.

-V|VERSION

Displays the NetView SNMP command version information. If other options are specified, they are ignored.

Sets the encryption algorithm that is used for SNMPv3 messages. The default value is DES. When DES encryption is used, the `-x` parameter can be omitted. A value of AES indicates that AES 128-bit encryption is used; the synonym AES128 can be used as a valid keyword for the `-x` parameter.

Sets the privacy pass phrase used for encrypted SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.

Consider the following when using the SNMP command:

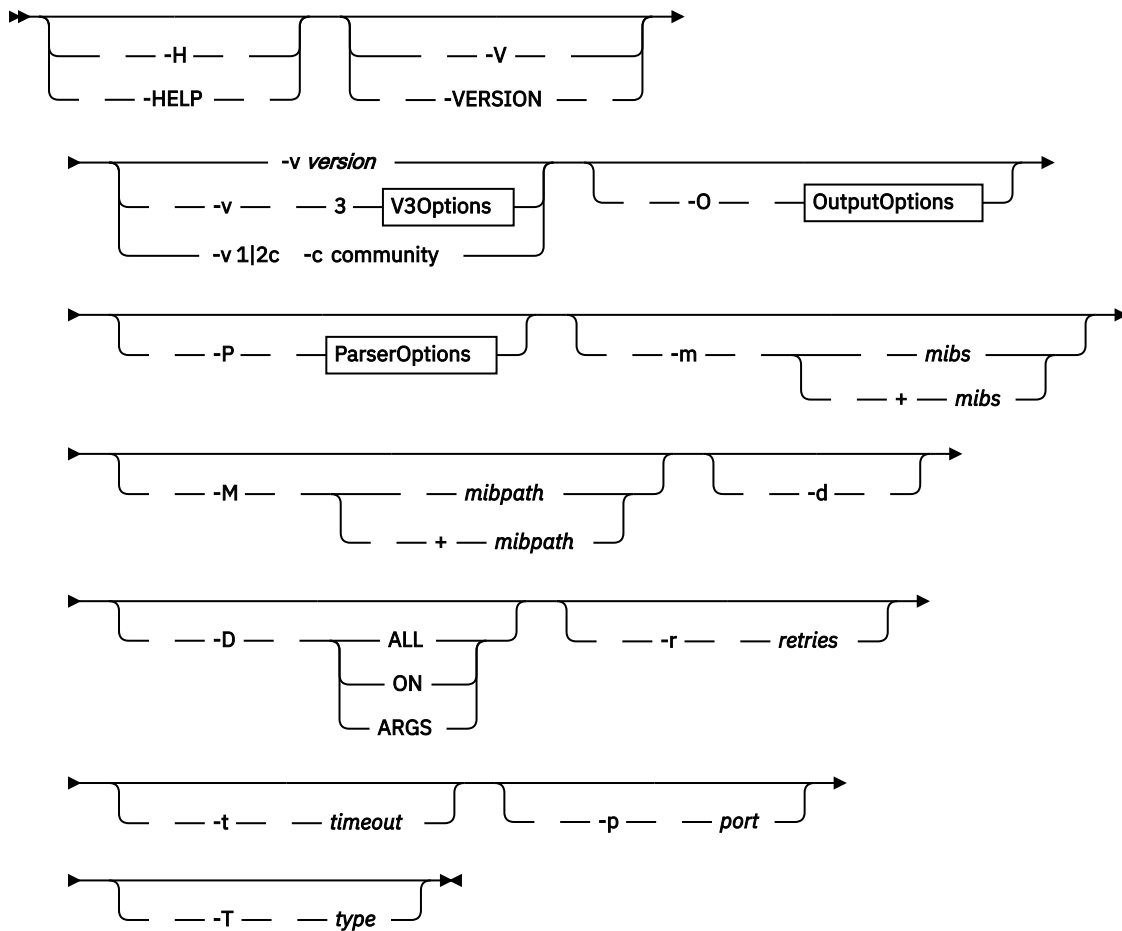
- Processing ended abnormally.

The following example sends a TRAP PDU request:

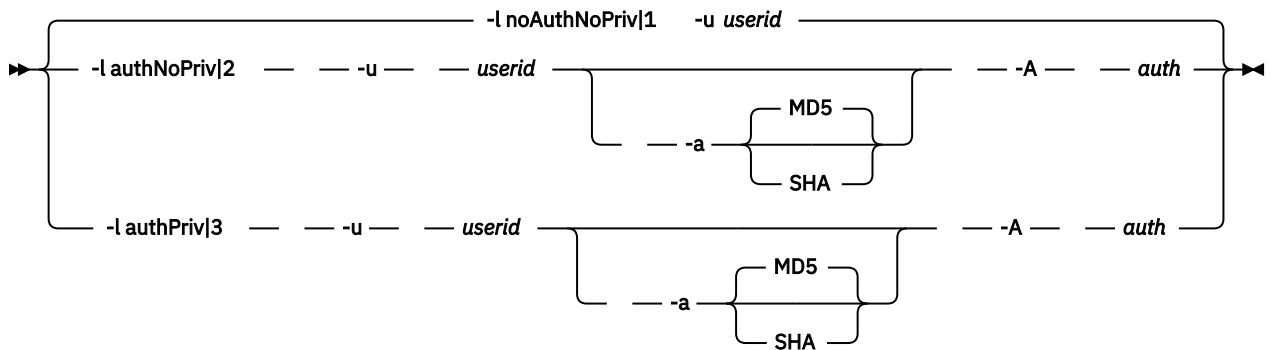
You receive a response similar to the following:

SNMP WALK (NCCF; CNMESNMP)

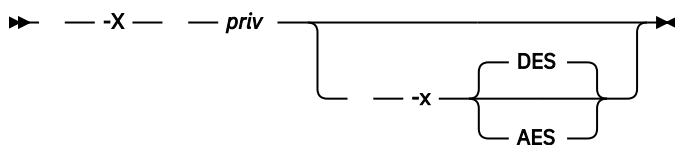
CommonOptions



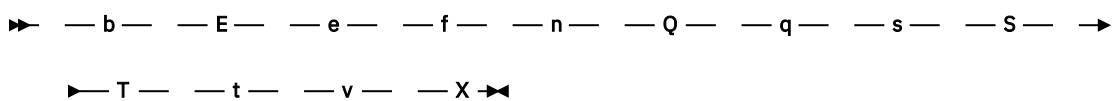
V3Options



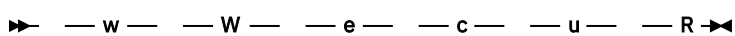
V3Pr



OutputOptions



ParserOptions



Purpose of Command

The SNMP WALK command enables you to retrieve the values for all of the MIB variables in a specified branch.

Operand Descriptions

-C

i

Specifies to include the requested OID in the results.

p

Specifies to print the number of variables found during the walk.

-h

Optionally specifies the destination IP host. When -h is specified for an IP host name, name server resolution to an IP address is performed in a separate process.

host

Specifies the destination IP host.

oid

Specifies the object ID (OID) of the MIB variable.

The following parameter descriptions are for the common options:

-a

Sets the authentication protocol used for authenticating SNMPv3 messages. The default value is MD5. When MD5 is used, the -a MD5 parameter can be omitted.

-A

Sets the authentication pass phrase used for authenticated SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.

-c

Specifies the community name. This parameter is required for SNMPv1 and SNMPv2c.

-d

Specifies to include the contents of the input and output data packets with the output.

-D

Used for debugging purposes.

ON

Turns on debugging of the command line interface.

ALL

Is the same as ON but includes debugging of command processing.

ARGS

Traces argument handling.

-H|HELP

Displays the SNMP Command Menu. If other options are specified, they are ignored.

-l

Specifies the level of authentication and encryption.

noAuthNoPriv|1

No authentication and no encryption. The -u parameter is required.

authNoPriv|2

MD5 or SHA1 authentication, with no encryption. The -u and -A parameters are required. MD5 authentication is the default. SHA1 authentication requires the -a SHA parameter.

authPriv|3

MD5 or SHA1 authentication and DES or AES encryption. The -u and -A and -X parameters are required. If the -a parameter is not specified, MD5 is the default value for authentication. If the -x

parameter is not specified, DES is the default value for encryption. SHA1 authentication requires the **-a** SHA parameter. AES encryption requires the **-x** AES parameter.

-m

Specifies the MIBs to parse for symbolic names. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.

-M

Specifies the directories in the UNIX System Services HFS, from where the NetView program is to search for MIB source files. You can optionally specify a plus sign (+) to prepend the specified values to the default values or those specified in COMMON.CNMSNMP.MIBPATH.

-O

Specifies the output options:

b

Prevents attempts to resolve index elements to names.

E

Modifies the input strings to include a backslash (\) before the quotation mark.

e

Removes any symbolic labels from values.

f

Prints the complete OID.

n

Prints the OID in its fully-specified numeric form.

Q

Removes the type information but keeps the equal sign (=).

q

Removes the equal sign (=) and type information.

s

Prints only the last symbolic part of the OID.

S

Prints only the last symbolic part of the OID and precedes it with the name of the MIB which defines the object.

T

Prints any printable characters enclosed in brackets [] after the hexadecimal encoding.

t

Prints timeticks values as raw numbers.

v

Prints only the value or values.

X

Indicates the index components of OIDs with brackets [].

-p

Specifies the port on the SNMP agent to send this request. The default value is 161 for all requests except INFORM and TRAP, which default to 162.

-P

Specifies the parser options:

w

Displays warning messages while parsing MIB source files.

W

Displays additional warning messages while parsing MIB source files.

e

Displays MIB errors.

- c**
Disables ASN.1 comments to extend to the end of the MIB source comment lines.
- u**
Enables underscores in symbols.
- R**
Replaces MIB objects using the MIB file that was read last.

-r
Specifies the number of retries. The default value is 1.

-t
Specifies the timeout value between retries. The default value is 1 second.

-T
Specifies the transport type. These are valid values:

- UDP
- UDP6
- UDPV6
- UDPIPV6
- TCP
- TCP6
- TCPV6
- TCPIPV6

Notes:

1. If IPv6Env=MIXED was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address. To use a host name that resolves to an IPv6 address, specify a transport type appropriate for IPv6 networking (any of the types that end with 6).
2. If IPv6Env=ONLY was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP6. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP6 transport type, namely, an IPv6 address.
3. If IPv6Env=NONE was defined for the NetView program and no transport type was specified, the SNMP command assumes a transport type of UDP. If a host name was specified, then the SNMP command expects to resolve that host name to an IP address appropriate for the UDP transport type, namely, an IPv4 address.

-u
The security name. This option is only valid for SNMPv3.

-v
Specifies which SNMP Protocol version to use. Valid values are 1, 2c, and 3. The default value is 1.

Note: For SNMP BULKWALK, SNMP GETBULK, and SNMP INFORM, valid values are 2c and 3.

-V|VERSION

Displays the NetView SNMP command version information. If other options are specified, they are ignored.

-x
Sets the encryption algorithm that is used for SNMPv3 messages. The default value is DES. When DES encryption is used, the -x parameter can be omitted. A value of AES indicates that AES 128-bit encryption is used; the synonym AES128 can be used as a valid keyword for the -x parameter.

-X
Sets the privacy pass phrase used for encrypted SNMPv3 messages. This pass phrase must be at least eight characters in length. Pass phrases are case-sensitive.

Usage Notes

Consider the following when using the SNMP command:

- The command-line interface is case-sensitive. Use the NETVASIS command or set OVERRIDE NETVASIS to YES.
- Do not run SNMP commands in NetView preinitialization command lists.
- Various NetView SNMP defaults can be modified with the COMMON.CNMSNMP definitions in the CNMSTYLE member. See the instructions in the CNMSTYLE member for more information.
- If AES encryption is being used for SNMPv3 requests, the z/OS Cryptographic Services Integrated Cryptography Service Facility (ICSF) must be configured and running on the z/OS host on which the NetView program is also running.

Return Codes

0

Processing ended without errors.

1

Processing ended with unrecoverable errors.

2

Processing ended with general errors.

5

Processing ended abnormally.

Sending an SNMP WALK request

The following example sends an SNMPv2c WALK system request to an IP host name (nmp130) with a community name of public2c, and returns the actual MIB in which the value is found and does not allow conversion of timeticks from raw data:

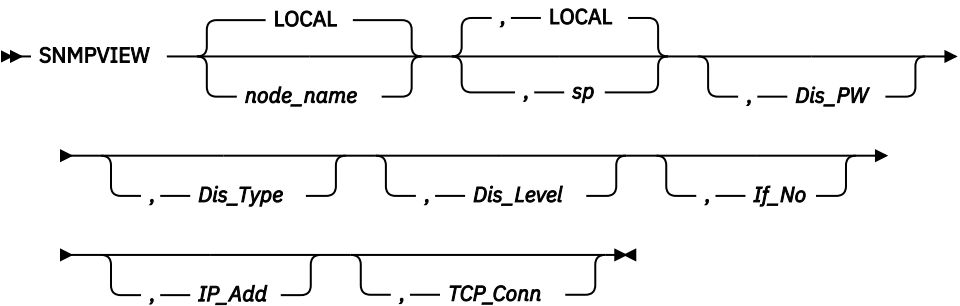
```
snmp walk -c2v -c public2v -h nmp130 system
```

You receive a response similar to the following:

```
CNM005I SNMPv2-MIB::sysDescr.0 = STRING: Sysname: OS/390 Nodename: NMP130
                                     Release: 19.00   Version: 03 Machine: 2094
CNM005I SNMPv2-MIB::sysObjectID.0 = OID: SNMPv2-SMI::enterprises.2.3.13
CNM005I SNMPv2-MIB::sysUpTime.0 = Timeticks: (23705200) 2 days, 17:50:52.00
CNM005I SNMPv2-MIB::sysContact.0 = STRING: SNMPBASE - Unspecified
CNM005I SNMPv2-MIB::sysName.0 = STRING: SNMPBASE - Unspecified
CNM005I SNMPv2-MIB::sysLocation.0 = STRING: SNMPBASE - Unspecified
CNM005I SNMPv2-MIB::sysServices.0 = INTEGER: 0
CNM005I SNMPv2-MIB::sysORLastChange.0 = Timeticks: (2100) 0:00:21.00
CNM005I SNMPv2-MIB::sysORID.1 = OID: SNMPv2-SMI::enterprises.2.11.7.1
CNM005I SNMPv2-MIB::sysORID.2 = OID: SNMPv2-SMI::enterprises.2.11.7.2
CNM005I SNMPv2-MIB::sysORDescr.1 = STRING: z/OS SNMP Agent
CNM005I SNMPv2-MIB::sysORDescr.2 = STRING: z/OS TCP/IP SNMP Subagent
CNM005I SNMPv2-MIB::sysORUpTime.1 = Timeticks: (0) 0:00:00.00
CNM005I SNMPv2-MIB::sysORUpTime.2 = Timeticks: (2100) 0:00:21.00
```

SNMPVIEW (AON)

Syntax



IBM-Defined Synonyms

Command or Operand	Synonym
SNMPVIEW	SNMPV
SNMPVIEW	SV

Purpose of Command

The SNMPVIEW command can be used to collect logically grouped portions of MIB information about a resource and return this information in REXX variables to the calling program. For more information about using the SNMP views function, see *IBM Z NetView User's Guide: Automated Operations Network*.

Operand Descriptions

node_name

Specifies the host name or TCP/IP address of the host from where the information is to be gathered. The default is LOCAL, which will use the local IP stack.

sp

Specifies the MVS TCP/IP service point to which the command is sent. The default is LOCAL, which uses the local IP stack.

Dis_PW

The community name used for the MIB get requests.

Dis_Type

The type of resource. Valid values are:

MVS

MVS TCP/IP stack

IP

Generic IP resource

Dis_Level

The level of screen display. Valid values are:

SYS

System screen

IF

Interface list

IFD

Interface detail. If IFD is specified, a value for *If_No* is also required.

CONN

Connection list. If CONN is specified, a value for *IP_Add* is also required.

CONND

Connection detail. If CONND is specified, *Dis_Type*=MVS and a value for *TCP_Conn* are also required.

If_No

The interface number for data collection. This is required if a screen display level of IFD is specified.

IP_Add

The IP address for data collection. This is required if a screen display level of CONN is specified.

TCP_Conn

The TCP/IP connection. This is required if a screen display level of CONND is specified. The connection types consist of the following:

- Local IP address
- Local Port
- Remote IP address
- Remote Port

and are specified in the following format:

```
LocalIpAddr.LocalPort.RemIpAddr.RemPort.
```

Restrictions

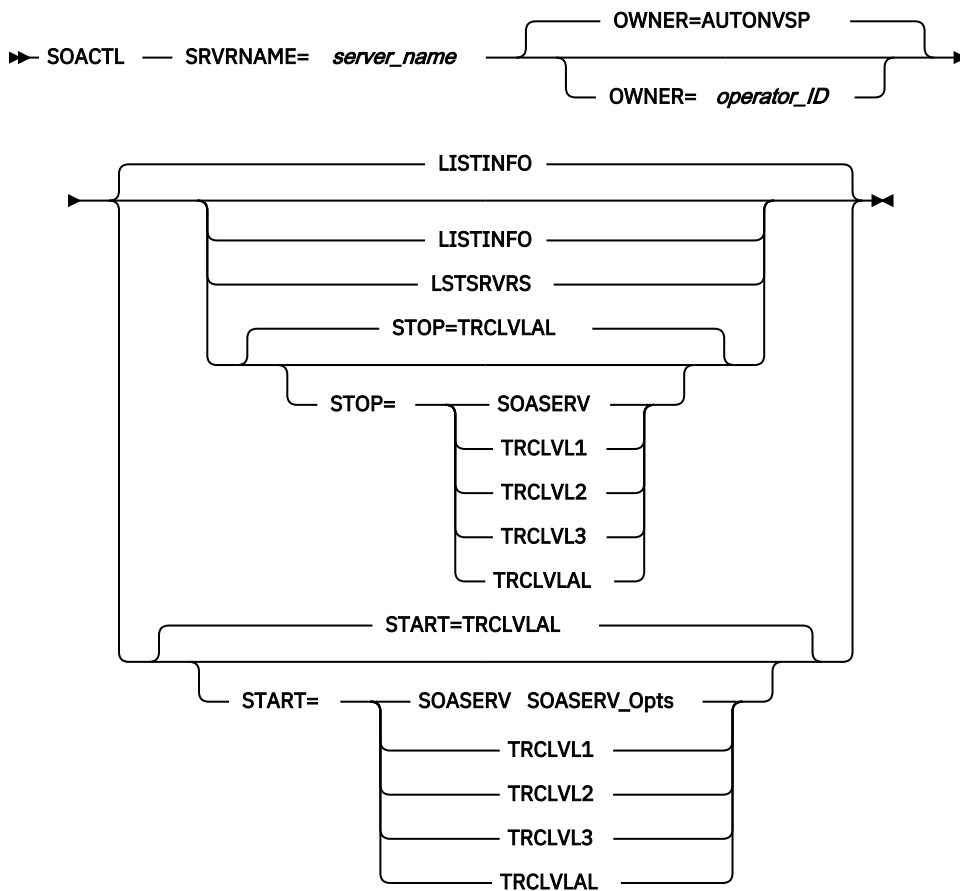
The following restrictions apply to the SNMPView command:

- The AON tower and the TCP subtower must be enabled in the CNMSTYLE member to successfully run this command.
- The SNMPView function does not support IPv6 addressing.

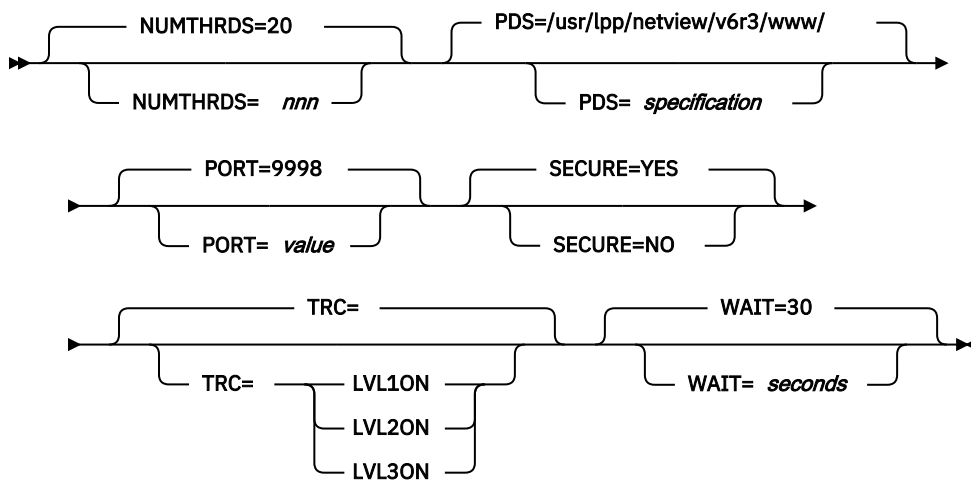
SOACTL (NCCF)

Syntax

NCCF SOACTL



SOASERV_Opts



Purpose of Command

You can use the SOACTL command to control Web Services server operations.

Operand Descriptions

LISTINFO

Displays detailed information about the specified Web Services server.

LSTSRVRS

Displays the status of the Web Services servers.

NUMTHRDS=*nnn*

Specifies the number of threads that are created to service incoming and outgoing IP connections. The minimum value is 1. Values greater than 2147483647 default to 2147483647.

The value for NUMTHRDS is assigned in the following order:

1. The NUMTHRDS value specified with the SOACTL command
2. The NVSP.*srvrname*.NUMTHRDS variable in the CNMSTYLE member
3. The default value is 20.

OWNER=*operator_ID*

Specifies the operator ID under which the Web Services server was started. This is the owner task. The Web Services server runs on a VOST attached to this task. The default value for the OWNER operand is the autotask name specified on the `fuction.autotask.NVSOATSK` CNMSTYLE statement.

When starting multiple Web Services servers, make sure that the OWNER task is unique and that a prior Web Services server is not running on this task.

The OWNER keyword is valid only with the START=SOASERV option.

PDS=*specification*

Specifies the PDS name or USS directory where server initialization files are located.

Specify the MVS-based PDS as `/'base-pds'`. To use gif or jpeg images in an html file, store the image as `base-pds.GIF(imagename)` or `base-pds.JPEG(imagename)`.

An example follows:

- PDS specification on the SOACTL command or on the NVSP.*srvrname*.PDS statement: `// 'USER . INIT'`
- HTML file location: `USER . INIT (MYHTML)`
- GIF image location: `USER . INIT . GIF (MYGIF)`
- HTML source (image URL address reference): `GIF (MYGIF)`

The value for PDS is assigned in the following order:

1. The PDS or directory value specified with the SOACTL command
2. The NVSP.*srvrname*.PDS variable in the CNMSTYLE member
3. The default value of `/usr/lpp/netview/v6r3/www/`

If you change the PDS default subdirectory, copy all the files in the existing subdirectory to the new location, keeping the same file permissions.

PORT=*value*

Specifies the port where the Web Services server is listening for connection requests. The minimum value is 0. Values greater than 2147483647 default to 2147483647.

The value for PORT is assigned in the following order:

1. The PORT value specified with the SOACTL command
2. The NVSP.*srvrname*.PORT variable in the CNMSTYLE member
3. The default value is 9998.

SECURE=NO|YES

Specifies whether the transport is secured with SSL encryption.

The value for SECURE is assigned in the following order:

1. The SECURE value specified with the SOACTL command
2. The NVSP.*srvrname*.SECURE variable in the CNMSTYLE member
3. The default value is YES.

SOASERV

Specifies the Web Services server.

SRVRNAME=server_name

Specifies the name of the target server. This is a required keyword, except with the LSTSRVRS option.

START

Starts the Web Services server or the specified trace.

STOP

Stops the Web Services server or the specified trace.

TRC=option

Specifies the initial trace level values. You can specify the following options:

LVL1ON

Sets the level 1 debug trace on.

LVL2ON

Sets the level 2 debug trace on.

LVL3ON

Sets the level 3 debug trace on.

The value for TRC is assigned in the following order:

1. The TRC value specified with the SOACTL command
2. The NVSP.*srvrname*.TRC variable in the CNMSTYLE member
3. The default value is blank, which is all trace levels turned off.

TRCLVLAL

Starts all debug trace levels. This is the default value.

TRCLVL1

Starts the level 1 debug trace.

TRCLVL2

Starts the level 2 debug trace.

TRCLVL3

Starts the level 3 debug trace.

WAIT=seconds

Specifies the number of seconds to wait for a command response. You can specify a value in the range of 1 - 10000000. Values greater than 10000000 default to 10000000. If you specify 0 (no wait), the default value of 30 seconds is used.

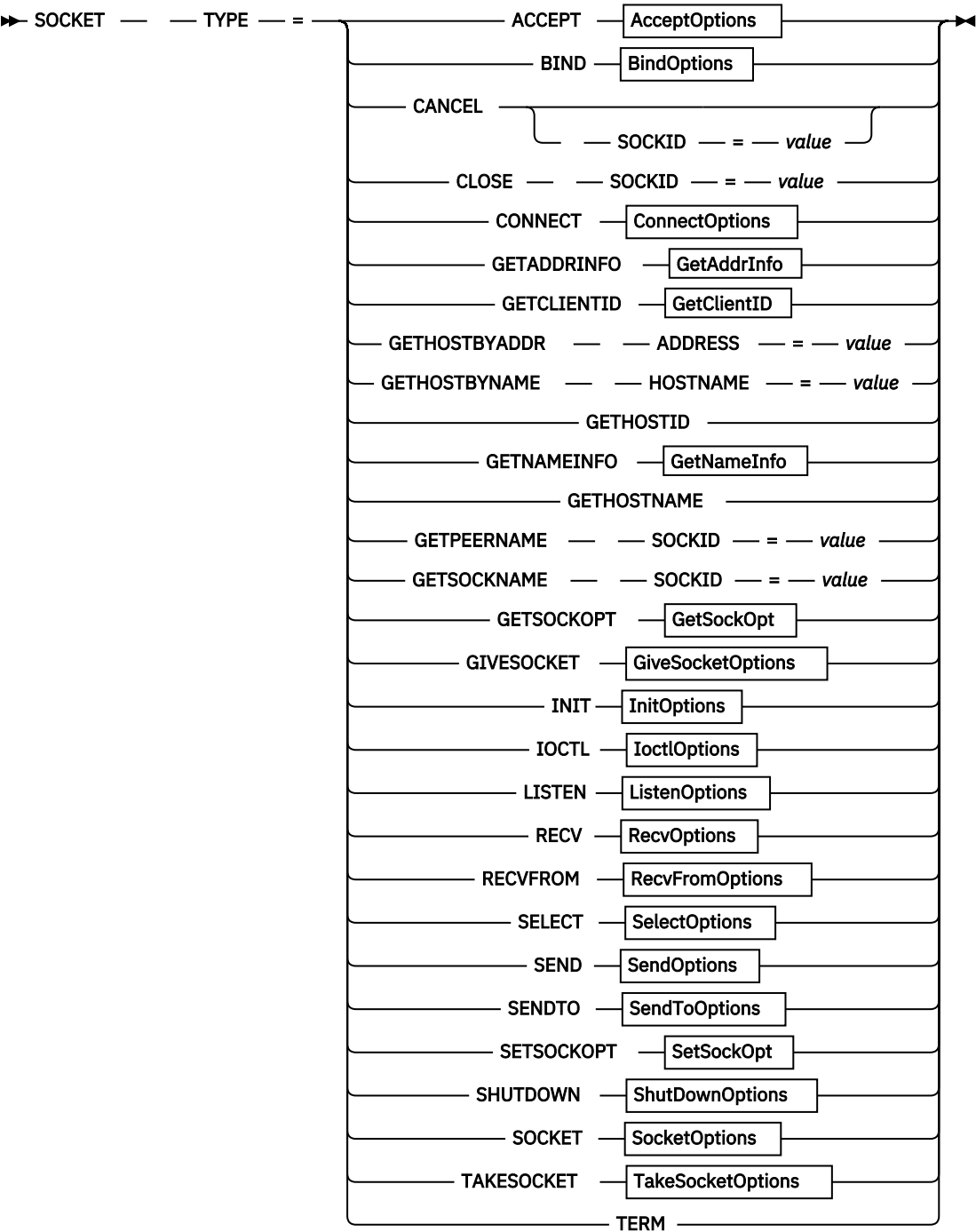
The value for WAIT is assigned in the following order:

1. The WAIT value specified with the SOACTL command
2. The NVSP.*srvrname*.WAIT variable in the CNMSTYLE member
3. The default value is 30.
4. The minimum value is 1. Values greater than 10000000 default to 10000000.

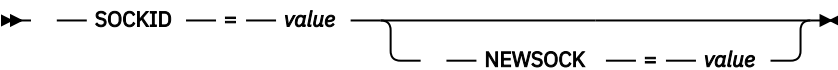
SOCKET (NCCF)

Syntax

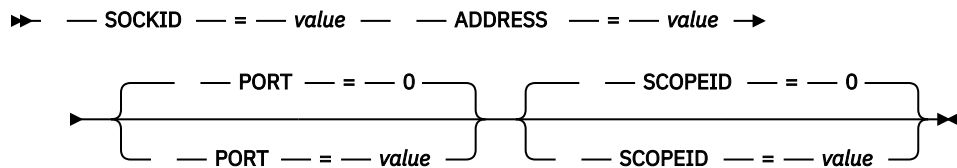
SOCKET



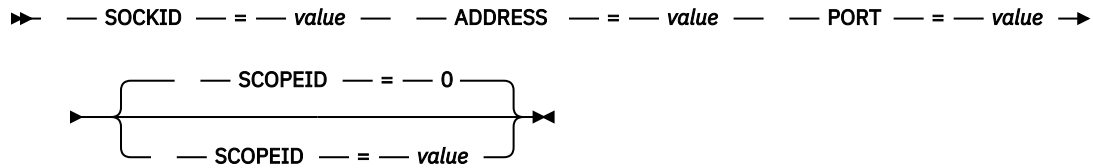
AcceptOptions



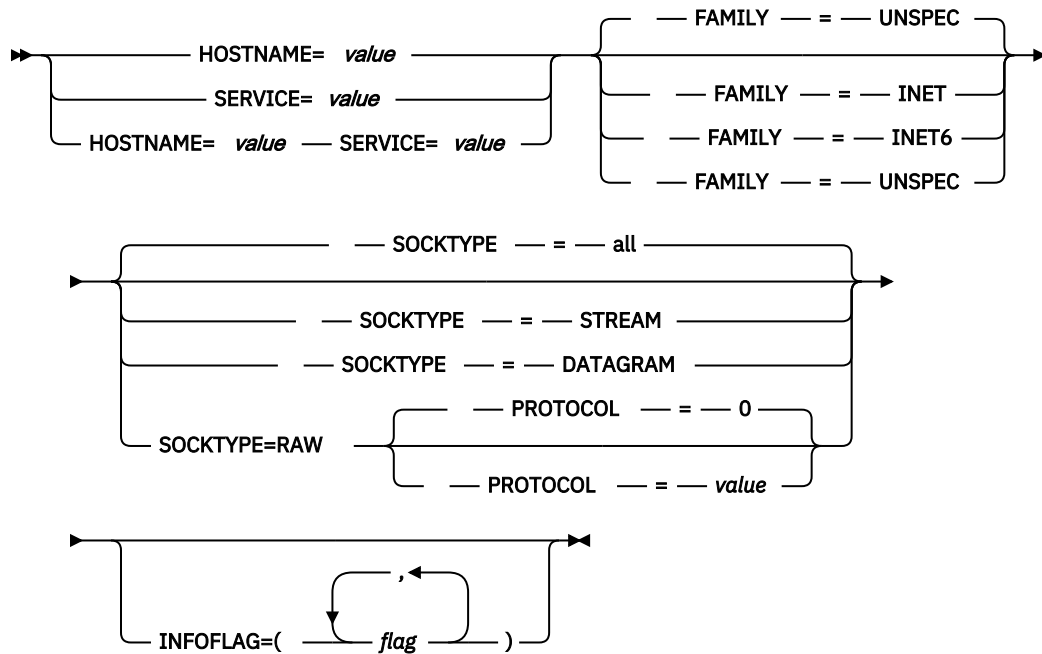
BindOptions



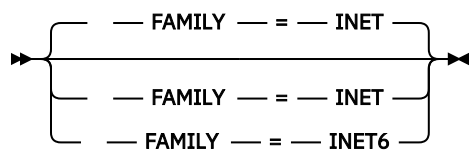
ConnectOptions



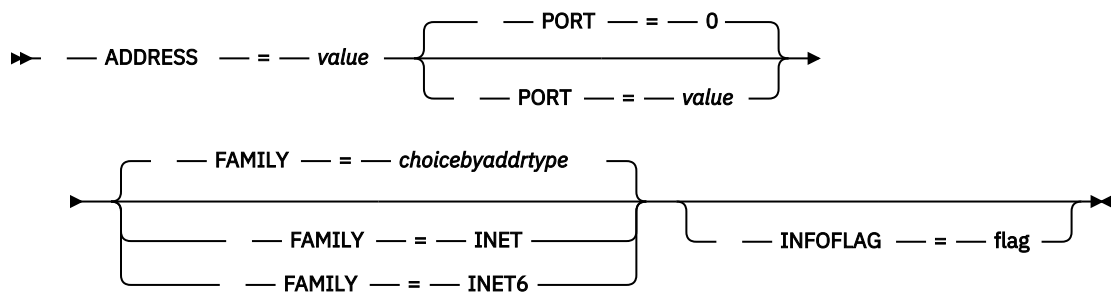
GetAddrInfo



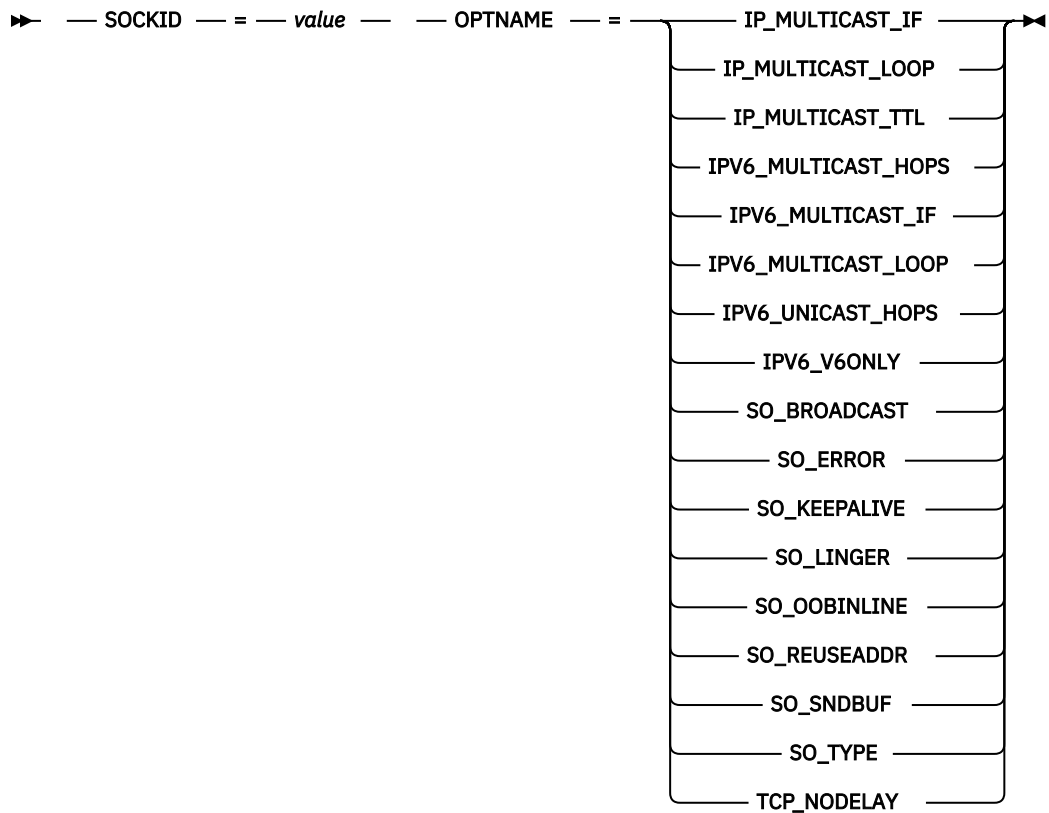
GetClientID



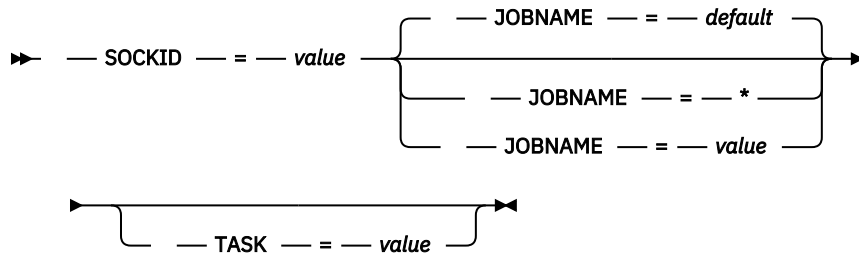
GetNameInfo



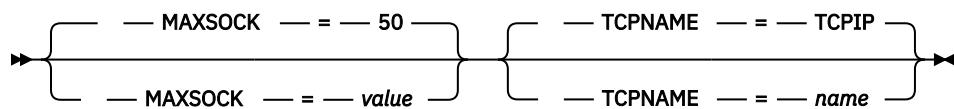
GetSockOpt



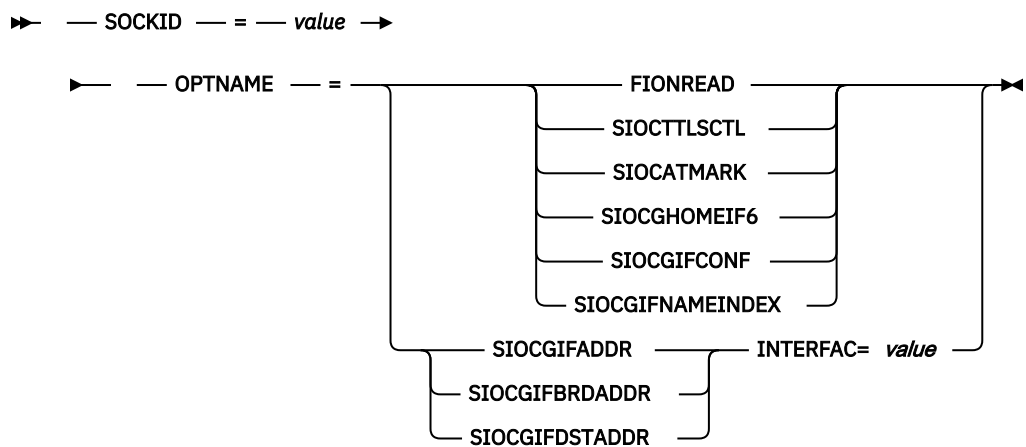
GiveSocketOptions



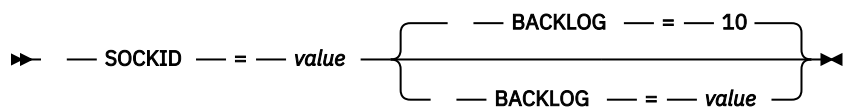
InitOptions



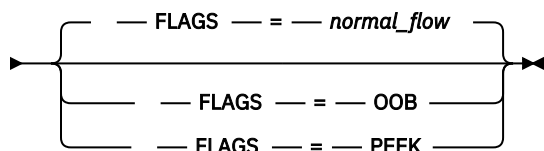
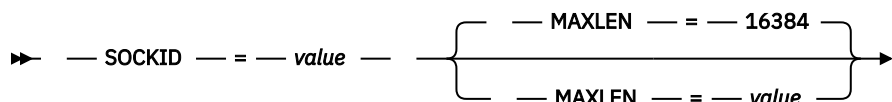
IoctlOptions



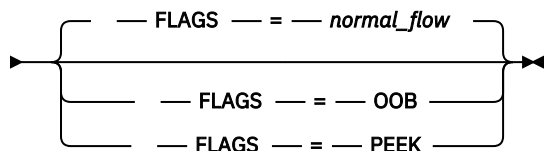
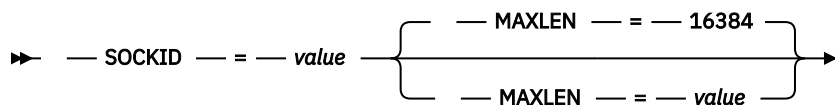
ListenOptions



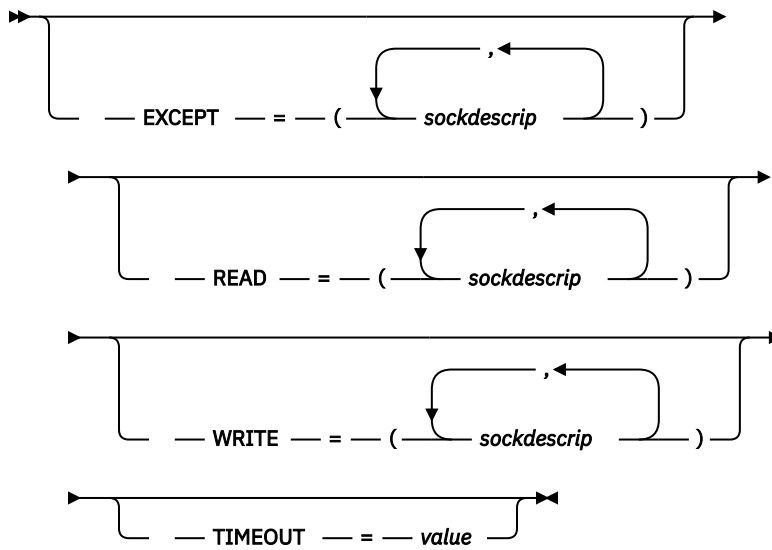
RecvOptions



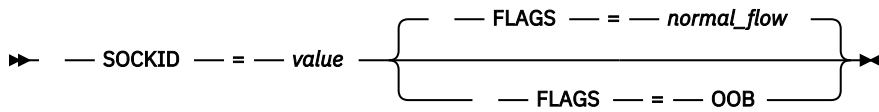
RecvFromOptions



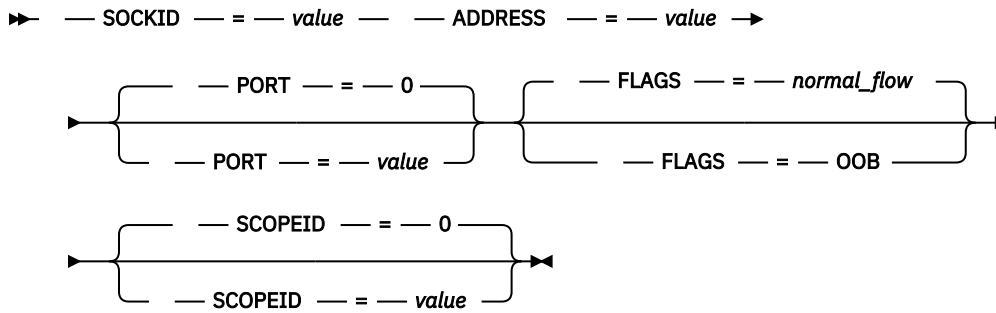
SelectOptions



SendOptions

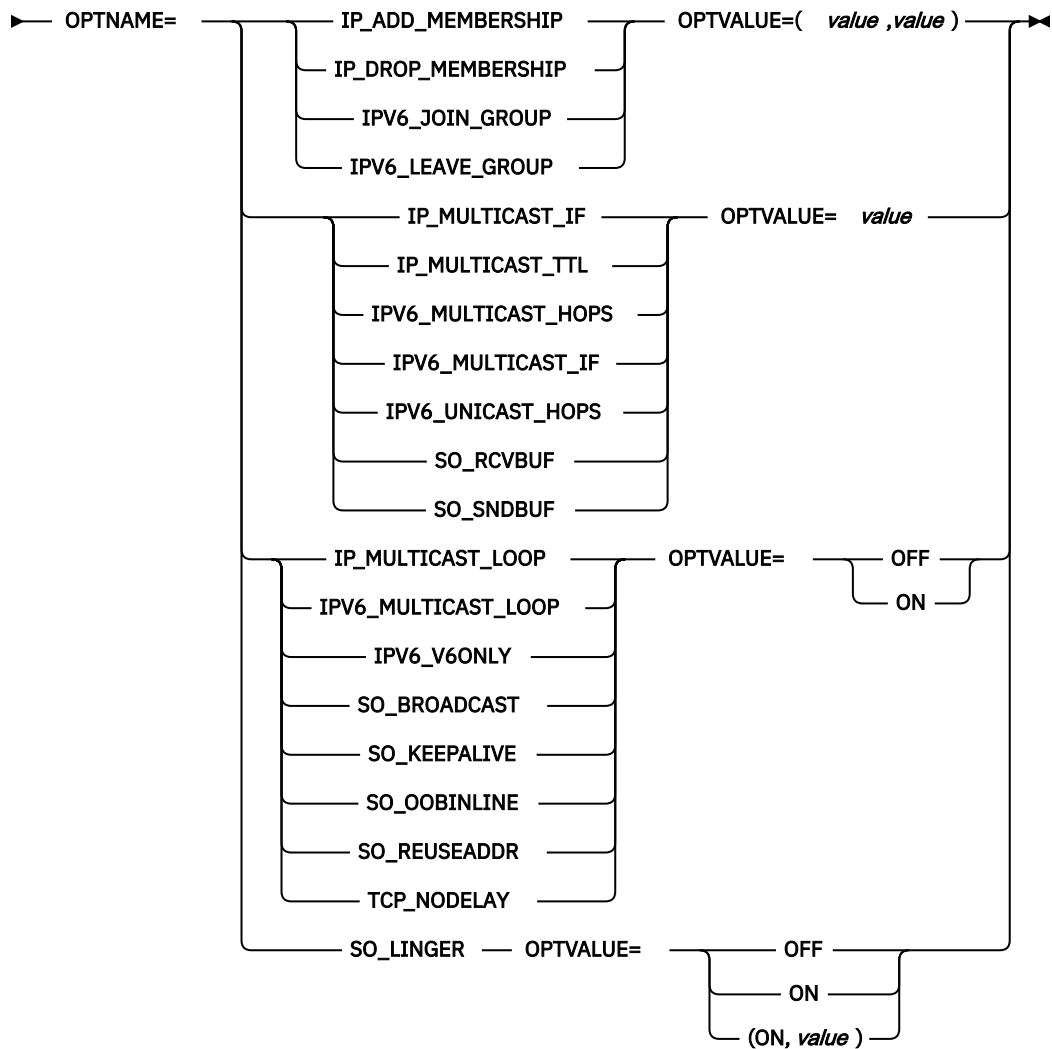


SendToOptions

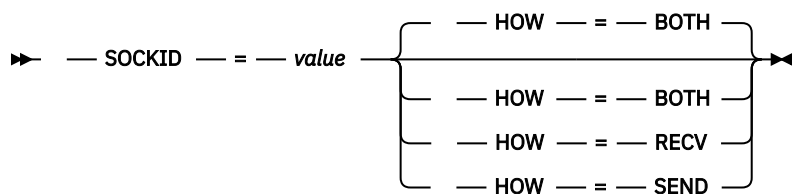


SetSockOpt

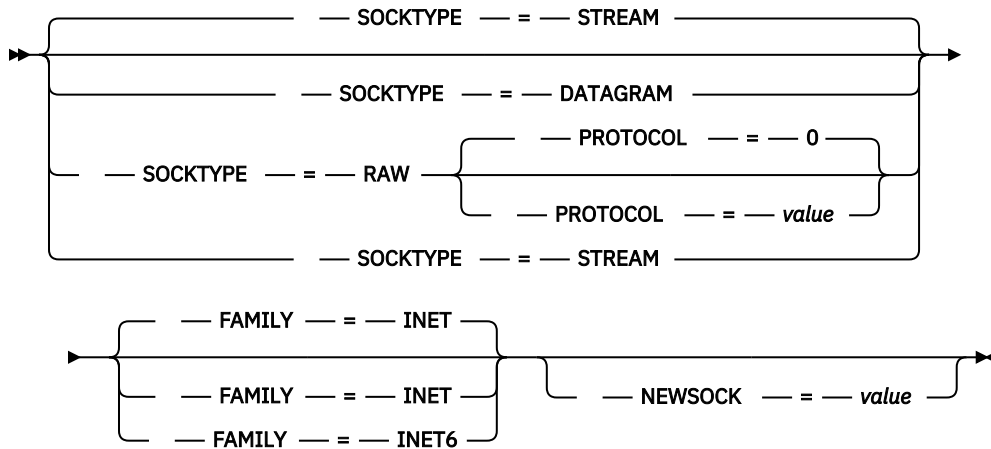
➤ SOCKID= *value* ➔



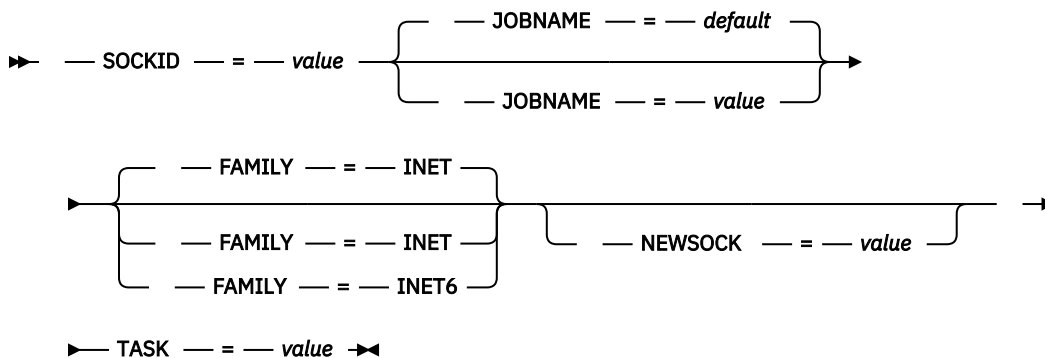
ShutDownOptions



SocketOptions



TakeSocketOptions



Purpose of Command

The SOCKET command is for requesting TCP/IP services, whether they are for information about the TCP/IP stack being used or for managing client or server applications. It begins with the TYPE keyword, which identifies the TCP/IP request, while the other keywords are used to provide additional information for carrying out the TCP/IP request.

Operand Descriptions

The TYPE keyword determines which TCP/IP service is being requested by the SOCKET command interface. This is a required keyword. Valid values are:

ACCEPT

Accepts a pending connection request.

BIND

Binds a socket to a specific address and port.

CANCEL

Cancels an outstanding asynchronous TCP/IP request.

CLOSE

Closes a socket.

CONNECT

Establishes a connection between sockets.

GETADDRINFO

Obtains host address and service characteristics, given the identification of a host (either by name or by address), the identification of a service (either by name or by port number), or the identification of both a host and a service.

GETCLIENTID

Obtains identification of the SOCKET command interface from TCP/IP.

GETHOSTBYADDR

Given the address of a host, it returns information about that host.

GETHOSTBYNAME

Given the name of an IPv4 host, it returns information about that host.

GETHOSTID

Obtains the host identifier from TCP/IP.

GETHOSTNAME

Obtains the host name from TCP/IP.

GETNAMEINFO

Obtains host name and service name information, given a host address, and possibly a port number.

GETPEERNAME

Obtains address and port information for a socket's connection peer.

GETSOCKNAME

Obtains address and port information for a socket.

GETSOCKOPT

Queries options for a socket.

GIVESOCKET

Gives a socket to another task or address space on the same TCP/IP stack.

INIT

Initialization of the SOCKET command interface with TCP/IP.

IOCTL

Queries the operating characteristics for a socket. The IOCTL function in TCP/IP enables you to set and query a fairly wide range of information about a socket.

LISTEN

Makes a socket passive, listening for connection requests.

RECV

Receives data.

RECVFROM

Receives data (includes source address information).

SELECT

Waits for events on zero or more sockets.

SEND

Sends data.

SENDTO

Sends data (includes destination address information).

SETSOCKOPT

Changes options for a socket.

SHUTDOWN

Ends communication on a socket.

SOCKET

Retrieves a socket from TCP/IP.

TAKESOCKET

Takes a socket in the same address family from another task or address space on the same TCP/IP stack.

TERM

Ending of the SOCKET command interface with TCP/IP.

The other keywords and values, and also when they can be used, are as follows.

ADDRESS

This keyword provides an IP address for the TCP/IP requests that require them. The value is an IP address in presentation form. This keyword is required when the value of TYPE is one of the following.

- BIND
- CONNECT
- SENDTO
- GETHOSTBYADDR
- GETNAMEINFO

This keyword cannot be specified when the TYPE keyword has any other value.

BACKLOG

This keyword provides the number of connection requests that can be outstanding for a passive socket. The value has the range of 0 - 10, inclusive. BACKLOG is only valid when the TYPE keyword has the value LISTEN. When the TYPE keyword has the value LISTEN and BACKLOG is not specified, then BACKLOG has the default value of 10.

EXCEPT

This keyword provides a list of one or more socket descriptors for which exception events are being tested. This keyword is valid only when the value of the TYPE keyword is SELECT, and this keyword is optional at that time. When more than one socket descriptor appears in the list, the list must be enclosed in parentheses, and each entry in the list must be separated by a blank or a comma.

FAMILY

This keyword provides the address family, which determines the capabilities of a socket. It also can be used to limit returned host information to one address family. The value can be one of the following:

INET

IP address family (also referred to as IPv4)

INET6

Internet Protocol version 6 (IPv6) address family

UNSPEC

Unspecified address family

The FAMILY keyword can be used only when the TYPE keyword has the following values:

- GETADDRINFO
- GETCLIENTID
- GETNAMEINFO
- SOCKET
- TAKESOCKET

The value UNSPEC can be used only when the value of the TYPE keyword is GETADDRINFO.

For socket capabilities, FAMILY indicates whether a socket can be used only to communicate in an IPv4 network (INET) or is capable of communicating in either an IPv4 or IPv6 network (INET6).

For returned host name information, FAMILY indicates whether the information is limited to the IP address family (INET), IPv6 address family (INET6), or not limited at all (UNSPEC). For a client identification request, FAMILY specifies the domain (address family) of the client.

FLAGS

This keyword provides options for sending or receiving out-of-band data, as well as peeking at available incoming data. The value can be one of the following:

OOB

Send or receive out-of-band data

PEEK

Receive data, but leave the data there for possible reception later

The **FLAGS** keyword can be used only when the **TYPE** keyword has one of the following values. Exactly which values of **FLAGS** are allowed with the chosen **TYPE** are shown in parentheses.

- **RECV** (OOB PEEK)
- **RCVFROM** (OOB PEEK)
- **SEND** (OOB)
- **SENDTO** (OOB)

When the **FLAGS** keyword is permitted, but has not been specified, it takes a default value of neither OOB nor PEEK (in other words, the user is sending or receiving data on the normal flow).

HOSTNAME

This keyword provides the name of a TCP/IP host. This keyword is required when the **TYPE** keyword has the value **GETHOSTBYNAME**. This keyword can be specified when the **TYPE** keyword has the value **GETADDRINFO**. It cannot be specified when the **TYPE** keyword has any other value.

HOW

This keyword provides the option for stopping data transfer on a network connection. The value can be one of the following:

SEND

End further send operations

RCV

End further receive operations

BOTH

End further send and receive operations

HOW can be specified only when the **TYPE** keyword has the value **SHUTDOWN**. When the value of the **TYPE** keyword is **SHUTDOWN** and the **HOW** keyword is omitted, **HOW** has the default value of **BOTH**.

INFOFLAG

This keyword is used to control the data returned by the **SOCKET** command when the **TYPE** keyword has the value of **GETADDRINFO** or **GETNAMEINFO**. For either value of **TYPE**, the default **INFOFLAG** setting is for no data controls. **INFOFLAG** cannot be specified when the **TYPE** keyword has any value other than **GETADDRINFO** or **GETNAMEINFO**.

When TYPE=GETADDRINFO

When the **TYPE** keyword has the value of **GETADDRINFO**, the **INFOFLAG** keyword can have one or more of the following values:

AI_PASSIVE

The returned address will be suitable for use in binding a socket for accepting connections for the specified service; that is, the returned address will be 0.0.0.0 (**INADDR_ANY** in IPv4) or :: (**INADDR6_ANY** in IPv6).

When this flag is not specified, then the returned address will be suitable for use in connecting (**SOCKET TYPE=CONNECT**) or sending datagrams (**SOCKET TYPE=SENDTO**); that is, it will be the default loopback address (127.0.0.1 in IPv4 or ::1 in IPv6).

If the **HOSTNAME** keyword is used, then this value will be ignored.

AI_CANONNAMEOK

If the **HOSTNAME** keyword is used, then this will request that TCP/IP return the canonical name for that host name.

AI_NUMERICHOST

Indicates that the value of the **HOSTNAME** keyword must be an IP address; otherwise, the host name will be rejected.

AI_NUMERICSERV

Indicates that the value of the **SERVICE** keyword must be a port number; otherwise, the service name will be rejected.

AI_V4MAPPED

Indicates that if FAMILY=INET6 or FAMILY=UNSPEC is specified and the TCP/IP stack is enabled for IPv6, then the user will accept IPv4-mapped IPv6 addresses. If AI_ALL is not also specified and there are no IPv6 addresses, then TCP/IP will query for IPv4 addresses. If any are found, they will be returned as IPv4-mapped IPv6 addresses. If this option is not used, then IPv4 addresses will not be presented as IPv4-mapped IPv6 addresses.

If FAMILY=INET is specified or FAMILY=UNSPEC is specified and the TCP/IP stack is not enabled for IPv6, then this option is ignored.

AI_ALL

Indicates that if FAMILY=INET6 is specified, then the user will see all addresses as IPv6 addresses (so IPv4 addresses will be shown as IPv4-mapped IPv6 addresses). If FAMILY=UNSPEC is specified, then the user will accept all IPv6 addresses and IPv4 address presentation will be governed by the AI_V4MAPPED option setting.

If FAMILY=INET is specified or FAMILY=UNSPEC is specified and the TCP/IP stack is not enabled for IPv6, then this option is ignored.

AI_ADDRCONFIG

Indicates that the configuration of interfaces on the TCP/IP stack will govern any host information that is returned.

If an IPv6 address exists for a specified host name, the query will return an IPv6 host address if *all* of these conditions exist:

- The TCP/IP stack is IPv6-enabled
- At least one IPv6 interface is defined
- Either FAMILY=INET6 or FAMILY=UNSPEC is specified

Otherwise, no IPv6 address will be returned.

If an IPv6 address is not returned for a specified host name, the query will return an IPv4 host address if *all* of these conditions exist:

- An IPv4 address exists for the host name
- At least one IPv4 interface is defined
- Either FAMILY=INET or FAMILY=UNSPEC is specified

Note: The loopback interfaces in IPv4 and IPv6 do not count as defined interfaces for this option.

When TYPE=GETNAMEINFO

When the TYPE keyword has the value of GETNAMEINFO, the INFOFLAG keyword can have one of the following values:

NI_DGRAM

Indicates that the query is for datagram service information. When this option is not set, stream service information is assumed.

NI_NAMEREQD

Indicates that the request fails if the host name cannot be located.

NI_NOFQDN

Indicates that just the name portion of the fully qualified domain name is returned.

NI_NUMERICHOST

Indicates that only the numeric form of the host address is returned.

NI_NUMERICSERV

Indicates that only the numeric form of the service address is returned.

INTERFAC

This keyword provides the name of a network interface. The value can be 1 - 16 characters in length, inclusive. INTERFAC can be used only when the value of the TYPE keyword is IOCTL and the value of the OPTNAME keyword is SIOCGIFADDR, SIOCGIFBRDADDR, or SIOCGIFDSTADDR.

JOBNAME

This keyword provides a z/OS job identifier when a socket is to be given (GIVESOCKET) or taken (TAKESOCKET). This keyword can be used only when the value of the TYPE keyword is GIVESOCKET or TAKESOCKET.

When the value of the TYPE keyword is GIVESOCKET, JOBNAME provides the identifier of a job that is allowed to take the socket and uses the same TCP/IP stack. The default value is the identifier of the job giving the socket. JOBNAME can also specify an asterisk (*) to signify that any address space using the same TCP/IP stack can take the socket that is being given.

When the value of the TYPE keyword is TAKESOCKET, JOBNAME provides the identifier of the job that gave the socket. The default value is the identifier of the job attempting the TAKESOCKET request.

MAXLEN

This keyword provides the number of bytes to be set aside for receiving data. The value can range 1 - 1048576. This keyword can be used only when the TYPE keyword has the value of RECV or RECVRFROM. If MAXLEN is not specified when the TYPE keyword has the value RECV or RECVRFROM, then MAXLEN takes a default value of 16384.

MAXSOCK

This keyword provides the maximum number of sockets to be allocated for the socket interface on a given task. This keyword is valid only when the value of the TYPE keyword is INIT, and this keyword is optional at that time. The range of the value is 50 - 2000, inclusive. When the value of the TYPE keyword is INIT and MAXSOCK is not specified, MAXSOCK takes a default value of 50.

NEWSOCK

This keyword provides a specific socket descriptor to be used by requests that obtain sockets. The requests that obtain sockets correspond to the following values of the TYPE keyword:

- SOCKET
- ACCEPT
- TAKESOCKET

Because TCP/IP limits an address space to 2000 sockets, the absolute range of values for NEWSOCK is 0 - 1999, inclusive. There is also a limitation of the range based upon the number of sockets requested by a task. For example, if SOCKET TYPE=INIT MAXSOCK=100 is issued by a NetView task, then the valid range for the NEWSOCK keyword becomes 0-99 (the number of sockets requested minus one). The NEWSOCK keyword cannot be used with any other value of the TYPE keyword. If the TYPE value is for a request that obtains a socket and NEWSOCK is not specified, then the socket descriptor for the newly obtained socket is assigned by TCP/IP.

OPTNAME

This keyword provides the name of an option that is to be set or queried for a socket. The OPTNAME keyword can be used (and is, in fact, required) only when the TYPE keyword has a value of SETSOCKOPT, GETSOCKOPT, or IOCTL. The values that OPTNAME allow depend upon which value of the TYPE keyword is being used, as shown in the following list:

- When TYPE=SETSOCKOPT, then the value of OPTNAME can be:
 - IP_ADD_MEMBERSHIP
Enable an IPv4 application to join a multicast group on a specific interface.
 - IP_DROP_MEMBERSHIP
Enable an IPv4 application to exit a multicast group.
 - IP_MULTICAST_IF
IPv4 interface address used for sending outbound multicast datagrams from the socket.
 - IP_MULTICAST_LOOP

Whether multicast datagrams are looped back for multicast datagrams sent to a group to which the sending host itself belongs. This is an IPv4-only option.

– IP_MULTICAST_TTL

Time-to-live for outbound multicast datagrams. This is an IPv4-only socket option

– IPV6_JOIN_GROUP

Enable an IPv6 application to join a multicast group.

– IPV6_LEAVE_GROUP

Enable an IPv6 application to leave a multicast group.

– IPV6_MULTICAST_HOPS

Hop limit for outbound multicast packets. This is an IPv6-only option.

– IPV6_MULTICAST_IF

Index of the IPv6 interface used for sending outbound multicast datagrams from the socket.

– IPV6_MULTICAST_LOOP

Whether multicast datagrams are looped back for multicast datagrams sent to a group to which the sending host itself belongs. This is an IPv6-only option.

– IPV6_UNICAST_HOPS

Hop limit for outbound unicast packets. This is an IPv6-only option.

– IPV6_V6ONLY

Whether the socket is restricted to sending and receiving only IPv6 packets.

– SO_BROADCAST

Message broadcast option for a socket. Applies only to datagram sockets.

– SO_KEEPALIVE

Periodic sending of packets on an otherwise idle connection. Applies only to stream sockets.

– SO_LINGER

Processing for data not yet transmitted when a socket has been closed. Applies only to stream sockets.

– SO_OOBINLINE

Ability to receive out-of-band data on a socket. Applies only to stream sockets.

– SO_RCVBUF

Size of the data portion of the receive buffer for a socket. When a value of 0 is specified, which disables this option, the default receive buffer data area size is governed by TCP/IP configuration parameters for stream and datagram sockets, and is 65535 for raw sockets.

– SO_REUSEADDR

Local address reuse option for a socket.

– SO_SNDBUF

Size of the data portion of the send buffer for a socket. When a value of 0 is specified, which disables this option, the default send buffer data area size is governed by TCP/IP configuration parameters for stream and datagram sockets, and is 65535 for raw sockets.

– TCP_NODELAY

Whether data sent over the socket is subject to the Nagle algorithm (see RFC 896).

- When TYPE=GETSOCKOPT, the value of OPTNAME can be any of those listed for TYPE=SETSOCKOPT except IP_ADD_MEMBERSHIP, IP_DROP_MEMBERSHIP, IPV6_JOIN_GROUP, and IPV6_LEAVE_GROUP. In addition, when TYPE=GETSOCKOPT, OPTNAME can be the following:

- SO_ERROR
Pending errors on a socket (and clears the error status).
- SO_TYPE
Socket type (stream, datagram, or raw).
- When TYPE=IOCTL, the value of OPTNAME can be:
 - FIONREAD
Number of immediately readable bytes on the socket.
 - SIOCTTLCTL
Query the Application Transparent Transport Layer Security (AT-TLS) status of a TCP/IP connection.
 - SIOCATMARK
Whether the current location in input data is pointing at out-of-band data.
 - SIOCGHOMEIF6
All IPv6 home interfaces.
 - SIOCGIFADDR
IPv4 network interface address for an interface name.
 - SIOCGIFBRDADDR
IPv4 network interface broadcast address for an interface name.
 - SIOCGIFCONF
IPv4 network interface configuration.
 - SIOCGIFDSTADDR
IPv4 network interface destination address for an interface name.
 - SIOCGIFNAMEINDEX
All interface names and indexes, including both IPv4 and IPv6 interfaces whether active or inactive

For more information about these options, see the EZASMI macro explanations for types SETSOCKOPT, GETSOCKOPT, and IOCTL in *z/OS Communications Server: IP Sockets Application Programming Interface guide*.

OPTVALUE

This keyword provides the new value for a socket option setting. This keyword can be used only when the value of the TYPE keyword is SETSOCKOPT and is required at that time. OPTVALUE can have one or two values, depending upon the socket option being set. When OPTVALUE has two values, they must be enclosed in parentheses and separated by either a blank or a comma. Here are the different values OPTVALUE can take, as determined by the socket option under consideration.

- When OPTNAME=IP_ADD_MEMBERSHIP, OPTVALUE can be:
 - (ipaddr1,ipaddr2)
ipaddr1 is the IPv4 multicast address and ipaddr2 is the IPv4 interface address.
- When OPTNAME=IP_DROP_MEMBERSHIP, OPTVALUE can be:
 - (ipaddr1,ipaddr2)
ipaddr1 is the IPv4 multicast address and ipaddr2 is the IPv4 interface address.
- When OPTNAME=IP_MULTICAST_IF, OPTVALUE can be:
 - ipaddr
ipaddr is an IPv4 interface address.

- When OPTNAME=IP_MULTICAST_LOOP, OPTVALUE can be
 - ON
Enables looping the datagrams back.
 - OFF
Disables looping the datagrams back
- When OPTNAME=IP_MULTICAST_TTL, OPTVALUE can be:
 - *n*
n is an integer 0 - 255, inclusive, providing the multicast datagram time to live.
- When OPTNAME=IPV6_JOIN_GROUP, OPTVALUE can be:
 - (ipaddr,*n*)
ipaddr is an IPv6 multicast address and *n* is an IPv6 interface index number.
- When OPTNAME=IPV6_LEAVE_GROUP, OPTVALUE can be:
 - (ipaddr,*n*)
ipaddr is an IPv6 multicast address and *n* is an IPv6 interface index number.
- When OPTNAME=IPV6_MULTICAST_HOPS, OPTVALUE can be:
 - -1
Use the TCP/IP stack's default hop limit.
 - *n*
n is an integer 0 - 255, inclusive, providing the hop limit.
- When OPTNAME=IPV6_MULTICAST_IF, OPTVALUE can be:
 - *n*
n is an integer 0 - 2147483647, inclusive, providing an IPv6 interface index number.
- When OPTNAME=IPV6_MULTICAST_LOOP, OPTVALUE can be:
 - ON
Enables looping the datagrams back
 - OFF
Disables looping the datagrams back.
- When OPTNAME=IPV6_UNICAST_HOPS, OPTVALUE can be:
 - -1
Use the TCP/IP stack's default hop limit.
 - *n*
n is an integer 0 - 255, inclusive, providing the hop limit.
- When OPTNAME=IPV6_V6ONLY, OPTVALUE can be:
 - ON
Enables restriction of sending and receiving only IPv6 datagrams.
 - OFF
Disables restriction of sending and receiving only IPv6 datagrams.
- When OPTNAME=SO_BROADCAST, OPTVALUE can be:
 - ON
Enables the ability to broadcast.

- OFF
Disables the ability to broadcast.
- When OPTNAME=SO_KEEPALIVE, OPTVALUE can be:
 - ON
Enable the use of the keep-alive packet sending mechanism.
 - OFF
Disable the use of the keep-alive packet sending mechanism.
- When OPTNAME=SO_LINGER, OPTVALUE can be:
 - (ON,*n*)
Enable the blocking of a CLOSE request for unsent data, and block until the data is sent or *n* seconds elapses, where *n* can range 0 - 2147483647, inclusive.
 - OFF
Disable the blocking of a CLOSE request for unsent data.
- When OPTNAME=SO_OOBINLINE, OPTVALUE can be:
 - ON
Enable placement of out-of-band data in the normal data flow.
 - OFF
Disable placement of out-of-band data in the normal data flow.
- When OPTNAME=SO_RCVBUF, OPTVALUE can be:
 - 0
Revert to the default size of the data portion of the receive buffer, that defaults based upon the protocol and TCP/IP configuration parameters for that protocol.
 - *n*
A positive integer that overrides the default size of the data portion of the receive buffer
- When OPTNAME=SO_REUSEADDR, OPTVALUE can be:
 - ON
Enable local address reuse.
 - OFF
Disable local address reuse.
- When OPTNAME=SO_SNDBUF, OPTVALUE can be:
 - 0
Revert to the default size of the data portion of the send buffer, that defaults based upon the protocol and TCP/IP configuration parameters for that protocol.
 - *n*
A positive integer that overrides the default size of the data portion of the send buffer.
- When OPTNAME=TCP_NODELAY, OPTVALUE can be:
 - ON
Allow TCP/IP to send small data packets on the socket before receiving acknowledgment for a previously sent data packet. In other words, disable small packet congestion control.
 - OFF
Have TCP/IP wait to send small data packets until receiving acknowledgment for a previously sent data packet. In other words, enable small packet congestion control.

PORT

This keyword provides the port number for a TCP/IP request. The value has the range of 0-65535, inclusive. This keyword is required when TYPE has the following value:

- CONNECT

PORT is optional and has a default value of 0 for the following values of TYPE:

- BIND
- GETNAMEINFO
- SENDTO

If TYPE=SENDTO is used with a socket that is not a raw socket, then you must specify PORT with the appropriate port number.

If TYPE=BIND and either the PORT keyword is omitted or PORT=0 is specified, then TCP/IP will assign a port number when binding the socket. The assigned port number can be determined by use of the GETSOCKNAME request for the socket.

PORT cannot be specified when the TYPE keyword has any other value.

PROTOCOL

This keyword provides the communication protocol number associated with a raw socket. This keyword is only valid when SOCKET=RAW is specified. The value of the PROTOCOL keyword is in the range of 0 - 2147483647, inclusive. The default value is 0.

Number assignments for IP-based protocols, such as the Internet Control Message Protocol (ICMP), are configured within TCP/IP.

READ

This keyword provides a list of one or more socket descriptors for which read events are being tested. This keyword is valid only when the value of the TYPE keyword is SELECT, and this keyword is optional at that time. When more than one socket descriptor appears in the list, the list must be enclosed in parentheses, and each entry in the list must be separated by a blank or a comma.

SCOPEID

This keyword is a decimal number identifying a set of interfaces appropriate for the scope of a supplied IP address. The following circumstance illustrates when the keyword is used.

An IPv6 address of link-local scope is supplied and you want to choose the appropriate set of interfaces by link index.

For all other address types, set the value of SCOPEID to 0, which is also the default value. The value of SCOPEID is ignored if the socket used for the request was obtained in the IP address family (FAMILY=INET).

This keyword can be used only when the TYPE keyword has one of the following values:

- BIND
- CONNECT
- SENDTO

SERVICE

This keyword provides the name or port number of a service for which information is being requested. The value can be from 1 to 32 characters in length, inclusive. This keyword can be used only when the TYPE keyword has the value GETADDRINFO.

SOCKID

This keyword provides the identifier of a socket to which a TCP/IP request applies. It is required for the following values of TYPE:

- ACCEPT
- BIND
- CLOSE

- CONNECT
- GETPEERNAME
- GETSOCKNAME
- GETSOCKOPT
- GIVESOCKET
- IOCTL
- LISTEN
- RECV
- RECVFROM
- SEND
- SENDTO
- SETSOCKOPT
- SHUTDOWN
- TAKESOCKET

Because TCP/IP allows at most 2000 sockets to be obtained by an address space, the value range for SOCKID is 0-1999, inclusive. This keyword is optional for TYPE=CANCEL. When used with TYPE=CANCEL, it identifies the socket whose asynchronous request is to be canceled. When omitted for TYPE=CANCEL, then the cancellation request is for an asynchronous request that is not specific to a socket. For TYPE=TAKESOCKET, this keyword provides the identifier of the socket that was used in a GIVESOCKET request by another task or address space on the same TCP/IP stack. This keyword cannot be specified when the TYPE keyword has any other values.

SOCKTYPE

This keyword provides the socket type, which is of interest either when a socket is being requested or when controlling host information being returned. As a result, this keyword can be specified only when TYPE=SOCKET or TYPE=GETADDRINFO. At those times, the SOCKTYPE keyword is optional. Its value can be one of the following:

DATAGRAM

Request is for a datagram socket (TYPE=SOCKET) or limit returned host information to datagram sockets (TYPE=GETADDRINFO).

RAW

Request is for a raw socket (TYPE=SOCKET) or limit returned host information to raw sockets (TYPE=GETADDRINFO).

STREAM

Request is for a stream socket (TYPE=SOCKET) or limit returned host information to stream sockets (TYPE=GETADDRINFO).

For TYPE=SOCKET, STREAM is the default value of SOCKTYPE. For TYPE=GETADDRINFO, all socket types (datagram, raw, and stream) are the default.

TASK

This keyword provides identification for a task that might take or has given a socket. The value can be 1 - 8 characters in length, inclusive. This keyword is valid only with the GIVESOCKET and TAKESOCKET values of the TYPE keyword. It is required with the TAKESOCKET value of the TYPE keyword. If the value of the TYPE keyword is GIVESOCKET and the TASK keyword is not specified, then any task in the address space identified with the JOBNAME keyword can take the socket. If a socket is being given to or given by a user of the SOCKET command interface, then SOCKET TYPE=GETCLIENTID can be used to obtain the identification of the task that will take or has given the socket.

TCPNAME

This keyword provides the name of the TCP/IP stack to be used by the socket interface on a given task. The value can be 1 - 8 characters in length, inclusive, and must be a valid z/OS job identifier. This keyword is valid only when the value of the TYPE keyword is INIT, and this keyword is optional at that

time. When the value of the TYPE keyword is INIT and TCPNAME is not specified, TCPNAME takes a default value of "TCP/IP".

TIMEOUT

This keyword provides the number of seconds that will elapse before a request times out. This keyword is valid only when the value of the TYPE keyword is SELECT, and this keyword is optional at that time. If not specified, then the applicable request will not timeout. The value of TIMEOUT can range 0 - 2147483647, inclusive.

WRITE

This keyword provides a list of one or more socket descriptors for which write events are being tested. This keyword is valid only when the value of the TYPE keyword is SELECT, and this keyword is optional at that time. When more than one socket descriptor appears in the list, the list must be enclosed in parentheses, and each entry in the list must be separated by a blank or a comma.

Usage Notes

- If a connection request was accepted and NetView security determined that the connection partner was not permitted to access the NetView system in this manner, messages BNH236E and BNH749I are issued and the socket associated with the connection (identified in BNH749I) is placed in suspended status.
- A security violation during a connection will result in the socket being placed in a suspended status. "Suspended status" means that while a connection still exists between this socket and the origin of the connection request, no communication requests involving the socket, other than SOCKET TYPE=CLOSE, will be allowed.
- Consider writing a NetView TCP/IP application based upon the NetView SOCKET command, instead of the REXX SOCKET function. The REXX SOCKET function can do hard waits, which can prevent a NetView task from processing the messages and DOMs that it receives. This can create a backlog and risk running the NetView system out of virtual storage.
- For information about return codes or error numbers found in messages produced by the SOCKET command, see *z/OS Communications Server: IP Sockets Application Programming Interface Guide*.

Return Codes

0

No error encountered - synchronous TCP/IP request completed successfully or asynchronous TCP/IP request scheduled successfully

4

Syntax error, authorization error, or a keyword conflict was detected by command processing

8

Command keywords and values are valid, but a failure occurred in TCP/IP request handling. The TCP/IP request failure is usually one of the following:

- The request was not valid given the current state of the socket or socket interface
- The request was passed to TCP/IP and failed (either a synchronous request failed or an asynchronous request failed acceptance processing)
- A request was attempted for a socket in suspended status and was disallowed.

12

Insufficient virtual storage for processing the request

16

Logic error. Message DWO050E, with more details regarding the error, are written to the network log.

Note that all of the above return codes are passed back immediately by SOCKET command processing. An asynchronous request which is accepted then fails has no mechanism for passing a command processing return code back to the SOCKET command user.

Example: Initializing the socket interface on a TCP/IP stack

To initialize the socket interface on a TCP/IP stack with the job identifier of TCP32, enter:

```
SOCKET TYPE=INIT TCPNAME=TCP32
```

Response

You receive the following message:

```
BNH600I SOCKET INTERFACE INITIALIZED WITH 50 SOCKETS ON TCP/IP TCP32
```

Example: Requesting a stream socket

To request a stream socket, enter:

```
SOCKET TYPE=SOCKET SOCKTYPE=STREAM
```

Response

You receive the following message:

```
BNH606I SOCKET REQUEST COMPLETED SUCCESSFULLY. SOCKET 0 HAS  
BEEN ALLOCATED
```

Example: Requesting a datagram socket

To request a datagram socket, enter:

```
SOCKET TYPE=SOCKET SOCKTYPE=DATAGRAM
```

Response

You receive the following message:

```
BNH606I SOCKET REQUEST COMPLETED SUCCESSFULLY. SOCKET 1 HAS  
BEEN ALLOCATED
```

Example: Binding a socket to a host address

To bind socket 0 to host 102.47.64.1 port 5000, enter:

```
SOCKET TYPE=BIND SOCKID=0 ADDRESS=102.47.64.1 PORT=5000
```

Response

You receive the following message:

```
BNH614I BIND REQUEST ON SOCKET 0 COMPLETED SUCCESSFULLY
```

Example: Connecting a socket to a host address

To connect socket 0 to host 99.47.64.2 port 6000, enter:

```
SOCKET TYPE=CONNECT SOCKID=0 ADDRESS=99.47.64.2 PORT=6000
```

Response

You receive the following message:

```
BNH611I SOCKET 0 CONNECTED TO 99.47.64.2 PORT 6000
```

Example: Sending a string over the connection

To send the string ABC over the connection that socket 0 has, enter:

```
PIPE LIT /ABC/  
| NETV SOCKET TYPE=SEND SOCKID=0  
| CORRWAIT 5  
| CONS
```

Response

You receive the following message:

```
BNH617I SOCKET 0 SENT 3 BYTES OF DATA
```

This particular example illustrates some important points regarding the sending of data and use of the SOCKET command within a PIPE. It is necessary to perform SOCKET TYPE=SEND and SOCKET TYPE=SENDTO within a PIPE or with message automation because of the way the SOCKET command collects and sends data. Also notice the use of CORRWAIT. When issuing the SOCKET command in a PIPE, it is usually necessary to use the CORRWAIT stage because most of the TCP/IP requests performed by the SOCKET command are asynchronous. For the following values of the TYPE keyword of the SOCKET command, use of the CORRWAIT stage is not necessary because the TCP/IP requests performed are synchronous:

- GETADDRINFO
- GETHOSTBYADDR
- GETHOSTBYNAME
- GETNAMEINFO
- INIT
- TERM

Example: Sending multiple data lines

This example illustrates the method by which data lines can be collected and sent by the SOCKET command in a PIPE. Consider the following segment of REXX code:

```
data.0 = 4  
data.1 = 'ABC'  
data.2 = 'DE'  
data.3 = 'F'  
data.4 = 'GHI'  
'PIPE STEM data.'  
| COLLECT'  
| NETV SOCKET TYPE=SEND SOCKID=0'  
| CORRWAIT 5'  
| CONS'
```

Response

Assuming that all of the data are sent at once and the SOCKET command completes before the 5 second wait given by the CORRWAIT stage ends, you see the following message:

```
BNH617I SOCKET 0 SENT 9 BYTES OF DATA
```

Note that the same data collection method applies to SOCKET TYPE=SENDTO.

Example: Waiting for read and exception events and timing out

To wait for read and exception events on sockets 0 and 2 and timeout after 10 seconds if neither socket becomes ready, enter:

```
SOCKET TYPE=SELECT READ=(0,2) EXCEPT=(0,2) TIMEOUT=10
```

Response

There is no message until at least one of the sockets becomes ready with a read or exception event or the SELECT request times out. If socket 0 becomes ready for the read event before the SELECT times out, then you see::

```
BNH610I SOCKET 0 READY FOR READ
```

If the SELECT request times out before any sockets become ready, then you see:

```
BNH609I SELECT REQUEST HAS TIMED OUT
```

Example: Giving a socket to a task in the address space for a specific job

To give socket 2 to any task in the address space for job CNMPROC2, enter:

```
SOCKET TYPE=GIVESOCKET SOCKID=2 JOBNAME=CNMPROC2
```

Response

You receive the following message:

```
BNH614I GIVESOCKET REQUEST ON SOCKET 2 COMPLETED SUCCESSFULLY
```

Note that because the TASK keyword was not specified, any task on the same TCP/IP stack within the address space for CNMPROC2 can take the socket.

Example: Requesting an IPv6-capable stream socket

To request an IPv6-capable stream socket, enter:

```
SOCKET TYPE=SOCKET SOCKTYPE=STREAM FAMILY=INET6
```

Response

You receive the following message:

```
BNH606I SOCKET REQUEST COMPLETED SUCCESSFULLY. SOCKET 0 HAS  
BEEN ALLOCATED
```

Example: Requesting a connection to a server in an IPv6 network

To connect to a server application in an IPv6 network at address 3FF6:98A0:1300:0A88:F234:9876:F0CA:530F and port 5000, enter:

```
SOCKET TYPE=CONNECT SOCKID=0  
ADDRESS=3FF6:98A0:1300:0A88:F234:9876:F0CA:530F PORT=5000
```

Response

You receive the following message:

```
BNH611I SOCKET 0 CONNECTED  
TO 3FF6:98A0:1300:0A88:F234:9876:F0CA:530F PORT 5000
```

Example: Limiting a socket to sending and receiving only IPv6 datagrams

To set the socket option which limits a socket to sending and receiving only IPv6 datagrams, enter:

```
SOCKET TYPE=SETSOCKOPT SOCKID=0 OPTNAME=IPV6_V6ONLY OPTVALUE=ON
```

Response

You receive the following message:

```
BNH614I SETSOCKOPT REQUEST ON SOCKET 0 COMPLETED SUCCESSFULLY
```


Example: Requesting host and service name information

To request host and service name information for IP address 3FF6:98A0:1300:0A88:F234:9876:F0CA:530F and port 23, enter:

```
SOCKET TYPE=GETNAMEINFO,  
ADDRESS=3FF6:98A0:1300:0A88:F234:9876:F0CA:530F,PORT=23
```

Response

You receive the following messages:

```
BNH639I HOST NAME IS myhost.mycity.mycompany.com  
BNH796I SERVICE NAME IS telnet
```

Example: Requesting host and service name address information

To request host and service address information for host name myhost.mycity.mycompany.com and service name telnet, enter:

```
NETVAVIS SOCKET TYPE=GETADDRINFO,  
HOSTNAME=myhost.mycity.mycompany.com,SERVICE=telnet
```

Response

You receive the following messages:

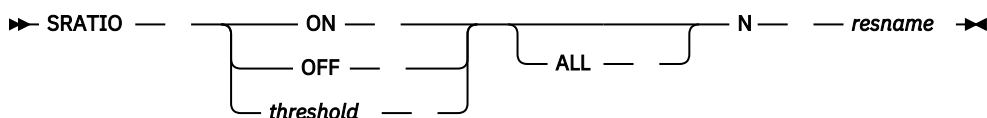
```
BNH641I HOST ADDRESS IS 3FF6:98A0:1300:0A88:F234:9876:F0CA:530F  
BNH791I ADDRESS FAMILY IS INET6  
BNH792I PORT NUMBER IS 23  
BNH793I SOCKET TYPE IS STREAM
```

Note that the service name supplied via the SERVICE keyword is case-sensitive.

SRATIO (NPDA)

Syntax

SRATIO



IBM-Defined Synonyms

Command or Operand	Synonym
SRATIO	SR

Purpose of Command

The SRATIO command:

- Enables or disables the generation of a performance event for a specified resource when an error-to-traffic (E/T) threshold is exceeded.
- Changes the threshold value that generates an alert for a specified resource.

Operand Descriptions

ALL

When specified in an environment other than an NPDA panel (such as a command list, autotask, PPT or NCCF console), specifies that the command is to take effect for all entries if multiple entries are found.

When specified from an NPDA panel, the ALL parameter has no effect.

ON

Specifies that the generation of performance events about the resource is to be enabled. This status is implied if you enter a threshold value. A default operand of ON is set for each threshold resource when it initially records data in the database. At the same time, if no threshold value for the resource was established during installation, the hardware monitor assigns the user-defined default error-to-traffic ratio to the resource. If the default error-to-traffic ratios are not user-defined, the hardware monitor assigns default error-to-traffic ratios as follows:

- Link-attached communications device, 3.0%
- Channel-attached communications device, 1.0%

OFF

Specifies that the generation of performance events about the resource is to be disabled.

threshold

Specifies the new threshold value. The value can have a range of 000–250 that is interpreted as 00.0–25.0 percent. The leading zeros are required.

N

Identifies the operand that follows as a resource name.

resname

Specifies the symbolic name of the resource. You can specify up to five resource names to fully qualify the resource for which data is to be displayed. The resource names that you can use with this command must have resource types that conform to the following conditions:

- In a second-level resource hierarchy, the only valid resource type for the second-level resources are CBUS, FRLY, and LAN.
- In a third-level resource hierarchy, all resource types are valid.
- In a fourth-level resource hierarchy, the fourth-level resource cannot have a resource type of LINE.
- In a fifth-level resource hierarchy, all resource types are valid.

Usage Notes

This command can be entered from the hardware monitor menu panel, a command list, an automated operator, or any NetView component:

- If you are issuing the command from within the hardware monitor and the name of the resource specified is not a unique resource configuration on the database, a selection panel is displayed on which you can choose the relevant configuration.
- If you are issuing the command from within a command list and the name of the resource specified is not a unique configuration on the database, message BNJ1963I is issued. Determine the unique resource and re-issue the command or use the ALL parameter to set all the configurations that match the specified resource.
- If you are issuing the command outside a command list in an environment other than the hardware monitor and the name of the resource specified is not a unique configuration on the database, the Hardware Monitor Multiple Entries panel are displayed. From this panel, select one or more configurations to display.

Restrictions

The following restrictions apply to the SRATIO command:

- No performance events are generated if the statistical record is the result of a permanent error or deactivation of a resource.
- This command cannot be run from a multiple-entries panel.

Return Codes

0

The command issued from a command list was successful for all entries of a multiple-entries panel.

2

The command issued from a command list did not specify the ALL parameter, but multiple entries were found.

4

The command issued from a command list encountered multiple entries and failed for one or more of the resource hierarchies found.

Example: Enabling error-to-traffic event generation for a specified PU

To enable error-to-traffic event generation for PU08, enter:

```
SRATIO ON N PU08
```

SREFRESH (STATMON)

Syntax

SREFRESH

➤ SREFRESH ➤

IBM-Defined Synonyms

Command or Operand	Synonym
SREFRESH	SR

Purpose of Command

The SREFRESH command switches the status monitor Domain Status Summary panel between dynamic and static states. The current setting of the SREFRESH state is indicated on the Domain Status Summary panel. In REFRESH=ON state, changes to the displayed resources are reflected dynamically on the panel as they occur. In REFRESH=OFF state, the panel is static.

Example: Switching the status monitor domain status summary panel refresh state

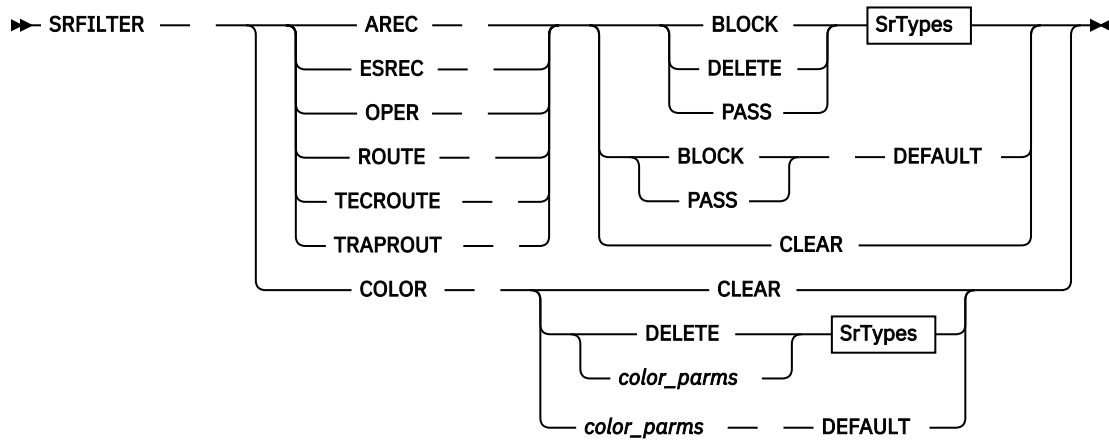
If you are displaying the status monitor Domain Status Summary panel, the refresh state indicator is located on the panel title line. To switch the state from REFRESH=ON to REFRESH=OFF or from REFRESH=OFF to REFRESH=ON, enter:

```
srefresh
```

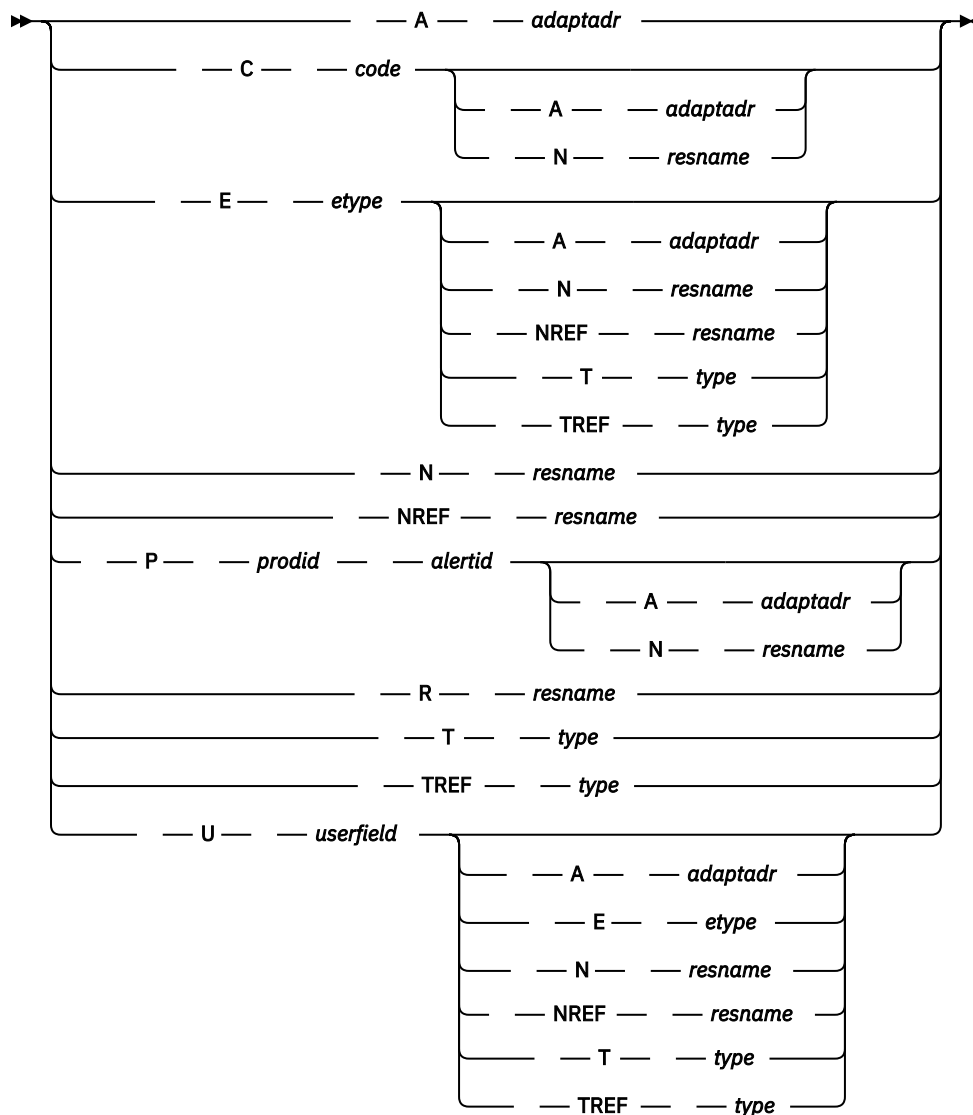
SRFILTER (NPDA)

Syntax

SRFILTER



SrTypes



IBM-Defined Synonyms

Command or Operand	Synonym
SRFILTER	SRF
ALARM	ALM
NOALARM	NOALM
HIGHINT	HIG
UNDERSCORE	UND
BLINK	BLI
REVERSE	REV
TURQUOISE	TUR
BLUE	BLU
GREEN	GRE
PINK	PIN
WHITE	WHI
YELLOW	YEL

Purpose of Command

The SRFILTER command establishes the conditions governing the recording of data in the hardware monitor database, the generation of messages to the authorized operator, the forwarding of alert data to a NetView focal point or to a designated event server, and the coloring of alerts on the Alerts panel.

Operand Descriptions

AREC

Sets a filter that controls whether alerts are to be recorded in the hardware monitor database. For alerts forwarded from entry-point NetView programs over the LUC or LU 6.2 transports, the AREC filters are ignored and AREC is set to PASS. For additional information about recording filters and forwarded alerts, see the *IBM Z NetView Automation Guide*.

ESREC

Sets a filter that controls whether events and statistics are to be recorded to the hardware monitor database. For alerts forwarded from entry-point NetView programs over the LUC or LU 6.2 transports, the ESREC filters are ignored and ESREC is set to PASS. *Alert-only* recording is performed for these alerts, but no event or statistical data is recorded to the database.

OPER

Set a filter for transmitting messages BNJ030I and BNJ146I to an authorized operator.

ROUTE

Sets a filter for routing alerts to the alert focal point (providing a focal point exists). An alert must pass the ESREC and AREC filters before the ROUTE filter is applied to the alert.

TECROUTE

Sets a filter for converting alerts to Event Integration Facility (EIF) events and forwarding the events to the designated event server. An alert must pass the ESREC and AREC filters before the TECROUTE filter is applied to the alert.

TRAPROUT

Sets a filter for converting alerts to SNMP traps and forwarding them to an SNMP manager. An alert must pass the ESREC and AREC filters before the TRAPROUT filter is applied to the alert.

BLOCK

Specifies that the data matching the conditions expressed in this filter element is to be blocked from the hardware monitor database, from the network operator, and from forwarding to the focal point or the designated event server, depending on the type of filter (AREC, ESREC, OPER, ROUTE, or TECROUTE).

DELETE

Specifies that a filter element in the filter table matching the conditions expressed in this filter element is to be deleted. You can specify the DELETE operand anywhere a BLOCK, PASS, or *color_parms* operand can be specified.

PASS

Specifies that the data matching the conditions expressed in this filter element is to be allowed through to the hardware monitor database and can also be passed to the network operator or forwarded to the focal point or the designated event server, depending on the type of filter (AREC, ESREC, OPER, ROUTE, or TECROUTE).

DEFAULT

Specifies that the default for the specified filter is to be overridden with a new default. Defaults are initially provided by the hardware monitor for each filter type. These defaults are effective when the specified filter elements fail to select a data record. You can change these defaults by setting a filter and specifying the DEFAULT keyword. You can specify the AREC, ESREC, OPER, ROUTE, and TECROUTE filter defaults as PASS or BLOCK.

The color filter default defines the color attributes for the first line of the rolling Alerts-Dynamic panel. The color default value is initialized to ALARM HIGHINT WHITE.

CLEAR

Specifies that all filter elements are to be removed and that the filters originally established by the NetView program will be created. Do not specify other operands, except the filter type, when using the CLEAR operand. The filters originally established by the NetView program are:

AREC filter

For alert recording filters, the following conditions are tested in the following order until a condition is satisfied:

BLOCK E HELD TREF CTRL

Blocks HELD event type records received from resources of type CTRL or from resources attached to a resource of type CTRL.

BLOCK E HELD TREF PUGW

Blocks HELD event type records received from resources of type PUGW or from resources attached to a resource of type PUGW.

BLOCK E HELD TREF LCTL

Blocks HELD event type records received from resources of type LCTL or from resources attached to a resource of type LCTL.

PASS E PERM TREF CTRL

Passes PERM event type records received from resources of type CTRL or from resources attached to a resource of type CTRL.

PASS E PERM TREF PUGW

Passes PERM event type records received from resources of type PUGW or from resources attached to a resource of type PUGW.

PASS E PERM TREF LCTL

Passes PERM event type records received from resources of type LCTL or from resources attached to a resource of type LCTL.

PASS E PERF TREF CTRL

Passes PERF event type records received from resources of type CTRL or from resources attached to a resource of type CTRL.

PASS E PERF TREF PUGW

Passes PERF event type records received from resources of type PUGW or from resources attached to a resource of type PUGW.

PASS E PERF TREF LCTL

Passes PERF event type records received from resources of type LCTL or from resources attached to a resource of type LCTL.

BLOCK E HELD

Blocks any HELD event type records.

PASS E INST

Passes any INST event type records.

PASS E NTFY

Passes any NTFY event type records.

PASS E PERF

Passes any PERF event type records.

PASS E PERM

Passes any PERM event type records.

PASS E RSLV

Passes any RSLV event type records.

PASS E SCUR

Passes any SCUR event type records.

PASS E UNKN

Passes any UNKN event type records.

PASS E USER

Passes any USER event type records.

BLOCK

Blocks all records not satisfying any of the previous conditions. The DEFAULT operand of the SRFILTER command affects only this entry of the AREC filters.

ESREC filter

For event and statistical recording filters, PASS passes all records to the events database.

OPER filter

For operator alert filters, BLOCK blocks all alert records from being transmitted to the authorized operator.

ROUTE filter

For route filters, PASS allows the alert to be transmitted to a focal point.

TECROUTE filter

For event server route filters, PASS allows the alert to be converted into an EIF event and forwarded to the designated event server.

TRAPROUT

For TRAPROUT filters, PASS allows the alert to be converted into an SNMP trap and forwarded to an SNMP manager.

COLOR filter

For color filters, the default color (ALARM HIGHINT WHITE) is used for the first line of the Alerts-Dynamic panel. This is used only when an alert does not match a color filter. When an alert using the default color rolls to the second line of the Alerts-Dynamic panel, the color defined by the ALD color map is used to color the alert. The color is turquoise if the color map is not changed.

Note: If you clear a filter, the default for that filter is reset to its initial default, not the default prior to the one you cleared. For example, the initial default for the AREC filter is BLOCK. If you changed the default for the AREC filter to PASS by entering SRFILTER AREC PASS DEFAULT, and you enter SRF AREC CLEAR, the default for AREC is changed to BLOCK (the initial default).

COLOR

Sets a filter defining the color in which an alert is displayed when the alert is presented on the Alerts-Dynamic, Alerts-Static, or Alerts-History panels.

color_parms

Specifies from one to four parameters associated with COLOR. You can specify up to four parameters, but you can select only one from each of the four groups. The four groups of parameters are:

ALARM|NOALARM

Specifies whether an alarm is to sound when an alert is received. The default value is ALARM. This parameter is ignored when the Alerts-History and Alerts-Static panels are built. It applies only when a new alert is rolled onto the Alerts-Dynamic panel.

HIGHINT

Specifies that text is to appear more intense on monochrome terminals.

UNDERSCORE|BLINK|REVERSE

Specifies whether the alert is to be underscored, to flash, or to be presented in reverse video.

TUR|BLUE|GREEN| PINK|RE D|WHITE|YELLOW

Specifies whether the alert is to be presented in turquoise, blue, green, pink, red, white, or yellow. TUR (turquoise) is the default.

A

Identifies the operand that follows as an adapter address.

adaptadr

Specifies the variable adapter address (one address of 12 hexadecimal digits). The A (adapter) address is not a valid option for a resource type of CBUS.

C

Identifies the operand that follows as an event (alert) descriptor identifying code for a problem record in a format other than the generic network management vector transport (NMVT) or management services unit (MSU) format.

code

Specifies the code that identifies a particular event or alert. You can determine this code by entering the appropriate SEL number (*nn*) plus the character C (for nongeneric alerts) on the command line of the Alerts-History, Alerts-Static, or Most Recent Events panels. The following reply is displayed:

```
BNJ962I AL/EV DESCRIPTION CODE FOR SELECTION nn IS bbbcc
```

where *bbb* is the block ID and *cc* is the action code.

E

Identifies the operand that follows as an event type.

etype

Specifies the event type on which the filter item is based. Event types are:

- AVAL
- BYPS
- CUST
- DLRC
- ENV
- HELD
- IMPD
- IMR
- INST
- INTV
- NTFY
- PAFF

- PERF
- PERM
- PROC
- REDL
- RSLV
- RSNT
- SCUR
- SNA
- TEMP
- UNKN
- USER

N

Identifies the operand that follows as a resource name or resource names.

The N keyword specifies that the resource names specified match the trailing names in the hierarchy. For example, if you enter:

```
SRFILTER AREC PASS N RES1 RES2
```

A record with the hierarchy RES4 RES3 RES1 RES2 or RES1 RES2 matches this filter. A record with the hierarchy RES4 RES1 RES2 RES3 does not match this filter because the names RES1 and RES2 do not appear at the end of the resource hierarchy.

The special character % means that an exact hierarchy match must occur. For example, if you enter:

```
SRFILTER AREC PASS N RES1 RES2 %
```

A record with the hierarchy RES1 RES2 matches this filter. A record with the hierarchy RES4 RES1 RES2 does not match this filter. To have an exact match, the number of resource names in the record must match the number in the filter statement.

resname

Specifies the symbolic name of the resource. You can specify up to five resource names to fully qualify the resource for which data is to be displayed.

You can use certain special characters (*, ?, and %) as part of the resource name and resource name hierarchy.

NREF

Identifies the operand that follows as a resource name or resource names.

The filter element is satisfied if the specified resource name or resource names are included in the resource hierarchy in the order stated. For example, if you enter:

```
SRFILTER AREC PASS NREF RES1 RES2
```

An alert with a hierarchy of RES4 RES1 RES2 RES3 matches this filter. Other hierarchies that match this filter are RES1 RES2, RES1 RES2 RES4, and RES3 RES1 RES2. A record with a resource hierarchy of RES1 RES3 RES2 does not match this filter. The requirement is that the names specified in the filter statement be in the hierarchy of the alert in the same order as specified in the filter.

You cannot use an asterisk by itself to represent a resource name that follows the NREF keyword.

P

Specifies the product and alert identifier pair for a problem record that is in generic NMVT or MSU format. You can determine the product set identifier and the alert ID by entering the appropriate SEL

number (*nn*) plus the character C on the command line of the Alerts-History, Alerts-Dynamic, or Most Recent Events panels. The following message is returned:

```
BNJ378I SELECTION nn FILTER CODE; PRODUCT ID pi ALERT ID ac
```

Where *pi* is the product set identifier and *ac* is the alert ID. This information is also available on the last panel of the hardware monitor Event Detail panel.

prodid

Specifies the variable product identifier (hardware or software) of the alert or event sender.

alertid

Specifies the variable alert ID number representing a specific alert or event description.

R

Identifies the operand that follows as a fully qualified resource name. You can code the R parameter only after the ESREC and DELETE operands. The R type filter is set only by the hardware monitor RATE function. The R filter is the highest priority filter. The RATE function is described in the *IBM Z NetView Administration Reference*.

T

Identifies the operand that follows as a resource type or resource types.

The T keyword specifies that the resource types specified match the trailing types in the hierarchy. For example, if you enter:

```
SRFILTER AREC PASS T TYP1 TYP2
```

A record with the hierarchy TYP4 TYP3 TYP1 TYP2 or TYP1 TYP2 matches this filter. A record with hierarchy TYP4 TYP1 TYP2 TYP3 does not match this filter because the types do not appear at the end of the types list.

The special character % specifies that an exact hierarchy match must occur. If you enter:

```
SRFILTER AREC PASS T TYP1 TYP2 %
```

A record with the hierarchy TYP1 TYP2 matches this filter. A record with TYP4 TYP1 TYP2 does not match this filter because it is not an exact match. To have an exact match, the number of resource types in the record must match the number in the filter statement.

type

Specifies the resource type. Examples of resource types are CHAN, COMC, processor, and LCTL. You can specify up to five resource types to fully qualify the resource for which data is to be filtered.

You can use certain special characters (* and %) as part of the resource type and the resource type hierarchy.

TREF

Identifies the operand that follows as a resource type or resource types.

The filter element is satisfied if the specified resource type or resource types are included in the resource hierarchy in the order stated. For example, if you enter:

```
SRFILTER AREC PASS TREF TYP1 TYP2
```

An alert with a hierarchy of TYP4 TYP1 TYP2 TYP3 matches this filter. Other hierarchies that match this filter are TYP1 TYP2, TYP1 TYP2 TYP4, and TYP3 TYP1 TYP2. The requirement is that the types specified in the filter statement be in the hierarchy of the alert in the same order as specified in the filter.

Do not use an asterisk (*) by itself to represent a resource type that follows the TREF keyword.

U

Specifies that user data follows. This allows filtering on the user data field (subvector X'33', subfield X'30') in an NMVT or MSU. You can determine the user data for an alert or event by entering the

appropriate selection number (SEL#) and the character U on the command line of the Alerts-History, Alerts-Static, or Most Recent Events panel. The following message is displayed:

```
BNJ980I SELECTION nn USER DATA: uuuuu
```

where *uuuuu* is the user data for the alert. The first five characters of user data are returned. For a generic alert, the entire user data field is displayed on the Event Detail panel.

Note: Regardless of the length, filtering is performed using only the first five characters of the data.

userfield

Specifies 1 - 5 characters of user data.

Restrictions

The following restrictions apply to the SRFILTER command:

- A record must pass both the ESREC and AREC recording filters to be recorded as an alert. Only alerts are processed by the ROUTE, TECROUTE, COLOR, TRAPROUT, and OPER filters. An alert that passes the ROUTE filter is processed by the hardware monitor for transmission to the alert focal point. An alert that passes the TECROUTE filter is converted to an EIF event and forwarded to the designated event server. An alert that passes the TRAPROUT filter is converted to an SNMP trap and sent to an SNMP manager. An alert passed by the OPER filter causes messages to be sent to the authorized operator.
- The complex nature of filter elements requires that certain items and combinations of items take priority over others. More specific items take precedence over less specific items. Elements of equal priority are processed in the order in which they are entered. Some examples include:

– Elements of equal priority:

```
SRFILTER AREC PASS NREF LINENAM1  
SRFILTER AREC BLOCK NREF NCPNAM1
```

This combination of filter elements passes records for the resource LINENAM1 and all attached resources. Records for NCPNAM1, and lines other than LINENAM1, are blocked. If the order of the two statements is reversed, all resources attached to the NCP are blocked.

– Elements of different priority:

```
SRFILTER ESREC BLOCK N CRTLNAM1  
SRFILTER ESREC PASS E TEMP N CTRLNAM1
```

These two elements pass only temporary records from the controller, and no statistical or event records other than temporary are passed. Because the second element is more detailed, and therefore of a higher priority, it makes no difference in which order the elements are entered.

– Elements that are identical except for PASS or BLOCK:

```
SRFILTER AREC BLOCK NREF LINENAM1  
SRFILTER AREC BLOCK NREF NCPNAM1  
SRFILTER AREC PASS NREF LINENAM1
```

This apparent conflict is resolved during the processing of the SRFILTER command. The hardware monitor searches for identical entries and changes PASS to BLOCK, or BLOCK to PASS, whichever was specified most recently. This example has the same result as the first example because the order of filter elements is not changed. Only the PASS or BLOCK status is changed.

Although these examples show recording filter elements, the priority rules are the same for the viewing filter elements.

- Conditions are tested to determine whether a data record matches the filter. The conditions are tested in the following order:
 1. A specific resource blocked by the hardware monitor RATE function using the R keyword
 2. An event (alert) description code or user field (keyword P or C) and an adapter address

3. An event (alert) description code or user field (keyword P or C) and a resource name
4. An event (alert) description code or user field (keyword P or C)
5. An event type or user data (keyword E or U) and an adapter address (keyword A)
6. An adapter address (keyword A)
7. An event type or user data (keyword E or U) and a resource name (keyword N)
8. A resource name (keyword N)
9. An event type or user data (keyword E or U) and a resource name reference (keyword NREF)
10. A resource name reference (keyword NREF)
11. An event type or user data (keyword E or U) and a resource type (keyword T)
12. A resource type (keyword T)
13. An event type or user data (keyword E or U) and a resource type reference (keyword TREF)
14. A resource type reference (keyword TREF)
15. A user data (keyword U) and an event type (keyword E)
16. An event type or user field (keyword E or U).

If a match occurs, the matching filter element action (PASS or BLOCK) is processed and further testing is suspended.

If the record fails to match any of the previously listed conditions, the record is processed according to the filter defaults. Initial filter defaults can cause the record to be passed for recording as an event or statistic, and blocked for recording as an alert, creating a message, or routing to another NetView system.

- You can set alert color for the Alerts-Dynamic, Alerts-History, and Alerts-Static panels by the following means:
 - SVFILTER command with COLOR keyword
 - SRFILTER COLOR filter
 - Color map

If you do not set a COLOR filter for an alert, the alert is displayed based on its appropriate color map.

If you set a COLOR recording filter for an alert, it overrides the color set in the color map. The recording filter color stays with the alert when the alert is logged to the hardware monitor database. If you change the color of a recording filter, it does not affect the color assigned to a previously logged alert.

- If you set the ESREC, AREC, OPER, ROUTE, or TECROUTE recording filters with the NetView automation table SRF action, the SRFILTER settings are overridden. If you set the COLOR recording filter with the NetView automation table COLOR, XHILITE, or BEEP actions, the SRFILTER COLOR command, and the color map are overridden. See *IBM Z NetView Automation Guide* for more information.
- If you set a COLOR viewing filter with the SVFILTER command, it overrides both the SRFILTER COLOR command and the color map. This allows you to override the general setting of colors for alerts for each operator.
- If the alert does not match the specific color filter, the color attributes from the default color filter are used on the first line of the Alerts-Dynamic panel when the alert is rolled onto the panel. When the alert rolls off the first line of the Alerts-Dynamic panel, the color map defines the color of the alert.
- If you forward an alert to a focal point, the alert color filters at the focal point determine its display color.
- In filtering, the hardware monitor treats the HELD event type as if it is a second alert or event type. This means that a HELD event type is always associated with another event type. The HELD event type has the same priority as all other event types. Therefore, the order in which you set the HELD type filter and any other event type filter is important. For example, the default SRF AREC filters are set as follows:

```
BLOCK E HELD
PASS E PERM
```

Therefore, a permanent alert that is also a HELD alert is blocked because the BLOCK filter is the first matching filter encountered.

- Whenever the owning domain and session domain do not match (distributed database retrieval is taking place), and selection SRF is a valid command from the current display, the command is processed in the owning domain, not the session domain. To clear the filters that were set, set up a cross-domain session with the owning domain using the SDOMAIN command and then issue the CLEAR command.

Note: Do not use the SRF option to control the routing of alerts to a focal point. If a routed alert is to be blocked, issue an SDOMAIN command and then enter the appropriate SRFILTER ROUTE command.

- Pattern matching support is provided for resource names, resource types, and resource name and resource type hierarchies using the special characters *, ?, and %. This support consists of the following:

When used by itself, this character represents any resource name or type. This usage applies to resource names and types associated with the keywords N and T.

When appended to a resource name, this character can represent zero, one, or more characters. Characters following an asterisk (*) are not allowed. This usage applies to resource names associated with the keywords N and NREF.

When you use an asterisk (*) by itself to represent a resource name, it does not count in processing priority. For example, if you use an asterisk (*) to represent resource names, the priority is the same as if you specified one resource name.

?

A placeholder that represents exactly one character anywhere in a resource name. You can use multiple ? characters in a resource name. This usage applies to resource names associated with the keywords N and NREF.

%

A trailing character indicating that an exact match on a hierarchy of resource levels and resource names or types is required. Specify this character last and following a resource name or type. This usage applies to resource names and types associated with the keywords N and T.

Example: Blocking information for a resource with an event and statistical filter

To block information for resource name PU3 with an event and statistical filter, enter:

```
SRFILTER ESREC BLOCK N PU3
```

Response

The usual response to an accepted command is as follows:

```
BNJ1341I SRF/SRFILTER COMMAND ACCEPTED
```

The following examples show the usage of * and ? characters in the SRFILTER command with specified resource names.

Example: Blocking all alerts for a specified resource

To block all alerts for the resource named RTP or for any resource whose name begins with the letters RTP, enter:

```
SRFILTER AREC BLOCK N RTP*
```

Example: Passing event or statistical information for resources beginning with the letters RTP

To pass any event or statistical information for any resource name with exactly four characters, starting with RTP, enter:

```
SRFILTER ESREC PASS N RTP?
```

Example: Deleting a previously entered SRFILTER command

To delete a previously entered SRFILTER command that was one of the following formats:

```
SRFILTER AREC PASS N RTP??*  
SRFILTER ESREC BLOCK N RTP??*
```

Enter:

```
SRFILTER ESREC DELETE N RTP??*
```

Example: Blocking alerts for a resource

To block alerts for any resource name with exactly five characters, starting with RTP, followed by any single character and ending with 0, enter:

```
SRFILTER AREC BLOCK N RTP?0
```

The following examples show the usage of * by itself and % as a trailing character in the SRFILTER command with specified resource names or types.

Example: Recording event or statistical information for a resource

To record any event or statistical information for a resource that matches on any hierarchy list of three or more resource names, ending with resource name RTP, enter:

```
SRFILTER ESREC PASS N * * RTP
```

Example: Blocking event or statistical information for a resource

To block any event or statistical information for any resource that matches on a hierarchy list of exactly two resource types, with COMC as the level 1 resource type and LINE as the level 2 resource type, enter:

```
SRFILTER ESREC BLOCK T COMC LINE %
```

Example: Blocking alerts for a resource

To block alerts for a resource that matches on any hierarchy list of exactly three resource names, ending with resource name RTP3, enter:

```
SRFILTER AREC BLOCK N * * RTP3 %
```

SRVRNV (NCCF; CNME0054)

Syntax

SRVRNV

➤ SRVRNV — — *cplu* — — *nvtask* — — *cmd* ➤

Purpose of Command

The SRVRNV command enables the NetView program to send a command to the NetView system at the network node server for a CP or LU resource. SRVRNV is used typically in an APPN environment when a

command (for example, **D NET,TOPO**) must be directed and the operator does not know the name of the intended NetView server.

Operand Descriptions

cplu

Specifies the CP or LU resource for which the NetView server is to be located. The resource name can be network-qualified.

nvtask

Specifies the NetView task on which the target command is to run. Use an asterisk (*) if the remote task name is to match the local task name.

cmd

Specifies the name of the command that is to run at the server NetView.

Restrictions

The VTAM command **D NET,DIRECTRY,ID=*cplu*,SCOPE=NS** is used to locate the server CP. The CPDOMAIN pipe stage is used to convert that CP name to a NetView domain name, which can then be used as a RMTCMD target. Therefore, SRVRNV is subject to the limitations and restrictions of these component commands. For example, the local VTAM must support APPN. The task name and the command must be valid at the target NetView system.

Return Codes

108

A timeout occurred.

112

An unexpected VTAM response was received.

116

The CPDOMAIN pipe stage returned a message other than DWO969I.

120

CP was not identified by VTAM.

other

A CPDOMAIN failure occurred. See the CPDOMAIN return codes in the *IBM Z NetView Programming: Pipes* for more information.

Example: Requesting topology information

The following example requests topology information from the server NetView for CD50S001:

```
SRVRNV CD50S001 * D NET,TOPO,LIST=EN,ID=S*
```

Response

If the SRVRNV command is successful, a response similar to the following is received:

```
- NTVAA      DWO969I DOMAIN FOR DEGN050N.DEGN050C IS NTV50
NTV50
IST097I  DISPLAY  ACCEPTED
IST350I  DISPLAY TYPE = TOPOLOGY
IST1295I CP NAME      NODETYPE ROUTERES CONGESTION CP-CP WEIGHT
IST1296I USIBMNT.SMONROE EN      *NA*      *NA*      YES   *NA*
IST1296I USIBMNT.SIMON  EN      *NA*      *NA*      YES   *NA*
IST1296I USIBMNT.STACEY EN      *NA*      *NA*      YES   *NA*
IST314I  END
```

STACK (NCCF)

Syntax

STACK

► STACK ◄

Purpose of Command

The STACK command suspends a command procedure while it is in pause or wait status so that commands or command lists can be entered. This command also causes commands and command lists that were stacked because of the original command procedure to run immediately and in the order entered.

Restrictions

The following restrictions apply to the STACK command:

- When a command procedure wait is followed by a STACK command, messages are still intercepted by the wait processing. The processing of messages is deferred until the command procedure is reinstated by the UNSTACK command. GO commands are rejected.
- The STACK command does not work in the session monitor or hardware monitor; it only works in the command facility.

Example: Beginning a STACK command

To begin a STACK command, enter:

```
STACK
```

Response

If the STACK command is successful, the following message is displayed:

```
DSI230I STACK STARTED
```

Example: Suspending a command list currently in a pause or wait state

To suspend a command list currently in a pause or wait state, enter:

```
STACK
```

Response

If the command is successful, the following message is displayed:

```
DSI230I STACK STARTED
```

If another command or command procedure is then issued, the newly entered command or command procedure then begins processing. Upon completion of the newly entered command, the following message is displayed:

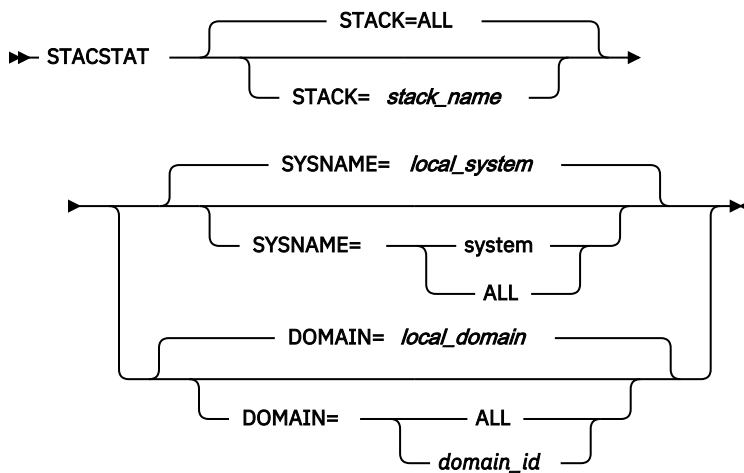
```
DSI588I COMMAND PROCEDURE commandname STACKED, ISSUE 'UNSTACK' TO RESUME
```

That message serves as a reminder that you used a STACK command and that the command procedure you have suspended has not completed.

STACSTAT (NCCF; CNME8230)

Syntax

NCCF STACSTAT



Purpose of Command

You can use the STACSTAT command to view configuration and status information about TCP/IP stacks. This information can be viewed from a 3270 console or from the Z NetView Enterprise Management Agent.

Note: This command is intended as a REXX interface. For end-user interfaces, use the Tivoli Enterprise Portal, the CNMSSTAC sample, or the NetView management console.

Operand Descriptions

STACK

The stack name for which data is requested. The default value is all TCP/IP stack names known to the DOMAIN or SYSNAME specified.

SYSNAME

The system name for the requested domain data. The default value is the local system. When the ALL value is in effect, the command is sent to all NetView programs known to the master NetView program.

Note: Use caution when you specify the ALL value because it can cause rediscovery to take place on multiple NetView programs.

DOMAIN

The domain to which the request is sent. The following are the valid options:

ALL

Specifies all domains in the sysplex. This value is valid only on the master NetView program.

Note: If you have a large number of stacks or systems in your sysplex, issuing ALL can cause slow response times and high processor utilization.

local_domain

If the DOMAIN keyword is not specified, the domain name of the local NetView system is used. This is the default.

domain_id

Specifies the ID of a specific domain. This option can be issued only from the master NetView program if the domain is not the local domain.

Usage Notes

If message DSI047E is received, contact your system programmer to enable the appropriate tower or subtower. For information about data collection and display of towers and subtowers, see *IBM Z NetView Installation: Configuring Additional Components*.

Return Codes

Return Code

Meaning

0

Successful.

2

Help was issued.

4

No data to display.

8 or above

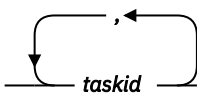
Failure, see the associated message.

START (EAS)

Syntax

EAS START

➤ MODIFY — — *procname* — , — START — — TASK — = — (— *taskid* —) — ➤



IBM-Defined Synonyms

Command or Operand

Synonym

MODIFY

F

Purpose of Command

The START command starts any Event/Automation Service task that is not already started.

Note: If an attempt is made to start a task that is already started, a warning console message is issued.

Operand Descriptions

procname

Specifies the Event/Automation Service job name.

TASK=taskid

Specifies the service tasks to be started. The *taskid* can have the following values:

ALERTA

The alert adapter service task

ALERTC

The confirmed alert adapter service task

MESSAGEA

The message adapter service task

MESSAGEC

The confirmed message adapter service task

EVENTRCV

The event receiver service task

TRAPALRT

The trap to alert conversion task

ALRTTRAP

The alert to trap conversion task

ALL

All service tasks

Restrictions

The following restrictions apply to the START command:

- You can specify only one TASK operand for each START command. If you want to specify more than one service task, separate each *taskid* with a comma and enclose the *taskids* string within parentheses.
- This command cannot be used to start the entire Event/Automation Service address space.

Example: Starting a service task

To start the alert adapter service task for the Event/Automation Service job named IHSAEVNT, enter:

```
F IHSAEVNT,START,TASK=ALERTA
```

Response

You receive the following response:

```
IHS0124I Alert Adapter task initialization complete.
```

Example: Starting multiple service tasks

To start the alert adapter and message adapter service tasks for the Event/Automation Service job named IHSAEVNT, enter:

```
F IHSAEVNT,START,TASK=(ALERTA,MESSAGEA)
```

Response

You receive the following response:

```
IHS0124I Alert Adapter task initialization complete.
IHS0124I Message Adapter task initialization complete.
```

START (GMFHS)

Syntax

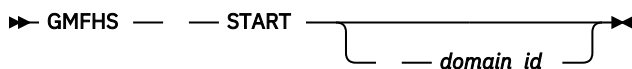
From an MVS console:

STARTGM

```
➤ START — — procname ————— ➤
      , — DOMAIN= — domain
```

From a NetView console:

GMFHS START



IBM-Defined Synonyms

Command or Operand	Synonym
MODIFY	F
START	S

Purpose of Command

The NetView GMFHS START command starts the GMFHS task. Use the MVS START command to start the GMFHS task from the MVS console.

You can enter the GMFHS START command from the MVS console or from a NetView terminal using the GMFHS command list.

Operand Descriptions

procname

Specifies the GMFHS MVS job name.

domain_id

Specifies the domain ID that GMFHS connects to when GMFHS starts. If *domain_id* is not specified, the default domain is determined by the domain specified in the GMFHS start up procedure.

Example: Starting a specified GMFHS job

To start a GMFHS job named GMFSBTSK.C from the MVS console, enter:

```
S GMFSBTSK.C
```

Response

A response similar to the following is displayed:

```
DUI4027I GMFHS MAIN TASK INITIALIZATION IS COMPLETE FOR DOMAIN = CNM01
DUI4003I GMFHS NETWORK CONFIGURATION INITIALIZED SUCCESSFULLY
```

Example: Starting GMFHS from a NetView terminal

To start GMFHS from a NetView terminal, enter:

```
GMFHS START
```

Response

A response similar to the following is displayed:

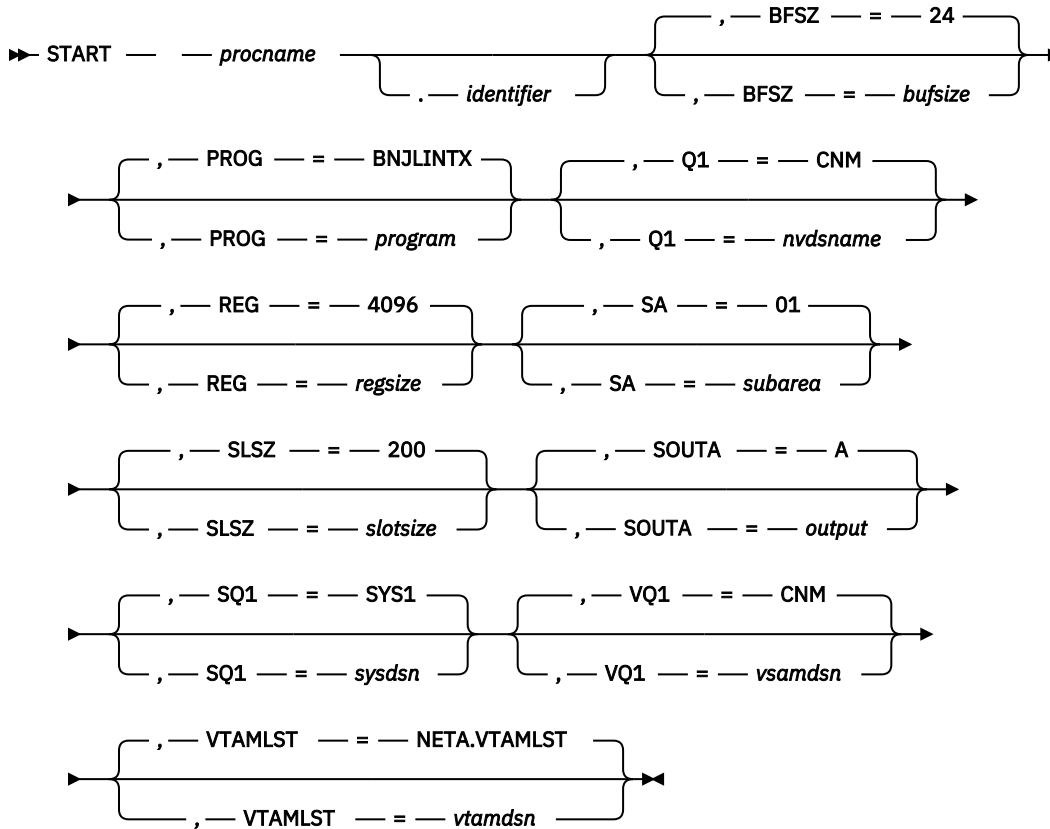
```
DSI013I COMMAND LIST GMFHS COMPLETE
```

START (MVS)

Syntax

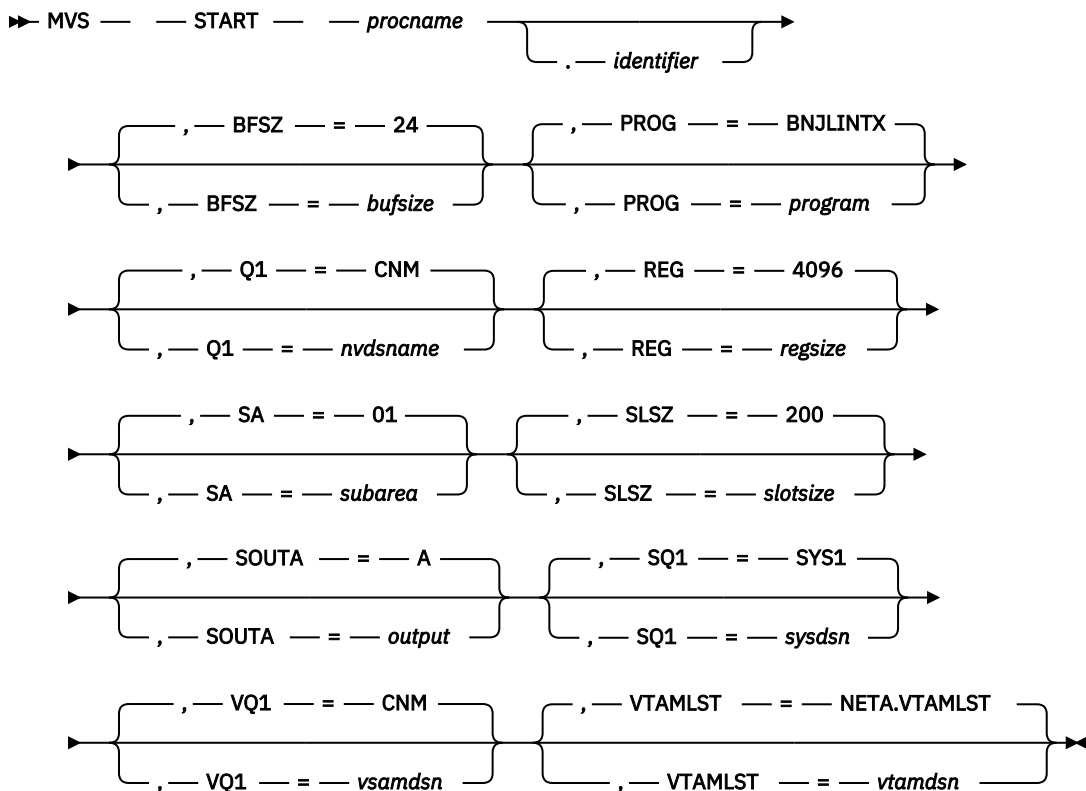
From an MVS console:

MVS START



From a NetView terminal:

MVS START



Purpose of Command

Both the NetView program and the NetView subsystem are initiated as started tasks in the system.

Operand Descriptions

procname

Specifies the NetView MVS job name.

identifier

Specifies the MVS job name identifying the task to be started.

BFSZ=bufsize

Specifies the buffer size, in kilobytes. The default value is 24K.

PROG=program

Specifies the program that starts the NetView program. The default value is BNJLINTX.

Q1=nvdsname

Specifies the high-level NetView data set name qualifier. The default value is CNM.

REG=regsize

Specifies the region size, in kilobytes, for the main task. The default value is 4096K.

SA=subarea

Specifies the subarea number. The default value is 01.

SLSZ=slotsize

Specifies the slot size, in kilobytes. The default value is 200K.

SOUTA=output

Specifies the defaulted printed output class. The default value is A.

SQ1=sysdsn

Specifies the high-level MVS data set name qualifier. The default value is SYS1.

VQ1=vsamdsn

Specifies the high-level VSAM data set name qualifier. The default value is CNM.

VTAMLST=vtamdsn

Specifies the high-level VTAM data set name qualifier. The default value is NETA.VTAMLST.

These names can vary in your installation; contact your system programmer for the current names.

You can also start the NetView program automatically by using the COMMNDnn member of SYS1.PARMLIB.

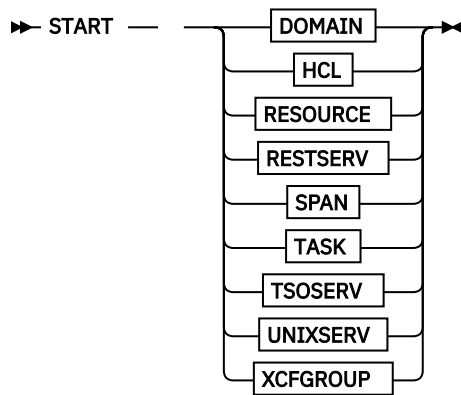
Restrictions

This command can be issued from a NetView operator ID (using the MVS command) or an MVS console.

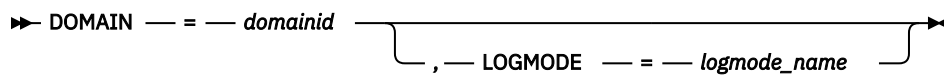
START (NCCF)

Syntax

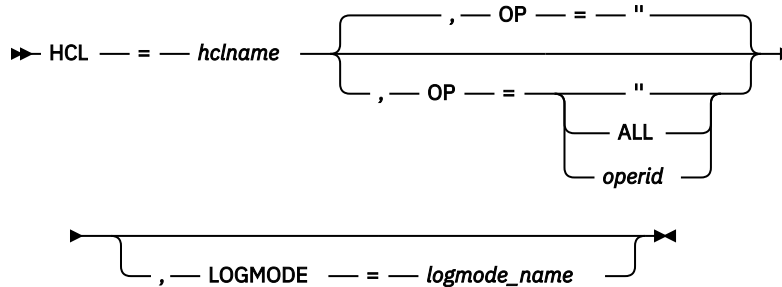
NCCF START



DOMAIN



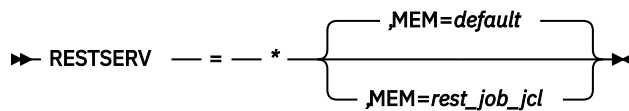
HCL



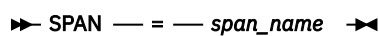
RESOURCE



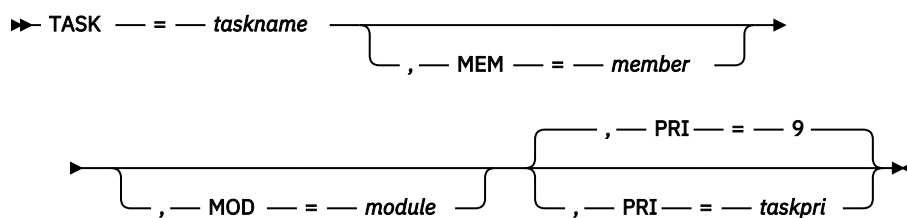
RESTSERV



SPAN



TASK



TSOSERV

HCL=hclname

Starts a hardcopy log task with the specified name for the specified operator or operators.

LOGMODE=logmode_name

Indicates the logmode entry to be used when binding the session. If you do not specify this operand, the logmode name defaults to the logmode name on the DLOGMOD entry coded on the APPL statement. If used with the hardcopy log, the *logmode_name* is the printer logmode.

MEM=

For TASK, uses the member of DSIPARM as a definition member. This operand is valid only for data services tasks. If this is a new data services task and is not defined in the CNMSTYLE member, the MEM operand is required. This operand mimics the function of the MEM keyword in the CNMSTYLE member. If not specified, *member* defaults to the last value entered for the MEM keyword, either from a previous START for this task or from the TASK definition in the CNMSTYLE member. For more information about the MEM operand, see the *IBM Z NetView Administration Reference*.

For TSOSERV, if the TSO server is to be started as a submitted job, the default value is CNMSJTSO, which is the name of the DSIPARM member that contains the JCL for the TSO server job. If the TSO server is to be started as a started task, the default value is CNMSSTSO, which is the name of the CNMSAMP member that contains the JCL for the TSO server job. CNMSSTSO must be installed in a system procedure library. See *IBM Z NetView Installation: Configuring Additional Components* for more information about the installation of CNMSSTSO. For more information about starting the TSO server as a started task or submitted batch job, see the DEFAULTS STRTSERV command.

For UNIXSERV, if the UNIX server is to be started as a submitted job, the default value is CNMSJUNX, which is the name of the DSIPARM member that contains the JCL for the UNIX server job. If the UNIX server is to be started as a started task, the default value is CNMSSUNX, which is the name of the CNMSAMP member that contains the JCL for the UNIX server job. CNMSSUNX must be installed in a system procedure library. See *IBM Z NetView Installation: Configuring Additional Components* for more information about the installation of CNMSSUNX. For more information about starting the UNIX server as a started task or submitted batch job, see the DEFAULTS STRTSERV command.

For the XCFGROUP option, specifies the name to use as the member name. The value is 1 - 16 characters in length. For DSIPLXnn groups, the member name must be the NetView domain name. If omitted, the MEM value defaults to the NetView domain.

For RESTSERV, the default value is EJN SSRST, which is the name of the DSIPARM member that contains the JCL for the REST API server started task. EJN SSRST must be installed in a system procedure library. For more information about the installation of EJN SSRST, see *IBM Z NetView Installation: Configuring Additional Components*. This value should be the same member name as was specified on the START RESTSERV=* command.

MOD=module

Specifies the load module name associated with the task that starts. If this is a new task and it is not defined in the CNMSTYLE member, the MOD keyword is required. If the task has previously been started in the NetView program, the MOD keyword can be used alone to change the module that is invoked when the task is started.

OP=

For HCL, specifies to start an operator's hardcopy log.

..

Specifies to start a session with the named hardcopy log for your operator ID. This is the default.

operid

Specifies to start a session with the named hardcopy log for the specified operator.

ALL

Specifies to start a session with the named hardcopy log for all operators.

For TSOSERV, specifies a valid logged on NetView operator ID or " (double quotation mark), indicating the NetView operator for whom this server is to be started, or NONE indicating that this server is a global server not associated with any specific operator and is not stopped as the result of any NetView operator logging off. The " indicates that the ID of the NetView operator issuing the START command

is to be used. A server that is not global ends when it is no longer associated with any NetView operators. Note that if the named server is already started, it is associated with this NetView operator (or given additional global status) without being restarted. The default value is "".

PRI=taskpri

Specifies the priority of the task. Valid values are 1 - 9. If this is a new task and it is not defined in the CNMSTYLE member, the PRI keyword is optional; if it is not specified, it defaults to 9. If the task has previously been started in the NetView program, the PRI keyword can be used alone to change the priority of the task. Automated operator tasks run at priority 5 and other operator tasks run at priority 4. If you specify a priority 1 - 4, you might impact other tasks. For more information about task priorities, see the *IBM Z NetView Administration Reference*.

RESOURCE=rname

The named resource (such as a terminal) is made available to span of control checking in this NetView system. When the NetView program is first started, all defined resources are available to the system.

You can use the START RESOURCE command to start a resource within a span. Because all resources in a span are started when the system is started, use the START RESOURCE command only if you used the STOP RESOURCE command first. You cannot use wildcard characters in the resource name. The resource name must match a resource name in a span of control definition.

RETSERV=*

Starts the REST Server. The REST Server will only start as a started task.

When specifying RETSERV, you must have the EJN SSRST sample installed in a system JCL procedure library. For more information about the installation of EJN SSRST, see *IBM Z NetView Installation: Configuring Additional Components*.

SPAN=span_name

The named span which is to become part of an operator's span of control. The spans that can be started by the START SPAN command are limited by either the SPAN or ISPAN statements in the operator profile or by the NETSPAN class in an SAF product, depending on the setting of the OPSPAN keyword on the OPTIONS statement or the last REFRESH command with the OPSPAN keyword. If you are logged on when a REFRESH command is issued to switch from OPSPAN=SAF to OPSPAN=NETV, you cannot start additional spans defined for your operator profile since NetView initialization until you log off and then log back on.

Note: Operators can start spans with OPSPAN=NETV only if they are logged on with OPSPAN=NETV. An operator can start any span that is defined in the operator's profile. If an operator logs on with OPSPAN=SAF and if OPSPAN is currently set to NETV, no spans can be started.

The span is started at the highest access level to which the operator is permitted. For more information about defining spans of control, see the *IBM Z NetView Security Reference*.

TASK=taskname

Activate the named optional task. These are the reserved task names: ALL, DPR, DST, HCL, HCT, LOG, MNT, NNT, OPT, OST, PPT, SYSOP, TCT, and any VOST name. A VOST *taskname* is of the form DSI#nnnn, where nnnn is in the range of 0000 - 9999.

TSOSERV=

Specifies a valid TSO user ID or "" (double quotation mark), indicating the NetView operator ID is to be used as the TSO user ID. This is the TSO user's ID and authority under which the TSO server job will run. The TSO server job will start as either a submitted batch job or a started task, depending upon the DEFAULTS STRTSERV command specification.

When specifying TSOSERV, you must have the CNMSSTSO sample installed if you have specified DEFAULTS STRTSERV=STRTPROC. See *IBM Z NetView Installation: Configuring Additional Components* for more information about the installation of CNMSSTSO.

UNIXSERV=*

Starts the UNIX server. The UNIX server job will start as either a submitted batch job or a started task, depending upon the DEFAULTS STRTSERV command specification.

When specifying UNIXSERV, you must have the CNMSSUNX sample installed if you have specified DEFAULTS STRTSERV=STRTPROC. See *IBM Z NetView Installation: Configuring Additional Components* for more information about the installation of CNMSSUNX.

USERFLD

Used to specify 32 bytes of user state field data for XCF group members. For user-defined groups, there is no predefined format for the field. You do not need to specify the USERFLD option if the command is issued for a DSIPLXnn group.

XCFGROUP

Specifies the name of the XCF group to join. The value is 1 - 8 characters in length. The DSIXCFMT task must be active if you specify this option.

The name of the NetView XCF group is DSIPLXnn, where nn is the value specified on the XCF.GROUPNUM statement in the CNMSTYLE member.

Restrictions

The following restrictions apply to the START command:

- If you start task AAUTSKLP using the START command, you must first start tasks DSIAMLUT and AAUTCNMI. To avoid unnecessary steps, use the STARTCNM command instead of the START command.
- To dynamically change the initialization member used by the task, the load module associated with the task, or the priority of a task, first stop the task, then restart the task using the START TASK command with the MEM, MOD, or PRI keyword.
- A new copy of the load module specified on the MOD keyword is loaded only if the following conditions exist:
 - The module was not already loaded by another task
 - The module was not specified on any task definitions in the CNMSTYLE member at NetView initialization
- Starting a session with the domain that you are currently in can result in a loop. For example, issuing START DOMAIN=domainid, where domainid is your current domain, can cause looping.
- The optional tasks that are supplied with the NetView program are meant to be run only under one task name. If you start a second task that uses the same load module as a currently running NetView-supplied optional task, unpredictable results might occur.
- The optional DSTs that are supplied with the NetView program and their individual initialization members are meant to be run under one task name. If you start a second DST that uses the same initialization member as a currently running NetView-supplied DST, unpredictable results might occur.
- You can define the task and DST names that are supplied with the NetView program as restricted values for the MOD keyword in CNMCMD. You can apply this restriction to any task that you do not want to run under multiple task names.
- The following tasks run continuously, therefore you do not need to start them unless they reach the MAXABEND limit:
 - DSIDCBMT
 - DSIHLLMT
 - DSILOGMT
 - DSISTMMT

The MAXABEND count for a task is reset to zero if the task has run for at least one hour since the lastabend.

- START DOMAIN and START HCL are not supported on a virtual OST (VOST).

Return Codes

0

Successful processing.

8

The command did not complete successfully. Check the accompanying messages for more information.

Example: Starting a session

To start a session with DOM1, enter:

```
START DOMAIN=DOM1
```

Example: Starting the hardcopy log for yourself

To start the hardcopy log for yourself, enter:

```
START HCL=LOG1
```

Example: Starting the hardcopy log for operator OP01

To start the hardcopy log for operator OP01, enter:

```
START HCL=LOG1,OP=OP01
```

Example: Activating the network log task

To activate the network log task, enter:

```
START TASK=DSILOG
```

Example: Starting a task and using a member as the definition member

To start task CNMTAMEL and use the member of DSIPARM called DUIISC as the definition member, enter:

```
START TASK=CNMTAMEL, MEM=DUIISC
```

Example: Starting a data services task

To start a data services task (DST) that is not defined in the CNMSTYLE member, that has a priority of 7, and has an initialization member of MYINIT, enter:

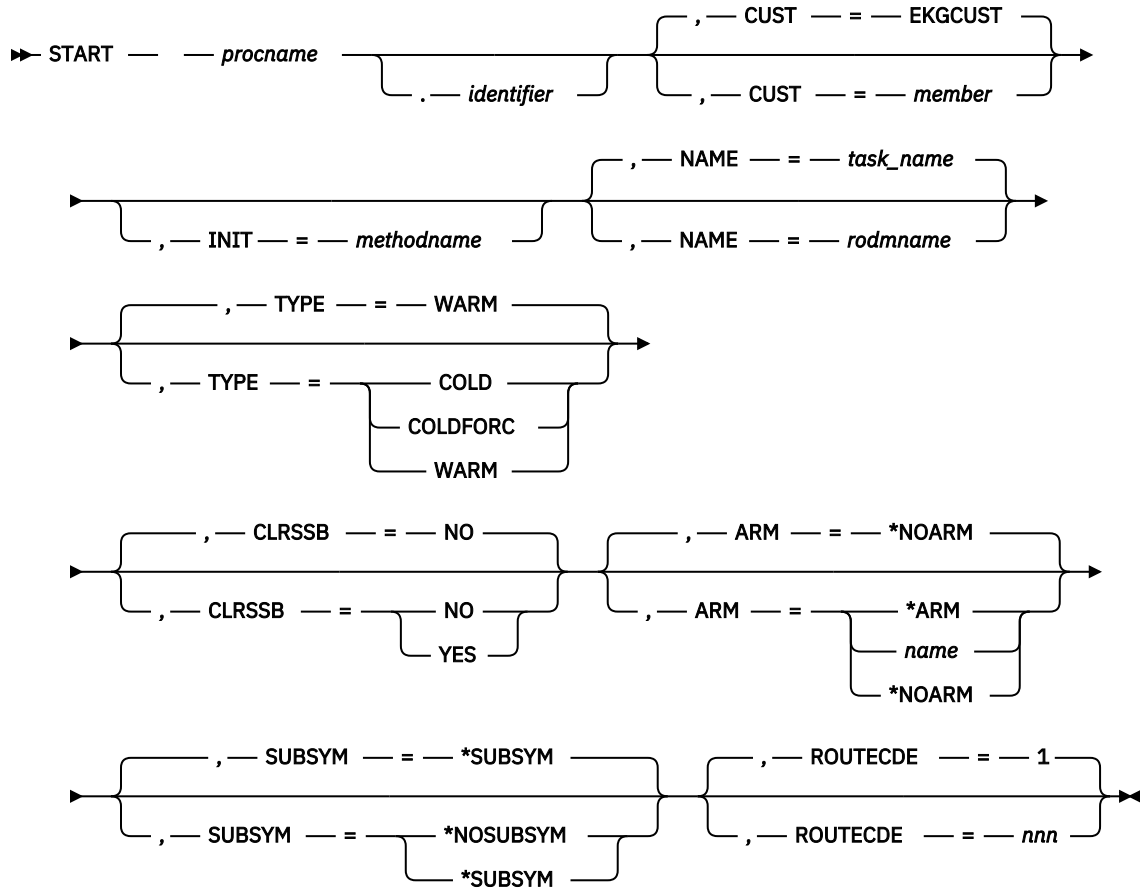
```
START TASK=MYTASK,MOD=DSIZDST, MEM=MYINIT, PRI=7
```

START (RODM)

Syntax

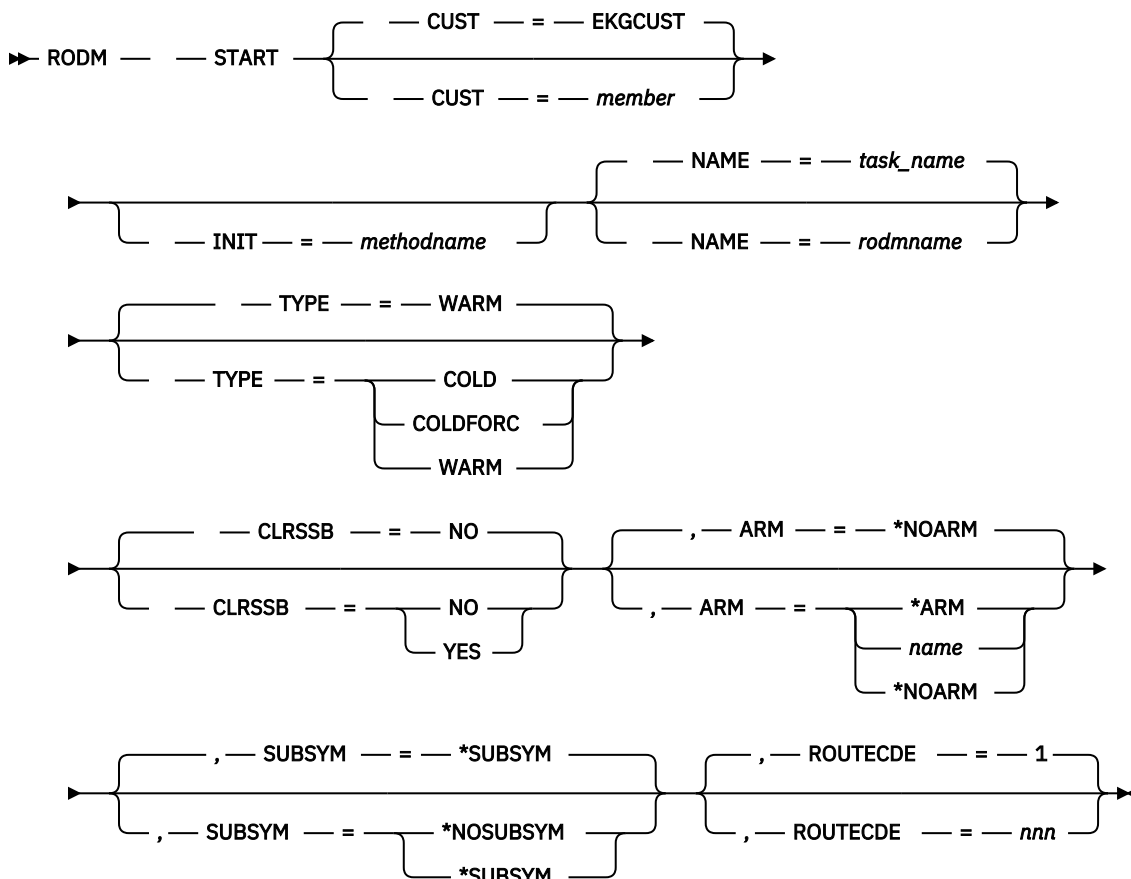
From an MVS console:

START



From a NetView terminal:

START



IBM-Defined Synonyms

Command or Operand	Synonym
START	S
COLD	C
WARM	W

Purpose of Command

RODM is initiated as a started task in the system. You can code the START operands in the job control language (JCL) or issue them with the RODM START command.

Operand Descriptions

procname

Specifies the name of the cataloged procedure that contains the JCL for RODM. The *procname* can also be the name of the started task.

identifier

Specifies the MVS job name identifying the task to be started. If you do not specify *identifier*, the *procname* is used as the name of your started task.

CUST=member

Specifies the name of the customization file to be loaded. This file must exist in the EKGCUST partitioned data set as specified in the RODM start JCL. The default value is EKGCUST.

INIT=methodname

Specifies the name of the RODM INIT method to receive control after RODM is started. The value COLD or WARM is passed to the method when the method receives control.

Two RODM INIT methods are supplied by IBM:

EKGINIT

Specifies the sample PL/I initialization method.

EKGLISLM

Specifies the RODM load function INIT method.

NAME=rodname

Specifies the name to be assigned to this RODM. This name is alphanumeric and can be up to 8 characters in length. The first character must be alphabetic. If you do not specify a name, the task name is assigned to this RODM.

TYPE

Indicates whether RODM is started warm or cold.

WARM

Indicates a warm start for RODM using data sets for which a checkpoint was previously taken. WARM is the default.

COLD

Indicates a cold start for RODM with an empty data cache. If you specify COLD, RODM prompts you to verify. If you indicate in the verification process that you do not want a cold start, RODM does not start. This happens to protect the checkpoint data from being overwritten if you intended to specify a warm start.

COLDFORC

Specifies cold start without issuing message EKG1918D, which requires operator intervention.

CLRSSB

Specifies whether MVS clears the system status block (SSB) in the common storage area (CSA) for *rodname*. Valid values are:

NO

Indicates the SSB for *rodname* is not cleared in the event of a catastrophic system abend. NO is the default.

YES

Indicates the SSB for *rodname* is cleared in the event of a catastrophic system abend. MVS does this because RODM was unable to recover sufficiently to clear the SSB itself.

Note: Use this parameter after a catastrophic RODM failure if you are unable to start a RODM with the same *rodname*. If you specify YES, you clear the SSB for any active RODM with the same *rodname*. After that, you will not be able to take a normal checkpoint for RODMs initialized with the same name. You might also have problems ending one of the RODMs.

ARM

Controls RODM registration with the MVS Automatic Restart Manager (ARM).

***ARM**

Register with ARM using a RODM-generated name, which is NETVIEW# concatenated with the RODM name.

name

Register with ARM using a user-specified name. A valid name has the following characteristics:

- Consists of 1–16 characters.
- The first character cannot be numeric.
- The remaining characters can be alphanumeric, or the following special characters: @, #, or \$.
- Alphabetic characters must be in uppercase.

***NOARM**

Do not register with ARM. This is the default value.

You can also use the MVS Automatic Restart Manager to group applications together by element type. The RODM element type is SYSNETV3. The element type cannot be changed.

SUBSYM

Specifies whether to use symbolic substitution for the RODM customization member.

***SUBSYM**

Enable symbolic substitution. This is the default value.

***NOSUBSYM**

Disable symbolic substitution.

ROUTECD

Specifies the route code to be used by RODM when sending messages to a z/OS console that are not command responses. This includes write to operator with reply messages. Valid values are in the range 1 – 128. The default value is 1.

Messages that can be issued before this parameter is processed will use the default route code 1, regardless of the value set here.

Example: Cold-starting RODM

To use the START command to cold-start RODM, enter the following from an MVS console:

```
S EKGXRODM,TYPE=C
```

Response

A response similar to the following is displayed:

```
$HASP373 EKGXRODM STARTED
IEF403I EKGXRODM - STARTED - TIME=13.32.40
EKG1906I EKGXRODM: THE RODM NAME IS EKGXRODM.
EKG1901I EKGXRODM: NO INIT METHOD IS SPECIFIED.
04 EKG1918D EKGXRODM: RODM EKGXRODM WILL COLD START. ENTER '1'
TO CONTINUE OR '2' TO TERMINATE.
r 04,1
IEE600I REPLY TO 04 IS;1
EKG0002I EKGXRODM: THE CURRENT ACTIVE LOG FILE IS NOW EKGLOGP
IEC161I 227-229,EKGXRODM,EKGXRODM,EKGD003
EKG5011I EKGXRODM: THE NUMBER OF CHECKPOINT FILES USED BY RODM
IS 2.
IEC070I 203-204,EKGXRODM,EKGXRODM,EKGD001,231,CPDLB2,
IEC070I VSAM.DIV.CHKPT001,VSAM.DIV.CHKPT001.DATA,
IEC070I CATALOG.CNMICF1.VCPDLB2
EKG1900I EKGXRODM: RODM EKGXRODM INITIALIZATION IS COMPLETE WITH LE/370.
```

Example: Cold-starting RODM and assigning a name

To use the START command to cold-start RODM and assign the name RODM1, enter the following from a NetView terminal:

```
RODM START TYPE=C,NAME=RODM1
```

Example: Starting a RODM task

To use the START command to cold-start the RODM task RODM2, enter the following from an MVS console:

```
S EKGXRODM.RODM2,TYPE=C
```

Response

A response similar to the following is displayed:


```

$HASP373 EKGXRODM STARTED
IEF403I EKGXRODM - STARTED - TIME=13.38.18
EKG1906I          RODM2: THE RODM NAME IS EKGXRODM.
EKG1901I          RODM2: NO INIT METHOD IS SPECIFIED.
05 EKG1918D          RODM2: RODM EKGXRODM WILL COLD START. ENTER '1' TO
CONTINUE OR '2' TO TERMINATE.
r 05,1
IEE600I REPLY TO 05 IS;1
EKG0002I          RODM2: THE CURRENT ACTIVE LOG FILE IS NOW EKGLOGP .
IEC161I 227-229,EKGXRODM,RODM2,EKGD003
EKG5011I          RODM2: THE NUMBER OF CHECKPOINT FILES USED BY RODM
IS 2.
IEC070I 203-204,EKGXRODM,RODM2,EKGD001,231,CPDLB2,VSAM.DIV.CHKPT001,
IEC070I VSAM.DIV.CHKPT001.DATA,CATALOG.CNMICF1.VCPDLB2
EKG1900I          RODM2: RODM EKGXRODM INITIALIZATION IS COMPLETE WITH LE/370.

```

Example: Cold-starting the RODM task RODM3 and assigning a name

To use the START command to cold-start the RODM task RODM3 and assign the name RODM4, enter the following from an MVS console:

```
S EKGXRODM.RODM3,TYPE=C,NAME=RODM4
```

Response

A response similar to the following is displayed:

```

$HASP373 EKGXRODM STARTED
IEF403I EKGXRODM - STARTED - TIME=13.41.21
EKG1906I          RODM3: THE RODM NAME IS RODM4 .
EKG1901I          RODM3: NO INIT METHOD IS SPECIFIED.
06 EKG1918D          RODM3: RODM RODM4 WILL COLD START. ENTER '1' TO
CONTINUE OR '2' TO TERMINATE.
r 6,1
IEE600I REPLY TO 06 IS;1
EKG0002I          RODM3: THE CURRENT ACTIVE LOG FILE IS NOW EKGLOGP .
IEC161I 227-229,EKGXRODM,RODM3,EKGD003
EKG5011I          RODM3: THE NUMBER OF CHECKPOINT FILES USED BY RODM
IS 2.
IEC070I 203-204,EKGXRODM,RODM3,EKGD001,231,CPDLB2,VSAM.DIV.CHKPT001,
IEC070I VSAM.DIV.CHKPT001.DATA,CATALOG.CNMICF1.VCPDLB2
EKG1900I          RODM3: RODM RODM4 INITIALIZATION IS COMPLETE WITH LE/370.

```

The following chart identifies the names in the preceding examples that become the RODM name and the MVS job name. An API request (such as connect) uses the RODM name, and the MVS MODIFY command uses the MVS job name.

Example Number	Procname	Identifier	NAME=rodmname	RODM Name	MVS Job Name
1	EKGXRODM			EKGXRODM	EKGXRODM
2	EKGXRODM		RODM1	RODM1	EKGXRODM
3	EKGXRODM	RODM2		RODM2	RODM2
4	EKGXRODM	RODM3	RODM4	RODM4	RODM3

A RODM warm start indicates that the data cache structure and the contents available to connected applications are initialized from whatever state is reflected in the checkpoint data sets. You can define these data sets to RODM through the DD statements EKGMAST, EKGTRAN, and EKGD001 through EKGD512 in the RODM system start JCL.

A RODM cold start implies that RODM is not restored from a checkpoint data set and is reset to contain the following RODM system classes and objects:

- UniversalClass
- EKG_SystemDataParent

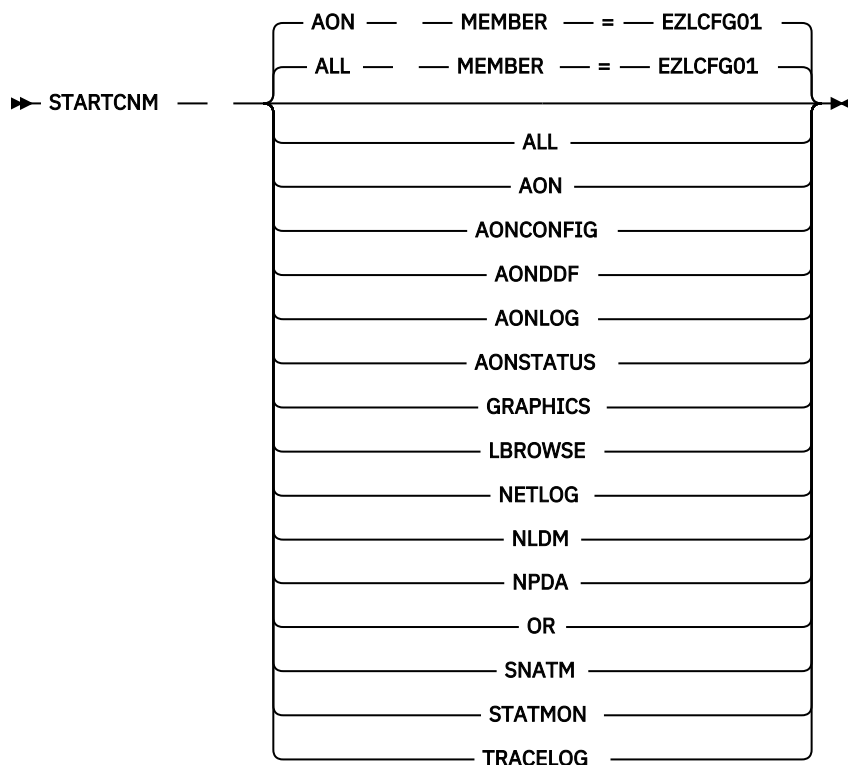
- EKG_System
- EKG_SystemObject
- EKG_NotificationQueue
- EKG_User
- EKG_Method

If the start operands specify an initialization method to be administered as part of the RODM initialization, that method can initialize the data cache structure and contents before the data cache becomes accessible to any application. The initialization method can issue a checkpoint request to cause the in-memory cache to be written to a data set on DASD. When this process is complete, the rest of the initialization process can complete.

STARTCNM (NCCF; CNME1015)

Syntax

STARTCNM



Note: Depending upon the TOWERs you have enabled in the CNMSTYLE member, not all operands and related tasks can apply to your NetView system. For example, NPDA and GRAPHICS only apply if those TOWERs are enabled. The ALL operand starts only those tasks which apply to your NetView system.

Purpose of Command

The STARTCNM command list starts the following tasks:

AON

If Automated Operations Network (AON) is installed, the STARTEZL ALL command is issued

AAUTCNMI

Session monitor

AAUTSKLP

Session monitor

BNJMNPD

Hardware monitor

CNMTAMEL

Communications between a NetView system and external components, for example, NMC and GMFHS

domid'VMT'

Status monitor

domid'BRW'

Log browse and status monitor

domid'LUC'

LUC communications (session monitor and hardware monitor)

DSI6DST

Management services transport task

DSIAMLUT

Session monitor

DSIATOPT

AUTOTEST command

DSICRTR

CNM router task

DSIGDS

Network product support

DSIHPDST

High performance transport task

DSIKREM

Central Site Control Facility (CSCF)

DSILOG

Network log task

DSIQTSK

RODM

DSITRACE

Trace log task

DSIUDST

RMTCMD command task

DUIDGHB

NETCONV IP connection

DUIFCSGW

COS gateway autotask

DUIFSSCO

Scope checker OPT task

FLBTOPO

SNA Topology Manager autotask

Operand Descriptions**ALL**

Starts all the tasks listed under the STARTCNM command. The default value is ALL.

When starting AON, the policy file or files that are loaded are determined by the CNMSTYLE member policy definitions.

AON

Issues the STARTEZL ALL command if AON is installed. The policy file or files that are loaded are determined by the CNMSTYLE member policy definitions.

AONCONFIG

Issues the STARTEZL CONFIG command if AON is installed. The policy file or files that are loaded are determined by the CNMSTYLE member policy definitions.

AONDDF

Issues the STARTEZL DDF command if AON is installed.

AONLOG

Issues the STARTEZL LOG command if AON is installed.

AONSTATUS

Issues the STARTEZL STATUS command if AON is installed.

GRAPHICS

Starts the CNMAMEL, DUIDGHB, DUIFSSCO, DUIFCSGW, and FLBTOPO tasks. For more information, see the *IBM Z NetView Installation: Configuring Graphical Components*.

LBROWSE

Starts the log browse task 'domid'BRW.

NETLOG

Starts the network log task DSILOG.

NLDM

Starts the tasks necessary for session monitor: DSIAMLUT, AAUTCNMI, AAUTSKLP, DSICRTR, and 'domid'LUC.

NPDA

Starts the tasks necessary for hardware monitor: BNJDSERV, BNJMPDA, DSICRTR, 'domid'LUC , and DSI6DST.

OR

Starts the RODM task DSIQTSK.

SNATM

Starts the FLBTOPO task.

STATMON

Starts the task necessary for status monitor: 'domid'VMT and 'domid'BRW.

TRACELOG

Starts the trace log task DSITRACE.

Example: Starting tasks

See the STARTCNM command to see a list of the tasks that can be started.

Note: The STARTCNM ALL command checks if AON is installed. If it is installed, it starts AON by issuing the STARTEZL ALL command.

Example: Starting hardware monitor tasks

To start hardware monitor tasks, enter:

```
STARTCNM NPDA
```

STARTDOM (NCCF; CNME7001)

Syntax

STARTDOM

➤ STARTDOM — — *domainid* — — *logmode* ➤

Purpose of Command

The STARTDOM command list starts cross-domain communication between the operator entering the command list and the domain entered. If you use STARTDOM to start a session over a switched line and the line has not been dialed, STARTDOM dials the domain before establishing the session.

Operand Descriptions

domainid

Specifies the domain name with which to start a session.

logmode

The logmode entry indicating the desired session bind operands. If you do not specify this operand, the logmode name defaults to the logmode name on the DLOGMOD entry coded on the APPL statement. If used with the hardcopy log, the *logmode* is the printer logmode.

Restrictions

The following restrictions apply to the STARTDOM command:

- The *domainid* must be a valid VTAM resource name.
- The *domainid* must be defined to the host at which the command list is run with a resource routing definition (RRD) statement.
- Provide the link station name and CDRM name as common global variables, if a dial is to be performed over a switched line. Use the SETADIAL command list, which must be run under the primary program operator interface task (PPT), to set these variables. The format of the SETADIAL command list is:

```
SETADIAL domainid linkname cdrmname
```

See *IBM Z NetView Installation: Configuring Additional Components* for more information about SETADIAL.

Example: Starting a session between your domain and domain CNM02

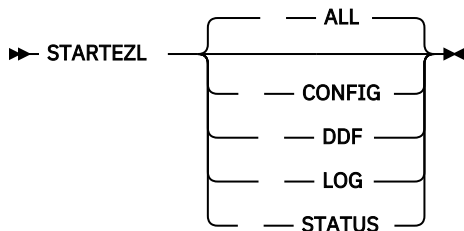
If you want to start a session between your domain and domain CNM02, enter:

```
STARTDOM CNM02
```

STARTEZL (AON)

Syntax

STARTEZL



Purpose of Command

The STARTEZL command starts AON components, which are tasks that drive the automation log, the control file, the status file, or DDF.

Operand Descriptions

ALL

Starts all of the components.

CONFIG

Starts the EZLTCFG task which is required for any policy function. You can substitute the synonyms CFG or EZLCFG for CONFIG. The CONFIG task loads the control file defined in the CNMSTYLE member. Any attempt to change the control file name is ignored.

DDF

Starts the Dynamic Display Facility component, the EZLTDDF task. You can substitute the synonym EZLDDF for DDF.

LOG

Starts the automation log component, the EZLTLOG task. You can substitute the synonym EZLLOG for LOG.

STATUS

Starts the status file component, the EZLTSTS task. You can substitute the synonyms STS or EZLSTS for STATUS.

Usage Notes

- The AON tower must be enabled in the CNMSTYLE member to successfully run this command.
- Before you issue this command, check with your system programmer to determine its impact on network automation.
- STARTEZL ignores the MEMBER=*parameter* when specified. The policy file or files that are loaded are read from the CNMSTYLE member definitions.
- After using STARTEZL to start a component or task, do not attempt to use STOPEZL to stop the component or task before receiving a message indicating that the component or task is ready and waiting for work.
- After issuing the STOPEZL CONFIG command, wait until the command has completely stopped, then issue STARTEZL CONFIG. Otherwise, all AON stops and remains inactive.
- The STARTEZL command can be issued in line mode. Therefore, you can issue it from within your own routines.

Examples

To restart all tasks, type:

```
STARTEZL ALL
```

Any tasks that were already started before you issue this command are not affected.

STARTGWY (AON)

Syntax

STARTGWY

► STARTGWY ◄

Purpose of Command

The STARTGWY command initializes all the gateway sessions for this NetView system as defined by the GATEWAY entries in the control file. When gateway sessions are active, you can send commands to other domains without having your own sessions active on those domains. Instead, the automation operators known as gateway operators log on to the other domains and handle communications for you. To use the STARTGWY command from the operator interface, see *IBM Z NetView User's Guide: Automated Operations Network*.

Usage Notes

- The AON tower must be enabled in the CNMSTYLE member to successfully run this command.
- When the gateways start successfully, the following command is displayed on the command facility:

```
EZL652I GATAN001 STARTING GATEWAY SESSIONS WITH TARGET DOMAINS
```

- For the gateway sessions to operate correctly, your system programmer must define the following entries in the control file for systems that have gateway connections:

```
AUTOOPS GATEWAY
GATEWAY domain_id
MONITOR domain_id
FORWARD
```

When the GATEWAY control file entries are correct, gateway sessions are established automatically when the NetView program and AON initialize.

- The STARTGWY command is useful if you are testing gateway sessions, or if you want to reestablish gateway sessions without waiting for AON automation to detect the broken connection and restart the sessions.

Examples

To start all gateway sessions, type:

```
STARTGWY
```

STATAPI (RODM)

Syntax

From an MVS console:

STATAPI

►► MODIFY — — *name* — , — STATAPI — — CLEAR — ►

From a NetView terminal:

STATAPI

►► RODM — — STATAPI — — CLEAR — ►

IBM-Defined Synonyms

Command or Operand	Synonym
MODIFY	F

Purpose of Command

The STATAPI command specifies that RODM writes the API statistics to the RODM log file as a type 8 record.

Operand Descriptions

name

Specifies the RODM MVS job name.

CLEAR

Clears the API statistics counters after writing the API statistics to the RODM log file as a type 8 record.

Restrictions

To write the API statistics to the RODM log, the RODM log must be active. You can query the RODM log with the RODM LOGQ command.

Example: Writing API statistics and clearing counters

To write the API statistics to the RODM log file as a type 8 record and clear the API statistics counters, enter the following from a NetView terminal:

```
RODM STATAPI,CLEAR
```

STATCELL (RODM)

Syntax

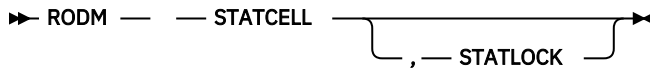
From an MVS console:

STATCELL

►► MODIFY — — *name* — , — STATCELL — — STATLOCK — ►

From a NetView terminal:

STATCELL



IBM-Defined Synonyms

Command or Operand	Synonym
MODIFY	F

Purpose of Command

The STATCELL command specifies that RODM writes the cell pool statistics to the RODM log file as a type 8 record.

Operand Descriptions

name

Specifies the RODM MVS job name

Example: Writing cell pool statistics to the RODM log file

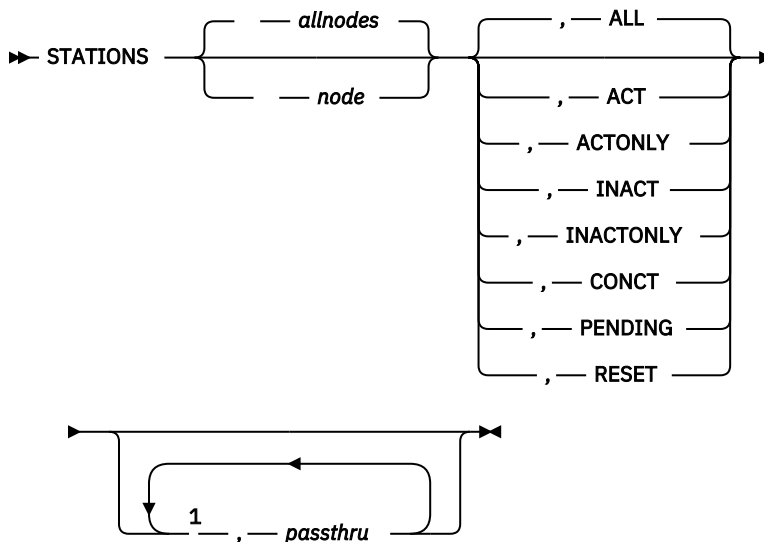
To write the cell pool statistics to the RODM log file as a type 8 record, enter the following from a NetView terminal:

```
RODM STATCELL
```

STATIONS (NCCF; CNME0033)

Syntax

STATIONS



Notes:

¹ If you do not specify a positional parameter, indicate the absence of the parameter by specifying a comma in its place.

IBM-Defined Synonyms

Command or Operand	Synonym
ACT	A
INACT	I

Purpose of Command

The STATIONS command list displays the status of all cross-subarea link stations within each node or for a specific node.

Operand Descriptions

node

Specifies the name of a node. If you omit this operand, information is displayed about all link stations in every active node.

ACT

Specifies that information is to be displayed about all active, pending, and connectable link stations within each node or specific node.

ACTONLY

Specifies that information is to be displayed about all link stations in an active state within each node or specific node. The display does not include link stations in pending or connectable states.

ALL

Specifies that information is to be displayed about all link stations (regardless of their status) within each node or specific node. ALL is the default.

CONCT

Specifies that information is to be displayed about all link stations in a CONCT (connectable) state within each node or specific node.

INACT

Specifies that information is to be displayed about all inactive link stations within each node or specific node.

INACTONLY

Specifies that information is to be displayed about all inactive link stations within each node or specific node. Resources in a RESET state are not included in the display.

PENDING

Specifies that information is to be displayed about all pending link stations within each node or specific node. A pending state is a transient state to or from the fully active state.

RESET

Specifies that information is to be displayed about all link stations in a RESET state within each node or specific node.

passthru

Specifies up to 6 parameters which are appended unchanged to the VTAM DISPLAY command issued by the STATIONS command. No validation for duplicate or conflicting parameters is performed.

Usage Notes

Consider the following when using the STATIONS command:

- If the status parameter (ALL, ACT, and so on) is omitted, and no *passthru* parameters are specified, then ALL is the default. However, if *passthru* parameters are specified and there is no status parameter specified, the NetView program does not include a SCOPE= keyword in the generated VTAM DISPLAY command. This enables you to include your own SCOPE= keyword using the *passthru* parameter.
- The valid values for the status parameter depend on the level of VTAM you are using.

Example: Displaying all the link stations

To display all the link stations, enter: STATIONS

A response similar to the following is displayed:

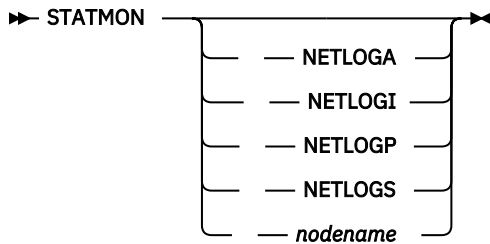
```
IST097I DISPLAY ACCEPTED
IST350I DISPLAY TYPE = STATIONS
IST393I PU T4/5 MAJOR NODE ISTDUS, SUBAREA=1
IST396I LNKSTA STATUS CTG GTG ADJNODE ADJSA NETID
IST397I 0CF-S ACTIV---I 1 1 NCPLOC1 107
IST610I LINE 0CF-L - STATUS ACTIV---I
IST314I END
```

The name of the physical unit is ISTDUS, the subarea address is 1, the link station name is 0CF-S, the status is ACTIV---I, the current transmission group (CTG) number is 1, and the defined transmission group number (GTG) is 1.

STATMON (STATMON)

Syntax

STATMON



Purpose of Command

The STATMON command invokes the status monitor full-screen mode. The status monitor dynamically collects information about SNA resources in the network and summarizes this information into a full-screen display. You can also use the status monitor to automatically reactivate specified failing resources and to browse the NetView log.

You can use the following commands while you are using the status monitor:

- BACK
- END
- FORWARD
- RETURN
- SCLIST
- SMENU
- SREFRESH
- SVTAM

Operand Descriptions

NETLOGA

The active network log.

NETLOGI

The inactive network log.

NETLOGP

The primary network log.

NETLOGS

The secondary network log.

nodename

The name of the node for which you want information.

Usage Notes

The following considerations apply to the STATMON command:

- For the STATMON command, if the common global variable CNMIMSTATMON contains a non-null value, the value is displayed at the bottom of the STATMON panel. This is useful for displaying the settings of your PF keys. You can set the value of this common global variable using the PFKDEF command. For more information, see the *IBM Z NetView Customization Guide*.
- If you specify only STATMON, the Domain Status Summary panel is displayed. If you enter a node name, the Node Status Detail panel is displayed for the specified node name.
- When you use this command, the status monitor component remains on the NetView component stack which is used with the ROLL command.
- The status monitor uses colors on color terminals or high and normal intensity on monochrome terminals to display information about different resource states. These states can be any of the following:

ACTIVE

Nodes that are active (shown in green or normal intensity)

PENDING

Nodes that are waiting to become active or inactive (shown in white or normal intensity)

INACT

Nodes that have been inactivated (shown in red or high intensity)

MONIT

Nodes that are inactive, but that the status monitor is automatically trying to reactivate (shown in turquoise or normal intensity)

NEVACT

Nodes that have never been in an active state (shown in turquoise or normal intensity)

OTHER

All other possible states (shown in turquoise or normal intensity)

When you first enter the status monitor, the status of the resources shown in the status monitor panels is refreshed automatically. You can then press the PF5 key to stop the resources from being refreshed automatically.

Table 6. Mapping VTAM States to Status Monitor States

VTAM Status Code	VTAM Status	Status Monitor Status	Notes
00xx	Inactive	Inactive (INACT)	The exceptions are: • 0000 (Reset) is mapped to OTHER. This is a substate of the VTAM Inactive status and is handled differently because of multiple ownership considerations. • 0002 (Released) is mapped to OTHER. This is a substate of the VTAM Inactive status and is handled differently because of multiple ownership considerations. • If the resource has been selected for re-activation by using the STATOPT statement, it is mapped to MONIT. • If the resource never reaches the active state since the resource has been known to VTAM, it is mapped to NEVACT. If the resource is released or reset, all the information associated with the resource is lost. Inactivating a major node causes all of the resources under it to be reset.
01xx	Pending Active	Pending (PENDING)	
02xx	Connectable	Other (OTHER)	
03xx	Reactivate		This VTAM status is changed to a VTAM Active or Inactive status after the resource it reactivated. Until then, this VTAM status is not mapped to a status monitor status.
04xx	Pending Active	Pending (PENDING)	
05xx	Active	Active (ACTIVE)	
06xx	Routable	Other (OTHER)	

STATS (NPDA; CNME3005)

Syntax

STATS

➤ STATS — — *resname* ➤

Purpose of Command

The STATS command list displays a list of the most recent statistics for the specified resource.

Operand Descriptions

resname

Specifies the symbolic name of the resource. You can specify up to five resource names to fully qualify the resource for which data is to be displayed. For more information about specifying resource names in a hierarchy, see *IBM Z NetView User's Guide: NetView*.

Restrictions

The following restrictions apply to the STATS command:

- This command list generates the NPDA MRECENT ST command.

Example: Generating the most recent statistics panel for a specified PU

To generate the Most Recent Statistics panel for PU08, enter:

```
STATS PU08
```

STATUS (GMFHS)

Syntax

GMFHS STATUS

➤ GMFHS STATUS ➤

Purpose of Command

The NetView GMFHS STATUS command provides a summary report showing the status of the GMFHS job.

You can enter the STATUS command from the MVS console using the MVS MODIFY command, or from a NetView terminal using the GMFHS command list.

Example: Displaying the status of the GMFHS job

To display the status of the GMFHS job, enter:

```
GMFHS STATUS
```

Response

A response similar to the following is displayed:

```
DUI4040I STATUS DISPLAY
DUI4041I RODM CONFIGURATION STATUS = COMPLETE
DUI4043I TYPE = GDS          STATUS = ACTIVE    SESSION = NTEFI016
DUI4042I TYPE = CNMTAMEL     STATUS = ACTIVE    SESSION = CNMTAMEL
PPIST = OK
DUI4042I TYPE = SCOPT        STATUS = ACTIVE    SESSION = DUIFSSCO
PPIST = OK
DUI4043I TYPE = RODM         STATUS = ACTIVE    SESSION = RODMNAME
DUI4037I END
```

This response contains the status of GMFHS sessions. For the graphic data server (GDS), this includes the logical unit (LU) name where the GDS resides. For CNMTAMEL and the scope checker (SCOPT), the response includes the status of the PPI.

STATUS (NCCF; CNME0034)

Syntax

NCCF STATUS

➤ STATUS — — code ➤

Purpose of Command

The NCCF STATUS command list displays information about VTAM status codes and status modifiers.

Operand Descriptions

code

Is the status code, or optionally, the status modifier.

Example: Receiving an explanation for status code STATUS=PCTD1

To receive an explanation for the status code STATUS=PCTD1, enter:

```
STATUS PCTD1
```

Example: Receiving an explanation for status code ACTIV-N---

To receive an explanation for the status code ACTIV-N---, enter:

```
STATUS ACTIV-N---
```

where -N--- is a modifier. When the code explanation displays, press the Help key to display help for the modifier.

STOP (EAS)

Syntax

EAS STOP

➤ MODIFY — — *procname* — , — STOP — — TASK — = — (— *taskid* —) ➤

IBM-Defined Synonyms

Command or Operand	Synonym
MODIFY	F

Purpose of Command

The STOP command stops any Event/Automation Service task that is not already stopped. Use the first form to stop only one service task and the second form to stop one or more service tasks.

Note: If an attempt is made to stop a task that is already stopped, a warning console message is issued.

Operand Descriptions

procname

Specifies the Event/Automation Service job name.

TASK=taskid

Specifies the service tasks to be stopped. The *taskid* can have the following values:

ALERTA

The alert adapter service task

ALERTC

The confirmed alert adapter service task

MESSAGEA

The message adapter service task

MESSAGEC

The confirmed message adapter service task

EVENTRCV

The event receiver service task

TRAPALRT

The trap to alert conversion task

ALRTTRAP

The alert to trap conversion task

ALL

All service tasks

Restrictions

The following restrictions apply to the STOP command:

- You can specify only one TASK operand for each STOP command. If you want to specify more than one service task, separate each *taskid* with a comma and enclose the *taskids* string within parentheses.
- This command cannot be used to stop the entire Event/Automation Service address space. If you want to stop the entire address space, see the TERM (EAS) command help.

Example: Stopping a service task

To stop the alert adapter service task for the Event/Automation Service job named IHSAEVNT, enter:

```
F IHSAEVNT,STOP,TASK=ALERTA
```

Response

You receive the following response:

```
IHS0118I Alert Adapter task has terminated.
```

Example: Stopping multiple service tasks

To stop the alert adapter and message adapter service tasks for the Event/Automation Service job named IHSAEVNT, enter:

```
F IHSAEVNT,STOP,TASK=(ALERTA,MESSAGEA)
```

Response

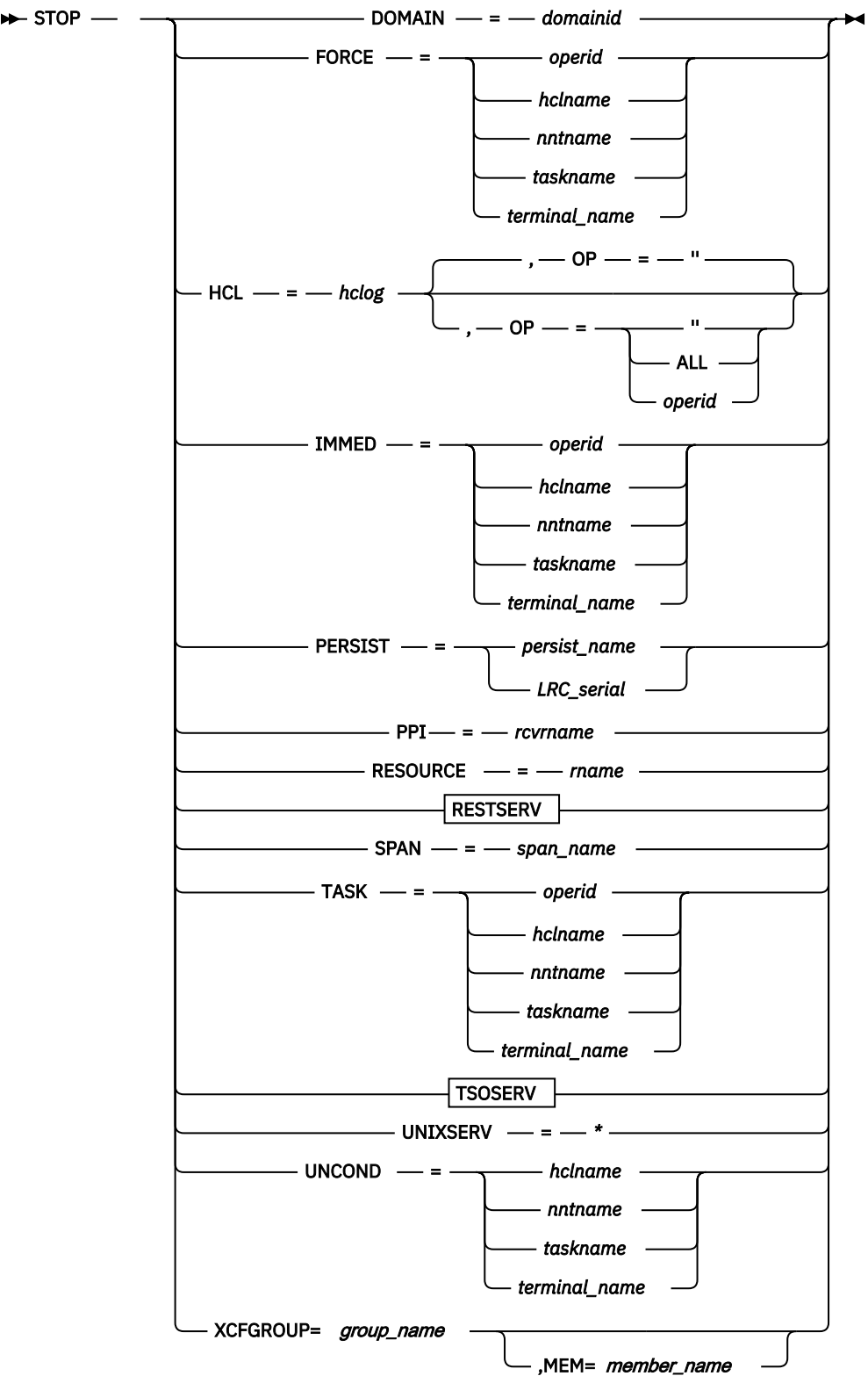
You receive the following response:

```
IHS0118I Alert Adapter task has terminated.  
IHS0118I Message Adapter task has terminated.
```


STOP (NCCF)

Syntax

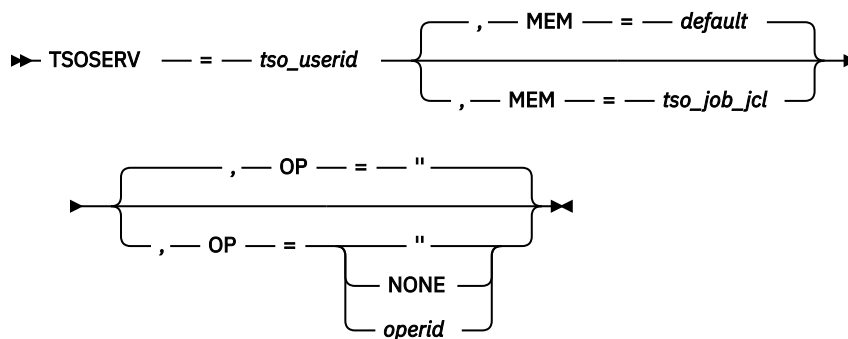
NCCF STOP



RESTSERV

►► RESTSERV — = — * ►►

TSOSERV



Purpose of Command

The NCCF STOP command stops:

- A session between the entering operator and another domain
- A hardcopy log (printer)
- Enabled PIPE PERSIST elements
- A resource within a span
- A span (deletes it from an operator's span of control)
- A task
- A process running on the target task
- A TSO command server task
- A UNIX command server task
- This NetView system's participation in an XCF group in the sysplex

Operand Descriptions

DOMAIN=*domainid*

Ends the cross-domain session between the specified domain and the local domain.

FORCE

Use STOP FORCE to rescue a task that appears unable to process normally. To end a task, use STOP TASK. If a task hangs when stopping it, the task can abend with a system code of X'EC4' with no memory dump. Note that if a system recovery routine is in control (for example, for the consoles component) a memory dump might be produced. If the target task is not an optional task, and is found to be ready and waiting for work (not hung or currently processing a command), message DSI530I is issued.

STOP FORCE processing always uses the least intrusive interruption consistent with the state of the target task. If the problem at the target task does not clear up within a minute or so, then repeating the STOP FORCE command results in a more aggressive interruption.

Attention: Optional tasks do not recover and continue when their functions are abnormally interrupted. You must issue a START command to reactivate optional tasks.

Optional tasks can take many minutes to end normally. When STOP FORCE is issued for a target task that is already in the process of ending, the action of STOP FORCE can be delayed by up to a minute. Remember, when you issue STOP FORCE before I/O activity ends you can cause data sets to be corrupted.

Valid parameters are:

hclname

Indicates which hardcopy log is to be deactivated.

nntname

Indicates which NetView-NetView task (NNT) is to be deactivated.

taskname

Indicates which task is to be deactivated.

terminal_name

Indicates which NetView terminal session is to be deactivated.

HCL=hcllog

Is the current name of the hardcopy log task you want to end.

IMMED

Unconditionally ends a task, log, or session. The parameters are the same as those for the STOP FORCE command.

The STOP IMMED command stops a task, including all usual cleanup procedures, but will not reinstate the task. Before using STOP IMMED always exhaust all other options. Be sure to make a note of the cautions listed under STOP FORCE. Collect documentation for a problem report for IBM Software Support before issuing STOP IMMED. STOP IMMED will immediately interrupt and end the target task. An X'EC4' abend will be reported.

Attention: The target task will lose storage and possibly other resources. Data sets can be corrupted.

MEM=

Specifies the name of the DSIPARM member that contains the JCL for the TSO server job. If the TSOSERV was started as a submitted job, the default value is CNMSJTST. If the TSOSERV was started as a started task, the default value is CNMSSTST. See the DEFAULTS STRTSERV command for more information.

For the XCFGROUP option, specifies the name to use as the member name. The value is 1 - 16 characters in length. For DSIPLXnn groups, the member name must be the NetView domain name. If omitted, the MEM value defaults to the NetView domain name.

For RESTSERV, the default value is EJNSSRST, which is the name of the CNMSAMP member that contains the JCL for the REST API server started task. EJNSSRST must be installed in a system procedure library. For more information about the installation of EJNSSRST, see *IBM Z NetView Installation: Configuring Additional Components*. This value should be the same member name as was specified on the START RESTSERV=* command.

OP=

For HCL, ends a session with a hardcopy log.

..

Specifies to end your hardcopy log session. This is the default.

ALL

Ends all connections with the named hardcopy log.

operid

Specifies to end a session with the hardcopy log and the specified operator.

For TSOSERV, specifies a valid NetView operator ID or " (double quotation mark), indicating the NetView operator for whom this server is to be stopped, or NONE indicating that this server is no longer considered a global server. " indicates that the ID of the NetView operator issuing the STOP command is to be used. Note that the server is disassociated with the specified operator, but is not stopped as long as it is still associated with another operator or considered global. The default value is "".

PERSIST=

Ends enabled PIPE PERSIST elements.

persist_name

Specifies the name of the PIPE enabling the PERSIST element. If more than one element has the same name, all the elements are ended. The name is case-sensitive.

LRC_serial

Specifies the long running command serial number.

Note: You can use the LIST PERSIST command to identify the *persist_name* or *LRC_serial* number.

PPI=rcvrname

This is the current Program-to-Program receiver name that must be released. Use this function for problem recovery only.

RESOURCE=rname

The named resource (such as a terminal) is made unavailable to span of control checking in this NetView system. You cannot use wildcard characters in the resource name. The resource name must exactly match a resource name in a span of control definition. When a resource is stopped and an exact match is found for the resource name in a span of control definition, span of control checking stops and resource access is denied. The only exception is for operators with CTL=GLOBAL, which bypasses span of control checking.

RESTSERV=*

Stops the REST Server. The REST Server will stop as a stopped task.

When specifying RESTSERV, you must have the EJN SSRST sample installed in a system JCL procedure library. For more information about the installation of EJN SSRST, see *IBM Z NetView Installation: Configuring Additional Components*.

SPAN=span_name

The span list (*span_name*) is taken out of an operator's control.

TASK=taskname

Use STOP TASK to cause another task to end normally. However, you cannot stop the PPT or any of these tasks:

- DSIDCBMT
- DSIHLLMT
- DSILOGMT
- DSIMONIT
- DSISTMMT
- DSITIMMT

Using the STOP TASK command to stop individual tasks can cause communication failures between active tasks. For example, if you stop task AAUTSKLP while using the command facility, and then enter the NLDM command, you might not get a response to the NLDM command, if the communication with task DSIAMLT had failed because task AAUTSKLP stopped. To prevent this situation from occurring, use the STOPCNM NLDM command so that all the NLDM tasks are stopped as one unit.

STOP TASK issues DSI660 when it initiates the stop process. You receive the DSI600 message when the target task is detached. You will not be able to restart the task until after the DSI600 is received. If you issued STOP TASK in a pipeline, use CORRWAIT to cause the pipeline to wait until the stop process is complete.

Optional tasks can take many minutes to end normally. Other tasks end promptly. For non-optional tasks only, if the ending task has not completed approximately one minute after the STOP TASK command, the NetView program will automatically issue STOP FORCE.

TSOSERV=

Specifies a valid TSO user ID or " (double quotation mark), indicating the NetView operator ID is to be used as the TSO user ID.

UNCOND

Unconditionally deactivates a task, log, or session. The parameters are the same as those for the STOP FORCE command.

The STOP UNCOND command causes storage owned by the target task to be lost. The BNH160 message is issued. Some definitions for the task are lost.

Attention: STOP UNCOND destroys important task control information in the NetView program. You might not be able to restart the NetView program until the next IPL of MVS. If you do use STOP UNCOND, do the following:

1. Note the cautions listed in STOP IMMED
2. Issue STOP IMMED
3. Collect documentation for IBM Software Support to determine the cause of the STOP IMMED failure
4. Be ready to close and restart the NetView program at your earliest opportunity.

UNIXSERV=*

Stops the UNIX server.

XCFGROUP

Specifies the name of the sysplex XCF group to leave. The value is 1 - 8 characters in length. The DSIXCFMT task must be active if you specify this option.

Restrictions

The following restrictions apply to the STOP command:

- If you issue the STOP IMMED, or STOP UNCOND command, transactions in progress might not finish writing VSAM data. For STOP FORCE, the I/O in process at the time of the abend is completed.
- Data services tasks are reinstated by ending any pending VSAM I/O and then ending the data services task. You receive the DSI600 message when the target task is detached. You will not be able to restart the task until after the DSI600 is received.
- You can restart data services tasks and hardcopy tasks using the START command.
- For data services tasks, issue STOP TASK so that the task can attempt a normal ending. If the task has not ended within a reasonable amount of time, issue STOP FORCE. Use STOP IMMED only as a last resort. The task hangs in the stopping processing, it can abend with a system abend of X'EC4' and no memory dump. Any transactions in progress might not finish writing VSAM data. Only the I/O in process at the time of the abend is completed. Note that if a system recovery routine is in control (for example, for the consoles component) a memory dump might be produced.
- If you stop task DSIAMLUT using STOP TASK, STOP FORCE, STOP IMMED, or STOP UNCOND, also stop task AAUTSKLP. Restart task AAUTSKLP after you restart task DSIAMLUT. You can issue STOP for AAUTSKLP followed by START for AAUTSKLP after DSIAMLUT ends if you want to recover AAUTSKLP. However, this procedure is not recommended. Use STOPCNM NLDM and STARTCNM NLDM.
- The following tasks run continuously, and are reinstated by the NetView program automatically if they fail, unless they reach the MAXABEND limit. If you attempt to stop them using the STOP TASK or STOP FORCE command, the tasks respond with message BNH110I and continue running. The tasks are:
 - DSIDCBMT
 - DSIHLLMT
 - DSILOGMT
 - DSIMONIT
 - DSISTMMT
 - DSITIMMT

You can recycle these tasks by using STOP IMMED. If you use STOP IMMED, these tasks abend and the NetView program automatically restarts them. This forced abend is counted against the MAXABEND limit for the task.

The MAXABEND count for a task is reset to zero if the task has run for at least one hour since the last abend.

- For STOP TASK, STOP FORCE, STOP IMMED, and STOP UNCOND *taskname* can be that of a virtual OST (VOST). VOST task names are of the form DSI#*nnnn* where *nnnn* is in the range of 0000 - 9999.
- For STOP TASK and LOGOFF OST and NNT tasks only, either command initiates monitoring of the task. If approximately one minute elapses and the task has not completed stopping, then the task suffers a system abend EC4.
- STOP XCFGROUP results in a change in the master NetView program in the sysplex if the local NetView program is the master NetView program.

Return Codes

0

Processing was successful.

8

An error occurred during processing.

Example: Stopping your hardcopy logging sessions

To stop your hardcopy logging sessions, enter:

```
STOP HCL=NRN1520A
```

Response

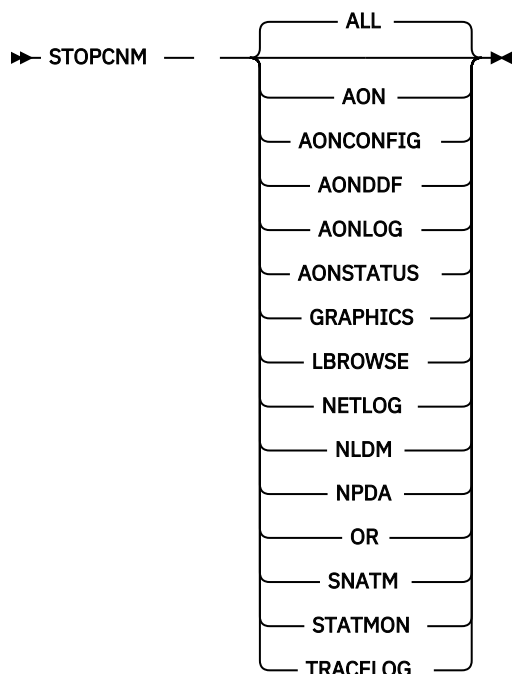
A message similar to the following is received.

```
DSI056I NRN1520A SESSION STOPPING FOR OPER1.
```

STOPCNM (NCCF; CNME1016)

Syntax

STOPCNM



Note: Depending upon the TOWERS you have enabled in the CNMSTYLE member, not all operands and related tasks can apply to your NetView system. For example, NPDA and GRAPHICS only apply if those TOWERS are enabled. The ALL operand starts only those tasks which apply to your NetView system.

Purpose of Command

The STOPCNM command list stops the following tasks:

AON

If Automated Operations Network (AON) is installed, the STOPEZL ALL command is issued

AAUTCNMI

Session monitor

AAUTSKLP

Session monitor

BNJMNPD

hardware monitor

CNMTAMEL

Communications between the NetView program and external components, for example, NMC and GMFHS

domid'VMT'

Status monitor

domid'BRW'

Log browse and status monitor

domid'LUC'

LUC communications (session monitor and hardware monitor)

DSI6DST

Management services transport task

DSIAMLUT

Session monitor

DSIATOPT

AUTOTEST command

DSICRTR

CNM router task

DSIGDS

Network product support

DSIHPDST

High performance transport task

DSIKREM

Central Site Control Facility (CSCF)

DSILOG

Network log task

DSIQTSK

RODM

DSITRACE

Trace log task

DSIUDST

RMTCMD command task

DUIDGHB

NETCONV IP connection

DUIFCSGW

COS gateway autotask

DUIFSSCO

Scope checker OPT task

FLBTOPO

SNA Topology Manager autotask

Operand Descriptions**ALL**

Stops all tasks previously started using the STARTCNM command. The default value is STOPCNM ALL.

AON

Issues the STOPEZL ALL command if AON is installed.

AONCONFIG

Issues the STOPEZL CONFIG command if AON is installed.

AONDDF

Issues the STOPEZL DDF command if AON is installed.

AONLOG

Issues the STOPEZL LOG command if AON is installed.

AONSTATUS

Issues the STOPEZL STATUS command if AON is installed.

GRAPHICS

Stops the CNMTAMEL, DUIDGHB, DUIFSSCO, DUIFCSGW, and FLBTOPO tasks.

LBROWSE

Stops the log browse task 'domid'BRW.

NETLOG

Stops the network log task DSILOG.

NLDM

Stops the session monitor tasks DSIAMLUT, AAUTCNMI, and AAUTSKLP.

NPDA

Stops the BNJDSERV and BNJMNPD hardware monitor tasks.

OR

Stops the RODM task DSIQTSK.

SNATM

Starts the SNATM task FLBTOPO.

STATMON

Stops the status monitor task 'domid'VMT.

TRACELOG

Stops the trace log task DSITRACE.

Restrictions

The following restrictions apply to the STOPCNM command:

- Check with your system programmer before using this command list.
- Use the STOPCNM command list when ending a VSAM DST (NETLOG or TRACELOG).
- The CNM router task (DSICRTR) is not affected by STOPCNM.
- Wait until you receive the DST IS READY message to run this command list.
- If you stop task DSIAMLUT by using STOP FORCE or STOP TASK, also stop task AAUTSKLP. Restart task AAUTSKLP after you restart task DSIAMLUT. Issue STOP for AAUTSKLP followed by START for AAUTSKLP after DSIAMLUT ends if you want to recover DSITSKLP. This procedure is not recommended. Use STOPCNM NLDM and STARTCNM NLDM.

- If you stop task DSIAMLUT by using STOP FORCE or STOP TASK, also stop task AAUTSKLP. Restart task AAUTSKLP after you restart task DSIAMLUT. Issue STOP for AAUTSKLP followed by START for AAUTSKLP after DSIAMLUT ends if you want to recover DSITSKLP. This procedure is not recommended.

Example: Stopping all tasks

Use the following command to stop any of these items:

- browse facility
- hardware monitor
- session monitor
- network log
- trace log
- network product support
- NetView management console
- SNA topology manager
- management services transport
- high performance transport
- RMTCMD command,
- RODM
- CSCF
- AON (if installed)

```
STOPCNM ALL
```

Note: The STOPCNM ALL command checks whether AON is installed. If it is installed, it stops AON by issuing the STOPEZL ALL command.

Example: Stopping the hardware monitor and support facility tasks

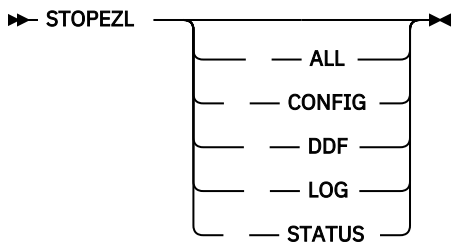
To stop the hardware monitor, enter:

```
STOPCNM NPDA
```

STOPEZL (AON)

Syntax

STOPEZL



Purpose of Command

The STOPEZL command stops AON components, which are tasks that drive the automation log, the control file, the status file, or DDF.

Operand Descriptions

ALL

Stops all the components.

CONFIG

Stops the control file component, the EZLTCFG task. You can substitute the synonyms CFG and EZLCFG for CONFIG.

DDF

Stops the Dynamic Display Facility component, the EZLTDDF task. You can substitute the synonym EZLDDF for DDF.

LOG

Stops the automation log component, the EZLTLOG task. You can substitute the synonym EZLLOG for LOG.

STATUS

Stops the status file component, the EZLTSTS tasks. You can substitute the synonyms STS and EZLSTS for STATUS.

Usage Notes

- The AON tower must be enabled in the CNMSTYLE member to successfully run this command.
- When the STOPEZL command stops the control file task, the information in the control file is no longer available and AON automation does not operate properly.
- The STOPEZL command does not stop AON activity completely and can cause errors in your NetView log. To stop AON automation activity completely, you must disable the AON base program by using the Enable/Disable panel under the AON: Support Functions panel.
- Before you issue STOPEZL, check with your system programmer to determine its impact on the network.

Note: At some locations, your system programmer might have added authorization protection for this command. Your system programmer can tell you which commands have authorization protection active.

- If you have issued STARTEZL to start a component or task, do not use the STOPEZL command to stop the task until after you receive the message:

```
DSI530I taskname: DST IS READY AND WAITING FOR WORK
```

- The STOPEZL command can be issued in line mode. Therefore, you can issue it from within your own routines.

Examples

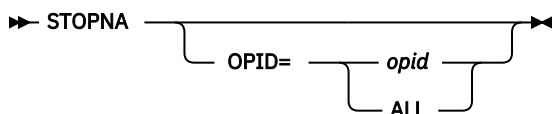
To stop the DDF component, type:

```
STOPEZL DDF
```

STOPNA (NCCF; CNME7201)

Syntax

NCCF STOPNA



Purpose of Command

The STOPNA command stops the long-running process started by the NACMD command.

Operand Descriptions

OPID

Specifies the operator ID under which the NACMD command is running. Specifying ALL has the same effect as specifying an operator ID because only one NACMD command can be active at a time. The operator under which the NACMD command is running can be determined by issuing the NACTL LISTCONN command.

SUBMIT (NCCF)

Syntax

SUBMIT

➤ SUBMIT — *data_set_name* — *dsiparm_member* —)

Purpose of Command

The SUBMIT command enters batch jobs into the system input stream to be processed.

Operand Descriptions

data_set_name

The name of the data set containing the jobs to be submitted. The data set must be cataloged. The data set can be sequential or a partitioned data set specified with a member. The data set name can be 1–44 characters for sequential data sets and 1–54 characters for a partitioned data set specified with a member.

(dsiparm_member)

The name of a member in the NetView DSIPARM data set. This member name must be in parentheses. The member name can be from 1 to 8 characters.

Restrictions

The following restrictions apply to the SUBMIT command:

- The SUBMIT command does not run on the primary program operator interface task (PPT).
- If you specify a data set name, the data set is dynamically allocated and then submitted to the internal reader.
- If a NetView DSIPARM member is submitted, the member is submitted to the internal reader.
- The SUBMIT command supports submitting batch jobs that have instream job control language (JCL), and the data set being submitted can have JCL DD data statements that have instream JCL. The SUBMIT command supports the JCL DLM operand.

Return Codes

0

Successful processing.

4

Not a valid command.

8

Error in processing. Check the accompanying DSI or CNM prefix message for more information.

Example: Submitting a job from the NetView DSIPARM data set

To submit a job from the NetView DSIPARM data set, member name DUMPSMF, enter:

```
SUBMIT (DUMPSMF)
```

Response

```
CNM279I DUMPSMF (JOB00357) SUBMITTED
```

Example: Submitting a job from a sequential data set

To submit a job from a sequential data set named RESTSMF.BATCH, enter:

```
SUBMIT RESTSMF.BATCH
```

Response

```
CNM279I RESTSMF (JOB00358) SUBMITTED
```

Example: Submitting a job from a partitioned data set

To submit a job from a partitioned data set named SYS1.PROCLIB, with member name INSTALL, enter:

```
SUBMIT SYS1.PROCLIB (INSTALL)
```

Response

```
CNM279I INSTALL (JOB00359) SUBMITTED
```

SUSPNRM (NCCF; CNME8602)

Syntax

➤ SUSPNRM ➤

Purpose of Command

The SUSPNRM command suspends NetView Resource Manager (NRM) processing.

SUSPTOPO (MSM)

Syntax

SUSPTOPO

➤ SUSPTOPO ➤

Purpose of Command

The SUSPTOPO command suspends processing of MultiSystem Manager topology requests (GETTOPO commands) for all NetView operators. If a GETTOPO command is issued while MultiSystem Manager processing is suspended, the GETTOPO command is ignored and message FLC045E is generated.

The status of the MultiSystem Manager must be *enabled* in order to suspend processing. After successful completion of this command, the status is set to *suspended*.

Usage Notes

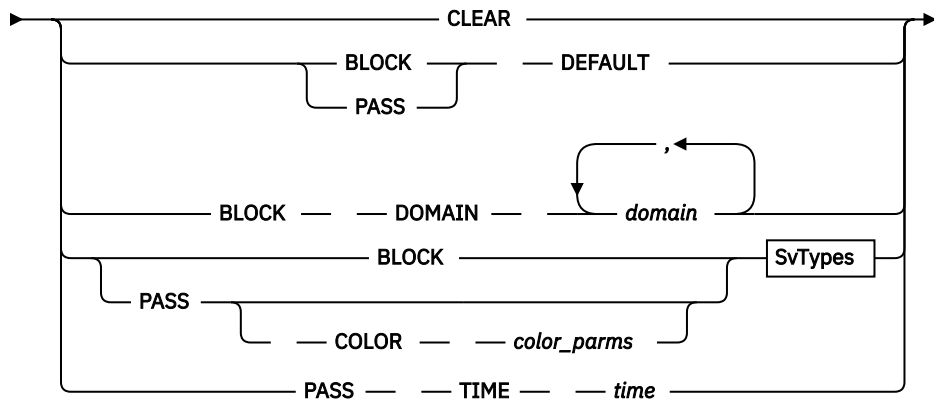
The MSM tower must be enabled in the CNMSTYLE member to successfully run this command.

SVFILTER (NPDA)

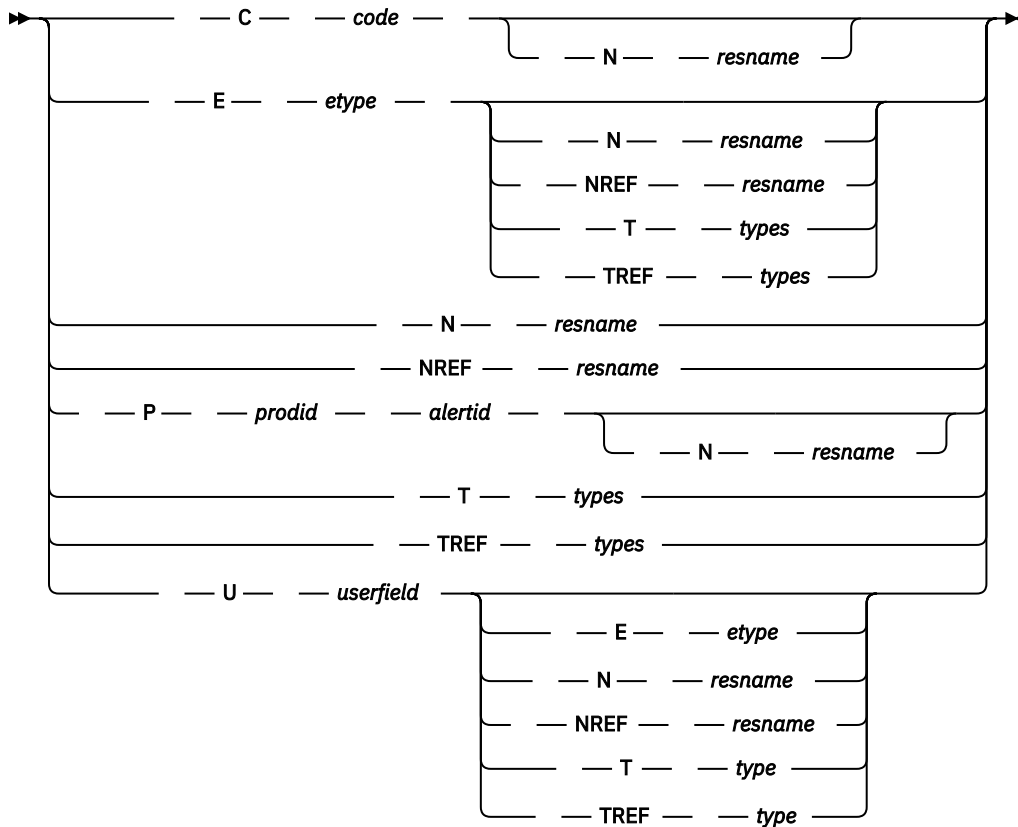
Syntax

SVFILTER

➤ SVFILTER — ➤



SvTypes



IBM-Defined Synonyms

Command or Operand	Synonym
SVFILTER	SVF
ALARM	ALM
NOALARM	NOALM
HIGHINT	HIG
UNDERSCORE	UND
BLINK	BLI
REVERSE	REV
TURQUOISE	TUR
BLUE	BLU
GREEN	GRE
PINK	PIN
WHITE	WHI
YELLOW	YEL

Purpose of Command

The SVFILTER command enables you to control which records are displayed at your terminal.

Operand Descriptions

CLEAR

Specifies that all viewing filter elements are to be removed and that the default condition for the viewing filter is to be reset to PASS. Do not specify other operands when using the CLEAR operand.

Note: If a filter is cleared, the default for that filter is reset to PASS, not to the default prior to clearing. For example, the initial default for the viewing filter is PASS. If the default was changed to BLOCK with the SVFILTER BLOCK DEFAULT command and you then enter the SVFILTER CLEAR command, the default for the viewing filter is changed to PASS (the initial default).

BLOCK

Specifies that the data matching the conditions expressed in this filter element is to be blocked from view.

PASS

Specifies that the data matching the conditions expressed in this filter element is to be allowed through viewing and displayed to the operator.

DEFAULT

Specifies that the current viewing filter default is to be overridden with a new default. Either PASS or BLOCK can be specified. If you specify BLOCK, no alerts are displayed. One or more SVFILTER PASS filters are required to display specific alerts.

BLOCK DOMAIN

Enables the focal point operator to block out alerts from specified distributed host domains. You can specify up to six domain names on one SVFILTER command. These domain names cannot be network-qualified; therefore, the BLOCK DOMAIN blocks alerts regardless of the network in which the alert originated.

Note: If you specify the DOMAIN keyword, you can specify only BLOCK. PASS is not allowed when you use the DOMAIN keyword.

TIME

Specifies that only the data recorded during the indicated time period, before the data request, is to be displayed. This specification is effective only while you are viewing the Total Events or Total Statistics panels. Also specify the PASS and *time* operands when you use the TIME operand.

time

Specifies the elapsed time in hours and minutes to be used with the TIME operand. The format of *time* is controlled by the setting of the time operands of the DEFAULTS and OVERRIDE commands.

COLOR

Sets a filter defining the color in which an alert is displayed when the alert is presented on the Alerts-Dynamic, Alerts-Static, or Alerts-History panels. COLOR cannot follow BLOCK.

color_parms

Specifies from one to four parameters associated with COLOR. You can specify up to four parameters, but only one from each of the four groups can be selected. The four groups of parameters are:

ALARM|NOALARM

Specifies whether an alarm is to sound when an alert is received. The default value is ALARM. This parameter is ignored when the Alerts-History and Alerts-Static panels are built. It applies only when a new alert is rolled onto the Alerts-Dynamic panel.

HIGHINT

Specifies that text is to appear more intense on monochrome terminals.

UNDERSCORE|BLINK|REVERSE

Specifies whether the alert is to be underscored, to flash, or to be presented in reverse video.

TUR|BLUE|GREEN| PINK|RED|WHITE|YELLOW

Specifies whether the alert is to be presented in turquoise, blue, green, pink, red, white, or yellow. TUR (turquoise) is the default.

C

Identifies the operand that follows as an event (alert) descriptor identifying code for a problem record in a format other than the generic network management vector transport (NMVT) or management services unit (MSU) format.

code

Specifies the code that identifies a particular event or alert. You can determine this code by entering the appropriate SEL number (*nn*) plus the character C (for nongeneric alerts) on the command line of the Alerts-History, Alerts-Static, or Most Recent Events panels. The following reply is displayed:

```
BNJ962I AL/EV DESCRIPTION CODE FOR SELECTION nn IS bbbcc
```

where *bbb* is the block ID and *cc* is the action code.

E

Identifies the operand that follows as an event type.

etype

Specifies the event type on which the filter item is based. Event types are:

- AVAL
- BYPS
- CUST
- DLRC
- ENV
- HELD
- IMPD
- IMR
- INST

- INTV
- NTFY
- PAFF
- PERF
- PERM
- PROC
- REDL
- RSLV
- RSNT
- SCUR
- SNA
- TEMP
- UNKN
- USER

N

Identifies the operand that follows as a resource name or resource names.

The N keyword specifies that the resource names specified match the trailing names in the hierarchy. For example, if you enter `SRFILTER AREC PASS N RES1 RES2`, a record with the hierarchy `RES4 RES3 RES1 RES2` or `RES1 RES2` matches this filter. A record with the hierarchy `RES4 RES1 RES2 RES3` does not match this filter because the names `RES1` and `RES2` do not appear at the end of the resource hierarchy.

The special character `%` means that an exact hierarchy match must occur. For example, if you enter `SRFILTER AREC PASS N RES1 RES2 %`, a record with the hierarchy `RES1 RES2` matches this filter. A record with the hierarchy `RES4 RES1 RES2` does not match this filter. To have an exact match, the number of resource names in the record must match the number in the filter statement.

resname

Specifies the symbolic name of the resource. You can specify up to five resource names to fully qualify the resource for which data is to be displayed.

You can use certain special characters (`*`, `?`, and `%`) as part of the resource name and resource name hierarchy.

NREF

Identifies the operand that follows as a resource name or resource names.

The filter element is satisfied if the specified resource name or resource names are included in the resource hierarchy in the order stated. For example, if you enter `SRFILTER AREC PASS NREF RES1 RES2`, an alert with a hierarchy of `RES4 RES1 RES2 RES3` matches this filter. Other hierarchies that match this filter are `RES1 RES2`, `RES1 RES2 RES4`, and `RES3 RES1 RES2`. A record with a resource hierarchy of `RES1 RES3 RES2` does not match this filter. The requirement is that the names specified in the filter statement be in the hierarchy of the alert in the same order as specified in the filter.

You cannot use an asterisk by itself to represent a resource name that follows the NREF keyword.

P

Specifies the product and alert identifier pair for a problem record that is in generic NMVT or MSU format. You can determine the product set identifier and the alert ID by entering the appropriate SEL number (*nn*) plus the character C on the command line of the Alerts-History, Alerts-Dynamic, or Most Recent Events panels. The following message is returned:

```
BNJ378I SELECTION nn FILTER CODE; PRODUCT ID pi
        ALERT ID ac
```


Where *pi* is the product set identifier and *ac* is the alert ID. This information is also available on the last panel of the hardware monitor Event Detail panel.

prodid

Specifies the variable product identifier (hardware or software) of the alert or event sender.

alertid

Specifies the variable alert ID number representing a specific alert or event description.

T

Identifies the operand that follows as a resource type or resource types.

The T keyword specifies that the resource types specified match the trailing types in the hierarchy. For example, if you enter `SRFILTER AREC PASS T TYP1 TYP2`, a record with the hierarchy `TYP4 TYP3 TYP1 TYP2` or `TYP1 TYP2` matches this filter. A record with hierarchy `TYP4 TYP1 TYP2 TYP3` does not match this filter because the types do not appear at the end of the types list.

The special character % specifies that an exact hierarchy match must occur. If you enter `SRFILTER AREC PASS T TYP1 TYP2 %`, a record with the hierarchy `TYP1 TYP2` matches this filter. A record with `TYP4 TYP1 TYP2` does not match this filter because it is not an exact match. To have an exact match, the number of resource types in the record must match the number in the filter statement.

type

Specifies the resource type. Examples of resource types are CHAN, COMC, processor, and LCTL. You can specify up to five resource types to fully qualify the resource for which data is to be filtered.

You can use certain special characters (* and %) as part of the resource type and the resource type hierarchy.

TREF

Identifies the operand that follows as a resource type or resource types.

The filter element is satisfied if the specified resource type or resource types are included in the resource hierarchy in the order stated. For example, if you enter `SRFILTER AREC PASS TREF TYP1 TYP2`, an alert with a hierarchy of `TYP4 TYP1 TYP2 TYP3` matches this filter. Other hierarchies that match this filter are `TYP1 TYP2`, `TYP1 TYP2 TYP4`, and `TYP3 TYP1 TYP2`. The requirement is that the types specified in the filter statement be in the hierarchy of the alert in the same order as specified in the filter.

Do not use an asterisk (*) by itself to represent a resource type that follows the TREF keyword.

U

Specifies that user data follows. This allows filtering on the user data field (subvector X'33', subfield X'30') in an NMVT or MSU. You can determine the user data for an alert or event by entering the appropriate selection number (SEL#) and the character U on the command line of the Alerts-History, Alerts-Static, or Most Recent Events panel. The following message will be displayed:

```
BNJ980I SELECTION nn USER DATA: uuuuu
```

Where *uuuuu* is the user data for the alert. The first five characters of user data are returned. For a generic alert, the entire user data field is displayed on the Event Detail panel.

Note: Regardless of the length, filtering is performed using only the first five characters of the data.

userfield

Specifies one to five characters of user data.

Restrictions

The following restrictions apply to the SVFILTER command:

- The viewing filters are stored under the BNJDSERV task. If BNJDSERV is stopped, all viewing filters are cleared. If an operator logs off, the viewing filters for that operator are not cleared.

- If NPDA commands that are subject to viewing filters (such as ALH) are issued under ATTACH processing, then any viewing filters issued by that virtual OST (VOST) are respected. If no viewing filters were issued by the VOST, then any viewing filters set by the owning operator are respected. For ATTACH requests issued on another NetView system, such as by an "ATTACH NPDA SDOMAIN . . ." command, then any viewing filters set by that VOST are respected, but any filters set by the owner are not respected. For more explanation about VOSTs and their owning operators, see the ATTACH command in the *IBM Z NetView Command Reference Volume 1 (A-N)*.
- Specify filter elements in the order shown in the syntax. Operators must specify their own viewing filters.
- Except for TIME, viewing filters apply only to the Alerts panels.
- You can use the SVFILTER command in a command list.
- The priority rules, as described for the SRFILTER command, also apply to the viewing filters for their common operands. The exception is the *domain* operand, which has the highest priority, and the R operand, which is not used.
- You can set alert color for the Alerts-Dynamic and Alerts-Static panels by the following means:
 - SVFILTER command with COLOR keyword
 - SRFILTER COLOR filter
 - Color map

If you do not set a COLOR filter for an alert, the alert is displayed based on its appropriate color map.

- Setting a COLOR recording filter for an alert overrides the color set in the color map. The recording filter color stays with the alert when the alert is logged to the hardware monitor database. If you change the color of a recording filter, it does not affect the color assigned to a previously logged alert.
- Setting a COLOR viewing filter with the SVFILTER command overrides both the SRFILTER COLOR command and the color map. This enables you to override the general setting of colors for alerts for each operator.
- If you set a default color filter, and if the alert does not match the specific color filter, the color attributes from the default color filter are used on the first line of the Alerts-Dynamic panel when the alert is rolled onto the screen. When the alert rolls off of the first line of the Alerts-Dynamic panel, the color map defines the color of the alert.
- In filtering, the hardware monitor treats the HELD event type as if it is a second alert or event type. This means that a HELD event type is always associated with another event type. The HELD event type has the same priority as all other event types; therefore, the order in which you set the HELD type filter and other event type filters is important. For example, the default SVF AREC filters are set as follows:

```
BLOCK E HELD
PASS E PERM
```

Therefore, a permanent alert that is also a HELD alert is blocked because the BLOCK filter is the first matching filter encountered.

- Pattern matching support is provided for resource names, resource types, and resource name and resource type hierarchies using the special characters *, ?, and %. This support consists of the following:

When used by itself, this character represents any resource name or type. This usage applies to resource names and types associated with the keywords N and T.

When appended to a resource or domain name, this character can represent any number of characters. Characters following an * are not allowed. This usage applies to resource names associated with the keywords N, NREF, and DOMAIN.

When you use * by itself to represent a resource name, it does not count in processing priority. For example, if you use * to represent resource names, the priority is the same as if you specified one resource name.

?

A placeholder that represents exactly one character anywhere in a resource or domain name. You can use multiple ?s in a name. This usage applies to resource names associated with the keywords N, NREF, and DOMAIN.

%

A trailing character that indicates that an exact match on a hierarchy of resource levels and resource names or types is required. Specify this character last, it must follow a resource name or type. This usage applies to resource names and types associated with the keywords N and T.

Examples

The format of times specified in the following examples assumes the default setting for time formats on the DEFAULTS and OVERRIDE commands.

Example: Setting a filter to view resources

To set a filter so only those resources with data less than 3 hours old, and those resources with attached resources less than 3 hours old, are viewed on the Total Events or Total Statistical Data panels, enter:

```
SVFILTER PASS TIME 3:00
```

Response

The usual response to an accepted command is as follows:

```
BNJ1355I SVF/SVFILTER COMMAND ACCEPTED
```

The following examples show the * and ? characters in the SVFILTER command with specified resource names or domain names.

Example: Blocking alerts from a resource

To block from view all alerts from any resource whose name begins with the letters RTP, enter:

```
SVFILTER BLOCK N RTP*
```

Example: Viewing alerts from a resource

To view all alerts from any resource name with exactly four characters, starting with RTP, enter:

```
SVFILTER PASS N RTP?
```

Example: Blocking alerts from view

To block from view all alerts from any resource name with five or more characters, starting with RTP, enter:

```
SVFILTER BLOCK N RTP??*
```

To block from view all alerts meeting the following conditions:

- The name of the domain from which the alert was received is five characters in length.
- The first three characters of the domain name are CNM.
- The last character of the domain name is 1.
- The network in which the alert originated does not matter.

enter:

```
SVFILTER BLOCK DOMAIN CNM?1
```

Example: Viewing alerts from resources

To view all alerts from any resource name that matches on any hierarchy list of three or more resource names, ending with resource name RTP, enter:

```
SVFILTER PASS N * * RTP
```

Example: Blocking alerts from resources

To block any alerts from any resource that matches on a hierarchy list of exactly two resource types, with COMC as the level 1 resource type and LINE as the level 2 resource type, enter:

```
SVFILTER BLOCK T COMC LINE %
```

Example: Viewing alerts from resources

To view alerts for a resource that matches on any hierarchy list of exactly three resource names, ending with resource name RTP3, enter:

```
SVFILTER PASS N * * RTP3 %
```

SVTAM (STATMON)

Syntax

SVTAM

▶▶ SVTAM ◀◀

IBM-Defined Synonyms

Command or Operand	Synonym
SVTAM	SV

Purpose of Command

The SVTAM command displays the VTAM commands that you can run against one or more of the displayed resources.

Example: Displaying allowed VTAM commands from a status monitor panel

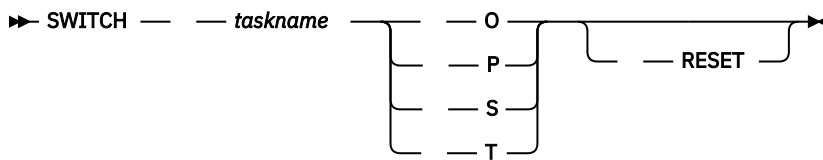
If you are displaying a status monitor panel containing resources, you can show a list of VTAM commands that you can run against one or more displayed resources by entering:

```
svtam
```

SWITCH (NCCF)

Syntax

SWITCH



Purpose of Command

The SWITCH command specifies or controls access to the primary network log, the secondary network log, the trace log, or the AON automation logs, or switches control of the data services tasks (DSTs).

Operand Descriptions

taskname

The 1-8 character name of the DST. Use the LIST command to obtain this name.

O

Specifies that the other VSAM file is to be active. If the primary file was previously active, the secondary file becomes active. If the secondary file was previously active, the primary file becomes active. If no VSAM file was previously active, the primary file becomes active. You can determine which data set has become active by examining response message DSI546I. For DSILOG and DSITRACE files, active requests are completed. New requests are sent to the newly opened file.

P

Specifies that the primary VSAM file is to be active. For DSILOG and DSITRACE files, active requests are completed. New requests are sent to the primary file.

S

Specifies that the secondary VSAM file is to be active. For DSILOG and DSITRACE files, active requests are completed. New requests are sent to the secondary file.

T

Specifies that access to the currently active VSAM file, primary or secondary, is to end after the current requests are complete. New requests are rejected.

RESET

RESET causes the target file to be reset. It is optional, but when specified, it must follow either P or S. The RESET option is particularly useful for the AON automation log or a VSAM file associated with a user-written DST. DSILOG or DSITRACE files are automatically reset whether RESET is specified.

Restrictions

The following restrictions apply to the SWITCH command:

- VSAM restrictions limit the RESET option to non-LSR (Locally Shared Resource) or non-GSR (Globally Shared Resource) data sets defined with the REUSE option. If you use the RESET option with LSR data sets or GSR data sets, OPEN errors will occur. SWITCH commands with RESET are rejected with message DSI163I for Session Monitor (NLDM) data sets, and Hardware Monitor (NPDA) data sets. For these tasks, either a RESETDB command or a DBAUTO command with the CLEAR operand can be used to clear an inactive data set before reissuing the SWITCH command.
- Do not specify RESET for VSAM files not defined with the REUSE option because OPEN errors can occur.
- The operator that issues the SWITCH command is notified of the VSAM data set OPEN or CLOSE completion by message DWO520I. The starter of the task or the authorized operator (if the starter of

the task is not available) is notified of the VSAM data set OPEN or CLOSE completion by message DSI556I. It is possible to receive both DSI556I and DWO520I, which contain similar information.

- When you issue the SWITCH command within a NetView pipeline, message DWO520I will correlate to the pipeline.
- If you are operating in an environment, such as SMS, in which the database can be moved to a different volume as a result of deleting and redefining, you can receive an open error in the form of message DSI556I with return code=X'08' and ACB error field=X'A8'. To delete and define a database in this type of environment, free the database before deleting it and reallocate the database after defining it. To do this, use the FREE command first and then the ALLOCATE command.
- For the database associated with the hardware monitor, use task name BNJDSERV.
- For the database associated with the session monitor, use task name AAUTSKLP. Switching the session monitor database between the primary and secondary VSAM files can cause the data of a session to become divided between the two VSAM files. Dividing the data can give unpredictable results when you try to display data that is not on the active VSAM file.
- If you specify T for the hardware monitor database, you will lose all incoming alerts.
- The SWITCH command issues the DSI545I, DSI546I, or DSI547I message to indicate which data set is active. If these messages are specified HOLD(Y) in the Automation Table, they will be automatically removed from the NetView screen when no longer applicable.

Example: Switching to the primary network log

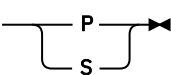
To switch to the primary network log, enter:

```
SWITCH DSILOG,P
```

SWLD (NLDM; CNME2002)

Syntax

SWLD

➡ SWLD — 

Purpose of Command

The SWLD command list switches the files used by the session monitor to store data on a VSAM database.

Operand Descriptions

P

Specifies the primary VSAM database.

S

Specifies the secondary VSAM database.

Restrictions

The following restrictions apply to the SWLD command:

- The session monitor logs session data to VSAM at various times during a session. Switching the session monitor database between the primary and secondary VSAM databases can cause a session's data to be divided between the two VSAM databases. Division of a session's data can provide unpredictable results when you try to display data that is not on the active VSAM database.
- If you are operating in an environment, such as SMS, in which the database can be moved to a different volume as a result of deleting and redefining, you can receive an open error in the form of message

DSI556I with return code=X'08' and ACB error field=X'A8'. To delete and define a database in this type of environment, free the database before deleting it and reallocate the database after defining it. To do this, use the FREE command first and then the ALLOCATE command.

- If session recording has been suspended, possibly because of an I/O error, and you switched databases, recording might need to be resumed with the NLDM SMDR START command.

Example: Changing the secondary database

To change to the secondary database if you are encountering errors on the primary database, enter:

```
SWLD S
```

Response

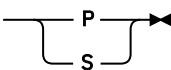
A message similar to the following is displayed:

```
DSI547I AAUTSKLP: SECONDARY VSAM DATA SET IS NOW ACTIVE
```

SWPD (NPDA; CNME3006)

Syntax

SWPD

➔ SWPD — 

Purpose of Command

The SWPD command list switches the primary and secondary databases associated with the hardware monitor BNJDSERV task. The files are used to record hardware monitor data.

Operand Descriptions

P

Specifies the primary database.

S

Specifies the secondary database.

Restrictions

The following restrictions apply to the SWPD command:

- Use this command list to make the primary or secondary database active.
- You cannot run this command list from a multiple-entries panel.
- If you are operating in an environment, such as SMS, in which the database can be moved to a different volume as a result of deleting and redefining, you can receive an open error in the form of message DSI556I with return code=X'08' and ACB error field=X'A8'. To delete and define a database in this type of environment, free the database before deleting it and reallocate the database after defining it. To do this, use the FREE command first and then the ALLOCATE command.

Example: Switching to the primary database

To switch to the primary database, enter:

```
SWPD P
```

Example: Switching to the secondary database

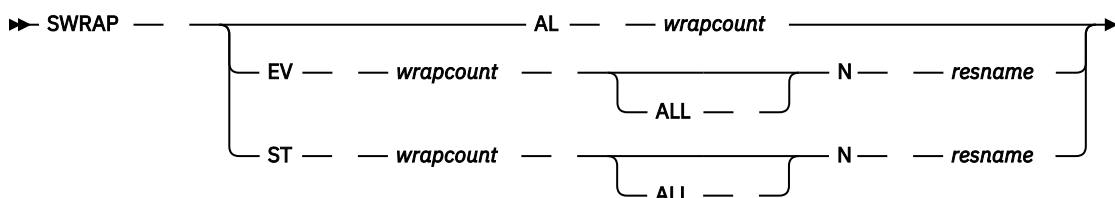
To switch to the secondary database, enter:

```
SWPD S
```

SWRAP (NPDA)

Syntax

SWRAP



IBM-Defined Synonyms

Command or Operand	Synonym
SWRAP	SW

Purpose of Command

The SWRAP command establishes the number of event or statistical records to be retained for a specified resource or the total number of alert records to be retained on the hardware monitor database. This command can be entered from the hardware monitor menu panel, a command list, an automated operator, or any NetView component.

When this command indicates a reduction in the wrap count value, the oldest records are deleted immediately. If the wrap count is very low, it can appear that the oldest record is not being wrapped off because the new record fits on the screen without deleting the old record from the screen. The oldest record is wrapped off the permanent database and is maintained on a temporary database until you return to the hardware monitor menu or enter an explicit hardware monitor command.

The default wrap counts are:

Record Type	Wrap Count
Event records per resource	
- Per resource	25
- Per resource of type LAN or type RING	100
Statistical records per resource	
- Per resource	25
- Per resource of type LAN or type RING	100
Alert records	100
SNA-unique data (RECFMS 01, 05) per resource (cannot be modified)	2

Operand Descriptions

AL

Specifies alert data.

ALL

When specified in an environment other than an NPDA panel (such as a command list, autotask, PPT or NCCF console), specifies that the command takes effect for all entries if multiple entries are found.

When specified from an NPDA panel, the ALL parameter has no effect.

wrapcount

Specifies the wrap count value in the range of 0–9999 for alerts and 0–450 for events and statistics. Note that these ranges differ for pre-V3R2 NetView programs.

EV

Specifies event data.

N

Specifies the operand that follows as a resource name.

resname

Specifies the symbolic name of the resource (for event and statistical data only). You can specify up to five resource names to fully qualify the resource for which data is to be displayed.

You can issue an SWRAP command only for resources against which data has been logged on the hardware monitor database.

ST

Specifies statistical data.

Restrictions

The following restrictions apply to the SWRAP command:

- Use this command carefully because you can inadvertently destroy error data.
- You cannot run this command from a multiple-entries panel.
- If you are issuing the command from within NPDA and the name of the resource specified is not a unique resource configuration on the database, a selection panel is displayed on which you can choose the relevant configuration.
- If you are issuing the command from within a command list and the name of the resource specified is not a unique configuration on the database, message BNJ1963I will be issued. Determine the unique resource and reissue the command or use the ALL parameter to set all the configurations that match the specified resource.
- If you are issuing the command outside a command list in an environment other than NPDA and the name of the resource specified is not a unique configuration on the database, the Hardware Monitor Multiple Entries panel will be displayed. From this panel, select one or more configurations to set.
- When the value specified on the SWRAP command is smaller than the wrap count had been prior to the SWRAP command, then if necessary records are purged from the hardware monitor database so that the total number of records retained matches the specified SWRAP value. To reclaim the free space made available by these purged records, use the "DBAUTO NPDA,REORG" command. If the REORG is not done, the free space cannot be reused.

Return Codes**Return Code****Meaning****0**

The command issued from a command list was successful for all entries of a multiple-entries panel.

2

The command issued from a command list did not specify the ALL parameter, but multiple-entries were found.

4

The command issued from a command list encountered multiple entries and failed for one or more of the resource hierarchies found.

Example: Setting the event wrap count to 100 for resource UNIT1

To set the event wrap count equal to 100 for the resource UNIT1, enter:

```
SWRAP EV 100 N UNIT1
```

Response

The usual response to this command is as follows:

```
BNJ325I WRAP EV COUNT IS 100 FOR UNIT1
```

TARGET (BROWSE CANZLOG)

Syntax

➤ TARGET ➤

Purpose of Command

The TARGET command displays a panel that shows the status of all NetView domains included in the current BROWSE session.

The BROWSE command can be used to browse Canzlog data from multiple NetView domains (for example, all domains in a sysplex, or all domains in a group defined using the ENT.GROUP CNMSTYLE statement). If you are browsing Canzlog data from multiple domains, you can use the TARGET command to see the current status of all of the domains included in the session.

The TARGET panel lists all of the included domains, and it shows the following information about each domain:

- The sysplex name, if the included domains are resolved from a sysplex
- The value of any recent search timeout
- Whether the domain is active or inactive

The information in the TARGET panel is color-coded:

- Green indicates that the domain is active.
- Red indicates that the domain is inactive.
- Pink indicates that the domain is active, but the connection failed.
- Yellow indicates a recent search timeout for the domain.

Usage Notes

Use the REFRESH subcommand to refresh the TARGET panel with current data.

TASK (GMFHS)

Syntax

TASK

➤ GMFHS TASK ➤

Purpose of Command

The TASK command displays a NetView GMFHS subtask status report.

You can enter the TASK command from the MVS console using the MVS MODIFY command or from a NetView terminal by using the GMFHS command list.

Example: Displaying the task status report for the GMFHS

To display the task status report for the GMFHS, enter:

```
GMFHS TASK
```

A response similar to the following is displayed:

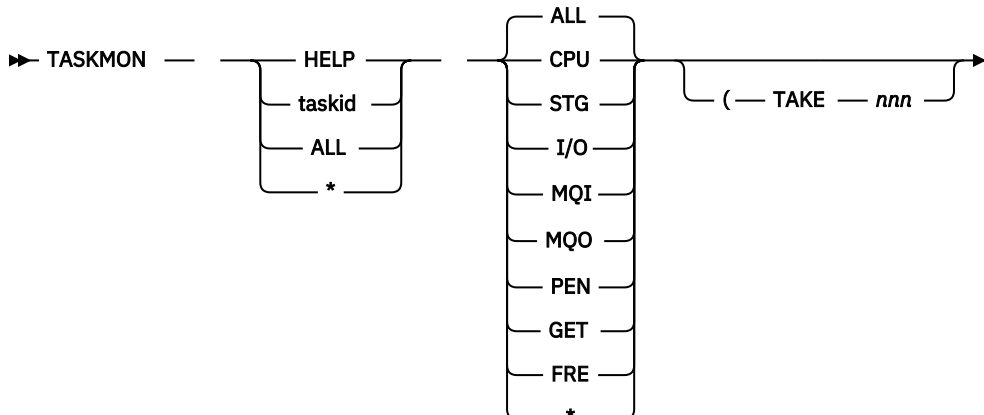
```
DUI4044I  GMFHS TASK DISPLAY
DUI4013I  TASK = IPC      STATUS = WAIT   QUEUE DEPTH = 0
DUI4045I  TASK = OPERIF   STATUS = ACTIVE  QUEUE DEPTH = 0
DUI4013I  TASK = VIEWMGR  STATUS = WAIT   QUEUE DEPTH = 0
DUI4013I  TASK = VSTATMGR STATUS = WAIT   QUEUE DEPTH = 0
DUI4013I  TASK = IRMGR    STATUS = WAIT   QUEUE DEPTH = 0
DUI4013I  TASK = RTMGR    STATUS = WAIT   QUEUE DEPTH = 0
DUI4013I  TASK = DBSERVER STATUS = WAIT   QUEUE DEPTH = 0
DUI4013I  TASK = EVENTMGR STATUS = WAIT   QUEUE DEPTH = 0
DUI4013I  TASK = NETCMD   STATUS = WAIT   QUEUE DEPTH = 0
DUI4013I  TASK = NETCON   STATUS = WAIT   QUEUE DEPTH = 0
DUI4037I  END
```

This example shows that all tasks, except the operator interface (OPERIF) task, are waiting and have no data queued. The operator interface task is active because it is processing the GMFHS TASK command.

TASKMON (CNME1100)

Syntax

TASKMON



Purpose of Command

The TASKMON command is a REXX procedure which gives you color-coded monitoring of all NetView tasks. The output under each group is sorted by the severity index. The first percentage column on the left represents a percentage of the maximum value allowed or, if no limit applies, the maximum value measured for any task monitoring is as follows:

Color codes:

White

SLOWSTG limit exceeded

Yellow

70% of limit for this line exceeded

Pink

80% of limit for this line exceeded

Red

90% of limit for this line exceeded

Operand Descriptions**ALL**

Requests data about all tasks.

taskid

The name of the task for which statistics are requested.

This is another way to specify all tasks.

ALL

Requests all statistics for a selected task or tasks. This is the default.

CPU

Requests CPU utilization statistics.

STG

Requests task storage managed by a NetView program.

I/O

Requests inputs and outputs of a task managed by a NetView system.

MQI

Requests the rate of messages coming into a task. This count will consist of only message queueing to task using NetView services DSIMQS, and only the buffers accounted for by DSIMQS.

MQO

Requests the rate of messages leaving a task. This count will consist of only message queueing to task using NetView services DSIMQS, and only the buffers accounted for by DSIMQS.

PEN

Request task penalty time statistics for all of the resource limits.

GET

Request statistics about the rate of storage obtained using the DSIGET macro (in kilobytes per minute).

FRE

Request statistics about the rate storage is released using the DSIFRE macro (in kilobytes per minute).

This is another way to request all statistics for all tasks.

TAKE

Include in the output only the specified number *nnn* of tasks nearest to their limit (as a percentage). If no limit has been applied, include only the specified number of tasks nearest the maximum value for any task of that category of data.

Usage Notes

You can use the TASKMON command to illustrate how DSITSTAT data can be interpreted by an automation procedure.

TASKMON provides statistics based on CPU percentage relative to a single processor. TASKUTIL provides statistics based on CPU percentage relative to the sum of processors.

Example: Issuing a TASKMON command

To issue a TASKMON command, enter:

```
TASKMON * *
```

A response similar to the following is displayed:

```
* NTV98      TASKMON * CPU PEN
| NTV98
TASKMON ---- START OF REPORT ----
Severity Index OPID          Current      Session      Maximum      Limit
-----CPU-----
      1.95% MAINTASK      1.85 %      2.18 %      10.15 %      95.00 %
      0.67% OPER4        0.64 %      0.80 %      3.68 %      95.00 %
      0.02% AAUTSKLP      0.02 %      0.36 %      2.83 %      95.00 %
      0.02% DSIDCBMT      0.02 %      0.30 %      2.90 %      95.00 %
      0.01% NTV98PPT      0.01 %      0.53 %      5.83 %      95.00 %
      0.01% CNMTAMEL      0.01 %      0.32 %      1.60 %      95.00 %
      0.01% FLBTOPO       0.01 %      0.27 %      1.84 %      95.00 %
      0.01% DSIMONIT      0.01 %      0.02 %      0.04 %      95.00 %
      0.01% DSITIMMT      0.01 %      0.01 %      0.04 %      95.00 %
| NTV98
Task Causing OPID          Current      Session      Seconds Left Total Seconds
-Penalty Time-
DSITIMMT- NTV98VMT      15.84 %      12.01 %      0.00 S      6.79 S
| NTV98
Severity Index
Inbound Wait OPID          Session
Total Seconds-
100.00% DSITIMMT      10.29 S
| NTV98      TASKMON ---- END OF REPORT ----
-----
```

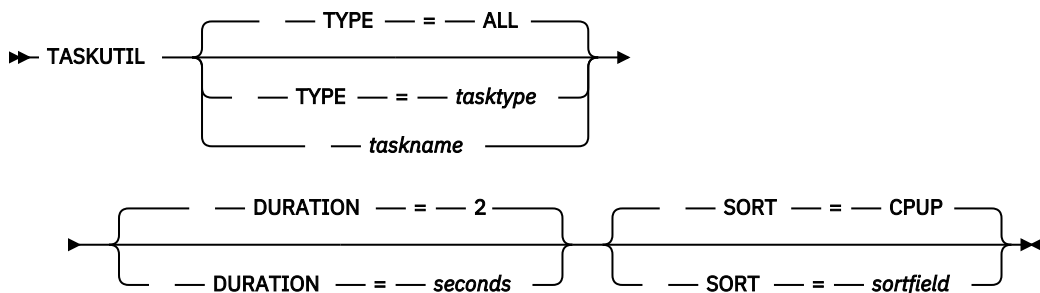
Note:

1. Notice in the example that the output under each group is sorted by the Severity Index. The first percentage column on the left represents a percentage of the maximum value.
2. DSITIMMT is the task causing a penalty. DSITIMMT is causing other tasks to wait (Inbound Wait) because of the MAXMQIN limits.
3. WINDOW TASKMON * * (TAKE 4 produces a panel effect, and displays the top four tasks in each of the measured categories. The WINDOW refresh PF key can be used to see updated values.
4. TASKMON also illustrates how DSITSTAT data can be interpreted by an automation procedure.
5. If TASKMON is used with no operands, the syntax diagram is displayed. Asterisk (*) is an abbreviation for ALL. The left parenthesis is required if (TAKE nnn is used.

TASKUTIL (NCCF)

Syntax

TASKUTIL



Purpose of Command

The TASKUTIL command displays central processing unit (CPU) utilization and storage use for NetView tasks.

Note: Use this command for NetView diagnosis and tuning purposes only.

If you do not specify parameters on the TASKUTIL command, information about all active NetView tasks is displayed in descending order by CPU utilization.

Trends in high CPU percentage utilization from repeatedly issuing the TASKUTIL command are more important than a high CPU percentage utilization for a particular task from a single issue of the TASKUTIL command. Similarly, constant growth in a task's message queue or storage use are more indicative of a problem than a high result from issuing a single TASKUTIL command.

By default, the output from the TASKUTIL command is placed in the network log and displayed on the operator's console. TASKUTIL output (message DWO022I) in the network log can be used for historical comparisons to help identify problem areas.

For more information about using the TASKUTIL command for tuning, see the *IBM Z NetView Tuning Guide* for more information.

Operand Descriptions

TYPE=tasktype

The type of NetView task. Valid tasks are:

ALL

All active NetView tasks. ALL is the default.

AUTO

NetView automation operator station tasks started with the AUTOTASK command. This does not include operator station tasks (OSTs) or distributed automation tasks (DISTs).

DIST

NetView distributed automation tasks started with the RMTCMD command. This does not include OSTs or autotasks.

DST

NetView data services tasks (DSTs). This does not include optional tasks (OPTs).

HCT

NetView hardcopy log tasks.

MNT

NetView main task.

NNT

NetView-NetView tasks.

OPT

NetView optional tasks. This does not include DSTs.

OST

NetView operator station tasks. This does not include autotasks or DISTs.

PPT

NetView primary program operator interface task (PPT).

VOST

Virtual operator station tasks (VOSTs).

taskname

The name of the NetView task whose CPU utilization and storage use you want to display. For operator tasks, this is the operator ID. If you specify two single quotation marks (") as the *taskname*, the task running the command is used.

DURATION=seconds

Specifies the length of the measurement (in seconds) over which utilizations are calculated. The valid range is 1-60 seconds. The default is 2 seconds.

SORT=sortfield

Specifies how the output is sorted. The valid values for *sortfield* are:

CPUP

Specifies to sort tasks in descending order by CPU utilization during the measurement. CPUP is the default.

CMDL

Specifies to sort tasks alphanumerically by active command list.

CPUT

Specifies to sort tasks in descending order by total CPU time.

MQUE

Specifies to sort tasks in descending order by the number of buffers on the task's public message queue or queues.

NAME

Specifies to sort tasks alphanumerically by task name.

PRTY

Specifies to sort tasks in descending order by MVS system dispatching priority.

STOR

Specifies to sort tasks in descending order by queued storage use (in kilobytes).

TYPE

Specifies to sort tasks alphabetically by task type.

Usage Notes

1. The task running the command is in a wait state for the measurement and is unable to process commands or messages until the TASKUTIL command completes.
2. As the number of available processors on a system increases, a small value for DURATION (such as 1 second) can magnify any data errors caused by the timing between the NetView program referencing information for the processors and the system updating the information for the processors. This type of error can produce invalid data for TASKUTIL command calculations. If the TASKUTIL command discovers such an error, it attempts to collect the data again, up to a maximum of 10 attempts.
3. If Workload Management (WLM) enclave support is enabled in the NetView program, the entry in the multiline DWO022I message with NETVIEW in the TASKNAME column and TOTL in the TYPE column, representing the NetView program, may have a value in the CPU-TIME column smaller than the sum of CPU-TIME values for the NetView tasks, since processor time consumed by active enclave tasks will not be included in the job step processor time used to calculate CPU-TIME for the NetView program.

Restrictions

The TASKUTIL command cannot be issued from the PPT.

Return Codes**Return Code****Meaning**

- | | |
|-----------|-----------------------------------|
| 0 | Processing was successful. |
| 8 | Not a valid parameter. |
| 12 | The task specified is not active. |
| 16 | Storage request failed. |
| 20 | TASKUTIL cannot run on the PPT. |

24

Operator is not authorized to use a keyword or value.

28

TASKUTIL was unable to gather valid processor information within 10 attempts.

Example: Displaying CPU and storage use

To display CPU utilization and storage use for the NetView tasks, enter:

```
TASKUTIL
```

Response

A response similar to the following is displayed:

```
DW0022I
TASKNAME  TYPE  DPR      CPU-TIME  N-CPU%  S-CPU%  MESSAGEQ  STORAGE-K  CMD
-----
NTV98PPT  PPT   255      3.88    49.68    0.08      0         122  **NONE**
OPER4     OST   251      0.42    31.65    0.05      0         80   **NONE**
NTV98VMT  OPT   250      1.38    8.19     0.01      0         56   N/A
DSIHLLMT  OPT   255      0.00    0.00     0.00     N/A        0   N/A
DSISTMT   OPT   255      0.00    0.00     0.00     N/A        0   N/A
SYSOP     OPT   255      0.00    0.00     0.00     N/A        0   N/A
NTV98     OPT   255      0.02    0.00     0.00     N/A        0   N/A
DSILOGMT  OPT   255      0.03    0.00     0.00     N/A        0   N/A
DSILOG    DST   254      0.23    0.00     0.00      0         12   N/A
CNMCSSIR  OPT   250      0.53    0.00     0.00      0          4   N/A
CNMCALRT  OPT   249      0.15    0.00     0.00     N/A        0   N/A
DSISVRT   DST   253      0.07    0.00     0.00      0        107   N/A
DSIELTSK  DST   253      0.02    0.00     0.00      0         15   N/A
DSI#0094  VOST  250      0.36    0.00     0.00      0         24   NLDM
SYSOP     MNT   255      0.01    0.00     0.00      0        764   N/A
NTV98BRW  OPT   250      0.09    0.00     0.00      0          4   N/A
DSIDCBMT  OPT   255      0.25    0.00     0.00     N/A        0   N/A
AUTO1     AUTO  250      0.51    0.00     0.00      0         48   **NONE**
AUTO2     AUTO  250      0.22    0.00     0.00      0         36   **NONE**
NETVIEW   OTHR  N/A      N/A     0.00     0.00     N/A       N/A   N/A
NETVIEW   SRB   N/A      1.89   10.49    0.02     N/A       N/A   N/A
NETVIEW   TOTL  33      14.46  100.00   0.15      0       1248   N/A
SYSTEM    TOTL  N/A      N/A     N/A     4.34     N/A       N/A   N/A
END DISPLAY
```

The meanings of the displayed operands are as follows:

TASKNAME

The name of the NetView task.

TYPE

The type of the NetView task.

In addition to the CPU utilization and storage use for the tasks specified, the following information is displayed:

NETVIEW OTHR

Specifies the task control block (TCB) utilization not attributable to active tasks. This includes processing for tasks that become inactive during the measurement.

NETVIEW SRB

Specifies the NetView address space system request block (SRB) total time and CPU utilization during the measurement.

NETVIEW TOTL

Specifies the NetView address space total CPU time, CPU utilization during the measurement, and storage use.

SYSTEM TOTL

Specifies the total system CPU utilization during the measurement.

DPR

Is the MVS task dispatching priority for the NetView task. This field is not applicable for NETVIEW OTHR, NETVIEW SRB, or SYSTEM TOTL.

CPU-TIME

Is the accumulated MVS task control block (TCB) time, in seconds, for the task from the time it was started. If a task is stopped and then restarted, the CPU time used by the task before the task was stopped is not included.

The CPU-TIME for NETVIEW TOTL can be larger than the total of all active tasks. It includes the TCB time of tasks which are currently inactive, but have been active since the NetView program was started.

This field is not applicable for NETVIEW OTHR or SYSTEM TOTL.

N-CPU%

Is the relative percentage of the task's NetView CPU utilization, based upon a maximum of 100%. This field is not applicable for SYSTEM TOTL.

S-CPU%

Is the task's contribution to the total system CPU utilization during the measurement, based on a maximum of 100%. The system CPU utilization is the average processor utilization percentage for all processors currently online.

MESSAGEQ

Is the number of messages currently backed up on the task's public message queue or queues. For NETVIEW TOTL, this is the total number of messages currently backed up on the public message queues of all active NetView tasks.

This field is not applicable for NETVIEW OTHR, NETVIEW SRB, or SYSTEM TOTL, or type OPT when TVB374CT is off. For information about TVB374CT, see the *IBM Z NetView Customization Guide*.

STORAGE-K

Is the amount of pooled and non-pooled storage, in kilobytes, currently being used by the task. This is queued storage acquired by DSIGET Q=YES requests. For the MNT task (SYSOP), this storage includes the total queued and non-queued CPOOL as well as the queued and non-queued non-CPOOL storage.

For NETVIEW TOTL, this is the amount of pooled and non-pooled storage, in kilobytes, currently being used by the NetView program. This includes the total amount of queued storage acquired by the tasks (DSIGET Q=YES requests) and non-queued storage (DSIGET Q=NO) requests. The total might not reflect the totals of all the separate tasks. This total storage can be used as an estimate and not as an exact figure. This total does not include storage acquired by GETMAIN macros and storage used to contain NetView load modules. To display total virtual storage used by the NetView program, use the NetView RESOURCE command.

This field is not applicable for NETVIEW OTHR, NETVIEW SRB, or SYSTEM TOTL.

CMD

Is the current active command list running on the task, if any. If the task is an independent VOST, this is the name used on the ATTACH command that created the VOST. This field is not applicable for NETVIEW OTHR, NETVIEW SRB, NETVIEW TOTL, or SYSTEM TOTL.

Example: Displaying CPU utilization and storage use

To display CPU utilization and storage use for all NetView tasks using a measurement duration of 8 seconds and sorting the output by total CPU time, enter:

```
TASKUTIL DURATION=8 SORT=CPUT
```

Response

A response similar to the following is displayed:

```
DW0022I
TASKNAME TYPE DPR      CPU-TIME N-CPU% S-CPU% MESSAGEQ STORAGE-K  CMDLIST
-----
```

AUTONCCF	AUTO	250	109.14	83.57	3.57	0	669	**NONE**
OPER3	OST	251	49.19	0.00	0.00	6	204	**NONE**
DSILOG	DST	254	16.66	0.00	0.00	0	24	N/A
AUTO1	AUTO	250	7.93	0.00	0.00	0	545	**NONE**
AUTOVTAM	AUTO	250	6.63	0.00	0.00	0	22	**NONE**
CNM01PPT	PPT	255	2.50	0.00	0.00	0	87	**NONE**
SYSOP	MNT	255	1.98	0.00	0.00	0	1188	N/A
AAUTSKLP	DST	247	1.86	0.00	0.00	0	1388	N/A
BNJDSERV	DST	249	1.42	0.00	0.00	0	101	N/A
CNMCSSIR	OPT	250	0.81	0.00	0.00	0	12	N/A
DSILCOPR	AUTO	250	0.66	0.00	0.00	0	27	**NONE**
CNM01LUC	DST	248	0.51	0.00	0.00	0	27	N/A
DSICRTR	DST	249	0.39	0.00	0.00	0	31	N/A
CNM01VMT	OPT	250	0.34	0.00	0.00	0	33	N/A
CNM01BRW	OPT	250	0.28	0.00	0.00	0	4	N/A
DSI#0094	VOST	250	0.36	0.00	0.00	0	24	NLDM
AUTOMVS	AUTO	250	0.28	0.00	0.00	0	60	**NONE**
DSIAMLUT	DST	248	0.24	0.00	0.00	0	26	N/A
DSIHPDST	DST	250	0.20	0.00	0.00	0	32	N/A
MVSCONS	AUTO	250	0.20	0.00	0.00	0	27	**NONE**
DSISVRT	DST	253	0.08	0.00	0.00	0	95	N/A
DSI6DST	DST	250	0.05	0.00	0.00	0	26	N/A
AAUTCNMI	DST	249	0.04	0.00	0.00	0	146	N/A
BRIGOPER	AUTO	250	0.04	0.00	0.00	0	17	**NONE**
REMTBRIG	AUTO	250	0.04	0.00	0.00	0	16	**NONE**
SQLOGTSK	DST	253	0.03	0.00	0.00	0	47	N/A
OPER1	OST	250	0.03	0.00	0.00	0	11	**NONE**
OPER2	OST	250	0.03	0.00	0.00	0	13	**NONE**
AUTO2	AUTO	250	0.03	0.00	0.00	0	11	**NONE**
CNMTAMEL	DST	249	0.02	0.00	0.00	0	52	N/A
CNMCALRT	OPT	250	0.00	0.00	0.00	0	0	N/A
NETVIEW	OTHR	N/A	N/A	0.00	0.00	N/A	N/A	N/A
NETVIEW	SRB	N/A	21.12	1.39	0.59	N/A	N/A	N/A
NETVIEW	TOTL	251	238.38	100.00	4.27	6	5505	N/A
SYSTEM	TOTL	N/A	N/A	N/A	18.33	N/A	N/A	N/A
END DISPLAY								

Example: Displaying CPU utilization and storage use

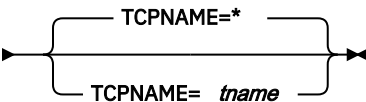
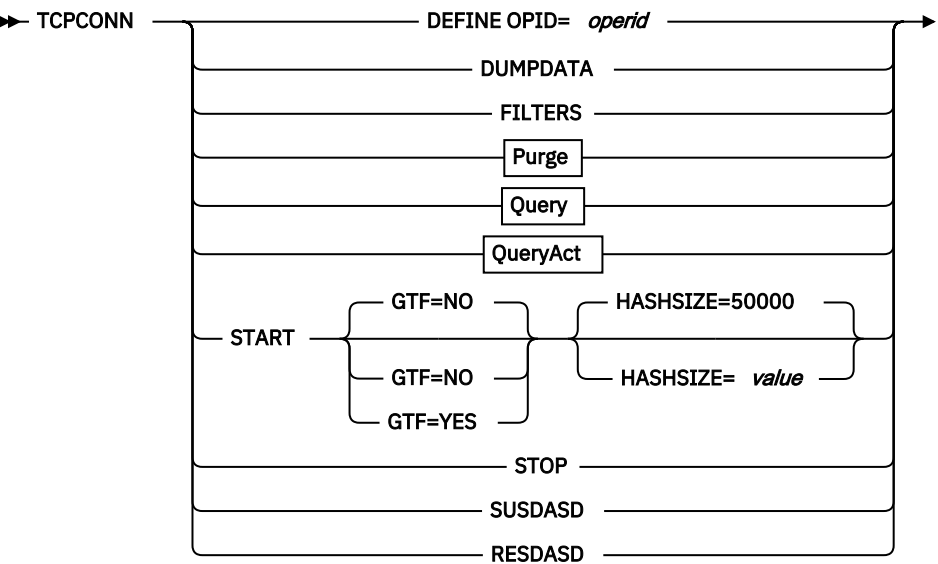
To display CPU utilization and storage usage for all NetView automation tasks started with the AUTOTASK command, enter:

```
TASKUTIL TYPE=AUTO
```

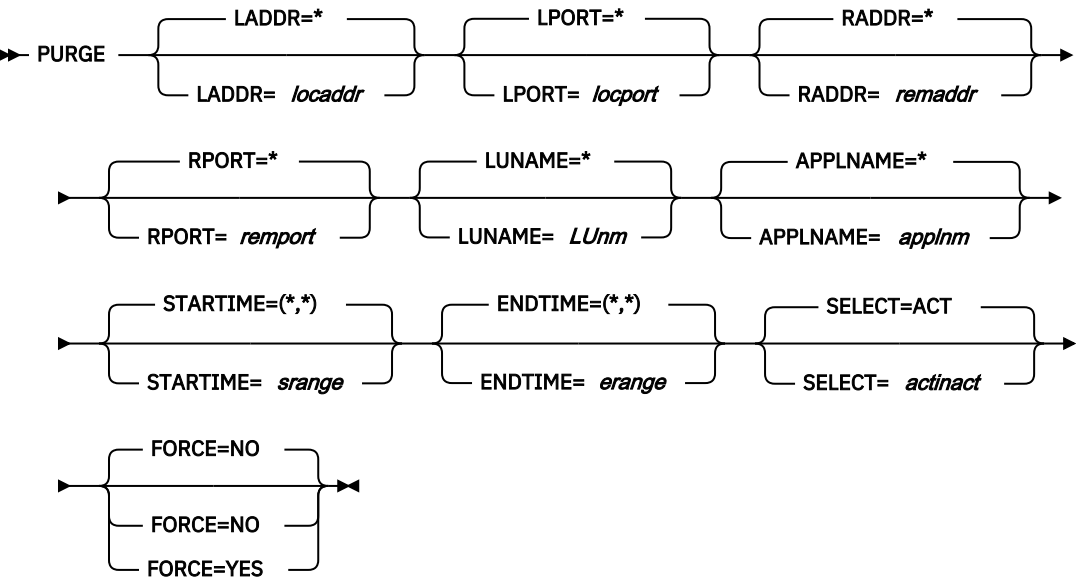
TCPCONN (NCCF)

Syntax

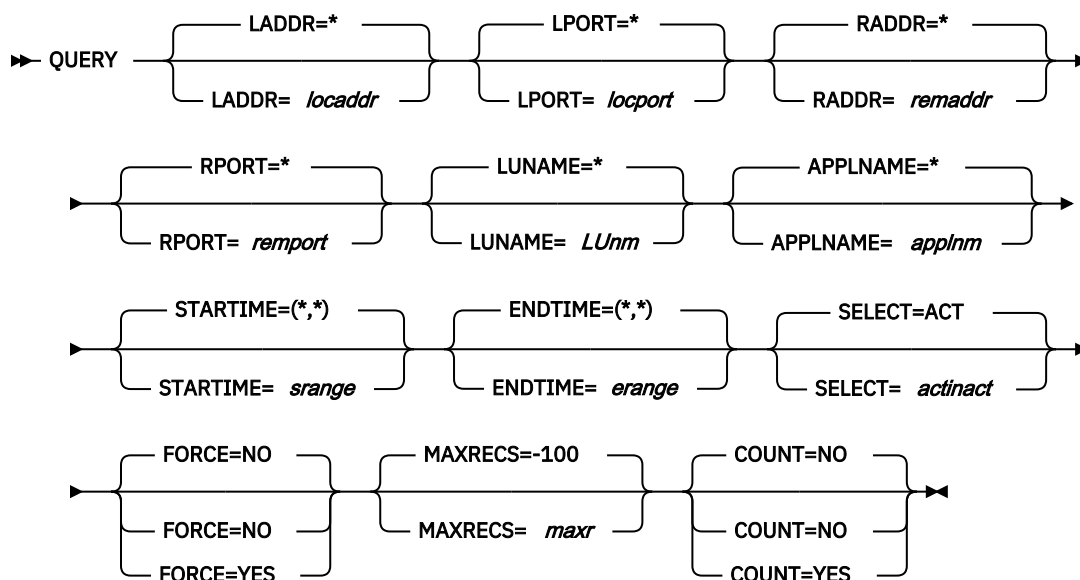
TCPCONN



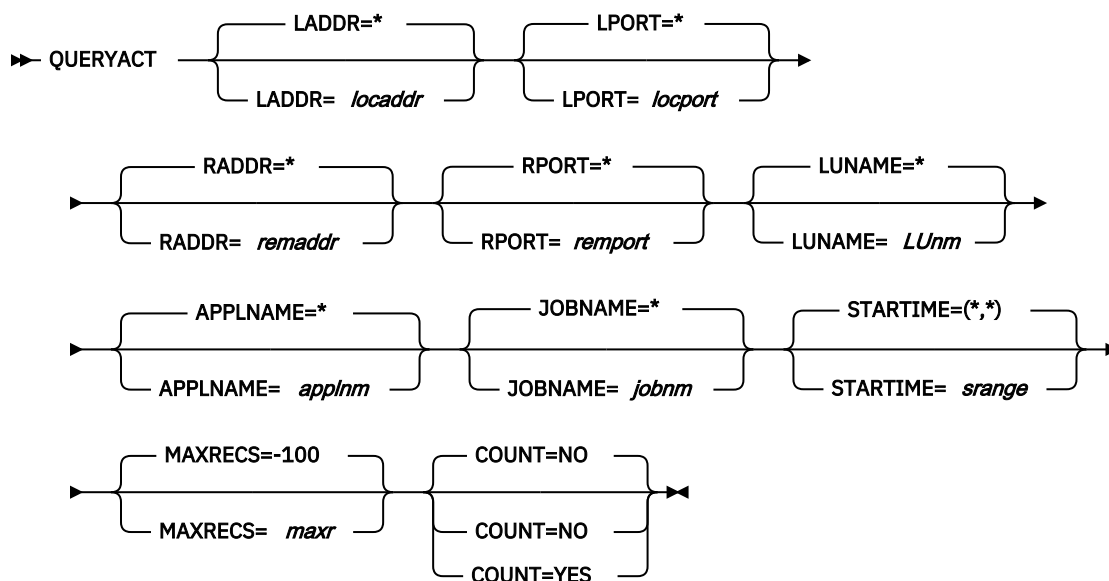
Purge



Query



QueryAct



Purpose of Command

The TCPCONN command is used to control the collection of TCP/IP connection data and to view the collected data.

Note:

1. The TCPCONN QUERYACT command uses a newer interface to the z/OS Communications Server, which does not require that the connections be collected by the NetView program. Use TCPCONN QUERYACT instead of TCPCONN QUERY for active connections.
2. The TCPCONN QUERY and TCPCONN QUERYACT commands are intended as REXX interfaces. For user interfaces, use the Tivoli Enterprise Portal or the CNMSTCPC sample.

Operand Descriptions

APPLNAME=*applnm*

Specifies the TN3270 application name.

Note: Because the application name is only available for inactive connections, the APPLNAME keyword cannot be specified with the SELECT=ACT or SELECT=ALL option.

COUNT=YES|NO

Specifies whether the response to the QUERY or QUERYACT option reflects the total number of connections even when this number exceeds the value specified by MAXRECS.

DEFINE

Associates a TCP/IP name with the name of an autotask that will collect packet data for the stack. (This is the same function performed by the FUNCTION.AUTOTASK.TCPCONN statement in the CNMSTYLE member.) TCPCONN START is then required to start the function. Data collection for the specified stack stops if it becomes defined on another autotask. The same OPID can only be defined to one TCPNAME, and to either TCPCONN or PKTS.

A specification of OPID=NONE undefines the specified TCPNAME.

DUMPDATA

Supports diagnostic information. Use this keyword when instructed to by IBM Software Support.

ENDTIME=*erange*

Specifies the range of end times for connections to be included in a QUERY or PURGE command. The *erange* consists of two values separated by commas and enclosed in parentheses; the first value specifies the beginning date and time for the range, and the second value specifies the ending date and time for the range. The following rules apply to these values:

- To specify a date and time, type the date followed by the time, separated by a blank. The formats of the date and time are controlled by the DEFAULTS and OVERRIDE commands.
- If either value consists only of a date, the current time for that date is included.
- If either value consists only of a time, the assumed date depends upon the specified time. If the time is later than the current time, yesterday's date is assumed; if the time is earlier than the current time, today's date is assumed.
- An asterisk before the comma will include all connections from the beginning of data collection. An asterisk after the comma will include all connections up to the present (possibly including active connections, depending on other options).
- Either value can also be specified as a 16-character hexadecimal store-clock value. This can be used to continue a previous QUERY by specifying a previously returned store-clock value from message BNH772I.

The following examples are valid values for *erange*:

- (04/05/04 10:00:00,04/06/04 17:00:00)
- (*,04/05/04)
- (B7ACDD41D4F00A01,*)

Note: Because end times are only available for inactive connections, the ENDTIME keyword cannot be specified with the SELECT=ACT or SELECT=ALL option.

FILTERS

Enables the TCPCONN.KEEP and TCPCONN.DASD statements in the CNMSTYLE member for the processes described above which are active processes (or stacks). These filters will also be in effect for all processes (or stacks) which start later.

Note: No TCPNAME values are allowed with Filters except a single asterisk (*).

FORCE=YES|NO

Specifies whether a search that might take a long time to complete is really intended. FORCE=YES is required if all of the following are specified or defaulted and LADDR and RADDR are non-specific, meaning that the value contains * (asterisk) or - (dash) and is not a host name.

- SELECT=INACT or SELECT=ALL
- LPORT=*
- ENDTIME=(*,*)

GTF=YES|NO

Specifies whether GTF tracing of incoming records and buffers occurs. If collection is already in progress for the specified stack, this keyword (with TCPCONN START) starts or stops GTF tracing without interrupting data collection.

The event identifier used for records written to GTF is X'5F8'.

HASHSIZE=value

This operand is used to improve performance. This specifies the approximate maximum number of active connections expected at a given time. The value cannot contain any spaces or commas. The default value is 50000.

JOBNAME=jobnm

Specifies socket application address space name criteria for the QUERYACT option. A question mark (?) can be used as a wildcard for a single character, and an asterisk (*) can be used with a zero (0) or for multiple characters. The asterisk must be the last character in the string. For example, a value of A?C* matches all names with a first character of A and a third character of C.

LADDR=locaddr

Specifies the local IP address (or set of addresses) for a QUERY or PURGE command. This address can be expressed in any of several possible formats:

- An IPv4 address in dotted-decimal format: ddd . ddd . ddd . ddd. Each ddd can be any of the following:
 - A decimal number from 0 to 255
 - A hyphen-separated range (such as 240-255)
 - An asterisk (*), representing the range 0 - 255

Leading zeros can be omitted. If the last ddd is an asterisk, and fewer than four ddd values are specified, the range 0 - 255 is assumed for each remaining ddd.

- An IPv6 address in colon-hexadecimal format: hhhh : hhhh : hhhh : hhhh : hhhh : hhhh : hhhh : hhhh, where each hhhh is a 0-4 digit hexadecimal number, or a hexadecimal range separated by a hyphen, such as FF00-FFFF. Consecutive groups of zeros can be replaced with a double colon (::). A double colon can be used to signify leading, trailing or embedded groups of zeros, and can be specified only once in an address. A single asterisk (*) can be used in place of hhhh to denote 0-FFFF. If the last hhhh is an asterisk and less than 8 hhhhs are specified, 0-FFFF is assumed for each remaining hhhh. If both an asterisk and a double colon are used in an address, the asterisk will only represent a single hhhh group regardless of its position.
- A single asterisk (*), representing all local IP addresses.
- A TCP/IP symbolic host name. The NetView program attempts to translate to an IP address using the TCPNAME operand, if the operand is specified without a wildcard. Otherwise, it uses the value for TCPNAME in the CNMSTYLE member. If a host name resolves to multiple IP addresses, the QUERY or PURGE command is attempted for all of these addresses.

Note: If an IPv6 address is specified in a mixed format (containing both colons and dots), then the NetView program inspects the high-order 12 bytes of the 16-byte address. If the first 12 bytes of the 16-byte address is a IPv4 migration value of 0:0:0:0:0:0:FFFF or 0:0:0:0:0:0:0, then the NetView program strips the high-order 12 bytes and processes the remaining 4 bytes as an IPv4 address of the dotted decimal format ddd . ddd . ddd . ddd. If anything other is specified in the high-order 12 bytes of the IPv6 address, such a specification is treated as a host name.

The following examples are processed as IPv4 addresses:

```
::*
::FFFF:*
9.42.44.52
::FFFF:9.42.*
00:000:0000:0:000:FFFF:9.42.44.52
::0:0:FFFF:9.42.44.52
```

The NetView program attempts to treat the following addresses as host name specifications:

```
myhostname-9-42-33-52 (valid host name specification)
::1:FFFF:9.42.44.52 (because this is not a valid IP address
specification, this is treated as a host name)
0:0:0:0-0:0:FFFF:9.42.33- (because this is not a valid IP address
specification, this is treated as a host name)
```

LPORT=locport

Specifies the local port number. *locport* can be either a decimal number or a single asterisk (*), representing all ports.

Note: QUERY and PURGE requests are optimized by local port. Whenever possible, specify a specific port rather than *.

LUNAME=LUnm

Specifies the TN3270 logical unit name.

Note: Because the logical unit name is only available for inactive connections, the LUNAME keyword cannot be specified with the SELECT=ACT or SELECT=ALL option.

MAXRECS=maxr

Specifies the maximum number of connection records to return for the QUERY or QUERYACT options. The value is a number between -9999999 and 9999999 (do not insert commas or periods). Connections are always listed in reverse chronological order. A positive value specifies the set of records ending with the oldest matching connection; a negative value specifies the set of records starting with the most recent matching connection. The default value is -100.

OPID=operid

Specifies the name of the autotask that collects connection information for the associated TCP/IP stack.

PURGE

Purges connection records matching the input criteria from storage or DASD. If the purge is successful, one or two BNH774I messages are returned (one for active connections and one for inactive connections).

Note: TCPCONN PURGE might take a long time to complete when purging inactive records from DASD. To avoid this, run this command on a timed basis as part of routine database maintenance.

RESDASD

Resumes the writing of connection records to DASD after being suspended with the SUSDASD keyword.

QUERY

Queries connection records, previously sent to the NetView system, matching the input criteria. If the query is successful, the BNH772I message is returned. See the description for the SELECT keyword.

QUERYACT

Queries the z/OS Communications Server for active connections matching the input criteria. If the QUERYACT query is successful, the BNH775I message is returned.

Note:

1. The TCPNAME value must contain a valid TCP/IP name with no wild cards when specified with QUERYACT.
2. QUERYACT uses the EZBNMIFR interface, as described in the *z/OS Communications Server: IP Programmer's Guide and Reference*. Some of the filtering criteria supported by this command are supported by the EZBNMIFR interface. Use these criteria, as listed below, whenever possible to limit the number of records returned by the interface. The other criteria are then applied to those records returned by the interface.

The following criteria are those supported by, and passed to, the EZBNMIFR interface:

TCPNAME

The stack name

LPORT and RPORT

The local and remote port

LADDR and RADDR

The local and remote IP address up to the first *ddd* or *hhhh* which is an asterisk or a range. See the description for the LADDR keyword.

JOBNAME

The job name

The following criteria are not supported by the EZBNMIFR interface and are applied to those connections returned by the interface:

STARTIME

The connection start time

LADDR and RADDR

The local and remote IP address after the first *ddd* or *hhhh* which is an asterisk or a range. See the description for the LADDR keyword.

APPLNAME

The application name

LUNAME

The LU name

MAXRECS

The maximum number of records

RADDR=*remaddr*

Specifies the remote IP address (or set of addresses) for a QUERY or PURGE. See the description of the LADDR keyword for information on how to specify an IP address.

RPORT=*remport*

Specifies the remote port number. *remport* can be either a decimal number or a single asterisk (*), representing all ports.

SELECT=*actinact*

Specifies active connections, inactive connections, or both. The valid values for *actinact* are:

- ACT
- INACT
- ALL

Note: TCPCONN QUERY with SELECT=ACT (or the active side of ALL) is deprecated. It still functions as before, but provides much less data about active connections than TCPCONN QUERYACT.

START

Starts a long-running process to collect connection start and stop data from the z/OS Communications Server for the specified TCP/IP stack. The data-collection process persists as an autotask defined by a previous TCPCONN DEFINE command or a FUNCTION.AUTOTASK.TCPCONN statement in the CNMSTYLE member.

If the autotask is already running, TCPCONN START can be used to start or stop GTF tracing without interrupting data collection.

The TCPCONN.KEEP and TCPCONN.DASD filter settings which are current in the CNMSTYLE member go into effect when TCPCONN.START is run.

STARTIME=*srange*

Specifies the range of start times for connections to be included in a QUERY, QUERYACT, or PURGE command. The *srange* consists of two values separated by commas and enclosed in parentheses; the first value specifies the beginning date and time for the range, and the second value specifies the ending date and time for the range. For more information and examples, see the description of the ENDTIME keyword.

STOP

Stops the collection of connection data for the specified stack. TCPCONN STOP is used to end the data collection started by TCPCONN START.

SUSDASD

Suspends the writing of connection records to DASD. Any connection records that would have been written to the database are discarded during this suspension. A TCPCONN database can fill up if routine cleanup is not performed. Avoid this if possible. However, when a full database condition occurs, and automation is set up to invoke the DBFULL (CNME2010) command list to perform a PURGE of the database, the DBFULL CLIST issues TCPCONN SUSDASD before it performs the PURGE and TCPCONN RESDASD after the PURGE is complete. This avoids writing to the database while the PURGE is occurring, which prevents I/O errors and allows the PURGE to complete faster.

TCPNAME=*tname*

For TCPCONN DEFINE, specifies the TCP/IP name to associate with an autotask that collects connection data for the stack (no wildcard characters).

For TCPCONN QUERYACT, specifies the TCP/IP name to query for active connection data (no wildcard characters).

For other requests, specifies the TCP/IP name for the request. *tname* must be a TCP/IP name specified by a previous TCPCONN DEFINE command or a FUNCTION.AUTOTASK.TCPCONN statement in the CNMSTYLE member. A single asterisk (*) can be used to specify all defined stacks. You can also use an asterisk as a wildcard at the end of the TCP/IP name; for example, ABC* matches any TCP/IP name beginning with ABC.

Restrictions

The following restrictions apply to the TCPCONN command:

- Collection of TCP/IP connection data requires a level of z/OS Communications Server which supports the NETMONITOR TCPCONN SERVICE profile statement. It also requires that all appropriate z/OS Communications Server definitions are in place. (See *IBM Z NetView Installation: Configuring Additional Components* for more information.) Without this support enabled, TCPCONN returns an error message.
- For any application that generates a high volume of connections (such as a Web server), filtering is enabled in z/OS Communications Server. Otherwise, TCPCONN START causes the NetView program to receive notification of every connection start and stop on the stack, which can cause performance problems.
- Use appropriate filtering in z/OS Communications Server (for example, the MINLIFETIME option) to minimize unnecessary connection data and avoid performance problems. Note that TCPCONN.KEEP statement in the CNMSTYLE member provides filtering only on the NetView side.

Return Codes

Return Code

Meaning

0

Successful processing.

4

The command did not complete successfully. Check the accompanying messages for more information.

TE (NCCF)

Syntax

TE

►► TE ◄◄

Purpose of Command

The TE (trace end) command stops all tracing of a REXX command list started by the TS command.

You can enter TE from a terminal, a REXX command list, or a NetView command list language procedure. TE is an immediate command if it is entered from a terminal.

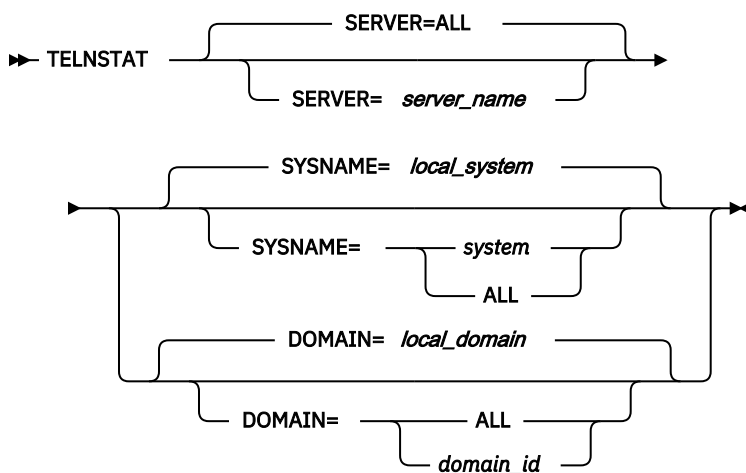
Restrictions

The TE command stops only tracing that was started by the TS command.

TELSTAT (NCCF; CNME8232)

Syntax

NCCF TELSTAT



Purpose of Command

You can use the TELSTAT command to view configuration and status information about Telnet servers. You can view this information from a 3270 console or from the Z NetView Enterprise Management Agent.

Note: This command is intended as a REXX interface. For end-user interfaces, use the Tivoli Enterprise Portal, the CNMSTNST sample, or the NetView management console.

Operand Descriptions

SERVER

Specifies the Telnet server for which data is requested. The default value is all Telnet servers known to the DOMAIN or SYSNAME specified.

SYSNAME

The system name for the requested data. The default value is the local system. When the ALL value is in effect, the command is sent to all NetView programs known to the master NetView program. If the command is issued on a non-master NetView program, only the local data is discovered.

Note: Use caution when you specify the ALL value because it can cause rediscovery to take place on multiple NetView programs.

DOMAIN

The domain to which the request is sent. The following are the valid options:

ALL

Specifies all domains in the sysplex. This value is only valid on the master NetView program.

Note: If you have many stacks or systems in your sysplex, issuing ALL can cause slow response times and high CPU utilization.

local

If the DOMAIN keyword is not specified, the domain name of the local NetView system is used. This is the default.

domain_id

Specifies the ID of a specific domain. This option can be issued only from the master NetView program if the domain is not the local domain.

domain

Specifies the ID of a specific domain. This option can be issued only from the master NetView program if the domain is not the local domain.

Usage Notes

If message DSI047E is received, contact your system programmer to enable the appropriate tower or subtower. For information about data collection and display towers and subtowers, see *IBM Z NetView Installation: Configuring Additional Components*.

Return Codes

Return Code

Meaning

0

The command is successful.

2

Help is issued.

4

There is no data to display.

8 or above

Failure, see the associated message.

TERM (EAS)

Syntax

EAS TERM

➤ MODIFY — — *procname* — , — TERM ➤

IBM-Defined Synonyms

Command or Operand	Synonym
MODIFY	F

Purpose of Command

The TERM command causes the NetView Event/Automation Service job to halt all activity and end normally.

Operand Descriptions

procname

Specifies the Event/Automation Service job name.

Usage Notes

The Event/Automation Service might take a number of minutes to end if any of its services are delayed as a result of TCP/IP connection problems. Use the EAS FORCE command if you want to end the Event/Automation Service without waiting for normal processing to complete.

Example: Ending the service job

To end the Event/Automation Service job named IHSAEVNT, enter:

```
F IHSAEVNT,TERM
```

Response

IHS0119I Event/Automation Service is terminating due to an operator request.

TERM (GMFHS)

Syntax

From an MVS console:

MVSTERM

►► F — — *procname* — , — TERM ◄◄

From a NetView terminal:

TERM

►► GMFHS TERM ◄◄

IBM-Defined Synonyms

Command or Operand	Synonym
MODIFY	F

Purpose of Command

The TERM command causes the NetView GMFHS job to halt all activity and end normally.

You can enter the TERM command from the MVS console or from a NetView terminal by using the GMFHS command list.

CHKPT

Specifies that the RODM checkpoint is taken. Processing continues after the checkpoint is taken.

Restrictions

The following restrictions apply to the TERM (RODM) command:

- When taking a checkpoint of RODM, ensure that the checkpoint data sets are kept together with the translation and master window data sets. If you do not keep these data sets together, you might experience problems warm starting RODM.
- You can use the MVS FORCE command in an MVS environment, although the MODIFY command might be preferable because RODM gets control and provides a smooth stop.
- In an MVS environment, RODM (under program control) makes its address space non-cancelable. This prevents you from issuing a CANCEL command to stop RODM.

Example: Ending a specified RODM without taking a checkpoint

To end RODM without taking a checkpoint, enter the following from a NetView terminal console:

```
RODM TERM
```

Response

A response similar to the following is displayed:

```
EKG1916I          EKGXRODM: RODM EKGXRODM TERMINATION IS IN PROGRESS.  
EKG1310I          EKGXRODM: THE LOG FLUSHING IS COMPLETED.  
EKG1917I          EKGXRODM: RODM EKGXRODM TERMINATION IS COMPLETE.  
      EKGXRODM      START      EKGTC000      0000  
IEF404I EKGXRODM - ENDED - TIME=13.49.18  
$HASP395 EKGXRODM ENDED
```

Example: Taking a checkpoint and ending a specified RODM

To take a checkpoint and end RODM, enter the following from a NetView terminal:

```
RODM TERM,CHKPT
```

Response

A response similar to the following is displayed:

```
EKG1302I          EKGXRODM: RODM EKGXRODM IS NOW CHECKPOINTING.  
EKG1115I          EKGXRODM: THE TRANSLATION WINDOW CHECKPOINT IS COMPLETE.  
EKG1303I          EKGXRODM: RODM EKGXRODM HAS COMPLETED CHECKPOINTING.  
EKG1916I          EKGXRODM: RODM EKGXRODM TERMINATION IS IN PROGRESS.  
EKG1310I          EKGXRODM: THE LOG FLUSHING IS COMPLETED.  
EKG1917I          EKGXRODM: RODM EKGXRODM TERMINATION IS COMPLETE.  
      EKGXRODM      START      EKGTC000      0000  
IEF404I EKGXRODM - ENDED - TIME=13.57.33  
$HASP395 EKGXRODM ENDED
```

TERMAMI (NCCF)

Syntax

TERMAMI

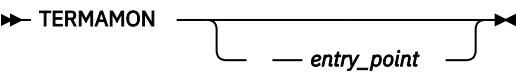
►► TERMAMI ►►

Purpose of Command

Stops the application management interface (AMI) instrumentation running on NetView/390.

TERMAMON (NCCF)

Syntax



Purpose of Command

The TERMAMON command stops the VTAM ACB Monitor or a specific ACB Monitor entry point. This command can only be issued on the ACB Monitor focal point NetView.

If the entry point is specified, ACB status reporting is stopped for the VTAM associated with that NetView system.

If the entry point is not specified, the DB2® database is cleaned up and ACB status reporting is turned off from the VTAM associated with this (focal point) NetView and from VTAMs associated with all of the ACB Monitor entry points.

Operand Descriptions

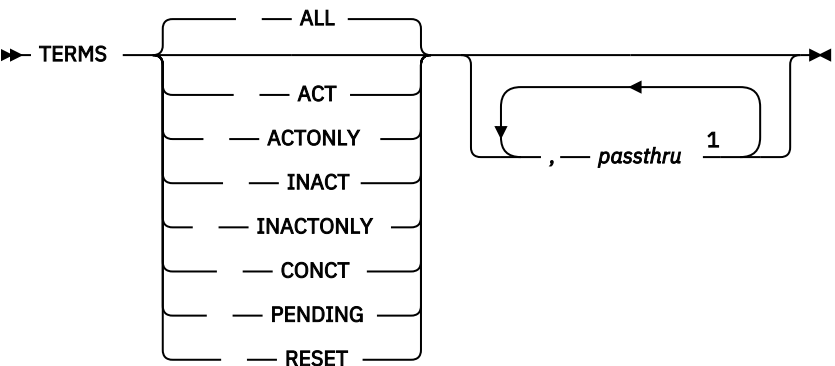
entry_point

Specifies the NetView domain name of an entry point to be stopped.

TERMS (NCCF; CNME0035)

Syntax

TERMS



Notes:

¹ If you do not specify a positional parameter, you must indicate the absence of the parameter by specifying a comma in its place.

IBM-Defined Synonyms

Command or Operand	Synonym
ACT	A
INACT	I

Purpose of Command

The TERMS command list displays the status of all device-type logical units (terminals) in active major nodes.

Operand Descriptions

ACT

Specifies that information is to be displayed about all active, pending, and connectable terminals within each major node.

ACTONLY

Specifies that information is to be displayed about all terminals in an active state within each major node. The display does not include terminals in pending or connectable states.

ALL

Specifies that information is to be displayed about all terminals (regardless of their status) within each major node. ALL is the default.

Note: Using the ALL operand (the default) in a domain that has many terminals results in an undesirably long display.

CONCT

Specifies that information is to be displayed about all terminals in a CONCT (connectable) state within each major node.

INACT

Specifies that information is to be displayed about all inactive terminals within each major node.

INACTONLY

Specifies that information is to be displayed about all inactive terminals within each major node. Resources in a RESET state are not included in the display.

PENDING

Specifies that information is to be displayed about all pending terminals within each major node. A pending state is a transient state to or from the fully active state.

RESET

Specifies that information is to be displayed about all terminals in a RESET state within each major node.

passthru

Specifies up to 6 parameters which are appended unchanged to the VTAM DISPLAY command issued by the TERMS command. No validation for duplicate or conflicting parameters is performed.

Usage Notes

Consider the following when using the TERMS command:

- If the status parameter (ALL, ACT, and so on) is omitted, and no *passthru* parameters are specified, then ALL is the default. However, if *passthru* parameters are specified and there is no status parameter specified, then the NetView system does not include a SCOPE= keyword in the generated VTAM DISPLAY command. This enables you to include your own SCOPE= keyword using the *passthru* parameter.
- The valid values of the status parameter depend on the level of VTAM you are using.

Example: Displaying all the inactive terminals

To display all the inactive terminals, use the following command:

```
TERMS INACT
```

Response

Information similar to the following is displayed:


```

IST350I DISPLAY TYPE = TERMS
IST354I PU T4/5 MAJOR NODE = ISTPUS
IST351I LOCAL 3270 MAJOR NODE = H21L
IST089I H21L420 TYPE = LOGICAL UNIT      , NEVAC      ,CUA=420
IST089I H21L42B TYPE = LOGICAL UNIT      , NEVAC      ,CUA=42B
IST352I LOCAL SNA MAJOR NODE = H21S

```

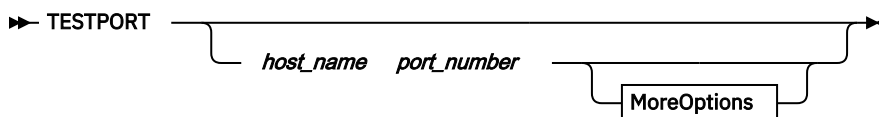
For each major node with terminals, the following is provided:

- The major node name (for example, ISTPUS, H21L, H21S)
- The line name and status (if the terminal is attached over a line)
- The name and status of the associated physical unit
- The name and status of the logical unit (for example, H21L420)

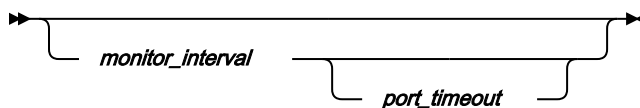
TESTPORT (NCCF; CNMETSTL)

Syntax

TESTPORT



MoreOptions



Purpose of Command

The TESTPORT command can be used to monitor a port that refuses a connection but appears to be normal when the NETSTAT command is issued. See *IBM Z NetView Command Reference Volume 2 (O-Z)* for more information about the NETSTAT command.

The NETSTAT command provides a display of all resources that are not currently active. However, the NETSTAT command can show a resource to be active when the port for that resource has refused a connection. The TESTPORT command provides additional function to monitor critical ports.

Operand Descriptions

host_name

The name of the host or the IP address that owns the port that is to be monitored.

port_number

The port that is to be monitored.

monitor_interval

The interval in a valid timer format.

port_timeout

The timeout value specified in seconds which, when it expires, ends the TCP/IP port connection request if it has not yet completed. The default for the NetView program is 300 seconds. The default value for TCP/IP might be less than 300 seconds and thus override the timeout value.

Usage Notes

- The TESTPORT command can be issued without any parameters. If the command is issued with no parameters, it reads definitions set by the COMMON.IPPORTMON statements described in the *IBM Z*

If *host_name* is specified, *port_number* must also be specified. If *port_number* is specified, *host_name* must also be specified. If both *port_number* and *host_name* are specified, *monitor_interval* and *port_timeout* are optional.

If parameters *host_name*, *port_number*, and *monitor_interval* are specified, TESTPORT sets a timer using the information provided. If *monitor_interval* is specified with a null value or a value of 0, the specified port is checked immediately and no timer is set. If *port_timeout* is specified, a null value for *monitor_interval* cannot be specified.

Operands must be entered in positional order, as shown in the syntax diagram.

IBM-Defined Synonyms

Command or Operand	Synonym
TIMER	TIMR, TIMERS

Purpose of Command

The TIMER command is a panel synonym that displays the Timer Management panel, which you use to display, add, change, or delete all scheduled timers.

Operand Descriptions

filter_criteria

Displays the timers that contain *filter_criteria* anywhere in the timer. If *filter_criteria* is not specified, all timers are displayed.

CATCHUP: YES

Displays the timers that were defined in a control file with CATCHUP specified.

COMMAND: *command*

Displays the timers that contain the command string beginning with *command*.

EC: YES

Displays the timers that were set with EVERYCON=YES.

ID: *timerid*

Displays the timers that contain *timerid* in the timer ID.

INTERVAL: *interval*

Displays the timers that contain *interval* as part of the interval.

OP: *opid*

Displays the timers that contain *opid* as part of the operator ID.

SAVE: SAVED

Displays the timers that were set with the SAVE parameter.

TARGET

Specifies the target of the operations performed by TIMER. The default is the system of the user. The user can specify a remote domain by specifying a *netid.domain*, or, if "SA" is specified for the COMMON.EZLRMTTIMER statement in the CNMSTYLE member, the user can specify a remote system by entering the name of the remote system.

TF: YES

Displays the timers that were set with TIMEFMSG=YES.

TIME: *date_time*

Displays the timers that are scheduled to run on the date and time that begin with *date_time*. The date and time are specified in the local NLS format separated by a space.

TIPADDR

Specifies the IP address or host name of the remote NetView program.

TOPID

Specifies the autotask to be used on the remote domain for processing the command. The default is your operator ID.

TPORT

Specifies the port number to be used for TCP/IP communications. The default is 4022.

TYPE: *timertype*

Displays the timers of type *timertype*. The valid types are EVERY, AT, AFTER, and CHRON.

Usage Notes

Consider the following when using the TIMER command:

- This command operates in full-screen mode only.

Operand Descriptions

host

Specifies the name of the remote host running the TN3270 server. It can be specified as a host name, or an IP address.

port

Specifies the port to use on the remote host. The default is 23.

rollkey

Specifies the key used while in the telnet session to roll to another NetView component on the roll stack. It is a character string specifying the key (for example, "PF6").

endkey

Specifies the key used while in the telnet session to end the telnet session. It is a character string specifying the key (for example, "PF3").

The supported values for rollkey and endkey are:

- ENTER
- CLEAR
- PA1 (endkey default)
- PA01
- PA2 (rollkey default)
- PA02
- PA3
- PA03
- PF1
- PF01
- PF2
- PF02
- PF3
- PF0
- PF4
- PF04
- PF5
- PF05
- PF6
- PF06
- PF7
- PF07
- PF8
- PF08
- PF9
- PF09
- PF10
- PF11
- PF12
- PF13
- PF14
- PF15

- PF16
- PF17
- PF18
- PF19
- PF20
- PF21
- PF22
- PF23
- PF24

Note: Be careful not to specify an *endkey* or *rollkey* that is needed for the telnet session. The default value for *rollkey* is PA2. The default value for *endkey* is PA1.

Usage Notes

The following restrictions apply to the TN3270 command:

- The keys selected for endkey and rollkey are not needed for use by the application started with telnet. For example, if TN3270 is used to start a TSO session, selecting PF3 as the endkey prevents the use of PF3 for navigating ISPF screens in TSO.
- If the operator rolls out of the telnet session, TCP/IP is no longer processing receives for data coming from the remote host. This can cause the TCP/IP connection to fail if the telnet session is not brought to the front soon enough.
- The command might hang because of TCP/IP not responding or a problem on the remote host. If that happens, RESET IMMED can be used to cancel the command. In some cases, it might be necessary to recycle the task.
- The TN3270 command is an interactive command and must not be scheduled to run by way of the TIMER command unless it runs on an operator's task with an operator present.

Example: Starting a TELNET session on a VM system

To telnet to a VM system with a TCP/IP host name of RALVM14 while accepting defaults for port, rollkey and endkey, enter:

```
TN3270 RALVM14
```

Example: Specifying a different port

To telnet to a system whose telnet server is listening on port 623, enter:

```
TN3270 RALVM14 -a 623
```

Example: Specifying a different Endkey

To telnet to a system and use PF20 as the endkey, enter:

```
TN3270 RALVM14 -e PF20
```

Example: Specifying rollkey and Endkey

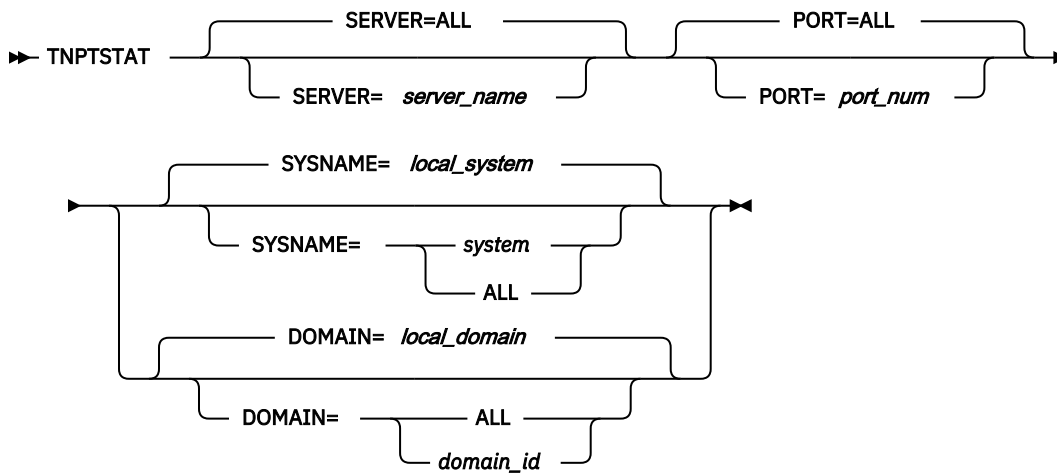
To telnet to a system and specify PF19 for rollkey and PF20 for endkey, enter:

```
TN3270 RALVM14 -r PF19 -e PF20
```

TNPTSTAT (NCCF; CNME8233)

Syntax

NCCF TNPTSTAT



Purpose of Command

You can use the TNPTSTAT command to view configuration and status information about Telnet server ports. You can view this information from a 3270 console or from the Z NetView Enterprise Management Agent.

Note: This command is intended as a REXX interface. For end-user interfaces, use the Tivoli Enterprise Portal, the CNMSTPST sample, or the NetView management console.

Operand Descriptions

SERVER

Specifies the Telnet server for which data is requested. The default value is all Telnet servers known to the DOMAIN or SYSNAME specified.

PORT

Specifies the Telnet server port for which data is requested. The default value is all Telnet server ports known to this NetView program.

SYSNAME

The system name for the requested data. The default value is the local system. When the ALL value is in effect, the command is sent to all NetView programs known to the master NetView program.

Note: Use caution when you specify the ALL value because it can cause rediscovery to take place on multiple NetView programs.

DOMAIN

The domain to which the request is sent. The following are the valid options:

ALL

Specifies all domains in the sysplex. This option is valid only on the master NetView program.

Note: If you have many stacks or systems in your sysplex, issuing ALL can cause slow response times and high CPU utilization.

local

If the DOMAIN keyword is not specified, the domain name of the local NetView system is used. This is the default.

domain_id

Specifies the ID of a specific domain. This option can be issued only from the master NetView program if the domain is not the local domain.

Usage Notes

If message DSI047E is received, contact your system programmer to enable the appropriate tower or subtower. For information on data collection and display towers and subtowers, see *IBM Z NetView Installation: Configuring Additional Components*.

Return Codes**Return Code****Meaning****0**

The command is successful.

2

Help is issued.

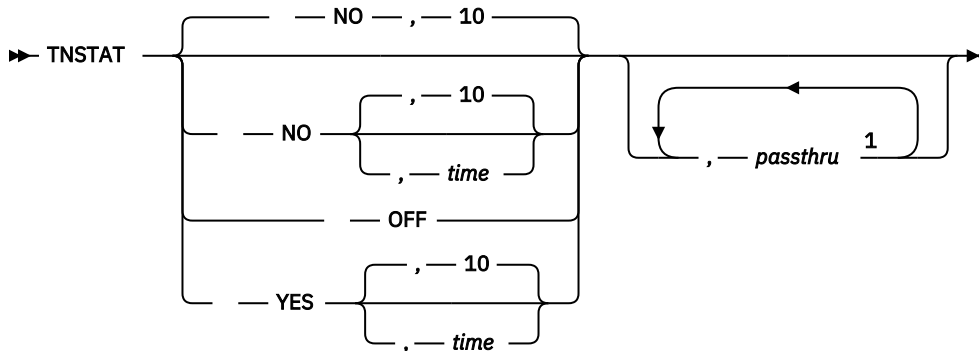
4

There is no data to display.

8 or above

Failure, see the associated message.

TNSTAT (NCCF; CNME0036)

Syntax**TNSTAT****Notes:**

¹ If you do not specify a positional parameter, you must indicate the absence of the parameter by specifying a comma in its place.

Purpose of Command

The TNSTAT command list changes, restarts, or stops the recording of tuning statistics. You can use tuning statistics to gather data for adjusting VTAM and network control program (NCP) variables to improve performance.

To use the tuning statistics facility, specify the TNSTAT start option when you start VTAM.

Operand Descriptions**NO**

Sends the tuning statistics to the system management facilities (SMF) log.

time

Specifies the number of minutes between tuning statistics recording events. This number can be in the range of 1–1440. The default is 10.

passthru

Specifies up to 6 parameters which are appended unchanged to the VTAM MODIFY command issued by the TNSTAT command. No validation for duplicate or conflicting parameters is performed.

Restrictions

If you omit a positional operand, indicate its absence with a comma. It is not necessary to specify trailing commas.

Example: Sending statistics to the SMF or external log

To send statistics from events recorded every 10 minutes to the SMF log enter:

```
TNSTAT
```

Example: Stopping the recording of running statistics

To stop recording tuning statistics, enter:

```
TNSTAT OFF
```

Example: Sending statistics to the system console

To send statistics from events recorded every 10 minutes to the system console and to the log, enter:

```
TNSTAT YES
```

Example: Sending statistics to the log

To send statistics from events recorded every 20 minutes to the log only (not to the system console), enter:

```
TNSTAT , ,20
```

Example: Sending statistics to the system console and to the log

To send statistics from events recorded every 20 minutes to the system console and to the log, enter:

```
TNSTAT YES,20
```

TOP

Syntax**TOP**

➡ TOP ⇐

IBM-Defined Synonyms

Command or Operand	Synonym
TOP	T

Purpose of Command

The TOP command displays the first page of a multipage panel.

Restrictions

If you enter this command for a single-page panel, no change occurs.

Example: Displaying the first page of a multipage panel

To display the first page of a multipage panel, enter either command:

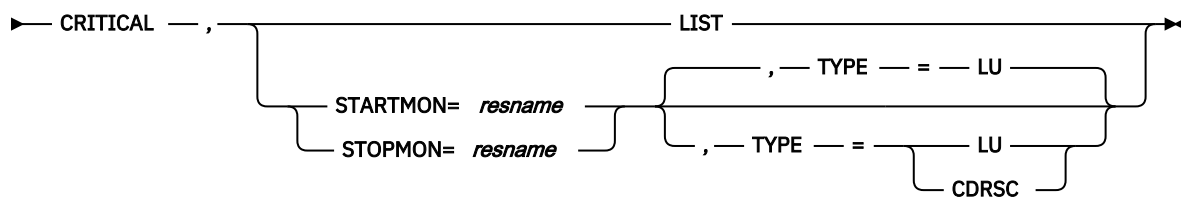
```
TOP
T
```

TOPOSNA CRITICAL (TOPOSNA)

Syntax

TOPOSNA CRITICAL

➤ TOPOSNA ➤



Purpose of Command

Ordinarily, when you use Locate Resource in NMC to request the SNA topology manager to locate and monitor a logical unit (LU) or cross domain resource (CDRSC), the SNA topology manager creates that resource in RODM and monitors it only while it is displayed in an NMC view. When the resource is no longer displayed in any view, the SNA topology manager stops monitoring it and removes it from RODM.

You can use the TOPOSNA CRITICAL command with the STARTMON keyword to monitor a critical LU or CDRSC continuously, regardless of its presence in an NMC view, and regardless of whether the SNA topology manager is monitoring it as part of an LU collection. Resources are created in RODM and are available for display in relevant views using a Locate Resource request or other navigation, but are not removed from RODM when you close the last view in which the resource is displayed. Monitoring continues until you issue the TOPOSNA CRITICAL command with the STOPMON keyword. You can also use the TOPOSNA CRITICAL command with the LIST keyword to list the LUs and CDRSCs that the SNA topology manager is currently monitoring continuously.

Operand Descriptions

LIST

Requests a list of the logical units (LUs) and cross domain resources (CDRSCs) currently being monitored continuously. Continuous monitoring of these resources can be requested in either of two ways:

- Using the TOPOSNA CRITICAL STARTMON command to start continuous monitoring of the resource immediately.
- Using RODM loader statements to create the resources directly in RODM with the monitorContinuously field set ON. In this case, continuous monitoring of the resource begins the next time SNA topology manager is warm started.

TYPE

Specifies the resource type.

LU

Indicates that the resource is a logical unit. This is the default.

CDRSC

Indicates that the resource is a cross domain resource.

STARTMON

Specifies that SNA topology manager is to begin the continuous monitoring of the specified resource.

STOPMON

Specifies that the SNA topology manager stops individual monitoring of the specified resource, and if not also monitoring it as part of an LU collection, remove it from RODM. If the SNA topology manager is monitoring the resource, and it is not also being monitored as part of an LU collection, monitoring ends and the resource is deleted from RODM regardless of whether it is currently displayed in a view.

Note: The specified resource must be one that is being monitored continuously. You can use the TOPOSNA CRITICAL LIST command to display the resources that are being monitored continuously.

resname

Specifies the resource name as *netid.cpname.resourceid* where:

netid

Is a 1- to 8-character name specifying the network identifier of the resource.

cpname

Is a 1- to 8-character name specifying the control point of the target resource.

resourceid

Is a 1- to 17-character name, which is the resource name. The form of this name depends on the resource type as follows:

logical unit

The name must be of the form *netid.luname*.

cross domain resource

If it is not a predefined alias, the name must be of the form *netid.cdrcsname*. If the resource is a predefined alias, the name must be of the form *cdrcsname* (*netid* is omitted). For more information about the naming of cross domain resources, refer to the appropriate VTAM manuals.

Note: Authorization checking for *resname* is done by verifying the *netid*, *cpname*, and *resourceid* values separately. Each value within *resourceid* is also verified separately.

Usage Notes

Samples FLBS8001 and FLBS8002 are provided in CNMSAMP. When installed, these provide a REFRESH command to enable multiple TOPOSNA CRITICAL commands to be issued against resources specified in a definition file. See the FLBS8001 sample for installation directions.

If a switched major node is deactivated, then any TOPOSNA CRITICAL continuous monitors that existed for LUs under the switched major node are ended. If the switched major node is later activated, then you must once again issue the TOPOSNA CRITICAL command to restart continuous monitoring for critical LUs.

Example: Continuously monitoring a critical LU

To start continuously monitoring logical unit LUA on control point CP5 in network SNANETA, enter:

```
TOPOSNA CRITICAL,STARTMON=SNANETA.CP5.SNANETA.LUA,TYPE=LU
```

Response

```
FLB595I REQUESTED MONITORING OF CRITICAL LU SNANETA.CP5.SNANETA.LUA
```

Example: Stopping the monitoring of a critical LU

For operator OPER1 to stop SNA topology manager's continuous monitoring of critical logical unit LUA on control point CP5 in network SNANETA, enter:

```
TOPOSNA CRITICAL,STOPMON=SNANETA.CP5.SNANETA.LUA,TYPE=LU
```

Response

```
FLB597I OPERATOR OPER1 STOPPED MONITORING OF CRITICAL LU
        SNANETA.CP5.SNANETA
```

Example: Listing the LUs and CDRSCs that are being monitored continuously

To list the logical units and cross domain resources that SNA topology manager is currently continuously monitoring, enter:

```
TOPOSNA CRITICAL,LIST
```

Response

```
- NTVCB      FLB590I SNA TOPOLOGY MANAGER CRITICAL LU LIST FOLLOWS:
+ NTVCB
FLB591I RESOURCE NAME                                RESOURCE TYPE MONITOR STATUS
FLB592I -----
FLB593I SNANETA.CP1.SNANETA.LU1                      LU             MONITORING
FLB593I SNANETB.CP2.SNANETB.CDRSC1                  CDRSC          REQUESTED
FLB593I SNANETA.CP2.SNANETA.LU2                      LU             FAILED
FLB593I SNANETA.CP3.SNANETA.CDRSC2                  CDRSC          INITIALIZED
FLB594I END OF CRITICAL LU LIST-----
- NTVCB      FLB411I TOPOSNA CRITICAL COMMAND COMPLETED SUCCESSFULLY
```

where:

MONITORING

Indicates that the SNA topology manager issued the monitor request and has received data in response.

REQUESTED

Indicates that the SNA topology manager has issued the monitor request but has not received a response.

FAILED

Indicates that the SNA topology manager attempted to monitor the resource but failed.

INITIALIZED

Indicates that the SNA topology manager is in the process of issuing the monitor request.

TOPOSNA LISTPOOL (TOPOSNA)

Syntax

TOPOSNA LISTPOOL

➤ TOPOSNA — LISTPOOL ➤

Purpose of Command

TOPOSNA LISTPOOL displays SNA topology manager storage pool statistics.

Operand Descriptions

LISTPOOL

Displays a matrix of storage types versus storage usage and allocation. For each storage pool allocated, a percentage used number is given. Use this operand to check the usage and fragmentation of the SNA topology manager internal storage pool.

The TOPOSNA LISTPOOL command is more useful for diagnostic purposes, rather than periodic internal-storage-pool monitoring.

Example: List SNA topology manager storage usage

To list SNA topology manager storage pool statistics, enter:

```
TOPOSNA LISTPOOL
```

Response

Information similar to the following is displayed:

FLB553I	SNA TOPOLOGY MANAGER STORAGE POOL STATISTICS FOLLOW						
FLB554I	STORAGE	SIZE	ALLOCATED	USED	ALLOCATED	USED	%
FLB555I	TYPE	IN BYTES	COUNT	COUNT	STORAGE-K	STORAGE-K	USED
FLB556I	-----	-----	-----	-----	-----	-----	-----
FLB557I	StatusEntry	228	642	542	143	121	84
FLB557I	LU_Aset	116	210	107	24	13	50
FLB557I	Node_Aset	84	485	404	40	34	83
FLB557I	Port_Aset	124	98	36	12	5	36
FLB557I	TG_Aset	114	214	178	24	20	83
FLB557I	TGC_Aset	80	101	77	8	7	76
FLB557I	Link_Aset	112	109	48	12	6	44
FLB557I	AgentEntry	0	0	0	0	0	0
FLB557I	Max_Aset	144	113	0	16	0	0
FLB557I	NTEnter	72	339	232	24	17	68
FLB557I	LUTEnter	0	0	0	0	0	0
FLB557I	ActionLU	184	176	107	32	20	60
FLB557I	CacheLU	136	180	107	24	15	59
FLB557I	GraphOb	32	252	183	8	6	72
FLB557I	LActLink	360	102	24	36	9	23
FLB557I	LActNode	192	106	8	20	2	7
FLB557I	LActPort	184	88	18	16	4	20
FLB557I	LActTG	236	103	9	24	3	8
FLB557I	NActCircuit	192	318	220	60	42	69
FLB557I	NActNode	324	101	75	32	24	74
FLB557I	CacheCirc	460	97	80	44	36	82
FLB557I	CacheLink	172	95	24	16	5	25
FLB557I	CacheNode	188	324	232	60	43	71
FLB557I	CachePort	124	98	18	12	3	18
FLB557I	HashEntry	12	2028	1971	24	24	97
FLB558I	END OF STORAGE POOL STATISTICS	-----	-----	-----	-----	-----	-----

TOPOSNA LISTREQS (TOPOSNA)

Syntax

TOPOSNA LISTREQS

➡ TOPOSNA — LISTREQS ➡

Purpose of Command

TOPOSNA LISTREQS displays the status of pending topology manager requests to its agents.

Operand Descriptions

LISTREQS

Displays the status of pending topology manager requests to its agents. For nodes being monitored, the messages display:

- The name of the agent node.
- The type of topology being obtained (network, local, or LU collection).
- The type of monitoring (continuous or timed). For timed monitors, the time remaining is shown.
- The stage of the monitoring:
 - Before any data is received (REQUESTED)
 - Before initial topology transfer is complete (INCOMPLETE)
 - After initial topology transfer is complete (COMPLETE)
 - Retrying is currently underway (RETRY)
- The local name (at the agent node) of the logical link or port

The messages are grouped as follows:

1. Network topology monitor operations (including retries)
2. Local topology monitor operations (including retries)
3. LU collection monitor operations (including retries)
4. Port/link control operations (activate, deactivate, recycle)

There is no specific ordering within each of these groups.

Restrictions

This command cannot be issued until you receive the following message:

```
FLB440I SNA TOPOLOGY MANAGER INITIALIZATION IS COMPLETE.
```

Example: Listing pending requests

To list pending topology requests, issue:

```
TOPOSNA LISTREQS
```

Response

```

FLB561I NODE          LOCAL  MONITOR MONITOR    MONITOR
FLB562I NAME          NAME   TYPE    STATUS     TIME
FLB563I -----
FLB564I SNANETA.A55M          NETWORK COMPLETE CONTINUOUS
FLB564I SNANETA.A01M          NETWORK COMPLETE   12 MINUTES
FLB564I SNANETA.A72M          NETWORK REQUESTED CONTINUOUS
FLB564I SNANETA.A61M          NETWORK RETRY    1635 SECONDS
FLB564I SNANETA.A57M          NETWORK INCOMPLETE CONTINUOUS
FLB564I SNANETA.A55M          LOCAL   COMPLETE CONTINUOUS
FLB564I SNANETA.A01M          LOCAL   COMPLETE   2 MINUTES
FLB564I SNANETA.A72M          LOCAL   REQUESTED CONTINUOUS
FLB564I SNANETA.A59M          LOCAL   RETRY    1227 SECONDS
FLB564I SNANETA.A57M          LOCAL   INCOMPLETE CONTINUOUS
FLB564I SNANETA.A99M          PWS438P LUCOL   COMPLETE CONTINUOUS
FLB564I SNANETA.A99M          PWS420P LUCOL   COMPLETE  120 MINUTES
FLB564I SNANETA.A01M          PWS442P LUCOL   REQUESTED CONTINUOUS
FLB564I SNANETA.A99M          PWS444P LUCOL   RETRY    54 SECONDS
FLB564I SNANETA.A99M          PWS433P LUCAL   INCOMPLETE CONTINUOUS
FLB565I END OF MONITOR REQUESTS-----
- A99NV  FLB411I TOPOSNA LISTREQS COMMAND COMPLETED SUCCESSFULLY

```

The example demonstrates all possible messages that can be issued. Note that messages can repeat.

TOPOSNA LISTRODM (TOPOSNA)

Syntax

TOPOSNA LISTRODM

➤ TOPOSNA — LISTRODM ➤

Purpose of Command

TOPOSNA LISTRODM displays RODM activity and object counts.

Operand Descriptions

LISTRODM

Displays a matrix of object types versus activity and object counts.

Example: List RODM activity and object counts

To list RODM activity and object counts, enter:

```
TOPOSNA LISTRODM
```

Response Information similar to the following is displayed:

```
FLB472I SNA TOPOLOGY MANAGER RODM ACTIVITY COUNTS FOLLOW:
FLB473I OBJ.
FLB474I TYPE   CREATE   DELETE   UPDATE   QUERY   LINK/  FLBTRST  RODM
FLB475I -----
FLB476I CDRM    11       0        0        0       11      11      11
FLB476I DefG    18       0       36        0       18      18      18
FLB476I EN      2        0        0        0        3        4        2
FLB476I ICN     6        2        3       12       21        7        4
FLB476I LEN     0        0        0        0        0        0        0
FLB476I MDH     1        0        3        2        9        1        1
FLB476I NN     159       0        5        7      205     160     159
FLB476I BrNN    0        0        0        0        0        0        0
FLB476I Sna     0        0        0        0        0        0        0
FLB476I T2.1    0        0        0        0        0        0        0
FLB476I T4     12        0       13        3       34        8       12
FLB476I T5      2        0        2        0        0        2        2
FLB476I VRN     3        0        0        0        3        3        3
FLB476I Link   1344       0        2        0     2804     1344     1344
FLB476I CDRS    0        0        0        0        0        0        0
FLB476I LU      3        0        0        0        6        3        3
FLB476I LUGr    0        0        0        0        0        0        0
FLB476I Port   262       0        0        0       576     262     262
FLB476I TG     530       0        5        9     1060     530     530
FLB476I ACir   424       0        3        7     2150     424     424
FLB476I SCir   21        0       21        2       104
FLB476I AggG   175       0     190       33       175
FLB476I AggC   395       0     394        0     1223
FLB476I TOTL  3369       2     678       78     8403     2777     2775
FLB477I END OF RODM ACTIVITY COUNTS-----
FLB411I TOPOSNA LISTRODM COMMAND COMPLETED SUCCESSFULLY
```

The first column contains an abbreviation of the RODM class name to which the counts apply. The abbreviations are:

CDRM

crossDomainResourceManager

DefG

definitonGroup

EN

appnEN

ICN
interchangeNode

LEN
lenNode

MDH
migrationDataHost

NN
appnNN

BrNN
appnBrNN

Sna
snaNode

T2.1
t2-1Node

T4
t4Node

T5
t5Node

VRN
virtualRoutingNode

Link
logicalLink

CDRS
crossDomainResource

LU
logicalUnit

LUGr
luGroup

Port
port

TG
appnTransmissionGroup

ACir
appnTransmissionGroupCircuit

SCir
subareaTransmissionGroupCircuit

AggG
aggregateGraph2

AggC
circuit2

SnaL
snaLocalTopology

The remainder of the columns contain the following activity and object counts:

CREATE
The number of EKG_CreateObject calls issued against OBJ TYPE.

DELETE
The number of EKG_DeleteObject calls issued against OBJ TYPE.

UPDATE
The number of EKG_ChangeMultipleFields calls issued against OBJ TYPE.

QUERY

The number of EKG_QueryField and EKG_QueryMultipleSubfields calls issued against OBJ TYPE.

LINK/UNLINK

The number of EKG_LinkTrigger and EKG_UnlinkTrigger calls issued against OBJ TYPE.

FLBTRST(STATUS)

The number of times the FLBTRST method was invoked for OBJ TYPE (indicates a status change occurred). There can be other non-objectlink and non-objectlinklist fields changed at the same time as the status changed; however these changes are performed with the same MAPI (method API) RODM calls as for the status change. This column contains a blank entry for aggregate object classes (SCir, AggG, AggC, and SnaL).

RODM COUNT

The current number of object instances of OBJ TYPE currently or previously known to the SNA topology manager since the SNA topology manager was last initialized. The SNA topology manager must have received a status report on an object, through an ongoing or previous topology monitor, for the object to be reflected in this count. This column contains a blank entry for aggregate objects (SCir, AggG, AggC, and SnaL).

TOPOSNA LISTSTOR (TOPOSNA)

Syntax

TOPOSNA LISTSTOR

➡ TOPOSNA — LISTSTOR ➡

Purpose of Command

TOPOSNA LISTSTOR displays storage usage counts for the SNA topology manager. It is not the total storage usage information for the SNA topology manager task. Use the TASKMON FLBTOPO STG command or the TASKUTIL FLBTOPO command to display the total storage being used by the SNA topology manager task.

Operand Descriptions

LISTSTOR

Displays a matrix of object types versus internal storage usage.

Example: List SNA topology manager storage usage

To list SNA topology manager storage usage, enter:

```
TOPOSNA LISTSTOR
```

Response

Information similar to the following is received:

```
TOPOSNA LISTSTOR
FLB577I SNA TOPOLOGY MANAGER STORAGE USAGE FOLLOWS:
FLB578I RESOURCE          CACHE1          CACHE2          TOTAL          TOTAL
FLB579I TYPE              COUNT1  STORAGE-K COUNT2  STORAGE-K STORAGE-K  MAXIMUM-K
FLB580I -----
FLB581I AttrSets          5170              859              859
FLB581I X-AttrSets        20                5                5
FLB581I CritLUs           0                 0                 0
FLB581I NodeTable         212               15               15
FLB581I RodmMain          1                 16               16
FLB581I StatHist          2775             593              593
FLB581I Heap              82                39               43
FLB581I IntTrace          1                 40               40
FLB581I CDRM              11                3                6
FLB581I DefGroup          0                 0                4                4
```

FLB581I	EN	3	1	2	1	2	2
FLB581I	ICN	7	2	4	1	3	3
FLB581I	LEN	0	0	0	0	0	0
FLB581I	MDH	0	0	1	1	1	1
FLB581I	NN	164	31	159	30	61	61
FLB581I	BrNN	0	0	0	0	0	0
FLB581I	SnaNode	0	0	0	0	0	0
FLB581I	T2.1	0	0	0	0	0	0
FLB581I	T4	11	3	12	3	6	6
FLB581I	T5	0	0	2	1	1	2
FLB581I	VRN	3	1	3	1	2	2
FLB581I	Link	1344	468	1344	216	684	684
FLB581I	LU	3	2	3	2	4	4
FLB581I	Port	262	47	262	31	78	78
FLB581I	TG/Circuit	514	404	445	202	606	606
FLB581I	IntDomCirc	0	0	387	11	11	11
FLB581I	IntNetCirc	0	0	7	1	1	1
FLB581I	nnDomain	0	0	166	5	5	5
FLB581I	nnDomNet	0	0	8	1	1	1
FLB581I	SnaLocal	0	0	1	1	1	1
FLB581I	TOTAL					3044	3049
FLB582I	END OF SNA TOPOLOGY MANAGER STORAGE USAGE-----						
FLB411I	TOPOSNA LISTSTOR COMMAND COMPLETED SUCCESSFULLY						

The first column contains an abbreviation of either an SNA topology manager internal storage class, RODM class name, or role of an object within an RODM class to which the storage usage applies.

The first set of abbreviations is for SNA topology manager internal storage classes. Only the COUNT1, TOTAL STORAGE-K, and TOTAL MAXIMUM-K columns contain counts. The abbreviations are:

AttrSets

Attribute Sets, an internal representation of the fields (attributes) of an RODM object. The COUNT column represents the current number of allocated Attribute Sets.

X-AttrSets

Extended Attribute Sets. Extra storage for fields, used when the standard (above) version of Attribute Sets is not large enough to hold all of an object's fields. The COUNT column represents the current number of allocated Extended Attribute Sets.

CritLUs

Critical LUs. The COUNT column represents the current number of Critical LUs being monitored.

NodeTable

Internal node lookup cache. The COUNT column represents the current number of node table entries.

RodmMain

RODM interface storage and is the size of the RODM response block. The COUNT column is always 1.

StatHist

Status History entries. The COUNT column indicates how many objects are currently maintaining a status history.

Heap

SNA topology manager C runtime heap utilization. The COUNT column indicates how many storage requests are currently allocated from the heap.

IntTrace

The amount of storage currently allocated to the internal trace buffer.

The second set of abbreviations is for RODM classes. Some entries represent multiple RODM classes. For example, the TG/Circuit entry represents TGs and circuits, both APPN and subarea versions. The abbreviations are:

CDRM

Cross Domain Resource Manager objects

DefGroup

Definition Group objects

EN

End node objects

ICN

Interchange Node objects

LEN

Low Entry Networking objects

MDH

Migration Data Host objects

NN

Network node objects

BrNN

Branch Network Node objects

SnaNode

SNA Node objects

T2.1

T2.1 objects

T4

T4 objects

T5

T5 objects

VRN

Virtual Routing Node objects

Link

Logical link objects

LU

Logical unit objects (includes Logical Unit, LU Group, and Cross Domain Resource RODM classes)

Port

Port objects

TG/Circuit

Transmission Group and Transmission Group Circuit objects (includes both Subarea and APPN flavors)

IntDomCirc

InterDomain Circuit objects

IntNetCirc

InterDomainNetwork Circuit objects

nnDomain

nnDomain objects

nnDomNet

nnDomainNetwork objects

SnaLocal

SnaLocal Topology objects

The remainder of the columns contain storage class counts, as follows:

COUNT1

For resources that do not correspond to RODM objects, represents the total count of that resource type. For resources that do correspond to RODM objects, represents the total count of that object currently active in CACHE1 (note that this value can differ from the COUNT2 field).

CACHE1 STORAGE-K

Amount of storage (in kilobytes) currently utilized by the RESOURCE TYPE in CACHE1.

COUNT2

For resources that do not correspond to RODM objects, no entry is made. For resources that do correspond to RODM objects, represents the total count of that object currently active in CACHE2 (note that this value can differ from the COUNT1 field).

CACHE2 STORAGE-K

Amount of storage (in kilobytes) currently utilized by the RESOURCE TYPE in CACHE2.

TOTAL STORAGE-K

The total amount of storage currently utilized by the RESOURCE TYPE (note that for resources that have CACHE1 and CACHE2 entries, this value is the total of the CACHE1 and CACHE2 values).

TOTAL MAXIMUM-K

The maximum amount of storage that this RESOURCE TYPE has utilized (the highwater mark).

Note:

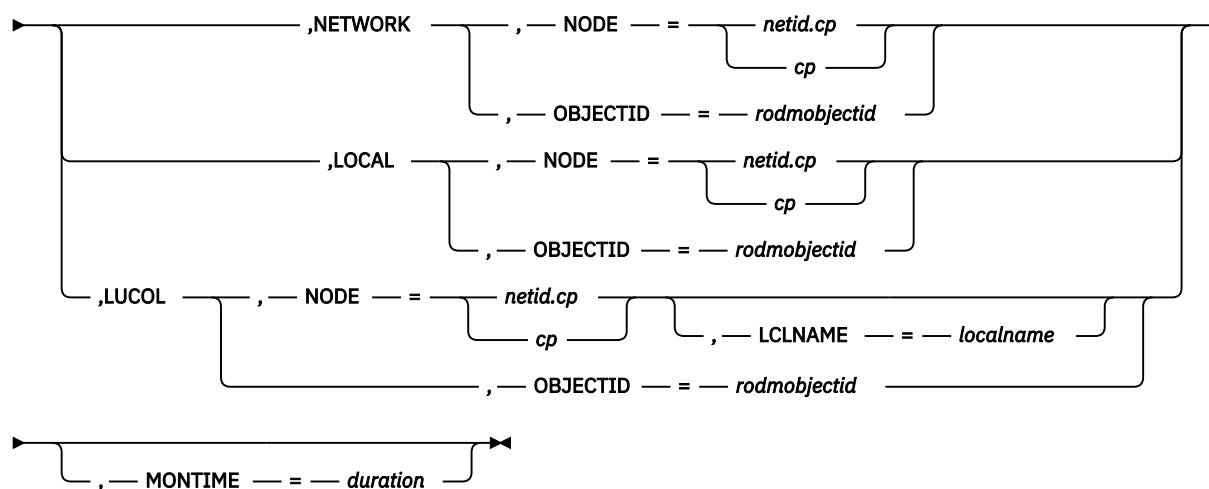
1. Storage values are rounded up to the nearest kilobyte before displaying, and are close approximations of the actual storage utilized at the instant the LISTSTOR command was issued.
2. CACHE1 versus CACHE2 - For resource types that have corresponding RODM objects, the SNA topology manager maintains a two-stage internal cache. Because some intermediate (temporary) storage objects are maintained, these numbers do not directly correspond to the actual numbers of objects of a given class in RODM. Use the TOPOSNA LISTRODM command for RODM object counts.

TOPOSNA MONITOR (TOPOSNA)

Syntax

TOPOSNA MONITOR

➤ TOPOSNA — — MONITOR ➤



Purpose of Command

TOPOSNA MONITOR starts the monitoring of local or network topology, or the collection of all LUs (including logical units, cross-domain resources, and LU groups) associated with a node. The agent at the monitored node first sends a copy of the appropriate data (such as network topology, local topology, or LU topology). As the data changes, the agent sends updates.

Operand Descriptions

MONITOR

Specifies that the topology manager is to begin monitoring the topology of an agent node.

NETWORK

Specifies network topology.

LOCAL

Specifies local topology.

LUCOL

Specifies monitoring of the collection of all LUs associated with the specified node.

NODE

Specifies the node, identified by network name, that the topology manager monitors.

netid.cp

The network qualified name of the control point of the agent node, where *netid* is the network identifier and *cp* is the control point name.

The *netid* and the *cp* name each have an 8-character maximum length; no blanks are included and the period between them is required.

Authorization checking of *netid* and *cp* is done separately; the 8-character authorization-checking limit is therefore not a limitation on checking the node name.

cp

The control point name of the agent node. The name consists solely of the control point name: no network identifier or period is included.

OBJECTID=*rodmobjectid*

The 8-byte RODM object identifier for the node (or logical link when monitoring an LU collection). The format is 16 hexadecimal digits (0-9, A-F). This parameter is intended to be used primarily by programs or command lists that have access to RODM rather than by operators. Authorization checking of *rodmobjectid* is done only on the first 8 characters.

LCLNAME=*localname*

The local name of the logical link. The LU collection is monitored using the agent specified by the NODE parameter and includes all LUs that reside at the logical link.

Note: Only one monitor for the LU collection is allowed for a logicalLink that can be reported by more than one SSCP.

MONTIME=*duration*

Specifies the length of time (in minutes) for monitoring. The time limit does not include the time required for the agent node to transfer the initial copy of its topology, but is the amount of time that the manager receives topology updates from the agent node.

Timing begins when all the initial topology data has been transferred. A completed transfer is indicated by message FLB406I for network topology, message FLB423I for local topology, or message FLB552I for LU collection. If you do not specify MONTIME, monitoring continues until an operator issues a STOP or STOPMGR request. When you specify MONTIME, the topology manager stops monitoring after the number of minutes specified by *duration*.

When an operator issues successive monitoring requests for the same agent node, the request for the longest monitor (including a continuous monitor request) overrides other requests. For example, a timed-monitor request cannot change continuous monitoring; it can reset the timing only on a previous, shorter timed-monitor request. A request for continuous monitoring overrides a previous timed-monitor request.

There is no default MONTIME value. The minimum value for MONTIME is zero (0); the maximum is 32000.

Restrictions

Before issuing a MONITOR request, consider these restrictions:

- This command cannot be issued until you receive the following message:

```
FLB440I SNA TOPOLOGY MANAGER INITIALIZATION IS COMPLETE.
```

If the topology manager was warm started, you do not receive this message until the topology manager has started all continuous monitor operations in effect at the time the last STOPMGR request was issued (or in effect at the time of the RODM checkpoint, if you load checkpoint data). Timed monitor

commands are not restarted. The messages for restarted monitor operations are sent to the authorized receiver.

- You can have only one active local topology monitor for each agent node, one active network topology monitor for each agent node, and one active LU collection monitor for a single agent node's entire LU collection. To have multiple, active, LU collection monitors for a single agent node, do the following:
 - Use the LCLNAME keyword to subset the agent node's LU collection by logicalLink (PU) under which the LUs are defined.
 - When you monitor a host node's LU collection, the VTAM agent does not include resources that are defined under major node ISTCDRDY in its initial topology report. However, the agent does subsequently send updates for dynamic real resources defined under this major node.
 - Use different values for the LCLNAME keyword to specify different subsets.

Even if several operators on the *same* NetView program issue monitor requests, they cannot have in effect monitor commands for different durations. Instead, the topology manager honors the request for the longest duration. However, operators on *different* NetView programs can issue monitor requests to an agent that differ in duration.

Multiple nodes can be monitored at the same time. The monitoring of local topology, network topology, and LU collection topology are independent of each other. For example, a STOP request or an end of a timed monitor for local topology at a node does not affect monitoring of network topology at that node.

- If you incorrectly request network topology from an end node other than a migration data host, an error message is not displayed; however, warning message FLB409W is displayed, indicating that the topology manager is retrying.
- Messages pertaining to each instance of the TOPOSNA MONITOR command are correlated with the command. This correlation continues until one of the following occurs:
 - The command completes successfully with no retry.
 - The command fails with no retry.
 - The command goes into retry after the default interval ends.

The time period after which the TOPOSNA MONITOR command goes into retry is determined by the default settings. You can display the default settings using the TOPOSNA QUERYDEF command. If the TOPOSNA START command goes into retry mode, you can issue the TOPOSNA LISTREQS command to see the status of the monitor. The MONITOR STATUS for the node the user tried to monitor is shown as RETRY. If the command fails all retries, TOPOSNA LISTREQS drops the node from the list of pending requests. If the command succeeds, TOPOSNA LISTREQS shows the MONITOR STATUS for that node as COMPLETE.

- You cannot change from continuous monitoring to timed monitoring. Use a STOP request to stop the continuous monitoring and then initiate the timed monitoring.
- You cannot reset a timed monitoring interval to a shorter interval. Use a STOP request to stop the timed monitoring and then initiate the timed monitoring with a shorter interval.

Example: Starting a continuous network monitor

A representative subset of possible response messages is shown for the examples.

To monitor network topology continuously from node APPNNET.NODE1, issue:

```
TOPOSNA MONITOR,NETWORK,NODE=APPNNET.NODE1
```

Response

```
FLB403I REQUESTED MONITORING OF SNA NETWORK TOPOLOGY FROM NODE
APPNNET.NODE1
FLB406I INITIAL TRANSFER OF SNA NETWORK TOPOLOGY FROM NODE
APPNNET.NODE1 IS COMPLETE
```

Example: Starting a timed local monitor

To start monitoring local topology at node APPNNET1.NODE1 for 10 minutes, after which time the monitoring automatically stops, enter:

```
TOPOSNA MONITOR,LOCAL,NODE=APPNNET1.NODE1,MONTIME=10
```

Response

```
FLB420I REQUESTED MONITORING OF SNA LOCAL TOPOLOGY FROM NODE
APPNNET1.NODE1
FLB423I INITIAL TRANSFER OF SNA LOCAL TOPOLOGY FROM NODE
APPNNET1.NODE1 IS COMPLETE
(10 minutes later)
FLB421I COMPLETED MONITORING OF SNA LOCAL TOPOLOGY FROM NODE
APPNNET1.NODE1
```

Example: Changing a timed monitor to a continuous monitor

To change the timed monitor from the previous example to a continuous monitor, enter:

```
TOPOSNA MONITOR,LOCAL,NODE=APPNNET1.NODE1
```

Response

```
FLB403I REQUESTED MONITORING OF SNA NETWORK TOPOLOGY FROM NODE
APPNNET.NODE1
FLB406I INITIAL TRANSFER OF SNA NETWORK TOPOLOGY FROM NODE
APPNNET.NODE1 IS COMPLETE
```

Example: Resetting a timed monitor to a longer time

If you have a timed monitor for node APPNNET1.NODE1 active with an interval of 5 minutes, you can change the interval to 10 minutes by entering:

```
TOPOSNA MONITOR,NETWORK,NODE=APPNNET1.NODE1,MONTIME=10
```

Response

```
FLB403I REQUESTED MONITORING OF SNA NETWORK TOPOLOGY FROM NODE
APPNNET.NODE1
FLB406I INITIAL TRANSFER OF SNA NETWORK TOPOLOGY FROM NODE APPNNET.NODE1
IS COMPLETE
```

The timed monitor interval is lengthened by 10 minutes. The same or different operators can issue the original command and the command in this example. The elapsed time from the first request is not subtracted from the second request.

Example: Automating monitoring

To specify the node with an RODM object identifier (intended for programs or command lists), use this form of the command:

```
TOPOSNA MONITOR,NETWORK,OBJECTID=0123456789ABCDEF,MONTIME=10
```

Example: Starting a continuous LU collection monitor

To monitor the LU collection under node NTCBMVS, issue:

```
TOPOSNA MONITOR,LUCOL,NODE=NTCBMVS
```

Response:

```

FLB540I REQUESTED MONITORING OF LU COLLECTION FROM SNANET.NTCBMVS
FLB552I INITIAL TRANSFER OF LU COLLECTION DATA FROM NODE
        SNANET.NTCBMVS IS COMPLETE

```

Example: Starting a timed LU collection monitor

To start a timed LU collection monitor at node NTCBMVS, enter:

```

TOPOSNA MONITOR,LUCOL,NODE=SNANET.NTCBMVS,MONTIME=10,LCLNAME=MVSLNK

```

Response:

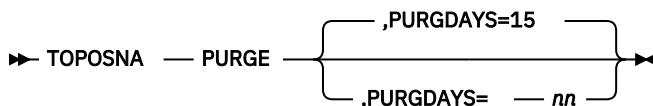
```

FLB540I REQUESTED MONITORING OF LU COLLECTION FROM SNANET.NTCBMVS
FLB552I INITIAL TRANSFER OF LU COLLECTION DATA FROM NODE
        SNANET.NTCBMVS IS COMPLETE

```

This starts monitoring of the node's LU collection, which will continue for 10 minutes, after which time the monitoring automatically stops and message FLB584I is issued.

TOPOSNA PURGE (TOPOSNA)

Syntax**TOPOSNA PURGE****Purpose of Command**

TOPOSNA PURGE deletes expired unreachable resources from the RODM data cache.

Operand Descriptions**PURGE**

Deletes from the RODM data cache all topology manager objects whose status fields (states fields in RODM) have a time stamp older than the value of PURGDAYS, are not currently being monitored by the SNA topology manager, and have the FLB_Creator field containing a value of FLB.

PURGDAYS=nn

The number of days that the PURGE request uses when determining what objects to delete from the RODM data cache.

The default value is 15 for PURGDAYS; *nn* can be an integer value in the range of 0–32767. Use values of 1–7 days with caution; small values of PURGDAYS increase the likelihood that a deleted object will be created again soon after purging. Even if their status has not changed, APPN network nodes automatically refresh their status every 5 or 7 days to prevent them from being erroneously purged from the APPN network topology database.

Note: A value of 0 means that the RODM data cache is purged of all resources that are not being monitored. Although a PURGE request with PURGDAYS=0 has the same effect as a cold start, use a cold start of the topology manager when the topology manager is stopped and later restarted. This provides better performance than starting the topology manager and then issuing the PURGE request.

Restrictions

Before issuing a PURGE request, consider these restrictions:

- A PURGE request does not delete resources unless these conditions are met:
 - The resource must not be currently monitored.
 - The resource must be at least the PURGDAYS value since its last update.
 - The FLB_Creator field contains a value of FLB.

Also, an aggregate object, such as nnDomain, is deleted when all its underlying objects are deleted.

For other objects that can be purged under specific conditions, see the *IBM Z NetView SNA Topology Manager Implementation Guide*.

- This command cannot be issued until you receive the following message:

```
FLB440I SNA TOPOLOGY MANAGER INITIALIZATION IS COMPLETE.
```

- Non-critical LUs are not purged when they are contained in open views.

Example: Purging resources

To remove resources from RODM that have not been updated for 12 days, enter:

```
TOPOSNA PURGE,PURGDAYS=12
```

Response

```
FLB411I TOPOSNA PURGE COMMAND COMPLETED SUCCESSFULLY
```

The purged resources will not be displayed in refreshed views.

TOPOSNA QUERYDEF (TOPOSNA)

Syntax

TOPOSNA QUERYDEF

➤ TOPOSNA — QUERYDEF ➤

Purpose of Command

TOPOSNA QUERYDEF displays the current settings for the topology manager values that can be set by a SETDEFS request.

Operand Descriptions

QUERYDEF

Requests that settings be displayed.

Restrictions

This command cannot be issued until you receive the following message:

```
FLB440I SNA TOPOLOGY MANAGER INITIALIZATION IS COMPLETE.
```

Example: Querying default values

The following example shows how the defaults are set to initial values, how some are reset to new values, and how the new set of defaults in effect is displayed.

```
TOPOSNA SETDEFS
TOPOSNA SETDEFS,AUTOMON=(NNLOCAL=YES),NETRETRY=(120),LCLRETRY=(NORETRY)
TOPOSNA QUERYDEF
```

```

FLB494I SNA TOPOLOGY MANAGER DEFAULT SETTINGS FOLLOW:
FLB495I MONITOR SNA LOCAL TOPOLOGY FOR NEW NETWORK NODES : YES
FLB496I MONITOR SNA LOCAL TOPOLOGY FOR NEW END NODES : NO
FLB650I MONITOR SNA NETWORK TOPOLOGY FOR NEW T5 NODES : NO
FLB651I MONITOR SNA LOCAL TOPOLOGY FOR NEW T5 NODES : NO
FLB497I SNA NETWORK TOPOLOGY IMMEDIATE RETRY INTERVAL : 12
FLB498I SNA NETWORK TOPOLOGY IMMEDIATE RETRY LIMIT : 5
FLB499I SNA NETWORK TOPOLOGY LONG-TERM RETRY INTERVAL : 1800
FLB500I SNA NETWORK TOPOLOGY LONG-TERM RETRY LIMIT : 48
FLB501I SNA LOCAL TOPOLOGY IMMEDIATE RETRY INTERVAL : NORETRY
FLB502I SNA LOCAL TOPOLOGY IMMEDIATE RETRY LIMIT : 5
FLB503I SNA LOCAL TOPOLOGY LONG-TERM RETRY INTERVAL : 1800
FLB504I SNA LOCAL TOPOLOGY LONG-TERM RETRY LIMIT : 48
FLB546I SNA LU COLLECTION IMMEDIATE RETRY INTERVAL : 60
FLB547I SNA LU COLLECTION IMMEDIATE RETRY LIMIT : 5
FLB548I SNA LU COLLECTION LONG-TERM RETRY INTERVAL : 1800
FLB549I SNA LU COLLECTION LONG-TERM RETRY LIMIT : 48
FLB493I ERROR RETRY LIMIT : 1
FLB520I RODM RETRY INTERVAL : 30
FLB521I RODM RETRY LIMIT : 1000
FLB528I CMIP SERVICES RETRY INTERVAL : 30
FLB529I CMIP SERVICES RETRY LIMIT : 1000
FLB411I TOPOSNA QUERYDEF COMMAND COMPLETED SUCCESSFULLY

```

TOPOSNA REFRESH (TOPOSNA)

[illegible]

¹ The parentheses are optional if only one class name is specified.

The TOPOSNA REFRESH command changes the initial default values that are provided by the NetView program for the Status Resolution table, the OSI-Display status table, and the Exception View table. When the command is invoked to refresh the Status Resolution table, the OSI-Display status table, or the Exception View table, the entire table is read and the updates take effect for subsequent SNA topology manager processing. When the TOPOSNA command is invoked to refresh the Exception View table and the CLASS keyword is specified, the resources in RODM that belong to the object classes specified on the TOPOSNA REFRESH command are updated to change the ExceptionViewList field to reflect the changes

made in the Exception View table. The topology manager resets the ExceptionViewList to the new value and does not append the new value to the existing value.

When refreshing the Exception View table and specifying the CLASS keyword, performance might be degraded because all of the resources in RODM for the object classes being refreshed are searched and modified according to the updates made in the table.

Operand Descriptions

REFRESH

Specifies that one or more of the tables will be refreshed.

ALLTABLS

Specifies that all of the customization tables will be refreshed.

RESOLVE

Specifies that the Status Resolution table will be refreshed.

OSIDISP

Specifies that the OSI-Display Status table will be refreshed.

EXVIEW

Specifies that the Exception View table be refreshed and that existing views/methods be modified based on the updated table.

CLASS

The object class or classes whose ExceptionViewList field is to be updated in RODM. Object classes to be updated can be specified in either of two ways:

classname

One or more specific object class names, separated by commas. Following is a list of possible CLASS values and their corresponding OBJECTCL entry values in FLBEXV:

- "APPNBRNN" — appnBrNN
- "APPNEN" — appnEN
- "APPNNN" — appnNN
- "APPNTRANSMISSIONGROUP" — appnTransmissionGroup
- "APPNTRANSMISSIONGROUPCIRCUIT" — appnTransmissionGroupCircuit
- "CROSSDOMAINRESOURCE" — crossDomainResource
- "CROSSDOMAINRESOURCEMANAGER" — crossDomainResourceManager
- "DEFINITIONGROUP" — definitionGroup
- "INTERCHANGENODE" — interchangeNode
- "LENNODE" — lenNode
- "LOGICALLINK" — logicalLink
- "LOGICALUNIT" — logical Unit
- "MIGRATIONDATAHOST" — migrationDataHost
- "PORT" — port
- "T2-1NODE" — t2-1Node
- "T4NODE" — t4Node
- "T5NODE" — t5Node
- "VIRTUALROUTINGNODE" — virtualRoutingNode

ALL

The ExceptionViewList field for all object classes in the table will be updated in RODM.

Restrictions

This command cannot be issued until you receive the following message:

```
FLB440I SNA TOPOLOGY MANAGER INITIALIZATION IS COMPLETE.
```

Example: Refresh the status resolution table

To refresh the status resolution table, enter:

```
TOPOSNA REFRESH,RESOLVE
```

Example: Refresh the OSI-Display status table

To refresh the OSI-display status table, enter:

```
TOPOSNA REFRESH,OSIDISP
```

Example: Refresh the exception view table

To refresh the exception view table, enter:

```
TOPOSNA REFRESH,EXVIEW
```

Example: Refresh both the OSI-Display status table and the exception view table

To refresh the OSI-display status table and the exception view table, enter:

```
TOPOSNA REFRESH,OSIDISP,EXVIEW
```

Example: Refresh all three of the tables

To refresh all SNA topology manager tables, enter:

```
TOPOSNA REFRESH,ALLTABLS
```

Example: Refresh the exception view table for specific object classes

To update the ExceptionViewList field in RODM for all resources associated with the port and logicalLink object classes, enter:

```
TOPOSNA REFRESH,EXVIEW,CLASS=(logicalLink,port)
```

Example: Refresh the exception view table for all exception view object classes

To update the ExceptionViewList field in RODM for all resources associated with all of the object classes that can appear in exception views, enter:

```
TOPOSNA REFRESH,EXVIEW,CLASS=(ALL)
```

Example: Refresh specific object class for all three tables

To update all SNA topology manager tables for all resources associated with the port object class:

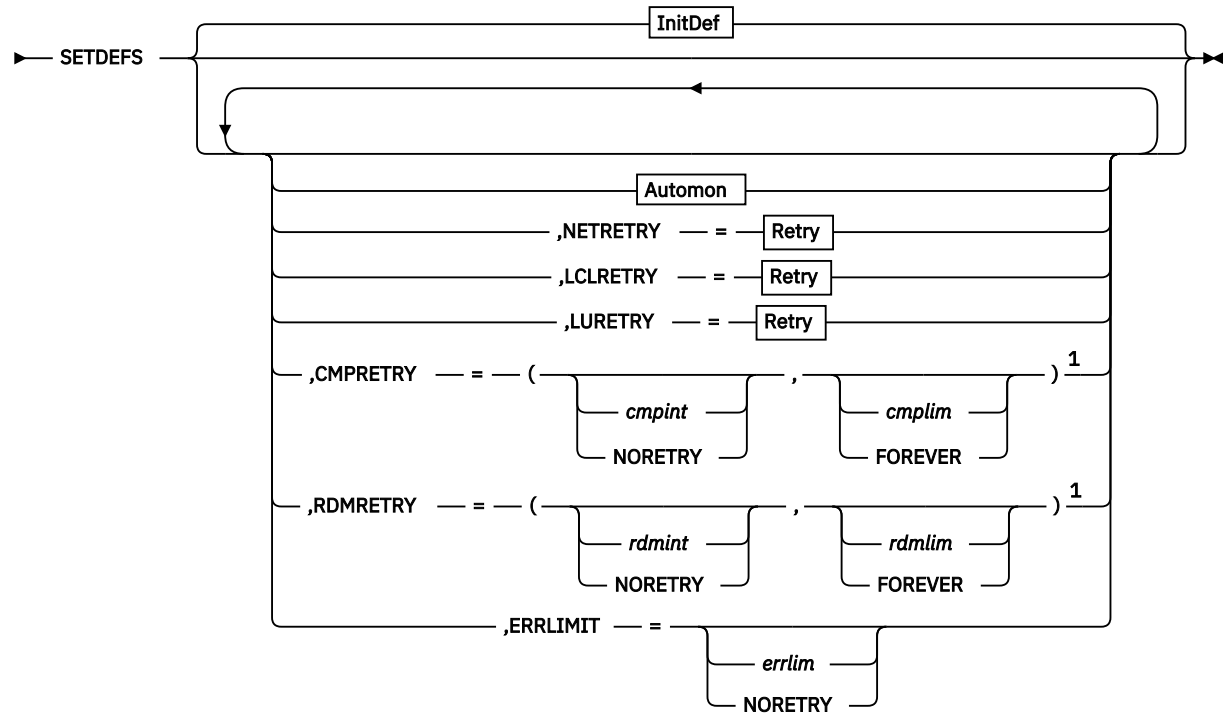
```
TOPOSNA REFRESH,ALLTABLS,CLASS=(port)
```

TOPOSNA SETDEFS (TOPOSNA)

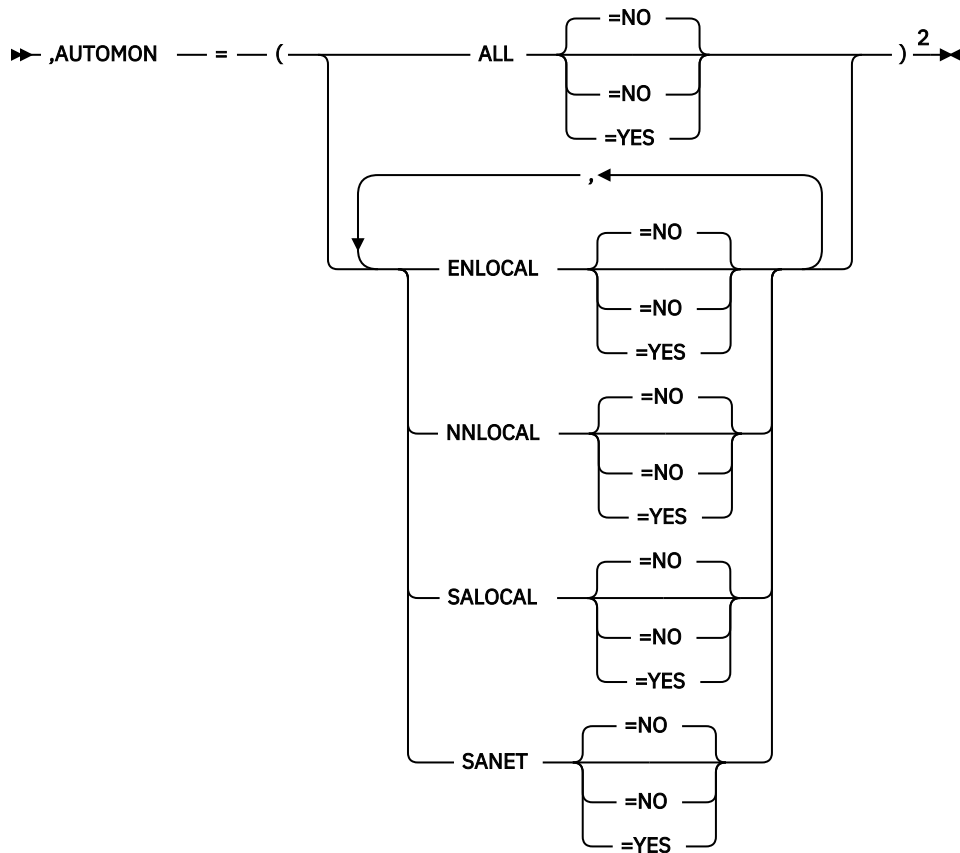
Syntax

TOPOSNA SETDEFS

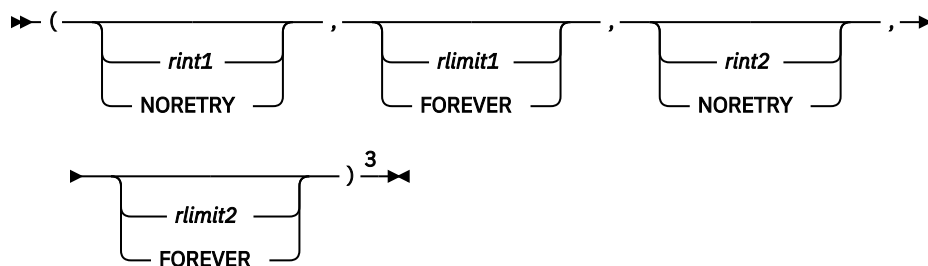
➤ TOPOSNA ➤



Automon



Retry



InitDef

```

➤ ,AUTOMON=(ALL=NO) — ,NETRETRY=(60,5,1800,48) — ,LCLRETRY=(60,5,1800,48) ➔
➤ ,LURETRY=(60,5,1800,48) — ,CMPRETRY=(30,1000) — ,RDMRETRY=(30,1000) ➔
➤ ,ERRLIMIT=0 ➤

```

Notes:

- ¹ At least one parameter must be specified, but a trailing comma (30,) is not allowed when the second value is omitted.
- ² The parentheses are optional.
- ³ At least one parameter must be specified, but trailing commas (,5,,) are not allowed when omitting values.

Purpose of Command

The TOPOSNA SETDEFS command modifies the defaults for the automatic monitoring of local and network topology at newly-discovered nodes, for reconnection to RODM and CMIP Services, and for the retry policy of other TOPOSNA commands.

Operand Descriptions

SETDEFS

Specifies that the policy values used by the topology manager be modified. When you change an initial default value (shown in the table below) with a SETDEFS request, the new value becomes the default. Any value that you do not specify on the new SETDEFS request takes its value from the last SETDEFS that changed it, or by using the initial default.

The topology manager default values are saved in the RODM data cache and remain in effect across warm and cold starts of the topology manager. Issue a SETDEFS request with no parameters or restart RODM to reset all defaults to the initial default values. The values in effect when the last STOPMGR request was issued or when the last RODM checkpoint was taken (if you have loaded RODM checkpoint data) are used when the topology manager is restarted. The parameter descriptions follow the table.

Table 7. Initial Settings for Defaults Set by Topology SETDEFS

Keyword	Parameter	Initial Default
AUTOMON	NNLOCAL	NO
NETRETRY	<i>rint1</i>	60
LCLRETRY	<i>rint1</i>	60
LURETRY	<i>rint1</i>	60
CMPRETRY	<i>cmpint</i>	30
RDMRETRY	<i>rdmint</i>	30
ERRLIMIT	<i>errlim</i>	0

AUTOMON

Performs either of the following:

- Specifies the automatic network topology monitoring action when the topology manager discovers a new t5Node, interchangeNode, migrationDataHost, or VTAM nodes represented by an active crossDomainResourceManager when the network ID is reported by an agent.
- Specifies the automatic local topology monitoring action when the topology manager discovers a new appnNN, appnEN, t5Node, interchangeNode, migrationDataHost, or a VTAM node represented by an active crossDomainResourceManager when the network ID is reported by an agent.

Note: AUTOMON does not apply to monitoring of LU collections.

A newly-discovered node is:

- A node reported for the first time after a cold start (and is thus any node, including the node named in a network monitoring command)
- A node that is reported after being purged from the RODM data cache by a TOPOSNA PURGE command
- A network node that was connected for the first time to a sub-network whose network topology is being monitored
- A node that was connected for the first time to a node whose local topology is being monitored

Valid parameters are:

ALL

Performs either of the following:

- Specifies network topology monitoring for all newly-discovered t5Node, interchangeNode, migrationDataHost, or VTAM nodes represented by an active crossDomainResourceManager when the network ID is reported by an agent.

- Specifies local topology monitoring for all newly-discovered appnNN, appnEN, t5Node, interchangeNode, migrationDataHost, or a VTAM node represented by an active crossDomainResourceManager when the network ID is reported by an agent.

Note: Automatic monitoring will be sent only to nodes whose network IDs are in the list of network IDs in the FLBSYSD initialization file (in the NETID_LIST category, using the keyword SNA_NETID).

Also, if automatic monitoring is set to YES, and if the first entry in the list of network IDs in the FLBSYSD initialization file is null, no automatic monitoring of topology will occur and message FLB464I is issued to the NetView log. For more information about FLBSYSD, see the *IBM Z NetView SNA Topology Manager Implementation Guide*.

ENLOCAL

Specifies local topology monitoring for newly-discovered end nodes.

NNLOCAL

Specifies local topology monitoring for newly-discovered network nodes.

SALocal

Specifies local topology monitoring for the following newly-discovered subarea resources:

- t5Nodes,
- interchangeNodes
- migrationDataHosts
- VTAM nodes represented by an active crossDomainResourceManager when the network ID is reported by an agent

SANET

Specifies network topology monitoring for the following newly-discovered subarea resources:

- t5Nodes,
- interchangeNodes
- migrationDataHosts
- VTAM nodes represented by an active crossDomainResourceManager when the network ID is reported by an agent

Valid values for each of the above parameters are:

YES

Start monitoring for the specified types of newly-discovered nodes.

NO

Do not start monitoring for the specified types of newly-discovered nodes. This is the default.

NETRETRY

Sets the retry limits and intervals to use when attempting to initiate a new network topology MONITOR operation. These parameters modify the sub-values of NETRETRY. Only the specified sub-values are changed.

Note that LCLRETRY and NETRETRY *each* have a separate set of retry values. Unrecoverable errors are not retried.

After the failure of an active monitor, the topology manager immediately retries the monitor once (except for unrecoverable errors), no matter what retry values are specified. The retry values are used to control what happens next. A message is issued when the active monitor fails. A second message is issued when retrying using the retry values begins. No further messages are issued until all retries fail or the transaction succeeds. No transactions except for monitoring are retried.

Retrying continues in most cases even if there is no agent active; the manager assumes that the agent will eventually become active. There are two sets of retry values: set 1 (*rint1* and *rlimit1*) and set 2 (*rint2* and *rlimit2*). Although there are no fundamental differences between the two sets, set 1 is used first. Think of a value from set 1 as an immediate retry value and a value from set 2 as a long-term retry value.

The time intervals between immediate retries are usually short (for example, 60 seconds). Immediate retries attempt to re-establish the transaction while minimizing the disruption to normal operation.

Long-term retries are used after the immediate retries have been exhausted. The time intervals between long-term retries are typically longer (for example, 1800 seconds). Long-term retries attempt to re-establish the transaction while minimizing the retry overhead.

You can specify 1 - 4 values for controlling retries, but trailing commas (5,3,,) are not allowed. Specifying (,,5,3) leaves the immediate retry values unchanged; all commas are required in this example because they hold positions.

rint1

The interval in seconds between immediate retries. A valid value is a number in the range of 0–3600 or NORETRY, which specifies that the failed MONITOR operation is not immediately retried.

rlimit1

The number of immediate retries. Specifies the initial number of times that the topology manager will retry a MONITOR operation. A valid value is a number in the range of 0–2147483647 (2**31-1), or FOREVER, which specifies that there is no limit to the number of immediate retries.

rint2

The interval in seconds between secondary retries. Secondary retries are attempted only after the topology manager has exhausted the immediate limit. A valid value is a number in the range of 0–86400 or NORETRY, which specifies that the failed MONITOR operation is not retried after the initial retries are exhausted.

rlimit2

The number of secondary retries. Specifies the initial number of times that the topology manager will retry a MONITOR operation after exhausting the immediate retry limit. A valid value is a number in the range of 0–2147483647 (2**31-1) or FOREVER, which specifies that there is no limit to the number of immediate retries.

LCLRETRY

Sets the retry limits and intervals to use when attempting to initiate a new local topology MONITOR operation. These parameters modify the sub-values of LOCAL-RETRY. The same four values (*rint1*, *rlimit1*, *rint2*, and *rlimit2*) as in NETRETRY are used, with the same rules and restrictions.

LURETRY

Sets the retry limits and intervals to use when attempting to initiate a new LU collection MONITOR operation. These parameters modify the sub-values of LU COLLECTION RETRY. The same four values (*rint1*, *rlimit1*, *rint2*, and *rlimit2*) as in NETRETRY are used, with the same rules and restrictions.

CMPRETRY

Sets the retry limits and intervals that are used when the SNA topology manager cannot connect to CMIP Services during reinitialization processing. These values determine how often to attempt to connect to CMIP Services and when to end the SNA topology manager if the connect attempts fail.

The CMPRETRY values are only used after a successful SNA topology manager initialization and connection to VTAM CMIP services. The CMIP_RETRY_INTERVAL and CMIP_RETRY_LIMIT keywords in the FLBSYSD initialization file specify the retry values used during SNA topology manager initialization.

You can specify 1 or 2 values for controlling retries, but a trailing comma (30,) is not allowed.

cmpint

The interval in seconds between CMIP Services connect retries. A valid value is a number in the range of 0–86400 or NORETRY, which specifies that the failed CMIP Services connect is not retried. When 0 (zero) or NORETRY is specified and the connection to CMIP services is lost, the SNA topology manager ends without trying to reestablish the connection. The shipped default value is 30.

cmplim

The number of times the SNA topology manager will attempt to connect to CMIP Services before ending. A valid value is a number in the range of 0–2147483647 (2**31-1) or FOREVER, which specifies that there is no limit to the number of retries. The shipped default value is 1000.

RDMRETRY

Sets the retry limits and intervals that are used when the SNA topology manager cannot connect to RODM during reinitialization processing. These values determine how often to attempt to connect to RODM and when to end the SNA topology manager if the RODM connect attempts fail.

The RDMRETRY values are only used after a successful SNA topology manager initialization and connection to RODM. The RODM_RETRY_INTERVAL and RODM_RETRY_LIMIT keywords in the FLBSYSD initialization file specify the retry values used during SNA topology manager initialization.

You can specify 1 or 2 values for controlling retries, but a trailing comma (30,) is not allowed.

rdmint

The interval in seconds between RODM connect retries. A valid value is a number in the range of 0–86400 or NORETRY, which specifies that the failed RODM connect is not retried. When 0 (zero) or NORETRY is specified and the RODM connection is lost, the SNA topology manager ends without trying to re-establish the connection. The shipped default value is 30.

rdmlim

The number of times the SNA topology manager will attempt connect to RODM before ending. A valid value is a number in the range of 0–2147483647 ($2^{31}-1$) or FOREVER, which specifies that there is no limit to the number of retries. The shipped default is 1000.

ERRLIMIT

Sets the error-retry limit for SNA topology manager monitor commands when there is a RODM request failure or internal failure. The command will be retried the number of times specified by the ERRLIMIT value. A valid value is a number in the range of 0–2147483647 ($2^{31}-1$) or NORETRY. When NORETRY or 0 (zero) is specified, the command will not be retried. The default value is 0.

Restrictions

Before issuing a SETDEFS request, consider these restrictions:

- When automatic monitoring is specified on the AUTOMON parameter using a value of YES for the ALL, ENLOCAL, NNLOCAL, SALocal, or SANET keyword, the network ID of the node must be present in the FLBSYSD initialization file. This network ID is specified in the NETID_LIST category using the SNA_NETID keyword. If the NETID_LIST category in the FLBSYSD file is null, no automatic monitoring of topology will occur regardless of the setting of AUTOMON.
- Values that you specify with SETDEFS become the new defaults, overwriting the existing defaults. SETDEFS values that you do not specify are unchanged and retained. Review the examples carefully.
- The initial default for the ENLOCAL, NNLOCAL, SALocal, and SANET parameters is NO. This is opposite to the parameter default (YES) that is used, for example, if you specify AUTOMON=ENLOCAL (without an =YES or =NO).
- Do not set ENLOCAL=YES unless you have agents on a significant percentage of the end nodes in the network. Local monitoring for any newly-discovered node that does not have the topology agent installed and active is retried according to the TOPOSNA SETDEFS LCLRETRY parameter values. Error messages are logged for each failed attempt. Setting AUTOMON=YES when you have only a few topology agents can generate a flood of error log entries.
- This command cannot be issued until you receive the following message:

```
FLB440I SNA TOPOLOGY MANAGER INITIALIZATION IS COMPLETE.
```

- Specify FOREVER as a retry limit cautiously. If you specify a numeric retry interval and FOREVER as a retry limit and if the retrying never succeeds, the operator only receives an initial message indicating the transaction is being retried. The operator will never receive messages FLB462E or FLB463E showing that the transaction failed, indicating a possible problem of obsolete or non-existent node names.

Example: Resetting defaults to initial values

A representative subset of possible response messages is shown for the examples.

To reset SNA topology manager defaults to their initial values, enter:

```
TOPOSNA SETDEFS
```

Response

```
FLB490I SNA TOPOLOGY MANAGER STORED DEFAULT VALUES IN RODM  
FLB411I TOPOSNA SETDEFS COMMAND COMPLETED SUCCESSFULLY
```

Example: Setting end node monitoring explicitly

To request end node monitoring for every newly-discovered end node, to retry all failed network topology transactions once a minute for the first five minutes, and then every five minutes thereafter, enter:

```
TOPOSNA SETDEFS,AUTOMON=(ENLOCAL=YES),NETRETRY=(60,FOREVER),  
LCLRETRY=(60,5,300)
```

Example: No default monitoring

To specify that there is to be no automatic monitoring of local topology or network topology, to retry all failed network topology transactions every 30 seconds, and not to retry failed local topology transactions, enter:

```
TOPOSNA SETDEFS,AUTOMON=(ALL=NO),NETRETRY=(30),LCLRETRY=(NORETRY)
```

Example: Setting monitoring using default parameter values

To set the startup defaults to turn on local topology for every newly-discovered end node and network node, to retry all failed network topology transactions once a minute for five minutes, and then retry every five minutes, enter:

```
TOPOSNA SETDEFS,AUTOMON=(ENLOCAL,NNLOCAL),NETRETRY=(60,5,300,FOREVER),  
LCLRETRY=(60,5,300,5),LURETRY=(60,5,300,5)
```

Example: Explicit and default parameter values for automatic monitoring

To turn on local topology monitoring for every newly-discovered end node, turn off local topology monitoring for every newly-discovered network node and VTAM subarea node, turn off network topology monitoring for every newly-discovered VTAM subarea node, and explicitly set all retry values, enter:

```
TOPOSNA SETDEFS,AUTOMON=(ENLOCAL,NNLOCAL=NO,SALocal=NO,SANET=NO),  
NETRETRY=(60,5,300,FOREVER),LCLRETRY=(60,5,300,5),  
LURETRY=(60,5,300,5)
```

Example: Setting network retrying off

To specify that network retrying is off, and that local retrying will skip the first set of retry values and then try every 300 seconds for 5 times, enter:

```
TOPOSNA SETDEFS,NETRETRY=(NORETRY,,NORETRY),LCLRETRY=(NORETRY,60,300,5)
```

Example: Setting command error limits

To set the SNA topology manager command error-retry limit to 3, enter:

```
TOPOSNA SETDEFS,ERRLIMIT=3
```

Example: Setting RODM connect retry and interval limits

To set the interval between failed RODM connect attempts to 40 seconds and to retry connects to RODM forever, enter:

```
TOPOSNA SETDEFS,RDMRETRY=(40,FOREVER)
```

Example: Setting CMIP services connect retry and interval limits

To set the interval between failed CMIP Services connect attempts to 20 seconds and to set the number of retries to 1600, enter:

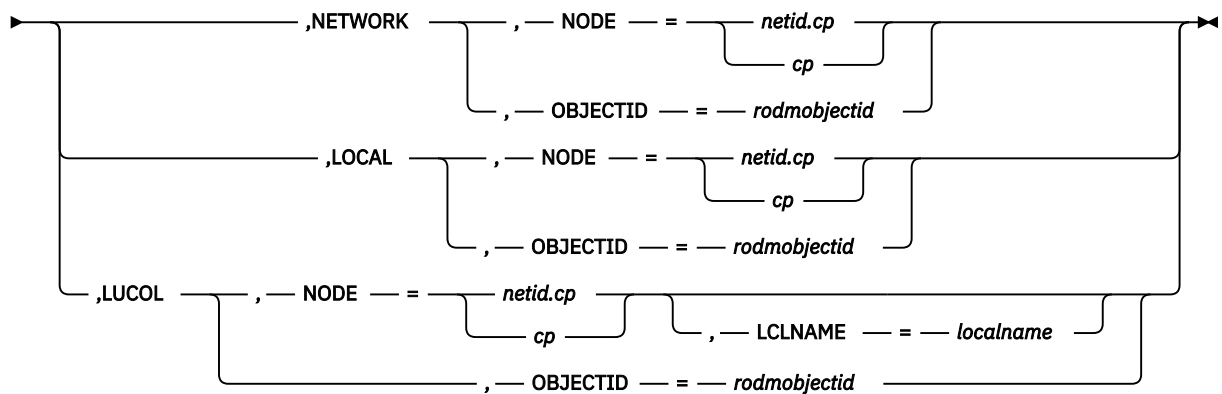
```
TOPOSNA SETDEFS,CMPRETRY=(20,1600)
```

TOPOSNA STOP (TOPOSNA)

Syntax

TOPOSNA STOP

➤ TOPOSNA — STOP ➔



Purpose of Command

TOPOSNA STOP stops the monitoring of local or network topology of an agent node, or of the collection of all LUs (including logical units, cross-domain resources, and LU groups) associated with the specified node.

A STOP request cancels both operator-issued monitor requests and monitor requests that are automatically issued during warm start of the topology manager.

Operand Descriptions

STOP

Specifies that the topology manager stop monitoring the topology of an agent.

Data already received is processed by the topology manager; later data is ignored. Because of the STOP, all affected resources have a status of unknown (default color=gray).

NETWORK

Specifies network topology.

LOCAL

Specifies local topology.

LUCOL

Specifies that monitoring of the collection of all LUs associated with the specified node be stopped.

NODE

Specifies the node, identified by network name, at which monitoring is to stop.

netid.cp

The network qualified name of the control point of the agent node, where *netid* is the network identifier and *cp* is the control point name.

The *netid* and the *cp* name each have an 8-character maximum length; no blanks are included and the period between them is required.

Authorization checking of *netid* and *cp* is done separately; the 8-character authorization-checking limit is therefore not a limitation on checking the node name.

cp

The control point name of the agent node. The name consists solely of the control point name, no network identifier or period.

OBJECTID=rodmbobjectid

The 8-byte identifier for the object in RODM representing the node or logical link. The format is the hexadecimal representation of the 8-byte object identifier in the form: 16 hexadecimal digits typed as characters (0 - 9, A-F). This parameter is intended to be used primarily by programs or command lists that have access to the RODM data cache, rather than by operators. Authorization checking of *rodmbobjectid* is done only on the first 8 characters.

LCLNAME=localname

The local name of the logical link for which monitoring is stopped, and only applies to the LUCOL operand.

Restrictions

Before issuing a STOP request, consider these restrictions:

- A single command cannot stop all monitoring at all nodes and of all types. Individual nodes and types of monitoring (local, network, LU collection) must be specified.
- Monitor requests do not record which operator issued them. If operator A and operator B both request the same monitor (the same type of monitoring at the same node), there is no problem. However, if either operator issues a STOP request for that monitor, monitoring stops for all operators.
- This command cannot be issued until you receive the following message:

```
FLB440I SNA TOPOLOGY MANAGER INITIALIZATION IS COMPLETE.
```

Example: Stopping network monitoring

Note: Only a representative subset of possible response messages is shown for the examples.

To stop network monitoring at APPNNET1.NODE1, enter:

```
TOPOSNA STOP,NETWORK,NODE=APPNNET1.NODE1
```

Response

```
FLB405W OPERATOR 'OPER1' STOPPED MONITORING SNA NETWORK
        TOPOLOGY FROM NODE APPNNET1.NODE1
FLB411I TOPOSNA STOP COMMAND COMPLETED SUCCESSFULLY
```

Operators who started network monitoring at node APPNNET1.NODE1 receive message FLB405W, which identifies the operator ID that stopped the network monitoring. The operator issuing the stop request receives message FLB411I.

Example: Stopping the monitoring of an LU collection

To stop monitoring of an LU collection by node NTCBMVS, enter:

```
TOPOSNA STOP,LUCOL,NODE=NTCBMVS
```

Response:

```
FLB541I OPERATOR NETOP1 STOPPED MONITORING LU COLLECTION FROM  
        SNANETA.NTCBMVS  
FLB411I TOPOSNA STOP COMMAND COMPLETED SUCCESSFULLY
```

Any operators who had started monitoring of LU collection by node NTCBMVS receive message FLB541I, which identifies the operator ID that stopped the monitoring. The operator who issued the stop request received message FLB411I.

TOPOSNA STOPMGR (TOPOSNA)

Syntax**TOPOSNA STOPMGR**

➤ TOPOSNA — STOPMGR ➤

Purpose of Command

TOPOSNA STOPMGR stops the topology manager task in an orderly fashion.

Operand Descriptions**STOPMGR**

Specifies that the topology manager task shuts down in an orderly fashion, by stopping pending monitor requests, finishing updates to the RODM data cache, then logging off the autotask. This command can be issued any time the topology manager has been started, even if initialization is not complete. To restart the topology manager, issue AUTOTASK OPID=FLBTOPO at the command line. Any continuous monitors that are active when the STOPMGR request is issued are restarted if the topology manager is warm started.

Restrictions

If possible, avoid deleting the topology manager autotask. Use TOPOSNA STOPMGR instead. Deleting the autotask while an RODM update is in progress can leave the topology data in the RODM data cache corrupted and unusable. If the STOPMGR request fails, try to stop all other active monitoring; then delete the autotask. If deleting the autotask is necessary and your data is corrupted, restart the topology manager.

Example: Stopping the topology manager

To stop the SNA topology manager, enter:

```
TOPOSNA STOPMGR
```

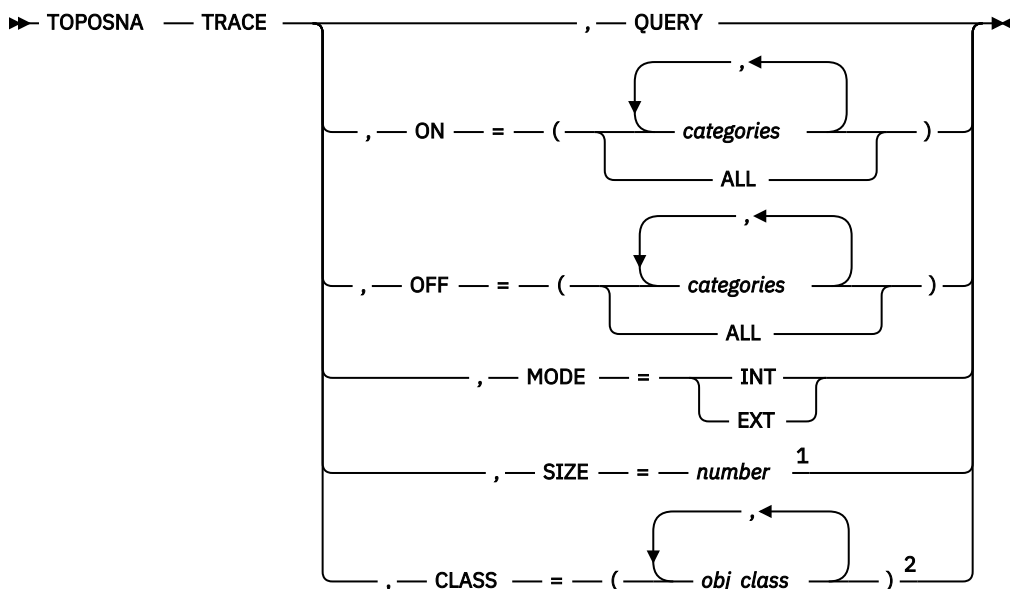
Response

```
FLB441I SNA TOPOLOGY MANAGER IS SHUTTING DOWN NORMALLY  
FLB443I SNA TOPOLOGY MANAGER SHUTDOWN IS COMPLETE  
FLB610I TASK FLBTOPO IS STARTING LOGOFF PROCESSING  
FLB611I TASK FLBTOPO HAS COMPLETED ITS LOGOFF PROCESSING
```

TOPOSNA TRACE (TOPOSNA)

Syntax

TOPOSNA TRACE



Notes:

¹ Parameter is not valid if MODE is EXT.

² Specifies the object classes to trace for RODM, SIGNALS, and UPDATE trace categories.

Purpose of Command

TOPOSNA TRACE starts, stops, or lists tracing in the topology manager.

Operand Descriptions

TRACE

Requests tracing changes or information for the topology manager. Trace records can be written to the generalized trace facility (GTF) or to an internal buffer that wraps trace data when filled to capacity. The effects of multiple TRACE requests are cumulative. Therefore, if a TOPOSNA TRACE command is issued when tracing is already active, the trace categories that are not specified remain unchanged. See the *IBM Z NetView Troubleshooting Guide* to decide which traces to use and how to interpret trace data.

QUERY

Lists all trace categories and object classes indicating for each whether it is turned on or off.

ON

Specifies the trace categories to be turned on. You can specify ALL to trace all categories of topology manager events, or you can specify one or more trace *categories*.

The trace *categories* are:

CMIP

Traces interactions between the SNA topology manager and CMIP services. This trace shows, for example, the CMIP data sent to and received from the agent nodes.

FSM

Traces finite state machine events.

LOG

Traces messages FLB600E, FLB601W, and FLB602I written to the NetView log by the topology manager.

MESSAGES

Traces messages issued by the SNA topology manager. These messages are also written to the NetView log. Placing them in the trace is sometimes useful when the sequence of events is important.

RODM

Traces interactions between the topology manager and the RODM data cache. This trace shows what RODM objects the topology manager created, referred to, updated, and deleted. See the CLASS keyword for added granularity of the RODM category trace information.

RODMDUMP

Traces interactions between the topology manager and the RODM data cache, but produces more information than when specifying the RODM trace category. Information, such as field values, is included.

RSM

Traces interactions between the SNA topology manager and the resource status manager. This trace shows requests to RSM to start or stop forwarding status updates for a resource.

SIGNALS

Traces the signals exchanged between the components within the topology manager. This category generates large amounts of trace information and is used primarily by IBM Software Support representatives when analyzing a problem. See the CLASS keyword for added granularity of the SIGNALS category trace information.

STORAGE

Traces SNA topology manager heap and internal storage pool information.

UPDATE

Traces resource updates. See the CLASS keyword for added granularity of the UPDATE category trace information.

OFF

Specifies the trace categories to be turned off. One or more *categories* or ALL can be specified. See the examples for this command for an effective way to combine ON and OFF in one TOPOSNA TRACE command.

MODE

With this optional keyword, you can select whether you want internal tracing or external tracing using the following values:

INT

Trace the data to an internal trace buffer.

EXT

Trace the data to the generalized trace facility (GTF).

The default is INT,SIZE=10.

SIZE

This optional keyword, which is closely associated with MODE=INT, specifies the size in pages of the internal trace buffer. MODE=INT is not explicitly required to be issued with the SIZE keyword. However, specifying SIZE=*number* without MODE=INT is only valid if the current MODE is INT.

The variable *number* is the number of pages (4096 bytes) of memory you want for the internal trace buffer. The allowable range is 10-999 pages with a default value of 10 pages. The SIZE keyword value cannot be set to a value that is less than 10, including 0 (zero).

CLASS

The CLASS keyword controls the object classes that are traced for the RODM, SIGNALS, and UPDATE trace categories. To add trace granularity for each of these categories, specify one or more of the following values for the variable *obj_class*.

CACHE

Cache manager object

COMMAND

Command objects and action handlers

DEFGROUP

Definition group object processing

GRAPH

Graph objects and View Manager

LINK

Action handler and cache link objects

LU

Action handler and cache LU objects

MISC

Miscellaneous objects

NODE

Action handler and cache node objects

NODETABL

The node table object

PORT

Action handler and cache port objects

RESOURCE

Synonym for DEFGRP, LINK, LU, NODE, PORT, and TG.

RODM

The RODM manager object

STACK

The stack manager and supportive objects

STATHIST

The Status history object

TG

Action handler and cache TG and TG Circuit objects

The eligible *obj_class* values for each trace category is displayed in the following table.

<i>Table 8. Trace Categories and obj_class Values</i>			
obj_class Value	RODM	SIGNALS	UPDATE
COMMAND		X	
DEFGROUP		X	
GRAPH	X	X	X
LINK		X	
LU	X	X	X
MISC	X	X	X
NODE	X	X	
NODETABL	X	X	X
PORT		X	
RESOURCE	X	X	X
RODM	X	X	X

<i>Table 8. Trace Categories and obj_class Values (continued)</i>			
obj_class Value	RODM	SIGNALS	UPDATE
STACK		X	
STATHIST		X	
TG		X	
X	X	X	

Do not specify the CLASS keyword, if you want to trace all the eligible object classes for the RODM, SIGNALS, or UPDATE trace category.

Restrictions

Before issuing a TRACE request, consider these restrictions:

- TRACE is the only TOPOSNA request you can issue before the topology manager task is started. Starting a trace before starting the topology manager task can help you diagnose problems in initialization. However, because tracing is independent of the topology manager task, stopping the topology manager does not stop tracing. If tracing is on when you stop the task, tracing resumes (with the same trace categories) when you start the task again.
- Before issuing a TRACE,MODE=EXT request, start the GTF and enable trace event X'5E8'.
- When tracing externally (MODE=EXT), stopping GTF stops logging of subsequent trace data, but tracing resumes with the same categories when GTF is started.
- The same trace category cannot be turned on and off in the same command. The ON and OFF parameters can be used once in each command.
- Specifying the CLASS keyword without the OFF or ON keyword, generates message FLB517I and no changes are made.
- The SIZE keyword is only valid when specified with MODE=INT or when the current trace mode is internal (INT). If the trace mode is external (EXT), message FLB525I is issued and the SIZE keyword is ignored.
- Using traces usually degrades performance, depending on which categories are enabled and the capacity of your system.

Example: Setting internal trace with 200 page buffer

A representative subset of possible response messages is shown for the examples.

To set the trace mode to internal with a 200 page internal trace buffer, enter:

```
TOPOSNA TRACE,MODE=INT,SIZE=200
```

Response

```
FLB532I SNA TOPOLOGY MANAGER INTERNAL TRACE TABLE SIZE CHANGED TO 200
      PAGES
FLB411I TOPOSNA TRACE COMMAND COMPLETED SUCCESSFULLY
```

Example: Tracing all categories except one

To start tracing for all categories except RODM, enter:

```
TOPOSNA TRACE,ON=(ALL),OFF=(RODM)
```

Response

```
FLB411I TOPOSNA TRACE COMMAND COMPLETED SUCCESSFULLY
```

Example: Trace categories are cumulative

To start tracing for all categories except RODM, and then stop tracing miscellaneous objects and resource objects (DEFGRP, LINK, LU, NODE, PORT, and TG) in the SIGNALS category, enter:

```
TOPOSNA TRACE,ON=(ALL),OFF=(RODM)
TOPOSNA TRACE,OFF=(SIGNALS),CLASS=(MISC,RES)
```

The trace commands are cumulative, all traces except RODM and the specific object classes of SIGNALS are turned on. To verify this, see the following example.

Example: Querying current trace settings

To query the status of the categories and classes being traced, enter:

```
TOPOSNA TRACE,QUERY
```

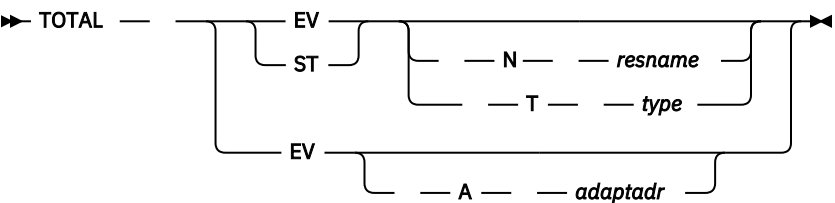
Response

```
FLB505I SNA TOPOLOGY MANAGER TRACE MODE IS INTERNAL
FLB506I SNA TOPOLOGY MANAGER INTERNAL TRACE BUFFER SIZE IS 200
PAGES
FLB516I -----
FLB509I SNA TOPOLOGY MANAGER TRACE CATEGORY SIGNALS IS ON
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS CACHE IS ON
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS COMMAND IS ON
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS DEFGRP IS OFF
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS GRAPH IS ON
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS LINK IS OFF
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS LU IS OFF
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS MISC IS OFF
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS NODE IS OFF
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS NODETABL IS ON
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS PORT IS OFF
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS RODM IS ON
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS STACK IS ON
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS STATHIST IS ON
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS TG IS OFF
FLB516I -----
FLB509I SNA TOPOLOGY MANAGER TRACE CATEGORY UPDATE IS ON
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS DEFGRP IS ON
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS LINK IS ON
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS LU IS ON
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS NODE IS ON
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS PORT IS ON
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS TG IS ON
FLB516I -----
FLB509I SNA TOPOLOGY MANAGER TRACE CATEGORY RODM IS OFF
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS DEFGRP IS OFF
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS LINK IS OFF
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS LU IS OFF
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS MISC IS OFF
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS NODE IS OFF
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS PORT IS OFF
FLB508I SNA TOPOLOGY MANAGER TRACE CLASS TG IS OFF
FLB516I -----
FLB509I SNA TOPOLOGY MANAGER TRACE CATEGORY CMIP IS ON
FLB509I SNA TOPOLOGY MANAGER TRACE CATEGORY FSM IS ON
FLB509I SNA TOPOLOGY MANAGER TRACE CATEGORY LOG IS ON
FLB509I SNA TOPOLOGY MANAGER TRACE CATEGORY MESSAGES IS ON
FLB509I SNA TOPOLOGY MANAGER TRACE CATEGORY RODMDUMP IS ON
FLB509I SNA TOPOLOGY MANAGER TRACE CATEGORY RSM IS ON
FLB509I SNA TOPOLOGY MANAGER TRACE CATEGORY STORAGE IS ON
FLB411I TOPOSNA TRACE COMMAND COMPLETED SUCCESSFULLY
```

TOTAL (NPDA)

Syntax

TOTAL



IBM-Defined Synonyms

Command or Operand	Synonym
TOTAL	TOT

Purpose of Command

The TOTAL command displays the count of event or statistical records for a specified resource or resource type. If you do not specify a resource name or type, a summary is displayed.

Operand Descriptions

EV

Specifies event records.

ST

Specifies statistical records.

N

Identifies the operand that follows as a resource name.

resname

Specifies the symbolic name of the resource. You can specify up to five resource names to fully qualify the resource for which data is to be displayed.

T

Identifies the operand that follows as a resource type.

type

Specifies resource type.

A

Identifies the operand that follows as an adapter address.

adaptadr

Specifies the 12-hexadecimal-digit adapter address. The A (adapter) address is not a valid option for a resource type of CBUS.

Restrictions

The following restrictions apply to the TOTAL command:

- If the name of the resource is not associated with a unique resource configuration on the database, a selection panel is displayed on which you can choose which configuration is relevant.

Example: Displaying total statistical records for a unit

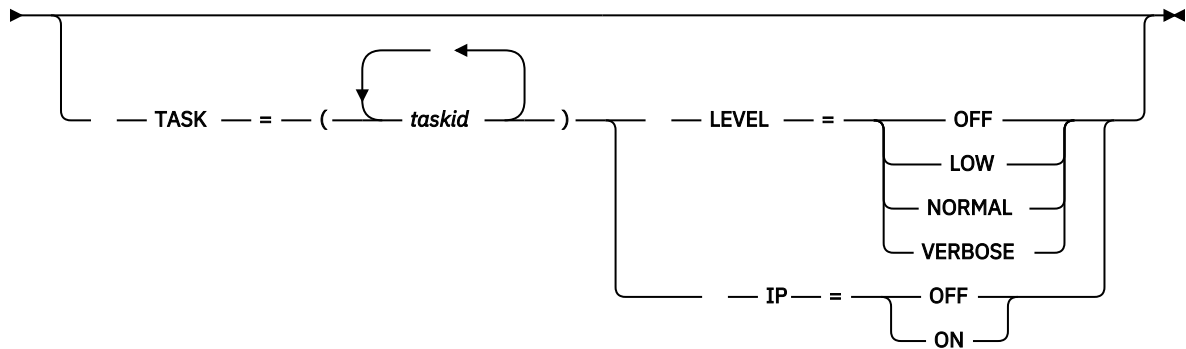
To display the total statistical records for UNIT1 enter:

TRACE (EAS)

Syntax

EAS TRACE

➤ MODIFY — — *procname* — , — TRACE ➔



IBM-Defined Synonyms

Command or Operand

Synonym

MODIFY

F

Purpose of Command

The TRACE command controls the level and content of the tracing performed by Event/Automation Service tasks.

You can enter the TRACE command without any operands to display the current Event/Automation Service trace parameters.

Operand Descriptions

procname

Specifies the Event/Automation Service job name.

TASK=taskid

Specifies the service tasks to be traced. If you specify TASK=taskid, you must also specify either the LEVEL parameter or the IP parameter. The *taskid* can have the following values:

ALERTA

The alert adapter service task

ALERTC

The confirmed alert adapter service task

MESSAGEA

The message adapter service task

MESSAGEC

The confirmed message adapter service task

EVENTRCV

The event receiver service task

TRAPALRT

The trap to alert conversion task

ALRTTRAP

Converts alerts to traps and sends them to SNMP

ALL

All service tasks

LEVEL=level

Specifies the Event/Automation Service data to be traced. LEVEL can have the following values:

OFF

Disable tracing for the task.

LOW

The lowest level of detail. Typically, this specification traces entry/exit functions.

NORMAL

An intermediate level of detail. Typically, this specifies the LOW level and additional event flow information.

VERBOSE

The highest level of detail. Typically, this specifies the NORMAL level and hexadecimal control block information.

IP=ON/OFF

Specifies whether the tracing of IP connection data is enabled or disabled.

Example: Tracing a service task

To trace the message adapter task for the Event/Automation Service job named IHSAEVNT at a VERBOSE tracing level, enter:

```
F IHSAEVNT,TRACE,TASK=MESSAGEA,LEVEL=VERBOSE
```

Response

You receive the following response:

```
IHS0076I TASK=MESSAGEA LEVEL=VERBOSE IP=OFF
```

Example: Displaying trace settings

To display the current trace settings for the Event/Automation Service job named IHSAEVNT, enter:

```
F IHSAEVNT,TRACE
```

Response

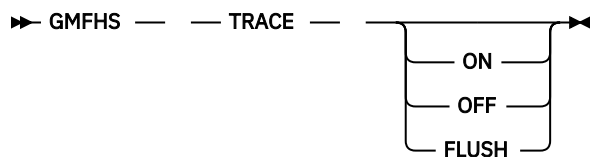
You receive a response similar to the following:

```
IHS0073I Current TRACE settings:
IHS0076I TASK=CONTROL LEVEL=OFF IP=OFF
IHS0076I TASK=ALERTA LEVEL=OFF IP=OFF
IHS0076I TASK=MESSAGEA LEVEL=OFF IP=OFF
IHS0076I TASK=EVENTRCV LEVEL=OFF IP=OFF
IHS0076I TASK=TRAPALRT LEVEL=OFF IP=OFF
IHS0076I TASK=ALRTTRAP LEVEL=OFF IP=OFF
IHS0076I TASK=MESSAGEC LEVEL=OFF IP=OFF
IHS0076I TASK=ALERTC LEVEL=OFF IP=OFF
```

TRACE (GMFHS)

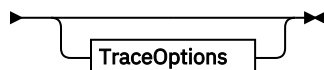
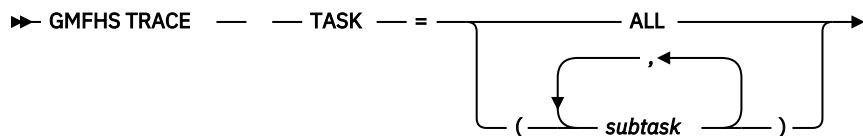
Syntax

To start or stop GMFHS tracing, display trace settings, or flush the GMFHS in-storage trace table:

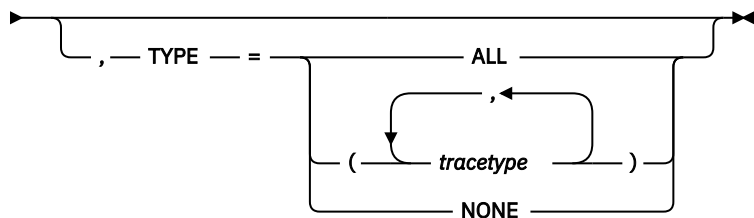
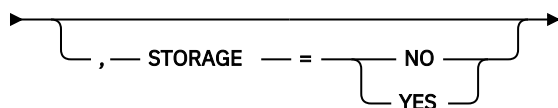
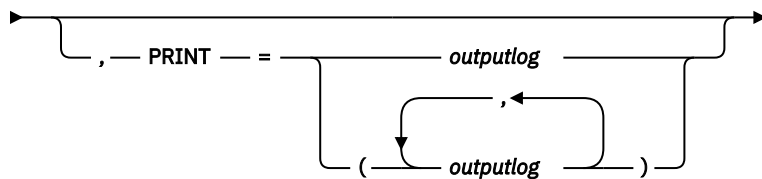
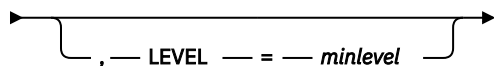
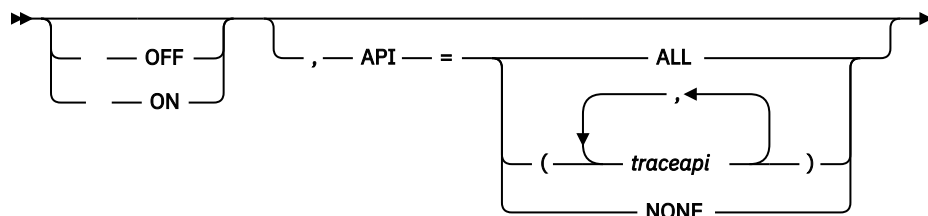


To set GMFHS trace options before or during tracing:

GMFHS TRACE



TraceOptions



Purpose of Command

The NetView GMFHS TRACE command controls the level and content of the tracing performed by GMFHS tasks.

You can enter the GMFHS TRACE command without any operands to display the current GMFHS trace parameters.

You can enter the GMFHS TRACE command from the MVS console using the MVS MODIFY command or from a NetView terminal using the GMFHS command list.

Operand Descriptions

ON

If you do not specify any other parameters, tracing is activated only for those tasks for which tracing has been specified. When used with the TASK keyword, indicates that the specified tasks are to be traced if tracing is active.

OFF

If you do not specify any other parameters, GMFHS stops all tracing. When used with the TASK keyword, indicates that the specified tasks are not to be traced if tracing is active.

Note: Specifying OFF without any other parameters does not affect the trace settings of individual GMFHS components.

TASK

Indicates the host subsystem task or tasks for which you want to change the trace options. TASK can be ALL or you can specify a *subtask*. If no other trace options are specified, the trace settings for the specified tasks are displayed.

ALL

Specifies all of the GMFHS tasks.

The *subtask* operand can have the following values:

DBSERVER

Specifies the database server task.

EVENTMGR

Specifies the event manager task.

IPC

Specifies the inter-process communications (IPC) task.

MAINTASK

Specifies the host subsystem main task.

NETCMD

Specifies the network command manager task.

NETCON

Specifies the network configuration manager task.

OPERIF

Specifies the operator interface task.

VIEWMGR

Specifies the view manager task.

RCMGR

Specifies the resource collection manager task.

RTMGR

Specifies the resource traits manager task.

IRMGR

Specifies the IPC-RODM manager task.

VSTATMGR

Specifies the view status manager task.

API

Specifies the application programming interfaces (APIs) to be traced.

ALL

Turns on all API tracing for the specified task or tasks.

NONE

Turns off all API tracing for the specified task or tasks.

If you do not want to turn on or off all API tracing for the specified task or tasks, you can specify one or more of the following values for *traceapi*:

IPC

Traces the IPC service requests for the specified tasks.

RODM

Traces the RODM user application programming interface (API) requests and responses.

PPI

Traces the requests made to the program-to-program interface for the NetView program by the IPC tasks that support program-to-program interface gateways only.

RCM

Traces the event flow of the Resource Collection Manager task.

LEVEL=*minlevel*

Specifies the level of tracing detail to be performed on the specified task or tasks. The valid range is from 0–99. In general, the *minlevel* value has the following meaning:

0–9

Specifies the least amount of detail.

10–19

Provides a trace of high-level program functions and events only.

20–29

Specifies a medium level of detail for functions and events.

30–59

Specifies a fine level of detail for functions and events.

60–69

Provides a method trace of high-level functions and events.

70–79

Provides a medium-level trace of functions and events.

80–89

Provides a fine-level method trace of functions and events.

90–99

Provides the most detailed trace.

Note: Method tracing is written to the RODM log.

PRINT

Specifies where trace entries for the specified task or tasks are issued. The *outputlog* operand can have the following values:

YES or FILE

Issued to the GMFHS output data sets. The output data sets are defined by GMFHS using the following DD statement in the GMFHS startup procedure:

CNMC

Network command manager (NETCMD)

CNMD

Database server (DBSERVER)

CNME

Event manager (EVENTMGR)

CNMF

Network configuration manager (NETCON)

CNMI

Interprocessor communication (IPC)

CNMM

GMFHS main (control) task (MAINTASK)

CNMN

Resource Collection Manager (RCMGR)

CNMO

Operator interface manager (OPERIF)

CNMP

IPC-RODM event manager (IRMGR)

CNMR

Resource traits manager (RTMGR)

CNMS

View status manager (VSTATMGR)

CNMV

View manager (VIEWMGR)

NO or INTERNAL

Issued to the GMFHS internal trace log.

GTF

Issued to GTF.

The event identifier (EID) used for the TRACE records written to GTF is X'5E2'.

For more information about GTF, see *MVS/ESA: Services Aids*.

STORAGE=NO|YES

Specifies whether get and free storage requests for specified task or tasks are to be traced.

TYPE

Specifies which IPC interfaces are traced. You can specify ALL, NONE, or *tracetype*.

ALL

Specifies to turn on all IPC API trace types.

NONE

Specifies to turn off all IPC API trace types.

If you do not want to turn on or off all IPC API trace types for the specified task or tasks, you can specify one or more of the following values for *tracetype*:

CNMTAMEL

Specifies to trace the messages exchanged between the IPC and the NetView CNMTAMEL task.

GDS

Specifies to trace the messages exchanged between the IPC and graphic data servers.

NOTIFY

Specifies to trace the IPC retrieval of notification blocks from the RODM notification queues.

PDU

Specifies to trace the protocol data units that are exchanged between GMFHS subtasks.

PPI

Specifies to trace all IPC requests to the program-to-program interface.

SCO

Specifies to trace the messages exchanged between the IPC and the scope checker optional task (DUIFSSCO) running in the NetView address space.

FLUSH

Indicates that the content of the internal trace log is written to the data set specified by CNMT DDNAME in the GMFHS startup procedure. The internal trace log is then reinitialized.

To prevent losing data when issuing a GMFHS TRACE FLUSH command, GMFHS allocates an internal trace log before it prints and releases the current log. See the *IBM Z NetView Administration Reference* for more information about the internal trace log.

Restrictions

The following restrictions apply to the TRACE command:

- Issuing the GMFHS TRACE command with the ON or OFF keyword along with the TASK keyword sets the tracing options before or during tracing. Issuing the GMFHS TRACE command with only the ON or OFF keyword starts or stops GMFHS tracing.
- See the *IBM Z NetView Administration Reference* for information about TRACE and TRACEPAGES.
- See the *IBM Z NetView Troubleshooting Guide* for more information about the GMFHS TRACE command.

Example: Setting trace parameters and starting the trace

To set the parameters for tracing the NETCMD and IPC tasks, and to start tracing, enter:

```
GMFHS TRACE TASK=(NETCMD,IPC),ON,LEVEL=30
GMFHS TRACE ON
```

Response

A response similar to the following is displayed:

```
DUI4060I CURRENT TRACE SETTINGS
DUI4068I TRACE COMMAND PROCESSED FOR NETCMD
DUI4091I NETCMD      1  LEVEL 30  PRINT F  API = (RODM,IPC,PPI)
  STORAGE 0  IPCAPI = (PDU,SCO,PPI,GDS,CNMTAMEL,NOTIFY)
DUI4068I TRACE COMMAND PROCESSED FOR IPC
DUI4091I IPC         1  LEVEL 30  PRINT F  API = (RODM,IPC,PPI)
  STORAGE 0  IPCAPI = (PDU,SCO,PPI,GDS,CNMTAMEL,NOTIFY)
DUI4037I END

DUI4056I TRACE IS ALREADY ACTIVE
```

Example: Stopping the tracing of all tasks

To stop tracing of all tasks, enter:

```
GMFHS TRACE OFF
```

Response

A response similar to the following is displayed:

```
DUI4058I TRACE HAS BEEN DEACTIVATED
```

Example: Displaying current trace settings

To display current trace settings, enter:

```
GMFHS TRACE
```

Response

A response similar to the following is displayed:

```
DUI4060I CURRENT TRACE SETTINGS
DUI4090I TRACING IS ON
DUI4091I MAIN      0  LEVEL 99  PRINT F  API = (RODM,IPC,PPI) STORAGE 0
  IPCAPI = (PDU,SCO,PPI,GDS,CNMTAMEL,NOTIFY)
DUI4091I IPC       0  LEVEL 99  PRINT F  API = (RODM,IPC,PPI) STORAGE 0
  IPCAPI = (PDU,SCO,PPI,GDS,CNMTAMEL,NOTIFY)
DUI4091I OPERIF    1  LEVEL 99  PRINT F  API = (RODM,IPC,PPI) STORAGE 0
  IPCAPI = (PDU,SCO,PPI,GDS,CNMTAMEL,NOTIFY)
DUI4091I VIEWMGR   0  LEVEL 99  PRINT F  API = (RODM,IPC,PPI) STORAGE 0
  IPCAPI = (PDU,SCO,PPI,GDS,CNMTAMEL,NOTIFY)
DUI4091I VSTATMGR  0  LEVEL 99  PRINT F  API = (RODM,IPC,PPI) STORAGE 0
  IPCAPI = (PDU,SCO,PPI,GDS,CNMTAMEL,NOTIFY)
DUI4091I RTMGR     0  LEVEL 99  PRINT F  API = (RODM,IPC,PPI) STORAGE 0
  IPCAPI = (PDU,SCO,PPI,GDS,CNMTAMEL,NOTIFY)
DUI4091I DBSERVER  0  LEVEL 99  PRINT F  API = (RODM,IPC,PPI) STORAGE 0
  IPCAPI = (PDU,SCO,PPI,GDS,CNMTAMEL,NOTIFY)
DUI4091I EVENTMGR  0  LEVEL 99  PRINT F  API = (RODM,IPC,PPI) STORAGE 0
  IPCAPI = (PDU,SCO,PPI,GDS,CNMTAMEL,NOTIFY)
DUI4091I IRMGR     0  LEVEL 99  PRINT F  API = (RODM,IPC,PPI) STORAGE 0
```

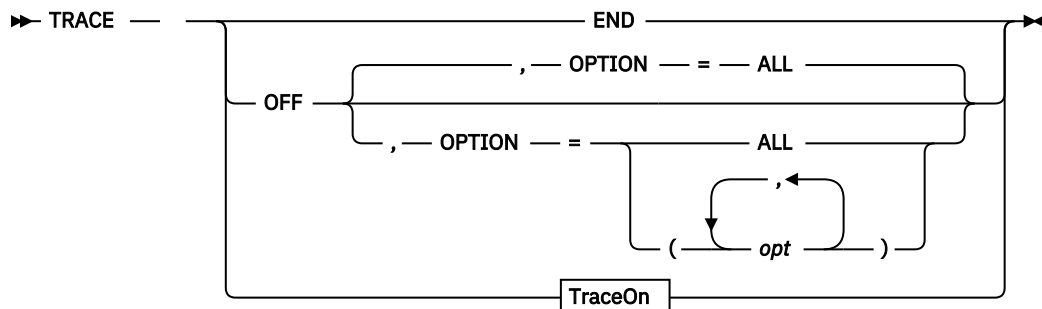
```

IPC API = (PDU,SCO,PPI,GDS,CNMTAMEL,NOTIFY)
DUI4091I NETCMD      0  LEVEL 99 PRINT F API = (RODM,IPC,PPI) STORAGE 0
IPC API = (PDU,SCO,PPI,GDS,CNMTAMEL,NOTIFY)
DUI4091I NETCON      0  LEVEL 99 PRINT F API = (RODM,IPC,PPI) STORAGE 0
IPC API = (PDU,SCO,PPI,GDS,CNMTAMEL,NOTIFY)
DUI4091I RCMGR       0  LEVEL 99 PPRINT F API = (RODM,IPC,PPI) STORAGE 0
IPC API = (PDU,SCO,PPI,GDS,CNMTAMEL,NOTIFY)
DUI4037I  END

```

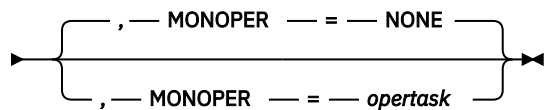
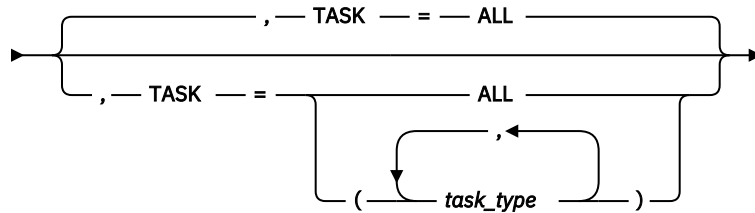
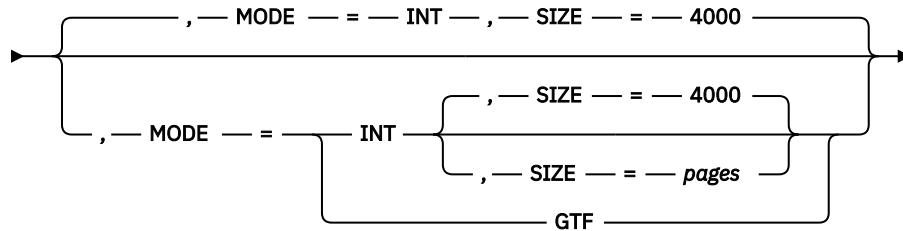
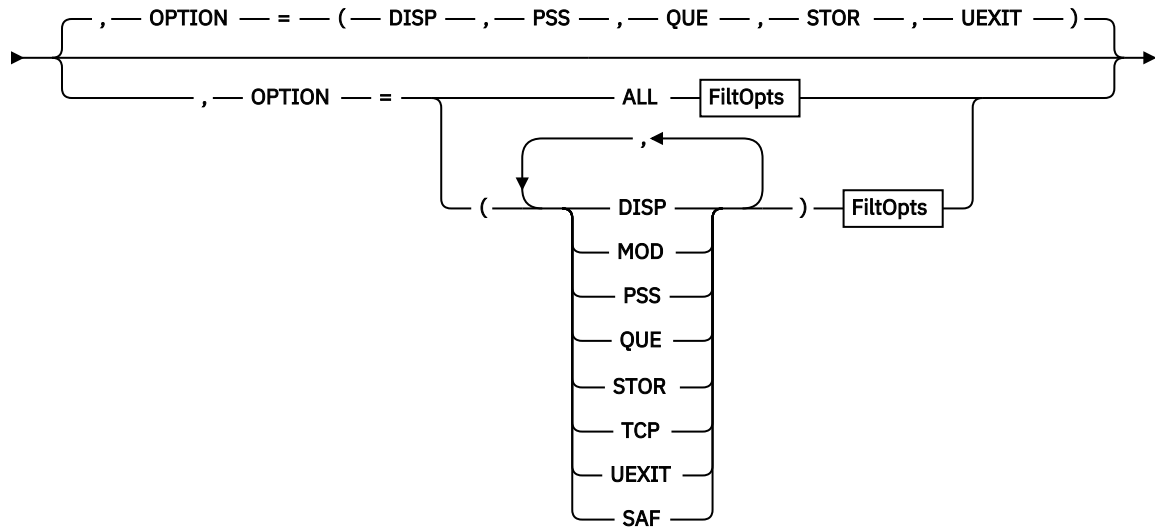
Syntax

NCCF TRACE



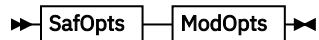
TraceOn

➤ ON ➤

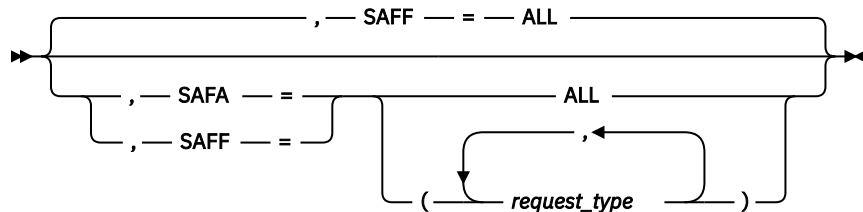


Note: SAF must be included in the list of options before SAFA or SAFF can be specified.

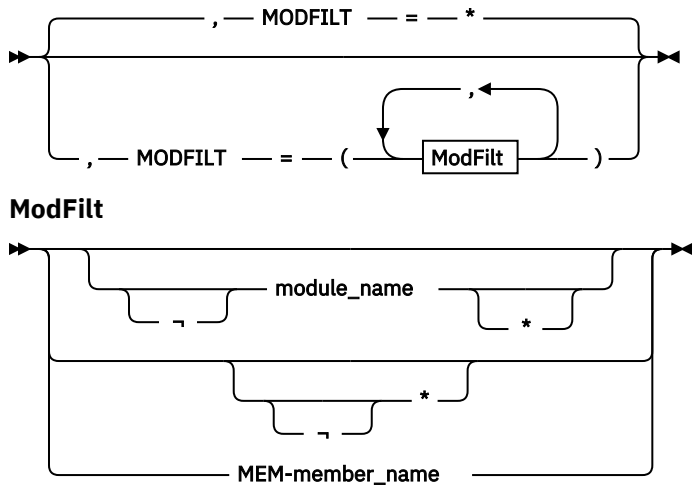
FiltOpts



SafOpts



ModOpts



IBM-Defined Synonyms

Command or Operand	Synonym
OPTION	OPT

Purpose of Command

The NCCF TRACE command initiates a sequence trace that records in virtual storage, in the DSITRACE VSAM data set, or in GTF, a sequence of NetView processing steps. These can help you solve problems you might encounter using the NetView program.

You can use the LIST TRACE command to get a list of the current trace settings.

Operand Descriptions

END

Indicates that all tracing is to stop and internal trace storage is to be freed. The data space that is being written to by the internal trace is deleted when tracing stops. You cannot specify other operands with the END operand. If you specify other operands, the command is rejected with an error message.

OFF

Turns off the indicated options.

OPTION

Indicates which options are to be traced. Each option identifies an internal event type that is to be traced.

The OPTION operand is optional.

If you do not specify OPTION on TRACE ON, the default options of QUE, PSS, DISP, STOR, and UEXIT are used.

If you do not specify OPTION on TRACE OFF, all the options that are currently being traced are turned off.

OPTION fields are not cumulative.

opt

The options are as follows:

ALL

Indicates all options.

Attention: Using OPTION=ALL severely degrades the performance of the system.

DISP

Indicates dispatching of tasks including waiting (DSIWAT), post (DSIPOS), and dispatch from a wait (resumption of processing from DSIWAT).

MOD

Indicates module entry and exit trace of a subset of NetView modules.

Attention: Using MOD severely degrades the performance of the system, therefore use MOD only to trap specified data.

PSS

Indicates presentation services, which involves input from and output to the terminal screen using DSIPSS.

QUE

Indicates inter-task queueing of buffers using DSIMQS.

STOR

Indicates getting and freeing of storage.

TCP

Indicates IP services-related calls.

UEXIT

Indicates installation exit calls for DSIEX01 through DSIEX19, CNM interface input exit (XITCI), CNM interface output exit (XITCO), DST initialization exit (XITDI), VSAM empty file exit (XITVN), VSAM input exit (XITVI), and VSAM output exit (XITVO).

SAF

Indicates calls made to an SAF product. The SAFA or SAFF keywords are used to specify the types of SAF requests to trace and when to trace them. If neither keyword is specified, SAFF=ALL is used as the default.

ON

Turns on the indicated options. ON is the only correct choice if the NetView trace is inactive.

MODE

Specifies in which area data is to be logged. Specify the MODE operand on the initial trace activation command only. If you specify MODE at other times, a warning or an error message is issued.

INT

Indicates to log the trace data in the internal table. INT is the default.

SIZE=pages

Indicates the number of pages of storage to allocate for in-storage trace table. This operand is optional. If MODE=INT, the default page size is 4000 pages (page size is 4k). The trace table is saved in a data space and the maximum page size value that can be specified is 524286 pages, the limit for a data space. If SIZE is specified with MODE=GTF, SIZE is ignored.

GTF

Indicates to log the trace data to the generalized trace facility (GTF). MODE=GTF is rejected if GTF is not active. The event records that are printed, such as MENT, MXIT, and PSS, will have the same format as specified in the *IBM Z NetView Troubleshooting Guide*. The event identifier (EID) used for the TRACE records written to GTF is X'5F6'. GTF also pads some records with zero (0) as needed.

TASK=task_type

Specifies a task name or a task type. Valid values for *task_type* are:

- ALL
- HCT
- MNT
- NNT
- OPT
- OST
- PPT

- VOST

The maximum number of task names or task types specified for one request is 25. TASK is allowed only for TRACE ON. The default is ALL.

SAFA=*request_type*

Indicates that all calls to an SAF product for the specified request type will be traced. This keyword is only valid when either SAF or ALL is specified on the OPTION statement. If you specify ALL, you cannot specify any other request types. Valid values for *request_type* are as follows:

- ALL
- AUTH
- EXTRACT
- FASTAUTH
- LIST
- STAT
- TOKENMAP
- TOKENXTR
- VERIFY

SAFA is valid only for TRACE ON.

Attention: Using SAFA can degrade the performance of the system.

SAFF=*request_type*

Specifies that unsuccessful calls to an SAF product for the specified request type will be traced. This keyword is only valid when either SAF or ALL is specified on the OPTION statement. If you specify ALL, you cannot specify any other request types. Valid values for *request_type* are as follows:

- ALL
- AUTH
- EXTRACT
- FASTAUTH
- LIST
- STAT
- TOKENMAP
- TOKENXTR
- VERIFY

SAFF is valid only for TRACE ON.

MODFILT

The MODFILT keyword has meaning only when OPTION=ALL or OPTION=MOD is specified, that is, when module entry/exit tracing is enabled. The MODFILT keyword is used to filter out unwanted modules from the trace. For some NetView program problems, IBM Software Support might ask you to specify certain MODFILT values so that only modules related to the problem are traced. Eliminating the tracing of modules not related to the problem can greatly reduce the likelihood of the trace wrapping.

Note:

1. You can use an asterisk as a wildcard with the MODFILT keyword, but a leading asterisk, as in *PIFRE, is not supported. The NetView program does not do syntax checking or validity checking of module names.
2. If the MODFILT keyword is specified with the TRACE ON command, then by default, all module entry traces and all module exit traces are blocked. You must use the filtering to unblock the trace.

Consider a MODFILT specification of:


```
MODFILT=(BNJCCCBA,~BNJCCC*,BNJ*,MEM-BATDSI)
```

Each of these MODFILT entries is explained below. They are processed in the order in which they are specified. A maximum of 200 entries can be specified.

BNJCCCBA

This is an example of a module name. All module entry/exit trace entries that are encountered for module BNJCCCBA are included in the trace and are not filtered out.

~BNJCCC*

The * is a wildcard character. The ~ is the not symbol, indicating that the module is not traced. Thus, modules that begin with the BNJCCC characters are not traced. If a module entry/exit trace entry for module BNJCCBB is encountered, it is filtered out and not included in the trace. Note that module BNJCCCBA are traced, because its entry was specified prior to the ~BNJCCC* entry.

BNJ*

All remaining BNJ* modules are traced.

MEM-BATDSI

The specified member name refers to a DSIPARM data set member. Note that the syntax requires the text **MEM** followed by a hyphen followed by the member name. For example, this is a valid specification that uses a member named BATDSI:

```
MODFILT=(MEM-BATDSI)
```

The member must contain one or more lines, where each line can contain the following:

- A blank line. Blank lines are ignored.
- A line that has the number sign (#) as its first non-blank character. The # is the comment character, and lines that begin with a # are comment lines and are ignored.
- One or more MODFILT filter entries, separated by either blanks or commas. An optional comment character # can follow the entries; all characters to the right of the # are ignored.

For example, the BATDSI member might contain the following lines:

```
# Trace desired DSI* modules.
DSIPITOS DSIPIFRE # We might want these two.
~DSIPI*           # We might not want the others.
DSI*              # All other DSI*'s are traced.
```

MEM-*member_name* entries are not permitted in a member. That is, nesting members within members is not permitted. The LIST TRACE command displays the filter entries that were read from a member. When the MODFILT keyword is specified, one or more filter values must be present. For example, if MODFILT=(MEM-BATDSI) is specified, then the BATDSI member must contain at least one filter entry.

MONOPER

The MONOPER keyword must be used only when IBM Software Support asks you to use it. If used incorrectly, the MONOPER keyword can cause the NetView program to run out of storage and end because of too many messages being queued to the monitoring operator task. Therefore, use the MONOPER keyword with extreme caution. It is a debugging aid and, even when used correctly, it can degrade performance.

When a monitoring operator task other than the default of NONE is specified, for example, when an *opertask* value of OPER2 is specified, then the NetView program sends messages to the specified operator task that display in real time the trace entries that are currently being traced by the TRACE command. The messages are written to the NetView log and displayed at the operator task. These trace entries assist IBM Software Support when debugging problems that have been encountered by customers. Tracing is disabled for the *opertask* task itself, and the *opertask* task cannot be present in the list of task values specified for the TASK keyword. For example, the

following command fails because NETOP1 is specified both as the MONOPER value and as a TASK list value:

```
TRACE ON OPTION=QUE TASK=(NETOP1,BNJDSERV) MONOPER=NETOP1
```

Use the *opertask* task only to monitor the trace entries displayed by the task. For example, do not enter commands at the *opertask* task, and do not perform any automation under this task. Use it only for monitoring the trace entries.

The default TASK keyword value of ALL is not permitted when the MONOPER keyword is specified. TASK=ALL must be explicitly specified when the value of ALL is used in conjunction with the MONOPER keyword. This restriction exists because TASK=ALL must typically not be used when the MONOPER keyword is specified. This combination can easily result in the NetView program running out of storage. The MONOPER keyword is usually used with a TASK value other than ALL.

See the *IBM Z NetView Troubleshooting Guide* to see an example of the trace entry messages that are displayed at the *opertask* when the MONOPER keyword is specified.

Restrictions

The following restrictions apply to the TRACE command:

- Do not use MODE=GTF and the SIZE operand together. SIZE is ignored if you specify MODE=GTF. See the *IBM Z NetView Troubleshooting Guide* for information about the trace record formats of records written to GTF.
- Using OPTION=ALL severely degrades the performance of the system.
- When you specify the OFF operand, all active tracing ceases. However, the SIZE and MODE settings remain effective. You can change these options after issuing the TRACE END command.
- When a trace is requested by specific task name, the ending of the task turns off the trace for that task. This task is not traced if it becomes active again.
- When a trace is requested by task type (including ALL), the ending of a task in the same task type will *not* turn off the trace for that task. When any task with the same task type becomes active, it is traced.

Return Codes

Return Code	Meaning
-------------	---------

0	The TRACE command was successful.
4	The TRACE command was not successful.

Example: Turning on the trace with default values

To turn on trace with the default values, enter:

```
TRACE ON
```

Response

Because no options are specified, options QUE, PSS, DISP, STOR, and UEXIT are traced for all the tasks. MODE defaults to INT and SIZE defaults to 4000 pages.

Example: Turning off indicated options

Assuming that trace was initially turned on with OPTION=ALL, to turn off certain options while continuing to trace all other active options, enter:

```
TRACE OFF OPT=(SAF,QUE,PSS)
```

Response

Trace continues for the remaining options DISP, MOD, STOR, UEXIT.

Example: Ending all tracing

To discontinue all tracing enter:

```
TRACE END
```

Example: Tracing non-zero return codes from specified SAF calls

To trace non-zero return codes from AUTH and VERIFY calls to SAF for all tasks, enter:

```
TRACE ON,OPTION=SAF,SAFF=(AUTH,VERIFY),TASK=ALL
```

Example: Tracing all SAF Calls for one operator

To trace all SAF calls on an active operator task OPER1 enter:

```
TRACE ON,OPTION=SAF,SAFA=ALL,TASK=OPER1
```

Example: Tracing HCT and OST tasks

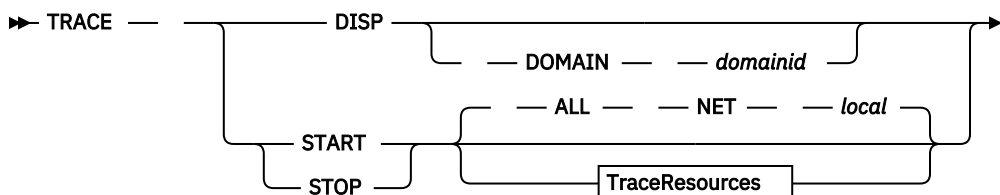
To trace all HCT and OST tasks, including autotasks, with the QUE option enter:

```
TRACE ON,OPTION=QUE,TASK=(HCT,OST)
```

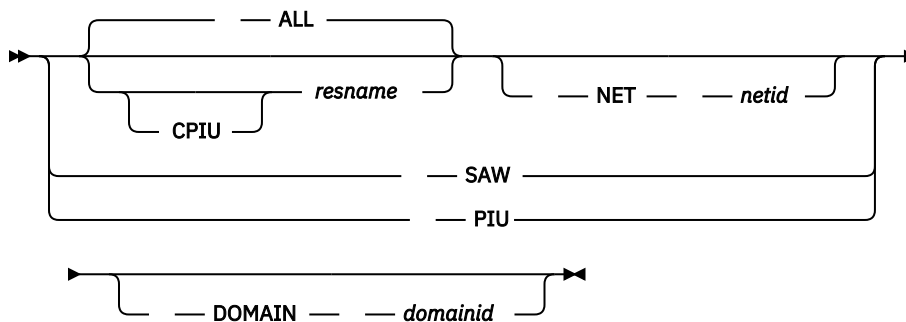
TRACE (NLDM)

Syntax

NLDM TRACE



TraceResources



Purpose of Command

The NLDM TRACE command starts or stops a session trace or displays resources that are being traced.

Operand Descriptions

DISP

Displays all traces that have been activated or deactivated explicitly.

DOMAIN *domainid*

Specifies the domain in which the TRACE command is processed.

START

Starts trace functions for specified resources.

STOP

Stops trace functions for specified resources.

ALL

Performs tracing for all network resources. ALL is the default.

PIU

Performs tracing for PIU trace data.

The PIU trace option is not affected by the ALL parameter.

SAW

Performs tracing for session awareness data.

The SAW trace option is not affected by the ALL parameter.

CPIU

Specifies that complete PIUs for a specified LU resource are to be received from VTAM and are not to be truncated. The CPIU parameter cannot be specified with the ALL, PIU, or SAW parameters.

If you specify CPIU when starting and then stopping a trace for the same resource, PIUs are traced, but they are truncated. This operand is ignored if the resource is not an LU.

The maximum length of PIUs for active sessions depends on the VTAM trace buffer size. You declare the trace buffer size at initialization in AAUPRMLP. The default is 4 kilobytes.

The maximum length of PIUs for inactive sessions is 921.

Note: The CPIU operand is valid only with VTAM Version 3 Release 4.1 or later releases.

resname

Specifies any valid network name (LU, PU, or SSCP name).

NET *netid*

Specifies the name of the network in which the specified resource resides. If you do not specify the *netid*, the operator's local *netid* is used.

The NET keyword cannot be specified with the PIU or SAW keywords.

Restrictions

The following restrictions apply to the TRACE command:

- PIU and SAW trace buffers are written to GTF.
The event identifier (EID) used for the TRACE records written to GTF is X'5F4' for SAW and X'5F5' for PIU. GTF also pads some records with 0 as needed.
- The PIU and SAW keywords are for debugging NetView and VTAM problems and should be used only when directed by IBM programming services.
- If collecting trace data has not been automated, you might be instructed to start data collection for some resources. The TRACE START command starts a session trace. It starts collecting session formation parameters, PIU trace data, and NCP trace data.
- The system programmer can start gathering trace data following session monitor initialization, and start tracing all resources. If this is done, you might be instructed to stop data collection for some resources.

- You can learn which devices are being traced by using the TRACE DISP command, which invokes panel display trace. The TRACE DISP command displays a list of resource names. Interpretation of that list depends on how tracing was started.
- If TRACE is started for all resources, the global trace field indicates ON and any resources that are displayed have had trace stopped.
- If TRACE is stopped for all resources, the global trace field indicates OFF and any resources that are being traced are displayed.
- To issue the TRACE command as a line-mode command, precede the TRACE command with NLDM.
- Both VTAM and the NCP are notified when an operator issues the trace explicit command. If the trace to VTAM is successful but the trace to NCP is rejected, the trace display indicates that the resource is being traced, but it will have incomplete trace data.

Example: Displaying a list of network names being traced for specific resources

If tracing has been started for specific resources, enter:

```
TRACE DISP
```

Displays a list of network names that are being traced.

Example: Starting a trace on a specific LU

To start tracing a specific LU, for example LCL3278A, which is in network A01M, enter:

```
TRACE START LCL3278A NET A01M
```

Example: Starting a trace for IMS1 sessions with terminal LUs

To start tracing for IMS1 sessions with terminal LUs that have also had a trace started for them in NETA, enter:

```
TRACE START IMS1 NET NETA
```

Example: Stopping all tracing in domain DOM1

To stop all tracing in domain DOM1, enter:

```
TRACE STOP ALL DOMAIN DOM1
```

Example: Displaying traces started in DOM1

To display traces started in DOM1, enter:

```
TRACE DISP DOMAIN DOM1
```

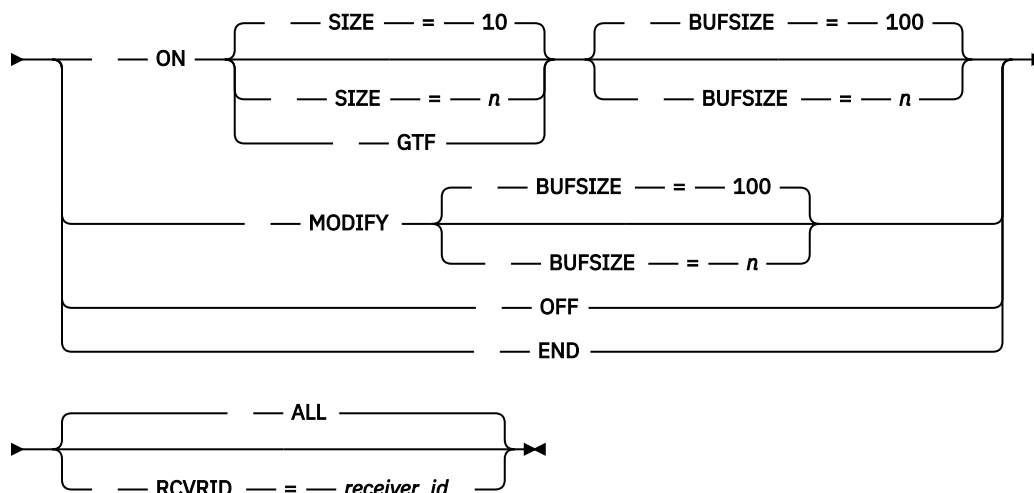
TRACEPPI (NCCF)

Syntax

From an MVS console:

TRACEPPI

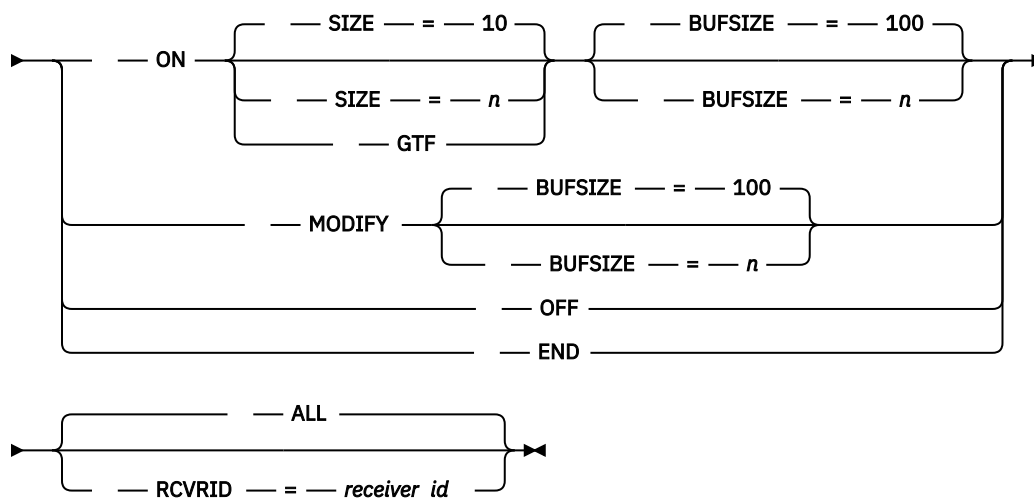
➤ MODIFY — — *ssiname* — , — TRACEPPI ➔



From a NetView terminal:

TRACEPPI

➤ MVS — — MODIFY — — *ssiname* — , — TRACEPPI ➔



\

IBM-Defined Synonyms

Command or Operand	Synonym
MODIFY	F

Purpose of Command

The TRACEPPI command starts, stops, modifies, or ends a trace for all program-to-program interface receivers or for a specified interface receiver.

This command can be issued from a NetView operator ID (using the MVS command) or an MVS console by using the modify function of the program-to-program interface. The system console operator receives a message indicating the success or failure of the command.

Operand Descriptions

ssiname

Specifies the MVS subsystem interface name.

ON

Turns on the program-to-program interface trace facility for all current and future program-to-program interface receivers or for the interface receiver specified in the RCVRID operand. If this is a new trace, the trace facility allocates the internal trace table. If you specify the Generalized Trace Facility (GTF) option, all program-to-program interface trace records are sent to GTF in a GTF user trace record.

MODIFY

Changes the BUFSIZE value for all current and future program-to-program interface receivers or for the receiver specified on the RCVRID operand.

OFF

Turns off the program-to-program interface trace facility for all current and future program-to-program interface receivers or for the receiver specified on the RCVRID operand, but does not free the trace table or clear any of the trace values.

END

Clears trace values and sets trace status to "not defined" for all program-to-program interface receivers or for the receiver specified on the RCVRID operand. If you specify END for all receivers, the trace facility also frees the trace table.

SIZE=*n*

Specifies the size in pages (4096 bytes) of the program-to-program interface internal trace table. The default value is 10 pages. This keyword is valid only with the ON keyword. This keyword is ignored if there is already a trace defined.

GTF

Sends the program-to-program interface trace records to GTF in a GTF user trace record. This keyword is valid only with the ON keyword. It is not valid with the SIZE keyword. If you specify GTF, the trace facility does not allocate a program-to-program interface internal trace table.

BUFSIZE=*n*

Specifies the number of bytes copied into a GTF or program-to-program interface trace record from a buffer being sent to a receiver. If you specify GTF, the buffer limit is 208 bytes. The BUFSIZE value cannot be larger than the program-to-program interface table; for example, if the trace table is defined as one page (4096 bytes), and the BUFSIZE is set at 5000 bytes, an error message will be sent to the console.

ALL

Specifies that the requested action applies to all current receivers as well as future receivers.

RCVRID =*receiver_id*

Identifies a single receiver to which the requested action applies.

Restrictions

If you turn the trace facility on, and then turn it off, the trace facility does not change any of the trace values. To change the BUFSIZE value of a particular receiver, issue a TRACEPPI MODIFY command for that receiver ID. To change the SIZE value, issue a TRACEPPI END command for all receivers, then issue a TRACEPPI ON command to start a new trace. The information in the last trace table is lost.

Example: Recording 256 bytes of all buffers

To record 256 bytes of all buffers being sent to or received from all receivers that are currently defined in the program-to-program interface, as well as all future receivers, enter either command:

```
F ssiname,TRACEPPI ON ALL SIZE=5 BUFSIZE=256
F ssiname,TRACEPPI ON SIZE=5 BUFSIZE=256
```

If no traces have been previously defined, the program-to-program interface trace facility allocates a 5-page trace table in the subsystem interface (SSI) address space. If a trace has been previously defined, the size parameter is ignored.

Example: Turning off the program-to-program interface trace

To turn off the program-to-program interface trace for all receivers, enter either command:

```
F ssiname,TRACEPPI OFF ALL
F ssiname,TRACEPPI OFF
```

The program-to-program interface internal trace buffer is not freed, and receivers' trace characteristics are not changed.

Example: Turning off the program-to-program interface trace for all receivers

To turn off the program-to-program interface trace for all receivers, free the program-to-program interface internal trace buffer, and clear all receivers' trace characteristics, enter either command:

```
F ssiname,TRACEPPI END ALL
F ssiname,TRACEPPI END
```

Example: Tracing activity for a receiver

To trace activity for receiver MYRCVR and record 256 bytes of all buffers sent or received by MYRCVR, enter:

```
F ssiname,TRACEPPI ON RCVRID=MYRCVR SIZE=5 BUFSIZE=256
```

If no traces have been previously defined, the program-to-program interface trace facility allocates a five page trace table in the SSI address space. If a trace has been previously defined, the size parameter is ignored.

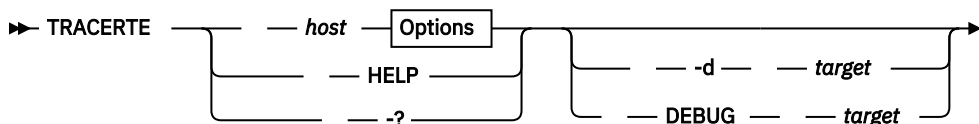
Example: Tracing activity for a receiver

To trace activity for receiver MYRCVR and record 100 bytes of all buffers sent or received by MYRCVR to GTF, enter:

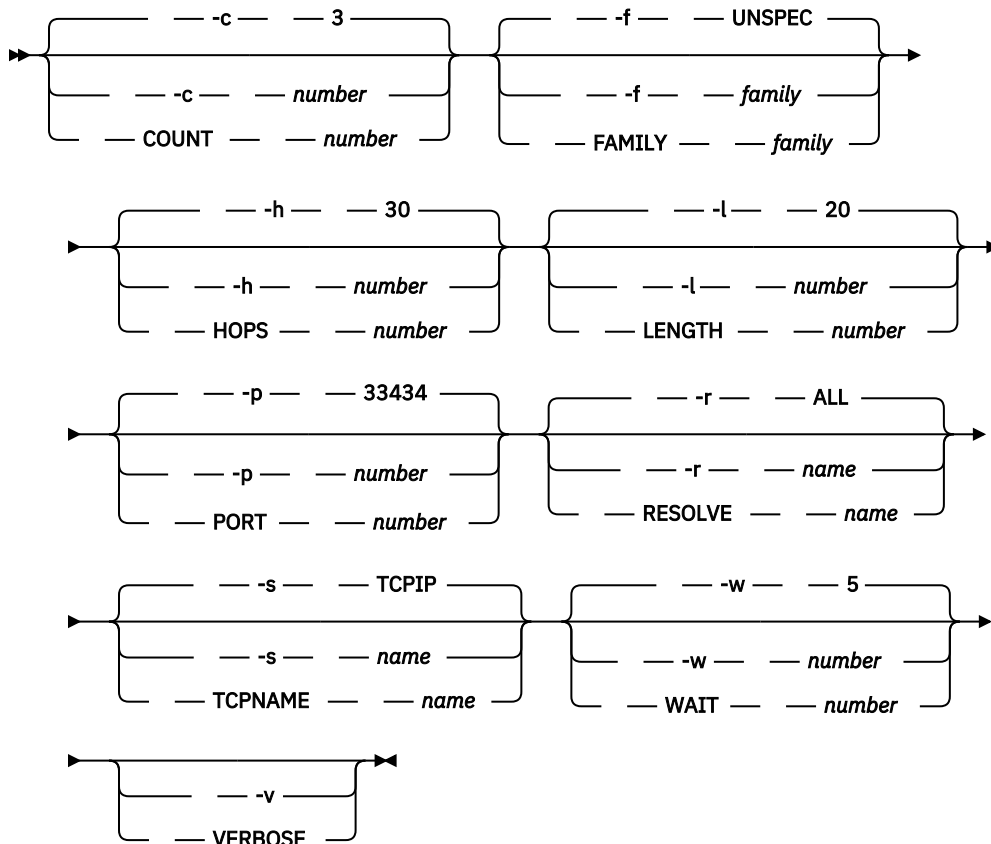
```
F ssiname,TRACEPPI ON RCVRID=MYRCVR GTF BUFSIZE=100
```

TRACERTE (NCCF; CNMETRTE)

Syntax



Options



IBM-Defined Synonyms

Command or Operand	Synonym
CNMETRTE	TRACERTE, TRACERT, TRCROUTE, TRCRTE, TRCRT

Purpose of Command

The TRACERTE command can be used to trace the routes of IP data packets to the specified host from the IP stack on the host on which the NetView program is running. This can be done to determine connectivity with a given endpoint, the actual routing to a given endpoint, and roundtrip times between the NetView program and the target host, and routers along the way.

Operand Descriptions

host

The IP host name or IP address of the target host of the command.

COUNT or -c

The number of probes to be sent on each hop. The default is 3 probes. The maximum value that can be specified is 20 probes.

DEBUG or -d

Turns tracing on, where *target* is as follows:

ALL or ON

Turns on debugging for all targets.

ARGS

Traces argument processing.

RESOLVE

Traces operations relating to Domain Name Server (DNS) resolution of IP addresses and host names.

-f or FAMILY

Specifies the address family, indicating the IP networks in which the host running the NetView program and the target host are expected to reside. Valid values are as follows:

UNSPEC

Unspecified, meaning the NetView system and the target host can be in either an IPv4 network, an IPv6 network, or both. The TRACERTE command uses the first IP address returned by the resolver. UNSPEC is the default value.

When an IP address is specified for the target host, the TRACERTE command, for IPv4 addresses and IPv4-mapped IPv6 addresses, only uses an IPv4 network interface for determining connectivity. For an IPv6 address, only an IPv6 network interface is used for determining connectivity.

INET

Use only an IPv4 network interface to determine connectivity. This value is ignored if an IP address is supplied as the target host.

Note:

1. If an IPv4 address or IPv4-mapped IPv6 address cannot be found for the specified host name, the PING request fails.
2. If DEFAULT.IPV6ENV=ONLY, the PING request fails. See the *IBM Z NetView Administration Reference* for additional information about the DEFAULT.IPV6ENV statement.

INET6

Use only an IPv6 network interface to determine connectivity. This value is ignored if an IP address is supplied as the target host.

Note:

1. If an IPv6 address or IPv4-mapped IPv6 address cannot be found for the specified host name, the PING request fails.
2. If default DEFAULT.IPV6ENV=NONE, the PING request fails. See the *IBM Z NetView Administration Reference* for additional information about the DEFAULT.IPV6ENV statement.

HELP or -?

Displays the online help for the TRACERTE command.

HOPS or -h

The maximum number of hops allowed to the Destination address. The default is 30 hops. The maximum value that can be specified is 255.

Note: If several consecutive attempts fail, this can indicate a mistyped argument and can be time-consuming. Output is ended early. In this case, specify the number of hops wanted instead of using the default value of 30 hops.

LENGTH or -l

The size of each probe sent. The default is 20 bytes. The maximum value that can be specified is 65495 bytes for IPv4 and 65515 bytes for IPv6.

PORT or -p

Specifies the starting port for the source and destination ports. For each hop, the source port is always the value specified, while the destination port is the source port plus the number of the current hop. The first hop is numbered as hop 1.

The range for valid values is 2048 - 60000. The default value is 33434.

RESOLVE or -r

Specifies the level of DNS resolution to be employed when the command is invoked. Valid values are as follows:

ALL

DNS resolution of all IP host names or addresses is attempted. This is the default value.

NONE

No DNS resolution is attempted on hops.

TCPNAME or -s

The IP stack name to use. The default stack name is TCPIP.

VERBOSE or -v

This option generates more output that can be useful when attempting to determine exactly why there are unexpected results from the TRACERTE command.

WAIT or -w

The maximum number of seconds to wait between probes. The default value is 5 seconds. The maximum value that can be specified is 255.

Usage Notes

When you issue the TRACERTE command without operands, a full-screen panel is displayed that you can use to specify options.

Tracing IP data packets

The following example traces the routes of IP data packets to the specified host, using all defaults:

```
TRACERTE EX1.EXAMPLE.COM
```

You will receive a response similar to the following:

```
BNH810I Tracing IP route to 6.52.8.129 max 30 hops
BNH811I 1: 7.52.40.121 (7.52.40.121) 2ms 2ms 1ms
BNH811I 2: 6.52.62.223 (6.52.62.223) 4ms 2ms 2ms
BNH811I 3: ex1-ex.example.com (6.52.0.1) 3ms 2ms 2ms
BNH811I 4: ex1.example.com (6.52.8.129) 1ms 4ms 2ms
```

TRANSMMSG (NCCF)

Syntax

```
➤ TRANSMMSG — — MEMBER — = — membername ➤
```

Purpose of Command

The TRANSMMSG command loads a set of message translation rules, defined in a member or members of DSIMSG, into the NetView program. These rules are used for globalization, including the formats used for displaying dates and times. You can issue this command successfully only once at each NetView initialization. Place this command in your NetView initial command list.

Operand Descriptions**MEMBER=*membername***

Is the name of the member of DSIMSG containing either message translation statements or %INCLUDE statements for members containing the translation statements.

Restrictions

The translation member the NetView program supplies is the CNMMSJPN member supplied for Japanese translations. Some English examples of translation are included in CNMSAMP members CNMMSENU and CNMTRXMP.

Return Codes

Return Code	Meaning
-------------	---------

0	Successful processing.
4	Unsuccessful processing.

Example: Loading the supplied CNMTRMSG translation member

To load translations specified by the CNMTRMSG sample that is supplied with the NetView program, enter the following command:

```
TRANMSG MEMBER=CNMTRMSG
```

Response

Message CNM257I, similar to the following, is displayed:.

```
CNM257I MESSAGE TRANSLATIONS HAVE BEEN LOADED FROM DSIMSG MEMBER
        CNMTRMSG
```

TS (NCCF)

Syntax

TS

➤ TS ➤

Purpose of Command

The TS (TRACE START) command causes interactive tracing of a REXX command list the next time the REXX interpreter processes a REXX clause. You can use this command to debug a new REXX command list.

You can enter this command from a terminal, a REXX command list, or a NetView command list language procedure. TS is an immediate command if it is entered from a terminal.

When TS is active, a pause occurs after each executable statement. For example, if the first executable command in a REXX command list is:

```
SAY 'THIS IS THE FIRST EXECUTABLE COMMAND IN THE SOURCE'
```

The NetView program responds with message CNM431I:

```
CNM431I REXX INTERACTIVE TRACE.  ENTER 'GO TRACE OFF' TO END TRACE,
ENTER 'GO' TO CONTINUE.
```

To continue the trace, enter the GO command.

The trace ends when:

- You enter TRACE OFF prefixed with GO
- You enter TE
- The REXX command list finishes processing

Reissue TS to trace another REXX command list. While a REXX command list is being traced interactively, you can run any REXX statement prefixed with GO at a breakpoint. This enables you to perform tasks such

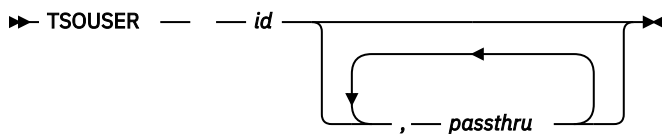
as displaying variables, changing variables, branching to a label, or exiting. For example, if you want the change the value of X to 5, enter:

G0 X=5

TSOUSER (NCCF; CNME0037)

Syntax

TSOUSER



Purpose of Command

The TSOUSER command list displays the status of a time-sharing option (TSO) user ID.

Operand Descriptions

id

Specifies the TSO user ID about which information is to be displayed

passthru

Specifies up to 6 parameters which are appended unchanged to the VTAM DISPLAY command issued by the TSOUSER command. No validation for duplicate or conflicting parameters is performed.

Example: Displaying the status of a TSO user

To display the status of TSO user TSO21, enter:

TS0USER TS021

Response

Information similar to the following is displayed:

```
IST097I DISPLAY ACCEPTED
IST350I DISPLAY TYPE = TSUSER
IST486I NAME= TS021, STATUS= DSCNT, DESIRED STATE=N/A
IST576I TS0 TRACE=OFF
IST262I APPLNAME = TS00001, STATUS = ACTIV
IST262I LUNAME = L3E0, STATUS= ACT/S
IST314I END
```

This response shows the TSO user ID as TSO21 with a status of DSCNT. The TSO trace is not active, the application name associated with the TSO user space is TSO0001, and the logical unit being used is L3E0.

Note: Certain VTAM message IDs are release dependent.

TUTOR (NCCF; CNME5003)

Syntax

TUTOR

▶▶ TUTOR — — *panelname* ◀◀

Purpose of Command

The TUTOR command list displays the panels that are in the CNMPNL1 library.

This command list is usually called from another command list to display online help panels.

Operand Descriptions

panelname

Specifies the panel to be displayed.

Return Codes

If you receive a nonzero return code, an error message is displayed.

Example: Viewing a screen in a library

If a system programmer has written a screen called CNMKNEE0 and placed it in the CNMPNL1 library, and you want to view the screen, enter:

```
TUTOR CNMKNEE0
```

UNMARK (AON)

Syntax

UNMARK

➤ UNMARK — — *root_comp* — . — *rv* — (— *operator_id* —) ➤

Purpose of Command

The UNMARK command removes a DDF assignment entry from the specified operator. The data used to identify the DDF entry as well as the ID of the operator to be removed must be supplied.

Operand Descriptions

root_comp

Defines the root component name as defined in the DDF entry to be unmarked.

rv

The resource name as it is displayed in the RefValue field of the DDF entry is unmarked.

operator_id

The operator ID assigned to this DDF entry.

Usage Notes

- The AON tower must be enabled in the CNMSTYLE member to successfully run this command.
- The *root* and *rv* parameters are required. If you do not specify the *operator_id* parameter, the UNMARK command uses your operator ID as the default.
- If you issue the UNMARK command for a DDF entry that is not assigned to an operator, the UNMARK command takes no action.
- Each automation component has its own syntax for specifying the name of the resource to be displayed.

Examples

To remove the operator ID, OPER1, from the DDF entry NCP001 under CNM01, type:

UNSTACK (NCCF)

Syntax

UNSTACK

➤ UNSTACK ➤

Purpose of Command

The UNSTACK command causes a command procedure suspended by STACK to continue processing in the wait or pause state.

Restrictions

When the STACK command is used for a command procedure in a timed wait, the timed wait continues. However, the command procedure cannot recognize a timeout and resume processing until the UNSTACK command is issued. If UNSTACK is issued *before* a timed wait expires, the timed wait continues as if it had never been interrupted. If UNSTACK is issued *after* a timed wait expires, the command procedure resumes processing as if the timeout had occurred at the time the UNSTACK command was issued.

Example: Resuming a command procedure suspended by the STACK command

To issue the UNSTACK command, enter:

```
UNSTACK
```

Response

If the UNSTACK command is successful, the following message is issued:

```
DSI586I COMMAND PROCEDURE clistname IS RESUMED
```

Example: Entering an UNSTACK command before entering a STACK command

If you enter an UNSTACK command without first entering a STACK command, or if you enter the UNSTACK command at a different linkage level from the corresponding STACK command, one of the following error messages is issued:

```
DSI231I NO STACK IS ACTIVE  
DSI233I STACK IS NOT ACTIVE AT THIS LEVEL
```

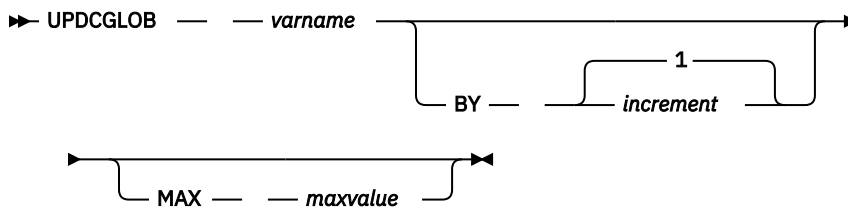
For example, suppose you enter a STACK command, and then enter another command procedure. Before you can enter an UNSTACK command, you might have to end the command procedure. If you are unsure of your level, enter UNSTACK anyway. If you are at the right level, the command is successful. If you are not at the right level, you get an error message. You can end the current command procedure and try again.

Note: The UNSTACK command works from all components that are provided by the NetView program.

UPDCGLOB (NCCF; CNME1080)

Syntax

UPDCGLOB



Purpose of Command

The UPDCGLOB command adds the value specified as *increment* to the current value of the common global variable, provided the result does not exceed the specified *maxvalue*. If the specified variable is not set (has a null value), it is treated as zero and the new value will be the value of the increment.

The UPDCGLOB command list serializes updates for common global variables by using PIPE VARLOAD to give the effect of a *compare and swap* logic. If serialization between tasks is not important, you can use the SETCGLOB command list instead.

Operand Descriptions

varname

Specifies the common global variable for the value that is updated. The name of the variable is given here without using the initial ampersand (&).

BY *increment*

Specifies a number added to the current value of the variable to determine a new value for the variable. If not specified, the *increment* defaults to 1.

MAX *maxvalue*

Specifies the maximum value you allow for the variable. If not specified, there is no maximum.

Restrictions

The use of UPDCGLOB is limited to the command procedure and automation environments (command must originate in REXX, HLL, automation, or an optional task). Use of UPDCGLOB directly from the operator's command line results in message DSI290I and return code 8.

The UPDCGLOB command list increments a common global variable that has a numeric or null value. UPDCGLOB increments in a serialized manner to function accurately even when a value is simultaneously incremented by several tasks. If serialization is not important, you can use the SETCGLOB command list.

Return Codes

- 0** The common global variable was set as requested.
- 4** The updated value exceeded the specified maximum.
- 8** The operator who issued the command list is not authorized to use UPDCGLOB.
- 20** No variable name was specified or no value was specified.

Example: Updating a common global variable

The following example requests the PPT to update the common global variable TASKCOUNT by 1. If TASKCOUNT was formerly set to 20, it is changed to 21.

```
UPDCGLOB TASKCOUNT BY 1 MAX 55
```

VARY (VTAM)

Syntax

➤ VARY — — *vtam_operands* ➤

IBM-Defined Synonyms

Command or Operand	Synonym
VARY	V

Purpose of Command

You can issue the VTAM VARY command from a NetView console to activate or deactivate resources. You must be authorized to issue this command.

Note: The VARY command is actually a z/OS command, which is used by the z/OS Communications Server and other products or system components in specific ways. This documentation refers only to the NetView front-end command, which passes the VARY command on to the VTAM program.

Operand Descriptions

vtam_operands

Enter the VARY command from the NetView console to recycle resources.

When you enter VARY from the VTAM system console, the format can differ depending on the operating system in use. See the z/OS Communications Server library for full details on operands and operating system dependencies.

Usage Notes

Consider the following when using the VARY command:

- NetView span checking will be performed on the value of the following keywords:
 - DVIPA
 - HOSTNAME (HN)
 - ID
 - IPADDR (IP)
 - LU1
 - LU2
 - PLU
 - SLU
 - TSouser (U)

Note: There is no checking on the ID keyword if TSouser is specified.

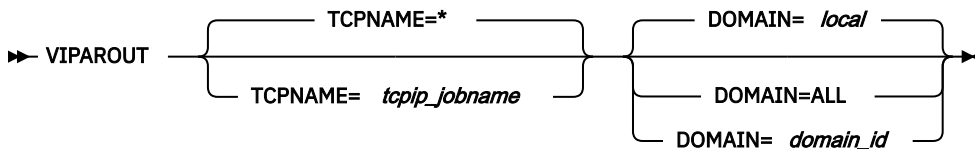
- You can protect any VTAM keywords and values using the NetView command authorization table or using an SAF product. All VTAM command synonyms and keywords must be defined to the table or SAF product.

- If you prefix the DISPLAY command with MVS, the same span and command authorization checking is done.
- See the *IBM Z NetView Security Reference* for more information and for details about the IP address format.

VIPAROUT (NCCF; CNME8216)

Syntax

NCCF VIPAROUT



Purpose of Command

You can use the VIPAROUT command to view status information about VIPA (virtual IP address) routes from a 3270 console or from the Tivoli Enterprise Portal using the Z NetView Enterprise Management Agent.

VIPA Routes data is returned in multilined message BNH824I. To see the format of the data returned by the BNH824I message, refer to the online help.

Note: This command is intended to be used as a REXX interface. For user interfaces, use the Tivoli Enterprise Portal or the CNMSVPRT sample.

Operand Descriptions

TCPNAME

The TCP/IP job name of the target stack. The default value is all stacks.

DOMAIN

The domain to which the request is sent. The following are the valid options:

ALL

Specifies all domains in the sysplex. This option can only be issued from the master NetView program.

Note: If you have many stacks or systems in your sysplex, issuing ALL can cause slow response times and high CPU utilization.

local

If the DOMAIN keyword is not specified, the domain name of the local NetView system is used. This is the default.

domain_id

Specifies the ID of a specific domain. This option can be issued only from the master NetView program if the domain is not the local domain.

Return Codes

Return Code

Meaning

0

The command is successful

2

Help is issued

- 3 There is no data to display
- 4 Required parameters are missing
- 5 There is a REXX NoValue error
- 6 There is a REXX syntax error
- 7 An internal command failed
- 8 The DVIPA tower is not enabled
- 9 The level of z/OS does not support this command
- 5 There is a signal halt

Usage Notes

- Any optional keyword followed by an equal sign (=) with no value is ignored rather than considered an error. The default values are used for keywords which have them.
- The DVIPA data collected is only collected from the local system except when you issue a DOMAIN value for a remote system from a master NetView system.
- If message DSI047E is received, contact your system programmer to enable the appropriate tower or subtower. For information about data collection and display towers and subtowers, see *IBM Z NetView Installation: Configuring Additional Components*.

Restrictions

The following restriction applies to the VIPAROUT command:

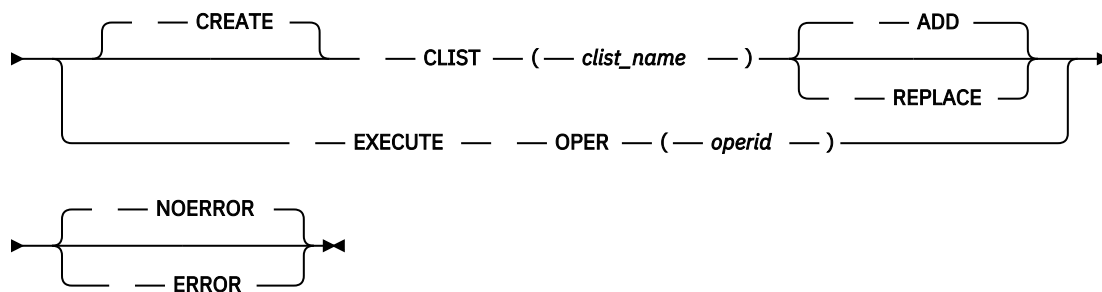
- The VIPAROUT command requires z/OS V1R11 (or later) z/OS Communications Server.

VPDALL (NCCF)

Syntax

VPDALL

➤ VPDALL — — CONFIG — (— *vtam_config_member* —) ➤



IBM-Defined Synonyms

Command or Operand	Synonym
EXECUTE	EXEC
OPER	OPERATOR

Purpose of Command

The VPDALL command creates commands to collect vital product data (VPD) and write it to the external log for PUs and link segments defined in the user's VTAM configuration definitions. The VPDALL command can either run these VPD commands as they are generated or create a command list containing the VPD commands that can be processed later.

The VPDALL command will perform system symbolic substitution on records read from the VTAM configuration member of the DSIVTAM data set. The &DOMAIN symbolic that is supplied with the NetView program is also included in the substitution process. The substitution is performed after comment removal but prior to record processing. This command also removes comments after substitution. Substitution is always performed on the &DOMAIN symbolic, unless substitution was disabled when the NetView program was started. For MVS and user-defined system symbolics, substitution is not performed if one of the following is true:

- You are not running on an MVS system
- You are running on an MVS system prior to MVS Version 5 Release 2
- Substitution was disabled when the NetView program was started
- You have not defined an MVS system symbolic on your MVS system.

Operand Descriptions

CONFIG(*vtam_config_member*)

Specifies the VTAM configuration member. The VTAM configuration member must contain a list of all VTAM major node definitions for which you collect VPD.

The VTAM configuration and major node definitions must be contained in the NetView DSIVTAM data set.

The *vtam_config_member* must be in parentheses.

CREATE

Creates a command list that can be processed at a later time. This command list contains VPD commands to collect VPD for the PUs and link segments specified in the VTAM major node definition. When you specify the CREATE option, VPDALL creates a NetView command list containing the VPD commands. CREATE is the default.

CLIST(*clist_name*)

Specifies the name of the command list to be created when you specify the CREATE option.

Specify a 1- to 8-character member name. The default is VPDACT.

The command list name must be in parentheses.

ADD

Specifies that the command list being created is to be added to the DSICLD data set. You can specify this operand as ADD or A. The default is ADD.

Note: If you attempt to add a command list that currently exists, the command list is not added, and message DWO029I is issued, indicating that the command list already exists. Reissue the command and specify either a different command list or the REPLACE option.

REPLACE

Specifies that the command list being created is to replace an existing command list in the DSICLD data set. You can specify this operand as REPLACE, REP, or R.

EXECUTE

Specifies that VPD commands are run as they are generated. The synonym for EXECUTE is EXEC.

OPER(*operid*)

Specifies the operator ID on which the VPD commands run when you specify the EXECUTE option. The operator ID must be in parentheses. The default is the invoking operator ID.

NOERROR

Specifies that a subtype record for errors is not written to the external logging facility. NOERROR is the default.

ERROR

Specifies that a subtype record for errors is written to the external logging facility. If you specify ERROR, a subtype record for errors is written to the log.

Restrictions

The following restrictions apply to the VPDALL command:

- You cannot specify CREATE and EXECUTE together. If you specify CLIST, the CREATE keyword is implied.
- Collect information about a network control program (NCP) directly from the primary host. However, you can indirectly obtain VPD from the secondary side by issuing the START DOMAIN command and then using the ROUTE command to route the VPD command to the host that owns the NCP on the primary side of the link. For more information, see the START and ROUTE commands.
- The VPDALL command generates VPD commands for resources specified in VTAM major node definitions for NCPs, local SNA devices, switched devices, and dynamic reconfiguration devices. If a major node other than these are found, processing ends and control is returned back to the VTAM configuration routine for another major node to process. Major nodes are recognized by the presence of the following keywords in the first non-comment statement:

PCCU	(NCP MAJOR NODE)
VBUILD TYPE=DR	(DYN RECONFIG MAJOR NODE)
VBUILD TYPE=LOCAL	(LOCAL SNA MAJOR NODE)
VBUILD TYPE=SWITCHED	(SWITCHED MAJOR NODE)

All other VTAM major node types are ignored. The VTAM configuration member that VTAM uses to bring up the network must also be used for the VPDALL command. If that member does not contain all of the VTAM major nodes, or contains VTAM major nodes that you want VPDALL to ignore, you can create a different VTAM configuration member for VPDALL to use.

- VPD commands to collect PU data are generated for NCPs and all PU type-2 controllers.
- The command list is placed in the NetView command list partitioned data set allocated to the DSICLD *ddname*. If the DSICLD file is a concatenation of partitioned data sets, the command list is placed in the first partitioned data set.

Example: Reading and parsing a VTAM configuration

To read and parse the VTAM configuration specified in member ATCCON01, enter:

```
VPDALL CONFIG(ATCCON01)
```

Response

This generates a command list called VPDACT that contains VPD commands for all PUs and link segments.

Example: Reading and parsing a VTAM configuration, and running VPD commands

To read and parse the VTAM configuration specified in member ATCCON02, and run VPD commands for all PUs and link segments on operator ID AUTO1, enter:

```
VPDALL CONFIG(ATCCON02) EXEC OPER(AUTO1)
```

Example: Reading and parsing a VTAM configuration, and generating a command list

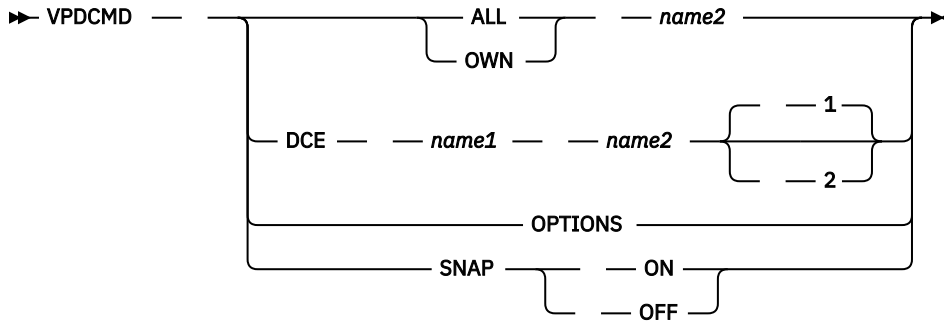
To read and parse the VTAM configuration specified in member ATCCON01, and generate a command list called DOM1VPD that contains VPD commands for all PUs and link segments, enter:

```
VPDALL CONFIG(ATCCON01) CREATE CLIST(DOM1VPD)
```

VPDCMD (NCCF)

Syntax

VPDCMD



Purpose of Command

The VPDCMD command retrieves vital product data (VPD) from supported devices. You can solicit data from the following devices:

- A specific PU
- A specific PU and its ports
- Data circuit-terminating equipment (DCE) between an NCP and a PU

To solicit data from a PU, network asset management uses the network management vector transport (NMVT) request/reply product set ID (PSID) mechanism. Only PUs that support NMVT request/reply PSID can be included in network asset management support. An attempt to solicit VPD from a device that does not support the request/supply PSID architecture can cause the keyboard to lock or extraneous data to display on the screen. Manual intervention, such as pressing the Reset key or clearing the screen, is required. This does not affect VPD collection. For more information, see the *IBM Z NetView Customization Guide*.

Note: Neither the NetView nor VTAM programs can predetermine whether a PU will respond correctly before a request is sent. The PU will either accept or reject the request.

Operand Descriptions

OWN

Indicates to solicit VPD from the specified PU only.

ALL

Indicates to solicit VPD from the specified PU and all the attached devices of the PU.

Note: The ALL option is not supported by NCP.

DCE

Indicates to solicit VPD from all the DCEs that exist in the path to the specified PU-pair.

name1

Indicates the name of an NCP.

name2

Indicates the name of a PU, including an NCP. For PU solicitation, this is the name of the PU being solicited for VPD. For DCE solicitation, this is the name of the PU just downstream from the link segments being solicited.

1|2

Specifies the starting link segment level (LSL) for which DCE data is being requested. If you specify a value of 1, the DCE data for LSL 1 and LSL 2 is displayed. If you specify a value of 2, only the DCE data for LSL 2 is displayed. Specifying a value is optional. The default is 1.

OPTIONS

Displays a list of initialization parameters of VPDTASK. See the VPDINIT statement in the *IBM Z NetView Administration Reference* for more details on the initialization parameters.

SNAP

Turns on a NetView trace with snap option. Use this operand when there is a problem with VPDCMD, and you want a problem determination aid. There is no default. Specify either ON or OFF. Issue the TRACE ON command for VPDTASK before issuing VPDCMD SNAP ON. If TRACE is off when VPDCMD SNAP ON is requested, an error message is issued. When TRACE for VPDTASK becomes active, the SNAP option is traced.

Restrictions

The following restrictions apply to the VPDCMD command:

- Collect information about a network control program (NCP) directly from the primary host. However, you can indirectly obtain VPD from the secondary side by issuing the START DOMAIN command and then using the ROUTE command to route the VPD command to the host that owns the NCP on the primary side of the link. Any attached devices in your network from which you want to collect VPD must have been turned on at least once. If the device has never been turned on before VPDCMD is issued, no VPD information about that device is stored in its PU. No VPD can be collected for that device.
- When this command is issued, the solicited VPD is displayed on your terminal and is not saved in storage. However, you can use a command list to automate the collection of VPD, and to write it to an external log.
- The VPDDCE and VPDPU command lists, which are provided by IBM, are used to solicit and log VPD from the network. These command lists are provided in the NetView sample library.
- See the *IBM Z NetView Administration Reference* for information about interpreting VPD messages, and the appropriate SNA manual for network Asset Management information.

Example: Requesting VPD from a PU and its ports

To request VPD from a PU and its ports, enter:

```
VPDCMD ALL L1PU1
```

Response

```
DW0009I REQUEST 'ALL' ACCEPTED FOR L1PU1 ,ID REQID = 0001
```

Example: Requesting vital product data from a PU

To request vital product data from PU N051F52, enter:

```
VPDCMD OWN N051F52
```

Example: Requesting vital product data from a PU and its attached devices

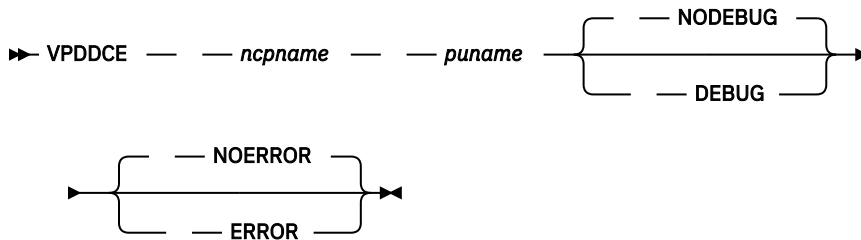
To request vital product data from PU H040PU and all devices attached to the PU, enter:

```
VPDCMD ALL H040PU
```

VPDDCE (NCCF; CNME0052)

Syntax

VPDDCE



Purpose of Command

The VPDDCE command list solicits and logs vital product data (VPD) from all the data circuit-terminating equipment (DCE) that is directly between a local communication controller and another PU.

This command list uses the VPDCMD command to solicit VPD, and the VPDLOG command to log the collected data to an external file. You can use this command list for a focal point collection of VPD in a distributed host environment.

VPD is collected from all the DCEs in the line that connects a communication controller and a PU, and is logged to an external file. Also, counters defined as task global variables are updated.

The VPDDCE command list builds a default record format that is written as a hardware monitor subrecord using the hardware monitor's SMF number, 37. The record type number (SMFVPD) is set to 37 as a common global variable in the CNMSTYLE member. However, this logic allows a record number in a user-defined range for convenience during customization.

Operand Descriptions

ncpname

Indicates the name of the communication controller that connects a local DCE. The maximum number of characters allowed is eight.

puname

Indicates the name of the PU at the other end of the line. The maximum number of characters allowed is eight.

NODEBUG

Specifies to suppress the VPD messages from the screen. NODEBUG is the default.

DEBUG

Specifies to show the VPD messages on the screen.

NOERROR

Specifies that a subtype record for errors is not written to the external logging facility when soliciting data from the device. NOERROR is the default.

ERROR

Specifies that a subtype record for errors is written to the external logging facility when soliciting data from the device.

Restrictions

The following restrictions apply to the VPDDCE command:

- Collect information about a network control program (NCP) directly from the primary host. However, you can indirectly obtain VPD from the secondary side by issuing the START DOMAIN command and

then using the ROUTE command to route the VPD command to the host that owns the NCP on the primary side of the link.

- If you specify two values for one operand, the last value for that operand is used. For example:

```
VPDDCE ncpname puname DEBUG NODEBUG
```

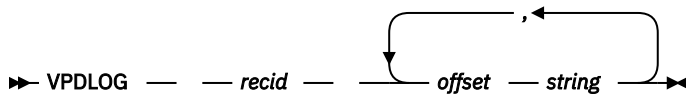
is equivalent to:

```
VPDDCE ncpname puname NODEBUG
```

VPDLOG (NCCF)

Syntax

VPDLOG



Purpose of Command

The VPDLOG command requests an external logging facility to put the collected vital product data (VPD) into an external file.

Operand Descriptions

recid

Is a record identifier. If system management facilities (SMF) are used, this is the number that is assigned to the SMF record. This number must be 37 or in the range of 128–255.

Note: If the default format that is provided by IBM is used, the record identifier is 37.

offset

Indicates a position where the following string must be placed in the record.

string

Indicates a string from VPD that must be written into the record. The maximum length of one string is 255 characters.

Note: Strings within single quotation marks are supported.

Restrictions

The following restrictions apply to the VPDLOG command:

- Collect information about a network control program (NCP) directly from the primary host. However, you can indirectly obtain VPD from the secondary side by issuing the START DOMAIN command and then using the ROUTE command to route the VPD command to the host that owns the NCP on the primary side of the link.
- If DSIELTSK (external logging facility task) fails, it issues an error message only once, at the time of the first failure. On subsequent issuances of the command, you do not receive another failure message. Check the external file and make sure that the data is logged. If it is not found, check the network log for message DSI170I.
- The strings must not overlap each other and must be in ascending order. The total length of a single record to be logged cannot exceed 500 characters. The total length does not include the header length.
- If you plan to use your records with programs or command lists that always expect fields with the same length, always put the pair (position, string) with the same position and same length as the last pair.

- If you plan to restrict access to this command, the command lists VPDDCE, VPDLOGC, VPDPU, and VPDXDOM also need to be restricted.

Example: Recording fields in a record

If you want to record *name1*, *name2*, or *linename* with the record ID 200, and the program that uses this record expects each of the fields to be eight characters long, enter:

```
VPDLOG 200 1 N1PU1 9 N1LINE1 17 N1NCP1 25 anything
```

The string *anything* at position 25 guarantees that the NCP name field starting at position 17 will be eight characters long.

Example: Indicating a quoted string from VPD with a record ID

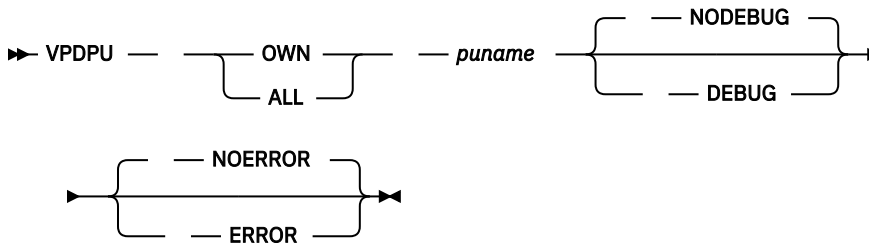
To indicate a quoted string from VPD with the record ID 130, enter:

```
VPDLOG 130 1 DATA 10 'IT'S A QUOTED STRING' 30 END
```

VPDPU (NCCF; CNME0051)

Syntax

VPDPU



Purpose of Command

The VPDPU command list solicits and logs vital product data (VPD) from a specified PU and, if specified, from all of its ports. It uses the VPDCMD command to solicit VPD, and the VPDLOG command to log the collected data to an external file.

You can use this command list for a focal point collection of VPD in a distributed host environment.

VPD is collected from the PU (and optionally from its ports), and is logged to an external file. Also, counters defined as task global variables are updated.

The VPDPU command list builds a default record format that is written as a hardware monitor subrecord using hardware monitor's SMF number, 37. The record type number (SMFVPD) is set to 37 as a common global variable in the CNMSTYLE member. However, this logic allows a record number in a user-defined range for convenience during customization.

Operand Descriptions

OWN

Solicits data from the node only.

ALL

Solicits data from the node and from all the attached devices.

puname

Indicates the name of the PU to be solicited. The maximum number of characters allowed is eight.

NODEBUG

Specifies to suppress the VPD messages from the screen. NODEBUG is the default.

DEBUG

Specifies to show the VPD messages on the screen.

NOERROR

Specifies that a subtype record for errors is not written to the external logging facility when soliciting data from the device. NOERROR is the default.

ERROR

Specifies that a subtype record for errors is written to the external logging facility when soliciting data from the device.

Restrictions

The following restrictions apply to the VPDPU command:

- Collect information about a network control program (NCP) directly from the primary host. However, you can indirectly obtain VPD from the secondary side by issuing the START DOMAIN command and then using the ROUTE command to route the VPD command to the host that owns the NCP on the primary side of the link.
- If you specify two values for one operand, the last value for that operand is used. For example:

```
VPDPU OWN puname DEBUG NODEBUG
```

is equivalent to:

```
VPDPU OWN puname NODEBUG
```

VRST (NCCF; CNME0038)

Syntax**VRST**

➤ VRST — — *status* ➤

Purpose of Command

The VRST command list displays the meaning of the virtual route status.

Operand Descriptions***status***

Is the virtual route status code

Example: Displaying the meaning of a virtual route status

To display the meaning of virtual route status BLCKD, enter:

```
VRST BLCKD
```

VSAMPOOL (NCCF)

Syntax

VSAMPOOL

➤ VSAMPOOL ➤

Purpose of Command

The VSAMPOOL command displays statistics on NetView VSAM resource pool utilization when the NetView program has been defined to use local shared resources (LSRs) or deferred writing of records (DFRs).

The VSAMPOOL command displays statistics from the LSR resource pool. The LSR resource pool is subdivided into buffer pools determined by control interval sizes. You define the LSR resource pool and buffer pools with the DSIZVLSR module. The LSR resource pool can be further subdivided into DATA and INDEX buffer pools. The VSAMPOOL command displays the statistics for the DATA and INDEX buffer pools separately when separate INDEX buffers have been defined.

For more information about the DSIZVLSR module, see *IBM Z NetView Installation: Getting Started*.

The following information is displayed for each active buffer pool in the LSR resource pool:

CINV

Control interval size of buffer pool

BUFNO

Number of buffers in the buffer pool

BFRFND

Number of VSAM retrieve requests that were satisfied by a record in a buffer

BUFRDS

Number of VSAM retrieve requests that were not satisfied by a record in a buffer and that required I/O

NUIW

Number of write I/Os that VSAM had to perform because there were no buffers available to do a read

UIW

Number of write I/Os that were not deferred

ERCT

The number of write errors that have occurred

The VSAMPOOL command is helpful in tuning NetView VSAM use. You can determine a more optimal number of VSAM LSR buffers to allocate. Underallocating or overallocating buffers can degrade performance by increasing storage use and paging. You can obtain additional information about VSAM data sets from the LISTCAT command. For additional information about improving VSAM performance, see the *IBM Z NetView Tuning Guide*.

Example: Displaying information about buffer pools

To display information about all VSAM LSR or DFR buffer pools, enter:

```
VSAMPOOL
```

Response

Information similar to the following is displayed:

```
CNM260I VSAM LSR/DFR RESOURCE POOL STATISTICS
BNH091I BUFFER TYPE = DATA
CNM948I  CINV  BUFNO  BFRFND    BUFRDS    NUIW    UIW    ERCT
CNM261I  7168    4      17       32        0     29     0
CNM261I  8192    4        0       11        4      3     0
```

CNM261I	16384	4	0	0	0	0	0
CNM261I	18432	16	0	0	0	0	0
CNM261I	24576	13	0	0	0	0	0
BNH090I	BUFFER TYPE = INDEX						
CNM948I	CINV	BUFNO	BFRFND	BUFRDS	NUIW	UIW	ERCT
CNM261I	512	3	0	0	0	0	0
CNM261I	1536	5	0	0	0	0	0
CNM261I	2048	3	11	1	0	1	0
CNM261I	2560	10	0	0	0	0	0
CNM261I	4096	4	46	5	0	2	0
CNM262I	END OF DISPLAY						

VTAMCMD (AON)

Syntax

VTAMCMD

➤ VTAMCMD ➤

Purpose of Command

The VTAMCMD command provides a full-screen interface to issue common commands.

Usage Notes

When you enter the VTAMCMD command, a full-screen panel is displayed showing the last commands entered from this panel. Move your cursor to the command you want to issue and press Enter, or you can type a new command to be issued.

Restrictions

- The AON tower and the SNA subtower must be enabled in the CNMSTYLE member to successfully run this command.
- This command operates in full-screen mode only.

VTAMOPT (AON)

Syntax

VTAMOPT

➤ VTAMOPT ➤

Purpose of Command

The VTAMOPT command provides a full-screen interface to display and change VTAM options.

Usage Notes

When you enter the VTAMOPT command, a full-screen panel is displayed with the current VTAM option settings. From the panel, you can type over the existing entry and press Enter to make needed changes.

Restrictions

- The AON tower and the SNA subtower must be enabled in the CNMSTYLE member to successfully run this command.
- This command operates in full-screen mode only.

WHAT (CANZLOG only)

Syntax

WHAT

➤ WHAT ➤

Purpose of Command

When browsing the Canzlog log, the WHAT subcommand displays information about the filtering for the current browse session:

- The name of the filter, if any.
- Information from the remark field of the filter, if any.
- The filter specifications, as resolved into keyword and value format.

When the Enter key is used to start the WHAT command, the information is always displayed in the immediate message area and truncated if necessary. However, when WHAT is typed or started via PF or PA key (PA2 is the default), then the information is displayed in the immediate message area if less than 80 characters, but is displayed on the NCCF screen otherwise.

WHENCE (CANZLOG only)

Syntax

WHENCE

➤ WHENCE ➤

Purpose of Command

The WHENCE command displays immediate message DWO672I with information about the selected message in the CANZLOG display. If no message is selected, the first message on the page is assumed.

WHO (NCCF; CNME1019)

Syntax

WHO

➤ WHO ➤

Purpose of Command

The WHO command list displays the status of all operator terminals, NetView-NetView task (NNT) sessions requested by other NetView systems, and information about your session.

After entering the WHO command list, you see information similar to the following:

```
* CNM01   WHO
C CNM01   LIST STATUS=OPS
- CNM01   OPERATOR: OPER1      TERM: A01A701  STATUS: ACTIVE
- CNM01   OPERATOR: AUTO1      TERM: AUTO1    STATUS: ACTIVE
- CNM01   OPERATOR: AUTO2      TERM: AUTO2    STATUS: ACTIVE
- CNM01   END OF STATUS DISPLAY
C CNM01   LIST STATUS=NNT
- CNM01   MAX SESS: 00000005
```

```

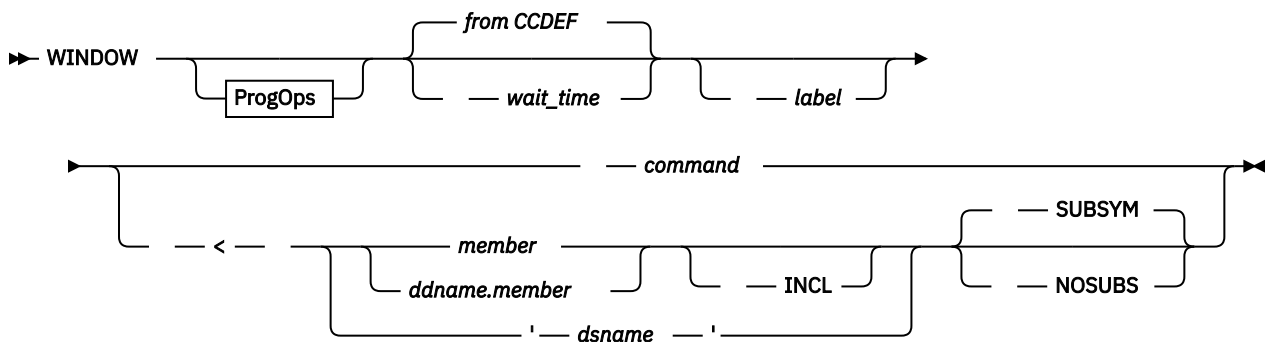
- CNM01 NO ACTIVE NCCF TO NCCF SESSIONS FOUND
C CNM01 LIST OPER1
- CNM01 STATION: OPER1 TERM: A01A701
- CNM01 HCOPY: NOT ACTIVE PROFILE: DSIPROFA
- CNM01 STATUS: ACTIVE
- CNM01 AUTHRCVR: NO CONTROL: GLOBAL
- CNM01 NGMFADMN: NO DEFAULT MVS CONSOLE NAME: NONE
- CNM01 OP CLASS LIST: 2
- CNM01 DOMAIN LIST: CNM01 (I) CNM02 (I) CNM99 (I) B01NV (I)
- CNM01 ACTIVE SPAN LIST: NONE
- CNM01 END OF STATUS DISPLAY

```

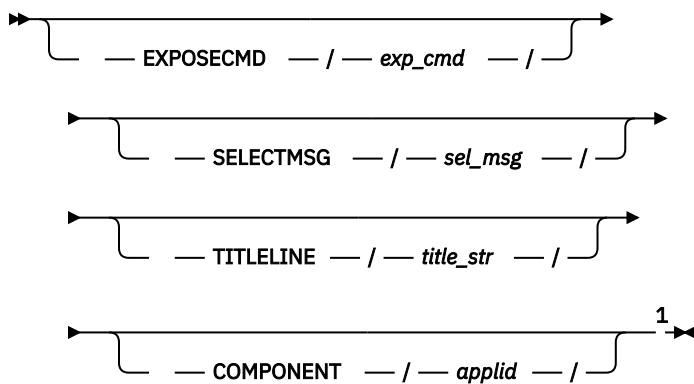
WINDOW (NCCF; CNME1505)

Syntax

WINDOW



ProgOps



Notes:

¹ These parameters are typically used by programmers.

IBM-Defined Synonyms

Command or Operand	Synonym
SUBSYM	SUBS
WINDOW < args	< args

Purpose of Command

The WINDOW command is a full screen application that captures and displays data from other commands that usually display messages. The WINDOW command facilitates searching the captured data and enables you to scroll forward and back, as well as left and right. WINDOW is also ROLLable.

Operand Descriptions

wait_time

Specifies the maximum time in seconds that WINDOW will wait for a response from *command*. Valid values are 1–10000000. Default value is taken from the CCDEF command definitions.

label

Specifies the target task or system where the command is to be run. Command prefix labels can also include a timeout value as part of the label syntax. For more information about syntax and usage of labels, see *IBM Z NetView User's Guide: NetView*.

EXPOSECMD

The delimited string *exp_cmd* consists of an external REXX subroutine name followed by one or more function names. Function names are used by the WINDOW command as subcommands and can override the existing WINDOW subcommands. This operand is generally used when WINDOW is invoked from within a command list. Sample CNME1096 contains an example of EXPOSECMD usage. Also, see sample CNMEXEC.

SELECTMSG

The delimited string *sel_msg* specifies a string that causes the WINDOW command to detect commands embedded in the data to be displayed. This operand is generally used when WINDOW is invoked from within a command list.

TITLELINE

The delimited string *title_str* specifies a string that is displayed as the title of the panel. The string has no effect on the operation of the WINDOW command or the function of any subcommand. If TITLELINE is not specified, the value used for the title is taken from the value of the *command* variable. This operand is generally used when WINDOW is invoked from within a command list.

COMPONENT

The variable *applid* is a 1- to 8-character component name. This name can be used to set or view the PF key definitions used with the panel displayed by WINDOW. When COMPONENT is specified, the application does not inherit the WINDOW PF key definitions.

command

The command for which WINDOW is to collect data. The output of this command will not be exposed.

WINDOW does not support sending commands other than PIPE to remote NetView domains running NetView Version 2 Release 3. You can enclose any command inside a PIPE command. See the online help for additional information about PIPE NETVIEW, PIPE CORRWAIT, and PIPE CONSOLE.

<

Indicates the < (From disk) PIPE stage.

Enter < member for a shorthand method of specifying the PIPE < (From disk) stage command. The WINDOW command automatically encloses your argument in an appropriate pipeline specification. It also appends the data set number and the member associated with each line, starting in column 81. The data set number is defined as -1 for operator data set members (see the OVERRIDE command), or 0 for INSTORE members. Otherwise, the data set number is defined as a positive integer correlating to the order of the LISTA output for the *ddname* in which the member was found.

You can also specify < as a synonym for the WINDOW command to obtain the same function and avoid extra keystrokes.

ddname

Specifies the *ddname* from where to read the member. When *ddname* is not specified, the default is to search all DDs. When specifying *ddname*, a period (.) is used to separate it from the member name. Do not use spaces before or after the period.

The supported *ddnames* are those which the DSIDKS macro supports. Issue the BR ! command for a list of supported data sets. To access dynamically allocated data sets, use *dsname*.

member

Specifies the 1- to 8-character name of the member or file to be read (parameter synonyms are not supported). This is a member of the data set concatenation associated with the *ddname* being used.

dsname

Specifies the data set name to be read by the PIPE < stage. You can specify a member name in parentheses as part of the data set name. The *dsname* must be enclosed in single quotation marks. Using *dsname* enables you to browse data wider than 80 columns as well as data sets that are not included in the standard NetView DD names.

INCL

Specifies that %INCLUDE are expanded when the member or file is read.

SUBSYM

Specifies that if the *membername* contains MVS system symbolics, the system symbolics are substituted in the statements before they are displayed. The &DOMAIN symbolic that is supplied with the NetView program is also included in the substitution process. Substitution is always performed on the &DOMAIN symbolic, unless substitution was disabled when the NetView program was started. For MVS and user-defined system symbolics, substitution is not performed if you are not running on an MVS system, you are running on an MVS system prior to MVS Version 5 Release 2, substitution was disabled when the NetView program was started, or you have not defined an MVS system symbolic on your MVS system. SUBSYM is the default.

NOSUBS

Specifies that if the *membername* contains MVS system symbolics, the system symbolics are not substituted in the statements before they are displayed.

Usage Notes

Consider the following when using the WINDOW command:

- For the WINDOW command, if the global variable CNMIMWINDOW contains a non-null value, the value is displayed at the bottom of the WINDOW panel. This is useful for displaying the settings of your PF keys. You can set the value of this global variable using the PFKDEF command. For more information, see the *IBM Z NetView Customization Guide*.
- You can use the following commands (or their associated PF keys) while you are using the WINDOW command:
 - ALL
 - BACK
 - BOTTOM
 - END
 - FIND
 - FORWARD
 - LEFT
 - REFRESH
 - RETURN
 - RIGHT
 - TOP

Other subcommands can be defined using EXPOSECMD.

- To display the output of a mixed-case command you can prefix the WINDOW command with NETVASIS or you can enter OVERRIDE NETVASIS=YES.

You can also use NETVASIS from the WINDOW command line.

- While using the WINDOW command, you can use the HELP function to display a list of functions and subcommands available to you.
- To use a data set with records longer than 80 characters, use the *dsname* form of the WINDOW command. This avoids the data wrapping and a data set being unreadable.

Example: Displaying a data set wider than 80 characters

To display member XYZ of data set USER.LISTING which is greater than 80 characters wide, enter:

```
WINDOW < 'USER.LISTING(XYZ)'
```

Example: Run a command on a remote NetView system

To run a command on a remote NetView system and display the response in a window, enter:

```
WINDOW CNM03/OPER1: D NET,CDRMS
```

WINDOW transmits the command to CNM03 for autotask OPER1. WINDOW automatically determines proper wait time both for the operation of the DISPLAY command at the remote site and for the RMTCMD transfer (assuming CNM03 is running NetView V2R4 or higher).

Example: Display named safe contents with title

To display the contents of a named PIPE SAFE in a window with a title of My Title, enter:

```
WINDOW TITLELINE %My Title% PIPE SAFE anyname | CONSOLE
```

The issuing application has issued WINDOW after collecting data from one or more sources into a named safe. WINDOW displays the data collected but shows a title that is more meaningful to the operator. Note that the WINDOW REFRESH subcommand does not collect any new data in this case.

WRAP (NCCF; CNME1020)

Syntax

WRAP

►► WRAP ◄◄

Purpose of Command

The WRAP command list changes the setting of AUTOWRAP. If AUTOWRAP is on, this command list turns it off. If AUTOWRAP is off, this command list turns it on.

Restrictions

If you use the WRAP command list more than once, it alternates between turning AUTOWRAP on and turning AUTOWRAP off. When AUTOWRAP is off, WRAP restores the on value for AUTOWRAP to the one that was used previously.

WRITESEC (NCCF)

Syntax

WRITESEC

►► WRITESEC — — DD — = — *ddname* — — DSN — = — *dsname* ◄◄

Purpose of Command

The WRITESEC command checks an operator's authority to write to a VSAM data set or DD name and optionally checks a member.

Operand Descriptions

ddname

Specifies the DD name against which the authority to write is checked. For VSAM DD names, WRITESEC converts *ddname* to its underlying data set name, then checks for authority to write to that data set name.

dsname

Specifies the data set name against which authority to write is checked.

Usage Notes

Consider the following when using the WRITESEC command:

- The WRITESEC command acts as a central point to control write access by DSIVSMX.
- When the WRITESEC command is entered by an operator, it results in one of the following messages:
 - DSI633I for successful write access
 - DSI213I for unsuccessful write access

The data is not accessed nor its existence verified.

- When protecting specific data set names using CMDAUTH=TABLE or CMDAUTH=SAF, slashes (/) must be substituted for periods (.) in the data set name.

Restrictions

The following restrictions apply to the WRITESEC command:

- When the WRITESEC command is issued with a VSAM DD name, the command normally converts the DD name to the underlying data set name. However, WRITESEC cannot process VSAM DD names under the primary program operator interface task (PPT).

Return Codes

Return Code

Meaning

- | | |
|-----------|--|
| 0 | The authorization check was successful. |
| 4 | The operator is not authorized to issue the command. |
| 8 | A syntax error occurred. |
| 12 | The DD name is unknown. |
| 16 | Dynamic names are not allowed under the PPT. |

Example: Checking write access to VSAM data sets

To determine whether write access exists for a VSAM data set, enter:

```
WRITESEC DD=xxxVSAM
```

Where xxx are the first three characters in the VSAM data set name.

Response

If write access is allowed, the response is message DSI633I; otherwise, the response is message DSI213I.

X25MONIT (AON)

Syntax

X25MONIT

► X25MONIT ◄

Purpose of Command

The X25MONIT command displays all X.25 resources defined in the control file. From the list panel, you can add, change, and delete X.25 resources.

Restrictions

The AON tower and the SNA subtower must be enabled in the CNMSTYLE member to successfully run this command.

Appendix A. Command List Commands

This section describes NetView commands that can be entered only from command lists or command processors written in a high-level language.

DEGRANT (REXX)

Syntax

DEGRANT

➤ DEGRANT — — *command* ➤

Purpose of Command

The DEGRANT command issues the specified command without any privileged status and with the authority of the task from where the DEGRANT command is running. REXX users must run AUTOBYPAS to obtain privileged status before issuing the DEGRANT command.

Operand Descriptions

command

Specifies the command to be issued. The specified command can run only if the task where the DEGRANT command runs is authorized. The specified command is subject to authority checking using the user name of the task where DEGRANT was started.

Return codes

Return Code

Meaning

–4

Installation exit 03 generated USERDROP.

–9

Authorizing identity cannot be switched because of lack of privilege by caller.

–500 to –599

Failure attempting to call installation exit 03. Contact IBM Software Support.

–108

Command is Type=I or Type=P.

–112

Command search failure, usually because the command is too long.

–116

ACCESS is not authorized. Command authorization restrictions prevent processing.

–120

Command is Type=D.

Restrictions

This command cannot be issued from a command line.

Examples

The following example shows how to include the AUTOBYPAS function when issuing the DEGRANT command:

```

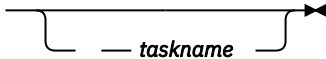
/* REXX */
Call AUTOBYPAS 'ON' /* obtain privileged status */
'EXCMD OPER1,DEGRANT SUBMIT (ABC)' /* Submit job ABC */
/* No authority check is performed for EXCMD, but SUBMIT is invoked
   only if OPER1 is authorized. */

```

DSITSTAT (REXX)

Syntax

DSITSTAT

➔ DSITSTAT 

Purpose of Command

The DSITSTAT command is coded in a REXX procedure to retrieve resource statistics for every task in a NetView system. The REXX procedure can take action automatically or display data using WINDOW, VIEW, or messages. The statistics are returned in message BNH159I. If DSITSTAT is not issued within a command procedure, message DSI290I is returned.

Operand Descriptions

taskname

The name of the NetView task for which statistics are retrieved. If *taskname* is not specified, statistics are retrieved for all tasks.

Example: Defining a data services task and its VSAM files

For an example of the use of DSIVSAM, refer to sample CNMS8019. This sample is a CLIST that:

1. Creates a VSAM file by running the DEFVSAMS REXX procedure (sample CNMS8020).

The DEFVSAMS REXX procedure is an example of how to use the DSIVSMX IDCAMS command to create VSAM files without restarting the NetView program. Change the VOL(CPDLB2) to the volume name of your VSAM files. The other data set parameters are set to work with these examples; however, when you write applications, you can choose the parameters appropriate to your installation.

2. Allocates the files to DDs using the ALLOC command.
3. Starts Data Services Tasks that manage the VSAM file.

The MEM=*member_name* parameters contain Data Services Task definitions that refer to the DD names allocated in the previous step. When the task starts, the VSAM files are available to the DSIVSAM command.

Example: A DSITSTAT command response

This example shows a successful response to a DSITSTAT command:

Response:

```

BNH159I opername
taskname curcpu sesscpu maxcpu limcpu curget maxget limget slowget curmqi
sessmqi maxmqi limmqi curmqo sessmqo maxmqo limmqo curi/o sessi/o maxi/o limi/o
curpen sesspen pendpen totpen getkbbm getsess g24kbbm g24sess frekbbm fresess
f24kbbm f24sess pentime pentask

```

The BNH159I message is formatted without header lines and is in specific columns. This data is processed by a user-written REXX procedure. The output is one (non-MLWTO) line per task. All numbers are expressed in decimals with no punctuation. All data is aligned in columns to assist in parsing the data using a REXX procedure. Other than the *opername* and *taskname*, no columns are blank. Use the

SUBSTR() function when this data is parsed. [Table 9 on page 517](#) explains the output of a DSITSTAT command:

Table 9. Output of a DSITSTAT Command

Token	Position	Description
BNH159I	1-7	The message ID for this successful response.
	8	Blank
<i>opername</i>	9-16	The operator name or task ID (TVBOPID) of an active task.
	17	Blank
<i>taskname</i>	18-25	The LU name or task ID (TVBLUNAM) of an active task.
	26	Blank
<i>curcpu</i>	27-30	The current task utilization in 0.01% units. The range is 0 - 9999.
	31	Blank
<i>sesscpu</i>	32-35	The task utilization for the entire session in 0.01% units. The range is 0 - 9999.
	36	Blank
<i>maxcpu</i>	37-40	The highest task utilization for the entire session in 0.01% in units. The range is 0 - 9999.
	41	Blank
<i>limcpu</i>	42-45	The applicable processor limit from the OVERRIDE or DEFAULTS command in 0.01% units. The range is 0 - 9999.
	46	Blank
<i>curget</i>	47-52	The current DSIGET storage utilization in KB. The range is 0 - 999999.
	53	Blank
<i>maxget</i>	54-59	The maximum storage limit for this task in KB. The range is 0 - 999999.
	60	Blank
<i>limget</i>	61-66	The applicable maximum storage limit for this task in KB. The range is 0 - 999999.
	67	Blank
<i>slowget</i>	68-73	The applicable storage slowdown limit for this task in KB. The range is 0 - 999999.
	74	Blank
<i>curmqi</i>	75-83	The current rate of message traffic from other tasks to this task in KB per minute. The range is 0 - 999999999.
	84	Blank
<i>sessmqi</i>	85-93	The overall session rate of message traffic from other tasks to this task in KB per minute. The range is 0 - 999999999.
	94	Blank

Table 9. Output of a DSITSTAT Command (continued)

Token	Position	Description
<i>maxmqi</i>	95-103	The maximum measured limit for the rate of message traffic from other tasks to this task in KB per minute. The range is 0 - 999999999.
	104	Blank
<i>limmqi</i>	105-113	The applicable set limit for the rate of message traffic from other tasks to this task in KB per minute. The range is 0 - 999999999.
	114	Blank
<i>curmqo</i>	115-123	The current rate of message traffic from this task to other tasks in KB per minute. The range is 0 - 999999999.
	124	Blank
<i>sessmqo</i>	125-133	The overall session rate of message traffic from this task to other tasks in KB per minute. The range is 0 - 999999999.
	134	Blank
<i>maxmqo</i>	135-143	The maximum measured limit for the rate of message traffic from this task to other tasks in KB per minute. The range is 0 - 999999999.
	144	Blank
<i>limmqo</i>	145-153	The applicable set limit for the rate of message traffic from this task to other tasks in KB per minute. The range is 0 - 999999999.
	154	Blank
<i>curi/o</i>	155-163	The current number of I/Os per minute for this task. The range is 0 - 999999999.
	164	Blank
<i>sessi/o</i>	165-173	The number of I/Os per minute recorded for this task. The range is 0 - 999999999.
	174	Blank
<i>maxi/o</i>	175-183	The maximum number of I/Os per minute recorded for this task. The range is 0 - 999999999.
	184	Blank
<i>limi/o</i>	185-193	The applicable limit for the number of I/Os per minute for this task. The range is 0 - 999999999.
	194	Blank
<i>curpen</i>	195-198	The current penalty time ratio in 0.01%. The range is 0 - 9999.
	199	Blank
<i>sesspen</i>	200-203	The session penalty time ration in 0.01%. The range is 0 - 9999.
	204	Blank

Table 9. Output of a DSITSTAT Command (continued)

Token	Position	Description
<i>pendpen</i>	205-213	The current task delay time in 0.01 seconds. The range is 0 - 999999999.
	214	Blank
<i>totpen</i>	215-223	The total time spent in penalty wait in 0.01 seconds. The range is 0 - 999999999.
	224	Blank
<i>getkbm</i>	225-233	The current rate of DSIGET activity in KB per minute. The range is 0 - 999999999.
	234	Blank
<i>getsess</i>	235-243	The session rate of DSIGET activity in KB per minute. The range is 0 - 999999999.
	244	Blank
<i>g24kbm</i>	245-253	The current rate of DSIGET activity in KB per minute for only the storage in the 24-bit address area. The range is 0 - 999999999.
	254	Blank
<i>g24sess</i>	255-263	The session rate of DSIGET activity in KB per minute for only the storage in the 24-bit address area. The range is 0 - 999999999.
	264	Blank
<i>frekbm</i>	265-273	The current rate of DSIFRE activity in KB per minute. The range is 0 - 999999999.
	274	Blank
<i>fresess</i>	275-283	The session rate of DSIFRE activity in KB per minute. The range is 0 - 999999999.
	284	Blank
<i>f24kbm</i>	285-293	The current rate of DSIFRE activity in KB per minute for only the storage in the 24-bit address area. The range is 0 - 999999999.
	294	Blank
<i>f24sess</i>	295-303	The session rate of DSIFRE activity in KB per minute for only the storage in the 24-bit address area. The range is 0 - 999999999.
	304	Blank
<i>pentime</i>	305-313	The number of penalty seconds that this task has caused other tasks because of its MAXMQIN limit. The range is 0 - 999999999.
	314	Blank

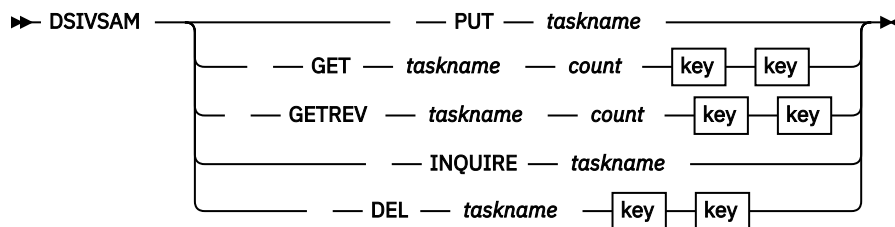
Table 9. Output of a DSITSTAT Command (continued)

Token	Position	Description
<i>pentask</i>	315-323	The first 8 bytes contain the name of the task that caused this task to slow down because of a MAXMQIN limit. The rightmost byte contains an indicator as follows: • + Indicates that the task is currently causing a penalty. • - Indicates that the task is not currently causing a penalty, but was the last task to do so.

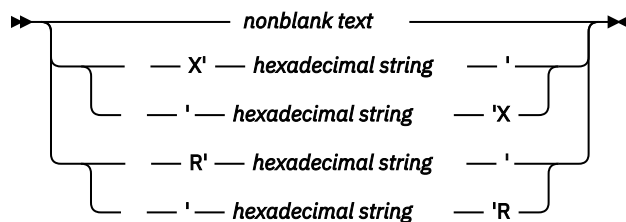
Note: If *taskname* is not valid or inactive, no messages are produced and a return code of 8 is returned.

DSIVSAM (NCCF)

Syntax



key



Purpose of Command

The DSIVSAM command processor can access keyed VSAM files that are defined by NetView data services tasks, such as DSILOG. This allows for implementation of all kinds of VSAM applications, including end-use application development in REXX (with the pipeline facility) and intensive VSAM diagnostics.

The DSIVSAM command runs where regular commands are supported, but not from PPT task, to access any keyed VSAM file on any data services task that supports VSAM.

The DSIVSAM command has the following functions:

- Various VSAM functions that are intended to be used with NetView pipelines.
- Ability to write multiple records or one at a time.
- Ability to read a single record or a range of records.
- Ability to read using generic (short) keys.
- Ability to delete a single record or a range of records.
- Ability to display all VSAM characteristics of the active file (similar to the LISTCAT command).
- Ability to accept any length key in any position in the record. You must format the records properly when writing them to VSAM and put the key in the correct location within the data record.

- Ability to adapt to any logical record length file.

Samples CNMS8016 and CNMS8021 illustrate the use of the DSIVSAM command.

Usage Notes

Consider the following when using the DSIVSAM command:

- In general, the DSIVSAM command is intended for, and mostly work best in PIPEs.
- In general, all operands are delimited by at least one blank, and commas, periods, and equal signs are delimiters. Using blanks improves readability.
- You can control who is allowed to initiate actions on VSAM data sets with the command security provided for the DSIVSAM command. You can protect data sets for NetView Data Services Tasks (DST) by using the DATASET class of the SAF product.

For more information, refer to the *IBM Z NetView Administration Reference*.

- DSIVSAM is an asynchronous command processor, operating on both the task where the PIPE runs, and on the DST, which you specify as a command parameter. Use the CORRWAIT stage command to make the PIPE wait for the completion of the DST requests. Consider the following:

- Code a high value for the CORRWAIT timeout value when using DSIVSAM.

Regardless of the timeout value specified, the PIPE finishes as soon as the DST requests are done.

- DSIVSAM requests from the same task are run sequentially at the DST, regardless of how many DSRBs are specified there.
- DSIVSAM PUT or DEL requests that specify a large amount of data do not receive a completion message until the last I/O is completed. This is one reason for using the high time value on CORRWAIT.
- DSIVSAM GET requests return data one record at a time to help maximize the parallel processing of the PIPE and DST. A high value for the CORRWAIT time might be necessary if a DSIVSAM GET request is issued right after a DSIVSAM PUT or DSIVSAM DEL. The DSIVSAM GET does not read the first buffer until all the previous I/O has completed.
- The PIPE is notified if the DST is inactive or ends while a request is being processed. The CORRWAIT ends when the task ends, and you might not receive as many buffers as requested.
- A low CORRWAIT time value can result in erratic results.
- A high CORRWAIT time value times out if a DST is unresponsive. Otherwise, the CORRWAIT ends immediately if the task ends and the request is completed successfully.

Restrictions

The NetView log uses a single DSRB for all I/O operations to the NetView log. Consider the following before using DSIVSAM to access the NetView log:

- Use the DSIWLS macro to write to the NetView log, not DSIVSAM. The DSILOG task requires that no other means of writing to the log be used.
- Using DSIVSAM to read records from the active NetView log delays the logging of new data. If necessary, read only a few records and determine whether reading the records interferes with the DSILOG throughput or causes storage queuing backlogs.
- DSIVSAM provides easy access to VSAM files currently managed by Data Services Tasks. Do not use it to access IBM data sets for which the format of the data is not published. IBM reserves the right to change the format of VSAM files on tasks, for example BNJDSERV (NPDA) and AAUTSKLP (NLDM). In no case can software, except as shipped with the NetView product, be used to reference this data.
- The maximum logical record length supported by the DSIVSAM GET and DSIVSAM GETREV commands is 32000.

Example: Defining a data services task and its VSAM files

For an example of the use of DSIVSAM, refer to sample CNMS8019. This sample is a CLIST that:

1. Creates a VSAM file by running the DEFVSAMS REXX procedure (sample CNMS8020).

The DEFVSAMS REXX procedure is an example of how to use the DSIVSMX IDCAMS command to create VSAM files without restarting the NetView program. Change the VOL(CPDLB2) to the volume name of your VSAM files. The other data set parameters are set to work with these examples; however, when you write applications, you can choose the parameters appropriate to your installation.

2. Allocates the files to DDs using the ALLOC command.
3. Starts Data Services Tasks that manage the VSAM file.

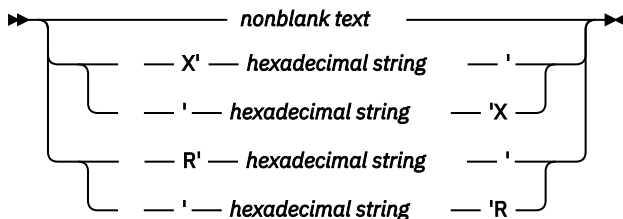
The MEM=*member_name* parameters contain Data Services Task definitions that refer to the DD names allocated in the previous step. When the task starts, the VSAM files are available to the DSIVSAM command.

DSIVSAM DEL (NCCF)

Syntax

➤ DSIVSAM — — DEL — — *taskname* — key — key ➤

key



Purpose of Command

The DSIVSAM DEL command deletes the record with the matching key or all records within the boundaries of the stated key range. Message DSI633I is issued when the request has completed successfully.

This command is similar to the DSIVSMX DEL command, which is illustrated in sample CNMS8017.

Operand Descriptions

taskname

The Data Services Task to be accessed.

key

The key range to delete, specified as one or two keys that determine the range of records to be deleted.

- For the first key specified, the options and padding rules are:
 - A single character string with no embedded blanks. The key is padded with hexadecimal nulls to the full key length and the characters are displayed left-aligned.
 - A hexadecimal character string in the form X'hexdigits' or 'hexdigits'X. The key is padded with hexadecimal nulls to the full key length and the 'hexdigits' are displayed left-aligned.
 - A hexadecimal character string in the form R'hexdigits' or 'hexdigits'R. The key is padded with hexadecimal nulls to the full key length and the 'hexdigits' are displayed right-aligned. This notation is useful if the VSAM file is intended to have numeric (binary) keys.

When a single key is used, only the record exactly matching the resulting full key value is deleted.

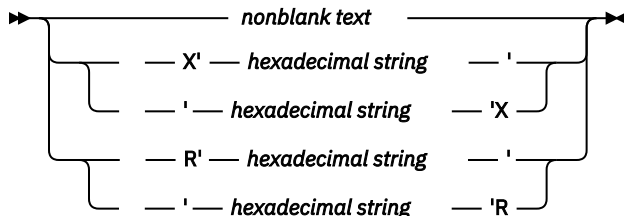
- For the second key specified, the options and padding rules are:
 - A single character string with no embedded blanks. The key is padded with hexadecimal ones to the full key length and the characters are displayed left-aligned.
 - A hexadecimal character string in the form X'hexdigits' or 'hexdigits'X. The key is padded with hexadecimal ones to the full key length and the 'hexdigits' are displayed left-aligned.
 - A hexadecimal character string in the form R'hexdigits' or 'hexdigits'R. The key is padded with hexadecimal nulls to the full key length and the 'hexdigits' are displayed right-aligned. This notation is useful if the VSAM file is intended to have numeric (binary) keys.
 - When two keys are used, all records in the resulting key range are deleted.

DSIVSAM GET (NCCF)

Syntax

➤ DSIVSAM — — GET — — *taskname* — — *count* — key — key ➤

key



Purpose of Command

The DSIVSAM GET command reads a single VSAM record or a range of records within the bounds of the specified key range. The records are returned as messages; data is returned as characters. The count value of *nn* limits the read to the first *nn* matching records. The maximum logical record length supported by the DSIVSAM GET command is 32000.

The first key specifies the starting key for the request. The value can be shorter than the full key length. The value is padded on the right to the full key length with binary zeros. The first record read that has a key greater than, or equal to, the starting key is returned with the key embedded in its physical location in the record text.

The second key is the optional ending key for the read. If the second key is omitted, the value defaults to the first key value, but is padded on the right with binary ones instead of zeros. If the value of the second key is specified, it is also padded with binary ones to the length of the VSAM key for the data set. For example, if a key of A is specified, the GET request finds the first record with an A in the first position of the key.

Note: The second key must resolve to an unsigned binary value that is equal to, or greater than, the first key. Otherwise, no data is read.

This command is similar to the DSIVSMX GET command, which is illustrated in samples CNMS8015 and CNMS8017.

Operand Descriptions

taskname

The Data Services Task to be accessed.

count

The maximum number of records to be read.

key

The key range to be read, which is specified as one or two keys that determine the range of records to be read.

For the first key specified, the options and padding rules are:

- A single character string with no embedded blanks. The key is padded with hexadecimal nulls to the full key length and the characters are displayed left-aligned.
- A hexadecimal character string in the form X'hexdigits' or 'hexdigits'X. The key is padded with hexadecimal nulls to the full key length and the 'hexdigits' are displayed left-aligned.
- A hexadecimal character string in the form R'hexdigits' or 'hexdigits'R. The key is padded with hexadecimal nulls to the full key length and the 'hexdigits' are displayed right-aligned. This notation is useful if the VSAM file is intended to have numeric (binary) keys.

For the second key specified, the options and padding rules are:

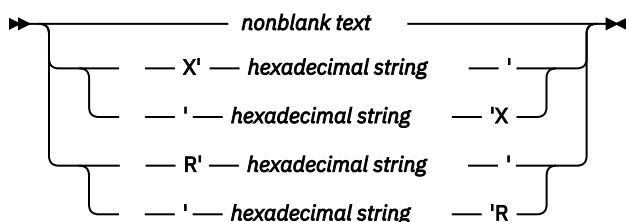
- A single character string with no embedded blanks. The key is padded with hexadecimal ones to the full key length and the characters are displayed left-aligned.
- A hexadecimal character string in the form X'hexdigits' or 'hexdigits'X. The key is padded with hexadecimal ones to the full key length and the 'hexdigits' are displayed left-aligned.
- A hexadecimal character string in the form R'hexdigits' or 'hexdigits'R. The key is padded with hexadecimal nulls to the full key length and the 'hexdigits' are displayed right-aligned. This notation is useful if the VSAM file is intended to have numeric (binary) keys.
- If the second key is omitted, the value for the second key is the original value of the first key, using the padding rules for the second key value.

DSIVSAM GETREV (NCCF)

Syntax

➤ DSIVSAM — — GETREV — — *taskname* — — *count* — key — key ➤

key



Purpose of Command

The DSIVSAM GETREV command is similar to DSIVSAM GET, except that VSAM records are read in reverse sequence. As with the GET command, the records are returned as a message. The first key is expected to resolve to an unsigned value that is greater than, or equal to, the second key. The first key is padded with binary ones and the second key is padded with binary zeros, if necessary. The second key defaults to a zero-padded equivalent of the first key.

For each byte of data in the record, there is one character in the output. The maximum logical record length supported by the DSIVSAM GETREV command is 32000.

Note: The first key must resolve to an unsigned binary value that is equal to, or greater than, the second key. Otherwise, no data is read.

This command is also similar to the DSIVSMX GET command, which is illustrated in samples CNMS8015 and CNMS8017.

DSIVSAM INQUIRE (NCCF)

Syntax

➤ DSIVSAM — — INQUIRE — — *taskname* ➤

Purpose of Command

The DSIVSAM INQUIRE command displays the data set characteristics for the active VSAM file on the task *taskname*.

This command is similar to the DSIVSMX INQUIRE command, which is illustrated in sample CNMS8016.

Operand Descriptions

TASKID

The operator ID or task ID where the INQUIRE command ran.

DDNAME

The name of the DD statement that identifies the data set.

DSN

The name of the VSAM file allocated to the DDNAME.

VOLUME(S)

The list of the volume names where the VSAM data set resides.

TYPE

The type of VSAM object.

BASE

The base cluster of a VSAM file.

PATH

The path used to access the VSAM file using an alternate key.

AIX®

The alternate index accessed directly (not usually done).

AUTOTOKE

A 16-character time value when the data set was opened. The value is the processor store clock value in hexadecimal.

PRISEC

Indicates whether the Data Services Task is to use the primary or secondary VSAM file.

NSR

Indicates whether the Data Services Task is to use NSR. The value is either Y or N.

LSR

Indicates whether the Data Services Task is to use LSR. The value is either Y or N.

DFR

Indicates whether the Data Services Task is to use DFR. The value is either Y or N.

ADR

Specify records by their address. The value is always N.

KEY

Specifies keyed access. The value is either Y or N.

SEQ

Specifies sequential access. The value is either Y or N.

DIR

Specifies direct access. The value is either Y or N.

IN

Specifies read-only data sets. The value is Y.

OUT

Specifies read/write data sets. The value is Y.

KSDS

Specifies keyed-sequential data sets. The value is Y.

OFLAG

Specifies if the data set is open. The value is either Y or N.

SHOWR15

If zero is returned, the operation was successful. Any other number that is returned is a SHOWCB error code from register 15.

SHOWR00

If zero is returned, the operation was successful. Any other number that is returned is a SHOWCB error code from register zero.

KEYLEN

The length of the key field for data records in the data component.

RKP

The displacement of the key field from the beginning of a data record.

STRNO

The number of requests for which VSAM is to remember its position in the data set.

BUFSP

The amount of space specified in the ACB or GENCB for I/O buffers.

DLRECL

The length of data records in the data component (maximum length for variable-length data records).

DCINV

The control interval size for the data component.

DBUFND

The number of I/O buffers to be used for data, as specified in the ACB or GENCB.

DBUFNO

The number of data component I/O buffers in use.

DNEXT

The number of extents now allocated to the data component. The maximum that can be allocated is 123.

DFS

The number of free control intervals per control area in the data component.

DNCIS

The number of control intervals that have been split in the data component.

DNSSS

The number of control areas that have been split in the data component.

DNEXCP

The number of EXCP macros that VSAM has issued for access to the data component.

DNLOGR

The number of records in the data component.

DNRETR

The number of records that have been retrieved from the data component.

DNINSR

The number of records that have been inserted into, or added to, the data component.

DNUPDR

The number of records in the data component that have been updated.

DNDEL

The number of records that have been deleted from the data component.

DAVSPAC

The amount of available space in the data component, in bytes.

STRMAX

The maximum number of strings concurrently active.

BSTRNO

The number of strings initially allocated for access to the base cluster by a path.

DENDRBA

The ending relative byte address (RBA) of the space used by the data component of the last used byte in the data set (high-used RBA).

DHALCRBA

The RBA of the end of the data component (high-allocated RBA).

ILRECL

The length of index records in the index component (control interval length minus 7).

ICINV

The control interval size for the index component.

IBUFNI

The number of I/O buffers to be used for index entries, as specified in the ACB or GENCB.

IBUFND

The number of I/O buffers in use for the index component.

INEXT

The number of extents now allocated to the index component. The maximum that can be allocated is 123.

INIXL

The number of levels in the index component.

INEXCP

The number of EXCP macros that VSAM has issued to access the index component.

INLOGR

The number of records in the index component.

IAVSPAC

The amount of available space in the index component, in bytes.

IENDRBA

The ending relative byte address (RBA) of the space used by the index component of the last used byte in the data set (high-used RBA).

IHALCRBA

The RBA of the end of the index component (high-allocated RBA).

DSIVSAM PUT (NCCF)

Syntax

➤ DSIVSAM — — PUT — — *taskname* ➤

Purpose of Command

The DSIVSAM PUT command creates or replaces the record with the specified exact key with the specified data. This command produces variable length VSAM records and sends message DSI633I to indicate the request has completed successfully.

Samples CNMS8016 and CNMS8021 illustrate the use of the DSIVSAM PUT command.

Operand Descriptions

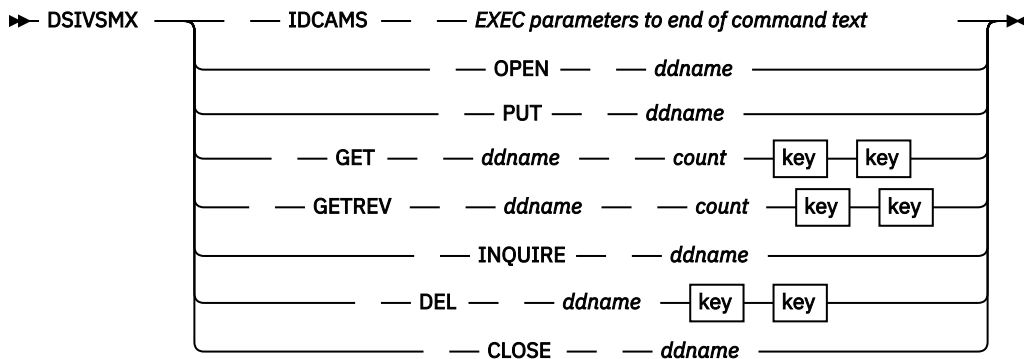
taskname

The Data Services Task to be accessed.

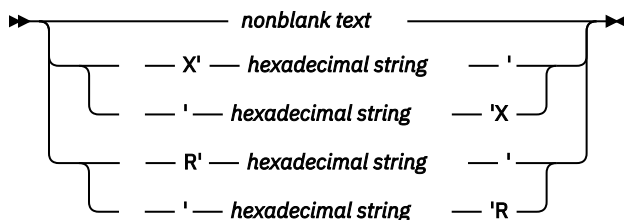
Separate the command parameters by one or more blanks. Other standard delimiters are treated as blanks by DSIVSAM.

DSIVSMX (NCCF)

Syntax



key



Purpose of Command

The DSIVSMX command processor can define, read, and write keyed VSAM files directly from where regular commands are supported, but not from the PPT task, without data services tasks. This enables the implementation of VSAM applications, including end-use application development in REXX (with the pipeline facility), and intensive VSAM diagnostics.

The DSIVSMX command provides access to keyed VSAM files and to IDCAMS, the VSAM Access Method Services utility.

The DSIVSMX command has the following functions:

- Various VSAM functions that are intended to be used with NetView pipelines
- IDCAMS access for defining, deleting, and maintaining VSAM files
- Open and close any number of VSAM files
- Write one record or multiple records with one request
- Read a single or range of records
- Read records in forward or reverse
- Read using generic (short) keys
- Delete a single record or range of records
- Display VSAM characteristics of the active file
- Adapt to any length key in any position in the record

- Adapt to any logical record length file
- Access alternate indexes

Operand Descriptions

ddname

Specifies a 1- to eight-character DDNAME. This label is on a DD statement in the NetView JCL or is a DDNAME defined by a NetView ALLOC command.

count

Specifies a decimal number. The maximum number of records to be read.

key

Specifies the keys of the VSAM records for the request.

If one key is specified, it is used for both a low range value and a high range value. If the length of the key used is smaller than the length of the key of the VSAM file, the key is padded (extended). Padding rules differ for the low and high values, based on which function is being performed.

If two keys are specified, the values are treated as follows:

- For GET and DEL requests, the first key is the starting (low) value and the second key, if any, is the ending (high) value.
- For GETREV requests, the first key is the starting (high) value and the second key, if any, is the ending (low) value.

In general, low key values are padded with binary zeros (X'00') and high key values are padded with binary ones (X'FF').

Keys are padded (extended) if the number of characters in the key you used is less than the physical key size. See the individual functions for specific rules, which differ by functions.

Note: For PUT requests, include the key in the data records as they are written. Ensure that your record is long enough to include the entire key. Generic keys are not valid for PUT.

EXEC parameters to end of command text

Data is passed to the IDCAMS VSAM utility as if it were coded on the EXEC card in MVS JCL. Refer to the MVS/DFP library or the DFSMS/MVS library for the description of IDCAMS.

Usage Notes

Consider the following when using the DSIVSMX command:

- In general, the DSIVSAM command is intended for, and mostly work best in PIPEs.
- In general, all operands are delimited by at least one blank. Commas, periods, and equal signs are also delimiters. Commands are easier to read when you use blanks.
- You can use DSIVSMX with alternate index VSAM files. Refer to *IBM Z NetView Programming: Pipes* for more information.
- The maximum logical record length supported by the DSIVSMX GET and DSIVSMX GETREV commands is 32000.
- You can control who accesses data sets by the following means:
 - The command security provided for the DSIVSMX command
 - The READSEC and WRITESEC commands
 - The DATASET class of the SAF product.

For more information, refer to the *IBM Z NetView Security Reference*.

DSIVSMX CLOSE (NCCF)

Syntax

➤ DSIVSMX — — CLOSE — — *ddname* ➤

Purpose of Command

The DSIVSMX CLOSE command closes the file associated with *ddname*, which ensures all buffered records are written to the disk. This command is issued when the operator logs off.

DSIVSMX CLOSE sends message DSI633I to indicate the request has completed successfully.

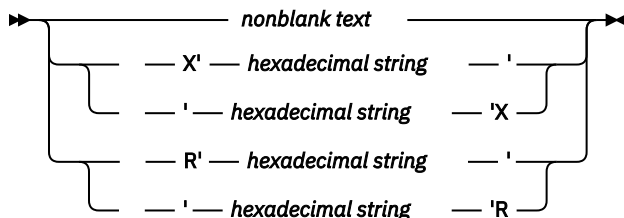
Samples CNMS8014, CNMS8015, and CNMS8020 illustrate the use of the DSIVSMX CLOSE command.

DSIVSMX DEL (NCCF)

Syntax

➤ DSIVSMX — — DEL — — *ddname* — key — key ➤

key



Purpose of Command

The DSIVSMX DEL command deletes the record with the exact matching key or all records within the boundaries of the stated key range. Message DSI633I is sent to indicate the request has completed successfully.

Sample CNMS8017 illustrates the use of the DSIVSMX DEL command.

Operand Descriptions

ddname

Specifies the DDNAME of a VSAM file. If the file to be deleted is not open, an attempt is made to open the file.

key

Specifies the key range to be deleted, which is specified as one or two keys.

1. For the first key specified, the options and padding rules are:

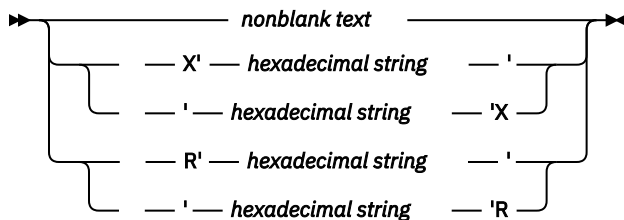
- A single character string with no embedded blanks. The key is padded with hexadecimal nulls to the full key length and the characters are displayed left-aligned.
- A hexadecimal character string in the form X'hexdigits' or 'hexdigits'X. The key is padded with hexadecimal nulls to the full key length and the 'hexdigits' are displayed left-aligned.

- A hexadecimal character string in the form R'hexdigits' or 'hexdigits'R. The key is padded with hexadecimal nulls to the full key length and the 'hexdigits' are displayed right-aligned. This notation is useful if the VSAM file is intended to have numeric (binary) keys.

For the second key specified, the options and padding rules are:

DSIVSMX GET (NCCF)

►► DSIVSMX — — GET — — *ddname* — — *count* — key — key ►►



The DSIVSMX GET command reads a record or sequence of records from a VSAM file. The output is returned in a piped output stream. If the file is not open, an attempt is made to open the file before the GET is started.

The second key is the optional ending key for the read. If the second key is omitted, the value defaults to the first key value, but is padded on the right with binary ones instead of zeros. If the value of the second key is specified, it is also padded with binary ones to the length of the VSAM key for the data set. For example, if a key of A is specified, the GET request finds the first record with an A in the first position of the key.

Note: The second key must resolve to an unsigned binary value that is equal to, or greater than, the first key. Otherwise, no data is read.

Operand Descriptions

ddname

Specifies the DDNAME of a VSAM file. If the file is not open, the system attempts to open the file before GET is issued.

count

Specifies the maximum number of records you want to retrieve.

keys

Specifies the key range to read, specified as one or two keys that determine the range of records to read.

For the first key specified, the options and padding rules are:

- A single character string with no embedded blanks. The key is padded with hexadecimal nulls to the full key length and the characters are displayed left-aligned.
- A hexadecimal character string in the form X'hexdigits' or 'hexdigits'X. The key is padded with hexadecimal nulls to the full key length and the 'hexdigits' are displayed left-aligned.
- A hexadecimal character string in the form R'hexdigits' or 'hexdigits'R. The key is padded with hexadecimal nulls to the full key length and the 'hexdigits' are displayed right-aligned. This notation is useful if the VSAM file is intended to have numeric (binary) keys.

For the second key specified, the options and padding rules are:

- A single character string with no embedded blanks. The key is padded with hexadecimal ones to the full key length and the characters are displayed left-aligned.
- A hexadecimal character string in the form X'hexdigits' or 'hexdigits'X. The key is padded with hexadecimal ones to the full key length and the 'hexdigits' are displayed left-aligned.
- A hexadecimal character string in the form R'hexdigits' or 'hexdigits'R. The key is padded with hexadecimal nulls to the full key length and the 'hexdigits' are displayed right-aligned. This notation is useful if the VSAM file is intended to have numeric (binary) keys.
- If the second key is omitted, the value for the second key is the original value of the first key, using the padding rules for the second key value.

Usage Notes

- Hexadecimal notation must be used with DSIVSMX when using special characters, including comma(,), space(), period(.), and equal sign(=). For example, the following does not work:

```
'PIPE NETV MOE DSIVSMX GET taskid 1 19991207-13:23:59.012345 | stem X.'
```

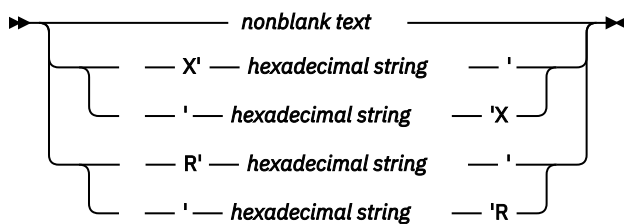
- To resolve the problem caused by using special characters, issue GET as follows, paying special attention to single quotation marks and double quotation marks: "PIPE NETV MOE DSIVSMX GET taskid 1 X'"C2X(19991207-13:23:59.012345)"' | STEM x."
- The maximum logical record length supported by the DSIVSMX GET command is 32000.

DSIVSMX GETREV (NCCF)

Syntax

➡ DSIVSMX — — GETREV — — *ddname* — — *count* — key — key ➡

key



Purpose of Command

The DSIVSMX GETREV command is similar to DSIVSMX GET, except that records are read in reverse sequence. The first key is expected to resolve to an unsigned value that is greater than, or equal to, the second key. The first key is padded with binary ones, and the second key is padded with binary zeros, if needed. The second key defaults to a zero padded equivalent of the first key.

For each byte of data in the record, there is one character in the output. The maximum logical record length supported by the DSIVSMX GETREV command is 32000.

Note: The first key must resolve to an unsigned binary value that is equal to, or greater, than the second key. Otherwise, no data is read..

This command is also similar to the DSIVSMX GET command, which is illustrated in samples CNMS8015 and CNMS8017.

DSIVSMX IDCAMS (NCCF)

Syntax

►► DSIVSMX — — IDCAMS — — EXEC parameters to end of command text ◄◄

Purpose of Command

The DSIVSMX IDCAMS command calls the IDCAMS utility to perform data set maintenance. The IDCAMS control statements are read from the message stream that started the DSIVSMX command. Typically, this is used within a PIPE with a STEM variable as the input stream.

Samples CNMS8013, CNMS8014, CNMS8015, and CNMS8020 illustrate the use of the DSIVSMX IDCAMS command.

DSIVSMX INQUIRE (NCCF)

Syntax

➤ DSIVSMX — — INQUIRE — — *ddname* ➤

Purpose of Command

The INQUIRE command retrieves VSAM data set characteristics for the file named by the DDNAME *ddname*.

Following are valid output values:

TASKID

The operator ID or task ID where the INQUIRE command ran.

DDNAME

The name of the DD statement that identifies the data set.

DSN

The name of the VSAM file allocated to the DDNAME.

VOLUME(S)

The list of the volume names where the VSAM data set resides.

TYPE

The type of VSAM object.

BASE

The base cluster of a VSAM file.

PATH

The path used to access the VSAM file using an alternate key.

AIX

The alternate index accessed directly. BASE and PATH are used most often. You do not see AIX unless you are performing a task such as low-level diagnostics.

AUTOTOKE

A 16-character time value when the data set was opened. The value is the processor store clock value in hexadecimal.

PRISEC

This value is always blank. DSIVSMX does not have a primary or secondary concept. DSIVSMX can service many DDNAME parameters.

NSR

This value is always Y. DSIVSMX does not support LSR and DFR specification.

LSR

This value is N. DSIVSMX does not support LSR specification.

DFR

This value is N. DSIVSMX does not support DFR specification.

ADR

Specify records by their address. The value is always N.

KEY

Specifies keyed access. The value is either Y or N.

SEQ

Specifies sequential access. The value is either Y or N.

DIR

Specifies direct access. The value is either Y or N.

IN

Specifies read-only data sets. The value is Y.

OUT

Specifies read/write data sets. The value is Y.

KSDS

Specifies keyed-sequential data sets. The value is Y.

OFLAG

Specifies if the data set is open. The value is either Y or N.

SHOWR15

If zero is returned, the operation was successful. Any other number that is returned is a SHOWCB error code from register 15.

SHOWR00

If zero is returned, the operation was successful. Any other number that is returned is a SHOWCB error code from register zero.

KEYLEN

The length of the key field for data records in the data component.

RKP

The displacement of the key field from the beginning of a data record.

STRNO

The number of requests for which VSAM is to remember its position in the data set.

BUFSP

The amount of space specified in the ACB or GENCB for I/O buffers.

DLRECL

The length of data records in the data component (maximum length for variable-length data records).

DCINV

The control interval size for the data component.

DBUFND

The number of I/O buffers to be used for data, as specified in the ACB or GENCB.

DBUFNO

The number of data component I/O buffers in use.

DNEXT

The number of extents now allocated to the data component. The maximum that can be allocated is 123.

DFS

The number of free control intervals per control area in the data component.

DNCIS

The number of control intervals that have been split in the data component.

DNSSS

The number of control areas that have been split in the data component.

DNEXCP

The number of EXCP macros that VSAM has issued for access to the data component.

DNLOGR

The number of records in the data component.

DNRETR

The number of records that have been retrieved from the data component.

DNINSR

The number of records that have been inserted into, or added to, the data component.

DNUPDR

The number of records in the data component that have been updated.

DNDEL

The number of records that have been deleted from the data component.

DAVSPAC

The amount of available space in the data component, in bytes.

STRMAX

The maximum number of strings concurrently active.

BSTRNO

The number of strings initially allocated for access to the base cluster by a path.

DENDRBA

The ending relative byte address (RBA) of the space used by the data component of the last used byte in the data set (high-used RBA).

DHALCRBA

The RBA of the end of the data component (high-allocated RBA).

ILRECL

The length of index records in the index component (control interval length minus 7).

ICINV

The control interval size for the index component.

IBUFNI

The number of I/O buffers to be used for index entries, as specified in the ACB or GENCB.

IBUFND

The number of I/O buffers in use for the index component.

INEXT

The number of extents now allocated to the index component. The maximum that can be allocated is 123.

INIXL

The number of levels in the index component.

INEXCP

The number of EXCP macros that VSAM has issued to access the index component.

INLOGR

The number of records in the index component.

IAVSPAC

The amount of available space in the index component, in bytes.

IENDRBA

The ending relative byte address (RBA) of the space used by the index component of the last used byte in the data set (high-used RBA).

IHALCRBA

The RBA of the end of the index component (high-allocated RBA).

Sample CNMS8016 illustrates the use of the DSIVSMX INQUIRE command.

DSIVSMX OPEN (NCCF)

Syntax

➤ DSIVSMX — — OPEN — — *ddname* ➤

Purpose of Command

The DSIVSMX OPEN command opens a VSAM file locally to the task that issues the command. Access remains open until a DSIVSMX CLOSE command is issued or the task is logged off.

DSIVSMX OPEN sends message DSI633I to indicate the request has completed successfully.

DSIVSMX automatically opens the VSAM file for any request if a DSIVSMX OPEN command is not issued first.

DSIVSMX OPEN opens the VSAM file in read mode. If the data set is empty, the open request fails. In this case, use the DSIVSMX PUT command to put an initial record in the file.

Avoid using DSIVSMX on VSAM files of Data Services Tasks (DST), especially if they have the REUSE attribute. If you OPEN one of these files and the DST attempts a RESET (such as DSILOG switching between primary and secondary logs), the SWITCH fails. To access VSAM files of DST tasks, use the DSIVSAM command processor.

Samples CNMS8014 and CNMS8015 illustrate the use of the DSIVSMX OPEN command.

Operand Descriptions***ddname***

Specifies the DDNAME of a VSAM file.

DSIVSMX PUT (NCCF)

Syntax

►► DSIVSMX — — PUT — — *ddname* ◄◄

Purpose of Command

The DSIVSMX PUT command expects a piped input stream of record images. The image format is identical to the output format of DSIVSMX GET. The key is specified within the text of the data records. This is useful in preloading a data set from data built into stem variables. Each byte of input data represents 1 byte of output data in the identical buffer position in the VSAM record.

DSIVSMX PUT produces variable length VSAM records.

DSIVSMX PUT sends message DSI633I to indicate the request has completed successfully.

Samples CNMS8014, CNMS8015, CNMS8017, CNMS8018, and CNMS8020 illustrate the use of the DSIVSMX PUT command.

Operand Descriptions

ddname

Specifies the DDNAME of a VSAM file. If the file is not open, the system attempts to open the file before PUT is issued.

DUIFECMV (RODM)

Syntax

DUIFECMV

► DUIFECMV — DOMAIN = *domname*

GMFHSDOM= *gmfhsgomname*

GMFHSDOM= *local*

► CLASS = *classname* OBJNAME = *objname* ►

► INDICAT = *indicator* ¹ STATUS = *status* ◄◄

Notes:

¹ STATUS is required if INDICAT is 2 or 4.

Purpose of Command

The DUIFECMV command processor is started when an alert from the NetView program is passed through the automation table. The DUIFECMV command processor sends information to GMFHS and initiates GMFHS processing of the alert. Refer to the *IBM Z NetView Resource Object Data Manager and GMFHS Programmer's Guide* for more information.

Operand Descriptions

DOMAIN

Specifies the name of the domain.

GMFHSDOM

Specifies the GMFHS domain to which DUIFECMV forwards alerts. The default value is the local domain.

CLASS

Specifies the class name of the object against which to log the alert.

OBJNAME

Specifies the name of the object against which to log the alert.

INDICAT

Specifies the status indicator. Valid values are:

0

Log the alert but do not update the status.

1

Update the status based on the alert type table and log the alert.

2

Update the status based on the status parameter and log the alert.

3

Update the status based on the alert type table, but do not log the alert.

4

Update the status based on the status parameter, but do not log the alert.

When a value of 1 or 3 is used, refer to the *IBM Z NetView Resource Object Data Manager and GMFHS Programmer's Guide* for more information about the alert type table.

STATUS

Specifies the status to set against the object if INDICAT=2 or INDICAT=4 is specified.

Restrictions

The DUIFECMV command processor must run under the autotask DUIFEAUT.

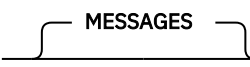
Example: DUIFECMV Automation Table entry

The following is an example of an automation table entry:

```
IF (MSUSEG(0000) ^= ' ' | MSUSEG(0002) ^= ' ') & HIER ^= ' '
  THEN EXEC(CMD(' DUIFECMV
    DOMAIN=SPNAME
    GMFHSDOM=CNM01
    CLASS=1.3.18.0.0.3315.0.3.1
    OBJNAME=2.9.3.2.7.4=SPNAME,1.3.18.0.0.3519=APPLNAME
    INDICAT=2
    STATUS=131'))
    ROUTE(ONE DUIFEAUT)) CONTINUE(Y) ;
```

FLUSHQ (REXX)

Syntax**FLUSHQ**

➔ ' — FLUSHQ —  ' ➔

Purpose of Command

The FLUSHQ instruction is used to remove all trapped messages from the message queue that have not been removed using MSGREAD.

Return codes

FLUSHQ sets the value of the return code RC to indicate the results of processing as follows:

Return Code

Meaning

-1

Syntax error

0

Successful completion

Usage Notes

Consider the following when using the FLUSHQ instruction:

- Because the TRAP instruction does not clear the queue of messages trapped by a previous TRAP, issue a FLUSHQ instruction between multiple TRAP instructions coded in the same command list.
- You can code the FLUSHQ instruction in both a nested REXX command list and the initial REXX command list. If you use the REXX CALL instruction to start a nested command list, all trapped messages are removed from the initial REXX command list trap message queue. If you start the nested list without the CALL instruction, only the trapped messages for the nested REXX command list are removed.

Restrictions

For REXX, the instruction is enclosed in single quotation marks to prevent variable substitution.

GETM commands

Introduction

Some commands return a reply that appears to be a sequence of separate messages, when the reply is a single multiline write-to-operator (MLWTO) message. The NetView program treats an MLWTO message as a single message. Only the first nonblank message identifier that appears as part of an MLWTO message is made available to satisfy a REXX TRAP instruction, or a NetView command list language &WAIT command. The first nonblank line of a multiline message is also the only line used for comparisons in NetView automation tables.

When the NetView automation table starts a command list as a result of automating an MSU, the MSU data and hardware monitor resource hierarchy data are available to the command list. The MSU data is contained in one buffer, and the hardware monitor resource hierarchy data is contained in a second buffer.

Figure 5 on page 539 shows an example MLWTO message. The message is in response to a LIST KEY=PF1 command. The MLWTO message appears to be a sequence of several separate messages, but the single quotation mark (') that appears as the message type identifier identifies the message as a single MLWTO message from the NetView program. A double quotation mark (") identifies a multiline message from an IBM or Tivoli program other than the NetView program. An equal sign (=) identifies a user-written multiline message.

```
NCCF          NETVIEW   NCF01 OPER1   10/13/19 11:44:23
* NTVAALIST KEY=PF1
' NTVAALIST KEY=PF1
DSI606I DISPLAY OF PF/PA KEY SETTINGS FOR NCCF
DSI607I KEY  ----TYPE-----COMMAND-----SET-APPL
DSI608I PF1  IMMED,APPEND  HELPNETVIEW
```

Figure 5. Example Multiline Message

TRAP, &WAIT, and NetView automation table processing use only the first nonblank line of a multiline message.

NetView provides commands that enable you to work with multiline messages and management services unit (MSU) buffers in a command list. These commands enable you to work with information in each individual line of a multiline message or data buffer of an MSU. The commands are as follows:

GETMLINE

Assigns the text of a specific line of a multiline message or a specific MSU data buffer to a specified variable.

GETMPRES

Retrieves the message presentation attributes of the text buffer for each line of a message.

GETMSIZE

Determines the number of lines of a multiline message. GETMSIZE also determines the number of MSU data buffers.

GETMTFLG

Retrieves the message type flags from the text buffer of any message or MSU.

GETMTYPE

Determines the line type of a specific line in a multiline message, or identifies the type of MSU data.

Note: The GETM commands pre-date NetView pipelines. You can use the PIPE command to solve these types of problems. For more information about the PIPE command and its stages, refer to *IBM Z NetView Programming: Pipes*.

GETMLINE (REXX)

Syntax

GETMLINE

➤ GETMLINE — — *variable_name* — — *line_number* ➤

Purpose of Command

Use the GETMLINE command within a command list to assign the text of an individual line of a multiline message to a specified variable. GETMLINE can also be run for a buffer containing MSU type data in a NetView command list environment. Use this command in a command list that is driven by NetView automation or that has processed a message using MSGREAD (REXX) or &WAIT (NetView command list language).

Operand Descriptions

variable_name

Identifies a command list variable coded in this command. GETMLINE assigns the value of the line you specify to this variable name. If you write the command list in the NetView command list language, do not use the ampersand (&) in the variable name.

The maximum character length of a variable name depends on the following:

11

When started from a command list written in the NetView command list language.

250

When started from a REXX-format command list.

line_number

Identifies the number of the line in the multiline message or the specific data buffer in the MSU from which you want to obtain the value. Blank lines in a multiline message are counted as lines.

When you code a number for *line_number* in REXX command lists, put the number within single quotation marks that enclose GETMLINE. For example:

```
'GETMLINE LINE1 2'
```

When you code a variable for *line_number* in REXX command lists, leave a blank after *variable_name* and close the quotation marks. Code the name of the variable that contains the value for *line_number* outside the quotation marks. For example, if the value of *line_number* is contained in a variable named NUM1, code:

```
'GETMLINE LINE1 'NUM1
```

Return codes

GETMLINE sets the return code (RC or &RETCODE) to indicate the processing results, as follows:

Return Code	Meaning
-------------	---------

0	Processing was successful.
8	Storage is not available to continue processing.
100	There are not enough parameters, or they are not issued from a command list.
104	The <i>line_number</i> length is not valid.
108	The <i>line_number</i> value is not valid.
116	The requested line number does not exist.
120	There is no multiline message or MSU.
124	There is no message or MSU to process.
128	The command list dictionary update failed.
132	Returned data has been truncated.
136	The command parameter is too long.

Usage Notes

In a NetView command list environment, if MSU data is greater than 255 characters, GETMLINE completes with a return code of 132 indicating that the 255 character limit for command list variables was exceeded. If this happens, the data is truncated at 255 characters before it is placed in the command list variable.

GETMLINE runs successfully for a buffer containing HIER type data because the data is not greater than 255.

For example, an automation task command list written in the NetView command list language contains the following two lines:

```
&NUM = 2  
GETMLINE SECOND &NUM
```

These two lines in a command list place the text of the second line of a multiline message into the command list variable &SECOND.

See “GETMTYPE (REXX)” on page 547 for example command lists that show how GETMLINE can be used with MLWTO messages.

GETMPRES (REXX)

Syntax

GETMPRES

➤ GETMPRES — — *variable_name* — — *line_number* ➤

Purpose of Command

You can use the GETMPRES command in command lists to retrieve the 4-byte message presentation value from the text buffer corresponding to each line of text for a message or MSU. If the message has no presentation attribute characters specified, the value of GETMPRES is null. If the value is *not* null and GETMTFLG bit 16 is set on, GETMPRES values are used for message and MSU presentation. These GETMPRES values are taken from one of the following sources:

- The presentation overrides specified in the message data buffer (MDB)
- The presentation overrides as specified by a previous automation table action

When one or more presentation attributes are set by the automation table (with the COLOR, HIGHINT, or XHILITE actions) all four of the presentation attributes for the message or MSU are copied to the GETMPRES fields and used to display that message or MSU. Attributes that are not set by the automation table are taken from either MDB override fields, the fields in MSGGFGPA, or the values specified with the OVERRIDE or DEFAULTS SCRNFMT commands.

If GETMPRES is null, the presentation attributes of the message or MSU are taken from *one* of the following:

- The fields in MSGGFGPA
- The values specified with the OVERRIDE or DEFAULTS SCRNFMT command
- For MSUs, the hardware monitor defaults

When GETMPRES is null, other presentation attributes apply to this message when it is displayed. When GETMPRES is null, you can check the fields in MSGGFGPA for presentation attributes.

Before using GETMPRES, you can determine whether the presentation vectors exist and are valid by issuing a GETMTFLG and checking bit 16.

Operand Descriptions

variable_name

Indicates the name of a command list variable, without the ampersand (&), that is used to store the returned data. The maximum character length of a variable name depends on the following:

11

When started from a command list written in the NetView command list language.

250

When started from a REXX-format command list.

The data represented by the variable name consists of a string of four characters of presentation attribute data. The four characters have the following meanings and possible values:

Byte

Description

1

Control field

00

MVS alarm-off indicator

80

MVS alarm-on indicator

Note: This is an MVS indicator which the NetView program does not use. The NetView indicators that actually control this alarm are in IFRAUTA1 bits 13, 14, and 30. The IFRAUTA1 indicators are also set by the automation table.

2

Color field

00

For messages, the value is determined from the MSGGFGPA setting or NetView OVERRIDE or DEFAULTS SCRNFMT settings, respectively. For MSUs, the hardware monitor-established defaults are used.

F0

Presentation background. Black on display, white on printer.

F1

Blue

F2

Red

F3

Pink (magenta)

F4

Green

F5

Turquoise (cyan)

F6

Yellow

F7

Presentation neutral. White on display, black on printer.

The color field is also set by the COLOR action in the automation table.

3

Highlighting field

00

No highlighting

F1

Blinking

F2

Reverse video

F4

Underscore

The highlighting field is also set by the XHILITE action in the automation table.

4

Intensity field

00

For messages, the value is determined from the MSGGFGPA setting or NetView OVERRIDE or DEFAULTS SCRNFMT settings. For MSUs, the hardware monitor-established defaults are used.

E4

Normal intensity

E8

High (bright) intensity

The intensity field is also set by the HIGHINT action in the automation table.

line_number

Indicates the number of the line for which data is requested.

Return codes

GETMPRES sets the return code to indicate the processing results, as follows:

Return Code**Meaning****0**

Processing was successful.

100

There are not enough parameters, or they were not issued from a command list.

104

The *line_number* length is not valid.

108

The *line_number* value is not valid.

116

The requested line number does not exist.

120

There is no multiline message or MSU.

124

There is no message or MSU to process.

128

The command list dictionary update failed.

132

The command parameter is too long.

Examples

Assume that the following statement is coded in a REXX or NetView command list language command list,:

```
'GETMPRES ATTRS 1'
```

```
&X = 1
GETMPRES ATTR2 &X
```

This coding causes the NetView program to update the specified variable with the presentation attributes of the first line of the message being processed.

The presentation data is considered null and a zero length is returned under the following conditions:

- The IFRAUTA1 indicator 48 is zero.
- You are using &IFRAUTA1 (NetView command list language) or IFRAUTA1() (REXX).
- The buffer header vectors are incorrect. For example, they have the incorrect length or key fields.

GETMSIZE (REXX)

Syntax

GETMSIZE

➤ GETMSIZE — — *variable_name* ➤

Purpose of Command

You can use the GETMSIZE command in command lists to determine the number of lines in a multiline message. GETMSIZE can also be run for a buffer string containing management services unit (MSU) data. The size returned is 1 or 2, depending on whether HIER data exists. Use this command in a command list started by NetView automation or that processes a message using MSGREAD (REXX) or &WAIT (NetView command list language).

Operand Descriptions

variable_name

Identifies a command list variable coded in this command. If you write the command list in the NetView command list language, do not use the ampersand (&) in the variable name. GETMSIZE sets the value of the variable to the number of lines in the multiline message or the number of associated MSU buffers. If a message is a single-line instead of a multiline message, the variable value is set to 1. This number includes any blank lines in the multiline message.

The maximum character length of a variable name depends on the following:

11

When started from a command list written in the NetView command list language.

250

When started from a REXX-format command list.

Return codes

GETMSIZE sets the return code (RC or &RETCODE) to indicate the processing results, as follows:

Return Code

Meaning

0

Processing was successful.

8

Storage is not available to continue processing.

100

There are not enough parameters, or they were not issued from a command list.

104

The internal numeric conversion failed.

108

The command list dictionary update failed.

112

The command parameter is too long.

For example, assume that the following statement is coded in an automation command list:

```
GETMSIZE NUMLINES
```

If the command list containing this command is triggered by the message in the following example, the variable &NUMLINES is set to the value of 7:

```

IEE104I 13.02.15 90.89 ACTIVITY 607 C
JOBS      M/S      TS USERS      SYSAS      INITS      ACTIVE/MAX VTAM
00000    00007    00000    00008    00001    00000/00300
LLA      LLA      LLA      NSW S    JES2JES2    IEFPROC    NSW S
SYSLOG   621      IEFPROC    OWT S    BASENET    BASENET    VTAM    NSW S
TSO      TSO      TCAS      OWT S    NETVREL2    NETVREL2    NETVIEW    NSW S
NETV2    NETV2    NETVIEW    NSW S

```

Figure 6. IEE104I Message to Trigger an Automation Command List

GETMTFLG (REXX)

Syntax

GETMTFLG

➔ GETMTFLG — — *variable_name* — — *line_number* ➔

Purpose of Command

You can use the GETMTFLG command in command lists to retrieve the presentation override flag for each line of text for a message or MSU. The value returned by this command processor is the 16-bit HDRTLNTY field from the buffer header. Bit 16 is the presentation override flag.

The GETMTFLG command returns a null value if the buffer being evaluated does not have an extended buffer header. Messages received from MVS through extended multiple console support (EMCS) consoles have extended buffer headers with all of the GETMTFLG information. Other messages might have an extended buffer header, or might not have all of the information. For line-type information, use the GETMTYPE command.

Operand Descriptions

variable_name

Indicates the name of a command list variable, without the ampersand (&), that is used to store the returned data. The maximum character length of a variable name depends on the following:

11

When started from a command list written in the NetView command list language.

250

When started from a REXX-format command list.

The data represented by the variable name consists of a string of 16 bits represented as EBCDIC ones and zeros. The bits have the following meaning corresponding to the field HDRTLNTY in NetView macro DSITIB:

16

Text object presentation attribute field overrides general object presentation attribute field. This bit indicates that presentation information is available for messages or MSUs.

Usage notes:

1. Other bits can be tested, but have no use.
2. Bits 1-5 contain line-type information for MVS messages. This line-type information can be of historical or diagnostic use, but is not constant from message to message. Therefore, NetView processes the buffers based on the value returned by GETMTYPE, rather than the value in the GETMTFLG bits 1-5.
3. You can determine whether the extended buffer header exists by checking bit position 48 of IFRAUTA1.

line_number

Indicates the number of the line for which data is requested.

Return codes

GETMTFLG sets the return code to indicate the processing results, as follows:

Return Code

Meaning

- | | |
|------------|---|
| 0 | Processing was successful. |
| 100 | There are not enough parameters, or they were not issued from a command list. |
| 104 | The <i>line_number</i> length is not valid. |
| 108 | The <i>line_number</i> value is not valid. |
| 116 | The requested line number does not exist. |
| 120 | There is no multiline message or MSU. |
| 124 | There is no message or MSU to process. |
| 128 | The command list dictionary update failed. |
| 132 | The command parameter is too long. |

Examples

Assume that the following statement is coded in a NetView command list language command list:

```
&X = 1  
GETMTFLG FLAG2 &X
```

The NetView program updates the specified variable with the presentation override flag data for the first line of the message being processed.

GETMTYPE (REXX)

Syntax

GETMTYPE

➤ GETMTYPE — — *variable_name* — — *line_number* ➤

Purpose of Command

You can use the GETMTYPE command in command lists to determine the line type of an individual line in a multiline message. GETMTYPE can also be run for a buffer string containing MSU data. Use this command in a command list that is driven by NetView automation or that processes a message using MSGREAD (REXX) or &WAIT (NetView command list language).

Operand Descriptions

variable_name

Identifies a command list variable coded in this command. If you write the command list in the NetView command list language, do not use the ampersand (&) in the variable name.

GETMTYPE sets the value of this variable as one of the following line types:

blank

Single-line message

C

Control line

L

Label line

D

Data line

DE

Combined data and end line

E

End-line without data

H

HIER data

M

Management services unit (MSU) data

The maximum character length of a variable name depends on the following:

11

When started from a command list written in the NetView command list language.

250

When started from a REXX-format command list.

line_number

Specifies the number of the line in the multiline message for which you want line type information, or the number of the buffer in the MSU that you want. Blank lines in the multiline message are counted as lines. For single-line messages, the value of *line_number* must be 1.

When you code a number for *line_number* in REXX command lists, put the number within single quotation marks that enclose GETMTYPE. For example:

```
'GETMTYPE TYPE1 3'
```

When you code a variable for *line_number* in REXX command lists, leave a blank after *variable_name* and close the quotation marks. Code the name of the variable that contains the value for *line_number* outside the quotation marks. For example, if the value of *line_number* is contained in a variable named NUM1, code:

```
'GETMTYPE TYPE1 'NUM1
```

Return codes

GETMTYPE also sets the return code (RC or &RETCODE) to indicate the processing results, as follows:

Return Code

Meaning

0

Processing was successful.

8

Storage is not available to continue processing.

100

There are not enough parameters, or they were not issued from a command list.

104

The *line_number* length is not valid.

108

The *line_number* value is not valid.

116

The requested line number does not exist.

120

There is no multiline message or MSU.

124

There is no message or MSU to process.

128

The command list dictionary update failed.

132

The command parameter is too long

Usage Notes

You can use GETMTYPE along with GETMSIZE in a command list to test each line of a multiline message for the type. If you use a variable name for the *line_number* field in the GETMTYPE command, you can test each line. Using GETMSIZE, you can have your command list test the correct number of lines for each message it receives.

If the command list receives the message illustrated in [Figure 6 on page 546](#), the line types are as follows:

- First line equals C
- Second and third lines equals L
- Fourth through the sixth lines equals D
- Seventh line equals DE

Usage notes:

1. MSU data typically consists of one or two buffers. The first buffer is the entire MSU buffer and the second buffer, if present, is the hardware monitor resource hierarchy buffer.
2. The GETMTYPE command predates NetView pipelines. You can use the PIPE command to solve this type of problem. For more information about the PIPE command and its stages, refer to *IBM Z NetView Programming: Pipes*.

Example: Command lists processing MLWTO messages

The command lists in [Figure 7 on page 550](#) and [Figure 8 on page 551](#) are examples of how you can code your command list to process based on the information in the individual lines of a multiline message. [Figure 7 on page 550](#) is written in REXX. [Figure 8 on page 551](#) is written in the NetView command list language.

```

/*****
/* SESSCNT  COMMAND LIST                               */
/* -----                                           */
/* FUNCTION:   This command list counts the number of OPCTL */
/*             sessions and FLSCN sessions that are active. */
/*             */
/* INPUT PARMS: None                                   */
/*             */
/* OUTPUT:     Two informational messages that display the number of */
/*             OPCTL and FLSCN sessions to the operator. */
*****/
OPCTLCNT = 0                                           /* init OPCTL counter */
FLSCNCNT = 0                                           /* init FLSCN counter */
'TRAP AND SUPPRESS MESSAGES *'                       /* TRAP the MLWTO msg */
'LISTSESS'                                           /* issue the command */
'WAIT 5 SECONDS FOR MESSAGES'                       /* WAIT for the msg */
SELECT                                              /* SELECT an EVENT */
  WHEN EVENT() = 'M' THEN                          /* message received */
    DO                                              /* process the message */
      'MSGREAD'                                     /* read the message in */
      'GETMSIZE  NUMLINES'                         /* get number of lines */
      DO CNTR = 4 TO NUMLINES                      /* loop thru MLWTO buf */
        'GETMLINE LINE' CNTR                      /* get a line in buf */
        'PARSEL2R LINE APPLNM SRCLUNM SESSNM TYPE RESTLINE' /* parse the line out */
        IF TYPE = 'OPCTL' THEN                    /* OPCTL session ? */
          OPCTLCNT = OPCTLCNT + 1                 /* yes,increment count */
        ELSE                                       /* must be FLSCN sess */
          FLSCNCNT = FLSCNCNT + 1                 /* increment count */
        'TRAP NO MESSAGES'                       /* end message trapping*/
        'FLUSHQ MESSAGES'                       /* flush message queue */
      END                                          /* loop thru MLWTO buf */
    END                                          /* process the message */
  OTHERWISE                                       /* event not a message */
    SAY 'ERROR PROCESSING LISTSESS COMMAND'       /* issue an error msg */
END                                              /* SELECT an EVENT */
/* issue messages to operator with count of FLSCN & OPCTL sessions */
SAY 'YOU HAVE ' OPCTLCNT ' OPCTL SESSIONS ACTIVE AND'
SAY ' ' FLSCNCNT ' FLSCN SESSIONS ACTIVE'

```

Figure 7. Command List Using Multiline Messages—REXX Example


```

&CONTROL ERR
* * * * *
*
* COMMAND LIST AUTHTOTE
*
* THIS COMMAND LIST DISPLAYS A COUNT OF HOW MANY MESSAGES (OR GROUPS
* OF MESSAGES, USING THE * SUFFIX) HAVE BEEN ASSIGNED TO BE
* ROUTED TO SPECIFIC AUTHORIZED RECEIVERS.
*
* IT IS CALLED AS FOLLOWS:  AUTHTOTE
*
*                               AUTHTOTE DISPLAY
*
*                               AUTHTOTE SUPPRESS
*
* THE FIRST TWO FORMS WILL DISPLAY THE MESSAGES RECEIVED ("DISPLAY"
* IS THE DEFAULT) AND A COUNT OF THE TOTAL NUMBER OF ASSIGNMENTS
* MADE, AND THE LAST FORM WILL SIMPLY DISPLAY THE TOTAL NUMBER
* OF ASSIGNMENTS.
*
* * * * *
* *
* SAVE THE INPUT PARAMETER
&OPT = &1
* INITIALIZE THE COUNTER FOR THE NUMBER OF MESSAGES ASSIGNED TO
* AUTHORIZED RECEIVERS.
&COUNT = 0
* SET THE COUNTER FOR THE NUMBER OF THE MESSAGE LINE TO BE
* PROCESSED TO 1.  THE WAIT STATEMENT PROCESSES THE FIRST MESSAGE LINE.
&MSZI = 0
* ISSUE THE COMMAND AND ENTER WAIT STATE
&WAIT ENDWAIT &OPT
&WAIT 'LIST MSG=AUTH',DSI636I=-COUNTER,DSI643I=-NOEXIT, +
      *60=-TIMEOUT,*ENDWAIT=-GOIN
-NXTLINE
* ADD 1 TO THE COUNTER FOR NUMBER OF LINES PROCESSED
&MSZI = &MSZI + 1
* IF THE NUMBER OF THE LINE TO BE PROCESSED IS GREATER THAN THE
* NUMBER IF LINES IN THE MESSAGE, THEN DISPLAY THE RESULTS
&IF &MSZI = &NUMLINES &THEN &GOTO -ENDIT
* ASSIGN THE TEXT OF THE NEXT LINE OF THE MESSAGE TO MSGTXT
GETMLINE MSGTXT &MSZI
* PARSE MSGTXT SO THAT THE MESSAGE ID OF THE LINE IS PLACED IN
* THE MSGID VARIABLE.

PARSEL2R MSGTXT MSGID MSGSTR
&IF &MSGID = DSI642I &THEN &GOTO -ENDIT
&IF &MSGID = DSI643I &THEN &GOTO -NOEXIT
&IF &MSGID = DSI636I &THEN &GOTO -COUNTER
&GOTO -NXTLINE
-NOEXIT
* THIS ENSURES THAT THE OPERATOR IS INFORMED OF THE RESULTS OF
* THE COMMAND WHEN NO MESSAGES HAVE BEEN ASSIGNED, BUT THE
* MESSAGE INFORMING THE OPERATOR HAS BEEN SUPPRESSED
&IF .&OPT = .SUPPRESS &THEN &GOTO -ENDIT
&EXIT
-COUNTER
* IF THE FIRST MESSAGE LINE IS BEING PROCESSED, DETERMINE THE NUMBER
* OF LINES FOR THE MESSAGE
IF &MSZI = 1 &THEN &GOTO -LINES
-COUNTER1
* TOTAL THE NUMBER OF MESSAGES ASSIGNED TO PRIMARY RECEIVERS
&COUNT = &COUNT + 1
&GOTO -NXTLINE
-LINES
* ASSIGN THE NUMBER OF LINES IN THE MESSAGE TO THE VARIABLE NUMLINES
GETMSIZE NUMLINES
&GOTO -COUNTER1
-GOIN
&WRITE COMMAND LIST AUTHTOTE ENDED BY "GO" COMMAND -- COUNT INCOMPLETE
&EXIT
-TIMEOUT
&WRITE NO RESPONSE WITHIN 60 SECONDS. COMMAND LIST TERMINATING
&EXIT
-ENDIT
* DISPLAY THE COUNT OF MESSAGES ASSIGNED TO AUTH RECEIVERS
&WRITE &COUNT MESSAGES ASSIGNED TO SPECIFIC AUTH RECEIVERS
&EXIT

```

Figure 8. Command List Using Multiline Messages—NetView Command List Language Example
Appendix A. Command List Commands **551**

GLOBALV (REXX)

Introduction

The GLOBALV command enables you to define, put, and get global variables in REXX and NetView command lists. The GLOBALV command also enables you to save global variables in a VSAM database. You can restore saved global variables if the NetView program or a task is stopped and restarted, or erase (purge) saved global variables from external storage. The GLOBALV command enables you to automate or trace global variable value changes. Global variables enable multiple command procedures, regardless of their language, to share a common set of values.

When used with SAVE, RESTORE, and PURGE, the GLOBALV command is a long-running command processor that you enter from any command list. When GLOBALV is a long-running command processor, you cannot use it in an installation exit or user-written exit.

There are two types of global variables:

- Task
- Common

Task global variables are accessible only to the NetView task under which they were defined or set. *Common* global variables are accessible to any NetView task.

When you *define* a variable as global with the GLOBALV command, the NetView program makes the locally assigned variable globally available to command processors executing from the command list. You must use GLOBALV to define a global variable before some command processors (such as VIEW) can use the global variable value. For example, when you use GLOBALV to define a global variable, the VIEW command processor can put or get values for the variable.

Note: For more information about the VIEW command, refer to the *IBM Z NetView Customization Guide*.

When you *put* a global variable using the GLOBALV command, the NetView program places that variable in a global variable dictionary. If the variable is a task global variable, it is stored in the global variable dictionary for the particular task under which the command list is running. Only command procedures running under the same task can access that task's global variable dictionary.

When you *get* a global variable using the GLOBALV command, the NetView program gets the global variable from the appropriate dictionary and places it in a REXX or NetView command list language variable of the same name. You can then perform arithmetic operations or other manipulations on the variable to suit the needs of your command list without affecting the value of the global variable in the dictionary.

When you *save* a global variable using the GLOBALV command, the NetView program retrieves the global variable from the appropriate dictionary and places it in external storage. NetView or the task can then end without affecting the global variable in external storage.

When you *restore* a global variable using the GLOBALV command, the NetView program retrieves the variable value from external storage. When the variable value is returned, it is placed in the appropriate dictionary.

When you *purge* a global variable using the GLOBALV command, the NetView program purges the variable from external storage. The GLOBALV purge does not affect the variable in the global dictionary.

When you automate a global variable using the GLOBALV command, the NetView program automates message DWO994I which contains the variable name and its new value. Automation table entries can be coded to process this message.

When you trace a global variable using the GLOBALV command, the NetView program writes message DWO990I, DWO991I, or DWO992I to the Netlog. Which message is logged depends on the environment in which the global variable was changed. These messages contain the global variable name, its value, and information about what program changed it.

Return codes

Table 10. GLOBALV Return Code Summary

Return Code	Meaning
-5	Operator canceled or reset the procedure. This applies only if the command processor was started by a REXX procedure.
0	Command completed successfully.
4	Not called from a command list, REXX, or high-level language procedure.
20	No trace list or automation list was found or no entries were found for the requested group ID.
24	Nonzero return code from macro DSIGET. Refer to <i>IBM Z NetView Programming: Assembler</i> for more information about the DSIGET macro.
52	Function is not valid. Check the spelling of the function and synonym.
88	Incorrect length for variable name.
104	Duplicate name. This could be a duplicate group ID or a duplicate variable name.
128	Option or subfunction is not valid. Check the spelling of the option or subfunction.
144	No function or variable defined.
156	Local dictionary not found or no user variables are defined.
176	Bad parameter combination.
268	None of the variables requested to be restored matched what is in the VSAM directory.
292	Not called by operator station task (OST), primary POI task (PPT), or NetView-to-NetView task (NNT).
304	Reset request from user.
308	The DST ended with an error.
540	Nonvalid character in variable name.
544	Odd count in double-byte character set (DBCS) shift-out and shift-in character.
1000 + x	Nonzero return code of x from DSIMQS. Refer to <i>IBM Z NetView Programming: Assembler</i> for more information about the DSIMQS macro.
4000 + x	Nonzero return code of x from DSIPUSH. Refer to <i>IBM Z NetView Programming: Assembler</i> for more information about the DSIPUSH macro.
11000 + x	Nonzero return code of x from DSIKVS. Refer to <i>IBM Z NetView Programming: Assembler</i> for more information about the DSIKVS macro.
13000 + x	Nonzero return code of x from DSILCS. Refer to <i>IBM Z NetView Programming: Assembler</i> for more information about the DSILCS macro.
14000 + x	Nonzero return code of x. Return code values are: • 4 Invalid variable name • 12 Insufficient storage • 20 Value length limit exceeded • 28 No command procedure related to current action • 32 Data was truncated

Table 10. GLOBALV Return Code Summary (continued)

Return Code	Meaning
16000 + x	Nonzero return code of x from DSIPAS. Refer to <i>IBM Z NetView Programming: Assembler</i> for more information about the DSIPAS macro.
25000 + x	Nonzero return code of x from DSIPRS. Refer to <i>IBM Z NetView Programming: Assembler</i> for more information about the DSIPRS macro.
25600 + a	$((100 + \text{Major RC}) * 256) + \text{Minor RC}$

Note: Major return codes (RCs) and minor return codes are returned when there is a VSAM failure. Refer to the authorized operator for messages AAU020I, AAU022I, and CNM475I.

Considering the following when using the GLOBALV command:

- In REXX and high-level languages, the global variable name can be 1-250 alphanumeric characters. Valid alphanumeric characters are:

```
A-Z 0-9 . # @ ¢ $ _ ! ?
```

The first character cannot be a number or a period.

Usage notes:

- GLOBALV does not support lowercase characters in variable names. The only exception is when lowercase characters are specified after the first period in a stem variable. Lowercase characters after the first period are supported.

When passing stem variables to GLOBALV, make sure the stem itself is in uppercase within the quotation marks and that the GLOBALV command is run in the NETVASIS address command environment.

By default, the NetView program translates all commands, keywords, and values to uppercase. For more information, refer to the *IBM Z NetView Programming: REXX and the NetView Command List Language*.

- If you want global variables you create in a REXX command list to be accessible to command lists written in the NetView command list language, do not use the following characters in the global variable name because the NetView command list language does not allow these characters:

```
. _ ¢ ! ?
```

- In NetView command list language, the global variable name can be 1-11 alphanumeric characters. Valid alphanumeric characters are:

```
A-Z 0-9 . # @ $
```

The first character cannot be a number or a period.

- When using SAVE, RESTORE, or PURGE, you can use an asterisk (*) at the end of the character string to include all variables beginning with the same characters. For example, the following string includes all variables beginning with ABC, such as ABC1, ABC2, and ABCXYZ:

```
ABC*
```

To use SAVE, RESTORE, and PURGE to access the VSAM database, the DSISVRT DST task must be properly defined and started. Otherwise, you receive return code 1008.

Usage notes:

- Refer to [“Using the asterisk \(*\)” on page 555](#) for information about preventing the use of a wildcard.
- For information about defining DSISVRT DST, refer to the *IBM Z NetView Administration Reference*.

- If you specify more than one global variable on the GLOBALV command, the variable names must be delimited by commas or blanks.
- Although the length limits for task and common global variable names and values are 250 and 31000 bytes, respectively, the Save/Restore database (managed by the DSISVRT task) limits are lower. Refer to the GLOBALV SAVE command for more detailed information. For double-byte character sets (DBCS), the maximum number of double-byte characters between the shift-out (X'0E') and shift-in (X'0F') control characters is 15499.
- You can give global variables a numeric value between -2147483647 and 2147483647.

If you go outside these limits:

- For REXX, they are converted to an exponential notation, making them unusable by NetView command list language command lists.
- For NetView command list language, they are treated as character strings.

Using the asterisk (*)

GLOBALV keywords AUTOC, AUTOT, PURGEC, PURGET, RESTOREC, RESTORET, SAVEC, SAVET, TRACEC, and TRACET all support the use of the asterisk (*) wildcard with a variable. However, there is a potential for severe system performance degradation or unintentional loss of data when the asterisk is specified. To minimize that potential, the NetView program provides a means of restricting usage of the asterisk to pre-authorized users. A command authorization check of the authority level of the user is performed whenever a single asterisk is encountered with one of the GLOBALV keywords listed previously, as shown in the following example:

```
GLOBALV SAVEC *
```

The check involves the asterisk symbol (*) mapping to the keyword ASTERISK, and then determining whether the user is authorized to use the ASTERISK keyword.

To prevent operators from using the wildcard, protect the word ASTERISK as a keyword on the GLOBALV command with your command security.

Note: For information about command security, refer to the *IBM Z NetView Security Reference*.

Command List examples of Put, Get, Save, Restore, Purge, and Update

Figure 9 on page 556 and Figure 10 on page 557 are examples of REXX command lists that show how to put, get, save, restore, purge, and update a task global variable. The first command list is named GLOBV1, and it runs the nested command list UPDATE1.

```

/* GLOBV1 Command List */
/* OP1TSK1 - OPER1's Task Variable 1 */
/* OP1COM1 - OPER1's Common Variable 1 */
/* The following assignment statements give the REXX */
/* variables, OP1TSK1 and OP1COM1 initial values. */
OP1TSK1 = '5'
OP1COM1 = '1'
/* The following statement PUTS a task global variable */
/* named OP1TSK1. The value of the task global variable */
/* is 5 because the value of the REXX variable OP1TSK1 */
/* is 5. */
'GLOBALV PUTT OP1TSK1'
/* The following statement PUTS a common global variable */
/* named OP1COM1. The value of the common global variable */
/* is 1 because the value of the REXX variable OP1COM1 */
/* is 1. */
'GLOBALV PUTC OP1COM1'
/* The following statement executes a nested Command List */
/* named UPDATE1. */
UPDATE1
/* The following statement GETS the value of task global */
/* variable OP1TSK1 and places it in a REXX variable */
/* named OP1TSK1. */
'GLOBALV GETT OP1TSK1'
/* The following statement GETS the value of common global */
/* variable OP1COM1 and places it in a REXX variable */
/* named OP1COM1. */
'GLOBALV GETC OP1COM1'
/* The following statements write out the values of the */
/* REXX variables OP1TSK1 and OP1COM1. OP1TSK1 will be 10 */
/* and OP1COM1 will be 6. */
SAY 'OP1TSK1 = 'OP1TSK1
SAY 'OP1COM1 = 'OP1COM1
/* The following statement will change the value of the */
/* REXX variables OP1TSK1 and OP1COM1 to 0. */
OP1TSK1 = '0'
OP1COM1 = '0'
/* The following statement will save the task global */
/* variable OP1TSK1 to the VSAM Save/Restore data base */
/* with DST, DSISVRT. OP1TSK1 will be saved to VSAM with */
/* a value of 10, since the task global variable value of */
/* OP1TSK1 is 10. */

```

Figure 9. GLOBV1 Command List

```

'GLOBALV SAVET OP1TSK1'
/* The following statement will save the common global */
/* variable OP1COM1 to the VSAM Save/Restore data base. */
/* OP1COM1 will be saved to VSAM with a value of 6, since */
/* the common global variable value of OP1COM1 is 6. */
'GLOBALV SAVEC OP1COM1'
/* The following statement will again put the task global */
/* variable name OP1TSK1. The value of OP1TSK1 is 0 */
/* because the REXX variable OP1TSK1 is 0. */
'GLOBALV PUTT OP1TSK1'
/* The following statement will again put the common */
/* global variable name OP1COM1. The value of OP1COM1 is */
/* 0 because the REXX variable OP1COM1 is 0. */
'GLOBALV PUTC OP1COM1'
/* The following statements will restore the task and */
/* common global variables OP1TSK1 and OP1COM1. The */
/* restore will change the global variables, but not the */
/* REXX variable values. */
'GLOBALV RESTORET OP1TSK1'
'GLOBALV RESTOREC OP1COM1'
/* The following statements will GET the values of the */
/* task and common global variables and place the values */
/* in the REXX variables OP1TSK1 and OP1COM1. */
'GLOBALV GETT OP1TSK1'
'GLOBALV GETC OP1COM1'
/* The following statements write out the values of the */
/* REXX variables OP1TSK1 and OP1COM1. OP1TSK1 will be 10 */
/* and OP1COM1 will be 6. */
SAY 'OP1TSK1 = 'OP1TSK1
SAY 'OP1COM1 = 'OP1COM1
/* The following statements will purge the entry in the */
/* VSAM data base for task global variable OP1TSK1 and */
/* command global variable OP1COM1. REXX variable OP1TSK1 */
/* and OP1COM1 will not be affected. Global dictionary */
/* variables OP1TSK1 and OP1COM1 will not be affected */
/* either. */
'GLOBALV PURGET OP1TSK1'
'GLOBALV PURGEC OP1COM1'
EXIT

```

```

/* UPDATE1 Command List */
/* The following statement gets the value of task global */
/* variable OP1TSK1 and puts it into a REXX variable named */
/* OP1TSK1. */
'GLOBALV GETT OP1TSK1'
/* The following statement gets the value of common global */
/* variable OP1COM1 and puts it into a REXX variable named */
/* OP1COM1. */
'GLOBALV GETC OP1COM1'
/* The following statement checks the value of the REXX */
/* variables OP1TSK1 and OP1COM1 to determine if a variable */
/* exists. If no variable exists a NULL value is returned. */
IF OP1TSK1 = '' THEN OP1TSK1 = 0
/* Increment the REXX variable by 5. */
ELSE
  OP1TSK1 = OP1TSK1 + 5

IF OP1COM1 = '' THEN OP1COM1 = 0

/* Increment the REXX variable by 5. */
ELSE
  OP1COM1 = OP1COM1 + 5
/* The following statements PUT the values of the task and */
/* common global variables OP1TSK1 and OP1COM1. */
'GLOBALV PUTT OP1TSK1'
'GLOBALV PUTC OP1COM1'
EXIT

```

Figure 10. UPDATE1 Command List

Command list GLOBV1 creates two REXX variables. OP1TSK1 has a value of 5 and OP1COM1 has a value of 1. A GLOBALV PUTT instruction is issued to set a task global variable named OP1TSK1, and a GLOBALV PUTC is issued to set a common global variable named OP1COM1. Because a REXX variable exists with the name of OP1TSK1 and has a value of 5, task global variable OP1TSK1 is also set to a value of 5. The same is true of the REXX variable OP1COM1, and the common global variable OP1COM1 is set to the value of 1. GLOBV1 then activates a nested command list named UPDATE1.

Command list UPDATE1 issues a GLOBALV GETT instruction to get the current value of task global variable OP1TSK1 into a REXX variable named OP1TSK1. UPDATE1 issues a GLOBALV GETC instruction to get the current value of common global variable OP1COM1 into a REXX variable named OP1COM1. The values of OP1TSK1 and OP1COM1 are checked to determine if they exist by testing for null values. If the value returned is null, the variable value is set to 0. The value of REXX variable OP1TSK1 is 5, and it is incremented by 5, making its current value 10. The value of REXX variable OP1COM1 is 1, and it is incremented by 5, making its current value 6. The GLOBALV PUTT instruction updates the value of task global variable OP1TSK1 in the task global variable dictionary. The GLOBALV PUTC instruction updates the value of common global variable OP1COM1 in the common global variable dictionary.

When UPDATE1 completes execution, control is returned to the GLOBV1 command list. GLOBV1 contains a GLOBALV GETT and a GLOBALV GETC instruction to get the current values of task global variable OP1TSK1 and common global variable OP1COM1, and places the values in the REXX variables OP1TSK1 and OP1COM1. The value of REXX variable OP1TSK1 is 10 and the value of OP1COM1 is 6. The SAY instruction displays the current values of the REXX variables OP1TSK1 and OP1COM1.

The REXX variables OP1TSK1 and OP1COM1 are then set to 0. This does not change the values of the task global variable OP1TSK1 or the common global variable OP1COM1. The GLOBALV SAVET instruction writes the value of the task global variable OP1TSK1 to the VSAM Save/Restore database. The current value of task global variable OP1TSK1 is 10, which is stored to VSAM. The GLOBALV SAVEC instruction writes the value of the common global variable OP1COM1 to the VSAM Save/Restore database. The current value of common global variable OP1COM1 is 6, which is stored to VSAM.

The task and command global variables OP1TSK1 and OP1COM1 are set by issuing the GLOBALV PUTT OP1TSK1 and the GLOBALV PUTC OP1COM1 instructions. Currently both the task global variable OP1TSK1 and the common global variable OP1COM1 have a value of 0. The GLOBV1 command list now restores the values for OP1TSK1 and OP1COM1 that were stored to the VSAM database. GLOBALV RESTORET OP1TSK1 takes the value of task global variable OP1TSK1 that was stored to VSAM previously and replaces the current value of task global variable OP1TSK1, leaving the REXX variable OP1TSK1 unchanged. GLOBALV RESTOREC OP1COM1 takes the value of common global variable OP1COM1 that was stored to VSAM previously and replaces the current value of common global variable OP1COM1, leaving the REXX variable OP1COM1 unchanged.

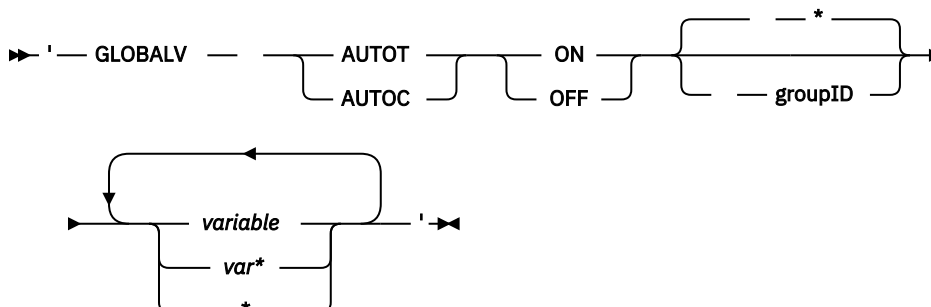
GLOBV1 issues a GLOBALV GETT instruction to get the current value of task global variable OP1TSK1 into a REXX variable named OP1TSK1. If the value of OP1TSK1 is equal to null, there was no entry in the VSAM database for task global variable OP1TSK1. The same is true for common global variable OP1COM1.

The variables OP1TSK1 and OP1COM1 are deleted from the VSAM database by issuing the GLOBALV PURGET and GLOBALV PURGEC instruction. The GLOBALV PURGET and GLOBALV PURGEC instructions have no effect on the REXX variables OP1TSK1 and OP1COM1, the task global variable OP1TSK1, or the command global variable OP1COM1.

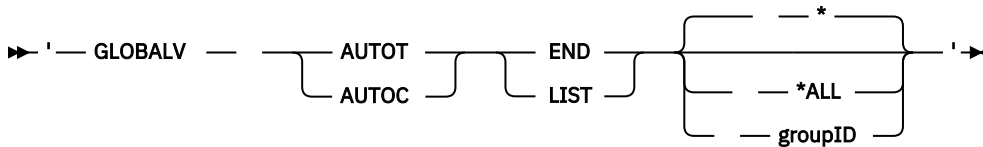
GLOBALV AUTO (REXX)

Syntax

GLOBALV AUTOT|AUTOC



GLOBALV AUTOT|AUTOC



Purpose of Command

Use the GLOBALV AUTOT or AUTOC command to turn on or off the automation of global variable value changes by a REXX or NetView command list, an HLL command, a PIPE, or a user written command that uses DSIVARS to change global variables. It can also be used to list the variable names being automated.

A global variable does not need to exist when automation for it is turned on or off. Automation for a global variable begins as soon as its name is added to the list of names to be automated using the ON option and ends as soon as it is removed from the list using the OFF or END options.

An asterisk (*) causes all common or task global variables to be automated. To avoid performance degradation if you are tracing many variables, consider how much processor time is required for tracing. Since the GLOBALV AUTO option can be restricted through security, it is assumed that only those users understanding the performance implications are allowed to use any AUTO options. AUTO is intended for use as a systems programmer diagnostic tool and not for general use.

When you automate a global variable, the DWO994I message is issued whenever the values of the global variable change. The DWO994I message contains the variable name and its new value. Automation table entries can be coded to process this message. See the examples that are provided. By default, this message will not be displayed or logged unless modified by an automation table entry.

Operand Descriptions

AUTOT or AUTOC

Indicates that the automation setting for task or common global variables with the specified variable names are to be set. Or a list of the global variable names currently being automated can be listed.

ON

Indicates that the automation setting for the specified global variables is to be turned on and automation for these variables is started.

OFF

Indicates that the automation setting for the specified global variables is to be turned off and automation for these variables is stopped.

END

Indicates that the automation setting for the specified group(s) of global variable names are to be turned off and internal storage is freed if this is the last group with automation entries.

LIST

Indicates that the automation setting for the specified group(s) of global variable names are to be listed.

groupID

This is the 1-8 character group identifier with which the specified global variable names are associated. '*' indicates that the NetView task ID under which the GLOBALV command is running is used as the group identifier. For END and LIST, '*ALL' indicates all groups are to be processed.

variable

For REXX, specifies the 1-250 character name or names of the task or common global variables to be automated. For NetView command list language, specifies the 1-11 character name or names of the task or common global variables to be automated. Refer to the GLOBALV command for a list of the characters you can use for a variable name. Variables can be separated by either a space or a comma. Variable names ending with an asterisk cause any global variable beginning with the preceding characters to have its automation setting changed.

Restrictions

For REXX, the instruction is enclosed in single quotation marks to prevent variable substitution.

Example: Coding variables for GLOBALV AUTO

The following examples show how you can code variables for the GLOBALV AUTOT or AUTOC commands.

Use either of the following examples to automate all variables in the task or common global variable dictionary:

```
'GLOBALV AUTOT ON * *'
```

```
'GLOBALV AUTOC ON * *'
```

Use either of the following examples to automate all variables beginning with VAR1 and LU and the variable ABC in the task or common global variable dictionary and assign them to automation group GROUP1:

```
'GLOBALV AUTOT ON GROUP1 VAR1* ABC LU*'
```

```
'GLOBALV AUTOC ON GROUP1 VAR1* ABC LU*'
```

Use either of the following examples to stop automation of variables VAR1, VAR2, and VAR3 in the task or common global dictionary:

```
'GLOBALV AUTOT OFF * VAR1 VAR2 VAR3'
```

```
'GLOBALV AUTOC OFF * VAR1 VAR2 VAR3'
```

Use either of the following examples to stop automation of variables assigned to group GROUP1:

```
'GLOBALV AUTOT END GROUP1'
```

```
'GLOBALV AUTOC END GROUP1'
```

Use either of the following examples to list all the variables being automated:

```
'GLOBALV AUTOT LIST *ALL'
```

```
'GLOBALV AUTOC LIST *ALL'
```

Coding automation table entries for automated global variables

The following examples show how you can code automation table entries for global variables that have automation turned on. Use the following example to run command **MYCMD** when any of the automated global variable values changes. By using this example the DWO994I message is neither logged nor displayed:

```
IF MSGID = 'DWO994I' THEN  
  EXEC(CMD('MYCMD') ROUTE(ONE MYAUTO *));
```

Use the following example to run command **MYCMD** when any of the automated global variable values changes. By using this example the DWO994I message is both logged and displayed.

```
IF MSGID = 'DWO994I' THEN  
  EXEC(CMD('MYCMD') ROUTE(ONE OPER1 *)) DISPLAY(Y)  
  NETLOG(Y);
```

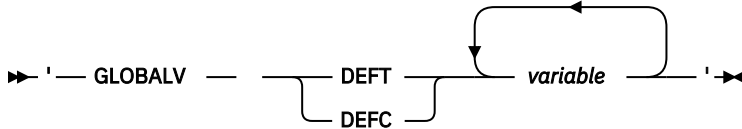
Use the following example to run command **MYCMD** whenever the value of global variable XYZ changes. By using this example the DWO994I message is only logged.

```
IF MSGID = 'DWO994I' & TOKEN(5) = 'XYZ' THEN  
  EXEC(CMD('MYCMD') ROUTE(ONE MYAUTO *)) NETLOG(Y);
```

GLOBALV DEF (REXX)

Syntax

GLOBALV DEFT|DEFC



Purpose of Command

Use the GLOBALV DEFT or DEFC command to define a task or common global variable from a REXX or NetView command list.

The GLOBALV DEFT or DEFC command makes the specified task or common global variable available to command processors executing from the command list. For example, when you use GLOBALV DEFT or DEFC to define a task or common global variable, the VIEW command processor can put or get values for the variable.

Note: For more information about the VIEW command, refer to the *IBM Z NetView Customization Guide*.

For NetView command list language, the GLOBALV DEFT or DEFC command is equivalent to &TGLOBAL and &CGLOBAL when you use it in a command list. After you issue a GLOBALV DEFT or DEFC, the command list references and sets the global variable. Refer to the *IBM Z NetView Programming: REXX and the NetView Command List Language* for more information about &TGLOBAL and &CGLOBAL.

In REXX command lists, the DEFT or DEFC operations creates an entry for the variable, but the value for that global variable is not changed by the actions of the REXX exec unless you issue a GLOBALV GET or PUT.

Operand Descriptions

DEFT or DEFC

Indicates that the specified variable name or names are to be defined as task (DEFT) or common (DEFC) global variables. Any prior definitions of the specified variable names are overridden, but variable values are not affected.

variable

For REXX, specifies the 1-250 character name or names of the task or common global variables to be defined. For NetView command list language, specifies the 1-11 character name or names of the task or command global variables to be defined. See “GLOBALV (REXX)” on page 552 for a list of the characters you can use for a variable name. Variables can be separated by either a space or a comma.

Restrictions

For REXX, the instruction is enclosed in single quotation marks to prevent variable substitution.

Example: Coding variables for GLOBALV DEF

Use any of the following examples to code variables for the GLOBALV DEFT or DEFC command:

Note: As these examples show, if you specify more than one global variable on the GLOBALV command, the variable names must be delimited by commas or blanks.

```
'GLOBALV DEFT VAR1 VAR2 VAR3'
```

```
'GLOBALV DEFT VARY1,VARY2,VARY3'
```

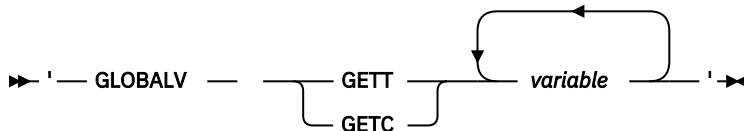
```
'GLOBALV DEFC VARIABLE_A VARIABLE_B VARIABLE_C'
```

```
'GLOBALV DEFC VARIABLE_X,VARIABLE_Y,VARIABLE_Z'
```

GLOBALV GET (REXX)

Syntax

GLOBALV GETT|GETC



Purpose of Command

Use the GLOBALV GETT or GETC command to access a task or common global variable in a REXX or NetView command list.

When a GLOBALV GETT or GETC command is processed, the NetView program gets the current value of the specified global variable from the task or common global variable dictionary and places the value in a REXX or NetView command list language variable of the same name. If a variable with the same name does not exist, the variable is created and its value is set to the task value currently assigned to the global variable.

If no task or common global variable with the specified variable name exists, the variable is not created. Any attempt to retrieve the variable causes a null value to be returned.

Refer to the *IBM Z NetView Programming: REXX and the NetView Command List Language* for examples about using common global variables. You can also use the &TGLOBAL(*name*) function to get task global variables and the &CGLOBAL(*name*) function to get common global variables.

The function of the GLOBALV GET command is influenced by any previous &TGLOBAL, &CGLOBAL, or GLOBALV defined command issued within the same NetView command list for the same named variable. For example, if a variable has been defined as a task global variable (&TGLOBAL *variable*), a subsequent GLOBALV GETC *variable* copies the common global value to the task global variable instead of to a NetView command list language variable. Refer to the *IBM Z NetView Programming: REXX and the NetView Command List Language* for more information about &TGLOBAL and &CGLOBAL.

The command lists in [“GLOBALV \(REXX\)” on page 552](#) show how to put, get, and update a task global variable.

Operand Descriptions

GETT or GETC

Indicates that the task or common global variables with the specified variable names is retrieved.

variable

For REXX, specifies the 1-250 character name or names of the task or common global variables to be retrieved. For NetView command list language, specifies the 1-11 character name of the task or

common global variables to be retrieved. See [“GLOBALV \(REXX\)” on page 552](#) for a list of the characters you can use for a variable name. Variables can be separated by either a space or a comma.

Restrictions

For REXX, the instruction is enclosed in single quotation marks to prevent variable substitution.

Example: Coding variables for GLOBALV GET

Use any of the following examples to code variables for the GLOBALV GETT or GETC command:

Note: As these examples show, if you specify more than one global variable on the GLOBALV command, the variable names must be delimited by commas or blanks.

```
'GLOBALV GETT VAR1 VAR2 VAR3'
```

```
'GLOBALV GETC VARY1,VARY2,VARY3'
```

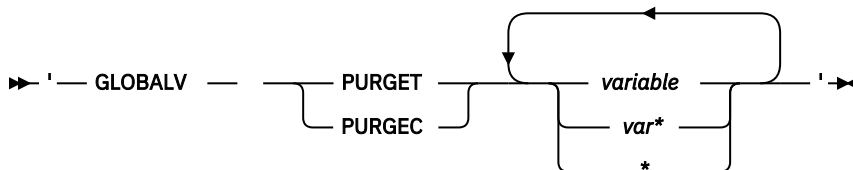
```
'GLOBALV GETT VARIABLE_A VARIABLE_B VARIABLE_C'
```

```
'GLOBALV GETC VARIABLE_X,VARIABLE_Y,VARIABLE_Z'
```

GLOBALV PURGE (REXX)

Syntax

GLOBALV PURGET|PURGEC



Purpose of Command

Use the GLOBALV PURGET or PURGEC command to purge task or common global variables from external storage.

When you use an asterisk symbol, the entire global dictionary is purged from external storage. If you are purging many variables, consider how much processor time is required for the I/O. See [“Using the asterisk \(*\)” on page 555](#) for information about preventing operators from using the wildcard.

Operand Descriptions

PURGET or PURGEC

Indicates that the task or common global variables with the specified variable types or names are purged from external storage.

variable [...]

For REXX, specifies the 1-250 character name or names of the task or common global variables to be purged from external storage. For NetView command list language, specifies the 1-11 character name or names of the task or common global variables to be purged from external storage. See [“GLOBALV \(REXX\)” on page 552](#) for a list of the characters you can use for a variable name. Variables can be separated by either a space or a comma.

Restrictions

For REXX, the instruction is enclosed in single quotation marks to prevent variable substitution.

Example: Coding variables for GLOBALV PURGE

The following examples show how you can code variables for the GLOBALV PURGET or PURGEC command:

Use either of the following examples to purge all variables from external storage:

```
'GLOBALV PURGET *'
```

```
'GLOBALV PURGEC *'
```

Use either of the following examples to purge all variables beginning with VAR1 and LU and the variable ABC from external storage:

```
'GLOBALV PURGET VAR1* ABC LU*'
```

```
'GLOBALV PURGEC VAR1* ABC LU*'
```

Use either of the following examples to purge all of the listed variables from external storage:

Note: As these examples show, if you specify more than one global variable on the GLOBALV command, the variable names must be delimited by commas or blanks.

```
'GLOBALV PURGET VAR1 VAR2 VAR3'
```

```
'GLOBALV PURGEC VAR_A,VAR_B,VAR_C'
```

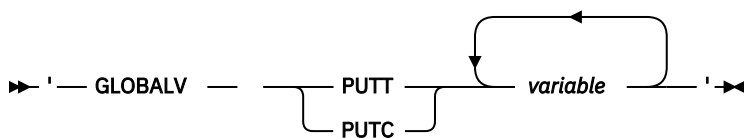
```
'GLOBALV PURGET VARIABLE_A VARIABLE_B VARIABLE_C'
```

```
'GLOBALV PURGEC VARIABLE_X,VARIABLE_Y,VARIABLE_Z'
```

GLOBALV PUT (REXX)

Syntax

GLOBALV PUTT|PUTC



Purpose of Command

Use the GLOBALV PUTT or PUTC command to set a task or common global variable from a REXX or NetView command list. The GLOBALV PUTT or PUTC command creates a task or common global variable with the specified variable name and places it in the task or common global variable dictionary.

When a GLOBALV PUTT or PUTC command is processed, the NetView program determines if a REXX or NetView command list language variable exists with the specified variable name. If a variable with the same name exists, the task or common global variable is created and its value is set to the value currently assigned to the variable.

To delete the value of a task or common global variable, set the value of the variable name to NULL (") and use the GLOBALV PUT command to replace the value. A GLOBALV GET request for the variable causes a null value to be returned.

If a NetView command list language variable with the specified variable name does not exist, a task or common global variable with the specified name is not created. Any attempt to retrieve the variable causes a null value to be returned.

The function of the GLOBALV PUT command is influenced by any previous &TGLOBAL, &CGLOBAL, or GLOBALV defined command issued within the same NetView command list for the same named variable. For example, if a variable has been defined as common (&CGLOBAL *variable*), a subsequent GLOBALV PUT *variable* moves the common global value to the task global variable by that name instead of moving a value from the NetView command list language variable. Refer to the *IBM Z NetView Programming: REXX and the NetView Command List Language* for more information about &TGLOBAL and &CGLOBAL.

If a task or common global variable exists with the specified variable name, the value of the task or common global variable is updated in the global variable dictionary.

Note: If you have more than one command list running under different tasks and accessing the same common global variable, the last value to which the variable is set is the value that is retrieved by any command list accessing the variable.

For example, a command list accesses a common global variable and, before that command list updates the variable, another command list running under a different task accesses the variable. If both command lists update the variable, the variable assumes the value given to it by the command list that updates it last.

To avoid problems with having a common global variable updated by different command procedures at the same time, you can use PIPE VARLOAD, which can compare and set a variable at one time. See the section about UPDCGLOB (CNME1080) for an example use of VARLOAD.

Refer to *IBM Z NetView Programming: REXX and the NetView Command List Language* for examples about using common global variables. Refer to *IBM Z NetView Programming: Pipes* for usage of VARLOAD.

The command lists in [“GLOBALV \(REXX\)” on page 552](#) show how to put, get, and update a task global variable.

Operand Descriptions

PUTT or PUTC

Indicates that task or common global variables with the specified variable names are put into the task or common global variable dictionary.

variable

For REXX, specifies the 1-250 name or names of the task or common global variables to be put. For NetView command list language, specifies the 1-11 character name or names of the task or command global variables to be put. See [“GLOBALV \(REXX\)” on page 552](#) for a list of the characters you can use for a variable name. Variables can be separated by either a space or a comma.

Restrictions

For REXX, the instruction is enclosed in single quotation marks to prevent variable substitution.

Example: Coding variables for GLOBALV PUT

Use any of the following examples to code variables for the GLOBALV PUTT or PUTC command:

Note: As these examples show, if you specify more than one global variable on the GLOBALV command, the variable names must be delimited by commas or blanks.

```
'GLOBALV PUTT VAR1 VAR2 VAR3'
```

```
'GLOBALV PUTC VARY1,VARY2,VARY3'
```

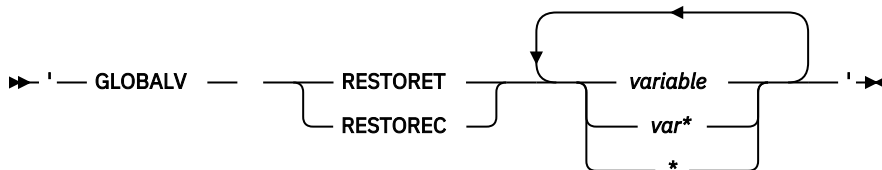
```
'GLOBALV PUTT VARIABLE_A VARIABLE_B VARIABLE_C'
```

```
'GLOBALV PUTC VARIABLE_X,VARIABLE_Y,VARIABLE_Z'
```

GLOBALV RESTORE (REXX)

Syntax

GLOBALV RESTORET|RESTOREC



Purpose of Command

Use the GLOBALV RESTORET or RESTOREC command to restore a task or common global variable in a REXX or NetView command list.

If you indicate RESTORET or RESTOREC for a variable that does not exist in external storage, the variable is restored in the global dictionary with a null value. This condition applies only if you specify the variables explicitly, not when you list all variables implicitly using an asterisk (*).

When you use an asterisk symbol, the entire global dictionary is restored from external storage or the VSAM file. If you are restoring many variables, consider how much processor time is required for the I/O. See [“Using the asterisk \(*\)” on page 555](#) for information about preventing operators from using the wildcard.

Operand Descriptions

RESTORET or RESTOREC

Indicates that the task or common global variables with the specified variable types or names are restored from external storage to the global dictionary.

Note: You can code RESTORET and RESTOREC using parameter synonyms RESTT or RESTC, respectively.

variable

For REXX, specifies the 1-250 character name or names of the task or common global variables to be restored to the global dictionary. For NetView command list language, specifies the 1-11 character name or names of the task or common global variables to be restored to the global dictionary. See [“GLOBALV \(REXX\)” on page 552](#) for a list of the characters you can use for a variable name. Variables can be separated by either a space or a comma.

Restrictions

For REXX, the instruction is enclosed in single quotation marks to prevent variable substitution.

Example: Coding variables for GLOBALV RESTORE

The following examples show how you can code variables for the GLOBALV RESTORET or RESTOREC commands.

Use either of the following examples to restore all variables in the task or common global variable dictionary from external storage:


```
'GLOBALV RESTORET *'
```

```
'GLOBALV RESTOREC *'
```

Use either of the following examples to restore all variables beginning with VAR1 and LU and the variable ABC in the task or common global variable dictionary from external storage:

```
'GLOBALV RESTORET VAR1* ABC LU*'
```

```
'GLOBALV RESTOREC VAR1* ABC LU*'
```

Use any of the following examples to restore all of the listed variables in the task or common global variable dictionary from external storage.

Note: As these examples show, if you specify more than one global variable on the GLOBALV command, the variable names must be delimited by commas or blanks.

```
'GLOBALV RESTORET VAR1 VAR2 VAR3'
```

```
'GLOBALV RESTOREC VARY1,VARY2,VARY3'
```

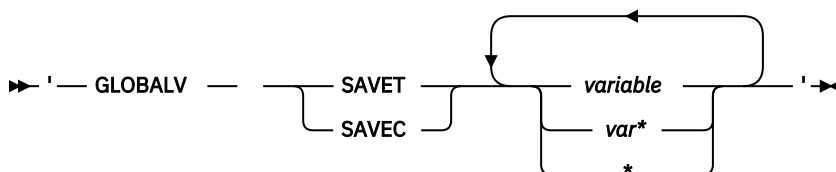
```
'GLOBALV RESTORET VARIABLE_A VARIABLE_B VARIABLE_C'
```

```
'GLOBALV RESTOREC VARIABLE_X,VARIABLE_Y,VARIABLE_Z'
```

GLOBALV SAVE (REXX)

Syntax

GLOBALV SAVET|SAVEC



Purpose of Command

Use the GLOBALV SAVET or SAVEC command to save a task or common global variable in a REXX or NetView command list, or VSAM database.

If you indicate SAVET or SAVEC for a variable that does not exist in the global dictionary, the variable is saved in external storage with a null value. This condition applies only if you specify the variables explicitly, not when you list all variables implicitly using an asterisk (*).

An asterisk (*) causes the global directory to be saved in external storage or to the VSAM file. To avoid performance degradation if you are saving many variables, consider how much storage space and processor time are required for the input/output (I/O). Refer to the GLOBALV command for information about preventing operators from using the wildcard.

Operand Descriptions

SAVET or SAVEC

Indicates that the task or common global variables with the specified variable names are saved from the global dictionary to external storage.

variable

For REXX, specifies the 1-250 character name or names of the task or common global variables to be saved to external storage. For NetView command list language, specifies the 1-11 character name or names of the task or common global variables to be saved to external storage. See [“GLOBALV \(REXX\)” on page 552](#) for a list of the characters you can use for a variable name. Variables can be separated by either a space or a comma.

Restrictions

For REXX, the instruction is enclosed in single quotation marks to prevent variable substitution.

The largest supported combined length for a variable name and its value is 4000 characters. Longer strings are not saved; they are output in a BNH917I message that can be automated, as in sample CNMS8053.

Because of the structure of long global variable names in the Save/Restore database, data corruption can result from specifying a wildcard for the SAVEC command when there are matching saved global variables. To avoid this scenario, delete the matching variables by running either the GLOBALV PURGE command with the wildcard name, or the DBAUTO SAVE CLEAR command to clear the entire database. Regardless of the length of the variable names that use the wildcard, it is good practice to clear the database before performing the SAVEC command. However, performing the purge and then the save concurrently on different tasks can also result in data corruption, so limit your usage of the GLOBALV SAVEC command.

Example: Coding variables for GLOBALV SAVE

The following examples show how you can code variables for the GLOBALV SAVET or SAVEC command.

Use either of the following examples to save all variables in the task or common global variable dictionary to external storage:

```
'GLOBALV SAVET *'
```

```
'GLOBALV SAVEC *'
```

Use either of the following examples to save all variables beginning with VAR1 and LU and the variable ABC in the task or common global variable dictionary to external storage:

```
'GLOBALV SAVET VAR1* ABC LU*'
```

```
'GLOBALV SAVEC VAR1* ABC LU*'
```

Use any of the following examples to save the specified variables in the task or common global variable dictionary to external storage:

Note: As these examples show, if you specify more than one global variable on the GLOBALV command, the variable names must be delimited by commas or blanks.

```
'GLOBALV SAVET VAR1 VAR2 VAR3'
```

```
'GLOBALV SAVEC VARY1,VARY2,VARY3'
```

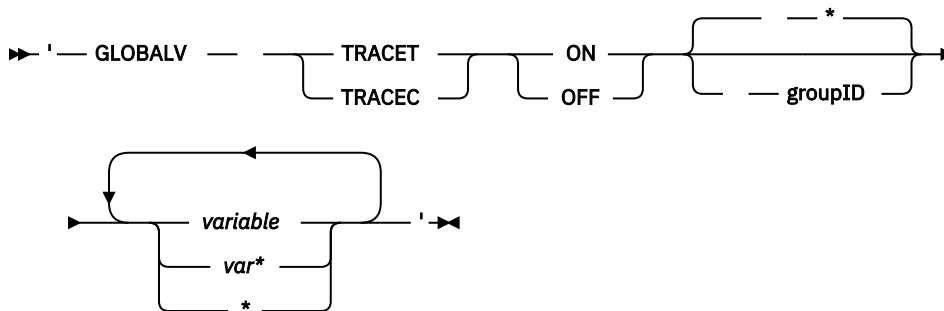
```
'GLOBALV SAVET VARIABLE_A VARIABLE_B VARIABLE_C'
```

```
'GLOBALV SAVEC VARIABLE_X,VARIABLE_Y,VARIABLE_Z'
```

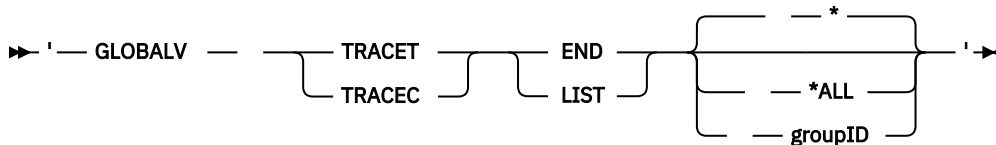
GLOBALV TRACE (REXX)

Syntax

GLOBALV TRACET|TRACEC



GLOBALV TRACET|TRACEC



Purpose of Command

Use the GLOBALV TRACET or TRACEC command to turn on or off the tracing of global variable value changes by a REXX or NetView command list, an HLL command, a PIPE, or a user written command that uses DSIVARS to change global variables. It can also be used to list the variable names being traced.

A global variable does not need to exist when tracing for it is turned on or off. Tracing for a global variable begins as soon as its name is added to the list of names to be traced using the ON option and ends as soon as it is removed from the list using the OFF or END options.

An asterisk (*) causes all common or task global variables to be traced. To avoid performance degradation if you are tracing many variables, consider how much processor (processor) time is required for tracing. Since the GLOBALV TRACE option can be restricted through security, it is assumed that only those users understanding the performance implications are allowed to use any TRACE options. TRACE is intended for use as a systems programmer diagnostic tool and not for general use.

Operand Descriptions

TRACET or TRACEC

Indicates that the trace setting for task or common global variables with the specified variable names are to be set. Or a list of the global variable names currently being traced can be listed.

ON

Indicates that the trace setting for the specified global variables is to be turned on and tracing for these variables is started.

OFF

Indicates that the trace setting for the specified global variables is to be turned off and tracing for these variables is stopped.

END

Indicates that the trace setting for the specified group(s) of global variable names are to be turned off and internal storage is freed if this is the last group with trace entries.

LIST

Indicates that the trace setting for the specified group(s) of global variable names are to be listed.

groupID

This is the 1-8 character group identifier with which the specified global variable names are associated. An '*' indicates that the NetView task ID under which the GLOBALV command is running is used as the group identifier. For END and LIST, '*ALL' indicates all groups are to be processed.

variable

For REXX, specifies the 1-250 character name or names of the task or common global variables to be traced. For NetView command list language, specifies the 1-11 character name or names of the task or common global variables to be traced. Refer to the GLOBALV command for a list of the characters you can use for a variable name. Variables can be separated by either a space or a comma. Variable names ending with an asterisk cause any global variable beginning with the preceding characters to have its trace setting changed.

Restrictions

For REXX, the instruction is enclosed in single quotation marks to prevent variable substitution.

Example: Coding variables for GLOBALV TRACE

The following examples show how you can code variables for the GLOBALV TRACET or TRACEC commands.

Use either of the following examples to trace all variables in the task or common global variable dictionary to the Netlog.

```
'GLOBALV TRACEC ON * *'
```

Use either of the following examples to trace all variables beginning with VAR1 and LU and the variable ABC in the task or common global variable dictionary and assign them to trace group GROUP1:

```
'GLOBALV TRACET ON GROUP1 VAR1* ABC LU*'
```

```
'GLOBALV TRACEC ON GROUP1 VAR1* ABC LU*'
```

Use either of the following examples to stop tracing the variables VAR1, VAR2, and VAR3 in the task or common global dictionary:

```
'GLOBALV TRACET OFF * VAR1 VAR2 VAR3'
```

```
'GLOBALV TRACEC OFF * VAR1 VAR2 VAR3'
```

Use either of the following examples to stop tracing the variables assigned to group GROUP1:

```
'GLOBALV TRACET END GROUP1'
```

```
'GLOBALV TRACEC END GROUP1'
```

Use either of the following examples to list all the variables being traced:

```
'GLOBALV TRACET LIST *ALL'
```

```
'GLOBALV TRACEC LIST *ALL'
```

MSGREAD (REXX)

Syntax

MSGREAD

➤ ' — MSGREAD — ' ➤

Purpose of Command

The MSGREAD instruction causes the NetView program to read a trapped message from the message queue. This new message becomes the current message for purposes of REXX functions that refer to the current message. If MSGREAD finds no message on the trapped message queue, then there is no current message following the MSGREAD operation. The command list then takes action based on the message. Refer to the TRAP and WAIT instructions for information about using these with the MSGREAD instruction.

When the MSGREAD instruction is issued, the oldest message in the queue is read and removed. Refer to the *IBM Z NetView Programming: REXX and the NetView Command List Language* for information about using TRAP in nested REXX command lists and the functions set by MSGREAD.

Note: The MSGREAD command pre-dates NetView pipelines. You can use the PIPE command to solve this type of problem. For more information about the PIPE command and its stages, refer to *IBM Z NetView Programming: Pipes*.

Return codes

MSGREAD sets the value of the return code (RC) to indicate the results of processing, as follows:

Return Code

Meaning

-2	Syntax error
-1	DSIGET failure
0	Successful completion
4	No messages in queue

Restrictions

For REXX, the instruction is enclosed in single quotation marks to prevent variable substitution.

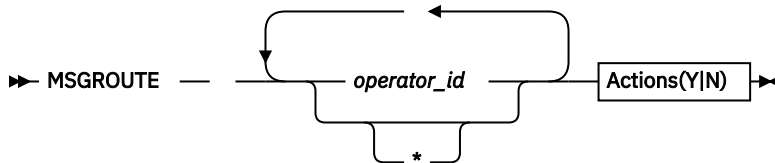
Using the MSGREAD instruction

The following is an example of a REXX command list named ACTLU. ACTLU issues a VTAM command to activate a node specified by the user. Messages sent as the result of the activated command are trapped, and the operator is notified of the result of the command list. The comments in the example explain how the command list works:

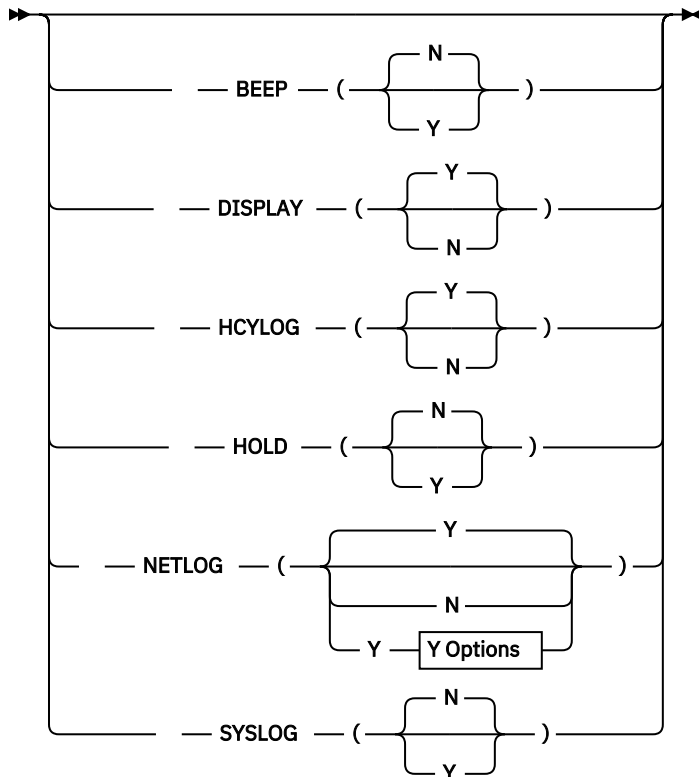
MSGROUTE (REXX)

Syntax

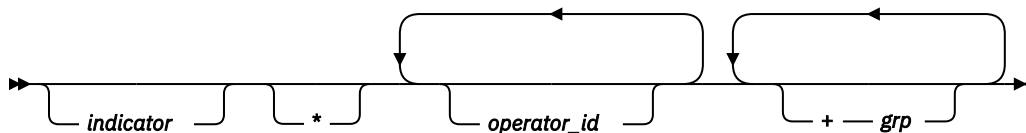
MSGROUTE



Actions(Y|N)



Y Options



Purpose of Command

The MSGROUTE command routes its current message to specified operators or operator groups. Some message attributes can be changed. The "current message" might be a message that caused the automation table to drive the command list that contains the MSGROUTE command, or it might be a message which flowed to MSGROUTE in a NetView pipeline.

After MSGROUTE routes a message, NetView automation processes the message unless it was previously automated. Messages sent by the MSGROUTE command are queued to the normal priority message queue of the target task or tasks.

Note: When you route a copy of a message to a cross-domain NetView system, the message drives the message table on the receiving system. That is, the copies work the same as the copies you create using the ASSIGN command.

Operand Descriptions

BEEP

Determines whether an audible alarm is sounded when the message is displayed. The default is BEEP(N).

DISPLAY

Determines whether the message is displayed. The default is DISPLAY(Y).

HCYLOG

Determines whether the message is placed in the hardcopy log. The default is HCYLOG(Y).

HOLD

Determines whether the message is held on the operator's panel after it is displayed. The default is HOLD(N).

NETLOG

Determines whether the message is placed in the NetView log and whether the message activates a status monitor important message indicator for specified operators or groups of operators. The default is NETLOG(Y).

Specifying NETLOG(Y) can include the following variables:

Indicates that the *cmdstring* or message is routed to the current operator task (the task where the message is intercepted for automation checking). If the operator task is CNMCSSIR, message routing can differ. Refer to the *IBM Z NetView Administration Reference* for more information about CNMCSSIR and NetView automation.

+grp

Specifies the group identifier of the groups of operators for whom the message is logged as important. You can code as many group identifiers as needed. The maximum length of a group identifier is eight characters, and it must begin with a plus sign (+). Define group identifiers with the ASSIGN command. Refer to the NetView online help for more information about the ASSIGN command.

indicator

Identifies the status monitor important message indicator.

operator_id

Specifies the operator identifier of the operators for whom the message is logged as important. The maximum length of an operator identifier is eight characters. You can code as many operator identifiers as needed.

An asterisk (*) can be specified for the *operator_id*. If MSGROUTE is issued on a virtual OST (VOST), the asterisk (*) is interpreted as the owning task's operator identifier. On any other task, the asterisk (*) is interpreted as the operator identifier of the issuing task. For more information, refer to the *IBM Z NetView Administration Reference*.

If the operator is not in the status monitor or a log browse but is logged on, message CNM039I is displayed:

```
CNM039I AN IMPORTANT MESSAGE HAS BEEN LOGGED -  
PLEASE BROWSE THE NETVIEW LOG.
```

If you specify only an *indicator*, the message is logged as important for the authorized receiver. The following example shows how NETLOG is coded with only an *indicator-number*:

```
MSGROUTE OPER1 NETLOG(Y 2)
```

The message is routed to OPER1. The message is also placed in the NetView log and is logged as an important message with a status monitor important message *indicator-number* of 2.

If you specify an *indicator-number* and a list of operators or groups of operators, the message is logged as important for the operators and groups of operators listed. The following example shows how a message is logged when you specify an *indicator-number* and a list of operators and groups of operators:

```
MSGROUTE OPER4 NETLOG(Y 2 * OPER1 +GRP5 OPER6)
```

The message is routed to OPER4. The message is also placed in the NetView log and is defined as an important message with a status monitor important message *indicator-number* of 2. The message activates a status monitor important message indicator for OPER1, OPER6, all of the operators assigned to group +GRP5, and the current operator. If operators OPER1 and OPER6 are also assigned to group +GRP5, each operator receives only one copy of message CNM039I if they are not in the status monitor.

Note: The NETLOG option cannot be specified without an operand between the parentheses. In other words, you cannot specify NETLOG().

operator_id

Is the operator identifier of the operators to whom the message is routed. The maximum length of an operator identifier is eight characters. You can code as many operator identifiers as needed.

You can also specify group identifiers for the groups of operators to whom the message is routed. You must define the group identifier to the NetView program with the ASSIGN command. The maximum length of a group identifier is eight characters, and the group identifier must begin with a plus sign (+).

An asterisk (*) can be specified for the *operator_id*. If MSGROUTE is issued on a virtual OST (VOST), the asterisk (*) is interpreted as the operator identifier of the issuing task. On any other task, the asterisk (*) is interpreted as the operator identifier of the issuing task.

SYSLOG

Determines whether the message is placed in the system log. The default value is SYSLOG(N).

Return codes

The return code (RC or &RETCODE) indicates the processing results as follows:

Return Code

Meaning

- | | |
|-----------|---|
| 8 | Operator or group identifier is not specified, or is greater than 8 characters. |
| 12 | The value is not valid for message action. |
| 16 | MSGROUTE was not entered from a REXX or NetView command list language command list. |
| 20 | MSGROUTE was not issued from a message-driven REXX or NetView command list language command list, or from the NetView automation table. |
| 24 | Operator or group identifier, or message action is not within the authority of the authority. |
| 28 | The storage request failed. |
| 32 | There is no active operator to which to route messages. |
| 36 | MSGROUTE was issued for a buffer with HDRMTYPE X'10' (MSU data). MSGROUTE does not support routing of MSU data. |

Usage Notes

Consider the following when using the MSGROUTE command:

- The DEFAULTS and OVERRIDE settings do not affect the settings in MSGROUTE.
- Many of the functions of MSGROUTE can also be performed by PIPE stages; see the EDIT and ROUTE stages, especially.

PARSEL2R (REXX)

Syntax

PARSEL2R

➤ PARSEL2R — — *source_variable* — — *parsing_template* ➤

Purpose of Command

The PARSEL2R command enables you to extract data from the character-string value of a variable and assign the extracted data to one or more variables using a set of rules called a *parsing template*. To parse variables in REXX, use either PARSEL2R or the REXX PARSE instruction. To parse variables with the NetView command list language, use the PARSEL2R command.

Note: Refer to the REXX library for more information about the PARSE instruction.

Operand Descriptions

source_variable

Identifies a command list variable.

In REXX command lists, code *source_variable* within single quotation marks.

In command lists written in the NetView command list language, you must code the *source_variable* without an ampersand (&). PARSEL2R extracts data from the value of the variable you named as the *source_variable*.

The maximum character length of a source variable name depends on the following:

11

When started from a command list written in the NetView command list language.

250

When started from a REXX-format command list.

parsing_template

Specifies a list of symbols, patterns, or character selectors, or a combination of these, separated by blanks. PARSEL2R uses this list as a template when parsing the source variable.

Symbols are command list variable names. The rules governing the length of the symbol names in the parsing template are the same as those for the length of the following *source_variable*. In REXX command lists, code command list variable names within single quotation marks. In command lists written in the NetView command list language, code command list variable names without an ampersand (&).

Patterns are coded using slashes (/) as delimiters. A pattern is the part of the source variable that you want to match.

A character selector is coded using an asterisk (*) for each single character you want to extract from the source variable.

Return codes

PARSEL2R sets the return code (RC or &RETCODE) to indicate the processing results as follows:

Return Code Meaning

- 0**
Processing was successful.
- 8**
Storage is not available to continue processing.
- 100**
There are not enough parameters, or not issued from a command list.
- 104**
Input buffer is blank.
- 108**
The command list dictionary lookup of source variable failed.
- 112**
Hexadecimal data in the template is not valid.
- 116**
The command list dictionary update failed.
- 120**
The trailing slash (/) is missing.

Using symbols in a parsing template

The symbols in the parsing template identify command list variables.

In REXX command lists, code command list variables within quotation marks.

In command lists written in the NetView command list language, code command list variables without the ampersand (&). If only symbols appear in the parsing template, the source variable data is assigned token-by-token from left to right. Tokens are defined as a string of nonblank characters. Tokens in the source variable are separated by one or more blanks. The tokens are assigned to the command list variables you identified with symbols in the parsing template.

The following examples show three lines that use a parsing template containing only symbols:

For REXX:

```
TITLE = 'PROCEDURE/ACTION NOT SUPPORTED: SENSE = X'087D'  
'PARSEL2R TITLE A1 A2 A3 A4 A5 A6 A7 A8'  
'PARSEL2R TITLE B1 B2 B3'
```

For command list:

```
TITLE = 'PROCEDURE/ACTION NOT SUPPORTED: SENSE = X'087D'  
'PARSEL2R TITLE A1 A2 A3 A4 A5 A6 A7 A8'  
'PARSEL2R TITLE B1 B2 B3'
```

The resulting values of the variables show how the token-by-token assignment from left to right works.

The following table shows the resulting values for the REXX and NetView command list language variables:

REXX Variable	NetView Variable	Value
A1	&A1	PROCEDURE/ACTION
A2	&A2	NOT
A3	&A3	SUPPORTED:
A4	&A4	SENSE
A5	&A5	=

REXX Variable	NetView Variable	Value
A6	&A6	X'087D'
A7	&A7	null
A8	&A8	null
B1	&B1	PROCEDURE/ACTION
B2	&B2	NOT
B3	&B3	SUPPORTED: SENSE = X'087D'

Note:

1. The value of B3 or &B3 is not a single token, but the remainder of the source variable after PARSEL2R has parsed it into the first two symbols.
2. Except for the last variable, leading blanks and trailing blanks are removed from each token in the string before it is assigned to a variable. Variable B3 or &B3 have leading or trailing blanks if TITLE contained extra blanks before SUPPORTED, or after X'087D'.

Using patterns in a parsing template

A pattern is a character or string of characters expected to appear within the source variable. Code patterns in the PARSEL2R parsing template using slashes (/) as delimiters. Use patterns within a parsing template to divide the source variable into segments.

When a pattern that you coded in the parsing template occurs in the source variable, the preceding portion of the source variable is treated as a segment. The symbols you defined in the parsing template preceding the pattern are used to parse the tokens in the corresponding segment of the source variable.

Note: If you want to use a slash (/) as a part of a pattern, code 2 consecutive slashes within the delimiter slashes. PARSEL2R reads this as one slash to be matched in the source variable. Two consecutive slashes by themselves (outside of delimiters) ends the parse.

The following examples show how you can use a parsing template containing patterns and symbols:

For REXX:

```
PARSIT = 'PROCEDURE/ACTION NOT SUPPORTED: SENSE = X'087D'
'PARSEL2R PARSIT A1 A2 A3 //// B1 B2 B3 /:/ C1 C2 C3 /=/ D1 D2 D3'
```

For command list:

```
&PARSIT = 'PROCEDURE/ACTION NOT SUPPORTED: SENSE = X'087D''
PARSEL2R PARSIT A1 A2 A3 //// B1 B2 B3 /:/ C1 C2 C3 /=/ D1 D2 D3
```

The following table shows the resulting values for the REXX and NetView command list language variables:

REXX Variable	NetView Variable	Value
A1	&A1	PROCEDURE
A2	&A2	null
A3	&A3	null
B1	&B1	ACTION
B2	&B2	NOT
B3	&B3	SUPPORTED
C1	&C1	SENSE

REXX Variable	NetView Variable	Value
C2	&C2	null
C3	&C3	null
D1	&D1	X'087D'
D2	&D2	null
D3	&D3	null

The following examples are of a parsing template containing patterns and symbols:

For REXX:

```
PARSIT = 'PROCEDURE/ACTION NOT SUPPORTED: SENSE = X'087D'
'PARSEL2R PARSIT A1 A2 A3 /:/ A4'
```

For command list:

```
&PARSIT = 'PROCEDURE/ACTION NOT SUPPORTED: SENSE = X'087D'
PARSEL2R PARSIT A1 A2 A3 /:/ A4
```

The following table shows the resulting values for the REXX and NetView command list language variables:

REXX Variable	NetView Variable	Value
A1	&A1	PROCEDURE/ACTION
A2	&A2	NOT
A3	&A3	SUPPORTED
A4	&A4	SENSE = X'087D'

Note: Because the variable just before a pattern (A3 or &A3 in the previous examples) is treated as the last variable in a segment, the leading and trailing blanks are not removed.

To remove the blanks from the A3 or &A3 variable and make the AX or &AX variable null, specify the parsing template as follows:

```
PARSEL2R PARSIT A1 A2 A3 AX /:/ A4
```

You can use variables as part of a pattern (between the slashes). When a variable is part of a pattern, code it outside of the quotation marks in a REXX command list.

The following example shows three lines from a REXX command list that uses a parsing template with a pattern containing a variable:

```
A0 = SENSE
PARSIT = 'PROCEDURE/ACTION NOT SUPPORTED: SENSE = X '087D'
'PARSEL2R PARSIT A1 A2 A3 /'A0' = / B1'
```

Because A0 is outside quotation marks, its value is used as part of the pattern. The pattern becomes SENSE =.

When a variable is part of a pattern, code it with an ampersand (&) in command lists written in the NetView command list language. The following example shows the NetView command list language equivalent of the REXX example:

```
&A0 = SENSE
&PARSIT = 'PROCEDURE/ACTION NOT SUPPORTED: SENSE = X'087D'
PARSEL2R PARSIT A1 A2 A3 /&A0 = / B1
```

The following table shows the resulting values for the REXX and NetView command list language variables:

REXX Variable	NetView Variable	Value
A1	&A1	PROCEDURE/ACTION
A2	&A2	NOT
A3	&A3	SUPPORTED:
B1	&B1	X'087D'

You can also use hexadecimal codes in the parsing template pattern. Code a hexadecimal pattern using an X before the slashes. PARSEL2R matches the hexadecimal code in the template with the character in the source variable that corresponds to your system.

The following examples show how to code a parsing template containing a hexadecimal pattern, where X'E7' is an EBCDIC letter X:

For REXX:

```
PARSIT = 'PROCEDURE/ACTION NOT SUPPORTED: SENSE = x'087D'
'PARSEL2R PARSIT A1 A2 X/E7/ A3'
```

For command list:

```
&PARSIT = 'PROCEDURE/ACTION NOT SUPPORTED: SENSE = X'087D'
PARSEL2R PARSIT A1 A2 X/E7/ A3
```

The following table shows the resulting values for the REXX and NetView command list language variables:

REXX Variable	NetView Variable	Value
A1	&A1	PROCEDURE/ACTION
A2	&A2	NOT SUPPORTED: SENSE
A3	&A3	= '087D'

Using patterns and symbols in a parsing template gives you a powerful tool to use when coding your command lists. For example, you can use PARSEL2R to code a source variable in your command list that contains a small table. You can also use the combination of symbols and patterns to search the source variable and assign a token to a variable, based on the matching pattern. This table can contain a string of alternating variables and labels. You can then use PARSEL2R to match a variable with a label to define the flow of logic within your command list.

Using character selectors in a parsing template

Character selectors in a PARSEL2R are coded as one or more asterisks (*), indicating that you must assign the preceding symbol one or more characters from the source variable. If a character selector does not follow a symbol in the template (coded at the beginning of the template or following a pattern), PARSEL2R skips that number of characters.

For example, if the source variable is:

```
DSI039I MSG FROM CNM01PPT: COMMON GLOBAL VARIABLES HAVE BEEN SET
```

And the parsing template is:

```
/FROM / DOMNAM ***** TASK /:/
```

The following values are assigned:

```
DOMNAM oī &DOMNAM = CNM01
TASK oī &TASK = PPT
```

Character selectors are usually used to break up a single token into multiple variables.

Figure 12 on page 581 and Figure 13 on page 581 show a parsing template using character selectors to change the value of a REXX message variable or a NetView command list language control variable. &MCSFLAG, or the value returned by the MCSFLAG() function, is set to 00000110 when the command list is entered. The command list statements set bit 6 to zero (0).

```
MCSBITS = MCSFLAG()
'PARSEL2R MCSBITS BITS1_5 ***** BIT6 * BITS7_8 **'
BIT6 = 0
MCSFLAG = BITS1_5 || BIT6
MCSFLAG = MCSFLAG || BITS7_8
```

Figure 12. REXX PARSEL2R Example Using Character Selectors

```
PARSEL2R MCSFLAG BITS1T05 ***** BIT6 * BITS7T08 **
&BIT6 = 0
&MCSFLAG = &CONCAT &BITS1T05 &BIT6
&MCSFLAG = &CONCAT &MCSFLAG &BITS7T08
```

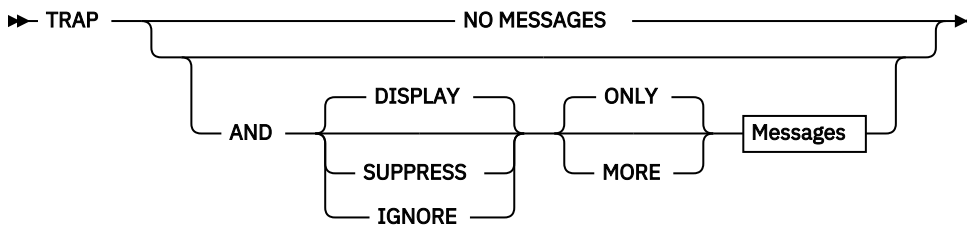
Figure 13. NetView Command List Language PARSEL2R Example Using Character Selectors

After the command list statements in the preceding figures are run, the value of MCSFLAG or &MCSFLAG is 00000010.

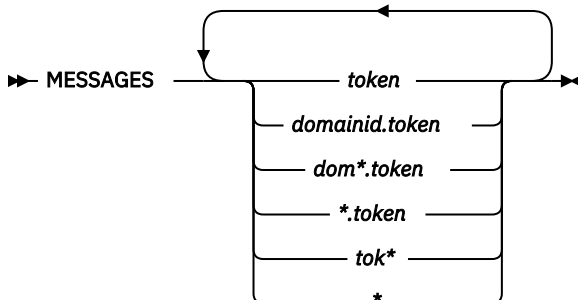
TRAP (REXX)

Syntax

TRAP



Messages



Purpose of Command

Use the TRAP instruction to define messages that are to be trapped and to specify whether the messages are displayed to the operator when they are trapped. You can also use the TRAP instruction to remove all messages from the list of messages to be trapped. TRAP can be used only for messages routed to the operator.

For REXX:

The TRAP instruction causes the NetView program to monitor the operator task for specified messages. If the messages occur, they are trapped and added to the message queue. Trapped messages can then be read using the MSGREAD instruction and can satisfy a WAIT instruction. Refer to the MSGREAD and WAIT (REXX) instructions for information about using these with the TRAP instruction.

If the trapped message satisfies the wait state, processing of the waiting command procedure resumes. If you do not suppress the message, it continues with the message flow. However, if you suppress the message, the NetView program marks it for deletion. In this case, automation-table processing does not occur and the NetView program does not display or log the message.

Note: For information about using TRAP in nested REXX command lists, refer to *IBM Z NetView Programming: REXX and the NetView Command List Language*. For more information about message flow, refer to the *IBM Z NetView Automation Guide*. Refer to the *IBM Z NetView Customization Guide* for an example about using the IGNORE option.

For HLL:

The TRAP command lets you specify message trapping criteria for HLL and REXX command procedures. When the TRAP command is issued, all subsequent messages that match the conditions defined by the trapping criteria are added to the message queue (TRAPQ).

When used with the WAIT FOR MESSAGES command and the CNMGETD service routine, the TRAP command allows you to code command procedures to intercept and process, (for example, Automate, Display, and Suppress certain messages. The message trapping criteria specified in the TRAP command define the set of conditions that satisfy subsequent WAIT FOR MESSAGES commands. TRAP can be issued only from command procedures written in an HLL or REXX.

NetView NNTs, PPTs, OSTs, and autotasks do not process any commands or messages queued to the low priority queue of a task that is running any command list or command processor. Because of this, only messages that are queued to the high or normal priority message queue of a waiting task are checked for matches to satisfy the WAIT condition.

Operand Descriptions

AND

Specifies to make the TRAP command more readable. AND can be used between:

- TRAP and DISPLAY
- TRAP and SUPPRESS
- TRAP and IGNORE

DISPLAY

Indicates that any message matching a specified *token* is displayed on the operator's screen when received by the NetView program. DISPLAY is the default.

IGNORE

Indicates that any message matching a specified *token* is neither displayed on the operator's screen or added to the message queue when received by the NetView program. Because an ignored message is not queued, it does not satisfy any outstanding WAIT and is not available to MSGREAD.

MORE

Indicates that the specified *tokens* are added to the list of *tokens* that were specified on a previous TRAP command. Any PERSIST TRAP that is enabled remains active.

Note: Each message in the resulting list retains its individual setting of the SUPPRESS|DISPLAY option. This enables some messages in the list to be suppressed while others are displayed.

NO MESSAGES

Indicates that the list of *tokens* that was specified on all previous TRAP commands is removed. The TRAP NO MESSAGES option clears the list of messages to trap. No other operands are valid with NO MESSAGES. Any PERSIST TRAP that was enabled by the calling procedure is canceled.

ONLY

Indicates that the specified *tokens* replace the list of *tokens* specified on all previous TRAP instructions in the called REXX command list. Any PERSIST TRAP that was enabled by the calling procedure is canceled. ONLY is the default value.

SUPPRESS

Indicates that messages matching a specified *token* are neither displayed on the operator's screen nor logged when received by the NetView program.

Usage notes:

1. When using TRAP and SUPPRESS, messages are not logged or automated.
2. If you route a command list to another domain using the ROUTE command, and your command list contains a TRAP AND SUPPRESS instruction, messages resulting from your command list are logged under the NNT in the receiving domain, but are suppressed in your domain.

MESSAGES

Indicates that the trapped items are messages. The 1 - 8 character *domainid* is the domain ID of the message or messages to be trapped. The 1 - 10 character *token* identifies the first token of the message or messages to be trapped. Most special characters are valid for *token*. There is no limit on the number of *tokens* that you can specify.

You can use a trailing asterisk (*) in the *domainid* or *token* fields as a wildcard character. You can also use an asterisk (*) as a character in a *token*. If the * is followed by a character, it is considered to be part of the *token*. For example, TRAP MESSAGES A*B* traps on *tokens* beginning with A*B followed by any characters.

Following are examples of how you can specify the messages you want to trap:

domainid.token

The command list traps any message whose domain identifier matches the 1 - 5 character (for REXX) or 1 - 8 character (for HLL) *domainid* and whose first token matches *token*.

dom*.token

The command list traps any message whose domain identifier matches the partial domain identifier specified by *dom** and whose first token matches *token*. For example, NCCF*.DSI463I means trap a DSI463I message from any domain with an identifier that starts with NCCF, such as NCCFA or NCCFB.

****.token***

The command procedure traps any message whose first token matches *token*. The message can be from any domain.

token

The command list traps any message whose first token matches *token*. The message can be from any domain.

tok*

The command list traps any message whose first token matches the partial token specified by *tok**. For example, DSI* means trap any messages whose first token begins with DSI, such as DSI463I or DSI386I. The message can be from any domain.

Specifies that the command list traps all messages routed to the operator.

Note: Using a single asterisk, or with SUPPRESS, can cause all messages to be trapped, including those that are not a result of issued commands.

Note: If you specify a token that contains a special character such as a comma, period, asterisk, or most other nonnumeric and nonalphabetic characters, use the DOMAIN.TOKEN format. The NetView program does not accept commas (,) or blank spaces when you specify a token because these characters are reserved as NetView default delimiters.

Return codes

TRAP sets the value of the return code (RC) to indicate the processing results, as follows:

Return Code Name	Value	Description
CNM_GOOD	0	Everything is OK.
CNM_BAD_INVOCATION	4	TRAP was not issued from an HLL or REXX command procedure.
CNM_BAD_SYNTAX	12	Syntax error. If DSI028I is issued, check <i>domainid.token</i> syntax. If DSI604I is issued, check the format of the TRAP command according to the operand specified.
CNM_BAD_COMMAND	144	TRAP was not issued from a command procedure running under an OST or NNT.
CNM_BAD_MRBLD + X	18000 + X	Nonzero return code X. See values for X in the following table.

Values for X:

Return Code Value	Value for X	Description
18000 + X	8	Request for storage has failed.

Usage Notes

Consider the following when using the TRAP command for REXX:

- Multiline messages, such as multiline write-to-operator (MLWTO), are treated as one message. Therefore, only the message identifier of the first nonblank message in a multiline message is available to the TRAP, and the TRAP can be satisfied only by that message identifier. Use GETMSIZE, GETMTYPE, and GETMLINE to access the other messages in a multiline message. Refer to the *IBM Z NetView Programming: REXX and the NetView Command List Language* for an example of the TRAP instruction in a REXX command list. Refer to the GETM commands for more information about multiline messages.
- When using a *token* event, messages not related to the command issued by the TRAP statement can be matched to the event and, depending on the options on the TRAP statement, can be suppressed with SUPPRESS when specifying a domain identifier or token. If the command list is suspended and the SUPPRESS option is in effect on the TRAP statement, any messages the task receives are suppressed before the command list is resumed. This includes any REXX trace output and messages if you are using the REXX TRACE instruction to debug the command list.
- Because NetView NNT, PPT, OST, and autotasks do not process any commands or messages queued to the low priority queue of a task that is running any command (assembler command or HLL, REXX, or NetView command list language command procedure), only messages that are queued to the high or normal priority queue of a waiting task are checked for matches to satisfy the WAIT condition.
- Normally, messages queued to tasks with `assign . . . copy= processing` can satisfy an outstanding trap. However, a message is not sent to the trapping task with `assign . . . copy= processing` if the message has a message automation table entry specifying DISPLAY(N). The `assign . . . copy= processing` requires a displayed message, but DISPLAY(N) specifies "no display", which prevents that processing. The message is not passed on and, therefore, cannot satisfy the TRAP condition.
- The TRAP ONLY and TRAP NO MESSAGES options cause any pending PIPE PERSIST TRAP commands to end without delivering the text that is specified with the *tText* option.
- If you are using the PIPE NETVIEW stage to invoke a command procedure that uses the TRAP command, make sure you allow the trapping to occur. You can use the EXPOSE stage after the NETVIEW stage to allow the trap exit to detect messages for trapping.

Consider the following when using the TRAP command for HLL:

- The TRAP command can be issued only from a command procedure running under an OST or NNT.
- All operands are order-dependent.

- Each TRAP command without the MORE operand cancels and replaces the previous TRAP command. To add tokens to a previous TRAP command, use the MORE operand.
- Issuing a TRAP command does not clear the queue of messages trapped by the previous TRAP command. To clear the message queue, issue a FLUSHQ.
- If you specify the same number in the TRAP command in both the waiting and nested command procedure, the message satisfies the TRAP in the nested command procedure. If the procedure ends before the message satisfies the TRAP, the message is queued for the waiting command procedure without ending the TRAP (used with WAIT FOR MESSAGES) or ending the waiting command procedure. The message queue continues to grow and the NetView program might run out of storage.
- Immediate messages are not trapped and they are not added to the message queue (TRAPQ).
- TRAP AND SUPPRESS MESSAGES * is not the same as TRAP NO MESSAGES. TRAP AND SUPPRESS MESSAGES * traps all messages but does not display them on the operator's console.
- Message trapping (TRAP command) takes precedence over NetView automation.
- When trapping TAF messages, the domain ID is the session ID for that TAF session.

Restrictions

The following restrictions apply to the TRAP command:

- Operands must be entered in the order shown in the syntax diagram.
- Messages issued by commands within NetView pipelines are not exposed to external interfaces, including trap processing. Therefore, make sure any procedure issuing the TRAP command is not itself being issued within a pipeline. For more information about restrictions to TRAP processing, see the TRAP() REXX function and the PIPE EXPOSE stage.
- For REXX, the instruction is enclosed in single quotation marks to prevent variable substitution.
- The TRAP instruction does not clear the queue of messages trapped by the previous TRAP. To clear the message queue, issue a FLUSHQ instruction. Refer to the FLUSHQ instruction for more information.
- Messages issued by DSIPSS with TYPE=FLASH cannot satisfy an outstanding TRAP.

Using the TRAP command

The following is an example of a REXX command list named ACTLU. ACTLU issues a VTAM command to activate a node specified by the user. Messages sent as the result of the activated command are trapped, and the operator is notified of the result of the command list. The comments in the example explain how the command list works:

```

/*****
/* ACTLU COMMAND LIST
/*
/* FUNCTION : TO ACTIVATE A VTAM NODE.
/* INPUT : 1 PARAMETER, THE NAME OF THE NODE.
/*****
SIGNAL ON ERROR /* SIGNAL IF ERROR OCCURS
IF MSGVAR(1) = '' THEN /* NO FIRST PARAMETER ?
DO /* THEN ISSUE REQUEST
SAY 'PLEASE ENTER "GO NODENAME"', /* REQUEST NODENAME FROM USER
'TO CONTINUE OR "GO STOP" TO', /* OR, ALLOW USER TO STOP
'STOP'
PARSE PULL NODE /* NODE = NODENAME OR STOP
END /* THEN ISSUE REQUEST
ELSE /* FIRST PARAMETER EXISTS
NODE = MSGVAR(1) /* ASSUME IT IS A NODE NAME
IF NODE-='STOP' THEN /* IF USER DID NOT CHOOSE STOP
DO /* PROCESS NODENAME
'TRAP AND SUPPRESS ONLY MESSAGES IST* ' /* TRAP ALL VTAM MSGS
'V NET,ACT,ID='NODE /* ISSUE VTAM ACTIVATE FOR NODE
RCSAVE=RC /* SAVE RETURN CODE IN RCSAVE
'WAIT 30 SECONDS FOR MESSAGES' /* WAIT FOR 30 SECONDS
SELECT
WHEN (EVENT()='M') THEN /* OUT OF WAIT - IS THERE A MSG?
DO /* PROCESS TRAPPED MESSAGE
'MSGREAD' /* READ IN 1ST MESSAGE
DO WHILE (RC=0) /* IF RC=0 THEN NO MORE MSGS
SELECT /* DETERMINE WHICH MESSAGE HIT
WHEN (MSGID() = 'IST061I') THEN /* NODE NOT FOUND
SAY '==> LU UNKNOWN', /* INFORM USER
'TO YOUR VTAM <==',
WHEN (MSGID() = 'IST093I') THEN /* NODE NOW ACTIVE
SAY '==> TERMINAL 'MSGVAR(1)' NOW', /* INFORM USER
MSGVAR(2) '<==',
OTHERWISE /* IGNORE THE VTAM MESSAGE
IF RCSAVE=0 THEN
'WAIT CONTINUE' /* CONTINUE WAITING
ELSE
DO
SAY 'ERROR PROCESSING', /* ERROR ENCOUNTERED ?
'ACTIVATE COMMAND' /* INFORM USER
SAY MSGSTR() /* DISPLAY MESSAGE TEXT
END
END /* OF SELECT FOR IST061I/IST093I
'MSGREAD' /* READ IN THE NEXT MESSAGE
END /* DO WHILE RC=0, LOOP BACK
END /* PROCESS TRAPPED MESSAGE DO
/* OUT OF DO WHILE

WHEN (EVENT()='E'|RCSAVE-='0') THEN /* CHECK FOR ERROR OR
/* TIME-OUT EVENTS
SAY 'ERROR PROCESSING', /* ERROR ENCOUNTERED?
'ACTIVATE COMMAND' /* INFORM USER
WHEN (EVENT()='T') THEN /* WAIT TIME-OUT ENCOUNTERED?
SAY 'NO RESPONSE TO', /* INFORM USER
'ACTLU CLIST FOR 'NODE
OTHERWISE; /* NO-OP
END /* OF SELECT FOR ERROR/TIME-OUT
END /* IF NODE-='STOP' PROCESSING

EXIT 0;
ERROR:
IF RC = 4 THEN
EXIT
ELSE
SAY 'ERROR OCCURRED. RETURN CODE IS ' RC
EXIT -1;
/*****

```

Figure 14. Sample Command List Using TRAP, WAIT, MSGREAD, and WAIT CONTINUE

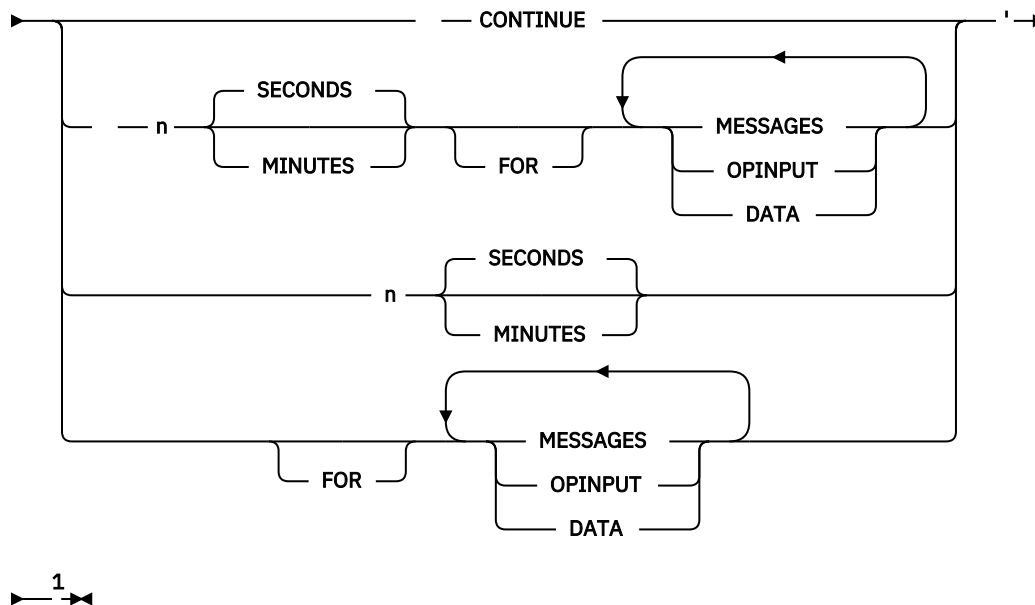
Refer to the *IBM Z NetView Programming: REXX and the NetView Command List Language* for more examples of REXX command lists for the NetView program.

WAIT (REXX)

Syntax

WAIT

➤ ' — WAIT ➤



Notes:

¹ OPINPUT and DATA are only valid for HLL.

Purpose of Command

When issued from a command procedure, the WAIT command temporarily suspends processing of that command procedure until a specified event occurs. For an HLL command procedure, the event can be one or more messages, operator input, data, a certain period of time, or any combination of these. The first occurrence of one of these events satisfies the wait and processing is resumed.

When the NetView program encounters a WAIT instruction in a REXX command list, the letter W is displayed in the upper right corner of the current command facility panel if the screen is refreshed because a message is received or the ENTER key is pressed. This W notifies the operator that the command list has halted its processing and is waiting for a message or group of messages or for a specific period of time. The first of these events that occurs satisfies the WAIT.

WAIT can be used in nested REXX command lists. For more information, refer to the *IBM Z NetView Programming: REXX and the NetView Command List Language*.

Operand Descriptions

CONTINUE

Specifies to continue waiting for additional messages, operator input, or data before command procedure processing is resumed.

The options specified on the previous TRAP and WAIT instructions remain in effect for the WAIT CONTINUE instruction. When processing resumes, the next instruction after the WAIT CONTINUE is issued.

For example, if you code the following instructions in a command list and one of the messages specified on the TRAP instruction is received in 5 seconds, the WAIT instruction is satisfied:

```
'TRAP AND SUPPRESS MESSAGES MSG1, MSG2, MSG3'
'WAIT 20 SECONDS FOR MESSAGES'
'MSGREAD'
.
.
.
'WAIT CONTINUE'
'MSGREAD'
```

The WAIT CONTINUE instruction waits up to 15 seconds to receive one of the messages specified on the TRAP instruction before resuming command list processing. (Fifteen seconds is the difference between the 20 seconds specified on the WAIT instruction and the 5 seconds already used to satisfy the WAIT instruction.) When processing resumes, the MSGREAD instruction following WAIT CONTINUE is issued.

DATA (HLL only)

The command procedure waits for data sent by CNMSMSG with message type of DATA.

FOR

Can be used to make the WAIT instruction syntax more readable.

MESSAGES

Specifies the command list waits for a trapped message to be added to the message queue before resuming processing. You define the specific messages for which the command list waits using the TRAP instruction. If you also specify a time interval, the command list waits for up to that amount of time and then resumes, even if one of the specified messages is not received. If one of the specified messages is already on the message queue, the command list resumes without waiting. Refer to the TRAP instruction for more information.

MINUTES

Specifies the command list waits *n* minutes before resuming processing.

n

Specifies the number of SECONDS or MINUTES that the command procedure waits for receipt of the messages specified on the TRAP command, for receipt of operator input, or for receipt of data before processing is resumed. The command procedure can also wait only for the specified time (no specified events) before processing is resumed.

Valid numbers for *n* are in the range of 0–2678400 seconds and 0–44640 minutes. The equivalent of 2678400 seconds or 44640 minutes is 31 days.

OPINPUT (HLL only)

The command procedure waits for operator input. Data from the QUEUE or GO command can satisfy this wait.

SECONDS

Specifies the command list waits *n* seconds before resuming processing. This is the default.

Return codes

WAIT sets the value of the return code (RC) to indicate the results of processing.

For REXX:

Return Code

Meaning

-1

There was a DSIGET failure.

0

Processing completed successfully.

4

WAIT is valid only from HLL or REXX command lists.

8

There are too many operands.

12

A syntax error occurred.

144

Not in OST or NNT.

152

WAIT was issued without a previous TRAP.

248

WAIT CONTINUE was issued without a previous valid WAIT.

For HLL:

Return Code Name	Value	Description
CNM_BAD_INVOCATION	4	WAIT was not issued from an HLL or REXX command procedure.
CNM_TOO_MANY	8	Too many operands.
CNM_BAD_SYNTAX	12	Syntax error.
CNM_BAD_COMMAND	144	WAIT was not issued from a command procedure running under an OST or NNT.
CNM_NO_TRAP	152	WAIT was issued but TRAP was not issued. You must issue valid TRAP before issuing WAIT for messages.
CNM_NO_PREV_WAIT	248	No previous WAIT; WAIT CONTINUE is not valid.
CNM_STRG_FAILURE	616	DSIGET failure in a service routine.
CNM_TIME_OUT_WAIT	224	WAIT timed out.
CNM_GO_ON_WAIT	228	GO satisfied wait.
CNM_MSG_ON_WAIT	232	Message received during wait.
CNM_OPINPUT_ON_WAIT	236	OPINPUT received during wait.
CNM_DATA_ON_WAIT	240	DATA received during wait.

Note: Return codes 224–240 are issued when a wait is satisfied. They are generated in place of a 0 return code to inform the operator which event satisfied the wait.

Usage Notes

Consider the following when using the WAIT command:

- The WAIT command can be issued only from an HLL or REXX command procedure running under an OST or NNT.
- A REXX WAIT and an HLL WAIT have some syntactical differences.
- Unlike REXX, an HLL command procedure that issues the WAIT CONTINUE command can continue waiting for operator input and data. As shown in this example, operator input received within 12 seconds satisfies the first wait and the command procedure continues to wait for operator input for the remaining 18 seconds. The same is true when waiting for data.

```

CNMCOMMAND DATA('WAIT 30 SECONDS FOR OPINPUT');
/* Wait up to 30 seconds for operator input */
CNMGETDATA
FUNC(FLUSHQ) QUEUE(OPERQ);
/* Remove the operator input from the queue */
CNMCOMMAND DATA('WAIT CONTINUE');
/* Continue waiting for more operator input */

```

- A WAIT CONTINUE command satisfies the previous valid WAIT command. Here the command procedure continues to wait using the wait criteria from the initial WAIT command because the second WAIT command is not valid. If the initial WAIT command is satisfied after 3 seconds, the command procedure continues to wait for messages for the remaining 7 seconds (because the WAIT 20 SECONDS FOR DATAA was not valid).

```
CNMCOMMAND DATA('TRAP MESSAGES *');
/* Trap all messages */
CNMCOMMAND DATA('WAIT 10 SECONDS FOR MESSAGES');
/* Wait up to 10 seconds for the first trapped message */
CNMGETDATA FUNC(FLUSHQ) QUEUE(TRAPQ);
/* Remove messages from the message queue */
CNMCOMMAND DATA('WAIT 20 SECONDS FOR DATAA');
/* A not valid wait for data (misspelled data) */
CNMGETDATA FUNC(FLUSHQ) QUEUE(DATAQ);
/* Remove data from the data queue */
CNMCOMMAND DATA('WAIT CONTINUE');
/* Continue waiting for the next trapped message */
```

- CNMGETD must successfully complete (HLBRC = 0) before a WAIT CONTINUE processes correctly.
- Refer to the NetView online help for a full description of the REXX WAIT command.
- The NetView operator's screen is refreshed whenever a message arrives or the ENTER key is pressed. If the screen is refreshed while an HLL command procedure is in a wait state, the pause and wait status indicators (P and W) are displayed in the upper right corner of the current command facility panel. The W indicator notifies the operator that the command procedure has halted its processing and is waiting for one or more messages, data, or a specified period of time. The P indicator notifies the operator that the command procedure has halted its processing and that one of the events for which it is waiting is operator input. When the wait is satisfied, processing starts again and the indicators are cleared from the screen.
- Flush the operator input queue (FLUSHQ) after a WAIT command is satisfied with operator input. Otherwise, subsequent WAIT FOR OPINPUT commands are satisfied by data already in the operator input queue. Refer to the NetView online help for more information about FLUSHQ.
- When an HLL command processor is in a wait or pause state, operator commands that are entered can be deferred. The commands are deferred based on the NetView DEFAULTS, OVERRIDE, and CMD commands. Refer to the NetView online help for information about these commands. If a command is not deferred, and it specifies the same message number on a TRAP instruction as the waiting command, the message satisfies the WAIT in the nondeferred command.
- The WAIT command with no operands is not valid.
- The following example illustrates the use of the TRAP and WAIT commands. If the initial WAIT command is satisfied after five seconds, the WAIT CONTINUE command gets control and the command procedure continues to wait for the remainder of the time specified (15 seconds) in the previous WAIT command.

```
CNMCOMMAND DATA('TRAP MESSAGES MSG1 MSG2 MSG3');
/* Trap messages with tokens MSG1, MSG2, MSG3 */
CNMCOMMAND DATA('WAIT 20 SECONDS FOR MESSAGES');
/* Wait up to 20 seconds for the first trapped message */
CNMGETDATA FUNC(FLUSHQ) QUEUE(TRAPQ);
/* Remove messages from the message queue */
CNMCOMMAND DATA('WAIT CONTINUE');
/* Continue waiting for the next trapped message */
```

Restrictions

The following restrictions apply to the WAIT instruction:

- WAIT cannot be used for PPT tasks.
- You must enter the operands in the order shown in the syntax diagram.
- You must code a time interval, MESSAGES, or both.
- For REXX, the instruction is enclosed in single quotation marks to prevent variable substitution.

The following is an example of a REXX command list named ACTLU. ACTLU issues a VTAM command to activate a node specified by the user. Messages sent as the result of the activate command are trapped, and the operator is notified of the result of the command list. The comments in the example explain the command list:

Figure 15. Sample Command List Using TRAP, WAIT, MSGREAD, and WAIT CONTINUE

Refer to the *IBM Z NetView Programming: REXX and the NetView Command List Language* for more examples of REXX command lists for the NetView program.

Example: Checking the result of a WAIT instruction

The NetView event that satisfied the WAIT is determined by the value of the REXX EVENT() function. The REXX command list can use the EVENT() function to set a variable and take appropriate action based on the set value. The possible returned values from EVENT() are as follows:

Value

Meaning

M

The message for which the command list is waiting has arrived. The message can be read using the MSGREAD instruction.

T

The time period for which the command list was waiting has expired, and processing is resumed.

G

You entered the GO command, and processing is resumed.

E

You did not code the WAIT or TRAP instructions correctly. For example, you entered the operands in the incorrect order or issued a WAIT for messages instruction without a matching TRAP instruction. The command list resumes processing.

X

A PERSIST command using the TRAP option has expired. This code is received when the correlated commands managed by the PERSIST command have completed.

If you do not issue a WAIT instruction in a command list, the value of the EVENT() function is replaced with a value of null.

Example: Processing one message

The following is an example of a skeleton command list showing how you can use the WAIT instruction to process one message at a time:

```
/* Command List Comments - Include message ID and text */
'TRAP AND SUPPRESS MESSAGES ..... '
/* issue command here */
'WAIT nn SECONDS FOR MESSAGES'
SELECT
  WHEN (EVENT()='M') THEN
    DO
      'MSGREAD'
      /* process message */
    END
  WHEN (EVENT()='T') THEN
    /* process WAIT time-out */
  WHEN (EVENT()='E') THEN
    /* process WAIT error */
  WHEN (EVENT()='G') THEN
    /* process GO entered */
  OTHERWISE
    /* process anything else */
END
'TRAP NO MESSAGES'
'FLUSHQ MESSAGES'
```

Example: using NetView commands with WAIT

The following is an example of a skeleton command list that processes several messages:

```

/* REXX Command List Comments - Include message ID and text */
END_OF_WAIT = 'NO'
'TRAP AND SUPPRESS MESSAGES ..... '
/* issue command here */
IF RC = 0 THEN
DO
  'WAIT nn SECONDS FOR MESSAGES'
  DO WHILE END_OF_WAIT = 'NO'
    'MSGREAD'
    SELECT
      WHEN (EVENT()='M') THEN
      DO
        /* process messages */
        /* call another routine to process the messages */
        CALL PROCESS_MSG
      END
      WHEN (EVENT()='T') THEN
      DO
        /* process WAIT time-out */
        END_OF_WAIT = 'YES'
      END
      WHEN (EVENT()='E') THEN
      DO
        /* process WAIT error */
        END_OF_WAIT = 'YES'
      END
      WHEN (EVENT()='G') THEN
      DO
        /* process GO entered */
        END_OF_WAIT = 'YES'
      END
      OTHERWISE
        /* process anything else */
    END
  END
END
END

```

Figure 16. Using NetView Commands with WAIT

```

'TRAP NO MESSAGES'
'FLUSHQ MESSAGES'
PROCESS_MSG:
/*****
/* PROCESS_MSG ROUTINE
/* This skeleton routine can be used to identify what types
/* of messages can occur and select an action based on those
/* messages.
/*
/* This skeleton shows how to handle the following conditions:
/* (1) a message that you want to act upon
/* (see MSGID()='xxxxxxx')
/* (2) a message that you want to ignore
/* (see MSGID()='yyyyyyy')
/* (3) a message that you did not plan for in your logic
/* (see OTHERWISE statement)
/*
/* You might have multiples of the above conditions based on what
/* you're doing with the command list and the messages.
/*
*****/
SELECT
  WHEN (MSGID()='xxxxxxx') THEN
    DO
      /* this is a message that you want to act upon and then
      end the TRAP/WAIT for any further messages
      /* insert message processing here */
      END_OF_WAIT = 'YES'
    END
  WHEN (MSGID()='yyyyyyy') THEN
    DO
      /* this is a message that you want to ignore and have suppressed
      by the TRAP command. You only need to continue
      waiting for the next message...
      'WAIT CONTINUE'
    END
  OTHERWISE
    DO
      /* process any other message that is on the trapped message
      queue. You can choose one of 2 options:
      (1) continue the wait using a 'WAIT CONTINUE' statement
      (2) end the wait by coding END_OF_WAIT='YES'
      In either case, recommend issuing an error message to the
      task running this command list using the SAY command.
    END
END

```

When a REXX command list is in a wait or pause state, operator commands that are entered can be delayed. Whether the commands are delayed is based on the NetView DEFAULTS, OVERRIDE, and CMD commands. Refer to the NetView online help for information about these commands.

If a command is not delayed, and it specifies the same message number on a TRAP command as the waiting command list, the message satisfies the WAIT in the nondelayed command.

The GO, STACK, UNSTACK, and RESET commands affect the processing of command lists in a wait state as follows:

GO

Ends a WAIT.

STACK

Suspends command list processing and causes any commands that are later queued to be processed. You can enter any command or command list for normal processing while a command list is suspended. The WAIT is not suspended, and events are still matched as they occur. The W does not remain in the upper right corner of the NetView panel. The GO command is rejected until the command list resumes processing.

UNSTACK

Resumes command list processing. The WAIT resumes processing events that were matched while the command list was suspended.

RESET

Ends a command list and all related nested command lists. RESET also starts HALT when you code SIGNAL ON HALT.

For more information about the GO, STACK, UNSTACK, and RESET commands, refer to the NetView online help.

Sending messages to the z/OS console

Introduction

You can use the following NetView commands to send messages to, and remove messages from, one or more z/OS system consoles:

WTO (write-to-operator)

Send messages to one or more consoles.

WTOR (write-to-operator with reply)

Send messages to the operator console and wait for replies.

DOM

Remove WTO or NetView held messages. The DOM command can also be used to remove NetView held messages from NetView terminals, even when they are not MVS messages.

For REXX:

When a command list is started from a NetView automation table, specific system message information is available as returned values when their respective REXX message functions are started. This specific system message information is also assigned when an MSGREAD instruction is issued. These assignments occur after every MSGREAD.

The WTO and WTOR commands use the values of the message functions as input when the commands are processed. However, before issuing a WTO or WTOR command, you can override the system values with your own user-assigned values.

You cannot assign a value to a REXX function. You must assign a value to a variable with the same name as the function, but without the parentheses at the end. For example, an MSGREAD instruction reads a message which assigns MCSFLAG() a value of 10000100. To change the value before issuing a WTO command, use the following assignment statement to change the value of MCSFLAG:

```
MCSFLAG = '01000100'
```

Starting a REXX message function does not set the variable of the same name to a value. These REXX variables are null until a user assignment is made to them. They are not affected by their respective REXX message functions.

If you assign values to any variables that are input to the WTO or WTOR commands, the values you assign are used instead of system values of the REXX functions.

The MSGREAD instruction does not change the user-assigned variable values. If you want to return to the system values that MSGREAD assigns to the functions, use the REXX DROP instruction to drop the variables before issuing the WTO or WTOR command, for example, DROP MCSFLAG. Refer to the appropriate manual in the REXX library for information about the DROP instruction.

For NetView command list language:

When a command list is started from a NetView automation table, specific system message information is assigned to NetView command list language message control variables when the command list starts. This specific system message information is also assigned when an &WAIT control statement is satisfied within the command list. These assignments occur after every &WAIT.

The WTO and WTOR commands use the values of the control variables as input when the commands are processed. However, before issuing a WTO or WTOR command, you can override the system values with your own user-assigned values.

If you assign values to any variables that are input to the WTO or WTOR commands, the values you assign are used instead of system values of the control variables.

With the NetView command list language, you can use assignment statements to directly change the values of the user variables. For example, use the following assignment statement to change the value of &MCSFLAG:

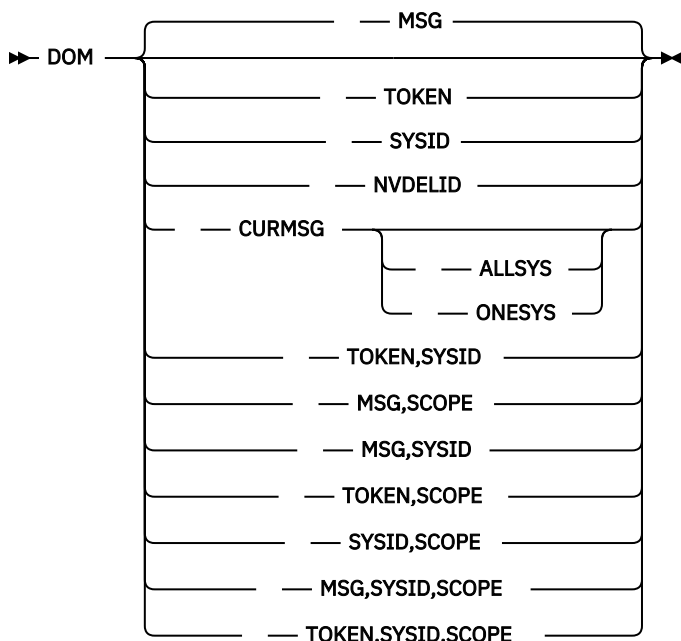
```
&MCSFLAG = 01000100
```

When you have assigned a user-defined value to the control variable, any subsequent references to the control variable substitutes the user-defined value. There is no mechanism to return to the system value within the same NetView command list language command list.

DOM (REXX)

Syntax

DOM



Purpose of Command

The delete operator message (DOM) command is used in a command list to remove a WTO or WTOR message from one or more MVS consoles, including MVS consoles that are in use by the NetView program. You can delete NetView held messages from NetView terminals using the NVDELID or CURMSG options. This is similar to the DOM function of MVS, and is useful for messages that were held by automation, and not issued from MVS as WTOs or WTORs.

Use the DOM command to remove action messages, or non-action messages marked with the DOMACTION(DELMMSG) or DOMACTION(AUTOMATE) automation action, after verifying that the action denoted by the message was completed. The DOM command uses the command options MSG, TOKEN, SYSID, SCOPE, NVDELID, or CURMSG to determine which messages must be removed. You can set a value for these variables as input to the DOM command, as shown for SMSGID in the following examples:

For REXX:

```
SMSGID = value
```

For command list:

```
&SMSGID = value
```

Parentheses are not used in either the NetView command list language or REXX versions.

If a value is not set for the variables that are used as input to the DOM command, the current system values are used by default.

For REXX, the current system values reside in the SMSGID(), MSGTOKEN(), MSGCSYID(), and NVDELID() functions.

For NetView command list language, the current system values reside in the &SMSGID, &MSGTOKEN, &MSGCSYID, and &NVDELID control variables. If your command list was started by automation or issued a WAIT that was satisfied by a message, then the current system value of REXX message functions and command list control variables such as SMSGID are derived from the driving message.

Operand Descriptions

CURMSG

The NetView deletion ID in the current message identifies which message is to be deleted. This option is useful for removing held messages that are not WTOs or WTORs. It is also useful in a REXX procedure when you want to delete the message that is currently being automated. For example, the message whose values are set by MSGREAD.

If no parameter is specified, the message is deleted, within the NetView program, without issuing an MVS system DOM signal. Optionally, you can specify one of the following parameters:

ALLSYS

Generates an MVS system DOM signal to all systems.

ONESYS

Generates an MVS system DOM signal to a single system.

DOM

When specified without parameters, DOM provides compatible function to DOM MSG.

MSG

The &SMSGID value determines the message to be deleted.

The &SCOPE value defaults to SYSTEMS.

MSG,SCOPE

The &SMSGID value determines the message to be deleted.

The &SCOPE value determines whether messages are deleted from only the issuing processor or all communicating MVS systems (JES3 only).

MSG,SYSID

The &SMSGID and &MSGCSYID values determine the message to be deleted.

The &SCOPE value defaults to SYSTEMS.

MSG,SYSID,SCOPE

The &SMSGID and &MSGCSYID values determine the message to be deleted.

The &SCOPE value determines whether messages are deleted from only the issuing processor or all communicating MVS systems (JES3 only).

NVDELID

The NetView deletion ID for the NVDELID REXX variable or &NVDELID NetView command list variable identifies which messages are deleted. This option provides a NetView console equivalent to the MVS DOM function. It is useful for removing held messages that are not WTOs or WTORs. The message is removed from all NetView tasks receiving the message. Use this option to delete messages for which the NVDELID value was saved, for example, in a global variable. This option does not use the MVS DOM function within the NetView program.

SYSID

All messages issued by system &MSGCSYID are deleted.

SYSID,SCOPE

All messages issued by system &MSGCSYID are deleted.

The &SCOPE value determines whether messages are deleted from only the issuing processor or from all communicating MVS systems (JES3 only).

TOKEN

The &MSGTOKEN value determines the message to be deleted.

The &SCOPE value defaults to SYSTEMS.

TOKEN,SCOPE

The &MSGTOKEN value determines the message to be deleted.

The &SCOPE value determines whether messages are deleted from only the issuing processor or all communicating MVS systems (JES3 only).

TOKEN,SYSID

The &MSGTOKEN and &MSGCSYID values determine the message to be deleted.

The &SCOPE value defaults to SYSTEMS.

TOKEN,SYSID,SCOPE

The &MSGTOKEN and &MSGCSYID values determine the message to be deleted.

The &SCOPE value determines whether messages are deleted from only the issuing processor or all communicating MVS systems (JES3 only).

Command List Variables

Table 11 on page 598 lists the DOM command variables:

Table 11. DOM Command List Variable Reference Table

DOM Macro Keyword	Function Variable Name	Usage	Applicable DOM command parameters
COUNT	Not required	Not used	None
DOMCBLK	Not required	Not used	None
TOKEN	MSGTOKEN() &MSGTOKEN	1- to 10-character decimal value, with a maximum value of 4294967295.	DOM TOKEN “1” on page 599 DOM TOKEN,SYSID DOM TOKEN,SCOPE “2” on page 599 DOM TOKEN,SYSID,SCOPE “2” on page 599
SCOPE “2” on page 599	(none)	Send DOM to SYSTEM or SYSTEMS	DOM MSG,SCOPE DOM TOKEN,SCOPE DOM SYSID,SCOPE DOM MSG,SYSID,SCOPE DOM TOKEN,SYSID,SCOPE
MSG MSGLIST “3” on page 599	SMSGID() &SMSGID	Value used on DOM of a message	DOM DOM MSG DOM MSG,SYSID DOM MSG,SCOPE “2” on page 599 DOM MSG,SYSID,SCOPE “2” on page 599
REPLY=YES	Not required	Not used	None

Table 11. DOM Command List Variable Reference Table (continued)

DOM Macro Keyword	Function Variable Name	Usage	Applicable DOM command parameters
SYSID	MSGCSYID() &MSGCSYID	MVS system identifier	DOM SYSID DOM MSG,SYSID DOM TOKEN,SYSID DOM SYSID,SCOPE ² on page 599 DOM MSG,SYSID,SCOPE DOM TOKEN,SYSID,SCOPE
(none)	NVDELID() &NVDELID	24-character values saved from a message (NetView deletion ID)	DOM NVDELID
(none)	(none)	NetView deletion ID for current message	DOM NVDELID

Table :

1. Do not use DOM by TOKEN with a TOKEN value of zero. See [“Usage Notes” on page 599](#) for more information.
2. JES3 only
3. Not required

Usage Notes

Consider the following when using the DOM command:

- There is neither a SCOPE() function nor an &SCOPE control variable.
- After you issue a WTO or WTOR command, the SMSGID value, corresponding to the message issued, is available in the SMSGID or &MSGID command list variables.

Consider the following when using DOM TOKEN:

- You can use TOKEN to group a series of WTOs and issue a DOM TOKEN to delete the group of messages. However, use caution when selecting the token value for the WTOs that are to be deleted with DOM TOKEN. If possible, plan the use of TOKENs so that the TOKEN used by your command procedure does not intersect with the TOKEN used by other command procedures in your system.
- WTOs issued without specifying TOKEN defaults to a value of zero (0). Do not use a DOM TOKEN with MSGTOKEN equal to zero because it can delete many messages that did not originate from your command processor.
- In a sysplex environment, a DOM with a TOKEN issued by one system in the complex is forwarded to all of the communicating MVS systems. If there is not a corresponding message to be deleted on the receiving systems (other than the system that issued the DOM), MVS might apply the DOM to a subsequent message that arrives with a valid TOKEN and ASID.

To ensure that token values are as close to unique as possible, they must be planned or generated. The example REXX EXEC shown in [Figure 17 on page 600](#) can be used to generate a TOKEN value from a store clock value. If you use the NetView command list language, you can call a REXX EXEC similar to this example from your command list to return the generated token value in a global variable.

```

/*****
/* Generate a 1-10 digit decimal number from the store clock value.*/
/*
/* This method can be used in REXX to generate suitable values
/* for TOKENs used in the WTO and DOM command processors.
/*
/*
NUMERIC DIGITS 20          /* STCKGMT converts to 20 digits
STCKVAL = STCKGMT()        /* Obtain the store clock value
NUMSTCK = C2D(STCKVAL)     /* Convert store clock to decimal
TOKVAL = NUMSTCK // 4294967295 /* Reduce to 4 byte number
SAY 'Example of value which could be used for token: ' TOKVAL
EXIT
*****/

```

Figure 17. REXX EXEC to Generate a Token Value from a Store Clock Value

- DOM TOKEN does not delete operator messages from the MVS subsystem allocatable consoles obtained by the NetView program if the subsystem interface is being used for MVS messages.
DOM TOKEN deletes operator messages from the EMCS consoles obtained by the NetView program.
- If DOM is issued without arguments and the current message is a NetView internal action message (IFRAUNVD), then the message is treated as for option CURMSG.

Return codes

The return code (RC or &RETCODE) indicates the processing results as follows:

Return Code

Meaning

- 4**
There is a syntax error.
- 8**
Storage is not available to continue processing.
- 100**
MSGID length is not valid.
- 104**
MSGID value is not valid.
- 108**
Not started from a command procedure.
- 112**
MSGTOKEN length is not valid.
- 116**
MSGTOKEN value is not valid.
- 120**
MSGCSYID length is not valid.
- 124**
MSGCSYID value is not valid.
- 128**
SCOPE length (JES3 only) is not valid.
- 132**
SCOPE value (JES3 only) is not valid.
- 136**
Specified DOM option is not valid.
- 140**
NVDELID length is not valid.
- 144**
NVDELID value is not valid.

148

NetView DOM routing failed.

152

There is no current message from which to copy NVDELID.

Example: Issuing DOM commands

The following example illustrates how to issue DOM commands:

```

/*****
/*  WTOEXAM  COMMAND LIST                                */
/*  -----                                */
/*
/*  FUNCTION: THIS NETVIEW REXX COMMAND LIST DEMONSTRATES THE ABILITY
/*            TO ISSUE WTO, WTOR, AND DOM COMMANDS.  IT SHOWS HOW
/*            TO CONTROL VARIOUS CONTROL VARIABLES THAT
/*            AFFECT ATTRIBUTES OF THE WTO.
/*
/*  STEPS:
/*    WTO A SERIES OF MESSAGES TO THE SYSTEM CONSOLE THAT HAVE
/*    DIFFERENT ATTRIBUTES, AS DEFINED IN MVS/XA SYSTEM
/*    MACROS AND FACILITIES VOLUME 2 UNDER THE
/*    WTO MACRO.
/*    SAVE THE MESSAGE IDS FOR THE WTO'ED MESSAGES.
/*    DELETE THE MESSAGES FROM THE CONSOLE.
/*    ISSUE AND RETRIEVE THE RESPONSE TO A WTOR COMMAND.
/*
/*****
/*****
/*  ISSUE SOME WTOS
/*****
ROUTCDE = '(1)'                                /* MASTER CONSOLE ACTION */
DESC = '(1)'                                  /* SYSTEM FAILURE        */
WTO.TEXT = 'This message is urgent!'          /* SET the text          */
'WTO'                                          /* ISSUE THE MESSAGE      */
/* Note: text is delivered as typed. No case change.
MSGID_FOR_FIRST_MESSAGE=MSGID                /* SAVE THE MESSAGE ID   */
ROUTCDE = '(2)'                                /* MASTER CONSOLE INFO   */
DESC = '(3)'                                  /* EVENTUAL ACTION REQUIRED */
'WTO This message can be handled later'       /* ISSUE THE MESSAGE      */
/* text is upper case, because ADDRESS NETVIEW is in effect.
MSGID_FOR_SECOND_MESSAGE=MSGID                /* SAVE THE MESSAGE ID   */
/*****
/*  ISSUE A MLWTO
/*****
ROUTCDE = '(2)'                                /* MASTER CONSOLE INFO   */
DESC = '(3)'                                  /* EVENTUAL ACTION REQUIRED */
LINETYPE='C'                                /* THIS IS A CONTROL LINE */
'WTO THE FIRST LINE OF A MLWTO'              /* ISSUE THE FIRST LINE   */
LINETYPE='D'                                /* THIS IS A DATA LINE   */
'WTO THE SECOND LINE OF A MLWTO'             /* ISSUE THE SECOND LINE  */
LINETYPE='DE'                               /* THIS IS THE LAST DATA LINE */
/*
'WTO THE THIRD LINE OF A MLWTO'              /* ISSUE THE THIRD LINE   */
MSGID_FOR_THIRD_MESSAGE=MSGID                /* SAVE THE MESSAGE ID   */
/*****
/*  DEMONSTRATE THE ABILITY TO DELETE OPERATOR MESSAGES (DOM)
/*****
'WAIT 5 SECONDS'                              /* LEAVE MESSAGE FOR 5 SECS */
MSGID=MSGID_FOR_FIRST_MESSAGE                /* RESTORE THE MESSAGE ID  */
'DOM'                                          /* DELETE THE MESSAGE      */
'WAIT 5 SECONDS'                              /* LEAVE MESSAGE FOR 5 SECS */
MSGID=MSGID_FOR_SECOND_MESSAGE               /* RESTORE THE MESSAGE ID  */
'DOM'                                          /* DELETE THE MESSAGE      */
'WAIT 5 SECONDS'                              /* LEAVE MESSAGE FOR 5 SECS */
MSGID=MSGID_FOR_THIRD_MESSAGE                /* RESTORE THE MESSAGE ID  */
'DOM'                                          /* DELETE THE MESSAGE      */
/*****
/*  ASK THE OPERATOR FOR INFORMATION VIA A WTOR
/*****
'WTOR ENTER SOME DATA:'                      /* ISSUE THE WTOR         */
SAY 'THE DATA ENTERED AT THE CONSOLE WAS: "WTOREPLY"'
EXIT

```

WTO (REXX)

Syntax

WTO



Purpose of Command

The write-to-operator (WTO) command sends a message to one or more MVS consoles.

Operand Descriptions

message_text

Specifies the message to send. This parameter is required unless WTO.TEXT is specified.

Command List Variables

Many other parameters and controls for the WTO are specified by setting local variables in your program. Some, but not all, of these parameters inherit values from the current message, if any. [Table 12 on page 602](#) shows these variables, the equivalent WTO macro keywords, and whether the value is inherited from the current message. In cases where there is both a REXX stem (from the WTO. root) and another variable name, if the REXX stem has a value, it is used in preference to the other variable. In many cases, the variable and REXX function have equivalents in the NetView Command List Language.

Refer to the *IBM Z NetView Programming: REXX and the NetView Command List Language* for more information about the NetView command list language control variables and REXX functions. Refer to the MVS library for more information about the MVS WTO macro.

In [Table 12 on page 602](#), the WTO command uses the values of the REXX functions or NetView command list language control variables as input. To override the REXX function value, set a REXX variable with the same name as the function, but without using the parentheses.

Table 12. WTO REXX Variable Reference Table

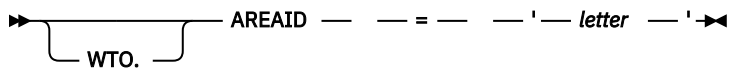
Variable Name	WTO Macro Keyword	Inheritance	Usage
WTO.AREAID AREAID	AREAID	Yes	One-letter console panel area.
WTO.CART CART	CART	Yes	8-byte command and response token (CART)
WTO.DESC DESC	DESC	Yes	Descriptor codes
WTO.JOBNAME	JOBNAME*	No	MVS job name
WTO.JOBNUM	JOBID*	No	MVS job ID
WTO.KEY KEY	KEY	Yes	Eight-byte character value. Use only printable characters and do not include commas or embedded blanks and do not use a single asterisk.

Table 12. WTO REXX Variable Reference Table (continued)			
Variable Name	WTO Macro Keyword	Inheritance	Usage
WTO.LINETYPE LINETYPE	LINETYPE	Yes	One- or two-byte MLWTO line type C, D, DE, E, H, L, M, or a blank
WTO.MCSFLAG MCSFLAG	MCSFLAG	Yes	Multiple console support flags
WTO.MRT	N/A	Yes	MRT message processing
WTO.MSGTOKEN MSGTOKEN	TOKEN	Yes	Decimal token, also used with DOM
WTO.MSGTYP MSGTYP	MSGTYP*	Yes	SESS, JOBNAMES, STATUS routing
WTO.ROUTCDE ROUTCDE	ROUTCDE	Yes	Routing codes
WTO.SMSGID SMSGID	CONNECT*	Yes	Value used to connect a multiline message and a DOM of an action message. In the NetView program, specify the correct linetype to indicate that you want to build an MLWTO.
WTO.CONSNAM SYSCONID	CONSNAM	Yes	Target console name
WTO.TEXT	TEXT	No	Message text
Note: - Macro keywords marked with an asterisk are documented only in the <i>MVS Programming: Authorized Assembler Services Reference</i>. - In all cases except WTO.CONSNAM and WTO.TEXT, there is a REXX function whose name is the same as the stem name. Use SYSCONID() to obtain the current console name. Use PIPE EDIT to obtain MRT and text. See the <i>IBM Z NetView Programming: REXX and the NetView Command List Language</i> for function descriptions. - Inheritance occurs even for messages originating at a remote system. Use care when issuing a WTO command using these inherited values, as they might not be valid on the system on which the message is being issued. - Values for any variables that indicate an Inheritance value of Yes are retrieved from the current message. The current message can be set as follows: - By automation table processing when control is passed to a command that is specified on the CMD parameter in an EXEC statement - By PIPE processing when control is passed to a command that is specified on a NETVIEW stage that has an input stream Be aware of the attributes that might be associated with the current message and how the inherited values can affect the outcome of the WTO command.			

AREAID, WTO.AREAID

Is a one-letter (A - J, Z) identifier for the area on the console panel that displays the message.

AREAID

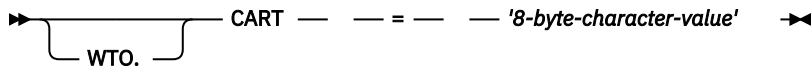


Note: MVS respects the AREAID specification only for MLWTOs on multiple console support consoles.

CART, WTO.CART

Is an eight-character command and response token (CART) represented by 8 bytes of hexadecimal or EBCDIC characters in the following form:

CART



The following example uses the hexadecimal notation:

```
WTO.CART = 'C1C2C3C4F0F1F2F3'X
```

The following example uses EBCDIC character notation:

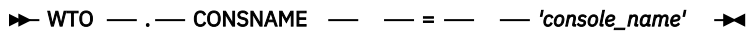
```
WTO.CART = 'ABCD1234'
```

Note: All 8 bytes must be specified.

WTO.CONSDNAME

The destination console name for the WTO.

CONSDNAME



Usage notes:

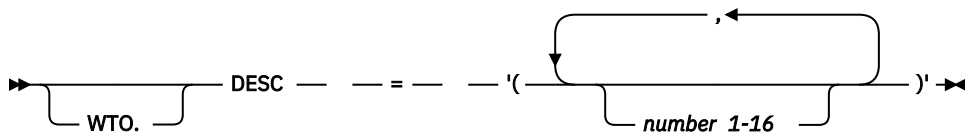
1. The console name you specify must be active. If it is not, a "nonvalid CONSDNAME" return code is received.
2. MVS reserves some console names. Currently, the following names are reserved:
 - HC
 - INSTREAM
 - INTERNAL
 - INTIDS
 - UNKNOWN

These reserved names cannot be used as CONSDNAME values for input to the WTO command.

DESC, WTO.DESC

The descriptor codes for the WTO, specified as a list of numbers in parentheses:

DESC



You can also specify specific numbers and a range of numbers, for example:

```
WTO.DESC = '(3,5-7)'
```

Usage notes:

1. For compatibility with previous releases of the NetView program, DESC can also be specified as a string of ones and zeros. For example, to turn on descriptor codes 3, 5, and 7:

```
DESC = '001010100000'
```

When specifying a string of ones and zeros, omit the opening and closing parentheses.

2. In general, do not use descriptor codes that are reserved in MVS. Descriptor code 13 is used by the NetView program to flag messages that have already been automated on the z/OS system.

Setting descriptor code 13 causes your messages to be ineligible for automation or display by the NetView program on your system. In a sysplex, however, these messages might still be eligible for automation and display on the other systems.

WTO.JOBNAME

The MVS job name from which it appears that the WTO has been issued.

JOBNAME

➤ WTO — . — JOBNAME — — = — — 'job_name' ➤

Note: If WTO.JOBNAME is set to a value of blanks, the resulting job name is blank. If WTO.JOBNAME is not set or is set to null, the value of the NetView job name is used..

WTO.JOBNUM

The MVS job number of the job from which the WTO appears to have been issued.

JOBNUM

➤ WTO — . — JOBNUM — — = — — 'job_number' ➤

Note: If WTO.JOBNUM is set to a value of blanks, the resulting job number is blank. If WTO.JOBNUM is not set or is set to null, the value of the NetView job number is used.

KEY, WTO.KEY

Specifies a retrieval key represented by 8 bytes of hexadecimal or EBCDIC characters in the following form:

KEY

➤ ————— KEY — — = — — '8-byte-character-value' ➤
 WTO.

The following example uses the hexadecimal notation:

```
WTO.KEY = 'C1C2C3C4F0F1F2F3'X
```

The following example uses character notation:

```
KEY = 'ABCD1234'
```

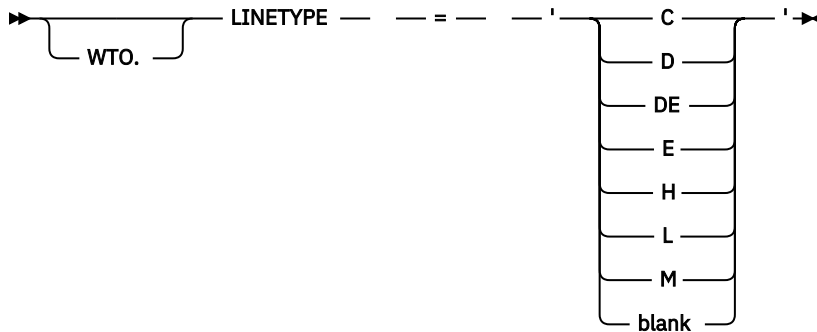
Usage notes:

1. All 8 bytes must be specified.
2. To enable use of the MVS display R with a value for the KEY keyword, use only printable characters and do not include commas or embedded blanks and do not use a single asterisk.

LINETYPE, WTO.LINETYPE

Provides the multiline write-to-operator (MLWTO) line type in the following form:

LINETYPE



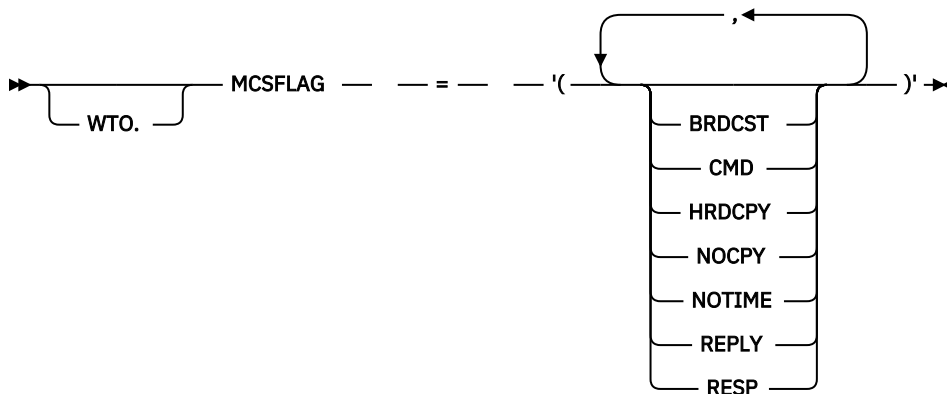
Usage notes:

1. When you use the WTO command to issue multiline messages, assign LINETYPE and issue a WTO command for each line.
2. Issue the WTO commands for all lines before the command list completes execution. The NetView program saves the lines until an end line is issued, or purges an unfinished multiline WTO when the command list ends. This helps prevent console hangs because of incomplete multiline WTOs.
3. If the M or H value is inherited or set in the LINETYPE variable, it is treated the same as a blank LINETYPE value (a single-line message type).
4. If the current message is an MLWTO message and the LINETYPE variable is not set, the inherited LINETYPE value is that of the first line in the MLWTO message. Thus, the first WTO message that is issued using the inherited LINETYPE value starts a new MLWTO message. Because a separate WTO command is required for each line of a message, use care when issuing a WTO command with an inherited LINETYPE value, as a WTO command with a LINETYPE value of E or DE is needed to end a MLWTO message.

MCSFLAG, WTO.MCSFLAG

The MCSFLAGs to be associated with the WTO, specified as a list of keywords in parentheses:

MCSFLAG



Usage notes:

1. More than one keyword can be specified. The delimiter is a comma or blank.
2. For compatibility with previous releases of NetView, MCSFLAG can also be specified as a string of ones and zeros.

For example, to set MCSFLAG to NOCPY:

```
MCSFLAG = '00000001'
```

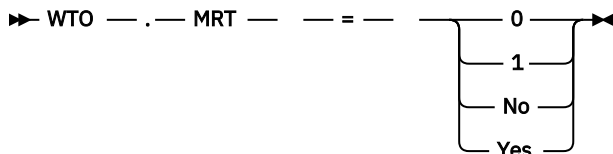
Note: The QREG0 and REG0 bits and keywords are ignored.

When specifying a string of ones and zeros, omit the opening and closing parentheses.

WTO.MRT

Signifies whether the message is considered by the Z NetView program to have been previously processed by the Message Revision Table (MRT). If this variable is not set, the current message, if any, is examined to determine if MRT processing has already occurred for it. If there is a current message, and it has been processed by an MRT, then the default is to bypass MRT processing for the message again. Otherwise, the default is to allow MRT processing. To determine the state of the current message regarding MRT processing, use the MRT input order on an EDIT PIPE stage. To indicate that the message to be issued has already been processed by the MRT, and therefore should not be processed by the MRT again, set the WTO.MRT variable to 1 or Yes. To indicate that the message to be issued is to be considered a new message which has not been processed by the MRT, and therefore should be processed by the MRT, set the WTO.MRT variable to 0 or No.

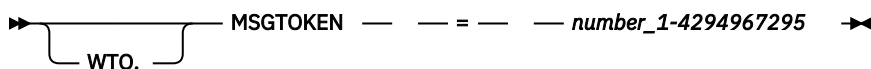
MRT



MSGTOKEN, WTO.MSGTOKEN

Is a numeric token associated with the WTO in the following form:

MSGTOKEN

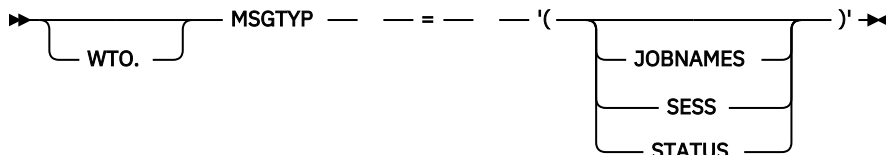


This token can be used to delete the operator message using the DOM command with the TOKEN option. See [“Usage Notes” on page 599](#) for more information.

MSGTYP, WTO.MSGTYP

The system message type flags for the WTO, specified as a list of keywords in parentheses:

MSGTYP



Usage notes:

1. More than one keyword can be specified. The delimiter is a comma or blank.
2. For compatibility with previous releases of NetView, MSGTYP can also be specified as a string of ones and zeros.

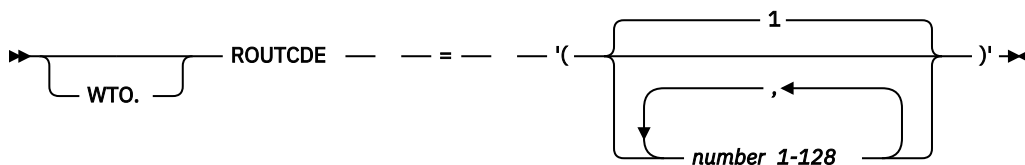
For example:

```
MSGTYP = '010'
```

ROUTCDE, WTO.ROUTCDE

The system routing codes:

ROUTCDE



The default routing code provided by the NetView program is 1 if the following conditions are true:

- No route code is set or inherited.
- No console name set or inherited.

If specified here, it is specified as a list of numbers in parentheses.

You can also specify numbers and a range of numbers, for example:

```
WTO.ROUTCDE = '(2,4-7)'
```

For compatibility with previous releases of the NetView program, ROUTCDE can also be specified as a string of ones and zeros. For example, to turn on codes 3, 5, and 7:

```
WTO.ROUTCDE = '001010100000'
```

When specifying a string of ones and zeros, omit the opening and closing parentheses.

SYSCONID

The destination console name for the WTO.

SYSCONID

➤ SYSCONID — — = — — 'console_name' ➤

Usage notes:

1. The console name you specify must be active. If it is not, a "nonvalid SYSCONID" return code is received.
2. MVS reserves some console names. Currently, the following names are reserved:
 - HC
 - INSTREAM
 - INTERNAL
 - UNKNOWN

These reserved names cannot be used as SYSCONID values for input to the WTO command.

WTO.TEXT

If the *message_text* is not supplied with the WTO command invocation, then the command looks for the message text in the WTO.TEXT variable.

TEXT

➤ WTO — . — TEXT — — = — — 'message_text' ➤

Usage Notes

Consider the following when using the WTO command:

- The values of these variables determine how the WTO command is processed. The variables provide the same input as the keywords on an MVS WTO macro. If a command list does not set the variables before issuing the WTO command, their values default to the current system values.
- For REXX command lists, the current system values are contained in the functions that correspond to the variable names. For example, the current system value for the REXX SYSCONID variable is contained in the SYSCONID() function.
- The WTO command returns values in the following REXX variables:
 - RC
 - SMSGID
- The WTO command returns values in the following NetView command list language control variables:

- RC, &RETCODE
- MSGID, &MSGID

Return codes

The return code (RC) indicates the processing results as follows:

Return Code	Meaning
4	WTO was sent with truncated text.
8	Storage is not available to continue processing.
16	Internal processing failed.
100	AREAID is not valid.
112	SYSCONID length was not valid.
116	SYSCONID value is not valid, or the console specified in SYSCONID is not currently active.
120	Internal decimal conversion failed.
124	The command list dictionary update failed.
132	The command is not allowed under PPT.
136	LINETYPE is not valid.
138	Not started from a command procedure.
140	KEY length is not valid.
148	MSGTOKEN length is not valid.
152	MSGTOKEN value is not valid.
156	CART length is not valid.
164	ROUTCDE value is not valid.
168	MSGTYP value is not valid.
172	MCSFLAG value is not valid.
176	DESC value is not valid.
180	JOBNAME value is not valid.

A return code with a value greater than 200 indicates that the return code was passed from the MVS WTO macro. Subtract 200 from the value of the return code. The new value corresponds to the return code that

was passed from the MVS WTO macro. For example, if you receive a return code of 208, refer to the MVS library for the meaning of return code 8 from the MVS WTO macro.

Example: Issuing WTO commands

For an example on how to issue WTO commands, see [“Example: Issuing DOM commands” on page 601.](#)

WTOR (REXX)

Syntax

WTOR

➔ WTOR 

Purpose of Command

The write-to-operator with reply (WTOR) command sends a message to one or more MVS consoles and requests a reply. Command lists that use the WTOR command do not complete until an operator replies. Therefore, use WTOR with care. Because command list processing is suspended while waiting for the reply from the WTOR, other command lists, including command lists that are run with a low priority, can be scheduled and run before the original command list completes.

When the operator replies to the WTOR message, the reply text is stored in the WTO.REPLY REXX variable (or &WTOREPLY if the command list is written in the NetView command list language). Also, the ID of the system console that replied is stored in the SYSCONID REXX variable (or &SYSCONID for the NetView command list language) unless the ID of the replying console is a reserved console name, in which case the SYSCONID variable is not altered from its previous value. Because SYSCONID is changed in this way, any subsequent WTO or WTOR command that is issued by the same command list is issued to the console that replied to the WTOR, unless SYSCONID is altered by the command list after it is modified by the results of the WTOR.

Operand Descriptions

message_text

Specifies the message to send. This parameter is required unless WTO.TEXT is specified.

Command List Variables

Many other parameters and controls for the WTOR are specified by setting local variables in your program. Some, but not all, of these parameters inherit values from the current message, if any. [Table 13 on page 611](#) shows these variables, the equivalent WTOR macro keywords, and whether the value is inherited from the current message. In cases where there is both a REXX stem (from the WTOR . root) and another variable name, if the REXX stem has a value, it is used in preference to the other variable. In many cases, the variable and REXX function have equivalents in the NetView Command List Language.

Refer to the *IBM Z NetView Programming: REXX and the NetView Command List Language* for more information about the NetView command list language control variables and REXX functions. Refer to the MVS library for more information about the MVS WTOR macro.

In [Table 13 on page 611](#), the WTOR command uses the values of the REXX functions or NetView command list language control variables (respectively) as input. To override a REXX function, set a REXX variable with the same name as the function, but without the parentheses.

Table 13. WTOR REXX Variable Reference Table

Variable Name	WTOR Macro Keyword	Inheritance	Usage
WTO.CART CART	CART	Yes	8-byte command and response token (CART)
WTO.DESC DESC	DESC	Yes	Descriptor codes
WTO.JOBNAME	JOBNAME*	No	MVS job name
WTO.JOBNUM	JOBID*	No	MVS job ID
WTO.KEY KEY	KEY	Yes	Eight-byte character value. Use only printable characters and do not include commas or embedded blanks and do not use a single asterisk.
WTO.MCSFLAG MCSFLAG	MCSFLAG	Yes	Multiple console support flags
WTO.MRT	N/A	Yes	MRT message processing
WTO.MSGTOKEN MSGTOKEN	TOKEN	Yes	Decimal token, also used with DOM
WTO.MSGTYP MSGTYP	MSGTYP*	Yes	SESS, JOBNAMEs, STATUS routing
WTO.ROUTCDE ROUTCDE	ROUTCDE	Yes	Routing codes
WTO.CONsNAME sYsCONID	CONsNAME	Yes	Target console name
WTO.TEXT	TEXT	No	Message text
WTO.REPLY	N/A	N/A	Variable that returns an operator's reply to a WTOR

Table 13. WTOR REXX Variable Reference Table (continued)

Variable Name	WTOR Macro Keyword	Inheritance	Usage
Note: - Macro keywords marked with an asterisk are documented only in the <i>MVS Programming: Authorized Assembler Services Reference</i>. - In all cases except WTO.CONSNAM and WTO.TEXT, there is a REXX function whose name is the same as the stem name. Use SYSCONID() to obtain the current console name. Use PIPE EDIT to obtain MRT and text. See the <i>IBM Z NetView Programming: REXX and the NetView Command List Language</i> for function descriptions. - Inheritance occurs even for messages originating at a remote system. Use care when issuing a WTOR command using these inherited values, as they might not be valid on the system on which the message is being issued. - Values for any variables that indicate an Inheritance value of Yes are retrieved from the current message. The current message can be set as follows: - By automation table processing when control is passed to a command that is specified on the CMD parameter in an EXEC statement - By PIPE processing when control is passed to a command that is specified on a NETVIEW stage that has an input stream Be aware of the attributes that might be associated with the current message and how the inherited values can affect the outcome of the WTO command.			

CART, WTO.CART

Is an eight-character command and response token (CART) represented by 8 bytes of hexadecimal or EBCDIC characters in the following form:

CART

The following example uses the hexadecimal notation:

```
WTO.CART = 'C1C2C3C4F0F1F2F3'X
```

The following example uses EBCDIC character notation:

```
WTO.CART = 'ABCD1234'
```

Note: All 8 bytes must be specified.

WTO.CONSNAM

The destination console name for the WTOR.

CONSNAM

Usage notes:

- The console name you specify must be active. If it is not, a "nonvalid CONSNAM" return code is received.
- MVS reserves some console names. Currently, the following names are reserved:
 - HC
 - INSTREAM

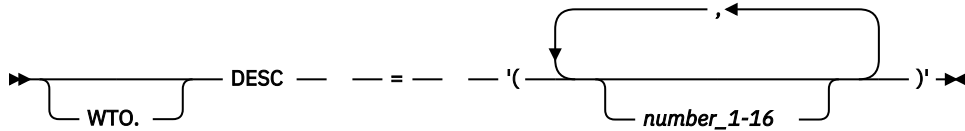
- INTERNAL
- UNKNOWN

These reserved names cannot be used as CONSNAMES for input to the WTOR command.

DESC, WTO.DESC

The descriptor codes for the WTOR, specified as a list of numbers in parentheses:

DESC



You can also specify specific numbers and a range of numbers, for example:

```
WTO.DESC = '(3,5-7)'
```

Usage notes:

1. For compatibility with previous releases of the NetView program, DESC can also be specified as a string of ones and zeros. For example, to turn on descriptor codes 3, 5, and 7:

```
DESC = '001010100000'
```

When specifying a string of ones and zeros, omit the opening and closing parentheses.

2. In general, do not use descriptor codes that are reserved in MVS. Descriptor code 13 is used by the NetView program to flag messages that have already been automated on the z/OS system.

Setting descriptor code 13 causes your messages to be ineligible for automation or display by the NetView program on your system. In a sysplex, however, these messages might still be eligible for automation and display on the other systems.

WTO.JOBNAME

The MVS job name from which it appears that the WTOR has been issued.

JOBNAME

```
WTO . JOBNAME = 'job_name'
```

Note: If WTO.JOBNAME is set to a value of blanks, the resulting job name is blank. If WTO.JOBNAME is not set or is set to null, the value of the NetView job name is used.

WTO.JOBNUM

The MVS job number of the job from which the WTOR appears to have been issued.

JOBNUM

```
WTO . JOBNUM = 'job_number'
```

Note: If WTO.JOBNUM is set to a value of blanks, the resulting job number is blank. If WTO.JOBNUM is not set or is set to null, the value of the NetView job number is used.

KEY, WTO.KEY

Specifies a retrieval key represented by 8 bytes of hexadecimal or EBCDIC characters in the following form:

KEY

```
WTO. KEY = '8-byte-character-value'
```

The following example uses the hexadecimal notation:

```
WTO.KEY = 'C1C2C3C4F0F1F2F3'X
```

The following example uses character notation:

```
KEY = 'ABCD1234'
```

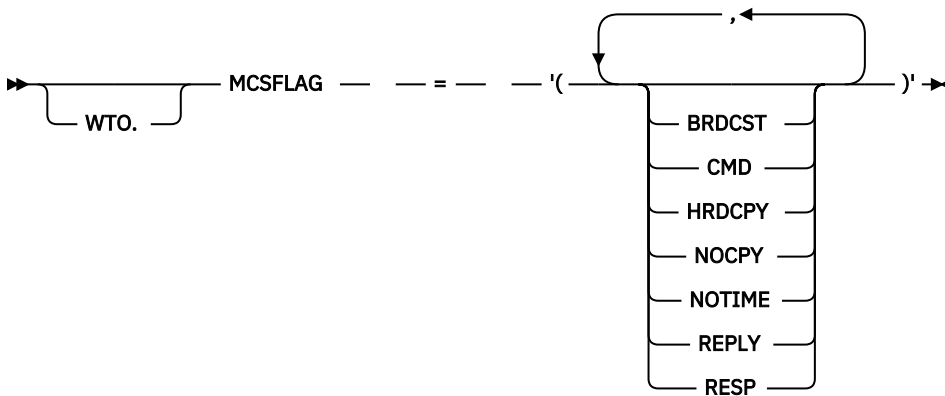
Usage notes:

1. All 8 bytes must be specified.
2. To enable use of the MVS display R with a value for the KEY keyword, use only printable characters and do not include commas or embedded blanks and do not use a single asterisk.

MCSFLAG, WTO.MCSFLAG

The MCSFLAGS to be associated with the WTOR, specified as a list of keywords in parentheses:

MCSFLAG



Usage notes:

1. More than one keyword can be specified. The delimiter is a comma or blank.
2. For compatibility with previous releases of NetView, MCSFLAG can also be specified as a string of ones and zeros.

For example, to set MCSFLAG to NOCPY:

```
MCSFLAG = '00000001'
```

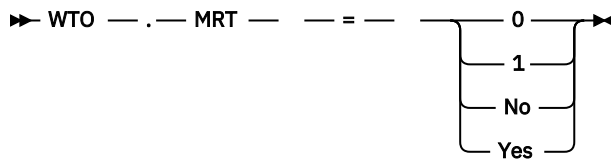
Note: The QREG0 and REG0 bits and keywords are ignored.

When specifying a string of ones and zeros, omit the opening and closing parentheses.

WTO.MRT

Signifies whether the message is considered by the Z NetView program to have been previously processed by the Message Revision Table (MRT). If this variable is not set, the current message, if any, is examined to determine if MRT processing has already occurred for it. If there is a current message, and it has been processed by an MRT, then the default is to bypass MRT processing for the message again. Otherwise, the default is to allow MRT processing. To determine the state of the current message regarding MRT processing, use the MRT input order on an EDIT PIPE stage. To indicate that the message to be issued has already been processed by the MRT, and therefore should not be processed by the MRT again, set the WTO.MRT variable to 1 or Yes. To indicate that the message to be issued is to be considered a new message which has not been processed by the MRT, and therefore should be processed by the MRT, set the WTO.MRT variable to 0 or No.

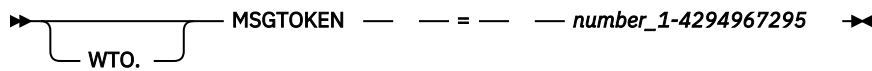
MRT



MSGTOKEN, WTO.MSGTOKEN

Is a numeric token associated with the WTOR in the following form:

MSGTOKEN

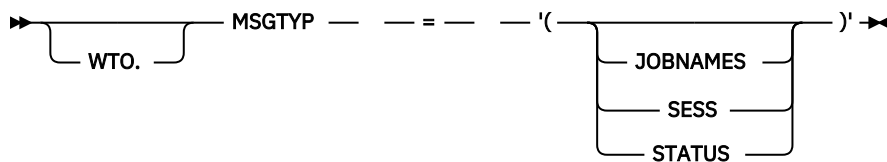


This token can be used to delete the operator message using the DOM command with the TOKEN option. See [“Usage Notes” on page 599](#) for more information.

MSGTYP, WTO.MSGTYP

The system message type flags for the WTOR, specified as a list of keywords in parentheses:

MSGTYP



Usage notes:

1. More than one keyword can be specified. The delimiter is a comma or blank.
2. For compatibility with previous releases of the NetView program, MSGTYP can also be specified as a string of ones and zeros.

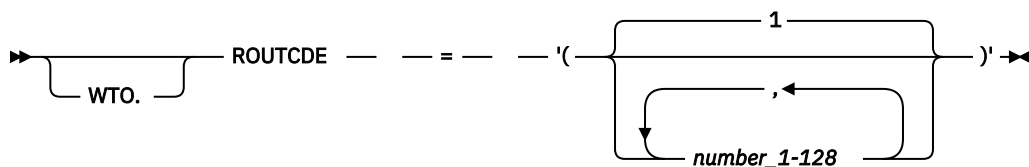
For example:

```
MSGTYP = '010'
```

ROUTCDE, WTO.ROUTCDE

The system routing codes:

ROUTCDE



The default routing code provided by the NetView program is 1 if the following conditions are true:

- No route code is set or inherited.
- No console name is set or inherited.

If specified here, it is specified as a list of numbers in parentheses.

You can also specify specific numbers and a range of numbers, for example:

```
WTO.ROUTCDE = ' (2,4-7) '
```

For compatibility with previous releases of the NetView program, ROUTCDE can also be specified as a string of ones and zeros. For example, to turn on codes 3, 5, and 7:

```
WTO.ROUTCDE = '001010100000'
```

When specifying a string of ones and zeros, omit the opening and closing parentheses.

SYSCONID

The destination console name for the WTOR.

SYSCONID

➤ SYSCONID — — = — — 'console_name' ➤

Usage notes:

1. The console name you specify must be active. If it is not, a "nonvalid SYSCONID" return code is received.
2. MVS reserves some console names. Currently, the following names are reserved:
 - HC
 - INSTREAM
 - INTERNAL
 - UNKNOWN

These reserved names cannot be used as SYSCONID values for input to the WTOR command.

WTO.TEXT

If the *message_text* is not supplied with the WTOR command invocation, then the command looks for the message text in the WTO.TEXT variable.

TEXT

➤ WTO — . — TEXT — — = — — 'message_text' ➤

Usage Notes

Consider the following when using the WTOR command:

- The values of these variables determine how the WTOR command is processed. The variables provide the same input as the keywords on an MVS WTOR macro. If a command list does not set the variables before issuing the WTOR command, their values default to the current system values.
- For REXX command lists, the current system values are contained in the functions that correspond to the variable names. For example, the current system value for the REXX SYSCONID variable is contained in the SYSCONID() function.
- The WTOR command returns values in the following REXX variables:
 - RC
 - WTO.SMSGID
 - WTO.CONSNAM or SYSCONID, depending on which variable you set.

Note: Beginning with z/OS v1r8, use WTO.CONSNAM.

 - WTO.REPLY
- The WTOR command returns values in the following NetView command list language control variables:
 - &RETCODE
 - &SMSGID
 - &SYSCONID
 - &WTOREPLY

Return codes

The return code (RC or &RETCODE) indicates the processing results as follows:

Return Code	Meaning
-------------	---------

4	The message was sent with truncated text.
8	Storage is not available to continue processing.
16	Internal processing failed.
100	There was no message text, or the command was running under a PPT.
104	SYSCONID length was not valid.
108	SYSCONID value is not valid, or the console specified in SYSCONID is not currently active.
112	The task is posted to end.
116	The WTO.REPLY command list dictionary update failed.
120	An internal decimal conversion failed.
124	The command list dictionary update failed.
138	The command was not started from a command procedure.
140	KEY length is not valid.
148	MSGTOKEN length is not valid.
152	MSGTOKEN value is not valid.
156	CART length is not valid.
164	ROUTCDE value is not valid.
168	MSGTYP value is not valid.
172	MCSFLAG value is not valid.
176	DESC value is not valid.
180	JOBNAME value is not valid.

A return code with a value greater than 200 indicates that the return code was passed from the MVS WTOR macro. Subtract 200 from the value of the return code. The new value corresponds to the return code that was passed from the MVS WTOR macro. For example, if you receive a return code of 208, refer to the MVS library for the meaning of return code 8 from the MVS WTOR macro.

Example: Issuing WTOR commands

For an example on how to issue WTOR commands, see [“Example: Issuing DOM commands” on page 601](#).

Appendix B. Event Types

Alerts are events (including resolutions) that require attention. As soon as an event record is created, the hardware monitor checks the current state of its recording filters to see if this event qualifies for alert status.

Events are classified by type. [Table 14 on page 619](#) provides a list of event types and their corresponding abbreviations and codes.

Table 14. Event Types with Abbreviations and Codes

Abbr.	Event Type	Description	Code
AVAL	Availability	The availability status of the reported resource has changed.	09
BYPS	Alert bypass	A loss of availability has been circumvented to allow the resource or an alternative resource to be used. The original problem still exists and you might not notice recovery. The recovery can be accomplished by intervention, either internal or external to the reporting product.	14
CUST	Customer application generated	A program that does not have an IBM order number generated the problem record.	05
DLRC	Delayed recovery	The sender is reporting a previously detected alertable condition that prevented reporting when detected, or the sender is reporting recovery from a condition that occurred earlier.	0F
ENV	Environment	A physical environmental problem has occurred.	0B
HELD	Held alert flag	An error condition was detected earlier, but the record was not sent at the time because there is no session available to send it. In filtering, the hardware monitor treats the HELD flag as if it was a second alert or event type. This means that a HELD flag is always associated with another event type. The HELD event type has the same filter priority as all other event types.	--
IMPD	Impending problem	Availability to the user is about to be lost.	11
IMR	Intensive Mode Recording	An error record resulted from the user invoking intensive mode recording, a feature of the NCP. When IMR is invoked, an error record is generated each time the NCP goes through an error retry.	08
INST	Installation	A system definition or an incompatibility between components has been reported.	0C
INTV	Intervention required	Intervention of a human operator is needed for corrective action.	04

Table 14. Event Types with Abbreviations and Codes (continued)

Abbr.	Event Type	Description	Code
NTFY	Notification of status change	Availability to the user is about to be lost. An important change of component or system/network status requiring operator notification is required.	0A
PAFF	Permanently affected resource	The originator of this alert has determined that the target resource is lost because of a persistent error in a resource other than the target.	10
PERF	Performance	A recognized measurement of performance, such as response time, has exceeded a determined threshold.	03
PERM	Permanent error	Availability to the user is lost unless there is external intervention to the reporting product.	01
PROC	Operation/procedure	A requested function cannot be performed because of an operational or procedural error.	0D
REDL	Redundancy lost	Redundant hardware or software is provided to ensure continued operation in the event of a failure or malfunction. As a result, failure of the remaining operational hardware or software results in a loss of corresponding services.	15
RSLV	Resolve major vector	The resolve major vector provides notification of the resolution of a previously reported problem. It contains an identification of the type of problem resolution and an identification of the failing resource.	--
RSNT	Resent alert flag	The alert has been resent, providing additional information about the original problem. In filtering, the hardware monitor treats the resent flag as if it were a second alert or event type.	--
SCUR	Security	A report of an incident that can indicate a possible security violation has been detected.	0E
SNA	SNA summary	A record containing SNA summary error counters. The record is normally the result of a NetView hardware monitor solicitation.	07
TEMP	Temporary or recoverable error	A momentary loss of availability is noticeable by the user, but is recovered from without intervention external to the reporting product.	02
USER	End user generated	A problem record initiated by a terminal operator.	06
UNKN	Unknown	The severity of the alert cannot be assessed.	12

Table 14. Event Types with Abbreviations and Codes (continued)

Abbr.	Event Type	Description	Code
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Note: BYPS, IMPD, PAFF, PERF, PERM, REDL, and TEMP are supported as part of the generic alert architecture.

In certain instances, the definitions of alert or event types used by nongeneric alert records differ from the current architected generic definitions.

You can use event types in filter-setting commands to control the types of data recorded in the hardware monitor's database or viewed by a NetView operator. See the SRFILTER and SVFILTER commands for more information about filter setting.

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